

Time Reversal of High-Frequency Pressure Waves for Defect Detection in Water Pipelines

by

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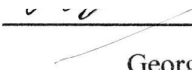
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*“As you set out for Ithaka
hope your road is a long one,
full of adventure, full of discovery.”*

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Georgios GRIGOROPOULOS

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Time Reversal of High-Frequency Pressure Waves for Defect Detection in Water Pipelines

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Abstract

Time reversal (TR) of waves is an intriguing wave property that has been found to successfully address the inverse problem in other scientific fields, such as physics and medicine, and deliver ground-breaking applications. Waves contain information about their sources and the media through which they propagate. Thus, TR of measured wave signals has the potential of localizing and characterizing wave sources and scatterers, as well as inferring the properties of the medium. This thesis presents the first-ever experimental proof of the time reversibility of acoustic waves propagating in water-filled viscoelastic HDPE pipes and develops a TR-based high-resolution methodology for defect detection in pipes, where the probing pressure waves with wavelengths comparable to the cross-sectional dimensions are employed (i.e., High Frequency Waves (HFW)). Despite offering superior resolution by virtue of their short wavelength, HFW in pipes are dispersed and strongly interact with the pipe wall, thus their implementation is not straightforward. Wave time reversal provides an elegant way to resolve these intricacies and permits defect localization with a resolution in the order of the probing wavelength. The proposed TR-based pipe diagnostic method comprises of two distinct steps: (i) an experimental or physical step that entails the active probing of the defected pipe system to extract a useful response, and (ii) a theoretical step where the experimentally obtained

response is used as input to an inverse problem to detect and localize the defect. Each of these two steps, that have never been attempted before using HFW in pipes, present their own set of challenges that this thesis effectively addresses. For instance, real-world water supply pipes are largely inaccessible; hence, to execute the experimental step a novel lab setup is assembled where high-frequency pressure transducers are placed within the pipe cavity. In the theoretical step, the defects are identified and localized by using TR to match the experimentally obtained response to the analytical model. However, this is a highly ill-posed problem due to the modelling uncertainties and measurement noise. To minimize modelling errors, this thesis proposes an impedance-based model where the elastic pipe wall behaviour is approximated using shell theory assumptions. This model presents a manageable closed-form dispersion relationship for the coupled fluid/pipe waveguide that affords intuitive analysis, encapsulates the essential physics, and can be adjusted to a pressure wave solution. The approximate impedance-based model is validated experimentally on an elastic thin-walled copper and a viscoelastic thick-walled High-Density Polyethylene (HDPE). For the purpose, the two water-filled pipes are subjected to a wideband in-pipe acoustic source, and the response is evaluated in both the fluid and structure domains. Subsequently, the dispersion relationship of the approximate impedance-based model is checked against both the experimentally derived dispersion curves and the predictions of a state-of-the-art general model for the two pipes. Besides, wall displacement solutions are extracted from the impedance-based model and are used in conjunction with the accelerometer measurements to explain the frequency intervals where the pipe wall participates the most. Appropriate simplifications to the characteristic equation of the impedance-based model yield well-known wave speed relations (e.g., Korteweg's equation) that are accurate only under particular frequency ranges. Furthermore, analysis shows the water hammer wave to be plane only at frequencies 20% of the pipe ring frequency f_{ring} . At frequencies that approach f_{ring} and beyond, the wavefront is no longer plane. Instead, the pipe wall resonates and executes a “breathing” motion, giving rise to a surface wave in the fluid that propagates along a small region adjacent to the pipe wall. Lastly, the proposed impedance-based model is adapted to a pressure wave solution and successfully tested against the experimental system response to narrowband probing pulses, hence substantiating its versatility.

To complete the theoretical step of the inverse problem, wave TR presents as an inversion technique that, if proven, enables the localization of defects. Water supply pipes, however, present a challenging environment for TR with considerable damping, instabilities, and

irregularities. The two mechanisms that are known to restrict time reversal, namely the stability of wave paths to perturbations and damping, are investigated analytically. Perturbations are found to grow slowly in time ($\sim t^{1/2}$) and are not a limiting factor for the time reversal of waves. To evaluate the effect of damping, an order of magnitude analysis on the non-reversible terms of the coupled waveguide momentum equations is performed and a dimensionless time reversal parameter TR is derived to show that damping develops linearly with time (i.e., $TR \sim t$). The TR parameter is applied to the presented experiments as well as relevant experimental proofs from the literature to find that the time reversal of waves only holds for $TR \sim 1$ or less; hence providing a general criterion to estimate the range over which time reversal-based wave techniques and methodologies are valid. A TR-based pulse-echo defect detection technique is proposed to provide a general inverse solution to the theoretical step. The developed closed-form high-frequency wave-leak interaction solution is formed on the basis of Neumann series and is experimentally verified on a real leak and small blockage, both localized with unprecedented accuracy. In fact, it is shown that the developed TR-based technique identifies defects with a resolution of the order of the minimum wavelength of the injected signal. To investigate the robustness of the proposed TR-based defect detection method, experiments are performed under conditions that fundamentally violate traditional time reversal and include turbulent flow ($Re = 40000$), or the introduction of inline junctions in the backpropagation wave path. It is found that flow causes a shift to the refocusing point that is linearly dependent on the probing frequency, while inline junctions exert severe damping to the propagating pressure wave. Finally, HF wave propagation over pipe bends is tested experimentally to reveal that the dispersion relationship is altered substantially compared to an equivalent straight pipe segment.

Keywords: Wave time reversal, pipe defect detection, fluid-structure interaction, wave-defect interaction, pressure transients, wave dispersion

Nomenclature

List of symbols

The following is a list of acronyms used in this thesis. These acronyms are also defined at their first place-of-use in the text.

$\mathbf{a}$:	acceleration vector (m/s^2)
A :	surface area (m^2)
a^2 :	inertia distribution in the coupled fluid filled pipe waveguide
A_L :	leak area (m^2)
A_L^* :	dimensionless leak area
a_n :	n^{th} root of the 1 st order Bessel function of the first kind ($n = 0, 1, 2, \dots$)
A_{obj} :	objective function
b :	relative bandwidth
C :	proportionality factor
c :	wave speed (m/s)
c^* :	dimensionless phase velocity
C_d :	leak discharge coefficient
c_f :	compressional wave speed in the fluid (m/s)
$c_{gr,n}$:	group velocity of the n^{th} mode (m/s)
C_n :	integration constants ($n = 1, 2, 3, \dots$)
c_p :	compressional wave speed in the pipe medium (m/s)
c_{ph} :	phase velocity (m/s)
$c_{ph,n}$:	phase velocity of the n^{th} mode (m/s)
d :	distance between scatterers (m)
D :	pipe diameter (m)
e :	acoustic energy density (kg/s^2)
E :	Young's modulus of elasticity (Pa)
E_a :	sound energy ($\text{kg}\cdot\text{m}^2/\text{s}^2$)

$E_{a,0}$:	initial sound energy ($\text{kg}\cdot\text{m}^2/\text{s}^2$)
$\mathbf{e}_i$:	perturbation of the position vector of the i^{th} scatterer (m)
e_k :	kinetic energy per unit volume ($\text{kg}/(\text{m}\cdot\text{s}^2)$)
e_p :	potential energy per unit volume ($\text{kg}/(\text{m}\cdot\text{s}^2)$)
$\mathbf{e}_r$:	radial unit vector
$\mathbf{e}_\theta$:	unit vector in the azimuthal coordinate
$\mathcal{F}^{-1}$:	theoretical inverse solution
$\mathcal{F}^\dagger$:	generalized inverse solution
$\mathcal{F}$:	theoretical system response
$\mathbf{F}$:	force vector acting on a body (N)
f :	wave frequency (1/s)
$f(t)$:	input pressure signal in time
$\hat{f}(\omega)$:	input pressure signal in frequency
f_c :	central frequency (1/s)
f_{DW} :	Darcy friction factor
$\mathbf{F}_{\text{ext}}$:	external forces to the C.V. (N)
$\mathbf{F}_{\text{ext,p}}$:	external forces to the pipe wall C.V. (N)
F_n :	n -axis external forces component on the pipe wall C.V. (N)
f_{ring} :	pipe ring frequency (1/s)
F_s :	expansion function (e.g., Taylor, Lagrange)
f_s :	function along the neutral fibre
$\mathbf{g}$:	gravity acceleration vector (m/s^2)
G :	Green's function (or impulse response) of the fluid-filled elastic pipe in the time domain
$\hat{G}$:	Green's function (or impulse response) of the fluid-filled elastic pipe in the frequency domain
$\widehat{G}_f^*$:	dimensionless Green's function (or impulse response) of the fluid-filled elastic pipe with respect to the fluid properties in the frequency domain
G_s :	shear modulus (Pa)
H :	hydraulic head (m)
h :	pipe wall thickness (m)

H_w :	water wave height (m)
h^* :	dimensionless pipe wall thickness
H_L :	pressure head at the leak (m)
$\mathbf{I}$:	energy flux (kg/s^3)
i :	imaginary unit
$\mathbf{l}$:	unit tensor ($\text{kg}/(\text{m}\cdot\text{s})$)
$\mathbf{i}, \mathbf{j}, \mathbf{k}$:	unit vectors along the x, y, z coordinates
I_0 :	0 th order modified Bessel function of the first kind
I_m :	moment of inertia of the cross-sectional area of the pipe element about an axis that passes through r_0 and coincides with the θ -axis (m^4)
$\mathfrak{I}_{actual}$:	actual leak/defect properties
$\mathfrak{I}_{image}$:	imaged leak/defect properties
$J(t)$:	time-dependent creep compliance ($\text{m}\cdot\text{s}^2/\text{kg}$)
J_0 :	0 th order Bessel function of the first kind
J_0 :	inverse modulus of elasticity ($\text{m}\cdot\text{s}^2/\text{kg}$)
J_1 :	1 st order Bessel function of the first kind
J_j :	creep coefficient of element j ($\text{m}\cdot\text{s}^2/\text{kg}$)
K :	shear constant
k :	wavenumber ($1/\text{m}$)
$\mathbf{k}$:	wavenumber vector ($1/\text{m}$)
K_f :	fluid bulk modulus (Pa)
$k_{\max}$:	wavelength associated with the largest frequency of the probing signal (m)
k_r :	radial wavenumber component ($1/\text{m}$)
$k_{r,n}$:	n^{th} mode radial wavenumber component ($1/\text{m}$)
k_x :	axial wavenumber component ($1/\text{m}$)
$k_{x,n}$:	n^{th} mode axial wavenumber component ($1/\text{m}$)
l :	pipe length (m)
L :	system length scale (m)
$\dot{m}$:	mass leak flowrate (kg/s)
m :	body mass (kg)
Ma :	Mach number

$\mathbf{M}_{\text{ext},p}$:	external torque applied to the pipe wall C.V. (Nm)
$\mathbf{N}$:	noise vector
$\mathbf{n}$:	normal vector to the C.S.
N :	measurement noise
N_{kv} :	number of Kelvin-Voigt elements
p :	pressure (Pa)
P :	pressure wave amplitude (Pa)
p^* :	dimensionless pressure
P^* :	dimensionless pressure wave amplitude
p' :	pressure perturbation (Pa)
$\bar{p}$:	pressure average over A (Pa)
$\hat{p}^*$:	dimensionless cyclic frequency with respect to the fluid properties in the frequency domain
p_0 :	ambient pressure along the pipe centreline
$\hat{p}^{\text{block}}$:	leak effect to the analytical pressure system response
$\hat{p}^I$:	incident pressure wave (frequency domain)
$\hat{p}^{\text{intact}}$:	analytical pressure response of an intact system
$\hat{p}^{\text{leak}}$:	leak effect to the analytical pressure system response
$p_r(r)$:	pressure function along the r -axis
$\hat{p}^R$:	reflected pressure wave (frequency domain)
p_{tr} :	time reversed pressure field (Pa)
$p_x(x)$:	pressure function along the x -axis
q :	external shear stresses on the $x\theta$ -plane of the pipe wall C.V. (Pa)
Q :	volumetric flowrate (m^3/s)
Q_L :	leak flowrate (m^3/s)
R :	pipe radius (m)
$\mathbf{r}$:	position vector (m)
r :	radial coordinate (m)
r_0 :	radial coordinate of the pipe wall C.V. centroid (m)
Re :	Reynolds number
Re_+ :	transient or unsteady Reynolds number

$\mathbf{r}_i$:	position vector of the i^{th} particle (m)
$\mathbf{r}_R$:	position vector of the receiver (m)
$\mathbf{r}_S$:	position vector of the source (m)
Rx :	receiver
$\hat{R}_{xy}$:	cross-correlation coefficient between signals x and y
s :	entropy (J/K)
S :	shear forces on the pipe wall per unit length (N/m)
S^* :	dimensionless shear forces
$\dot{S}$:	mass rate of a monopole acoustic source (kg/s)
$\hat{S}^*$:	dimensionless mass rate of a monopole acoustic source in the frequency domain
S_{ampl}^* :	dimensionless shear forces wave amplitude
S_L :	leak properties ($\text{m}^3/(\text{Pa}\cdot\text{s})$)
t :	time (s)
T :	time scale/ (s)
t^* :	dimensionless time
T_i :	peak time of pulse i (s)
TR :	time reversal parameter
TR_f :	time reversal parameter in fluid
TR_p :	time reversal parameter in pipe wall
T_s :	signal period (s)
T_{tot} :	total signal duration (s)
$T_{tot,i}$:	total recorded signal duration of the i^{th} measurement (s)
t_v :	pipe viscous time scale (s)
Tx :	transmitter
u :	axial fluid particle displacement (m)
u_p :	axial displacements of a pipe wall particle (m)
$u_{p,0}$:	axial displacement of the pipe wall element centroid (m)
$u_{p,0}^*$:	dimensionless axial displacement of the pipe wall element centroid
$U_{p,0}^*$:	dimensionless centroid axial displacements wave amplitude

$u_r(r)$:	axial fluid particle displacement function along the r -axis
$\bar{V}$:	cross-sectional average flow velocity (m/s)
$\mathbf{v}$:	flow velocity vector (m/s)
$\bar{\mathbf{v}}$:	mass-weighted average flow velocity over $\mathcal{V}$ (m/s)
V^* :	friction velocity (m/s)
$\mathcal{V}$:	fluid control volume (C.V.) (m^3)
$\mathbf{v}_i(T_{tot})$:	i^{th} particle velocity at $t = T_{tot}$
v_L :	flow velocity at the leak (m/s)
v_n^* :	mass-weighted average n -axis flow velocity component over A (m/s)
$\bar{v}_n^*$:	momentum-weighted n -axis velocity component average over A (m/s)
$\bar{v}_{p,n}$:	mass-weighted n -axis component of the pipe wall C.V. particles velocity across the pipe thickness (m/s)
$\mathcal{V}_p$:	pipe wall control volume (C.V.) (m^3)
$\mathbf{v}_p$:	velocity of the pipe wall C.V. particles of the C.V.
$v_{p,n}$:	n -axis component of the pipe wall C.V. particles velocity (m/s)
V_{rms} :	projector input voltage ($\text{kg}\cdot\text{m}^2/(\text{s}^3\cdot\text{A})$)
w :	radial fluid particle displacement (m)
$\mathbf{w}$:	velocity of the C.S. (m/s)
$\dot{W}_{Area}$:	work done by the pressure wave on the pipe cross-sectional area ($\text{J}/(\text{m}\cdot\text{s})$)
$\dot{W}_R$:	work done by the pressure wave on the pipe perimeter per unit time and length ($\text{J}/(\text{m}\cdot\text{s})$)
w_p :	radial displacements of a pipe wall particle (m)
W_p	radial pipe wall displacements wave amplitude (m)
$\mathbf{w}_p$:	velocity of the pipe wall C.V. boundaries (m/s)
w_p^* :	dimensionless radial displacements of a pipe wall particle
W_p^* :	dimensionless pipe wall centroid axial displacements wave amplitude
$w_{p,0}$:	radial displacement of the pipe wall element centroid (m)
$w_t(r)$:	radial fluid particle displacement function along the r -axis
$\mathbf{x}_i(T_{tot})$:	i^{th} particle position at $t = T_{tot}$
$\mathbf{x}_L$:	leak location (m)

$\mathbf{x}_m$:	measurement position (m)
$\mathbf{x}_s$:	source position (m)
Y_0 :	0 th order Bessel function of the second kind
Z :	pressure over radial displacements impedance condition at the fluid/pipe interface (Pa/m)
$z(t)$:	particle free-fall trajectory (m)
z_L :	elevation at the leak (m)
Z_L :	steady state acoustic impedance of the orifice (Pa·s/m)
Z_L^* :	dimensionless leak impedance
Z_f^* :	dimensionless impedance condition at the fluid/pipe interface with respect to the fluid properties
Z_s :	pressure over radial velocity impedance condition at the fluid/pipe interface (Pa/m)
Z_s^* :	dimensionless impedance condition at the fluid/pipe interface
α :	angular rotation of the pipe wall C.V. radial line (rad)
α_i :	scattering strength factor of the i^{th} scatterer
β :	particle instability growth factor
β_n :	n^{th} root of the 0 th order Bessel function of the first kind ($n = 1, 2, 3, \dots$)
γ_{att} :	attenuation factor
γ :	wave speed ratio
γ_{ij} :	shear strain component (i : direction of the face normal, j : direction of the component)
δ :	Gaussian modulated sine pulse duration (s)
δa :	straw opening area (m ²)
$\Delta \hat{\mathbf{p}}$:	leak effect in the experimental system response
δs :	puncture area on the C.S. (m ²)
$\delta \mathbf{v}$:	straw volume enclosed within the C.V. (m ³)
$\Delta x, \Delta y, \Delta z$:	infinitesimal C.V. dimensions in cartesian coordinates (m)
$\Delta \theta$:	infinitesimal azimuthal angle (rad)
$\Delta \Phi$:	phase difference between signals P_1 and P_2 (deg.)
ε :	modelling/epistemic uncertainties

ε_{ij} :	normal strain component (i : direction of the face normal, j : direction of the component)
η :	shear strains tracer
η_{air} :	air resistance coefficient
θ_{actual} :	actual system properties
$\theta_{estimate}$:	estimated system properties
$\theta_{k,n}$:	n^{th} mode wavenumber angle to the pipe centreline (deg.)
κ :	bulk fluid viscosity ($\text{kg}/(\text{m}\cdot\text{s}^2)$)
λ :	wavelength (m)
λ_{min} :	minimum probing wavelength (m)
μ :	dynamic fluid viscosity ($\text{N}\cdot\text{s}/\text{m}^2$)
ν :	kinematic fluid viscosity (m^2/s)
ν_p :	Poisson's ratio
ν_t :	eddy viscosity (m^2/s)
ζ :	pipe wall curvature correction
ρ :	fluid density (kg/m^3)
ρ' :	fluid density perturbation (kg/m^3)
ρ^* :	fluid density average over the integration surface A (kg/m^3)
$\bar{\rho}$:	fluid density average over the entire C.V. (kg/m^3)
ρ_0 :	reference (or ambient) fluid density (kg/m^3)
ρ_p :	pipe material density (kg/m^3)
$\bar{\rho}_p$:	average pipe wall material density across the thickness (kg/m^3)
σ :	standard deviation
σ_{ij} :	normal stress component on the pipe wall C.V. (i : direction of the face normal, j : direction of the component) (Pa)
$\bar{\sigma}_{ij}$:	average stress component on the pipe wall C.V. across the pipe thickness (i : direction of the face normal, j : direction of the component) (Pa)
σ_{tr} :	standard deviation of the refocusing error
$\mathbf{T}$:	fluid viscous stress tensor ($\text{kg}/(\text{m}\cdot\text{s}^2)$)
$\bar{\mathbf{T}}$:	average fluid viscous stress tensor over A ($\text{kg}/(\text{m}\cdot\text{s})$)
$\mathbf{T}_p$:	pipe wall viscous stress tensor ($\text{kg}/(\text{m}\cdot\text{s}^2)$)

$\mathbf{T}$:	stress tensor (kg/(m·s))
τ_{ij} :	shear stress component on the pipe wall C.V. (i : direction of the face normal, j : direction of the component) (Pa)
$\bar{\tau}_{ij}$:	average shear stress component on the pipe wall C.V. across the pipe thickness (i : direction of the face normal, j : direction of the component) (Pa)
τ_j :	relaxation time for element j ((m·s ²)/kg)
v_p :	azimuthal displacements of the pipe wall particle (m)
$v_{p,0}$:	azimuthal displacements of the pipe wall element centroid (m)
ω :	angular velocity (rad/s)
ω :	angular wave frequency (rad/s)
ω^* :	dimensionless cyclic frequency
ω_c :	central angular frequency (rad/s)
ω_f^* :	dimensionless cyclic frequency with respect to the fluid properties
$\omega_{n,\text{press. release}}$:	cut-off frequency of the n^{th} mode in a pressure release fluid-filled pipe (rad/s)
$\omega_{n,\text{rigid}}$:	cut-off frequency of the n^{th} mode in a rigid fluid-filled pipe (rad/s)
$(\diamond \circ)$:	(receiver location source location)
$(\rho\mathbf{v})^*$:	average linear momentum density over A (kg/(m ² ·s))
$ P(f) ^2$:	normalized spectrum amplitude

List of acronyms

The following is a list of acronyms used in this thesis. These acronyms are also defined at their first place-of-use in the text.

CCTV :	Closed-Circuit Television
FBW :	Frequency Bandwidth
FRM :	Frequency Response-based Method
FWTM :	Full Width at Tenth of Maximum
GPR :	Ground Penetrating Radars
HDPE:	High-Density Polyethylene
HF :	High Frequency

HK :	Hong Kong
ITM :	Inverse Transient Method
LF/LW :	Low Frequency/Long Wave
MFP:	Matched Field Processing
NSF :	National Science Foundation
PE :	Polyethylene
PIG :	Pipeline Inspection Gauges
PVC :	Polyvinyl Chloride
R&R :	Rehabilitation and Replacement
ReNUWIt :	Re-inventing the Nation's Urban Water Infrastructure
RGC :	Research Grants Council
SNR :	Signal-to-Noise Ratio
TBDDMs :	Transient-Based Defect Detection Methods
TDM :	Transient Damping-based Method
TR :	Time Reversal
TRM :	Transient Reflection-based Method
USA :	United States of America
UWSS :	Urban Water Supply Systems
WIN :	Water Intelligent Network
WSD :	Water Supplies Department

Chapter 1

Introduction

1.1 Motivation

Urban Water Supply Systems (UWSS) are undoubtedly essential to the support of economic development, human well-being, and quality of life. Regarding their physical characteristics, UWSS are extensive (e.g., in Hong Kong (HK) the overall length of the pipe network is ~ 8000 km [1]), complex, and largely inaccessible, thus difficult to inspect. Moreover, UWSS are continuously expanding (e.g., over 40000 km of pipes are installed each year in mainland China), ageing, subjected to large pressures (~ 6 to 10 bar in HK), and fraught with numerous invisible defects and operational inefficiencies. The resulting massive water losses and accompanying energy waste present inextricable challenges to making cities sustainable and carbon neutral. In addition, cracks and loose joints constitute intrusion points for pathogens in the system and increase the risk of water contamination. To put these claims into context, in HK for the year 2015 alone, water losses accounted for 15% of the total water supply (i.e., 150M m^3 of freshwater), corresponding to HK\$800M in financial losses [2]. These results follow an almost two-decade long, “brute force” rehabilitation and replacement (R&R) program that repaired 41% of the water distribution network. It also bears mentioning that, unless targeted to the faulty segments of the UWSS, R&R programs are particularly difficult to implement, costly, and fairly disruptive. Moreover, the UWSS contributes a large portion to the carbon footprint of the city of HK as the third largest energy user [2]. Issues with UWSS are not limited within the confines of HK. On a global scale, the volume of non-revenue water (NRW) is estimated around 126B m^3 per year, enough to support 130 cities the size of HK.

The equivalent financial cost, based on the (most often government subsidized) charges of utility companies per m³ of freshwater, is at the excess of US\$39B per year [3].

The severity of the issue has been widely recognised by government bodies. In HK, a number of directives, initiatives, and targets have been established to improve the operational efficiency of UWSS. These include the 2030 UWSS water-loss reduction strategy [4], the 2030+ climate action plan [5], the Water Intelligent Network (WIN) vision [6], and the 2050 net-zero targets [7]. The current USA administration pledged to invest about US\$111B in water infrastructure, of which US\$45B is specifically allocated to replace corroded, thus potentially toxic, lead pipelines [8]. Also, relevant research projects such as “Re-inventing the Nation’s Urban Water Infrastructure (ReNUWIt)” are established and funded by the National Science Foundation (NSF) of the USA [9, 10, 11]. The Smart Urban Water Supply Systems (Smart UWSS) [12] theme-based project, funded by the Research Grants Council (RGC) of HK, also aimed to develop innovative methodologies to assist the design, operation, monitoring, and diagnostic repair of UWSS. In mainland China, the betterment of water infrastructure has been designated a critical national issue and is key to meeting China’s 2030 “Three Red Lines” water limits and 2060 net-zero targets [13].

It is worth noting that similar problems are affecting oil and gas pipelines, despite being topologically simpler than water pipelines and being built to higher safety standards with greater funding availability. For example, over the past three decades, oil pipeline defects in the USA resulted in property damage of estimated worth around US\$7B, over 500 fatalities, and thousands of injuries [14, 15]. A single leak incident on November 22, 2013, in the Donghuang oil pipeline in Qingdao, Shandong Province, claimed 62 lives, injured 136, and amounted to over RMB700M in economic damages [16]. Notably, pipe corrosion, construction defects, and material/mechanical failure are among the main contributing factors behind the major accidents in oil and gas pipelines [17, 18]

1.2 Background

Naturally, and given the extend of the issue, there are various available technologies for the diagnosis and condition assessment of pipelines [19]. One class consists of intrusive methods that require the excavation of a pipe segment to (i) obtain a sample of the pipe for inspection, or (ii) create an insertion point for a CCTV camera, an acoustic sonar, SmartBall [20], or the

Sahara system [21]. Such pipeline inspection gauges (PIGs), that incorporate on-board optical, magnetic, and acoustic sensors, must be deployed via suitably sized ports into the pipeline most often attached to a tether. Besides, PIGs are not autonomous, hence carried by flow at advective speeds, and are generally large in dimensions and rigid, thus cannot be applied to pipelines that present tight bends. Oil and gas pipelines, where PIGs are primarily applied to, have long-radius bends, full-bore in-line valves, and single diameter pipelines (all of which allow free passage of in-line sensing tools). Contrary, UWSS are characterized by a network of pipes with numerous pipe-diameter changes, partial-bore inline valves and abrupt changes in direction both horizontally and vertically that do not permit the use of inline “smart PIGs”. Anecdotally, the Water Supplies Department (WSD) of HK conducted a trial run with an in-pipe sensor along a relatively simple transmission line and concluded that such technology is cost and labour intensive [22], unfolds at convective speeds (~ 0.3 m/s), requires the isolation of pipe side-branches (which disrupts normal service) to prevent straying of the inspection tool into other pipes (which sometimes results in an unwanted branch blockage and makes it difficult to retrieve the sensing tool), and traffic diversion. Above all, the exercise was inconclusive regarding the presence of leaks at two locations that happened to be close to side branch junctions despite the isolation of said branches.

The other class of methods includes fibre optic sensors [23, 24], ground penetrating radars [25, 26, 27] and acoustic noise correlators [28, 29]. Fibre optic sensors are installed parallel to the pipeline along its length and are sensitive to either the acoustic noise signal or the temperature gradient in the soil formed due to a leak presence. These effects alter the propagation characteristics for optic signals along the fibre and enable leak detection. In the case of acoustic leak signals, however, the method is susceptible to the environmental noise and the range is limited by the attainable signal-to-noise ratio (SNR). In fact, unless the fibre optic is sound insulated, the approach is inefficient [30]. Furthermore, research on detecting temperature changes adjacent to leaks using fibre optics is still theoretical, and issues regarding false positives due to external factors (e.g., rainfall) need to be addressed [23, 31]. Regardless, retrofitting a fibre optic leak detection system to existing pipelines is unfeasible and involves considerable costs [32]. Ground penetrating radars (GPR) operate on a similar principle to medical ultrasound, where a transmitting antenna emits high-frequency electromagnetic pulses to the ground to be partially reflected by buried objects or voids and later captured by a receiving antenna. Leaks create voids in the soil, or anomalies along the pipe that are potentially detectable by GPR [33]. However, the GPR approach is mainly applicable to metallic pipes

and has limited success identifying weakly reflective plastic pipes made from polyvinyl chloride (PVC) or polyethylene (PE) [25]. Moreover, reinforcement steel bars – usually present in the concrete pavement – shield wave transmission and reflection, while neighbouring underground facilities may further hinder detection [27]. Most importantly, GPR is costly [30], and is not practically applicable to the full extent of the water pipe network. Finally, acoustic noise correlation techniques apply cross-correlation between noise signals (acoustic and/or vibrational) obtained at two locations either side of a leak to determine the time delay in the arrival times. Knowledge of the wave speed allows for the subsequent estimation of the leak location. However, the technique is affected by traffic noise, has been proven problematic in plastic pipes, and its effectiveness depends on the sensors used and their mounting locations [34, 35].

Given the shortcomings of existing methods, it is imperative to identify the criteria for an ideal defect detection system. It must be fast to apply (i.e., provide real-time results, if possible), long range, non-intrusive, non-destructive, sensitive to defects (e.g., both leaks and blockages), accurate to an order that would enable cost-effective repairs, economical, reliable, and robust. Perhaps it is not surprising that most of these criteria (if not all) are fulfilled by active-probing wave-based defect detection methods. Certainly, it is a commonly accepted fact that wave-based diagnostic methods have revolutionized the medical field. Imaging techniques such as ultrasound and elastography offer real-time imaging of the human body by means of mechanical waves that travel at high speeds ($\sim$ km/s), are essentially non-intrusive, while also having minimal energy requirements. Nonetheless, direct transfer of technology from the medical field to UWSS is impossible; unlike the human body, UWSS (i) are inaccessible, (ii) have mechanical properties/topologies that are in most cases unknown, (iii) are large in scale, while (iv) most water utility companies have only scant knowledge of the administered UWSS. Evidently, it is impractical to apply classical ultrasound to an 8000 km long UWSS, even if the pipes were exposed (see GPR technology). However, acoustic waves that propagate along the pipeline, guided by its boundaries (hence, *guided waves*), unquestionably provide a real path for developing diagnostic systems for UWSS that meet the above stated criteria.

Indeed, a pioneering paper by Liggett & Chen [36] introduced the idea of utilizing pressure transients to probe the UWSS and, on that basis, a multitude of methodologies has emerged to form the class of Transient-Based Defect Detection Methods (TBDDMs). A wave-based methodology, much like TBDDMs, comprises of two steps: (i) an experimental and, (ii) a theoretical step, performed in that exact order. In the first – experimental – step, an injected

pressure signal is modified in accordance with both the physical characteristics of the system and potential defects as it propagates along the waveguide (i.e., water-filled elastic pipe). Subsequently, the recorded system response contains the defect signature(s) that, if properly analysed and processed, may identify the defect(s) in the system. Such analysis is carried out in the theoretical step. In summary, actively generated acoustic waves propagate in water supply pipes and interact with defects to induce useful reflections. The various methodologies developed in the framework of TBDDMs are grouped into (1) inverse transient methods (ITM) [36, 37, 38], (2) transient reflection-based methods (TRM) [39, 40, 41], (3) transient damping-based methods (TDM) [42], and (4) frequency response-based methods (FRM) [43, 44, 45, 46, 47, 48, 49, 50, 51].

These wave-based methods conform to their expected traits of being fast to carry out (wavespeed in water-filled elastic pipes $\sim$ km/s), long range ($\sim$ km), economical, non-intrusive, non-disrupting, and are also appropriate for all faults. Thus far, their application is based on low frequency transients (< 20 Hz) most often produced by the manual operation of relatively slow mechanical devices (e.g., valves). Since the wavelength $\lambda_f = c_f/f$ (where c_f is the wavespeed, and f is the probing frequency), low frequency (LF) implies long waves (LW); thus, the use of simple one-dimensional waterhammer models to model their propagation along water supply pipes is justified. A passive implementation of the LF methodology forms the basis of the Wireless Water Sentinel at Singapore (WaterWiSe@SG) project [52], where over 1000 sensors are used to “listen” to the system, measure the background pressure, and identify sudden events (e.g., pipe bursts). Nonetheless, a major limitation of LF/LW methods as a consequence of the long wavelength, is their poor localization resolution of defects and anomalies. That is because the attainable resolution approaches the diffraction limit ($\lambda_f/2$) at best. For example, in [53] sudden bursts could only be located to within ~ 45 m even though the sensor was only 20 m away from the fault, and leaks in Milan, Italy, were placed within an 800 m radius [54]. A conclusive theoretical proof explaining the low resolution of LF is provided in [55]. Therefore, a higher resolution in defect localization commands the use of high frequency (HF) waves that have lengths comparable to the pipe cross-sectional dimensions.

1.3 Objective

The Smart UWSS project [12], in order to augment the attainable resolution in defect detection, advocates the combined use of LF and HF waves, thus merging their strengths and addressing their weaknesses. At first, pressure waves in the LF regime with wavelengths much larger compared to the pipe diameter will propagate over long distances (in the order of km) and provide with a rough assessment of the water distribution network condition. Once a problematic segment is identified, HF pressure waves (from 10 to 70 kHz) will pinpoint the location of the defect within a few centimetres. However, probing UWSS with HF waves has never been attempted before, and specifically defines the knowledge gap that this thesis contributes to. In the following chapters, it becomes clear that implementing HF waves in pipes increases physical complexities and introduces considerable modelling challenges. Notably, this thesis aims to prove both experimentally and analytically the time reversal property of HF waves propagating in (viscous) water supply pipes. Once proven, wave time reversal presents a graceful process to resolve these intricacies and enables defect detection with HF waves, thus remarkable resolution.

The defect detection problem in a water-filled pipe is succinctly described by the equation:

$$\mathcal{F}(\mathfrak{S}_{image}, \omega, \boldsymbol{\theta}_{estimate}) - P(\mathfrak{S}_{actual}, \omega, \boldsymbol{\theta}_{actual}) = N + \varepsilon \quad (1.3.1)$$

or

$$\mathcal{F}(\mathfrak{S}_{image}, \omega, \boldsymbol{\theta}_{estimate}) = P(\mathfrak{S}_{actual}, \omega, \boldsymbol{\theta}_{actual}) + N + \varepsilon \quad (1.3.2)$$

where P corresponds to the experimentally obtained response of the system to a known probing signal, and $\mathcal{F}$ to a mathematical model that represents the theoretical system response to the same probing signal and depends on the physics that the modeller includes. Hence, wave-based diagnostic methods entail both an experimental and a theoretical step. To illustrate, consider the example of the water filled pipe in Figure 1.3.1i that contains a leak somewhere along its length. The experimental or physical step is applied first, where a known probing acoustic pulse $f(t)$ is injected in the system at point A. The created pressure wave propagates along the water-filled pipe, interacts with leak/defect, and is scattered off its surface. A receiver at point C records the pressure in the medium throughout this process. Naturally, the obtained system response P is a function of the leak characteristics $\mathfrak{S}_{actual}$, the actual mechanical and geometrical properties of the system $\boldsymbol{\theta}_{actual}$, and the frequency of the probing signal ω .

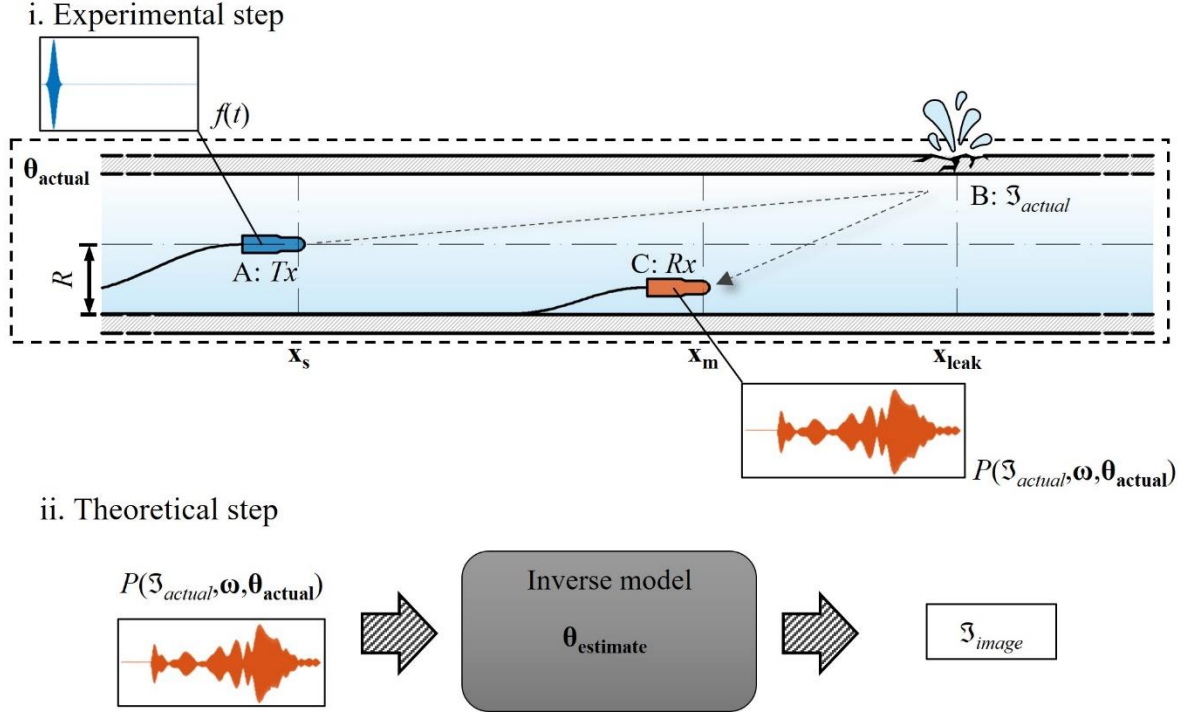


Figure 1.3.1: An illustration of a wave-based defect detection technique applied to a water filled pipe with a leak. In the experimental step, a short pulse $f(t)$ is injected into the system at A, interacts with the waveguide (θ_{actual}) and leak ($\mathfrak{S}_{\text{actual}}$) and yields response P at C. In the theoretical step, the experimentally obtained response P is input to an inverse model that approximately describes the real system (θ_{estimate}) to yield an image of the leak ($\mathfrak{S}_{\text{image}}$).

Next is the theoretical step, shown in Figure 1.3.1ii, where the experimentally obtained response P is used as input to an appropriate inverse model that is solved analytically/numerically. In essence, the objective is to minimize the left-hand side of Eq. (1.3.1), thus yield a model response $\mathcal{F}$ that best matches P . Since $\mathcal{F}$ is a function of the leak/defect image $\mathfrak{S}_{\text{image}}$, doing so would also characterize the real leak, $\mathfrak{S}_{\text{actual}}$. Both the experimental and theoretical steps present their unique set of challenges, especially when applied to a new, “uncharted” field that most often requires novel approaches.

Theoretical challenges

The direct (forward) problem answers the question: Given an input and a system, what is the output? On the other hand, the inverse problem attempts to reconstruct reality from projections, much like Plato’s cave allegory¹: Given a response, what is the system and/or input? How

¹ Plato’s allegory here is to be taken at face value, and not by its philosophical interpretation: The prisoners are recreating an imperfect image of the real world based on the shadows (i.e., projections) of the people and inanimate objects.

healthy is the pipe system given a measured response? Unfortunately, this inverse problem is highly ill-posed because of modelling and measurement errors, uncertainty about the “healthy” state of the system and limited data, that is represented in Eq. (1.3.1) by the right-hand side term $(N + \varepsilon)$. It is the presence of $(N + \varepsilon)$ and practically limited amount of acquired data that deems obtaining $\mathfrak{S}_{image}$ by a formal inversion of $\mathcal{F}$ impossible. Indeed, a non-zero $(N + \varepsilon)$ means that $P(\mathfrak{S}_{actual}, \omega, \theta_{actual}) + N + \varepsilon$ do not belong in the range of $\mathcal{F}$. To address the inverse problem, it is imperative to first address the modelling errors. Consequently, it is necessary to produce a theoretical model $\mathcal{F}$ that captures the essential physics of the UWSS waveguide with the least ε in the frequency range of interest. There are numerous models that describe the substantial fluid/structure interaction at high frequencies [56, 57, 58, 59, 60, 61]. None of these models, however, were developed with UWSS in mind, or with the intention of being used in an inverse problem solution, attempting to detect defects. Therefore, a fundamental objective of this thesis is to identify an analytical model among many in the literature to describe HF wave propagation in elastic pipes that captures the essential physics, is valid for frequencies up to 70 kHz, is applicable to a wide range of UWSS pipes (i.e., metallic *and* plastic), and has an expression simple enough to be adjustable to an inverse solution. Such model(s) must be thoroughly assessed and verified, based on comparisons with experimentally obtained results, as well as the predictions of more complete but otherwise cumbersome analytical models. As a welcome offshoot, this broad study would allow the different physical mechanisms that support HF wave propagation in fluid-filled pipes at different frequency intervals to be delineated and, hence, provide with useful insights. Specifically, Chapters 1, 2, 3, and 4 are devoted to deriving and validating an appropriate theoretical model $\mathcal{F}$.

Once the mathematical model is established, the research focuses on the development of a stable, high-resolution, and robust inversion technique of $\mathcal{F}$ for the localization and sizing of a defect. Time Reversal (TR) has been found to address such challenge in other fields and led to many ground-breaking applications in physics, medicine, and engineering [62, 63, 64, 65, 66, 67, 68]. The validity of TR under a particular environment must be investigated and proven; only then it may be concluded that the medium supports techniques such as Match Field Processing (MFP) or TR-based pulse-echo for defect detection. Chapter 5 presents the first-ever experimental and analytical proof of TR for HF waves in water-filled pipes and shows that the TR pressure field uniquely refocuses at the original source. Still, proving TR in pipes does not conclude the research, but generates additional conundrums. During experimental

demonstrations of TR in the lab, one of the fundamental questions is: How is HF-TR possible in a viscoelastic High-Density Polyethylene (HDPE) pipe that is known to exert considerable damping to pressure waves? Addressing this question allows to estimate the range over which TR-based defect detection methods are applicable. An extensive scale analysis over the damping-related non-reversible terms in the equations of motion of the examined water pipe system produces the dimensionless TR parameter that can be used to estimate the TR range, not only in UWSS pipes but hydraulic applications in general.

Formally, Eq. (1.3.2) is ill-posed and $\mathcal{F}^{-1}$ does not exist. Wave TR, introduced in Chapter 5, is used to establish a TR-based pulse-echo defect detection technique to yield a generalized inverse solution $\mathcal{F}^\dagger$ and realize the theoretical step. An analytical model that describes the interaction between HF waves and defects on the basis of Neumann series is derived. The developed TR-based method is tested experimentally and in two separate cases manages to localize a leak and a small (compared to the pipe cross-sectional area) blockage with unprecedented accuracy, shown to be proportional to the probing wavelength. The derivation of $\mathcal{F}^\dagger$ and its experimental validation using a real leak and blockage is discussed in Chapter 6.

Finally, to establish HF TR as a practical technique for detection of existing defects in a pipe, it is necessary to experiment under a variety of unfavourable environments. For instance, is HF TR in pipes still valid under turbulent flow conditions where damping is more pronounced? Is the analytical model for wave propagation in elastic models still applicable under flow conditions? What is the relationship between the observed pressure wave damping and flowrate? Does flow induce an error to the refocusing point of TR, and does this error depend on the probing wave frequency? How sensitive is TR to the presence of in-line pipe junctions? Besides, UWSS are not entirely constructed of straight pipe segments, but also include pipe bends. Hence, how do pipe bends affect HF wave propagation? Again, is the analytical model for wave propagation in elastic pipes still relevant in this case? The robustness of the proposed $\mathcal{F}^\dagger$ against damping, instabilities, and irregularities present in the waveguide is examined in Chapter 7.

Practical challenges

The experimental step carries its very own collection of challenges. The conventional approach of using manual actuation of valves can only generate LF waves, and a standardized technology to inject HF waves in water supply pipelines does not exist. Therefore, to perform the most

basic of tasks of the physical step – that of acoustic signal emission – this thesis proposes that waterproof piezoelectric transducers typically used in ocean acoustic applications are repurposed and inserted inside the pipe cavity. Finding an accessible real water supply pipe is yet another challenge; typical construction practice places UWSS out of sight and without provisions for access. Retrofitting existing pipeline installations with access points incurs considerable costs and disrupts normal operation. Consequently, the realization of the physical step necessitates the construction of an experimental setup in a lab environment.

High-frequency waves imply very short time scales, thus signal transmission and acquisition demand particularly high sampling rates ($\sim 1\text{M}$ Samples/s) and a high-order of synchronization accuracy. This can only be achieved through specialized programming and the development of software to control the accompanying instrumentation to the transducers, such as precision data acquisition devices, signal conditioners, and power amplifiers. This “laundry list” of issues that need to be addressed is by no means exhaustive, and there are myriad other questions to be answered. Some examples include: What type of signals are best for the problem at hand? What is an acceptable way to mount the transducers inside the pipe? How can the response of the pipe wall to an in-pipe acoustic source be evaluated? How is the not-so-linear transducer response considered, so that the results are as much valid and representative of reality? Is there a practical way to reverse the acquired response of the fluid/pipe system and then use it as a probing signal on-the-fly? These questions are principally addressed in Chapters 4 and 5, that describe the experimental validation of $\mathcal{F}$, and proof of TR in viscous pipes, respectively.

This thesis attempts to provide valid answers to the questions posed, and hopefully act as a vessel to introduce the intriguing method TR to the general hydraulic engineering audience, thus paving the way towards new and exciting applications not necessarily relevant to defect detection.

1.4 Thesis organization

The primary objective of this thesis is to develop a high-resolution defect detection method for water supply pipes based on HF probing waves with wavelengths comparable to the pipe cross-sectional dimensions. At these frequencies, waves in a bounded waveguide (i.e., fluid-filled pipe) are dispersed, and fluid-structure interaction is pronounced, so that defect detection is hindered. TR of waves is able to resolve these complications, and this thesis experimentally

and analytically proves its validity in viscous water supply pipes for the first time. Furthermore, a TR-based pulse echo defect detection technique is established based on an analytical wave/defect interaction solution, and the mechanisms that limit the applicability of TR are analytically and experimentally investigated. The thesis consists of seven chapters including the introduction, that are organized as follows:

Chapter 2 derives the equations that govern pressure wave propagation in a pipe under the effect of a leak (mass source/sink), while considering the length and time scales of the problem for the appropriate simplifications. Moreover, the equations of motion (i.e., both linear and angular momentum equations) for an elastic pipe wall are derived using a control volume (CV) approach to a pipe wall element of finite thickness. Applying the Kirchhoff assumptions for thin plates leads to the approximate shell theory equations that will be employed in the following Chapter 3.

Chapter 3 introduces the impedance-based model that describes the two-dimensional wave (i.e., axisymmetric) propagation in fluid filled elastic pipes by combining the pressure wave and pipe wall equation solutions at the fluid/pipe wall interface (i.e., pressure over radial pipe wall displacement, p/w). The ideal boundary conditions of a rigid, infinitely flexible, as well as empty pipe are also introduced to be used for comparison purposes with the case of a water-filled flexible pipe in Chapter 4. It is shown that the characteristic equation obtained from the impedance-based approach for a flexible pipe can afford intuitive analysis and the different mechanisms that support wave propagation in a fluid-filled flexible pipe can be delineated. In fact, a suite of well-known simpler models can be extracted from the complete expression following simplifications and term omissions. The pipe wall displacement solutions and pressure wave work equations are also provided; these will be used to provide valuable insight to the experimental waveguide dispersion curves developed in Chapter 4. Finally, the impedance relationship is introduced to an appropriate pressure wave solution expression to produce the waveguide's Green function (or impulse response) of the fluid-filled elastic pipe.

Chapter 4 verifies the approximate impedance-based model by experimental means. In particular, a lab setup that comprises of a thin-walled DN80 copper and a thick-walled DN80 HDPE is employed to study the physical step of the proposed technique and its features are discussed in detail. In essence, the *forward* part of the problem is realized i.e., Given an acoustic point source within the boundaries of a fluid-filled elastic pipe, what is the response of the system at a range x ? Notably, the response of the tested systems to the in-pipe wideband

acoustic source is evaluated at both the fluid (i.e., pressure field) and pipe wall (i.e., displacement field) through the combined use of hydrophones and accelerometers. Dispersion curves that fully describe wave propagation in the different waveguides tested are developed from measurements in both fluid and structure domains and compared to the predictions of both the impedance-based model and a general state-of-the-art model that forgoes the approximations of the shell theory. Through this comparison, it is demonstrated that the impedance-based model accurately describes the behaviour of the investigated waveguides in the frequency range of interest. A thorough discussion that employs the dispersion curves for the ideal cases of a rigid, infinitely flexible, and empty pipe provides further insight. In Chapter 3, a suite of simple models is derived following simplifications to the impedance-based characteristic equation; here, the frequency range of validity for each of these models is provided and discussed. Analytical pressure wave work plots indicate that the quasi-plane waterhammer wave behaves as a surface wave at frequencies close to the pipe ring frequency f_{ring} and beyond. Lastly, the derived Green's function is verified using the experimentally obtained response of the DN80 HDPE pipe to a narrowband Gaussian modulated sine pulse.

Chapter 5 proposes the use of TR to resolve multipath wave propagation effects in bounded waveguides, such as these present in fluid-filled elastic pipes under high-frequency excitation and thoroughly investigated in Chapter 4. Thus, this chapter attempts to establish the theoretical step of the proposed method based on TR: Given a dispersed signal at a range x , is it possible to uniquely identify the emanating source? Since waves contain information about their sources and the propagating media, time reversal of measured signals has the potential of localizing and characterizing wave sources (active or passive) and of inferring the properties of the medium. The validity of time reversal in a given environment, however, must be proven. In this chapter, the time reversibility of pressure waves is demonstrated experimentally in the water-filled viscoelastic HDPE pipes despite the considerable damping and using a single channel transducer setup. To better appreciate the surprising result, the two mechanisms that have the potential for “breaking” wave TR, namely damping and the stability of wave paths to perturbations, are investigated. Perturbations grow slowly in time ($\sim t^{1/2}$) and are not critical for the time reversal of waves. An extensive order of magnitude analysis is performed on the non-reversible terms of the governing equations for the coupled waveguide to derive a dimensionless TR parameter that shows damping evolving linearly in time ($\sim t$). The TR parameter is evaluated for the properties of the presented experimental setup as well as other relevant experimental proofs from the literature to find that time reversal of waves only holds

for $TR \sim 1$ or below. Consequently, the TR parameter provides a general criterion to estimate the range over which time reversal-based wave techniques and methodologies are valid in hydraulic-related applications. One such application is TR-based defect detection in water supply pipes.

Chapter 6 derives a closed-form HF wave-leak interaction solution on the basis of a Neumann series and makes use of the impedance-based wave propagation model to enable time reversal-based pulse-echo leak detection in fluid-filled elastic pipes. The method involves an experimental and an analytical part. The response of the defected system is obtained experimentally, and the reflected wave due to the defect is isolated. The isolated reflection is then convolved with the time-reversed (i.e., phase conjugated in the frequency domain) predictions of the analytical solution to localize the defect. The method is applied experimentally to localize a 2 cm diameter leak in a DN50 HDPE pipe and also a small piece of failed bitumen lining inserted as blockage in a DN80 HDPE pipe with accuracy of the order of the probing wavelength. The ability of high-frequency waves to localize defects with a resolution of the order of the minimum wavelength of the probing signal is also demonstrated.

Chapter 7 experimentally investigates the fidelity of time reversal against the inhibiting mechanisms of damping and wave path instabilities. Experiments presented in previous chapters are performed under quiescent water conditions that preclude the effects of turbulent flow and in-line pipe junctions. However, for high-frequency time reversal to be a practical method applicable to real-world water supply systems, it is necessary to evaluate its efficacy in a relevant environment. Hence, turbulent flow is introduced to the experimental setup, and different scenarios are tested to find a shift in the TR focal point. Interestingly, developed Lissajous curves reveal that the observed shift is linearly dependent on the probing wave frequency. Besides the shift to the TR focal spot, turbulent flow appears to induce considerable damping to the propagating pressure wave. To further investigate, the spectral attenuation for a wideband signal is assessed in space under various flowrates. Besides, the experimental dispersion curves under said conditions are also developed, to identify whether flow affects the transfer function of the water filled pipe waveguide. Non-reciprocal time reversal, a slight variation of traditional time reversal that yielded positive results in time varying media (e.g., ocean currents), is explored for the first time in pipes with flow. An interesting experiment involves performing time reversal tests between “sister” pipes: the forward step is performed along an intact DN80 HDPE pipe, while the TR step is conducted along a DN80 HDPE pipe with two in-line junctions, but otherwise identical to the original. Finally, experimental results

show that pipe bends present slightly different dispersion characteristics compared to equivalent straight pipe segments.

1.5 Related publications

The following is a list of the papers prepared during the course of the post-graduate studies:

Journal papers

- **Grigoropoulos, G.**, Ghidaoui, M.S., Louati, M. & Nasraoui, S. (2022), Time reversal of waves in hydraulics: experimental and theoretical proof with applications. *Journal of Hydraulic Research*, 60(1), 1-24
- **Grigoropoulos, G.**, Nasraoui, S., Louati, M. & Ghidaoui, M.S (2022), Pipeline leak detection using high-frequency acoustic wave time reversal. Part I: Experimental and analytical validation of an impedance-based wave propagation model in fluid-filled elastic pipes. *Mechanical Systems and Signal Processing*, (In preparation)
- **Grigoropoulos, G.**, Nasraoui, S., Louati, M. & Ghidaoui, M.S (2022), Pipeline leak detection using high-frequency acoustic wave time reversal. Part II: Time reversal-based technique for leak localization in viscoelastic HDPE pipes. *Mechanical Systems and Signal Processing*, (In preparation)
- Gozum, M.M., Nasraoui, S., **Grigoropoulos, G.**, Louati, M., Ghidaoui, M.S. (2022), A noise-based high-resolution time-reversal method for acoustic defect localization in water pipes, *The Journal of the Acoustical Society of America*, (Under Review)

Conference papers

- **Grigoropoulos, G.**, Louati, M., Ghidaoui, M.S., Dimitrakopoulos, I. (2018), Effect of structural vibration in the propagation of high-frequency waves through a fluid-filled elastic pipe, 13th International Conference on Pressure Surges, v. 1, 2018, p. 365-379
- **Grigoropoulos, G.**, Louati, M., Yang, T.C., Ghidaoui, M.S. (2019), Experimental evaluation of the fluid-structure interaction under high-frequency wave excitation in a fluid-filled HDPE pipe, The 38th International Association for Hydro-Environment Engineering and Research World Congress (IAHR 2019): 'Water – Connecting the World', Panama City, Panama., 1-6 September 2019

Patents

- **Grigoropoulos, G.,** Nasraoui, S., & Ghidaoui, M.S. (2021), Defect detection in pressurized pipelines using time reversal of high frequency waves. (*U.S. Patent No.-*). *U.S. Patent and Trademark Office.* (In preparation)

Chapter 2

Governing equations

2.1 Introduction

This chapter presents a mathematically rigorous derivation of the equations that govern the propagation of acoustic pressure waves along a cylindrical pipe in the presence of a leak. In particular, the supposed leak is modelled as a mass sink, in accordance with the excellent notes of Professor William G. Gray distributed as part of the CIVL 5550 – Modelling Fluid Systems course taught during the winter semester of 2020. In addition to the governing equations for the fluid, the equations of motion for the pipe wall are also derived. Although the produced momentum conservation equations for the pipe wall match those provided in [69], the derivation approach is markedly different. Specifically, instead of applying the different forces on a free-body diagram, the more elegant control volume (C.V.) approach is followed to reach an identical result. Great care has been taken to highlight the numerous assumptions, often omitted in relevant publications.

2.2 Fluid equations

2.2.1 Mass conservation equation

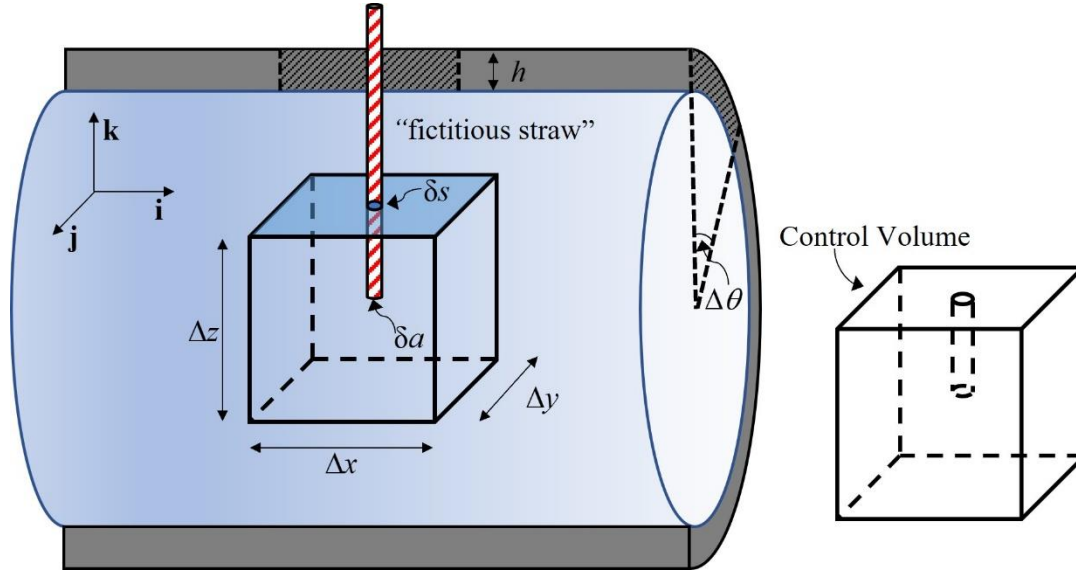


Figure 2.2.1: A small volume of fluid is considered inside a cylindrical pipe. This control volume includes a “fictitious straw” that is analogous to a point mass source/sink. Also depicted is the control volume for the pipe wall, contained within an angle $\Delta\theta$, pipe thickness h and length Δx .

We shall start from the mass conservation equation, that in its general integral form is:

$$\frac{d}{dt} \int_{\mathcal{V}} \rho d\mathcal{V} + \int_A \rho (\mathbf{v} - \mathbf{w}) \cdot \mathbf{n} dA = 0 \quad (2.2.1)$$

In Eq. (2.2.1), $\mathcal{V}$ is the control volume (C.V.) of interest, ρ is the fluid density, A is the area of the C.V. boundaries (identified as the control surface (C.S.)), $\mathbf{v}$ is the flow velocity, and $\mathbf{w}$ is the velocity of the C.S. The first term corresponds to the change of mass within the C.V. in time t , while the second term accounts for the net outflow mass rate exchanged with the surroundings of the C.V. through its C.S. From the general formulation for the mass conservation equation, we proceed to the appropriate considerations for our application. Figure 2.2.1 presents an “exposed” part of a cylindrical pipe. Within the fluid domain lays a small cubic-shaped volume containing a small location where fluid escapes (or, is injected) through a “fictitious straw”. For simplicity, the small cube is considered fixed in space, so that the velocity of its boundaries $\mathbf{w} = 0$. The straw is also immobile and has impenetrable walls. As

will be shown in mathematical terms later, the assumed control volume is that of the cuboid minus the volume occupied by the straw. Also, the corresponding control surface includes the walls of the straw, except for the area on the surface of the cuboid pierced by the straw.

The mass conservation equation for the considered C.V. is obtained from Eq. (2.2.1) after setting $\mathbf{w} = 0$:

$$\frac{\partial}{\partial t} \int_{\mathcal{V}} \rho d\mathcal{V} + \int_A \rho \mathbf{v} \cdot \mathbf{n} dA = 0 \quad (2.2.2)$$

Equation (2.2.2) presents the mass conservation equation for a fixed control volume in integral form that applies to a system of any size. It is also implied that neither control volume $\mathcal{V}$, nor control surface A change with time. We proceed to evaluate Eq. (2.2.2) on a term-by-term basis for our C.V., as depicted in Figure 2.2.1. The straw volume enclosed within the cuboid is equal to $\delta\mathcal{V}$, the straw opening area is δa , and the puncture area on the surface of the cuboid is δs . Moreover, the straw is oriented such that the normal to δa is the unit vector $\mathbf{k}$.

i. Rate of mass accumulation within the C.V.

$$\frac{\partial}{\partial t} \int_{\mathcal{V}} \rho d\mathcal{V} = \frac{\partial}{\partial t} [\bar{\rho} (\Delta x \Delta y \Delta z - \delta\mathcal{V})] \quad (2.2.3)$$

where $\bar{\rho}$ is the average fluid density value over the entire C.V. The mean value theorem suggests that there is a point $(x + \beta_x \Delta x, y + \beta_y \Delta y, z + \beta_z \Delta z)$ with $\beta_x, \beta_y, \beta_z \in [0, 1]$ within the integration domain such that:

$$\int_{\mathcal{V}} \rho d\mathcal{V} = \mathcal{V} \rho(x + \beta_x \Delta x, y + \beta_y \Delta y, z + \beta_z \Delta z) = \mathcal{V} \bar{\rho} \quad (2.2.4)$$

Hence, the average fluid density value evaluated over the C.V. is:

$$\bar{\rho} = \frac{\int_{\mathcal{V}} \rho d\mathcal{V}}{\int_{\mathcal{V}} d\mathcal{V}} = \rho(x + \beta_x \Delta x, y + \beta_y \Delta y, z + \beta_z \Delta z) \quad (2.2.5)$$

ii. Net mass outflow rate from the C.V.

$$\begin{aligned}
\int_A \rho \mathbf{v} \cdot \mathbf{n} dA = & \left(\rho^* v_x^* \Delta y \Delta z \right) \Big|_{x+\Delta x} - \left(\rho^* v_x^* \Delta y \Delta z \right) \Big|_x \\
& + \left(\rho^* v_y^* \Delta x \Delta z \right) \Big|_{y+\Delta y} - \left(\rho^* v_y^* \Delta x \Delta z \right) \Big|_y \\
& + \left[\rho^* v_z^* (\Delta x \Delta y - \delta s) \right] \Big|_{z+\Delta z} - \left(\rho^* v_z^* \Delta x \Delta y \right) \Big|_z \\
& + \left(\rho^* v_z^* \delta a \right) \Big|_{s_{hor}}
\end{aligned} \tag{2.2.6}$$

In Eq. (2.2.6), ρ^* is the density average over the integration surface A . Indicatively, we consider a surface on the yz -plane (where x is constant) to evaluate ρ^* . By the mean value theorem, there is a point on the surface $(x, y + \gamma_y \Delta y, z + \gamma_z \Delta z)$ with $\gamma_y, \gamma_z \in [0, 1]$ where:

$$\int_A \rho dA \Big|_x = \int_A \rho dy dz = \rho(x, y + \gamma_y \Delta y, z + \gamma_z \Delta z) \Delta y \Delta z = \rho^* \Big|_x \Delta y \Delta z \tag{2.2.7}$$

Hence, the average density over the considered surface on the yz -plane is:

$$\rho^* \Big|_x = \frac{\int_A \rho dA}{\int_A dA} \Big|_x = \rho(x, y + \gamma_y \Delta y, z + \gamma_z \Delta z) \tag{2.2.8}$$

In addition, v_x^* is the mass-weighted average flow velocity x -axis component, evaluated over the same surface on the yz -plane. Again, there exists a point $(x, y + \varepsilon_y \Delta y, z + \varepsilon_z \Delta z)$ with $\varepsilon_y, \varepsilon_z \in [0, 1]$ where:

$$\int_A \rho \mathbf{v} \cdot \mathbf{n} dA \Big|_x = \rho^* \int_A v_x dy dz = \rho^* v_x(x, y + \varepsilon_y \Delta y, z + \varepsilon_z \Delta z) \Delta y \Delta z = \rho^* v_x^* \Big|_x \Delta y \Delta z \tag{2.2.9}$$

Therefore, the mass-weighted velocity average v_x^* over the considered surface is calculated as:

$$v_x^* \Big|_x = \frac{\int_A \rho \mathbf{v} \cdot \mathbf{n} dA}{\int_A \rho dA} \Big|_x = v_x(x, y + \varepsilon_y \Delta y, z + \varepsilon_z \Delta z) \tag{2.2.10}$$

The average density and velocity component values over the remaining surfaces of the C.V. may be calculated in an analogous manner, but the expressions are skipped for brevity. The last term of Eq. (2.2.6) represents the flow through the opening of the fictitious straw. Summing the right-hand sides of Eqns. (2.2.3) & (2.2.6) to zero gives the complete expression of the mass conservation equation:

$$\begin{aligned}
\frac{\partial}{\partial t} \left[\bar{\rho} (\Delta x \Delta y \Delta z - \delta \Psi) \right] &+ \left(\rho^* v_x^* \Delta y \Delta z \right) \Big|_{x+\Delta x} - \left(\rho^* v_x^* \Delta y \Delta z \right) \Big|_x \\
&+ \left(\rho^* v_y^* \Delta x \Delta z \right) \Big|_{y+\Delta y} - \left(\rho^* v_y^* \Delta x \Delta z \right) \Big|_y \\
&+ \left[\rho^* v_z^* (\Delta x \Delta y - \delta s) \right] \Big|_{z+\Delta z} - \left(\rho^* v_z^* \Delta x \Delta y \right) \Big|_z \\
&+ \left(\rho^* v_z^* \delta a \right) \Big|_{s_{hor}} = 0
\end{aligned} \tag{2.2.11}$$

We divide all terms of Eq. (2.2.11) by the cuboid volume $\Delta x \Delta y \Delta z$ to obtain:

$$\begin{aligned}
\frac{\partial}{\partial t} \left[\bar{\rho} \left(1 - \frac{\delta \Psi}{\Delta x \Delta y \Delta z} \right) \right] &+ \frac{\rho^* v_x^* \Big|_{x+\Delta x} - \rho^* v_x^* \Big|_x}{\Delta x} + \frac{\rho^* v_y^* \Big|_{y+\Delta y} - \rho^* v_y^* \Big|_y}{\Delta y} \\
&+ \frac{\left[\rho^* v_z^* \left(1 - \frac{\delta s}{\Delta x \Delta y} \right) \right] \Big|_{z+\Delta z} - \rho^* v_z^* \Big|_z}{\Delta z} + \frac{(\rho^* v_z^* \delta a) \Big|_{s_{hor}}}{\Delta x \Delta y \Delta z} = 0
\end{aligned} \tag{2.2.12}$$

Imposing a condition where both the volume $\delta \Psi$ and cross-sectional area of the straw δs are much smaller than that of the infinitesimal cuboid yields:

$$\lim_{\Delta x, \Delta y, \Delta z \rightarrow 0} \frac{\delta \Psi}{\Delta x \Delta y \Delta z} \ll 1 \text{ and } \lim_{\Delta x, \Delta y \rightarrow 0} \frac{\delta s}{\Delta x \Delta y} \ll 1 \tag{2.2.13}$$

Therefore, $\Delta x, \Delta y, \Delta z \rightarrow 0$ is understood in the sense that the length scale of the control volume is negligibly small compared to the macroscale of the problem (pipe diameter), but much larger than the point source length scale. The crux of this idea, expressed mathematically in Eq. (2.2.13), allows us to write Eq. (2.2.12) as:

$$\begin{aligned}
\frac{\partial \bar{\rho}}{\partial t} + \frac{\rho^* v_x^* \Big|_{x+\Delta x} - \rho^* v_x^* \Big|_x}{\Delta x} &+ \frac{\rho^* v_y^* \Big|_{y+\Delta y} - \rho^* v_y^* \Big|_y}{\Delta y} + \frac{\rho^* v_z^* \Big|_{z+\Delta z} - \rho^* v_z^* \Big|_z}{\Delta z} \\
&+ \frac{(\rho^* v_z^* \delta a) \Big|_{s_{hor}}}{\Delta x \Delta y \Delta z} = 0
\end{aligned} \tag{2.2.14}$$

Although $\delta a \rightarrow 0$, the volumetric flowrate ($Q = v_z^* \delta a$) and the mass flowrate ($\dot{m} = \rho^* Q$) are different from zero. That is, as δa gets small, v_z^* gets large in such a way that:

$$\dot{m} = (\rho^* v_z^* \delta a) \Big|_{s_{hor}} = \rho^* Q \Big|_{s_{hor}} \neq 0 \tag{2.2.15}$$

Finally, we take the limit of Eq. (2.2.14) as the cuboid volume $\Delta x \Delta y \Delta z$ approaches zero:

$$\lim_{\Delta x, \Delta y, \Delta z \rightarrow 0} \left[\frac{\partial \bar{\rho}}{\partial t} + \frac{\rho^* v_x^*|_{x+\Delta x} - \rho^* v_x^*|_x}{\Delta x} + \frac{\rho^* v_y^*|_{y+\Delta y} - \rho^* v_y^*|_y}{\Delta y} + \frac{\rho^* v_z^*|_{z+\Delta z} - \rho^* v_z^*|_z}{\Delta z} + \frac{(\rho^* v_z^* \delta a)|_{s_{hor}}}{\Delta x \Delta y \Delta z} \right] = 0 \quad (2.2.16)$$

to yield:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v_x)}{\partial x} + \frac{\partial(\rho v_y)}{\partial y} + \frac{\partial(\rho v_z)}{\partial z} + \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) = 0 \quad (2.2.17)$$

Note that the overbar ($\bar{\cdot}$) and asterisk ($\cdot^*$) symbols are dropped in Eq. (2.2.17). This follows from the mean value theorem expressions (Eqns. (2.2.5), (2.2.8), and (2.2.10)), indicating that as Δx , Δy , and $\Delta z \rightarrow 0$ density and velocity are no longer average but point values:

$$\bar{\rho} \rightarrow \rho(x, y, z), \rho^* \rightarrow \rho(x, y, z), \text{ and } v_n^* \rightarrow v_n(x, y, z) \quad (2.2.18)$$

where n indicates the direction of the flow velocity vector. Moreover, the mass outflow term through the straw at the limit of Δx , Δy , and $\Delta z \rightarrow 0$ is evaluated as:

$$\lim_{\Delta x, \Delta y, \Delta z \rightarrow 0} \frac{(\bar{\rho} v_z \delta a)|_s}{\Delta x \Delta y \Delta z} = \lim_{\Delta x, \Delta y, \Delta z \rightarrow 0} \frac{\dot{m}}{\Delta x \Delta y \Delta z} = \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (2.2.19)$$

where $\delta_3(\mathbf{x} - \mathbf{x}_L)$ is the three-dimensional Dirac delta function (with dimensions L^{-3}), and $\mathbf{x}_L$ denotes the leak location. We combine Eqns. (2.2.17), (2.2.19) and adopt the vector notation for the divergence operator to obtain:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) + \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) = 0 \quad (2.2.20)$$

For steady-state conditions, the fluid flow quantities are time-independent:

$$\rho = \rho_0(\mathbf{x}), \mathbf{v} = \mathbf{v}_0(\mathbf{x}), \dot{m} = \dot{m}_0 \quad (2.2.21)$$

thus, Eq. (2.2.20) becomes:

$$\nabla \cdot (\rho_0 \mathbf{v}_0) + \dot{m}_0 \delta_3(\mathbf{x} - \mathbf{x}_L) = 0 \quad (2.2.22)$$

Conversely, for unsteady conditions it is convenient to write the fluid flow quantities as the sum of a steady-state (denoted by the subscript “0”) and a perturbation term (in prime notation), such that:

$$\rho = \rho_0(\mathbf{x}) + \rho'(\mathbf{x}, t), \mathbf{v} = \mathbf{v}_0(\mathbf{x}) + \mathbf{v}'(\mathbf{x}, t), \dot{m} = \dot{m}_0 + \dot{m}', \text{ and } p = p_0(x) + p'(x, t) \quad (2.2.23)$$

For example, pressure p in the transient state is defined as the sum of an ambient pressure p_0 and a perturbation term p' . Inserting the expressions of Eq. (2.2.23) (except for pressure, p) into Eq. (2.2.20) gives:

$$\frac{\partial \rho_0}{\partial t} + \frac{\partial \rho'}{\partial t} + (\dot{m}_0 + \dot{m}') \delta_3(\mathbf{x} - \mathbf{x}_L) + \nabla \cdot (\rho_0 + \rho')(\mathbf{v}_0 + \mathbf{v}') = 0 \quad (2.2.24)$$

Again, since $\partial \rho_0 / \partial t = 0$, we get:

$$\frac{\partial \rho'}{\partial t} + (\dot{m}_0 + \dot{m}') \delta_3(\mathbf{x} - \mathbf{x}_L) + \nabla \cdot (\rho_0 + \rho')(\mathbf{v}_0 + \mathbf{v}') = 0 \quad (2.2.25)$$

Performing the calculations (product rule) in Eq. (2.2.25) yields:

$$\begin{aligned} \frac{\partial \rho'}{\partial t} + (\dot{m}_0 + \dot{m}') \delta_3(\mathbf{x} - \mathbf{x}_L) + \rho_0 \nabla \cdot \mathbf{v}_0 + \rho_0 \nabla \cdot \mathbf{v}' + \rho' \nabla \cdot \mathbf{v}_0 + \rho' \nabla \cdot \mathbf{v}' \\ + \mathbf{v}_0 \cdot \nabla \rho_0 + \mathbf{v}_0 \cdot \nabla \rho' + \mathbf{v}' \cdot \nabla \rho_0 + \mathbf{v}' \cdot \nabla \rho' = 0 \end{aligned} \quad (2.2.26)$$

In Eq. (2.2.26) we identify the steady state equation terms (Eq. (2.2.22)) and, since these sum up to zero, the expression is simplified to:

$$\frac{\partial \rho'}{\partial t} + \dot{m}' \delta_3(\mathbf{x} - \mathbf{x}_L) + \nabla \cdot (\rho_0 \mathbf{v}') + \nabla \cdot (\rho' \mathbf{v}_0) + \nabla \cdot (\rho' \mathbf{v}') = 0 \quad (2.2.27)$$

At this point, we introduce the equation of state, $p = p(\rho, s)$, that relates pressure p with density ρ and entropy s . For acoustic wave propagation, the pressure variations are often so rapid that there is no heat exchange between the medium and the surrounding environment; thus, the process is adiabatic. Moreover, as will be shown in the upcoming scale analysis for the momentum conservation equation terms, friction forces are insignificant to our application. Consequently, both heat conduction and viscous dissipation are neglected, and the flow is described as isentropic to denote that the entropy s of a fluid particle remains constant. In this case, pressure is only a function of density, and the equation of state becomes $p = p(\rho)$, or in its differential form:

$$dp = \left(\frac{dp}{d\rho} \right)_s d\rho, \text{ where } s \text{ is constant} \quad (2.2.28)$$

where $dp/d\rho$ for constant entropy s equals to the speed of sound c_f raised to the power 2 (i.e., $dp/d\rho = c_f^2$). Before proceeding further, it is important to comprehend the order of magnitude of the change in density $d\rho$ in a UWSS environment. Assume a hypothetical change in head of

$H = 50$ m (which is highly excessive and discouraged in practice for such systems as to not compromise their structural integrity), a reference density $\rho_0 = 1000$ kg/m³, a wave speed $c_f = 1000$ m/s and the acceleration of gravity $\|\mathbf{g}\| = 10$ m/s. The associated change in density is calculated from Eq. (2.2.28) as follows:

$$d\rho = \rho_0 \|\mathbf{g}\| h / c_f^2 = 50 \times 10^4 / 10^6 = 0.5 \text{ kg/m}^3 \quad (2.2.29)$$

Notably, density only changes by 0.5 kg/m³ for a substantial head difference of 50 m. Pressure wave amplitudes generated by high-frequency transducers, which are the focus of this thesis, are in the order of 10^{-2} . Repeating the calculations for such pressure difference yields:

$$d\rho = \rho_0 \|\mathbf{g}\| h / c_f^2 = 10^{-2} \times 10^4 / 10^6 = 10^{-4} \text{ kg/m}^3 \quad (2.2.30)$$

In addition, a relationship between ρ' and p' may be obtained directly from Eq. (2.2.28) if the left-hand side is integrated from p_0 to $p_0 + p'$, and the right-hand side from ρ_0 to $\rho_0 + \rho'$:

$$\int_{p_0}^{p_0+p'} dp = c_f^2 \int_{\rho_0}^{\rho_0+\rho'} d\rho, \text{ or } p' = c_f^2 \rho' \quad (2.2.31)$$

where the fact that ρ' is small has been employed to assume that c_f^2 is constant over the interval $[\rho_0, \rho_0 + \rho']$. Equivalently, the expression of Eq. (2.2.31) may also be reached using a Taylor series expansion² to evaluate p' :

$$p' = p - p_0 = p(\rho) - p(\rho_0) = (\rho - \rho_0) \left. \frac{dp}{d\rho} \right|_{\rho_0} + (\rho - \rho_0)^2 \left. \frac{d^2p}{d\rho^2} \right|_{\rho_0} + \dots \quad (2.2.32)$$

According to Eq. (2.2.30), the changes in density studied in this thesis are $\rho' = \rho - \rho_0 \sim 10^{-4}$ kg/m³. Therefore, $(\rho - \rho_0)^2 \sim 10^{-8}$ kg/m³, and Eq. (2.2.32) rightfully reduces to:

$$p' \cong (\rho - \rho_0) \left. \frac{dp}{d\rho} \right|_{\rho_0} \text{ or } p' \cong c_f^2 \rho' \quad (2.2.33)$$

Finally, pressure p may be expressed in acceleration terms after making use of Newton's second law of motion stating that the vector sum of the forces $\mathbf{F}$ on a body is equal to its mass m multiplied by its acceleration $\mathbf{a}$:

$$\mathbf{F} = m\mathbf{a} \quad (2.2.34)$$

² $f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$

As a representative system, we shall consider a pipe characterized by a length scale L and cross-sectional area A , filled with continuous fluid of density ρ_0 . An order of magnitude of its mass is given by $m \sim \rho_0 AL$. Since acceleration $\mathbf{a}$ scales as $\|\mathbf{v}\|/T$, we may write:

$$\mathbf{F} \sim \rho_0 AL \frac{\|\mathbf{v}\|}{T} \quad (2.2.35)$$

For processes that evolve at an advective time scale, $T = L/\|\mathbf{v}\|$ and $p \sim \|\mathbf{F}\|/A$; hence, leading to $p \sim \rho_0 \|\mathbf{v}\|^2$, which is the classic Bernoulli pressure scale. On the other hand, when the time scale is $T \sim L/c_f$ (as is the case in this thesis):

$$p = \frac{\|\mathbf{F}\|}{A} \sim \rho_0 L \frac{\|\mathbf{v}\|}{(L/c_f)} = \rho_0 c_f \|\mathbf{v}\| \quad (2.2.36)$$

From Eqns. (2.2.33) & (2.2.36) we obtain:

$$\rho' \cong \rho_0 \frac{\|\mathbf{v}'\|}{c_f} \text{ or } \rho' \cong \rho_0 Ma \quad (2.2.37)$$

where Ma is the Mach number. Based on these defined scales for the various quantities, we cross-compare the terms appearing in the mass conservation equation (Eq. (2.2.27)), and these deemed insignificant are omitted from its final expression. Prior to that, the reader is reminded the fundamental scales for our problem:

- i. Length scale (L): is that of the probing wavelength. The focus of this thesis is on probing wavelengths comparable to the pipe diameter, D (i.e., $L \sim D$).
- ii. Time scale (T): is the length over the speed of sound.

The individual terms of the mass conservation equation (Eq. (2.2.27)) and their pertinent scales are:

$$\begin{array}{ll} \text{i.} & \frac{\partial \rho'}{\partial t} \sim \rho_0 \frac{\|\mathbf{v}\|}{c} \frac{c_f}{D} = \rho_0 \frac{\|\mathbf{v}\|}{D} \\ \text{ii.} & \nabla \cdot (\rho' \mathbf{v}_0) \sim \rho_0 \frac{\|\mathbf{v}\|}{D} \left(\frac{\|\mathbf{v}\|}{c_f} \right) = \rho_0 \frac{\|\mathbf{v}\|}{D} Ma \\ \text{iii.} & \nabla \cdot (\rho_0 \mathbf{v}') \sim \rho_0 \frac{\|\mathbf{v}\|}{D} \\ \text{iv.} & \nabla \cdot (\rho' \mathbf{v}') \sim \rho_0 \frac{\|\mathbf{v}\|}{D} \left(\frac{\|\mathbf{v}\|}{c_f} \right) = \rho_0 \frac{\|\mathbf{v}\|}{D} Ma \end{array}$$

In water supply pipe flows, the Mach number is usually of the order of 10^{-3} (i.e., $c_f \gg \|\mathbf{v}_0\|, \|\mathbf{v}'\|$); thus, terms that scale to that specific parameter (i.e., terms (ii) & (iv)) are excluded from the mass conservation equation (Eq. (2.2.27)). As a result, we obtain:

$$\frac{\partial \rho'}{\partial t} + \dot{m}' \delta_3(\mathbf{x} - \mathbf{x}_L) + \nabla \cdot (\rho_0 \mathbf{v}') = 0 \quad (2.2.38)$$

Further, we drop the prime notation ($'$) and use the fact that $\partial \rho / \partial p = c_f^{-2}$ (so that $\partial \rho / \partial t = (\partial \rho / \partial p) \partial p / \partial t$, or $\partial \rho / \partial t = c_f^{-2} \partial p / \partial t$) to obtain:

$$\boxed{\frac{1}{c_f^2} \frac{\partial p}{\partial t} + \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) + \nabla \cdot (\rho_0 \mathbf{v}) = 0} \quad (2.2.39)$$

Equation (2.2.39) gives the appropriate expression of the mass conservation principle for our problem. In the following subsection, we develop the momentum conservation equation on the same basis.

2.2.2 Momentum conservation equation

The general integral form of the momentum conservation equation that applies to any C.V. is:

$$\frac{d}{dt} \int_V \rho \mathbf{v} dV + \int_A \rho \mathbf{v} (\mathbf{v} - \mathbf{w}) \cdot \mathbf{n} dA = \sum \mathbf{F}_{\text{ext}} \quad (2.2.40)$$

Since the boundaries of our considered C.V. are immobile, we set $\mathbf{w} = 0$ in Eq. (2.2.40) to obtain:

$$\frac{\partial}{\partial t} \int_V \rho \mathbf{v} dV + \int_A \rho \mathbf{v} \mathbf{v} \cdot \mathbf{n} dA = \sum \mathbf{F}_{\text{ext}} \quad (2.2.41)$$

The individual terms of Eq. (2.2.41) are evaluated for our specific problem, as shown in Figure 2.2.1. The steps are similar to these followed to obtain the mass conservation equation (Eq. (2.2.20)). We start with the momentum accumulation within the C.V. (outmost left term in Eq. (2.2.41)):

- i. Rate of momentum accumulation within the C.V.

$$\frac{\partial}{\partial t} \int_V \rho \mathbf{v} dV = \frac{\partial}{\partial t} [\bar{\rho} \bar{\mathbf{v}} (\Delta x \Delta y \Delta z - \delta \mathbf{v})] \quad (2.2.42)$$

where $\bar{\rho}$ is the average fluid density evaluated over the C.V. (Eq. (2.2.5)), while $\bar{\mathbf{v}}$ is the mass-weighted average velocity over the C.V. According to the mean value theorem, there is a point within the volume $(x + \zeta_x \Delta x, y + \zeta_y \Delta y, z + \zeta_z \Delta z)$ with $\zeta_x, \zeta_y, \zeta_z \in [0, 1]$ such that:

$$\int_V \rho \mathbf{v} dV = \bar{\rho} \int_V \mathbf{v} dV = \bar{\rho} \mathbf{v} (x + \zeta_x \Delta x, y + \zeta_y \Delta y, z + \zeta_z \Delta z) V = \bar{\rho} \bar{\mathbf{v}} V \quad (2.2.43)$$

Therefore, the mass-weighted velocity average $\bar{\mathbf{v}}$ over the C.V. is:

$$\bar{\mathbf{v}} = \frac{\int_V \rho \mathbf{v} dV}{\int_V \rho dV} = \mathbf{v} (x + \zeta_x \Delta x, y + \zeta_y \Delta y, z + \zeta_z \Delta z) \quad (2.2.44)$$

ii. Net rate of momentum outflow from the C.V.

$$\begin{aligned} \int_A \rho \mathbf{v} \mathbf{v} \cdot \mathbf{n} dA = & \left[(\rho \mathbf{v})^* \bar{v}_x^* \Delta y \Delta z \right]_{x+\Delta x} - \left[(\rho \mathbf{v})^* \bar{v}_x^* \Delta y \Delta z \right]_x \\ & + \left[(\rho \mathbf{v})^* \bar{v}_y^* \Delta x \Delta z \right]_{y+\Delta y} - \left[(\rho \mathbf{v})^* \bar{v}_y^* \Delta x \Delta z \right]_y \\ & + \left[(\rho \mathbf{v})^* \bar{v}_z^* (\Delta x \Delta y - \delta s) \right]_{z+\Delta z} - \left[(\rho \mathbf{v})^* \bar{v}_z^* \Delta x \Delta y \right]_z \\ & + \left[(\rho \mathbf{v})^* \bar{v}_z^* \Delta x \Delta y \right]_s \end{aligned} \quad (2.2.45)$$

where $(\rho \mathbf{v})^*$ is the average linear momentum density over the integration surface A . To further illustrate, we consider a surface laying on the yz -plane (where x is constant) over which we shall evaluate the average linear momentum $(\rho \mathbf{v})^*$. As per the mean value theorem, there is a point $(x, y + \eta_y \Delta y, z + \eta_z \Delta z)$ with $\eta_y, \eta_z \in [0, 1]$ where:

$$\int_A \rho \mathbf{v} dA \Big|_x = \int_A \rho \mathbf{v} dy dz = (\rho \mathbf{v}) (x, y + \eta_y \Delta y, z + \eta_z \Delta z) \Delta y \Delta z = (\rho \mathbf{v})^* \Big|_x \Delta y \Delta z \quad (2.2.46)$$

hence, the average linear momentum over the considered surface is:

$$(\rho \mathbf{v})^* \Big|_x = \frac{\int_A \rho \mathbf{v} dA}{\int_A dA} \Big|_x = (\rho \mathbf{v}) (x, y + \eta_y \Delta y, z + \eta_z \Delta z) \quad (2.2.47)$$

On the other hand, $\bar{v}_x^*$ is the momentum-weighted x -axis velocity component average over the same surface. From the mean value theorem, there is a point on the surface $(x, y + \iota_y \Delta y, z + \iota_z \Delta z)$ with $\iota_y, \iota_z \in [0, 1]$, where:

$$\begin{aligned}
\left. \int_A (\rho \mathbf{v}) \mathbf{v} \cdot \mathbf{n} dA \right|_x &= (\rho \mathbf{v})^* \int_A v_x dy dz \\
&= (\rho \mathbf{v})^* v_x (x, y + \iota_y \Delta y, z + \iota_z \Delta z) \Delta y \Delta z = (\rho \mathbf{v})^* v_x^* \big|_x \Delta y \Delta z
\end{aligned} \tag{2.2.48}$$

Consequently, $\bar{v}_x^*$ is defined as:

$$\bar{v}_x^* \big|_x = \frac{\int_A (\rho \mathbf{v}) \mathbf{v} \cdot \mathbf{n} dA}{\int_A (\rho \mathbf{v}) dA} \bigg|_x = v_x (x, y + \iota_y \Delta y, z + \iota_z \Delta z) \tag{2.2.49}$$

Likewise, we may define the average linear momentum and momentum-weighted velocity components over the remaining surfaces of the C.V., but the expressions are skipped for brevity.

iii. External forces

The sum of the external forces $\sum F_{ext}$ in their general form are written as:

$$\sum \mathbf{F}_{ext} = - \int_A \mathbf{n} p dA + \int_A \mathbf{n} \cdot \boldsymbol{\tau} dA + \int_{\Omega} \rho \mathbf{g} d\Omega \tag{2.2.50}$$

In Eq. (2.2.50), $\mathbf{g}$ is the acceleration of gravity, p is the thermodynamic pressure for a non-stationary fluid, and $\boldsymbol{\tau}$ is the viscous stress tensor. Both p and $\boldsymbol{\tau}$ are components of the stress tensor $\mathbf{T}$, such that $\mathbf{T} = -p\mathbf{I} + \boldsymbol{\tau}$, where $\mathbf{I}$ represents a unit tensor. The stress tensor $\mathbf{T}$, accounts for forces acting on the surface of the C.V. and may be decomposed into a sum of pressure p (that acts equally at all directions) and friction effects $\boldsymbol{\tau}$ (that only exist when the C.V. is in motion). The first term on the right-hand side of Eq. (2.2.50) (pressure forces) is evaluated for the assumed control volume as:

$$\begin{aligned}
\int_A \mathbf{n} p dA &= \mathbf{i} \left[(\bar{p} \Delta y \Delta z) \big|_{x+\Delta x} - (\bar{p} \Delta y \Delta z) \big|_x \right] \\
&\quad + \mathbf{j} \left[(\bar{p} \Delta x \Delta z) \big|_{y+\Delta y} - (\bar{p} \Delta x \Delta z) \big|_y \right] \\
&\quad + \mathbf{k} \left\{ \left[\bar{p} (\Delta x \Delta y - \delta s) \right] \big|_{z+\Delta z} - (\bar{p} \Delta x \Delta y) \big|_z \right\} \\
&\quad + \mathbf{n} \left[\left(\int_0^{2\pi} p d\theta \right) \delta l \delta r \right] \bigg|_{s_{ver}} + \mathbf{k} (\bar{p} \delta a) \big|_{s_{hor}}
\end{aligned} \tag{2.2.51}$$

where $\bar{p}$ is the average pressure over the integration surface. Assuming a surface on the yz -plane and utilizing the mean value theorem for integrals, there is a point $(x, y + \kappa_y \Delta y, z + \kappa_z \Delta z)$ with $\kappa_y, \kappa_z \in [0, 1]$ such that:

$$\int_A p dA \Big|_x = p(x, y + \kappa_y \Delta y, z + \kappa_z \Delta z) dy dz = \bar{p}|_x dy dz \quad (2.2.52)$$

Thus, the average pressure $\bar{p}$ over the examined surface may be calculated as:

$$\bar{p}|_x = \frac{\int_A p dA}{\int_A dA} \Big|_x = p(x, y + \kappa_y \Delta y, z + \kappa_z \Delta z) \quad (2.2.53)$$

The second-last term in the right-hand side of Eq. (2.2.51) accounts for the surface pressure on the impenetrable vertical walls of the straw:

$$\mathbf{n} \left[\left(\int_0^{2\pi} p d\theta \right) \delta l \delta r \right] \Big|_{s_{ver}} = \mathbf{n} (\bar{p} 2\pi \delta r \delta l) \Big|_{s_{ver}} \quad (2.2.54)$$

The normal vector to the vertical straw wall surface, s_{ver} , is denoted as $\mathbf{n}$, the length of the straw that protrudes in the considered cuboid C.V. is equal to δl , and δr is the straw radius. The second term in the right-hand side of Eq. (2.2.50) that accounts for the shear forces on the surfaces of the C.V. is written as:

$$\begin{aligned} \int_A \mathbf{n} \cdot \boldsymbol{\tau} dA = & \mathbf{i} \cdot \left[(\bar{\boldsymbol{\tau}} \Delta y \Delta z) \Big|_{x+\Delta x} - (\bar{\boldsymbol{\tau}} \Delta y \Delta z) \Big|_x \right] \\ & + \mathbf{j} \cdot \left[(\bar{\boldsymbol{\tau}} \Delta x \Delta z) \Big|_{y+\Delta y} - (\bar{\boldsymbol{\tau}} \Delta x \Delta z) \Big|_y \right] \\ & + \mathbf{k} \cdot \left\{ \left[\bar{\boldsymbol{\tau}} (\Delta x \Delta y - \delta s) \right] \Big|_{z+\Delta z} - (\bar{\boldsymbol{\tau}} \Delta x \Delta y) \Big|_z \right\} \\ & + \mathbf{n} \cdot \left[\bar{\boldsymbol{\tau}} \left(\int_0^{2\pi} d\theta \right) \delta l \delta r \right] \Big|_{s_{ver}} + \mathbf{k} \cdot (\bar{\boldsymbol{\tau}} \delta a) \Big|_{s_{hor}} \end{aligned} \quad (2.2.55)$$

where $\bar{\boldsymbol{\tau}}$ is the average of shear stresses over the integration surface A . For a surface on the yz -plane, we employ the mean value theorem to define a point on the surface $(x, y + \xi_y \Delta y, z + \xi_z \Delta z)$ with $\xi_y, \xi_z \in [0, 1]$ where:

$$\int_A \boldsymbol{\tau} dA \Big|_x = \boldsymbol{\tau}(x, y + \xi_y \Delta y, z + \xi_z \Delta z) \Delta y \Delta z = \bar{\boldsymbol{\tau}}|_x \Delta y \Delta z \quad (2.2.56)$$

It follows that the average shear stress, $\bar{\boldsymbol{\tau}}$, over the assumed surface are:

$$\bar{\mathbf{T}}|_x = \frac{\int_A \mathbf{T} dA}{\int_A dA} \bigg|_x = \mathbf{T}(x, y + \xi_y \Delta y, z + \xi_z \Delta z) \quad (2.2.57)$$

It should be noted that $\mathbf{T}$ is a tensor of order 2. The shear forces are also evaluated at the rigid vertical side walls of the straw. As is the case for the pressure forces, the term is written in Eq. (2.2.55) as:

$$\mathbf{n} \cdot \left[\left(\int_0^{2\pi} \mathbf{T} d\theta \right) \delta l \delta r \right] \bigg|_{s_{ver}} = \mathbf{n} \cdot [\bar{\mathbf{T}} 2\pi \delta l \delta r] \big|_{s_{ver}} \quad (2.2.58)$$

Next, we consider the third term on the right-hand side of Eq. (2.2.50) that corresponds to the gravity forces applied on the C.V. domain:

$$\int_V \rho \mathbf{g} dV = [\bar{\rho} \mathbf{g} (\Delta x \Delta y \Delta z - \delta \Psi)] \quad (2.2.59)$$

where $\bar{\rho}$ is the average density of the C.V. The right-hand sides of Eqns. (2.2.42), (2.2.45), (2.2.51), (2.2.55) and (2.2.59) are summed up, and all terms of the occurring equation are divided by the cuboid volume $\Delta V = \Delta x \Delta y \Delta z$ to obtain:

$$\begin{aligned} \frac{\partial}{\partial t} \left[\bar{\rho} \mathbf{v} \left(1 - \frac{\delta \Psi}{\Delta x \Delta y \Delta z} \right) \right] &+ \frac{\left[(\rho v)_x^* \bar{v}_x^* \right]_{x+\Delta x} - \left[(\rho v)_x^* \bar{v}_x^* \right]_x}{\Delta x} + \frac{\left[(\rho v)_y^* \bar{v}_y^* \right]_{y+\Delta y} - \left[(\rho v)_y^* \bar{v}_y^* \right]_y}{\Delta y} \\ &+ \frac{\left[(\rho v)_z^* \bar{v}_z^* \left(1 - \frac{\delta s}{\Delta x \Delta y} \right) \right]_{z+\Delta z} - \left[(\rho v)_z^* \bar{v}_z^* \right]_z}{\Delta z} + \frac{\left[(\rho v)_z^* \bar{v}_z^* \delta a \right]_s}{\Delta x \Delta y \Delta z} = \\ &-\mathbf{i} \frac{\left[(\bar{p})_{x+\Delta x} - (\bar{p})_x \right]}{\Delta x} - \mathbf{j} \frac{\left[(\bar{p})_{y+\Delta y} - (\bar{p})_y \right]}{\Delta y} + \mathbf{n} \frac{\bar{p} (2\pi \delta r \delta l)_{s_{ver}}}{\Delta x \Delta y} \frac{1}{\Delta z} \\ &-\mathbf{k} \frac{\left\{ \left[\bar{p} \left(1 - \frac{\delta s}{\Delta x \Delta y} \right) \right]_{z+\Delta z} - (\bar{p})_z \right\}}{\Delta z} - \mathbf{k} \frac{\left(\bar{p} \frac{\delta a}{\Delta x \Delta y} \right)_{s_{hor}}}{\Delta z} \\ &+\mathbf{i} \cdot \frac{\left[(\bar{\mathbf{T}})_{x+\Delta x} - (\bar{\mathbf{T}})_x \right]}{\Delta x} + \mathbf{j} \cdot \frac{\left[(\bar{\mathbf{T}})_{y+\Delta y} - (\bar{\mathbf{T}})_y \right]}{\Delta y} + \mathbf{n} \cdot \frac{[\bar{\mathbf{T}} 2\pi \delta r \delta l]_{s_{ver}}}{\Delta x \Delta y} \frac{1}{\Delta z} \\ &+\mathbf{k} \cdot \frac{\left\{ \left[\bar{\mathbf{T}} \left(1 - \frac{\delta s}{\Delta x \Delta y} \right) \right]_{z+\Delta z} - (\bar{\mathbf{T}})_z \right\}}{\Delta z} + \mathbf{k} \cdot \frac{\left(\bar{\mathbf{T}} \frac{\delta a}{\Delta x \Delta y} \right)_{s_{hor}}}{\Delta z} \\ &+ \left[\bar{\rho} \mathbf{g} \left(1 - \frac{\delta \Psi}{\Delta x \Delta y \Delta z} \right) \right] \end{aligned} \quad (2.2.60)$$

We choose the dimensions of the straw much smaller in comparison to these of the cuboid. At the limit where δr , δa , and $\delta s \rightarrow 0$ (and, by extension, $\delta \mathbf{v} \rightarrow 0$), Eq. (2.2.60) becomes:

$$\begin{aligned}
& \frac{\partial(\bar{\rho}\bar{\mathbf{v}})}{\partial t} + \frac{\left[(\rho v)^* \bar{v}_x^*\right]_{x+\Delta x} - \left[(\rho v)^* \bar{v}_x^*\right]_x}{\Delta x} + \frac{\left[(\rho v)^* \bar{v}_y^*\right]_{y+\Delta y} - \left[(\rho v)^* \bar{v}_y^*\right]_y}{\Delta y} \\
& + \frac{\left[(\rho v)^* \bar{v}_z^*\right]_{z+\Delta z} - \left[(\rho v)^* \bar{v}_z^*\right]_z}{\Delta z} + \frac{\left[(\rho v)^* \bar{v}_z^* \delta a\right]_s}{\Delta x \Delta y \Delta z} = \\
& -\mathbf{i} \frac{\left[(\bar{p})\right]_{x+\Delta x} - (\bar{p})|_x}{\Delta x} - \mathbf{j} \frac{\left[(\bar{p})\right]_{y+\Delta y} - (\bar{p})|_y}{\Delta y} - \mathbf{k} \frac{\left[(\bar{p})\right]_{z+\Delta z} - (\bar{p})|_z}{\Delta z} \\
& + \mathbf{i} \cdot \frac{\left[(\bar{\mathbf{T}})\right]_{x+\Delta x} - (\bar{\mathbf{T}})|_x}{\Delta x} + \mathbf{j} \cdot \frac{\left[(\bar{\mathbf{T}})\right]_{y+\Delta y} - (\bar{\mathbf{T}})|_y}{\Delta y} + \mathbf{k} \cdot \frac{\left[(\bar{\mathbf{T}})\right]_{z+\Delta z} - (\bar{\mathbf{T}})|_z}{\Delta z} \\
& + \bar{\rho} \mathbf{g}
\end{aligned} \tag{2.2.61}$$

As the dimensions of the straw tend to zero, the terms

$$\frac{\delta s}{\Delta x \Delta y} \text{ and } \frac{\delta \mathbf{v}}{\Delta x \Delta y \Delta z} \tag{2.2.62}$$

that appear in Eq. (2.2.60) are omitted from Eq. (2.2.61). For the same reason, the pressure and fiction force terms evaluated at the vertical straw side walls:

$$\mathbf{n} \cdot \frac{\left[\bar{p}(2\pi\delta r\delta l)\right]_{s_{\text{ver}}}}{\Delta x \Delta y} \frac{1}{\Delta z} \text{ and } \mathbf{n} \cdot \frac{\left[\bar{\mathbf{T}}2\pi\delta r\delta l\right]_{s_{\text{ver}}}}{\Delta x \Delta y} \frac{1}{\Delta z} \tag{2.2.63}$$

become negligible as the straw dimensions tend to zero and are also excluded from Eq. (2.2.61). Nevertheless, the terms in Eq. (2.2.60) that relate to the straw opening δa require special attention; these are:

$$\frac{\left(\bar{\rho}\bar{\mathbf{v}}_z\delta a\right)|_s}{\Delta x \Delta y \Delta z}, \frac{\mathbf{k}\left(\bar{p}\frac{\delta a}{\Delta x \Delta y}\right)|_s}{\Delta z}, \text{ and } \frac{\mathbf{k}\cdot\left(\bar{\mathbf{T}}\frac{\delta a}{\Delta x \Delta y}\right)|_s}{\Delta z} \tag{2.2.64}$$

Previously (Eq. (2.2.15)), it is assumed Q and $\dot{m}$ are different from zero as $\delta a \rightarrow 0$. Contrary to the source/sink outflow, however, both $\bar{p}\delta a$ and $\bar{\mathbf{T}}\delta \mathbf{a}$ tend to zero as $\delta a \rightarrow 0$. Therefore, only the first among the terms in Eq. (2.2.64) remains. We proceed to take the limit of Eq. (2.2.61) as Δx , Δy , and Δz tend to zero. In such case, the momentum outflow term becomes:

$$\lim_{\Delta x, \Delta y, \Delta z \rightarrow 0} \frac{(\bar{\rho} \bar{\mathbf{v}} \bar{v}_z \delta a)|_s}{\Delta x \Delta y \Delta z} = \frac{\mathbf{v} \dot{m}}{\Delta x \Delta y \Delta z} = \mathbf{v} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (2.2.65)$$

Moreover, from the expressions of the mean value theorem (Eqns. (2.2.5), (2.2.44), (2.2.47), (2.2.49), (2.2.53) and (2.2.57)), at the limit of Δx , Δy , and $\Delta z \rightarrow 0$, the average values become point values:

$$\begin{aligned} \bar{\rho} &\rightarrow \rho(x, y, z), \bar{\mathbf{v}} \rightarrow \mathbf{v}(x, y, z), (\rho \mathbf{v})^* \rightarrow (\rho \mathbf{v})(x, y, z), \bar{v}_n^* \rightarrow v_n(x, y, z), \\ \bar{p} &\rightarrow p(x, y, z), \text{ and } \bar{\mathbf{T}} \rightarrow \mathbf{T}(x, y, z) \end{aligned} \quad (2.2.66)$$

thus, the overbar ($\bar{\cdot}$) and asterisk ($\cdot^*$) notations are dropped. Finally, adopting the vector notation for the divergence and gradient operators gives:

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \mathbf{v} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) + \nabla p - \nabla \cdot \mathbf{T} - \rho \mathbf{g} = 0 \quad (2.2.67)$$

Opening the first term of Eq. (2.2.67) yields:

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \frac{\partial \rho}{\partial t} + \mathbf{v} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) + (\rho \mathbf{v}) \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla (\rho \mathbf{v}) + \nabla p - \nabla \cdot \mathbf{T} - \rho \mathbf{g} = 0 \quad (2.2.68)$$

After rearranging and setting $\mathbf{v}$ as the common factor, we get:

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \left[\overbrace{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) + \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L)}^{=0 \text{ from continuity equation}} \right] + (\rho \mathbf{v}) \nabla \cdot \mathbf{v} + \nabla p - \nabla \cdot \mathbf{T} - \rho \mathbf{g} = 0 \quad (2.2.69)$$

The terms within the square brackets in Eq. (2.2.69) are equal to zero following from the continuity equation (Eq. (2.2.20)); thus, Eq. (2.2.69) becomes:

$$\rho \frac{\partial \mathbf{v}}{\partial t} + (\rho \mathbf{v}) \nabla \cdot \mathbf{v} + \nabla p - \nabla \cdot \mathbf{T} - \rho \mathbf{g} = 0 \quad (2.2.70)$$

We first consider a steady-state condition, where p and $\mathbf{T}$ are not a function of time i.e.:

$$p = p_0(\mathbf{x}), \mathbf{T} = \mathbf{T}_0(\mathbf{x}) \quad (2.2.71)$$

We substitute into Eq. (2.2.70) to get the momentum conservation equation at steady state:

$$(\rho_0 \mathbf{v}_0) \nabla \cdot \mathbf{v}_0 + \nabla p_0 - \nabla \cdot \mathbf{T}_0 - \rho_0 \mathbf{g} = 0 \quad (2.2.72)$$

Similarly, using Eq. (2.2.23), we proceed to write the momentum conservation equation for the transient state:

$$(\rho_0 + \rho') \frac{\partial(\mathbf{v}_0 + \mathbf{v}')}{\partial t} + [(\rho_0 + \rho')(\mathbf{v}_0 + \mathbf{v}')] \nabla \cdot (\mathbf{v}_0 + \mathbf{v}') + \nabla(p_0 + p') - \nabla \cdot (\boldsymbol{\tau}_0 + \boldsymbol{\tau}') - (\rho_0 + \rho') \mathbf{g} = 0 \quad (2.2.73)$$

or,

$$\begin{aligned} \rho_0 \frac{\partial \mathbf{v}'}{\partial t} + \rho' \frac{\partial \mathbf{v}'}{\partial t} + (\rho_0 \mathbf{v}_0) \nabla \cdot \mathbf{v}_0 + (\rho_0 \mathbf{v}_0) \nabla \cdot \mathbf{v}' + (\rho_0 \mathbf{v}') \nabla \cdot \mathbf{v}_0 \\ + (\rho_0 \mathbf{v}') \nabla \cdot \mathbf{v}' + (\rho' \mathbf{v}_0) \nabla \cdot \mathbf{v}_0 + (\rho' \mathbf{v}_0) \nabla \cdot \mathbf{v}' + (\rho' \mathbf{v}') \nabla \cdot \mathbf{v}_0 \\ + (\rho' \mathbf{v}') \nabla \cdot \mathbf{v}' + \nabla(p_0 + p') - \nabla \cdot (\boldsymbol{\tau}_0 + \boldsymbol{\tau}') - (\rho_0 + \rho') \mathbf{g} = 0 \end{aligned} \quad (2.2.74)$$

We simplify Eq. (2.2.74) using Eq. (2.2.72) to obtain:

$$\begin{aligned} \rho_0 \frac{\partial \mathbf{v}'}{\partial t} + \rho' \frac{\partial \mathbf{v}'}{\partial t} + (\rho_0 \mathbf{v}_0) \nabla \cdot \mathbf{v}' + (\rho_0 \mathbf{v}') \nabla \cdot \mathbf{v}_0 + (\rho_0 \mathbf{v}') \nabla \cdot \mathbf{v}' + (\rho' \mathbf{v}_0) \nabla \cdot \mathbf{v}_0 \\ + (\rho' \mathbf{v}_0) \nabla \cdot \mathbf{v}' + (\rho' \mathbf{v}') \nabla \cdot \mathbf{v}_0 + (\rho' \mathbf{v}') \nabla \cdot \mathbf{v}' + \nabla p' - \nabla \cdot \boldsymbol{\tau}' - \rho' \mathbf{g} = 0 \end{aligned} \quad (2.2.75)$$

The individual terms of Eq. (2.2.75) are examined in terms of scale:

$$\begin{aligned} \text{i.} \quad \rho_0 \frac{\partial \mathbf{v}'}{\partial t} &\sim \rho_0 \frac{\|\mathbf{v}\| c_f}{L} & \text{ii.} \quad \rho' \frac{\partial \mathbf{v}'}{\partial t} &\sim \rho_0 \frac{\|\mathbf{v}\|^2}{L} \\ \text{iii.} \quad (\rho_0 \mathbf{v}_0) \nabla \cdot \mathbf{v}' &\sim \rho_0 \frac{\|\mathbf{v}\|^2}{L} & \text{iv.} \quad \nabla \cdot \boldsymbol{\tau}' &\sim \mu \frac{\|\mathbf{v}\|}{L^2} \end{aligned}$$

Note: The remaining advective acceleration terms scale similarly to term (iii); thus, are not examined separately. Also, viscous term (iv) is scaled with L (i.e., the probing wavelength), and not e.g., Kolmogorov's scale, since pressure waves do not interact with features smaller than their wavelength.

The relative magnitude of the temporal acceleration (term (i)) to the viscous stresses (term (iv)) gives:

$$\frac{\rho_0 \frac{\partial \mathbf{v}'}{\partial t}}{\nabla \cdot \boldsymbol{\tau}'} \sim \frac{\rho_0 c_f L}{\mu} = \frac{L c_f}{\nu} = \text{Re}_+ = \text{Re} \cdot Ma \quad (2.2.76)$$

where Re_+ is the transient or unsteady Reynolds number that is equal to the product of the conventional Reynolds number Re with the Mach number Ma , μ is the dynamic viscosity and $\nu (= \mu/\rho_0)$ is the kinematic viscosity. The kinematic viscosity for water is of the order $\nu \sim 10^{-6} \text{ m}^2/\text{s}$, while a representative value for the wave speed is $c_f \sim 10^3 \text{ m/s}$. Viscous effects become

important when $\text{Re}_+ \sim 1$ or for $L \sim \text{Re}_+ \nu / c_f = 1 \times 10^{-6} / 10^{-3} = 10^{-9}$ m, a scale that is multiple orders smaller than the wavelengths employed in the framework of this thesis. Note that the inverse scale of Eq. (2.2.76) is revisited again in Chapter 5 in the context of the time reversibility of the governing equations. Another term that needs to be evaluated in comparison to the temporal acceleration term is that of the gravity forces:

$$\frac{\rho_0 \frac{\partial \mathbf{v}'}{\partial t}}{\rho' \mathbf{g}} \sim \frac{\rho_0 \frac{\|\mathbf{v}\| c_f}{L}}{\rho_0 \frac{\|\mathbf{v}\|}{c_f} \|\mathbf{g}\|} = \frac{c_f^2}{\|\mathbf{g}\| L} \quad (2.2.77)$$

Considering typical values from water supply pipes, $L \sim 100$ to 500 mm, $c_f \sim 300$ to 1500 m/s, then $c_f^2 / \|\mathbf{g}\| L$ ranges from 18 to 2.25×10^6 . Therefore, $c_f^2 / \|\mathbf{g}\| L \gg 1$, implying that $\|\rho' \mathbf{g}\| \ll \|\rho_0 \partial \mathbf{v}' / \partial t\|$. The acceleration of gravity $\mathbf{g}$ is a constant, while the product $\|\mathbf{g}\| L$ is comparatively much smaller than c_f^2 . Consequently, the effect of gravity is negligible. Finally, term (i) is larger than the other terms by $c_f^2 / \|\mathbf{v}\| = 1/Ma$ and remains, while all other terms (i.e., terms (ii) and (iii)) are neglected. The momentum conservation equation becomes:

$$\rho_0 \frac{\partial \mathbf{v}'}{\partial t} + \nabla p' = 0 \quad (2.2.78)$$

Finally, after dropping the prime notation ($\cdot'$):

$$\boxed{\frac{\partial(\rho_0 \mathbf{v})}{\partial t} + \nabla p = 0} \quad (2.2.79)$$

Equation (2.2.79) constitutes the momentum conservation equation for the stated problem. In the next section, the mass and momentum conservation equations are combined to produce the wave equation.

2.2.3 Wave equation

To derive the wave equation that includes the effect of the leak positioned at $\mathbf{x}_L$, we first take the time derivative of the mass conservation equation (Eq. (2.2.39)):

$$\frac{1}{c_f^2} \frac{\partial^2 p}{\partial t^2} + \frac{\partial \dot{m}}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_L) + \nabla \cdot \frac{\partial(\rho_0 \mathbf{v})}{\partial t} = 0 \quad (2.2.80)$$

Then, we take the divergence of the negated momentum conservation equation (Eq. (2.2.79)):

$$-\nabla \cdot \frac{\partial(\rho_0 \mathbf{v})}{\partial t} - \nabla^2 p = 0 \quad (2.2.81)$$

Equations (2.2.80) and (2.2.81) are added together to obtain:

$$\frac{1}{c_f^2} \frac{\partial^2 p}{\partial t^2} - \nabla^2 p = -\frac{\partial \dot{m}}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (2.2.82)$$

or,

$$\boxed{\frac{\partial^2 p}{\partial t^2} - c_f^2 \nabla^2 p = -c_f^2 \frac{\partial \dot{m}}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_L)} \quad (2.2.83)$$

In this thesis we employ wave time-reversal techniques to locate passive sources as represented by leaks. Hence, it is instructive to examine the form of Eq. (2.2.83) for $-t$:

$$\frac{\partial^2 p}{\partial (-t)^2} - c_f^2 \nabla^2 p = -c_f^2 \frac{\partial \dot{m}}{\partial (-t)} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (2.2.84)$$

or,

$$\frac{\partial^2 p}{\partial t^2} - c_f^2 \nabla^2 p = c_f^2 \frac{\partial \dot{m}}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (2.2.85)$$

Although the left-hand side of Eqns. (2.2.83) and (2.2.85) are identical, the right-hand sides differ by a minus sign; thus, Eq. (2.2.83) is not time reversible. However, the fact does not prohibit the use of time reversal techniques to localize leaks in water supply pipes. In such case, a known acoustic pulse is emitted from a point source, propagates along the pipe, and is dispersed due the pipe walls. As soon as the pulse meets the leak, it is scattered, essentially rendering the leak into a passive acoustic source. The recorded dispersed response due to the leak is then time-reversed (i.e., signals that arrive first are emitted last) and omitted from the same position. Since the wave equation is symmetric in time, except for the source/leak term, the time-reversed pulse will converge to the source location i.e., the leak. There, the source should also act in reverse to absorb the pulse but, instead, the focused pulse diverges again. Thus, the size of the focal spot when recreating the original pulse is bounded by the diffraction limit, defined as half the pulse wavelength. Regardless, when applying high-frequency probing waves with wavelengths comparable to the pipe diameter, the position of the leak can ideally be identified within a distance equal to the pipe radius.

2.2.4 Energy equation

In addition to the wave equation, we may obtain the acoustical mechanical energy equation after appropriately manipulating the mass and momentum conservation equations. The mass conservation equation (Eq. (2.2.39)) is multiplied with p/ρ_0 :

$$\frac{1}{\rho_0 c_f^2} \frac{\partial(p^2/2)}{\partial t} + \frac{p}{\rho_0} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) + \frac{p}{\rho_0} \nabla \cdot (\rho_0 \mathbf{v}) = 0 \quad (2.2.86)$$

while the momentum conservation equation (Eq. (2.2.79)) is multiplied with $\mathbf{v}$:

$$\rho_0 \frac{\partial(\mathbf{v}^2/2)}{\partial t} + \mathbf{v} \cdot \nabla p = 0 \quad (2.2.87)$$

Finally, Eqns. (2.2.86) and (2.2.87) are added together to obtain:

$$\boxed{\frac{1}{2\rho_0 c_f^2} \frac{\partial p^2}{\partial t} + \frac{\rho_0}{2} \frac{\partial \mathbf{v}^2}{\partial t} + \nabla \cdot (p\mathbf{v}) = -\frac{p}{\rho_0} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L)} \quad (2.2.88)$$

Alternatively, Eq. (2.2.88) may be written as:

$$\frac{\partial e}{\partial t} + \nabla \cdot \mathbf{I} = -\frac{p}{\rho_0} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (2.2.89)$$

where e is the acoustic energy density and $\mathbf{I}$ is the energy flux defined, respectively, as:

$$e = \overbrace{\frac{p^2}{2\rho_0 c_f^2}}^{\text{Potential energy density}} + \overbrace{\frac{\rho_0 \mathbf{v}^2}{2}}^{\text{Kinetic energy density}} \quad (2.2.90)$$

and

$$\mathbf{I} = p\mathbf{v} \quad (2.2.91)$$

The acoustical mechanical energy conservation law (Eq. (2.2.89)) may be evaluated for the fixed volume of cylindrical pipe with length l and radius R as:

$$\frac{\partial}{\partial t} \iiint_V e dV + \iiint_V \nabla \cdot \mathbf{I} dV = - \iiint_V \frac{p}{\rho_0} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) dV \quad (2.2.92)$$

Using the divergence theorem on Eq. (2.2.89) leads to:

$$\frac{\partial}{\partial t} \iiint_V e dV + \iint_A \mathbf{n} \cdot \mathbf{I} dA = - \iiint_V \frac{p}{\rho_0} m \delta_3(\mathbf{x} - \mathbf{x}_L) dV \quad (2.2.93)$$

Equation (2.2.93) is evaluated term-by-term, starting from the outmost left:

$$\begin{aligned} \frac{\partial}{\partial t} \iiint_V e dV &= \frac{\partial}{\partial t} \int_0^l \int_0^R \int_0^{2\pi} e r d\theta dr dx = \\ &= \frac{\partial}{\partial t} \int_0^L 2\pi \bar{e} r dr dx = \frac{\partial}{\partial t} \int_0^L \pi R^2 \tilde{\bar{e}} dx = (\pi L R^2) \frac{\partial \hat{\tilde{\bar{e}}}}{\partial t} \end{aligned} \quad (2.2.94)$$

where the overbar ($\bar{\cdot}$) denotes an average along θ , the tilde ($\tilde{\cdot}$) an average along R , and the hat ($\hat{\cdot}$) along L . For the second term of Eq. (2.2.93) we evaluate the fluxes across the surfaces of the considered C.V. These surfaces are the pipe cross-sections at $x = 0$ and $x = l$, as well as the cylindrical surface at $r = R$. In particular:

$$\iint_A \mathbf{n} \cdot \mathbf{I} dA = \int_{x=0} \mathbf{i} \cdot p \mathbf{v} dA - \int_{x=l} \mathbf{i} \cdot p \mathbf{v} dA + \int_{r=R} \mathbf{e}_r \cdot p \mathbf{v} dA \quad (2.2.95)$$

The radial unit vector $\mathbf{e}_r$ may be decomposed into the sum of the unit vectors along the y and x -axes, multiplied by $\cos\theta$ and $\sin\theta$, respectively:

$$\iint_A \mathbf{n} \cdot \mathbf{I} dA = \int_{x=0} \mathbf{i} \cdot p \mathbf{v} dA - \int_{x=l} \mathbf{i} \cdot p \mathbf{v} dA + \int_{x=0} (\mathbf{j} \cos\theta + \mathbf{k} \sin\theta) \cdot p \mathbf{v} dA \quad (2.2.96)$$

Calculating the vector dot products that appear in Eq. (2.2.96) yields:

$$\iint_A \mathbf{n} \cdot \mathbf{I} dA = \int_{x=0} p v_x dA - \int_{x=l} p v_x dA + \int_{r=R} p (v_y \cos\theta + v_z \sin\theta) dA \quad (2.2.97)$$

However, since $v_y = v_r \cos\theta$ and $v_z = v_r \sin\theta$, we obtain:

$$\iint_A \mathbf{n} \cdot \mathbf{I} dA = \int_{x=0} p v_x dA - \int_{x=l} p v_x dA + \int_{r=R} p v_r \left[(\cos\theta)^2 + (\sin\theta)^2 \right] dA \quad (2.2.98)$$

After evaluating the integrals and making the necessary simplifications:

$$\iint_A \mathbf{n} \cdot \mathbf{I} dA = \tilde{\bar{p}} \tilde{\bar{v}}_x A \Big|_{x=0} - \tilde{\bar{p}} \tilde{\bar{v}}_x A \Big|_{x=l} + \widehat{\bar{p}} \widehat{\bar{v}}_r A \Big|_{r=R} \quad (2.2.99)$$

Substituting the appropriate areas at each evaluated surface yields:

$$\iint_A \mathbf{n} \cdot \mathbf{I} dA = \pi R^2 \left[\left(\tilde{\bar{p}} \tilde{\bar{v}}_x \right) \Big|_{x=0} - \left(\tilde{\bar{p}} \tilde{\bar{v}}_x \right) \Big|_{x=l} \right] + 2\pi R L \widehat{\bar{p}} \widehat{\bar{v}}_r A \Big|_{r=R} \quad (2.2.100)$$

The right-hand term of Eq. (2.2.93) becomes:

$$\iiint_V \frac{p}{\rho_0} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) dV = \int_0^l \int_0^R \int_0^{2\pi} \frac{p}{\rho_0} \dot{m} \delta_3(\mathbf{x} - \mathbf{x}_L) r d\theta dr dx = - \frac{p \dot{m}}{\rho_0} \Big|_{\mathbf{x}=\mathbf{x}_L} \quad (2.2.101)$$

The final expression of Eq. (2.2.88) after Eqns. (2.2.94), (2.2.100) and (2.2.101) becomes:

$$\left(\pi l R^2 \right) \frac{\partial \hat{\bar{e}}}{\partial t} + 2\pi l R \hat{p} \hat{v}_r + \pi R^2 \left[\left(\hat{p} \hat{v}_x \right) \Big|_{x=l} - \left(\hat{p} \hat{v}_x \right) \Big|_{x=0} \right] = - \frac{p \dot{m}}{\rho_0} \Big|_{\mathbf{x}=\mathbf{x}_L} \quad (2.2.102)$$

In Eq. (2.2.102), the first, outmost left term, accounts for the energy accumulated in the system due to compressibility (in both the fluid and pipe wall), as well as the kinetic energy rate due to the flow velocity. The second term is the work rate done by the pressure wave on the pipe wall perimeter. More importantly, through this term the fluid equations are coupled with the pipe wall equations that are derived in the next section. The third term is the work rate is the rate of work done by the pressure wave over at the pipe cross-section. The right-hand side of Eq. (2.2.102) accounts to the rate of energy gained/lost in the system due to the mass source/sink at $\mathbf{x}_L$. Finally, it should be pointed out that Eq. (2.2.102) indicates no fluxes in the azimuthal direction, since the velocity θ -component does not cross the surfaces of the assumed C.V. Hence, the velocity θ -component follows a spiral motion along the pipe.

2.3 Pipe wall equations

2.3.1 Linear momentum equations for a pipe element of finite thickness

The general form of the momentum conservation equation for a material C.V., similar to that defined by the boundaries of the pipe wall element, is written as:

$$\frac{D}{Dt} \int_{V_p} \rho_p \mathbf{v}_p dV_p = \sum \mathbf{F}_{\text{ext},p} \quad (2.3.1)$$

where ρ_p is the density of the pipe wall material, V_p is the pipe wall control volume, $\mathbf{v}_p$ is the velocity of the pipe wall C.V. particles, and $\mathbf{F}_{\text{ext},p}$ are the external forces to the C.V. The considered pipe wall C.V. extends in the x -direction by Δx , has a thickness h in the r -direction and is contained within an angle $\Delta\theta$ in the θ -direction (Figure 2.2.1). Taking that into account, the integration over the volume of the C.V. on the left-hand side of Eq. (2.3.1) becomes:

$$\frac{D}{Dt} \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p \mathbf{v}_p r dr d\theta dx = \sum \mathbf{F}_{\text{ext},p} \quad (2.3.2)$$

In this thesis, solely axisymmetric vibrations are of interest; thus, we assume that azimuthal displacements are equal to zero. By extend, the velocity component in the θ -direction $v_{p,\theta} = 0$ and the velocity vector components become:

$$\mathbf{v}_p = (v_{p,r}, 0, v_{p,x}) = v_{p,r} \mathbf{e}_r + 0 \mathbf{e}_\theta + v_{p,x} \mathbf{i} \quad (2.3.3)$$

The momentum conservation constitutes a vector equation, and we proceed to derive its component in the r -direction first:

$$\frac{D}{Dt} \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p v_{p,r} \mathbf{e}_r r dr d\theta dx = \sum F_r \mathbf{e}_r \quad (2.3.4)$$

In principle, when considering a cylindrical coordinate system, the time derivative of the unit vector in the r -direction produces the unit vector in the θ -direction i.e., $d\mathbf{e}_r/dt = \mathbf{e}_\theta$. In this case, however, we already assumed that the displacement field is axisymmetric; thus, $d\mathbf{e}_r/dt = 0$. Since the unit vector $\mathbf{e}_r$ is not a function of time, it is excluded from the time derivative expression. In addition, we integrate across the pipe wall thickness to obtain:

$$\mathbf{e}_r \frac{D}{Dt} \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \bar{\rho}_p \bar{v}_{p,r} R h d\theta dx = \sum F_r \mathbf{e}_r \quad (2.3.5)$$

where $\bar{\rho}_p$ is the average pipe material density across the pipe wall element thickness. According to the mean value theorem, there is a point along the radius $(R + B_r h)$ with $B_r \in [-1/2, 1/2]$ where:

$$\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p r dr = \rho_p (R + B_r h) R h \quad (2.3.6)$$

It follows that the average density is defined as:

$$\bar{\rho}_p = \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r dr} = \rho_p (R + B_r h) \quad (2.3.7)$$

Moreover, $\bar{v}_{p,r}$ is the mass-weighted average radial velocity evaluated across the thickness of the pipe wall element. By the mean value theorem, there is a point along the radius $(R + \Gamma_r h)$ with $\Gamma_r \in [-1/2, 1/2]$ where:

$$\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p v_{p,r} r dr = \bar{\rho}_p \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} v_{p,r} r dr = \rho_p (R + B_r h) v_{p,r} (R + \Gamma_r h) R h \quad (2.3.8)$$

Thus, the mass-weighted average radial velocity along the radial direction $\bar{v}_r$, is expressed as:

$$\bar{v}_{p,r} = \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p v_{p,r} r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p r dr} = v_{p,r} (R + \Gamma_r h) \quad (2.3.9)$$

Using the mean value theorem, from Eq. (2.3.5) we obtain:

$$\mathbf{e}_r \frac{D}{Dt} (\bar{\rho}_p \bar{v}_{p,r} R \Delta r \Delta \theta \Delta x) = \sum F_r \mathbf{e}_r \quad (2.3.10)$$

We proceed to evaluate the right-hand side term of Eq. (2.3.10). Referring to the pipe wall C.V., there are the following force components applied along the r -direction:

- i. The τ_{xr} shear stress on the $r\theta$ -plane at two cross-sections: x and $x + \Delta x$.
- ii. The normal stress σ_{rr} on the $x\theta$ -plane, at the inner and outer pipe wall radii: $R - h/2$ and $R + h/2$ respectively.
- iii. The vertical component of the hoop stress $\sigma_{\theta\theta}$ on the rx -plane at θ and $\theta + \Delta\theta$. To obtain its radial component, $\sigma_{\theta\theta}$ should be multiplied by $\sin\Delta\theta/2$. For a small angle $\Delta\theta$, the value of the sin function is approximated by: $\sin\Delta\theta/2 \approx \Delta\theta/2$.

Before proceeding to evaluate the force term, it is useful to establish the expressions for the face areas on the considered pipe wall element C.V.:

- i. Face on the $r\theta$ -plane (x is constant; marked as ① in Figure 2.3.1):

$$\int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \int_{\theta}^{\theta+\Delta\theta} r d\theta dr = R \Delta \theta \Delta r \quad (2.3.11)$$

- ii. Face on the rx -plane (θ is constant; marked as ② in Figure 2.3.1):

$$\int_x^{x+\Delta x} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} dr dx = \Delta r \Delta x \quad (2.3.12)$$

- iii. Finally, for a face on the $x\theta$ -plane (r is constant; marked as ③ in Figure 2.3.1):

$$\int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} r d\theta dx = r \Delta \theta \Delta x \quad (2.3.13)$$

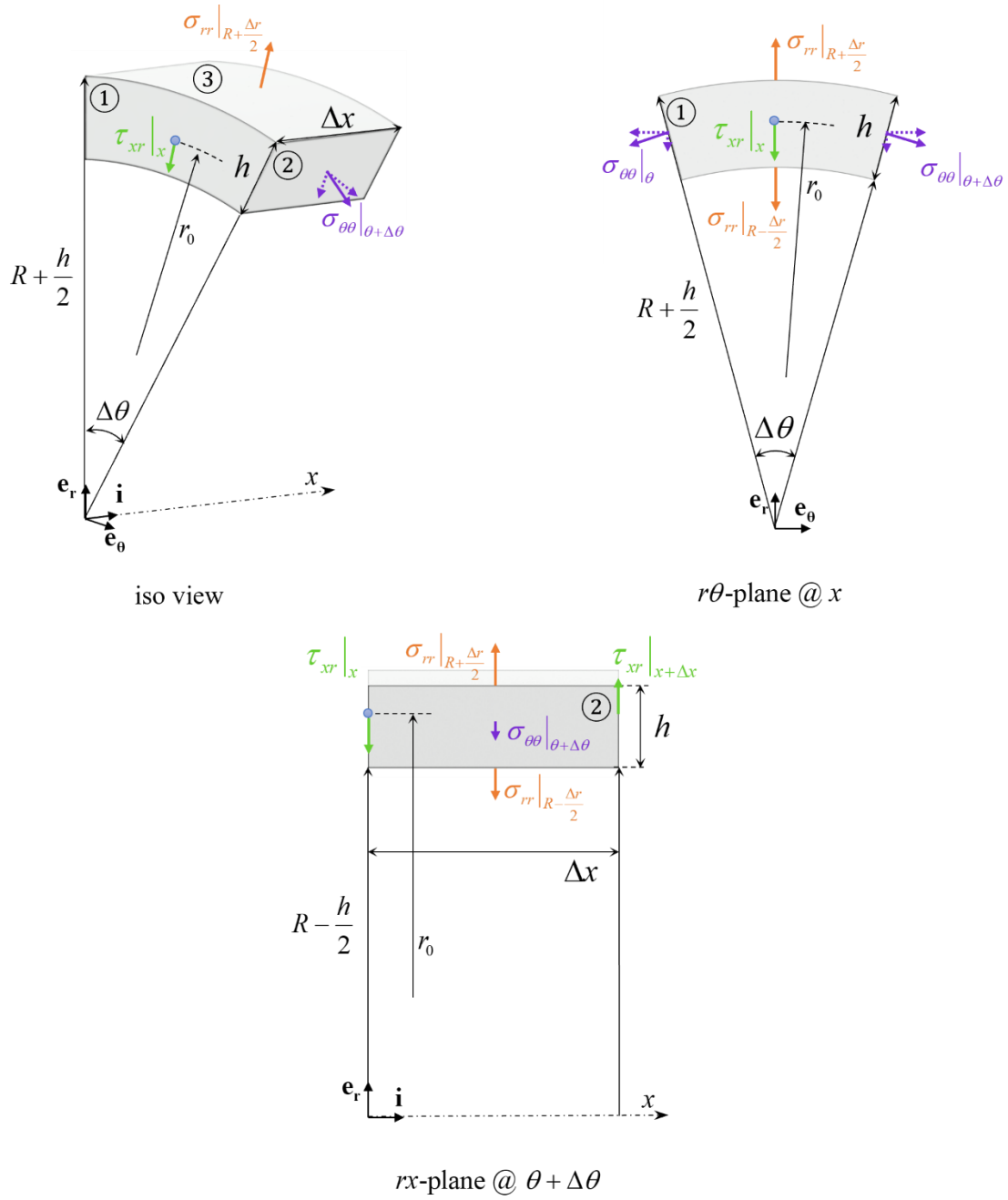


Figure 2.3.1: Stress components applied on the C.V. appearing in the linear momentum equation along the r -direction. Note that r_0 represents the radial coordinate of the pipe element centroid.

To evaluate the force applied on each face, we integrate the stresses over the area of the pertinent face. Therefore, the sum of forces in the radial direction is:

$$\begin{aligned} \Sigma F_r \mathbf{e}_r = & \left(\int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \tau_{xr} r dr d\theta \right) \Big|_{x+\Delta x} - \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \tau_{xr} r dr d\theta \Big|_x \\ & + \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \sigma_{rr} r d\theta dx \Big|_{R+\frac{h}{2}} - \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \sigma_{rr} r d\theta dx \Big|_{R-\frac{h}{2}} \\ & - \int_x^{x+\Delta x} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \sigma_{\theta\theta} dr dx \Big|_{\theta+\Delta\theta} - \int_x^{x+\Delta x} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \sigma_{\theta\theta} dr dx \Big|_{\theta} \Big) \mathbf{e}_r \end{aligned} \quad (2.3.14)$$

Making use of the area expressions (Eq. (2.3.11) to (2.3.13)), we proceed to evaluate the integrals in Eq. (2.3.14):

$$\begin{aligned} \Sigma F_r \mathbf{e}_r = & \left(\bar{\tau}_{xr} Rh \Delta\theta \Big|_{x+\Delta x} - \bar{\tau}_{xr} Rh \Delta\theta \Big|_x \right. \\ & + \bar{\sigma}_{rr} r \Delta\theta \Delta x \Big|_{R+\frac{h}{2}} - \bar{\sigma}_{rr} r \Delta\theta \Delta x \Big|_{R-\frac{h}{2}} \\ & \left. - \bar{\sigma}_{\theta\theta} \frac{\Delta\theta}{2} h \Delta x \Big|_{\theta+\Delta\theta} - \bar{\sigma}_{\theta\theta} \frac{\Delta\theta}{2} h \Delta x \Big|_{\theta} \right) \mathbf{e}_r \end{aligned} \quad (2.3.15)$$

where the overbars denote average stress quantities. For instance, according to the mean value theorem, there is a point across the pipe wall ($R + E_r h$) with $E_r \in [-1/2, 1/2]$ where:

$$\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \tau_{xr} r dr = \tau_{xr} (R + E_r h) Rh \quad (2.3.16)$$

Thus, the average $\bar{\tau}_{rx}$ shear stress across the pipe thickness is:

$$\bar{\tau}_{xr} = \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \tau_{xr} r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r dr} = \tau_{xr} (R + E_r h) \quad (2.3.17)$$

Similarly, for the $\sigma_{\theta\theta}$ hoop stress there is a point across the pipe thickness ($R + Z_r h$) with $Z_r \in [-1/2, 1/2]$ where:

$$\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \sigma_{\theta\theta} r dr = \sigma_{\theta\theta} (R + Z_r h) Rh \quad (2.3.18)$$

The average $\bar{\sigma}_{\theta\theta}$ hoop stress is, therefore, defined as:

$$\bar{\sigma}_{\theta\theta} = \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \sigma_{\theta\theta} r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r dr} = \sigma_{\theta\theta} (R + Z_r \Delta r) \quad (2.3.19)$$

Finally, for the σ_{rr} normal stresses there is a point $(r, \theta + H_\theta \Delta\theta, x + H_x \Delta x)$ with $H_\theta, H_x \in [0, 1]$ where:

$$\int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \sigma_{rr} r d\theta dx = \sigma_{rr} (r, \theta + H_\theta \Delta\theta, x + H_x \Delta x) r \Delta\theta \Delta x \quad (2.3.20)$$

Hence, the average $\bar{\sigma}_{rr}$ normal stress over the integration surface is:

$$\bar{\sigma}_{rr} = \frac{\int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \sigma_{rr} r d\theta dx}{\int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} r d\theta dx} = \sigma_{rr} (r, \theta + H_\theta \Delta\theta, x + H_x \Delta x) \quad (2.3.21)$$

Equation (2.3.10) is re-written after substituting its right-hand side with the result of Eq. (2.3.15):

$$\begin{aligned} \mathbf{e}_r \frac{D}{Dt} (\bar{\rho}_p \bar{v}_{p,r} R \Delta r \Delta \theta \Delta x) &= (\bar{\tau}_{xr} R h \Delta \theta \Big|_{x+\Delta x} - \bar{\tau}_{xr} R h \Delta \theta \Big|_x \\ &\quad + \bar{\sigma}_{rr} r \Delta \theta \Delta x \Big|_{R+\frac{h}{2}} - \bar{\sigma}_{rr} r \Delta \theta \Delta x \Big|_{R-\frac{h}{2}} \\ &\quad - \bar{\sigma}_{\theta\theta} \frac{\Delta \theta}{2} h \Delta x \Big|_{\theta+\Delta\theta} - \bar{\sigma}_{\theta\theta} \frac{\Delta \theta}{2} h \Delta x \Big|_\theta) \mathbf{e}_r \end{aligned} \quad (2.3.22)$$

We take the dot product of Eq. (2.22) with $\mathbf{e}_r$ and divide through by $\Delta\theta$ and Δx to obtain:

$$\begin{aligned} \frac{D}{Dt} (\bar{\rho}_p \bar{v}_{p,r} R \Delta r) &= \frac{\bar{\tau}_{xr} R h \Big|_{x+\Delta x} - \bar{\tau}_{xr} R h \Big|_x}{\Delta x} \\ &\quad + \sigma_{rr} r \Big|_{R+\frac{h}{2}} - \sigma_{rr} r \Big|_{R-\frac{h}{2}} \\ &\quad - \frac{\bar{\sigma}_{\theta\theta}}{2} h \Big|_{\theta+\Delta\theta} - \frac{\bar{\sigma}_{\theta\theta}}{2} h \Big|_\theta \end{aligned} \quad (2.3.23)$$

We take the limit as $\Delta x \rightarrow 0$, and the first term in the right-hand side of Eq. (2.3.23) becomes:

$$\lim_{\Delta x \rightarrow 0} \frac{\bar{\tau}_{xr} R h \Big|_{x+\Delta x} - \bar{\tau}_{xr} R h \Big|_x}{\Delta x} = \frac{\partial (\bar{\tau}_{xr} R h)}{\partial x} \quad (2.3.24)$$

Arising from the stress continuity condition at the fluid/pipe interface, the normal stress σ_{rr} , evaluated at the internal radius of the pipe element, is equal to the internal fluid pressure per

unit area, denoted as $-p$. On the other hand, we assume the pipe is externally surrounded by a much lighter fluid (e.g., air); thus, on that surface $\sigma_{rr} = 0$. In summary:

$$\sigma_{rr}\Big|_{R+\frac{h}{2}} = 0, \text{ and } \sigma_{rr}\Big|_{R-\frac{h}{2}} = -p \quad (2.3.25)$$

Due to symmetry, the hoop stresses $\sigma_{\theta\theta}$ at the opposing surfaces laying on the rx -planes an angle $\Delta\theta$ apart have equal magnitudes. In addition, their vertical components have the same direction. Therefore:

$$\frac{\bar{\sigma}_{\theta\theta}h}{2}\Big|_{\theta+\Delta\theta} = \frac{\bar{\sigma}_{\theta\theta}h}{2}\Big|_{\theta} \quad (2.3.26)$$

After considering Eqns. (2.3.23) to (2.3.26), the momentum conservation equation in the r -direction becomes:

$$\frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,r} Rh) = \frac{\partial(\bar{\tau}_{xr} Rh)}{\partial x} + p\left(R - \frac{h}{2}\right) - \bar{\sigma}_{\theta\theta}h \quad (2.3.27)$$

It is assumed that the shell thickness h remains constant during deformation; thus, it may be excluded from the derivative expressions. Furthermore, R is defined as the mean undeformed radius, which is independent of both t and x . Equation (2.3.27) is re-written as:

$$Rh \frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,r}) = Rh \frac{\partial \bar{\tau}_{xr}}{\partial x} + p\left(R - \frac{h}{2}\right) - \bar{\sigma}_{\theta\theta}h \quad (2.3.28)$$

We divide Eq. (2.3.28) through by R to obtain:

$$h \frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,r}) = h \frac{\partial \bar{\tau}_{xr}}{\partial x} + p\left(1 - \frac{h}{2R}\right) - h \frac{\bar{\sigma}_{\theta\theta}}{R} \quad (2.3.29)$$

Finally, for a thin-walled pipe $R \gg h$ and Eq. (2.3.29) becomes:

$$h \frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,r}) = h \frac{\partial \bar{\tau}_{xr}}{\partial x} + p - h \frac{\bar{\sigma}_{\theta\theta}}{R} \quad (2.3.30)$$

As in the r -direction, we shall derive the momentum conservation equation in the x -direction. We start from the general expression of the momentum conservation equation component in said direction:

$$\frac{D}{Dt} \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p v_{p,x} r dr d\theta dx = \sum F_x \mathbf{i} \quad (2.3.31)$$

Integrating across the pipe element thickness yields:

$$\frac{D}{Dt} \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \bar{\rho}_p \bar{v}_{p,x} \mathbf{i} R h d\theta dx = \sum F_x \mathbf{i} \quad (2.3.32)$$

where $\bar{\rho}$ is the average pipe material density evaluated across the pipe wall element thickness (see Eq. (2.3.7)) and $\bar{v}_{p,x}$ is the mass-weighted axial velocity component evaluated across the same line. The mean value theorem suggests that there is a point along the radius ($R + I_r h$) with $I_r \in [-1/2, 1/2]$ where:

$$\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p v_{p,x} r dr = \bar{\rho}_p \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} v_{p,x} r dr = \rho_p (R + I_r h) v_{p,x} (R + I_r h) R h \quad (2.3.33)$$

Thus, the mass-weighted axial velocity component $\bar{v}_x$ is expressed as:

$$\bar{v}_x = \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p v_{p,x} r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p r dr} = v_{p,x} (R + I_r h) \quad (2.3.34)$$

Using the mean value theorem, we obtain:

$$\frac{D}{Dt} (\bar{\rho}_p \bar{v}_{p,x} \mathbf{i} R \Delta r \Delta \theta \Delta x) = \sum F_x \mathbf{i} \quad (2.3.35)$$

We shall define the sum of forces in the axial direction $\sum F_x$. Referring to the C.V. of Figure 2.3.2, there are the following force components applied along the x -axis:

- i. The σ_{xx} normal stress on the $r\theta$ -plane at two cross-sections: x and $x + \Delta x$.
- ii. The τ_{rx} shear stress on the $x\theta$ -plane, at the inner and outer pipe radii: $R - h/2$ and $R + h/2$, respectively.

Using Eq. (2.3.11) to (2.3.13) for the definition of the pertinent areas where the stresses are applied to, we proceed to write the expression for the sum of forces in the x -direction:

$$\begin{aligned} \sum F_x \mathbf{i} = & \left(\int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \sigma_{xx} r dr d\theta \right) \Big|_{x+\Delta x} - \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \sigma_{xx} r dr d\theta \Big|_x \\ & + \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \tau_{rx} r d\theta dx \Big|_{R+\frac{h}{2}} - \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \tau_{rx} r d\theta dx \Big|_{R-\frac{h}{2}} \Big) \mathbf{i} \end{aligned} \quad (2.3.36)$$

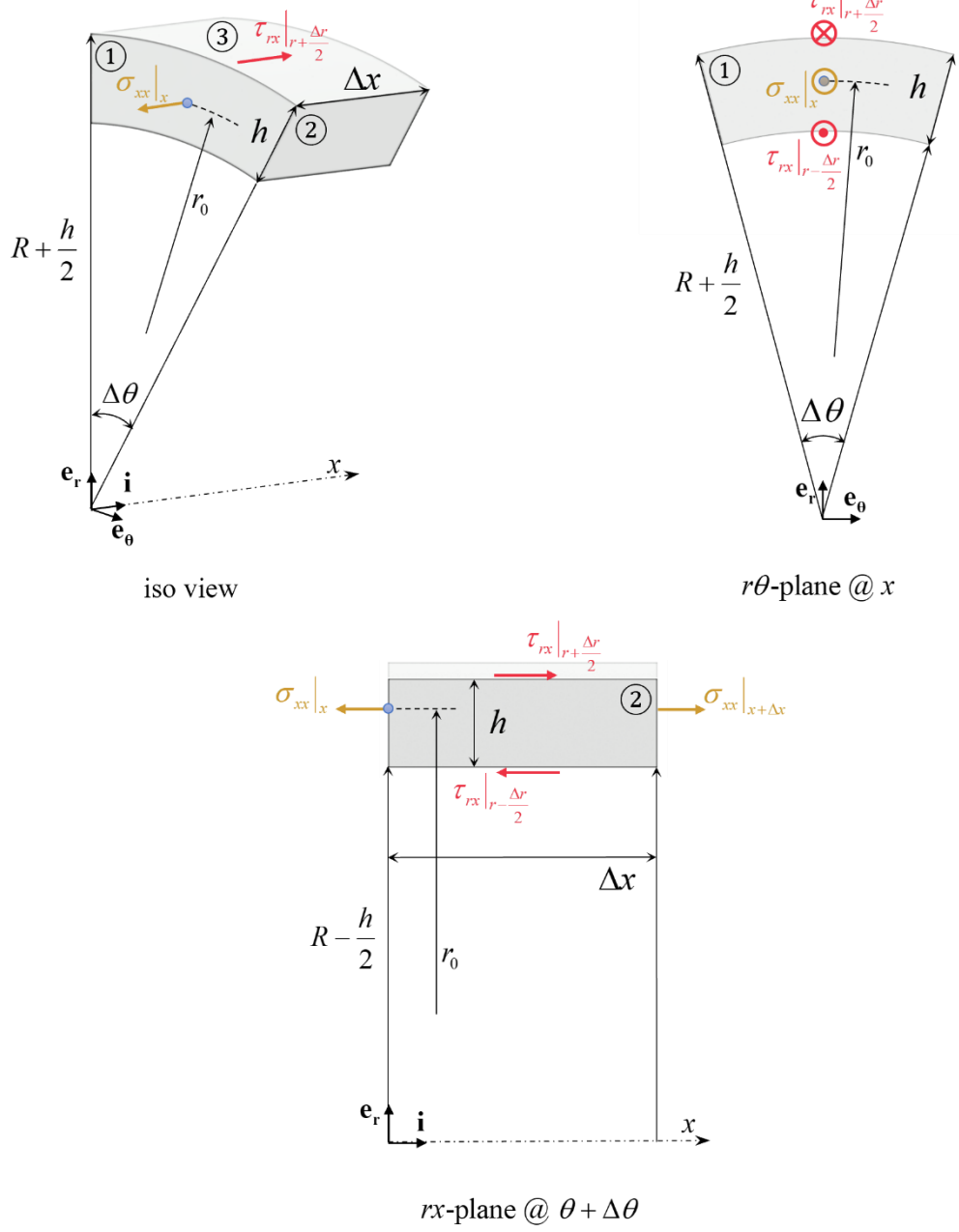


Figure 2.3.2: Stress components on the C.V. included in the linear momentum equation along the x -direction. Note that r_0 represents the radial coordinate of the pipe element centroid.

The integral calculations in Eq. (2.3.30) yield:

$$\Sigma F_x \mathbf{i} = \left(\bar{\sigma}_{xx} Rh \Delta \theta \Big|_{x+\Delta x} - \bar{\sigma}_{xx} Rh \Delta \theta \Big|_x + \bar{\tau}_{rx} r \Delta \theta \Delta x \Big|_{R+\frac{h}{2}} - \bar{\tau}_{rx} r \Delta \theta \Delta x \Big|_{R-\frac{h}{2}} \right) \mathbf{i} \quad (2.3.37)$$

where the overbars denote average stresses quantities. In particular, $\bar{\sigma}_{xx}$ is the average normal stress in the x -direction averaged across the pipe thickness. According to the mean value theorem, there is a point $(R + K_r h)$ with $K_r \in [-1/2, 1/2]$ where:

$$\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \sigma_{xx} r dr = \sigma_{xx} (R + K_r h) Rh \quad (2.3.38)$$

Thus, the average $\bar{\sigma}_{xx}$ normal stress across the pipe thickness is:

$$\bar{\sigma}_{xx} = \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \sigma_{xx} r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r dr} = \sigma_{xx} (R + K_r h) \quad (2.3.39)$$

Likewise, there is a point on the $x\theta$ -plane $(r, \theta + \Xi_\theta \Delta \theta, x + \Xi_x \Delta x)$ with $\Xi_\theta, \Xi_x \in [0, 1]$ where:

$$\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \tau_{rx} r dr = \tau_{rx} (r, \theta + \Xi_\theta \Delta \theta, x + \Xi_x \Delta x) Rh \quad (2.3.40)$$

Therefore, the average $\bar{\tau}_{rx}$ shear force applied on the integration surface is:

$$\bar{\tau}_{rx} = \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \tau_{rx} r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r dr} = \tau_{rx} (r, \theta + \Xi_\theta \Delta \theta, x + \Xi_x \Delta x) \quad (2.3.41)$$

We combine Eqns. (2.3.35) and (2.3.37) to obtain:

$$\mathbf{i} \frac{D}{Dt} (\bar{\rho}_p \bar{v}_{p,x} Rh \Delta \theta \Delta x) = \left(\bar{\sigma}_{xx} Rh \Delta \theta \Big|_{x+\Delta x} - \bar{\sigma}_{xx} Rh \Delta \theta \Big|_x + \bar{\tau}_{rx} r \Delta \theta \Delta x \Big|_{R+\frac{h}{2}} - \bar{\tau}_{rx} r \Delta \theta \Delta x \Big|_{R-\frac{h}{2}} \right) \mathbf{i} \quad (2.3.42)$$

Note that $d\mathbf{i}/dt = 0$, hence the unit vector on the left-hand side of Eq. (2.3.42) is excluded from the time derivative expression. Equation (2.3.42) is divided through by $\Delta \theta$ and Δx to yield:

$$\frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,x} Rh) = \frac{\bar{\sigma}_{xx} Rh|_{x+\Delta x} - \bar{\sigma}_{xx} Rh|_x}{\Delta x} + \bar{\tau}_{rx} r|_{R+\frac{h}{2}} - \bar{\tau}_{rx} r|_{R-\frac{h}{2}} \quad (2.3.43)$$

We take the limit as $\Delta x \rightarrow 0$, and the first term on the right-hand side of Eq. (2.3.43) becomes:

$$\lim_{\Delta x \rightarrow 0} \frac{\bar{\sigma}_{xx} Rh|_{x+\Delta x} - \bar{\sigma}_{xx} Rh|_x}{\Delta x} = \frac{\partial(\bar{\sigma}_{xx} Rh)}{\partial x} \quad (2.3.44)$$

The τ_{rx} shear terms are due to the shearing stresses exerted by the surrounding fluids on the pipe wall. We assume that such stress only exists in the internal surface of the pipe, since the external pipe surface is surrounded by a much lighter fluid, and is denoted per unit area as $-q$. From the continuity of stresses at the pipe/fluid interface, we obtain:

$$\bar{\tau}_{rx}|_{R+\frac{\Delta r}{2}} = 0, \text{ and } \bar{\tau}_{rx}|_{R-\frac{\Delta r}{2}} = -q \quad (2.3.45)$$

Applying the outcomes of Eqns. (2.3.44) & (2.3.45) to Eq. (2.3.43) results in:

$$\frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,x} Rh) = \frac{\partial(\bar{\sigma}_{xx} Rh)}{\partial x} + q \left(R - \frac{h}{2} \right) \quad (2.3.46)$$

Both the mean undeformed pipe radius R and pipe wall thickness h remain unaltered during loading, and may be excluded from the derivative expressions:

$$Rh \frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,x}) = Rh \frac{\partial \bar{\sigma}_{xx}}{\partial x} + q \left(R - \frac{h}{2} \right) \quad (2.3.47)$$

Equation (2.3.47) is divided through by R to obtain:

$$h \frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,x}) = h \frac{\partial \bar{\sigma}_{xx}}{\partial x} + q \left(1 - \frac{h}{2R} \right) \quad (2.3.48)$$

Finally, we assume a thin-walled pipe condition where $R \gg h$; thus:

$$h \frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,x}) = h \frac{\partial \bar{\sigma}_{xx}}{\partial x} + q \quad (2.3.49)$$

Following scale analysis that leads to Eq. (2.2.76), we show that fluid viscosity is inconsequential for our application. Hence, we set $q = 0$ and the momentum conservation equation in the x -direction becomes:

$$h \frac{D}{Dt}(\bar{\rho}_p \bar{v}_{p,x}) = h \frac{\partial \bar{\sigma}_{xx}}{\partial x} \quad (2.3.50)$$

The momentum conservation equation components (Eqns. (2.3.30) & (2.3.50)) may be further simplified if we consider the mass conservation equation for the pipe wall element C.V. The general form of the mass conservation equation for a material C.V. is:

$$\frac{D}{Dt} \int_{V_p} \rho_p dV_p = 0 \quad (2.3.51)$$

For the case of the curved pipe wall element C.V., Eq. (2.3.51) becomes:

$$\frac{D}{Dt} \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \rho_p r dr d\theta dx = 0 \quad (2.3.52)$$

We proceed to evaluate the integration over the pipe wall thickness:

$$\frac{D}{Dt} \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \bar{\rho}_p R h d\theta dx = 0 \quad (2.3.53)$$

where $\bar{\rho}_p$ is the average density across the thickness of the C.V. (Eq. (2.3.7)). By the mean value theorem, we get:

$$\frac{D}{Dt} (\bar{\rho}_p R h \Delta\theta \Delta x) = 0 \quad (2.3.54)$$

We divide Eq. (2.3.54) through by Δx and $\Delta\theta$ to obtain:

$$\frac{D}{Dt} (\bar{\rho}_p R h) = 0 \quad (2.3.55)$$

The pipe wall thickness h is assumed to remain constant during wave propagation. In addition, the undeformed pipe radius R is not a function of time t . As a result, both h and R are excluded from the material derivative expression:

$$R h \frac{D\bar{\rho}_p}{Dt} = 0, \text{ or } \frac{D\bar{\rho}_p}{Dt} = 0 \quad (2.3.56)$$

We combine the result of Eq. (2.3.56) with Eq. (2.3.30) and further simplify the momentum conservation equation in the r -direction to:

$$\bar{\rho}_p h \frac{D\bar{v}_{p,r}}{Dt} = h \frac{\partial \bar{\tau}_{rx}}{\partial x} + p - h \frac{\bar{\sigma}_{\theta\theta}}{R} \quad (2.3.57)$$

The material derivative is expanded, and Eq. (2.3.57) becomes:

$$\bar{\rho}_p h \left(\frac{\partial \bar{v}_{p,r}}{\partial t} + \mathbf{v} \cdot \nabla \bar{v}_{p,r} \right) = h \frac{\partial \bar{\tau}_{rx}}{\partial x} + p - h \frac{\bar{\sigma}_{\theta\theta}}{R} \quad (2.3.58)$$

We shall perform an order analysis on the terms that appear within round brackets in Eq. (2.3.58) to identify possible simplifications supported by the application presented in this thesis. The temporal acceleration term scales to:

$$\frac{\partial v_{p,r}}{\partial t} \sim \frac{\|\mathbf{v}_p\|}{T} = \frac{\|\mathbf{v}_p\|}{L} c_p \quad (2.3.59)$$

while for the advective acceleration term:

$$\mathbf{v}_p \cdot \nabla \bar{v}_{p,r} \sim \frac{\|\mathbf{v}_p\|^2}{L} \quad (2.3.60)$$

Thus, the relative magnitude of the advective over that of the temporal acceleration term is:

$$\frac{\mathbf{v}_p \cdot \nabla \bar{v}_{p,r}}{\frac{\partial \bar{v}_{p,r}}{\partial t}} \sim \frac{\frac{\|\mathbf{v}_p\|^2}{L}}{\frac{\|\mathbf{v}_p\|}{L} c_p} = \frac{\|\mathbf{v}_p\|}{c_p} = Ma \quad (2.3.61)$$

However, the Mach number Ma for the case of a wave propagating in a solid medium is negligibly small; thus, the advective acceleration terms are excluded from further consideration. As a result, the momentum conservation equation r -component becomes:

$$\bar{\rho}_p h \frac{\partial \bar{v}_{p,r}}{\partial t} = h \frac{\partial \bar{\tau}_{rx}}{\partial x} + p - h \frac{\bar{\sigma}_{\theta\theta}}{R} \quad (2.3.62)$$

Following a similar analysis, the momentum conservation equation x -component (Eq. (2.3.50)) is written as:

$$\bar{\rho}_p \Delta r \frac{\partial \bar{v}_{p,x}}{\partial t} = \Delta r \frac{\partial \bar{\sigma}_{xx}}{\partial x} \quad (2.3.63)$$

Expressing the velocity components in terms of displacements gives:

$$\bar{v}_{p,r} = \frac{\partial w_{p,0}}{\partial t}, \text{ and } \bar{v}_{p,x} = \frac{\partial u_{p,0}}{\partial t} \quad (2.3.64)$$

where $w_{p,0}$ and $u_{p,0}$ are the displacements of the pipe wall element centroid in the radial and axial directions, respectively. Replacing Eq. (2.3.64) into the momentum conservation equation components (Eqns. (2.3.62) & (2.3.63)) gives:

i. r -component

$$\rho_p h \frac{\partial^2 w_{p,0}}{\partial t^2} = h \frac{\partial \bar{\tau}_{rx}}{\partial x} + p - h \frac{\bar{\sigma}_{\theta\theta}}{R} \quad (2.3.65)$$

ii. x -component

$$\rho_p h \frac{\partial^2 u_{p,0}}{\partial t^2} = h \frac{\partial \bar{\sigma}_{xx}}{\partial x} \quad (2.3.66)$$

The overbar in density ρ_p is dropped if we assume a homogenous pipe material.

2.3.2 Angular momentum equation for a pipe wall of finite thickness

The general form of the angular momentum equation is:

$$\frac{d}{dt} \int_V \mathbf{r} \times \rho_p \mathbf{v}_p dV = \int_A \rho_p (\mathbf{r} \times \mathbf{v}_p) (\mathbf{v}_p - \mathbf{w}_p) dA + \sum \mathbf{M}_{\text{ext},p} \quad (2.3.67)$$

where $\mathbf{r}$ is a vector from some point to the particle in the C.V. with momentum per volume equal to $\rho_p \mathbf{v}_p$, $\mathbf{w}_p$ is the velocity of the C.V. boundaries, and $\mathbf{M}_{\text{ext},p}$ ($= \mathbf{r} \times \mathbf{F}_{\text{ext},p}$) is the external torque applied on the C.V. Applying this principle to the material pipe element C.V., where $\mathbf{w}_p = 0$, we obtain:

$$\frac{D}{Dt} \int_V \mathbf{r} \times \rho_p \mathbf{v}_p dV = \mathbf{r} \times \sum \mathbf{F}_{\text{ext},p} \quad (2.3.68)$$

Considering the dimensions of the C.V. as well as the mass conservation equation result (Eq. (2.3.56)), Eq. (2.3.68) is written as:

$$\bar{\rho}_p \frac{D}{Dt} \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \mathbf{v}_p r dr d\theta dx = \mathbf{r} \times \sum \mathbf{F}_{\text{ext},p} \quad (2.3.69)$$

where vector $\mathbf{r} = (r - r_0)\mathbf{e}_r$, and the velocity vector $\mathbf{v}_p = v_{p,r}\mathbf{e}_r + v_{p,\theta}\mathbf{e}_\theta$ since the velocity θ -component is omitted due to the axisymmetric assumption. For the pipe element, r_0 is the radial coordinate of the centroid of its face that lays on the $r\theta$ -plane. It is intuitive to consider that the pipe wall element centroid coincides with its neutral fibre that is free from bending stresses. The radial coordinate r_0 of the centroid with respect to the pipe centreline is:

$$r_0 = \frac{1}{A_{face}} \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r^2 dr d\theta = \frac{1}{R\Delta\theta h} \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r^2 dr d\theta = \frac{h^2}{12R} + R \quad (2.3.70)$$

Taking the cross-product and using the fact that $\mathbf{e}_r \times \mathbf{i} = -\mathbf{e}_\theta$ yields:

$$\bar{\rho}_p \frac{D}{Dt} \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \left\{ -\mathbf{e}_\theta \left[(r-r_0) v_{p,x} \right] \right\} r dr d\theta dx = \mathbf{r} \times \sum \mathbf{F}_{ext,p} \quad (2.3.71)$$

The velocity component along the x -axis may be expressed in terms of displacements as $v_x = \partial u / \partial t$, where u denotes the axial displacements. To estimate the axial displacements across the pipe thickness we employ a Taylor series expansion about the centroid r_0 , where the axial displacements are equal to u_0 :

$$u_p(r, x, t) = u_{p,0}(x, t) + (r-r_0) \left. \frac{\partial u_p}{\partial r} \right|_{r=r_0} + \frac{(r-r_0)^2}{2!} \left. \frac{\partial^2 u_p}{\partial r^2} \right|_{r=r_0} + \dots = \sum_{n=0}^{\infty} \frac{(r-r_0)^n}{n!} \left. \frac{\partial^n u_p}{\partial r^n} \right|_{r=r_0} \quad (2.3.72)$$

According to the Kirchhoff assumptions, lines normal to the neutral fibre (where $u = u_0$) may rotate but must remain straight. The rotation angle along the x -axis is defined as:

$$\alpha = \left. \frac{\partial u_p}{\partial r} \right|_{r=r_0} \quad (2.3.73)$$

and corresponds to the linear distribution of axial displacements across the pipe wall thickness. Since radial lines need to remain straight, higher-order terms are omitted, and Eq. (2.3.72) becomes:

$$u_p(r, x, t) = u_{p,0}(x, t) + (r-r_0) \alpha(x, t) \quad (2.3.74)$$

Essentially, Eq. (2.3.74) describes the distribution of axial displacements due to both an axial and a bending load. Taking the time derivative of Eq. (2.3.74) and replacing into Eq. (2.3.71) yields:

$$\bar{\rho}_p \frac{D}{Dt} \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{e}_\theta \left\{ (r-r_0) \left[\frac{\partial u_{p,0}}{\partial t} + (r-r_0) \omega \right] \right\} r dr d\theta dx = -\mathbf{r} \times \sum \mathbf{F}_{ext,p} \quad (2.3.75)$$

where $\omega (= \partial \alpha / \partial t)$ is the angular velocity. We proceed to evaluate the integral on the left-hand side of Eq. (2.3.75). To that end, the expression is first split in two terms:

$$\begin{aligned}
\bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{e}_\theta \left\{ (r-r_0) \left[\frac{\partial u_{p,0}}{\partial t} + (r-r_0) \boldsymbol{\omega} \right] \right\} r dr d\theta dx = \\
\bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{e}_\theta (r-r_0) \frac{\partial u_{p,0}}{\partial t} r dr d\theta dx + \\
\bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{e}_\theta (r-r_0)^2 \boldsymbol{\omega} r dr d\theta dx
\end{aligned} \tag{2.3.76}$$

For the first term on the right-hand side of Eq. (2.3.76) we get:

$$\begin{aligned}
\bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{e}_\theta (r-r_0) \frac{\partial u_{p,0}}{\partial t} r dr d\theta dx = \\
\bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \mathbf{e}_\theta \left[\frac{r^3}{3} - \frac{r^2}{2} r_0 \right]_{R-\frac{h}{2}}^{R+\frac{h}{2}} \frac{\partial u_{p,0}}{\partial t} d\theta dx
\end{aligned} \tag{2.3.77}$$

Replacing the expression for r_0 from Eq. (2.3.70) into Eq. (2.3.77) and evaluating the expression within the square brackets yields:

$$\begin{aligned}
\bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{e}_\theta (r-r_0) \frac{\partial u_{p,0}}{\partial t} r dr d\theta dx \\
= \bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \mathbf{e}_\theta \overbrace{\left[\frac{h^3}{12} + \Delta r R^2 - R h \left(\frac{h^2}{12R} + R \right) \right]}^{=0} \frac{\partial u_{p,0}}{\partial t} d\theta dx = 0
\end{aligned} \tag{2.3.78}$$

thus, the first term on the right-hand side of Eq. (2.3.76) is excluded from further consideration.

As for the remaining second term on the right-hand side of Eq. (2.3.76), we get:

$$\begin{aligned}
\bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{e}_\theta (r-r_0)^2 \boldsymbol{\omega} r dr d\theta dx \\
= \bar{\rho}_p \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \mathbf{e}_\theta \boldsymbol{\omega} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0)^2 r dr d\theta dx = \bar{\rho}_p I_m \int_x^{x+\Delta x} \int_\theta^{\theta+\Delta\theta} \mathbf{e}_\theta \boldsymbol{\omega} d\theta dx
\end{aligned} \tag{2.3.79}$$

where I_m is the moment of inertia of the cross-sectional area of the element about an axis that passes through the centroid of the area and coincides with the θ -axis. Specifically, the moment of inertia I_m is equal to:

$$I_m = \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0)^2 r dr = \frac{h^3 R}{12} - \frac{h^5}{144R} = I \left(1 - \frac{h^2}{12R^2} \right) \tag{2.3.80}$$

where $I = h^3 R/12$ and r_0 is given from Eq. (2.3.70). As a result, the angular momentum conservation equation (Eq. (2.3.75)) becomes:

$$\bar{\rho}_p I_0 \frac{D}{Dt} \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \mathbf{e}_\theta \omega d\theta dx = -\mathbf{r} \times \sum \mathbf{F}_{\text{ext,p}} \quad (2.3.81)$$

Performing the remaining integral calculations in the left-hand side of Eq. (2.3.81) yields:

$$\bar{\rho}_p I_0 \frac{D}{Dt} [\mathbf{e}_\theta \omega \Delta\theta \Delta x] = -\mathbf{r} \times \sum \mathbf{F}_{\text{ext,p}} \quad (2.3.82)$$

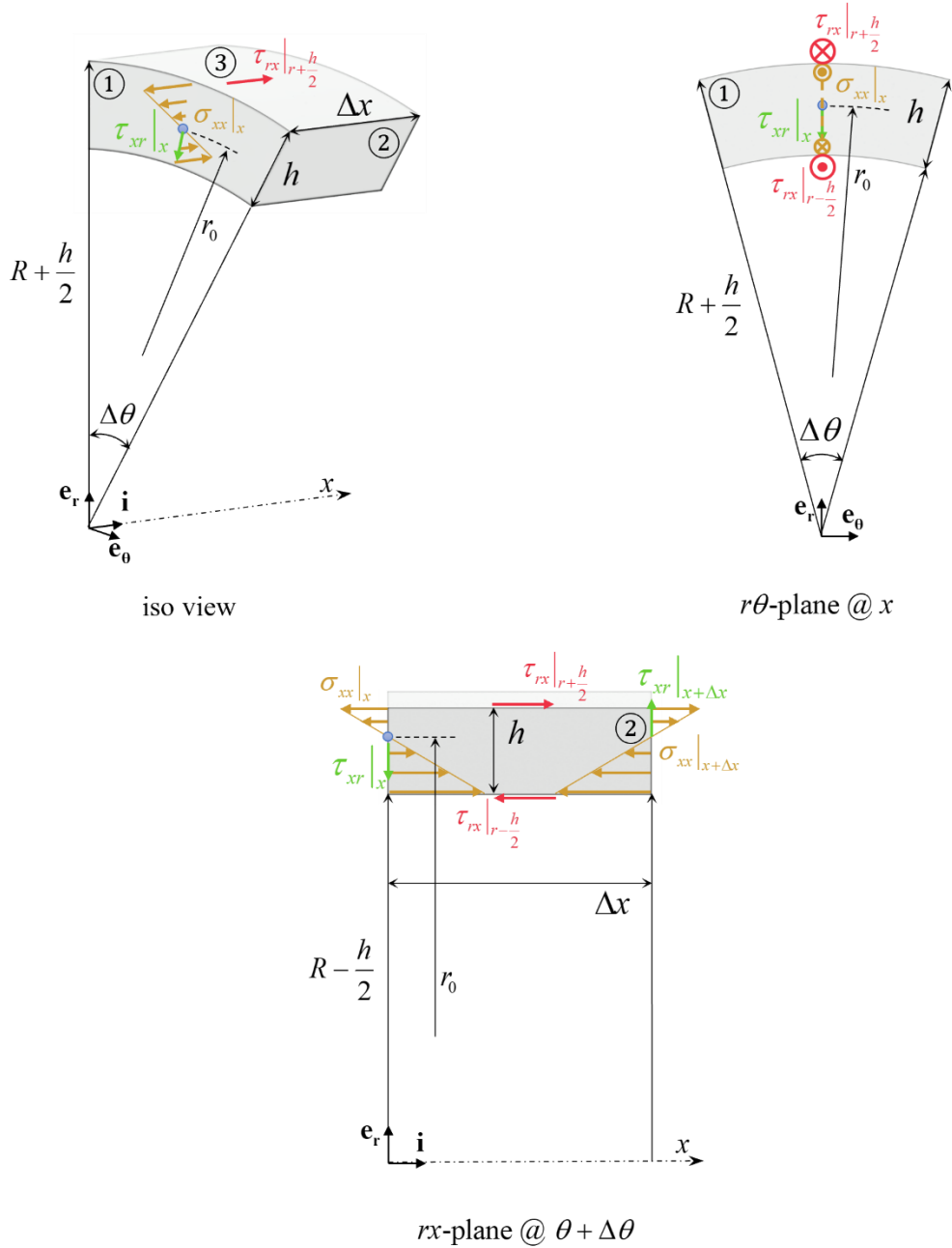


Figure 2.3.3: Stress components contributing to the angular momentum about the pipe element centroid and along the θ -axis.

We proceed to evaluate the right-hand side of Eq. (2.3.82). Referring to the C.V. in Figure 2.3.3, there are the following force components that create an angular rotation with respect to the pipe element centroid:

- i. The σ_{xx} normal stresses on the $r\theta$ -plane at two cross-sections: x and $x + \Delta x$. The position vector $\mathbf{r}$ lays along the r -axis and its length is evaluated from the face element centroid r_0 to the pipe wall particle that contributes to the angular momentum. Note the linear σ_{xx} stress distribution that is discussed later.
- ii. The τ_{xr} shear stresses on the $r\theta$ -plane at two cross-sections: x and $x + \Delta x$. The position vector $\mathbf{r}$ lays on the x -axis.
- iii. The τ_{rx} shear stresses on the $x\theta$ -plane, on the inner and outer pipe radii, $R - h/2$ and $R + h/2$, respectively. The position vector $\mathbf{r}$ lays on the r -axis.

Using Eqns. (2.3.11) to (2.3.13) for the definition of the pertinent areas the stresses are applied to, we proceed to write the expression for the angular momentum about the θ -axis:

$$\begin{aligned}
 \mathbf{r} \times \sum \mathbf{F}_{\text{ext}} = & \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \mathbf{r} \times \sigma_{xx} \mathbf{i} r dr d\theta \Big|_{x+\Delta x} - \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \mathbf{r} \times \sigma_{xx} \mathbf{i} r dr d\theta \Big|_x \\
 & + \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \mathbf{r} \times \tau_{xr} \mathbf{e}_r r dr d\theta \Big|_{x+\Delta x} - \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \mathbf{r} \times \tau_{xr} \mathbf{e}_r r dr d\theta \Big|_x \\
 & + \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \mathbf{r} \times \tau_{rx} \mathbf{i} r d\theta dx \Big|_{R+\frac{h}{2}} - \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \mathbf{r} \times \tau_{rx} \mathbf{i} r d\theta dx \Big|_{R-\frac{h}{2}}
 \end{aligned} \tag{2.3.83}$$

We shall start with the angular momentum about the centroid r_0 due to normal stresses σ_{xx} (i.e., the first row of Eq. (2.3.83)). Due to bending of the pipe wall element, normal stresses σ_{xx} are not applied uniformly across the pipe wall thickness, but we assume a linear distribution instead. We may expand the normal stresses σ_{xx} about the centroid r_0 as:

$$\sigma_{xx}(r, x, t) = \sigma_{xx,0}(x, t) + (r - r_0) \frac{\partial \sigma_{xx}}{\partial r}(x, t) \tag{2.3.84}$$

Equation (2.3.84) essentially mirrors the distribution of axial displacements across the thickness (Eq. (2.3.72)). In addition, the position vector $\mathbf{r} = (r - r_0)\mathbf{e}_r$. We get:

$$\begin{aligned}
& \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \sigma_{xx} \mathbf{i} r dr d\theta \right|_{x+\Delta x} - \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \sigma_{xx} \mathbf{i} r dr d\theta \right|_x \\
&= \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0) \mathbf{e}_r \times \left[\sigma_{xx,0} + (r-r_0) \frac{\partial \sigma_{xx}}{\partial r} \right] \mathbf{i} r dr d\theta \right|_{x+\Delta x} \\
& \quad - \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0) \mathbf{e}_r \times \left[\sigma_{xx,0} + (r-r_0) \frac{\partial \sigma_{xx}}{\partial r} \right] \mathbf{i} r dr d\theta \right|_x
\end{aligned} \tag{2.3.85}$$

We shall isolate the first term on the right-hand side of Eq. (2.3.85) and perform the calculations:

$$\begin{aligned}
& \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0) \mathbf{e}_r \times \left[\sigma_{xx,0} + (r-r_0) \frac{\partial \sigma_{xx}}{\partial r} \right] \mathbf{i} r dr d\theta \right|_{x+\Delta x} \\
&= - \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0) \sigma_{xx,0} \mathbf{e}_{\theta} r dr d\theta \right|_{x+\Delta x} - \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0)^2 \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_{\theta} r dr d\theta \right|_{x+\Delta x}
\end{aligned} \tag{2.3.86}$$

The first term on the right-hand side of Eq. (2.3.86) is:

$$- \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0) \sigma_{xx,0} \mathbf{e}_{\theta} r dr d\theta \right|_{x+\Delta x} = - \int_{\theta}^{\theta+\Delta\theta} \bar{\sigma}_{xx,0} \mathbf{e}_{\theta} \left[\frac{r^3}{3} - r_0 \frac{r^2}{2} \right]_{R-\frac{hr}{2}}^{R+\frac{h}{2}} d\theta \tag{2.3.87}$$

where r_0 is given by Eq. (2.3.70). Substituting into Eq. (2.3.87) yields:

$$\begin{aligned}
& - \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0) \sigma_{xx,0} \mathbf{e}_{\theta} r dr d\theta \right|_{x+\Delta x} = - \int_{\theta}^{\theta+\Delta\theta} \bar{\sigma}_{xx,0} \mathbf{e}_{\theta} \left[\frac{r^3}{3} - r_0 \frac{r^2}{2} \right]_{R-\frac{hr}{2}}^{R+\frac{h}{2}} d\theta \Big|_{x+\Delta x} \\
&= - \int_{\theta}^{\theta+\Delta\theta} \bar{\sigma}_{xx,0} \mathbf{e}_{\theta} \overbrace{\left[\frac{h^3}{12} + R^2 h - Rh \left(\frac{h^2}{12R} + R \right) \right]}^{=0} d\theta \Big|_{x+\Delta x} = 0
\end{aligned} \tag{2.3.88}$$

Equation (2.3.88) gives an expected result since a uniform stress distribution does not produce a moment about the centroid r_0 . We proceed to evaluate the second term on the right-hand side of Eq. (2.3.86) using the result of Eq. (2.3.80):

$$- \left. \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0)^2 \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_{\theta} r dr d\theta \right|_{x+\Delta x} = -I_m \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_{\theta} \Delta\theta \Big|_{x+\Delta x} \tag{2.3.89}$$

Similarly, for the second term on the right-hand side of Eq. (2.3.85), where the angular momentum due to σ_{xx} is evaluated at x , we get:

$$-\int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} (r-r_0) \mathbf{e}_r \times \left[\sigma_{xx,0} + (r-r_0) \frac{\partial \sigma_{xx}}{\partial r} \right] \mathbf{i} r dr d\theta \Big|_x = I_m \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_\theta \Delta\theta \Big|_x \quad (2.3.90)$$

Therefore:

$$\begin{aligned} & \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \sigma_{xx} \mathbf{i} r dr d\theta \Big|_{x+\Delta x} - \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \sigma_{xx} \mathbf{i} r dr d\theta \Big|_x \\ &= -I_0 \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_\theta \Delta\theta \Big|_{x+\Delta x} + I_m \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_\theta \Delta\theta \Big|_x \end{aligned} \quad (2.3.91)$$

The τ_{xr} shear stresses also produce angular momentum about the θ -axis. In this case, however, the position vector lays on the x -axis. The second row of Eq. (2.3.83) is isolated:

$$\begin{aligned} & \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \tau_{xr} \mathbf{e}_r r dr d\theta \Big|_{x+\Delta x} - \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \tau_{xr} \mathbf{e}_r r dr d\theta \Big|_x \\ &= \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \Delta x \mathbf{i} \times \tau_{xr} \mathbf{e}_r r dr d\theta \Big|_{x+\Delta x} - \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} 0 \times \tau_{xr} \mathbf{e}_r r dr d\theta \Big|_x \end{aligned} \quad (2.3.92)$$

We consider the centroid r_0 on the face of the pipe element on the $r\theta$ -plane at x . Consequently, the position vector $\mathbf{r}$ from the surface where τ_{rx} is applied at x is equal to zero. For the face of the element at $x + \Delta x$, however, the position vector is $\mathbf{r} = \Delta x \mathbf{i}$. Equation (2.3.92) becomes:

$$\int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \tau_{xr} \mathbf{e}_r r dr d\theta \Big|_{x+\Delta x} - \int_{\theta}^{\theta+\Delta\theta} \int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \mathbf{r} \times \tau_{xr} \mathbf{e}_r r dr d\theta \Big|_x = \bar{\tau}_{xr} \mathbf{e}_\theta R h \Delta\theta \Delta x \Big|_{x+\Delta x} \quad (2.3.93)$$

Finally, we shall examine the τ_{rx} shear stresses applied at the internal radius of the pipe wall due to the contained fluid i.e., the third row of Eq. (2.3.83). As shown in Eq. (2.3.45), $\tau_{rx} \Big|_{R+\frac{h}{2}} = 0$; thus,

$$\begin{aligned} & \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \mathbf{r} \times \tau_{rx} \mathbf{i} r d\theta dx \Big|_{R+\frac{h}{2}} \stackrel{=0}{=} - \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \mathbf{r} \times \tau_{rx} \mathbf{i} r d\theta dx \Big|_{R-\frac{h}{2}} \\ &= - \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} (r-r_0) \mathbf{e}_r \times \tau_{rx} \mathbf{i} r d\theta dx \Big|_{R-\frac{h}{2}} = \tau_{rx} (r-r_0) \mathbf{e}_\theta r \Delta\theta \Delta x \Big|_{R-\frac{h}{2}} \end{aligned} \quad (2.3.94)$$

Equation (2.3.94) for $r = R - h/2$ and r_0 from Eq. (2.3.70) becomes:

$$- \int_x^{x+\Delta x} \int_{\theta}^{\theta+\Delta\theta} \mathbf{r} \times \tau_{rx} \mathbf{i} r d\theta dx \Big|_{R-\frac{\Delta r}{2}} = -q \frac{h}{2} \left(1 + \frac{h}{6R} \right) \mathbf{e}_\theta \left(R + \frac{h}{2} \right) \Delta\theta \Delta x \quad (2.3.95)$$

where $\bar{\tau}_{rx} \Big|_{R-h/2} = 0$. The results of Eqns. (2.3.91), (2.3.93), and (2.3.95) are replaced into Eq. (2.3.83):

$$\begin{aligned}
\mathbf{r} \times \sum \mathbf{F}_{\text{ext}} = & -I_m \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_\theta \Delta \theta \Big|_{x+\Delta x} + I_m \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_\theta \Delta \theta \Big|_x \\
& + \bar{\tau}_{xr} \mathbf{e}_\theta R h \Delta \theta \Delta x \Big|_{x+\Delta x} \\
& - q \frac{\Delta r}{2} \left(1 + \frac{h}{6R} \right) \mathbf{e}_\theta \left(R + \frac{h}{2} \right) \Delta \theta \Delta x
\end{aligned} \tag{2.3.96}$$

and the angular momentum conservation equation (Eq. (2.3.82)) becomes:

$$\begin{aligned}
\bar{\rho}_p I_m \frac{D}{Dt} [\boldsymbol{\omega} \mathbf{e}_\theta \Delta \theta \Delta x] = & I_m \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_\theta \Delta \theta \Big|_{x+\Delta x} - I_m \frac{\partial \sigma_{xx}}{\partial r} \mathbf{e}_\theta \Delta \theta \Big|_x \\
& - \bar{\tau}_{xr} \mathbf{e}_\theta R h \Delta \theta \Delta x \Big|_{x+\Delta x} \\
& + q \frac{\Delta r}{2} \left(1 + \frac{\Delta r}{6R} \right) \mathbf{e}_\theta \left(R + \frac{\Delta r}{2} \right) \Delta \theta \Delta x
\end{aligned} \tag{2.3.97}$$

In general, when considering cylindrical coordinates, the time derivative of unit vector $\mathbf{e}_\theta$ yields $-\mathbf{e}_r$ i.e., $d\mathbf{e}_\theta/dt = -\mathbf{e}_r$. However, the axisymmetric displacement field assumption forces the displacements be equal to zero and the C.V. fixed along the θ -direction. As a result, $d\mathbf{e}_\theta/dt = 0$; thus, we may take $\mathbf{e}_\theta$ outside of the time derivative expression in Eq. (2.3.96). We combine this step with taking the dot product of Eq. (2.3.97) with $\mathbf{e}_\theta$ to obtain:

$$\begin{aligned}
\bar{\rho}_p I_m \frac{D}{Dt} [\boldsymbol{\omega} \Delta \theta \Delta x] = & I_m \frac{\partial \sigma_{xx}}{\partial r} \Delta \theta \Big|_{x+\Delta x} - I_m \frac{\partial \sigma_{xx}}{\partial r} \Delta \theta \Big|_x \\
& - \bar{\tau}_{xr} R h \Delta \theta \Delta x \Big|_{x+\Delta x} \\
& + q \frac{h}{2} \left(1 + \frac{h}{6R} \right) \left(R + \frac{h}{2} \right) \Delta \theta \Delta x
\end{aligned} \tag{2.3.98}$$

We divide Eq. (2.3.98) through by $R \Delta \theta \Delta x$:

$$\bar{\rho}_p \frac{I_m}{R} \frac{D\boldsymbol{\omega}}{Dt} = \frac{I_m}{R} \frac{\frac{\partial \sigma_{xx}}{\partial r} \Big|_{x+\Delta x} - \frac{\partial \sigma_{xx}}{\partial r} \Big|_x}{\Delta x} - \bar{\tau}_{xr} h \Big|_{x+\Delta x} + q \frac{h}{2} \left(1 + \frac{h}{6R} \right) \left(1 + \frac{h}{2R} \right) \tag{2.3.99}$$

and take the limit when $\Delta x \rightarrow 0$:

$$\bar{\rho}_p \frac{I_m}{R} \frac{D\boldsymbol{\omega}}{Dt} = \frac{I_m}{R} \frac{\partial^2 \sigma_{xx}}{\partial r \partial x} - \bar{\tau}_{xr} h + q \frac{h}{2} \left(1 + \frac{h}{6R} \right) \left(1 + \frac{h}{2R} \right) \tag{2.3.100}$$

For a thin wall pipe assumption, $R \gg h$, Eq. (2.3.100) becomes:

$$\bar{\rho}_p \frac{I_m}{R} \frac{D\boldsymbol{\omega}}{Dt} = \frac{I_m}{R} \frac{\partial^2 \sigma_{xx}}{\partial r \partial x} - \bar{\tau}_{xr} h + q \frac{h}{2} \tag{2.3.101}$$

The material derivative is expanded, and Eq. (2.3.101) becomes:

$$\bar{\rho}_p \frac{I_m}{R} \left(\frac{\partial \mathbf{\omega}}{\partial t} + \mathbf{v}_p \cdot \nabla \mathbf{\omega} \right) = \frac{I_m}{R} \frac{\partial^2 \sigma_{xx}}{\partial r \partial x} - \bar{\tau}_{xr} h + q \frac{h}{2} \quad (2.3.102)$$

As shown in Eq. (2.3.61), the relative magnitude of the advective over the temporal acceleration term is in the order of the Mach number, hence negligibly small. The non-linear advective term is omitted, and the angular momentum conservation equation becomes:

$$\boxed{\rho_p \frac{I_m}{R} \frac{\partial \mathbf{\omega}}{\partial t} = \frac{I_m}{R} \frac{\partial^2 \sigma_{xx}}{\partial r \partial x} - \bar{\tau}_{xr} h + q \frac{h}{2}} \quad (2.3.103)$$

2.3.3 Stress-strain relationships

To further simplify the momentum conservation component equations, it is helpful to write the general strain-displacement equations in cylindrical coordinates. When doing so, however, it should also be considered that in this thesis solely axisymmetric vibrations are of interest. Consequently, the azimuthal displacements v_p and all derivatives with respect to θ are taken equal to zero i.e.,

$$v_p = 0, \text{ and } \frac{\partial}{\partial \theta} = 0 \quad (2.3.104)$$

Following the above simplification, the generalized strain-displacement equations are:

$$\varepsilon_{rr} = \frac{\partial w_p}{\partial r} \quad (2.3.105)$$

$$\varepsilon_{\theta\theta} = \frac{1}{R} \frac{\partial v_p}{\partial \theta} + \frac{w_p}{R} \rightarrow \boxed{\varepsilon_{\theta\theta} = \frac{w_p}{R}} \quad (2.3.106)$$

$$\varepsilon_{xx} = \frac{\partial u_p}{\partial x} \quad (2.3.107)$$

$$\gamma_{xr} = \frac{\partial u_p}{\partial r} + \frac{\partial w_p}{\partial x} \quad (2.3.108)$$

$$\gamma_{r\theta} = \frac{\partial v_p}{\partial r} - \frac{v_p}{R} + \frac{1}{R} \frac{\partial w_p}{\partial \theta} \rightarrow \boxed{\gamma_{r\theta} = 0} \quad (2.3.109)$$

$$\gamma_{\theta x} = \frac{1}{R} \frac{\partial u_p}{\partial \theta} + \frac{\partial v_p}{\partial x} \rightarrow \boxed{\gamma_{\theta x} = 0} \quad (2.3.110)$$

where ε_{ij} and γ_{ij} are the normal and shear strains, respectively, with subscript i denoting the direction of the face normal and j the direction of the component. Displacements on the r -axis are represented as $w_p(r, \theta, x, t)$, while on the x -axis as $u_p(r, \theta, x, t)$, and the θ -axis as $v_p(r, \theta, x, t)$.

Strains are related to the applicable stresses through Hooke's law. The generalized strain-stress relationships in cylindrical coordinates are:

$$\varepsilon_{rr} = \frac{1}{E} [\sigma_{rr} - \nu_p (\sigma_{\theta\theta} + \sigma_{xx})] \quad (2.3.111)$$

$$\varepsilon_{\theta\theta} = \frac{1}{E} [\sigma_{\theta\theta} - \nu_p (\sigma_{rr} + \sigma_{xx})] \quad (2.3.112)$$

$$\varepsilon_{xx} = \frac{1}{E} [\sigma_{xx} - \nu_p (\sigma_{rr} + \sigma_{\theta\theta})] \quad (2.3.113)$$

$$\tau_{r\theta} = G_s \gamma_{r\theta}, \tau_{xr} = G_s \gamma_{xr}, \text{ and } \tau_{\theta x} = G_s \gamma_{\theta x} \quad (2.3.114)$$

where ν_p is Poisson's ratio, E is the Young's modulus of elasticity, and G_s is the shear modulus $[= E/(1 - \nu)]$. From Eqns. (2.3.109), (2.3.110) and (2.3.114) we extract that, for an axisymmetric-only displacement assumption, the shear stress components $\tau_{r\theta}$ and $\tau_{\theta x}$ are:

$$\tau_{r\theta} = \tau_{\theta x} = 0 \quad (2.3.115)$$

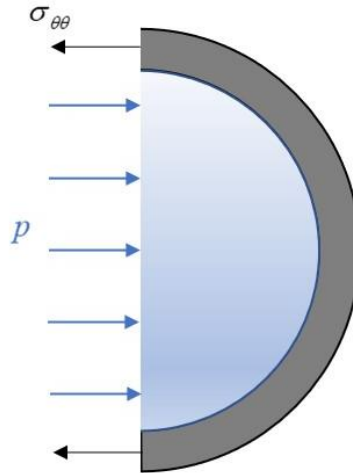


Figure 2.3.4: Pressure in the azimuthal direction is balanced by the azimuthal stress component in the pipe wall. The pipe extends to a length l towards the page.

Note that in Eqns. (2.3.111) to (2.3.113) all normal stress components appear. However, for a pipe with a thickness h much smaller than the radius R it is not necessary to solve the elasticity problem for all three dimensions. To support such claim, we consider a cylindrical pipe of

length l , wall thickness h , and radius R , filled with fluid under pressure p . To calculate the circumferential (or hoop) stresses $\sigma_{\theta\theta}$, the fluid-filled pipe is cut in half along its diameter and the free-body diagram that ensues is shown in Figure 2.3.4. Keeping in mind that the pipe extends to a length l , the force equilibrium as presented in Figure 2.3.4 is:

$$\bar{\sigma}_{\theta\theta} 2hl = p2Rl, \text{ or } \bar{\sigma}_{\theta\theta} = p \frac{R}{h} \quad (2.3.116)$$

In reality, hoop stress $\sigma_{\theta\theta}$ is a function of radius r , but the difference across the pipe thickness h is small for a thin-walled pipe assumption. It should be noted that the result of Eq. (2.3.116) represents the average hoop stress, as in Eq. (2.3.19). Considering the same fluid-filled pipe example, we take a section along a plane perpendicular to the longitudinal axis, as shown in Figure 2.3.5

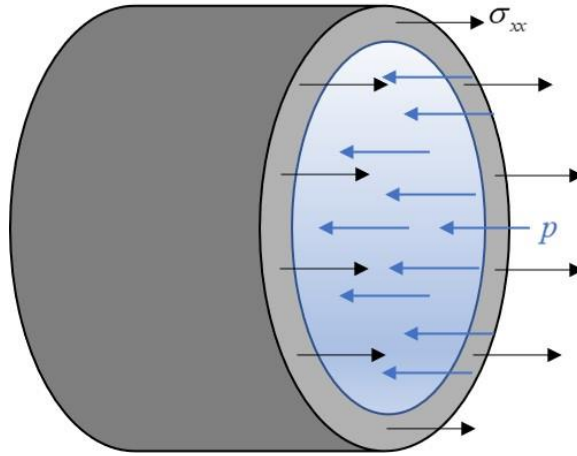


Figure 2.3.5: Pressure in the axial direction is balanced by the axial stresses in the pipe wall.

The force equilibrium from Figure 2.3.5 is written as:

$$\sigma_{xx} 2\pi R \Delta r = p\pi R^2, \text{ or } \sigma_{xx} = p \frac{R}{2\Delta r} \quad (2.3.117)$$

It should be noted that this simple analysis is only applicable far away from the pipe ends, and that Eq. (2.3.117) corresponds to the average axial stress across the pipe thickness. Finally, we examine the radial stress σ_{rr} . The radial stresses at the inner pipe radius $(R - h/2)$ and outer radius $(R + h/2)$ has been examined in Eq. (2.3.25) for a pipe assumed to be surrounded by vacuum, or a much lighter fluid (e.g., air) in comparison to that contained within. If we presume the radial stress to be uniformly distributed through the pipe thickness, the arithmetic mean is:

$$\sigma_{rr,\text{mean}} = -\frac{p}{2} \quad (2.3.118)$$

Assuming a thin-walled pipe ($R/h \geq 20$) under internal pressure p , a comparison between Eqns. (2.3.116), (2.3.117), and (2.3.118) shows that the hoop stress is twice the axial stress, and the absolute value of the radial stress is equal to 1/20 and 1/10 of the hoop and axial stresses, respectively. Thus, the stress state can be considered biaxial, and the radial stress average value across the thickness is negligible i.e., $\sigma_{rr} = 0$, and consideration of the normal radial stresses would only be worthwhile only in the close vicinity of concentrated loads.

For $\sigma_{rr} = 0$, the generalized stress-strain relationships (Eq. (2.3.111) to (2.3.113)) reduce to:

$$\varepsilon_{rr} = -\frac{\nu_p}{E}(\sigma_{\theta\theta} + \sigma_{xx}) \quad (2.3.119)$$

$$\varepsilon_{\theta\theta} = \frac{1}{E}(\sigma_{\theta\theta} - \nu_p \sigma_{xx}) \quad (2.3.120)$$

$$\varepsilon_{xx} = \frac{1}{E}(\sigma_{xx} - \nu_p \sigma_{\theta\theta}) \quad (2.3.121)$$

Since one of the shell dimensions (i.e., the thickness) is much smaller compared to the other two, a three-dimensional consideration for the displacement and stress fields is often not necessary. Instead, simplified two-dimensional theories are introduced, wherein an educated assumption is made as to the form of the fields in question. Assuming axisymmetric displacements has already allowed us to structure our problem in two dimensions. Regardless, further simplification is possible following the paradigm of the approximate one-dimensional beam theories, such as the Euler-Bernoulli, or Timoshenko beam theory.

A “neutral fibre” of the pipe wall element that extends along the x -axis crossing its centroid has already been considered in Eq. (2.3.70). This neutral fibre sets the reference frame for the element. A two-dimensional field function f is defined as:

$$f(r, x, t) = F_s f_s(x, t) \quad (2.3.122)$$

where F_s is an expansion function of any type (Taylor, Lagrange, harmonics, etc.), and $f_s(x, t)$ is the function along the neutral fibre. The simplest expansion would be:

$$f(r, x, t) = f_s(x, t) \quad (2.3.123)$$

where it is implied that f has a constant distribution across the thickness of the element but can vary from one section to another. If f represents the axial stress σ_{xx} , then a membrane stress

state is obtained. During bending, however, the axial stresses σ_{xx} vary linearly cross the thickness. Thus, to capture bending, the expansion must be improved by adding a linear term:

$$f(r, x, t) = f_s(x, t) + rf_s^*(x, t) \quad (2.3.124)$$

The specific form of the expansion function F_s depends on the kinematics of deformation to be included. For instance, the classic Euler-Bernoulli beam theory was derived from the following axiomatic assumptions:

- i. The cross-sections are rigid on their plane.
- ii. The cross-sections remain planar and normal to the deformed axis of the beam.

The extension of the Euler-Bernoulli beam theory to plates/shells is known as the Kirchhoff plate theory. In this case, the neutral fibre is defined as a middle surface instead. Otherwise, the Kirchhoff hypotheses mirror these of the Euler-Bernoulli beam theory:

- i. Lines normal to the middle surface (defined at half the plate thickness) remain straight after deformation.
- ii. The thickness of the plate/shell does not change during a deformation.
- iii. Lines normal to the middle surface rotate such that they remain perpendicular to the middle surface after deformation.

We return to our problem to consider Taylor series expansions of the displacements in the radial and axial directions about the neutral fibre of the pipe wall element. These expansions will be later simplified according to the Kirchhoff/Euler-Bernoulli assumptions. Starting with the radial displacements w_p , we obtain:

$$w_p(r, x, t) = w_{p,0}(x, t) + (r - r_0) \left. \frac{\partial w_p}{\partial r} \right|_{r=r_0} + \frac{(r - r_0)^2}{2!} \left. \frac{\partial^2 w_p}{\partial r^2} \right|_{r=r_0} + \dots = \sum_{n=0}^{\infty} \frac{(r - r_0)^n}{n!} \left. \frac{\partial^n w_p}{\partial r^n} \right|_{r=r_0} \quad (2.3.125)$$

where $w_{p,0}$ are the radial displacements of the neutral fibre as defined by the radial coordinate r_0 (Eq. (2.3.70)). However, the second of the Kirchhoff hypotheses requires that the thickness of the plate/shell remains unchanged during deformation, thus:

$$\varepsilon_{rr} = \frac{\partial w_p}{\partial r} \bigg|_{r=r_0} = 0 \quad (2.3.126)$$

which also implies that the radial displacement w_p is independent of r . Therefore, only the first order term of Eq. (2.3.125) is further considered. The validity of such assumption is limited to excitations where the probing wavelength (i.e., the length scale L) is an order of magnitude larger compared to the pipe wall thickness h or more. Eventually, the description for radial displacements becomes:

$$w_p(r, x, t) = w_{p,0}(x, t) \quad (2.3.127)$$

Similarly, axial displacements across the pipe thickness can also be estimated using a Taylor series expansion about r_0 , as already performed in Eq. (2.3.72) to (2.3.74). After introducing the axial and radial displacement equations (Eqns. (2.3.74) and (2.3.127)), the strain-displacements equations become:

$$\varepsilon_{\theta\theta} = \frac{w_{p,0}}{r} \quad (2.3.128)$$

$$\varepsilon_{xx} = \frac{\partial u_{p,0}}{\partial x} + (r - r_0) \frac{\partial \alpha}{\partial x} \quad (2.3.129)$$

$$\gamma_{xr} = \frac{\partial w_{p,0}}{\partial x} - \alpha \quad (2.3.130)$$

$$\varepsilon_{rr} = \gamma_{r\theta} = \gamma_{\theta x} = 0 \quad (2.3.131)$$

Note that the strains ε_{xx} and $\varepsilon_{\theta\theta}$ vary through the pipe wall thickness, while the transverse shear strains γ_{xr} are constant. So far, we have ignored the Kirchoff hypothesis that dictates normal lines to remain perpendicular to the middle surface. Expressed in mathematical terms, this assumption suggests that shear strains γ_{xr} are taken equal to zero so that:

$$\alpha = \frac{\partial w_{p,0}}{\partial x} \quad (2.3.132)$$

However, doing so limits the accuracy of the solution at high-frequency loads with wavelengths comparable to the shell thickness. If the normality condition is relaxed and allow for arbitrary but constant rotation of the transverse normal lines, we fall back on the hypotheses of the Timoshenko beam theory where shear deformations are accounted for. The equivalent theories in plates/shells are dubbed “first-order shear deformation plate theories” [70]. The discussed

shell theory in this thesis, as first derived by Lin & Morgan [71], also considers the effect of shear deformations, thus $\gamma_{xr} \neq 0$.

Solving Eqns. (2.3.120) and (2.3.121) for normal stresses yields:

$$\sigma_{\theta\theta} = \frac{E}{(1-\nu_p^2)} (\epsilon_{\theta\theta} + \nu_p \epsilon_{xx}) \quad (2.3.133)$$

$$\sigma_{xx} = \frac{E}{(1-\nu_p^2)} (\epsilon_{xx} + \nu_p \epsilon_{\theta\theta}) \quad (2.3.134)$$

while shear stresses are:

$$\tau_{xr} = KG_s \gamma_{rx} = K \frac{E}{1-\nu_p} \gamma_{xr} \quad (2.3.135)$$

The shear stresses predicted should be corrected, since the model yields a constant shear distribution across the wall thickness. This violates the condition where the inner and outer radii of the pipe are stress-free. To account for that, the shear correction factor K is introduced in Eq. (2.3.135). Factor K mainly relates to the cross-sectional geometry of the element, and in our case is taken equal to 8/9. We replace the strain-displacement equations (Eqns. (2.3.128) to (2.3.130)) into the stress-strain equations (Eqns. (2.3.133) to (2.3.135)) to obtain the following stress-displacement equations:

$$\sigma_{\theta\theta} = \frac{E}{(1-\nu_p^2)} \left\{ \frac{w_{p,0}}{R} + \nu_p \left[\frac{\partial u_{p,0}}{\partial x} + (r-r_0) \frac{\partial \alpha}{\partial x} \right] \right\} \quad (2.3.136)$$

$$\sigma_{xx} = \frac{E}{(1-\nu_p^2)} \left[\frac{\partial u_{p,0}}{\partial x} + (r-r_0) \frac{\partial \alpha}{\partial x} + \frac{\nu_p}{R} w_{p,0} \right] \quad (2.3.137)$$

$$\tau_{xr} = KG_s \left(\alpha + \frac{\partial w_{p,0}}{\partial x} \right) \quad (2.3.138)$$

The stress-displacement equations (Eqns. (2.3.136) to (2.3.138)) are substituted in the linear and angular momentum equations for the pipe wall element of finite thickness (Eqns. (2.3.65), (2.3.66), and (2.3.103)). However, in the momentum conservation equations, stresses are averaged across the pipe thickness. Therefore, before substituting the averaged normal and shear stress values must be evaluated. According to Eqns. (2.3.17) and (2.3.138) the averaged shear stresses $\bar{\tau}_{rx}$ in terms of displacements are:

$$\bar{\tau}_{xr} = \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} KG \left(\alpha + \frac{\partial w_{p,0}}{\partial x} \right) r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r dr} = KG_s \left(\alpha + \frac{\partial w_{p,0}}{\partial x} \right) \quad (2.3.139)$$

From Eqns. (2.3.19) and (2.3.136), the thickness-averaged hoop stresses $\bar{\sigma}_{\theta\theta}$ are:

$$\begin{aligned} \bar{\sigma}_{\theta\theta} &= \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \frac{E}{(1-\nu^2)} \left\{ \frac{w_{p,0}}{R} + \nu_p \left[\frac{\partial u_{p,0}}{\partial x} + (r-r_0) \frac{\partial \alpha}{\partial x} \right] \right\} r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r dr} \\ &= \frac{E}{(1-\nu^2)} \left(\frac{w_{p,0}}{R} + \nu_p \frac{\partial u_{p,0}}{\partial x} \right) \end{aligned} \quad (2.3.140)$$

And, lastly, from Eqns. (2.3.39) and (2.3.137) the averaged axial stresses $\bar{\sigma}_{xx}$ are:

$$\begin{aligned} \bar{\sigma}_{xx} &= \frac{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} \frac{E}{(1-\nu_p^2)} \left[\frac{\partial u_{p,0}}{\partial x} + (r-r_0) \frac{\partial \alpha}{\partial x} + \frac{\nu_p}{R} w_{p,0} \right] r dr}{\int_{R-\frac{h}{2}}^{R+\frac{h}{2}} r dr} \\ &= \frac{E}{(1-\nu_p^2)} \left(\frac{\partial u_{p,0}}{\partial x} + \frac{\nu_p}{R} w_{p,0} \right) \end{aligned} \quad (2.3.141)$$

Substituting the outcomes of Eqns. (2.3.139) to (2.3.141) into Eqn. (2.3.65), (2.3.66), and (2.3.103) yields the momentum conservation equations in terms of pipe wall element displacements. These are:

i. r -component

$$\boxed{\rho_p h \frac{\partial^2 w_{p,0}}{\partial t^2} = \frac{\partial S}{\partial x} - \frac{Eh}{(1-\nu_p^2)} \left(\frac{w_{p,0}}{R^2} + \frac{\nu_p}{R} \frac{\partial u_{p,0}}{\partial x} \right) + p} \quad (2.3.142)$$

ii. x -component

$$\boxed{\rho_p h \frac{\partial^2 u_{p,0}}{\partial t^2} = \frac{Eh}{(1-\nu_p^2)} \left(\frac{\partial^2 u_{p,0}}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_{p,0}}{\partial x} \right)} \quad (2.3.143)$$

iii. Angular momentum

$$\rho_p \frac{I_m}{R} \frac{\partial^2 \alpha}{\partial t^2} = \frac{I_m}{R} \frac{Eh}{(1-\nu_p^2)} \frac{\partial^2 \alpha}{\partial x^2} - S + q \frac{h}{2} \quad (2.3.144)$$

where S is defined as:

$$S = KG_s h \left(\alpha + \frac{\partial w_{p,0}}{\partial x} \right) \quad (2.3.145)$$

Eqns. (2.3.142) to (2.3.145) describe the motion of the pipe wall following the simplifications described in this subsection. Note that for inviscid fluid – an assumption shown valid in Eq. (2.2.76) – $q = 0$. In the following sections of this thesis, these equations form the basis of the impedance-based model for wave propagation in fluid-filled elastic pipes.

Chapter 3

Impedence-based model

3.1 Introduction

Chapter 2 derives in detail the governing equations for the two domains of a fluid-filled elastic pipe but does not define their in-between synergy. That is, the equation that describes wave propagation in the fluid is not tied to the equations of motion of the pipe wall. Nevertheless, as a wave propagates along a fluid-filled elastic pipe, there is a definite interaction between the two parts of the coupled system. Indeed, the influence of flexible pipe walls in the propagation of acoustic waves in fluid-filled pipes was recognized by researchers since the 19th century, when it was found experimentally that the wave speed is reduced as the pipe wall deforms [72]. To account for the pipe wall flexibility, Korteweg [73] set up a simple static model that is used by hydraulic engineers designing against transient phenomena to this day. There, the fluid pressure was simply balanced by the hoop forces in the pipe wall. The applicability of that model, however, is limited to waves with a wavelength much larger than the pipe diameter (i.e., waterhammer waves). Early researchers have showed that at high frequencies, where wavelengths are comparable to the pipe radius, wave propagation is indeed dispersive, which implies the existence of higher order modes [74, 75, 76, 56]. In effect, the wave speed is no longer constant, but changes with frequency, and the fluid pressure field varies along the axial, radial and azimuthal dimensions. Besides, different mechanisms progressively become important in order to adequately describe wave propagation, such as pipe wall inertia, axial continuity, and flexural forces in the pipe wall [77]. Fay et al. [78] set up a model based on the impedance condition at the pipe/fluid interface to study the applicability of a fluid-filled

impedance tube apparatus. However, their model failed to account for the axial stresses that develop in the pipe wall, and flexure was described according to the simplified assumptions of the Euler-Bernoulli that introduce errors at higher frequencies. Lin & Morgan [57] introduced a similar dispersion relationship to [78] from the impedance condition at the fluid/pipe wall interface, but in this case the pipe wall was described by a more complete self-derived shell theory model [71]. In fact, the same equations of motion for the pipe wall are also derived in Chapter 2 of this thesis using a control volume (C.V.) approach, instead of the force balance used in [71].

In this chapter, the pressure wave equation for an intact pipe is given first, adjusted from Chapter 2 where the effect of a leak is also included. A solution to the pressure wave is provided that is only usable once the fluid boundary is defined. Thus, the impedance condition at the fluid/pipe wall interface is established, and before applying the elastic pipe wall equations, two ideal conditions are discussed first, that of a rigid and pressure-release boundary. These two scenarios set the limits for the more realistic elastic pipe wall boundary and make for a better introduction to the behaviour of high frequency waves in a bounded waveguide. In the acoustic field, it is well-documented that high-frequency wave propagation through a complex waveguide with defined boundaries, such as a fluid-filled pipe, forms guided waves [79]. To illustrate, a narrowband probing signal/pulse that consists of multiple frequencies becomes dispersed and splits into individual wave packets that travel at different wave speeds (modal group velocities), a phenomenon that is described as multipath propagation [60]. The separated wave packets (modes) arrive at the receiver at different times, making the evaluation of the system response convoluted. To keep things simple, the mechanisms of multipath propagation are discussed first in the ideal rigid and pressure-release pipes before introducing the elastic pipe wall boundary conditions.

The equations of motion of the elastic pipe wall from Chapter 2 are then combined with the pertinent fluid equations through the impedance condition at the fluid/pipe interface to produce the characteristic equation of the coupled pipe waveguide. At low frequencies (i.e., very long waves), two modes propagate; one is water-borne and corresponds to the waterhammer wave, while the other is structure-borne and corresponds to a compressional wave in the pipe wall. At higher frequencies that exceed the pipe ring frequency, additional high-order modes cut-on (i.e., start propagating). The specific condition that is satisfied at the cut-on frequencies is provided, and so is the characteristic equation for an empty pipe wall. The latter, allows for the identification of the structural modes of the coupled waveguide. It is also shown that the

characteristic equation of [78] as well as Korteweg's equation [73] may be retrieved following simplifications (and, thus, are special cases) to the derived characteristic equation. For the first time, the pipe wall displacement solutions and the expressions for the work done by the pressure wave to the pipe wall and cross-sectional pipe area are developed. Finally, the impedance condition is implemented in an analytical pressure wave solution for a point acoustic source in a pipe [80], enabling the evaluation of the system response to any abstract probing signal.

As already noted in Chapter 2, the shell theory equations involve some simplifying assumptions regarding the displacement distributions in the pipe wall that are not valid when the probing wavelength is of the same order as the pipe wall thickness and, consequently, the presented solutions may introduce an error at very high frequencies and/or thick pipes. Since the work of Lin & Morgan [57], other researchers have attempted to overcome the limitations introduced by shell theory. Gazis [81, 70] considered the general equations of elasticity and potential expressions to describe the three-dimensional displacement field of a hollow pipe. Del Grosso [58] simplified Gazis' model to two dimensions and combined it with the wave equations for a contained fluid to predict the exact dispersion relationship for axisymmetric modes in a fluid-filled pipe. Recently, Baik et al. [59], extended the solution to all three dimensions.

Although the validity of these state-of-the-art acoustic models, like [59], is general (i.e., independent of frequency and pipe dimensions), their application is cumbersome. Their formulation is based on the exact equations of elastodynamics that describe the three-dimensional displacement field in both fluid and solid (pipe) domains. Alas, solving the occurring dispersion equation is susceptible to numerical instabilities. Moreover, it is difficult to identify physical mechanisms in the equations of such models and, hence, interpretations are far from the reach of most classically trained engineers. Finding the most accurate model to describe the pipe wall regardless of the excitation frequency and pipe wall thickness is outside the scope of this thesis. To quote Einstein, model must be as simple as possible but still capture the essential physics. Comparatively to intricate state-of-the-art models, the applied shell-theory based model is more robust and simpler in reaching a solution while, at the same time, offering better insight in the wave propagation physics. Most importantly, the impedance condition obtained is more versatile and may readily be applied in a pressure wave solution [80] and, therefore, used for defect detection in pipes (see Chapter 6).

3.2 Model equations

3.2.1 Wave equation solution for an intact fluid-filled pipe

We assume a quiescent inviscid fluid, contained within the boundaries of a cylindrical pipe. The fluid-filled pipe is intact; thus, the mass conservation equation developed in Chapter 2 (Eq. 2.2.39) is modified to forgo the term pertinent to the mass source/sink:

$$\frac{1}{c_f^2} \frac{\partial p}{\partial t} + \rho_0 \nabla \cdot \mathbf{v} = 0 \quad (3.2.1)$$

On the other hand, the momentum conservation equation retains its form from Chapter 2 (Eq. 2.2.79). The time derivative of the mass conservation equation (Eq. (3.2.1)) is subtracted from the divergence of the momentum conservation equation to obtain the wave equation for an intact pipe:

$$\frac{\partial^2 p}{\partial t^2} - c_f^2 \nabla^2 p = 0 \quad (3.2.2)$$

Let x , r and θ be the cylindrical coordinates, where the x -axis coincides with the longitudinal tube axis. Axisymmetric vibrations are merely of interest when the acoustic source is located along the pipe centreline, and pressure variations in the fluid are independent of the azimuthal coordinate, θ . Under these assumptions, the Laplace operator in the pressure wave equation (Eq. (3.2.2)) is expanded to:

$$\frac{\partial^2 p}{\partial t^2} = c_f^2 \left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{\partial p}{\partial r} \right) \quad (3.2.3)$$

A separable, time harmonic solution is assumed for pressure p , given by:

$$p(x, r, t) = p_r(r) p_x(x) \exp(-i\omega t) \quad (3.2.4)$$

where $p_x(x)$, $p_r(r)$ are pressure functions along the x and r axis, respectively, ω ($= 2\pi f$) is the cyclic frequency, and i ($= \sqrt{-1}$) is the imaginary unit. Equation (3.2.4) is inserted into (3.2.3) to yield:

$$k^2 + \frac{1}{p_x} \frac{d^2 p_x}{dx^2} = -\frac{1}{p_r} \left(\frac{d^2 p_r}{dr^2} + \frac{1}{r} \frac{dp_r}{dr} \right) = \text{const.} = k_r^2 \quad (3.2.5)$$

where k ($= \omega/c_f$) is the wavenumber. Solving for p_r yields:

$$r^2 \frac{d^2 p_r}{dr^2} + r \frac{dp_r}{dr} + r^2 k_r^2 p_r = 0 \quad (3.2.6)$$

that is a linear second-order ODE, called the Bessel differential equation. The solution to the particular ODE is [83]:

$$p_r(r) = C_1 J_0(k_r r) + C_2 Y_0(k_r r) \quad (3.2.7)$$

where J_0 , Y_0 are the zeroth order Bessel functions of the first and second kind, respectively, and C_1 , C_2 are constants. For the solution to remain physical at $r = 0$ (i.e., $p_r \neq \infty$), C_2 must be equal to zero. Therefore, the solution becomes:

$$p_r(r) = p_0 J_0(r k_r) \quad (3.2.8)$$

where p_0 is the arbitrary ambient pressure along the pipe centreline. Solving Eq. (3.2.5) for p_x gives:

$$\frac{d^2 p_x}{dx^2} + p_x(k^2 - k_r^2) = 0 \quad (3.2.9)$$

which is a second-order linear constant-coefficient ordinary differential equation, and the general solution is:

$$p_x(x) = C_3 \exp(ix\sqrt{k^2 - k_r^2}) + C_4 \exp(-ix\sqrt{k^2 - k_r^2}) \quad (3.2.10)$$

where constants C_3 and C_4 are allowed to be complex so that a real-valued solution is obtained. The pressure solution is produced after substituting Eqns. (3.2.9) and (3.2.8) into (3.2.4):

$$p = p_0 J_0(r k_r) \left[C_3 \exp(ix\sqrt{k^2 - k_r^2}) + C_4 \exp(-ix\sqrt{k^2 - k_r^2}) \right] \exp(-i\omega t) \quad (3.2.11)$$

Considering only the positive x direction for the propagating wave, and arbitrarily setting $C_3 = 1$ yields:

$$p(x, r, t) = p_0 J_0(k_r r) \exp \left[i \left(x\sqrt{k^2 - k_r^2} - \omega t \right) \right] \quad (3.2.12)$$

Similarly, the solutions for the fluid particle displacements (u , w) in the axial and radial coordinates (x , r) may be assumed to be of the form:

$$u(x, r, t) = u_r(r) J_0(k_r r) \exp \left[i \left(x\sqrt{k^2 - k_r^2} - \omega t \right) + \frac{i\pi}{2} \right] \quad (3.2.13)$$

$$w(x, r, t) = w_r(r) J_0(k_r r) \exp \left[i \left(x \sqrt{k^2 - k_r^2} - \omega t \right) \right] \quad (3.2.14)$$

Note that a phase difference equal to $\pi/2$ is introduced between the axial and radial displacements; the reason is revealed in an upcoming section where the pipe wall displacements are discussed. We wish to express the fluid displacement functions u_r and w_r in the radial direction in terms of pressure. The axial and radial components of the momentum equation (Eq. (2.2.79)) in terms of displacements u , w and pressure p are written as:

$$\rho_0 \frac{\partial^2 u}{\partial t^2} + \frac{\partial p}{\partial x} = 0 \quad (3.2.15)$$

$$\rho_0 \frac{\partial^2 w}{\partial t^2} + \frac{\partial p}{\partial r} = 0 \quad (3.2.16)$$

Substituting the solutions for pressure p (Eq. (3.2.12)) and axial fluid displacements u (Eq. (3.2.13)) into the axial momentum conservation equation (Eq. (3.2.15)) yields the axial fluid displacement function along the radial direction u_r :

$$u_r(r) = \frac{p_0 \sqrt{k^2 - k_r^2}}{\rho_0 \omega^2} J_0(r k_r) \quad (3.2.17)$$

For the radial fluid displacement function along the radial direction w_r , Eqns. (3.2.14) and (3.2.17) are replaced into the radial momentum conservation component equation Eq.(3.2.16) to obtain:

$$w_r(r) = -\frac{p_0 k_r}{\rho_0 \omega^2} J_1(r k_r) \quad (3.2.18)$$

Therefore, the fluid displacement solutions in the axial x and radial r directions are retrieved after replacing Eq. (3.2.17) into Eq. (3.2.13) and Eq. (3.2.18) into (3.2.14), respectively:

$$u(x, r, t) = \frac{p_0}{\rho_0 \omega c_{ph}} J_0(k_r r) \exp \left[i \left(k_x x - \omega t \right) + \frac{i\pi}{2} \right] \quad (3.2.19)$$

$$w(x, r, t) = -\frac{p_0 k_r}{\rho_0 \omega} J_1(k_r r) \exp \left[i \left(k_x x - \omega t \right) \right] \quad (3.2.20)$$

where $k_x (= ((\omega/c_f)^2 - k_r^2)^{1/2})$ is the axial wavenumber component, $c_{ph} (= \omega/k_x)$ is the phase velocity (i.e., the velocity at which the wave phase propagates along the waveguide), J_1 is the first order Bessel function of the first kind. To solve the wave equations (Eqns. (3.2.12), (3.2.19), and (3.2.20)) for a fluid-filled pipe waveguide, an additional equation that links the

axial wavenumber k_x (or radial wavenumber k_r) to the cyclic frequency ω is required. Such an equation is obtained after defining the boundary conditions at the fluid/pipe wall interface.

Note that the effect of pipe roughness is ignored in the derivation of the pressure wave solution. A turbulent pipe flow as a result of the pipe roughness would result in a reduced range i.e., pipe roughness essentially attenuates the propagating pressure wave as shown in [84], where the effect of the viscous boundary layer close to the pipe wall was considered analytically. In particular, [84] reported that the effect of the viscous boundary is only to reduce the propagating range and has no effect on the wave propagation characteristics of the waveguide (i.e., the real part of the – complex in this case – axial wavenumber k_x). Finally, the probing wavelengths in this thesis are much larger than the pipe roughness scale, hence do not interact. In case the attenuation characteristics were of interest, then the viscous term in the Navier-Stokes equation must be retained when producing the pressure wave equation. The wave equation solution would then be expressed in terms of a complex wavenumber, where the imaginary part relates to the wave attenuation.

Regarding the interaction of the pressure wave with flow present in the pipe, there are two scenarios. First, to include the effect of a *uniform mean flow*, the terms that scale to the Mach number in the continuity equation are retained, and so are the advective terms in the momentum conservation equation. A modified wave equation is produced where the time derivative moves with the mean flow. Subsequently, the wavenumber in the solution of the pressure wave becomes a function of the Mach number. Second, a *turbulent flow field* is modelled according to the “aero-acoustic analogy” of Lighthill, where the complex fluid mechanical process is described by an additional source in the wave equation, known as Lighthill’s stress tensor [80].

3.2.2 Ideal boundary conditions

The radial pipe boundaries are introduced through the impedance condition at the fluid/pipe interface. Such condition is defined equal to the ratio of pressure p over the radial fluid displacements at $r = R$:

$$Z(\omega, k_r) = \left. \frac{p(r)}{w(r)} \right|_{r=R} = -\frac{\rho_0 \omega^2}{k_r} \frac{J_0(k_r R)}{J_1(k_r R)} \quad (3.2.21)$$

Prior to discussing the boundary conditions of an elastic pipe wall, it is worthwhile to examine two extreme cases, these of (i) a rigid and (ii) pressure-release boundary for the fluid. These two cases will set the limits for the “intermediate” condition of the flexible pipe wall.

Rigid pipe

For a rigid pipe wall, it may be assumed that the radial fluid displacements at the fluid/pipe interface are equal to zero. Hence, the impedance equation is written as:

$$Z_{\text{rigid}}(\omega, k_r) = \left. \frac{p(r)}{w(r)} \right|_{r=R} = \infty \quad (3.2.22)$$

The condition of Eq. (3.2.22) is satisfied for $w(R) = 0$, or $J_1(k_r R) = 0$. That is, when the argument $k_r R$ of the first order Bessel function of the first kind coincides with the roots a_n of the said function:

$$k_r R = a_n, \text{ where } J_1(a_n) = 0 \text{ for } n = 0, 1, 2, \dots \quad (3.2.23)$$

The radial wavenumber component $k_r^2 = k^2 - k_x^2$, hence Eq. (3.2.23) is rewritten in terms of k_x as:

$$k_x = \sqrt{k^2 - \frac{a_n^2}{R^2}} \quad (3.2.24)$$

Under the condition that for a wave to propagate the axial wavenumber must be real, Eq. (3.2.24) implies that $k^2 \geq a_n^2/R^2$. Contrary, if k_x is imaginary, waves are *evanescent* and exponentially decaying from $x = 0$ (Eq. (3.2.12)). Note that the existence of multiple roots n indicates a pertinent number of waves/modes, that each propagates only when $k^2 \geq a_n^2/R^2$ and is evanescent otherwise. Since the wavenumber k is a function of the wave frequency, that condition specifies the – so-called – cut-off frequency above which a particular mode n propagates. The cut-off frequency for the n^{th} mode is retrieved from the condition $k^2 > a_n^2/R^2$ considering that $k = \omega/c_f$:

$$\omega_{n,\text{rigid}} = \frac{a_n c_f}{R} \quad (3.2.25)$$

The J_1 roots are $a_0 = 0$, $a_1 \approx 3.83$, $a_2 \approx 7.01$, $a_3 \approx 7.01$, etc. Hence, the 0^{th} mode propagates at all frequencies, the 1^{st} mode for $f \geq f_1 = 3.83c_f/(2\pi R)$, the 2^{nd} mode for $f \geq f_2 = 7.01c_f/(2\pi R)$, etc.

Note that the larger the pipe radius, the smaller the value of the cut-off frequency. The phase velocity for each mode as a function of frequency ω is given by:

$$c_{ph,n \text{ (rigid)}} = \frac{\omega}{k_{x,n}} = \frac{\omega}{\sqrt{\frac{\omega^2}{c_f^2} - \frac{a_n^2}{R^2}}} = \frac{c_f}{\sqrt{1 - \frac{c_f^2 a_n^2}{\omega^2 R^2}}} \geq c_f \quad (3.2.26)$$

Consequently, the phase velocity for the 0th mode is independent of frequency and equal to $c_{ph,0} = c_f$, whereas for all other modes c_{ph} is a function of frequency ω and smaller than the wave speed in fluid c_f . Finally, the group velocity i.e., the velocity at which wave energy propagates, is defined as:

$$c_{gr,n} = \frac{d\omega}{dk_{x,n}} = \frac{c_f^2}{c_{ph,n}} \leq c_f^2 \quad (3.2.27)$$

Therefore, for $n > 0$, the group velocity $c_{gr,n}$ is smaller compared to c_f , despite $c_{ph,n} > c_f$. That is because modes of order $n > 0$ do not propagate parallel to the pipe centreline but follow a, longer, zigzag pattern. The fact is revealed from the wavenumber vector relation:

$$\mathbf{k} = k_{x,n} \mathbf{i} + k_{r,n} \mathbf{j} \quad (3.2.28)$$

where $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors in the x and r directions, respectively. For the 0th mode, $k_{x,0} = k$ (Eq. (3.2.24)) and $k_{r,0} = 0$; thus, the wavefront is normal to the pipe centreline and wave propagation direction. Conversely, for modes of order $n > 0$, the radial wavenumber component $k_{r,n} \neq 0$ and the wavefront is no longer normal to the pipe centreline but forms an angle $\theta_{k,n}$, given by:

$$\theta_{k,n} = \arctan\left(\frac{k_{r,n}}{k_{x,n}}\right) = \arctan\left(\frac{k_{r,n}}{\sqrt{\frac{\omega^2}{c_f^2} - \frac{a_n^2}{R^2}}}\right) = \arctan\left(\frac{\sqrt{\frac{1}{c_f^2} - \frac{1}{c_{ph,n}^2}}}{\sqrt{\frac{1}{c_f^2} - \frac{a_n^2}{R^2 \omega^2}}}\right) \quad (3.2.29)$$

From Eq. (3.2.26) it is also apparent that as $\omega \rightarrow \infty$, $c_{ph,n} \rightarrow c_f$, and $k_x \rightarrow k$. Consequently, for a given mode n , the angle $\theta_{k,n}$ reduces with frequency ω (Eq. (3.2.29)). Besides, from the same equation, at the cut-off frequency $\omega_{n,\text{rigid}}$ the angle $\theta_{k,n} \rightarrow \pi/2$ i.e., the wavefront is parallel to the pipe centreline and the mode propagates purely radially ($k_x = 0$). It is also interesting to note that at the cut-off frequency $\omega_{n,\text{rigid}}$ Eq. (3.2.26) suggests that $c_{ph,n} \rightarrow \infty$. Therefore, at the cut-off frequencies, the pipe behaves as a radially oscillating cylinder. The presented analysis

demonstrates the mechanism of multipath high-frequency wave propagation in pipes. By definition, when the phase velocity is dependent on the wave frequency, propagation is dispersive. Next, the other extreme case of a pressure release boundary condition at the fluid/pipe interface is investigated.

Pressure-release pipe

Consider a fluid column where the pressure at its radial boundary is equal to zero. In such case, the impedance condition (Eq. (3.2.21)) becomes:

$$Z(\omega, k_r) = \frac{p(r)}{w(r)} \Big|_{r=R} = 0 \quad (3.2.30)$$

The condition is satisfied when $p(R) = 0$, or $J_0(k_r R) = 0$, or when the argument of the 0th order Bessel function of the first kind $k_r R$ is equal to the function roots β_n :

$$k_r R = \beta_n, \text{ where } J_0(\beta_n) = 0 \text{ for } n = 1, 2, \dots \quad (3.2.31)$$

The condition of Eq. (3.2.31) is rewritten in terms of k_x as:

$$k_x = \sqrt{k^2 - \frac{\beta_n^2}{R^2}} \quad (3.2.32)$$

Again, for a wave to propagate, the axial wavenumber must be real and, from Eq. (3.2.32), $k^2 \geq \beta_n^2/R^2$. If the condition is not satisfied, the wave is evanescent. Considering that the wavenumber k is a function of frequency, the particular condition defines the cut-off frequency of mode n as:

$$\omega_{n, \text{press. release}} = \frac{\beta_n c_f}{R} \quad (3.2.33)$$

and the phase velocity for each mode as a function of frequency ω is given by:

$$c_{ph, n \text{ (press. release)}} = \frac{\omega}{k_{x, n}} = \frac{\omega}{\sqrt{k^2 - \frac{\beta_n^2}{R^2}}} = \frac{c_f}{\sqrt{1 - \frac{c_f^2 \beta_n^2}{\omega^2 R^2}}} \geq c_f \quad (3.2.34)$$

The expression of Eq. (3.2.33) is identical to that of Eq. (3.2.25) except for the utilized roots. The J_0 roots are $\beta_0 \approx 2.40$, $\beta_1 \approx 5.52$, $\beta_2 \approx 8.65$, $\beta_3 \approx 11.79$, etc. Unlike the case of the rigid pipe, there is no mode that propagates at all frequencies, but all present distinct cut-off frequencies.

In terms of normal frequency f , the 1st mode propagates when $f \geq f_1 = 2.40c_f/(2\pi R)$, the 2nd mode when $f \geq f_2 = 5.52c_f/(2\pi R)$, etc. Note that for the same pipe radius R and wave speed c_f , the mode cut-off frequencies for a free-boundary fluid column appear earlier compared to the pertinent modes in the fluid-filled rigid pipe. Regardless, Eqns. (3.2.26) to (3.2.29) and consequent findings are valid for both rigid and free-boundary cases (where a_n should be replaced with β_n). In a realistic setting, however, the boundary conditions should describe a flexible pipe instead of a rigid or pressure-release wall. The next section employs the equations of motion for an elastic pipe wall derived in Chapter 1 to define an appropriate boundary condition at the fluid/pipe interface.

3.2.3 Elastic pipe wall boundary

In Chapter 2, the equations of motion for an elastic pipe wall are derived according to the shell theory introduced by Lin & Morgan [71]. It is reminded that these equations are applicable to a pipe of axial symmetry with a wall thickness h much smaller than the pipe radius R ($R \gg h$). In addition, the distribution of axial displacements is linear across the pipe thickness and the pipe thickness remains unchanged under excitation. Due to these assumptions, it is expected that the validity of the approximate shell theory model is limited to wave excitations where the wavelength is comparable to the pipe wall thickness or smaller. The different components of the momentum conservation equation for the pipe wall (Eqns. (2.3.14) to (2.3.145)) are combined together to a single equation, written in terms of radial displacements $w_{p,0}$ and pressure $p|_{r=R-h/2}$ evaluated at the fluid/pipe wall interface:

$$\left[1 + \frac{h^2}{6(1-\nu)K} \left(\frac{1}{c_p^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} \right) \right] \times \left[\left(\frac{R}{c_p^2} \frac{\partial^2}{\partial t^2} - R^2 \frac{\partial^2}{\partial x^2} \right) \left(\frac{R}{c_p^2} \frac{\partial^2 w_{p,0}}{\partial t^2} + \frac{w_{p,0}}{R} - \frac{R p|_{r=R-h/2}}{\rho_p c_p^2 h} \right) + \nu^2 R \frac{\partial^2 w_{p,0}}{\partial x^2} \right] - \frac{h^2}{12} \left(\frac{1}{c_p^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} \right) \left[\left(\frac{R^2}{c_p^2} \frac{\partial^2}{\partial t^2} - R^2 \frac{\partial^2}{\partial x^2} - 2\nu \right) R \frac{\partial^2 w_{p,0}}{\partial x^2} - \frac{w_{p,0}}{R} \right] = 0 \quad (3.2.35)$$

where $c_p (= E/\rho_p(1 - \nu_p^2)^{1/2})$ is the speed of sound in the pipe material. Harmonic solutions are assumed for the radial pipe displacements $w_{p,0}$ and pressure p , written as:

$$p = P \exp i(k_x x - \omega t) \quad (3.2.36)$$

$$w_p = W_p \exp i(k_x x - \omega t) \quad (3.2.37)$$

where P , W_p are the wave amplitudes, $k_x (= \omega/c_{ph})$ is the axial wavenumber, and i is the imaginary unit. On using Eqns. (3.2.35), (3.2.36) and (3.2.37) yields the following relationship between frequency ω and phase velocity $c_{ph} (= \omega/k_x)$:

$$\frac{p|_{r=R-\frac{h}{2}}}{w_p} = \frac{\rho_p c_p^2 h}{R^2} \left\{ \begin{array}{l} \text{Hoop stresses} \\ 1 \\ \text{Inertia forces} \\ - \frac{\omega^2 R^2}{c_p^2} \\ \text{Poisson effect} \\ \text{Axial stresses} \\ - \frac{c_p^2 v_p^2}{c_p^2 - c_{ph}^2} \\ \text{Flexure with shear strain} \\ + \frac{h^2 \omega^4 R (c_p^2 - c_{ph}^2)}{12 c_{ph}^2 \left[c_{ph}^2 c_p^4 + \frac{h^2 \omega^2}{6(1-\nu)K} (c_p^2 - c_{ph}^2) \right]} \end{array} \right\} \quad (3.2.38)$$

Equation (3.2.38) evaluates the ratio of pressure $p|_{r=R-h/2}$ applied on the pipe wall over the resulting radial displacements w_p ; hence, corresponding to the impedance condition at the pipe wall boundary. The form of the impedance equation is compact and offers an insight into the mechanisms that contribute into the wave propagation in the pipe wall, since each of the terms appearing on the right-hand side of Eq. (3.2.38) is of distinct nature. In particular, the first term corresponds to the static force balance between the applied pressure and hoop stresses, the second to the pipe wall inertia, the third is related to the Poisson effect that interweaves radial with axial displacements, while the last refers to the flexural displacements including the effect of shear deformations. The impedance equation (3.2.38) may be replaced into Eq. (3.2.21) as a boundary condition to the fluid to account for the flexibility of the pipe wall and, thus, obtain the characteristic equation for the waveguide:

$$\frac{J_0(k_r R)}{k_r J_1(k_r R)} = \frac{h \rho_p}{R \rho_0} \left\{ \begin{array}{l} \text{Hoop stresses} \\ 1 \\ \text{Inertia forces} \\ - \frac{c_p^2}{R^2 \omega^2} \\ \text{Poisson effect} \\ \text{Axial stresses} \\ + \frac{c_p^4 v_p^2}{R^2 \omega^2 (c_p^2 - c_{ph}^2)} \\ \text{Flexure with shear strain} \\ - \frac{h^2 \omega^4 (c_p^2 - c_{ph}^2)}{12 c_{ph}^2 \left[c_{ph}^2 + \frac{h^2 \omega^2}{6(1-\nu)K c_p^4} (c_p^2 - c_{ph}^2) \right]} \end{array} \right\} \quad (3.2.39)$$

The dimensionless parameters of Table 3.2.1 are introduced, and the characteristic equation of the coupled system (Eq. (3.2.39)) becomes:

$$\frac{a^2 J_0(k_r R)}{k_r R J_1(k_r R)} = 1 - \frac{1}{\omega^{*2}} + \frac{v_p^2}{\omega^{*2} (1 - c^{*2})} - \frac{\xi \omega^{*2} (1 - c^{*2})}{c^{*2} [c^{*2} + \xi \eta \omega^{*2} (1 - c^{*2})]} \quad (3.2.40)$$

The argument of the Bessel functions on the left-hand side of Eq.(3.2.40) is the product of the internal pipe radius R with the radial wavenumber k_r , where the latter can also be expressed as $(k^2 - k_x^2)^{1/2}$. In the dimensionless terms of Table 3.2.1, the Bessel function argument is written as:

$$\begin{aligned} k_r R &= R(k^2 - k_x^2)^{1/2} \\ &= R \left(\frac{\omega^2}{c_f^2} - \frac{\omega^2}{c_{ph}^2} \right)^{1/2} \\ &= \omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \end{aligned} \quad (3.2.41)$$

The final form of the characteristic equation expressed in dimensionless terms, and after combining Eqns. (3.2.40) and (3.2.41) becomes:

$$\begin{aligned} \frac{\alpha^2 J_0 \left[\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right]}{\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} J_1 \left[\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right]} = \\ 1 - \frac{1}{\omega^{*2}} + \frac{v_p^2}{\omega^{*2}(1 - c^{*2})} - \frac{\xi \omega^{*2}(1 - c^{*2})}{c^{*2} [c^{*2} + \xi \eta \omega^{*2}(1 - c^{*2})]} \end{aligned} \quad (3.2.42)$$

Table 3.2.1: List of dimensionless parameters used in the waveguide characteristic equation.

Frequency, ω^*	$R\omega/c_p$
Phase velocity, c^*	c_{ph}/c_p
Curvature correction, ξ	$h^2/(12R^2)$
Shear strains tracer, η	$2/[(1 - v_p)K]$
Wave speed ratio, γ	c_f/c_p
Inertia distribution, a^2	$\rho_0 R/(\rho_p h)$
Pressure, p^*	$Rp/(\rho_p c_p^2 h)$
Shear forces, S^*	$S/(\rho_p c_p^2 h)$
Time, t^*	$t/(R/c_p)$
Lengths, $u_{p,0}^*, w_p^*, x^*, r^*, h^*$	$u_{p,0}/R, w_p/R, x/R, r/R, h/R$

Note that Eq. (3.2.42) is valid when the Bessel function argument is real (or complex). Such condition corresponds to the dimensionless phase velocity c^* being larger than the wave speed

ratio γ ($c^* \geq \gamma$) or, equivalently, the phase velocity c_{ph} being larger than the wave speed in the fluid c_f ($c_{ph} \geq c_f$). When the opposite is true, the Bessel function argument becomes purely imaginary, and the modified Bessel functions I_a may be used instead. By definition, $I_a(x) = i^{-a} J_a(ix)$, and Eq. (3.2.42) becomes:

$$\frac{\alpha^2 I_0 \left[\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right]}{\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} I_1 \left[\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right]} = -1 + \frac{1}{\omega^{*2}} - \frac{v_p^2}{\omega^{*2} (1 - c^{*2})} + \frac{\xi \omega^{*2} (1 - c^{*2})}{c^{*2} [c^{*2} + \xi \eta \omega^{*2} (1 - c^{*2})]} \quad (3.2.43)$$

Equation (3.2.43) may be used when the dimensionless phase velocity c^* is smaller than the wave speed ratio γ ($c^* \leq \gamma$), or when $c_{ph} \leq c_f$. The characteristic equations (3.2.42) and (3.2.43) produce the dispersion relationship (phase velocity c^* vs frequency ω^*) for the fluid-filled elastic pipe waveguide in terms of 4 dimensionless parameters a , γ , ξ , and v_p . Put differently, it is possible to completely describe the coupled waveguide, as long as the inertia ratio a^2 and stiffness ratio γ between the fluid and structural domains, as well as the cross-sectional dimensions (i.e., radius R and thickness h) and material properties (i.e., Poisson ratio v_p) of the pipe are known.

To solve either Eqns. (3.2.42) or (3.2.43) the phase velocity values c^* that satisfy the characteristic equation for a specific frequency ω^* need to be found. In fact, multiple phase velocity values c^* may be valid, and each corresponds to a different propagating mode for the coupled fluid-filled elastic pipe waveguide. The characteristic equation (Eq. (3.2.42) or (3.2.43)) reduces to a quadratic equation in c^{*2} when $\omega^* \rightarrow 0$ (i.e., for very long waves), and it is straightforward to show that at least two modes propagate (i.e., the quadratic equation has always two roots). Of the two modes, one is the 0th mode that is primarily fluid-borne, plane-wave like and, thus, bears resemblance to the 0th mode for a fluid-filled rigid pipe. The other is the structure-borne 1st mode [57], described by a compressional wave that propagates axially along the pipe wall. To demonstrate, consider the first-order terms of the Taylor series expansion for Bessel functions J_0 and J_1 ; an approximation that is valid for small value arguments (i.e., at low frequencies):

$$J_0(x) \approx 1, \text{ and } J_1(x) \approx \frac{x}{2} \quad (3.2.44)$$

Taking the simplification of Eq. (3.2.44) into account and multiplying through with ω^{*2} , the characteristic equation (Eq. (3.2.42)) becomes:

$$\frac{2\alpha^2}{\left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}}\right)} = \omega^{*2} - 1 + \frac{v_p^2}{(1 - c^{*2})} - \frac{\xi\omega^{*4}(1 - c^{*2})}{c^{*2} \left[c^{*2} + \xi\eta\omega^{*2}(1 - c^{*2}) \right]} \quad (3.2.45)$$

At the limit of $\omega^* \rightarrow 0$, Eq. (3.2.45) yields (the last term tends to 0, given that ω^* appears in a higher order in the numerator than the denominator):

$$\frac{2\alpha^2}{\left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}}\right)} = -1 + \frac{v_p^2}{(1 - c^{*2})} \quad (3.2.46)$$

that, as noted, is a quadratic in c^{*2} . Indicatively, two cases are considered that reduce the quadratic to a simple equation of c^{*2} and derive the solutions for the 0th and 1st propagating modes. First, for a pipe wall with a Poisson ratio $\nu = 0$, Eq. (3.2.46) produces Korteweg's equation [73]:

$$c_0^{*2} = \frac{\gamma^2}{2\alpha^2\gamma^2 + 1}, \text{ or } c_{ph,0} = \sqrt{\frac{c_f^2}{2\frac{R}{h}\frac{K_f}{E} + 1}} \quad (3.2.47)$$

Equation (3.2.47) describes the fundamental (or 0th) mode at very low frequencies. The phase speed c_{ph} is equal to the wave speed in unbounded fluid c_f , but slightly reduced under the effect of the elastic pipe wall. Given the large wavelength, the wavefront remains plane and normal to the pipe centreline. It is evident that Korteweg's equation only considers the static equilibrium between the fluid pressure and pipe wall hoop stresses. On the other hand, for an inertia distribution ratio $\alpha^2 \rightarrow 0$, a condition that corresponds to a very light fluid in comparison to the pipe wall (or, an empty pipe), Eq. (3.2.46) yields:

$$c_1^{*2} = 1 - v_p^2, \text{ or } c_{ph,1} = \sqrt{c_p^2(1 - v_p^2)} \quad (3.2.48)$$

that describes a compressional wave in the pipe wall propagating at a wave speed c_p , somewhat reduced by $(1 - v_p^2)^{1/2}$. Hence, the particular solution of Eq. (3.2.48) is attributable to the 1st, structure-borne, mode. Note that Eqns. (3.2.47) and (3.2.48) only present two extreme

scenarios for the roots of the characteristic equation at very low frequencies (Eq. (3.2.46)). For a given set of parameters a^2 , γ , and ν that are positive and different than zero, the quadratic equation (Eq. (3.2.46)) must be solved to retrieve the phase speed c_0^* , and c_1^* of the 0th and 1st modes, respectively, when $\omega^* \rightarrow 0$.

At increased frequencies, the complete characteristic equation (Eq. (3.2.42)) affords multiple solutions in addition to the two found at low frequencies. Still, the smallest among the solution values belongs to the fundamental (or 0th) mode. An infinite number of radial modes may be calculated, propagating at frequencies higher than their respective cut-off frequencies. These cut-off frequencies may be calculated by the characteristic equation keeping in mind that the corresponding phase velocity c^* tends to infinity at these points (see the analysis in subsection 3.2.2). Hence, from Eq. (3.2.42), we take the limit for $c \rightarrow \infty$. The rightmost term of Eq. (3.2.42) may be shown to approach infinity since c appears in higher order in the denominator. Eventually, the radial mode cut-off frequencies are obtained after solving:

$$\frac{J_0(\omega^*/\gamma)}{J_1(\omega^*/\gamma)} = \frac{\omega^{*2} - 1}{a^2 \omega^* \gamma} \quad (3.2.49)$$

for ω^* . It is clearly inferred that at the cut-off frequencies, the defining mechanism is that of pipe wall inertia and hoop stresses since the values of the remaining terms of Eq. (3.2.42) approach zero. Intuitively, that is expected since at the cut-off frequency the axial wavenumber component is equal to zero ($k_x = 0$), and the coupled pipe waveguide performs a radially oscillating motion over its entire length.

Empty pipe

For an empty pipe, the pressure of the contained fluid is assumed to be equal to zero. Consequently, the left-hand side of the characteristic equation (Eq. (3.2.42)) that corresponds to the fluid loading is set equal to zero to obtain:

$$0 = 1 - \frac{1}{\omega^{*2}} + \frac{\nu_p^2}{\omega^{*2}(1 - c^{*2})} - \frac{\xi \omega^{*2}(1 - c^{*2})}{c^{*2} [c^{*2} + \xi \eta \omega^{*2}(1 - c^{*2})]} \quad (3.2.50)$$

that provides the dispersion relationship between phase velocity c^* and frequency ω^* for an empty pipe. Phase velocity c^* appears in Eq. (3.2.42) raised to the power of six; hence, for a given frequency ω^* , there are at most three positive real values of c^* . Therefore, this model

describes at most three propagating modes of vibration for the empty pipe wall. Equation (3.2.50) can also be written as a quadratic equation in ω^{*2} :

$$\begin{aligned} \xi(1-c^{*2})^2(1-\eta c^{*2})\omega^{*4} + c^{*2}(1-c^{*2})\left[(1-v_p^2)\xi\eta - (1+\xi\eta)c^{*2}\right]\omega^{*2} \\ + c^{*4}(1-v_p^2 - c^{*2}) = 0 \end{aligned} \quad (3.2.51)$$

From Eq. (3.2.51) some limiting values for the phase velocity c^* and frequency ω^* can be extracted. After dividing through by ω^{*4} and taking the limit as $\omega^* \rightarrow \infty$, the phase velocity c^{*2} must be equal to 1, 1, and $1/\eta$ for the equation to be satisfied i.e.,

$$c^{*2} = 1, 1, \text{ and } 1/\eta \text{ (} c_{ph} = c_p, c_p, \text{ and } c_p/\eta \text{), at } \omega^* \rightarrow \infty \quad (3.2.52)$$

That is, at high frequencies with wavelengths much smaller than the pipe wall thickness, the phase velocity for two of the three propagating modes approaches that of the wave speed in an unbounded pipe material medium. Conversely, the phase velocity of the third mode is reduced by η under the effect of flexure. Another interesting case is obtained by taking the limit of Eq. (3.2.51) at $c^* \rightarrow \infty$; thus, identifying the cut-off frequencies of the two high-order modes (note that the third mode propagates at all frequencies, and $c^{*2} = 1$ at $\omega^* \rightarrow 0$). Doing so, gives:

$$\omega^{*2} = 1, \text{ and } 1/\xi\eta, \text{ at } c^* \rightarrow \infty \quad (3.2.53)$$

Therefore, one empty pipe mode cut-offs at $\omega^{*2} = 1$, corresponding to the ring frequency where the pipe shell performs a radial “breathing motion”, and the other at $\omega^{*2} = 1/\xi\eta$. Given that we are limited to pipes where $R \gg h$, ξ is always smaller than 1 and $1/\xi\eta > 1$. The above is to demonstrate that such analysis is easily performed on an analytical model with an explicit characteristic equation. In contrast, more complete but also complex models like the ones based on the elastodynamic equations [81, 58, 61] to describe the pipe and fluid domains are difficult to physically interpret. On that note, additional special cases of Eq. (3.2.42) may be obtained after performing suitable approximations. These special cases are simpler in their formulation but are only valid at a lower range of excitation frequencies as will be shown in a following chapter.

3.2.4 Simplifications of the characteristic equation

Flexible pipe without shear deformations and axial stresses

As noted in Chapter 2, the flexural term of the Eq. (3.2.39) (i.e., the rightmost term) is derived on the basis that shear deformations are included in the expression of the angular rotation α , in accordance to the assumptions of the Timoshenko beam theory. Fay et al. [78], on the other hand, followed the assumptions of the Euler-Bernoulli beam theory that neglects such effect stating instead that $\alpha = \partial w_{p,0}/\partial x$, hence producing a simpler flexural term. Besides, in their derivation axial stresses were also ignored. Under these assumptions, their dispersion relationship is:

$$\frac{J_0(k_r R)}{(k_r R) J_1(k_r R)} = \frac{h \rho_p}{R \rho_0} \left(\begin{array}{c} \text{Hoop stresses} \quad \text{Inertial forces} \quad \text{Flexural term} \\ 1 \quad - \quad \frac{c_p^2}{\omega^2 R^2} \quad - \quad \frac{h^2 \omega^2 c_p^2}{12 c_{ph}^4} \end{array} \right) \quad (3.2.54)$$

Despite the similarities of Eq. (3.2.40) to (3.2.54), the most serious drawback of the simplified characteristic equation is the lack of a mechanism to support the axial wave propagation in the pipe wall. Consequently, at low frequencies, only the fundamental mode (0th) mode appears to propagate, while the 1st mode that comprises of a compressional wave in the pipe wall is disregarded. At higher frequencies, it is expected that propagating modes involving an axially propagating compressional wave in the pipe wall will be misrepresented. In addition, it is likely that these modes with a dominant flexural term will present an error, especially at higher frequencies where the effect of shear deformations becomes significant. The dependence of shear deformations on frequency is evident in the second term of the denominator of the flexural term in Eq. (3.2.39) that is absent from the pertinent term in Eq. (3.2.54).

Flexible pipe with non-negligible radial inertia

A further simplification is to completely drop the flexural term from the expression on Eq. (3.2.54) and only account for the pipe wall inertia and static hoop stresses. The dispersion equation in that case is written as:

$$\frac{J_0(k_r R)}{(k_r R) J_1(k_r R)} = \frac{h \rho_p}{R \rho_0} \left(\begin{array}{c} \text{Hoop stresses} \quad \text{Inertial forces} \\ 1 \quad - \quad \frac{c_p^2}{\omega^2 R^2} \end{array} \right) \quad (3.2.55)$$

Although Eq. (3.2.55) predicts multiple modes, it is anticipated that its validity is limited to low-frequency values below the ring frequency $\omega^* = 1$ and, thus, can only describe the

fundamental (0th) mode If only the first term of the Taylor series expansion for the Bessel functions is considered (see Eq. (3.2.44)), then Eq. (3.2.55) becomes:

$$\frac{\omega}{k_x} = \sqrt{\frac{c_f^2}{1 + \frac{2R}{h} \frac{K_f}{E} \frac{1}{1 - 4R^2 \frac{\omega^2}{4c_p^2}}}} \quad (3.2.56)$$

whose applicable frequency range for describing the fundamental mode is further reduced.

Flexible pipe but negligible radial inertia

Eliminating the frequency term (i.e., the inertial forces) from the denominator of the right-hand side of Eq. (3.2.56) yields the well-known Korteweg's formula (Eq. (3.2.57)). Korteweg [73] derived the equation based on a static setting for the force equilibrium in a pipe cross-section, balancing the internal fluid pressure with the hoop stresses of an elastic pipe. The wave speed prediction of Eq. (3.2.57) for the water hammer wave is only reliable for very long waves with frequencies approaching zero.

$$\frac{\omega}{k_x} = \sqrt{\frac{c_f^2}{1 + \frac{2R}{h} \frac{K_f}{E}}} \quad (3.2.57)$$

If the elastic pipe effect in reducing the wave speed is ignored, Eq. (3.2.57) yields the wave speed in a rigid pipe (Eq. (3.2.26) for $a_n = 0$). In conclusion, all models presented in this subsection are special cases of Eq. (3.2.39), valid only under a certain range of frequencies.

3.2.5 Pipe wall displacements

If one solves the characteristic equation and finds the valid (ω, c_{ph}) pairs for a given propagating mode, then the pressure of the contained fluid and fluid particle displacements may be evaluated from Eqns. (3.2.12) to (3.2.14) at all points of the fluid column. Similar solutions can also be retrieved for the displacements of the pipe wall. First, the equations of motion for the pipe wall are written according to the dimensionless parameters of Table 3.2.1:

i. r -component

$$\frac{\partial^2 u_{p,0}^*}{\partial t^{*2}} - \frac{\partial^2 u_{p,0}^*}{\partial x^{*2}} - v_p \frac{\partial w_p^*}{\partial x^*} = 0 \quad (3.2.58)$$

ii. x -component

$$\frac{\partial^2 w_p^*}{\partial t^{*2}} + w_p^* + v_p \frac{\partial u_{p,0}^*}{\partial x^*} - \frac{\partial S^*}{\partial x^*} - p^* = 0 \quad (3.2.59)$$

iii. Angular momentum

$$\xi \left(\frac{\partial^3 w_p^*}{\partial t^{*2} \partial x^*} - \frac{\partial^3 w_p^*}{\partial x^{*3}} \right) - \xi \eta \left(\frac{\partial^2 S^*}{\partial t^{*2}} - \frac{\partial^2 S^*}{\partial x^{*2}} \right) - S^* = 0 \quad (3.2.60)$$

where

$$S^* = \frac{KG}{\rho_p c_p^2} \left(\alpha + R \frac{\partial w_p^*}{\partial x^*} \right) \quad (3.2.61)$$

Harmonic solutions are assumed for the axial and radial pipe wall displacements, as well as for the external pressure and shear force, written as:

$$\begin{aligned} u_{p,0}^* &= U_{p,0}^* \exp \left[i\omega^* \left(\frac{x^*}{c^*} - t \right) \right], \quad w_p^* = W_p^* \exp \left[i\omega^* \left(\frac{x^*}{c^*} - t \right) \right], \\ p^* &= P^* \exp \left[i\omega^* \left(\frac{x^*}{c^*} - t \right) \right], \quad S^* = S_{ampl}^* \exp \left[i\omega^* \left(\frac{x^*}{c^*} - t \right) \right] \end{aligned} \quad (3.2.62)$$

where $U_{p,0}^*$, W_p^* , P^* , and S_{ampl}^* are the amplitudes of the pertinent waves. Substituting the solutions (Eq. (3.2.62)) into the dimensionless equations of motion for the pipe wall yields a system of equations with unknowns $U_{p,0}^*$, W_p^* , and S_{ampl}^* :

i. r -component

$$\omega^{*2} U_{p,0}^* \left(\frac{1}{c^{*2}} - 1 \right) - v_p i \frac{\omega^*}{c^*} W_p^* = 0 \quad (3.2.63)$$

ii. x -component

$$W_p^* (1 - \omega^{*2}) + i \frac{\omega^*}{c^*} (v_p U_{p,0}^* - S_{ampl}^*) - P^* = 0 \quad (3.2.64)$$

iii. Angular momentum

$$\xi i W_p^* \frac{\omega^{*3}}{c^*} \left(\frac{1}{c^{*2}} - 1 \right) - \left[\xi \eta \omega^{*2} \left(\frac{1}{c^{*2}} - 1 \right) - 1 \right] S_{ampl}^* = 0 \quad (3.2.65)$$

and the angular rotation α is:

$$\alpha = \frac{\rho_t c_p^2}{KG} S_{ampl}^* + iR \frac{\omega^*}{c^*} \quad (3.2.66)$$

Solving the system of equations (Eqns. (3.2.63) to (3.2.66)) produces the wave amplitudes in terms of frequency ω^* , phase velocity c^* and pressure P^* :

i. Radial displacements

$$W_p^* = - \frac{c^{*2} (c^{*2} - 1) \left[(c^{*2} - 1) \xi \eta \omega^{*2} + c^{*2} \right]}{\left\{ \begin{aligned} & - (c^{*2} - 1) c^{*2} \omega^{*2} \left[c^{*2} (\xi \eta - 1) + \xi \eta (v_p^2 - 1) \right] \\ & + (c^{*2} - 1)^2 \xi \omega^{*4} (c^{*2} \eta - 1) - c^{*4} (c^{*2} + v_p^2 - 1) \end{aligned} \right\}} P^* \quad (3.2.67)$$

ii. Axial displacements

$$U_{p,0}^* = \frac{ic^{*3} v_p \left[(c^{*2} - 1) \xi \eta \omega^{*2} + c^{*2} \right]}{\left\{ \begin{aligned} & - c^{*2} (c^{*2} - 1) \omega^{*3} \left[c^{*2} (\xi \eta - 1) + \xi \eta (v_p^2 - 1) \right] \\ & + (c^{*2} - 1)^2 \xi \omega^{*5} (c^{*2} \eta - 1) - c^{*2} \omega^{*5} (c^{*2} + v_p^2 - 1) \end{aligned} \right\}} P^* \quad (3.2.68)$$

iii. Shear forces

$$S_{ampl}^* = \frac{i (c^{*2} - 1)^2 \xi \omega^{*5}}{\left\{ \begin{aligned} & c^{*3} \left[\omega^{*2} (c^{*2} - c^{*2} \omega^{*2} + v_p^2 + \omega^{*2} - 1) \left(1 - \frac{\xi \eta \omega^{*2}}{c^{*2}} - \xi \eta \omega^{*2} \right) \right] \\ & + \frac{(c^{*2} - 1)^2 \xi \omega^{*6}}{c^*} \end{aligned} \right\}} P^* \quad (3.2.69)$$

Besides, from Eq. (3.2.63), the ratio of the radial displacement amplitude over that of the axial displacements may be obtained directly:

$$\frac{W_p^*}{U_{p,0}^*} = \frac{\omega^*}{c^* v_p} (c^{*2} - 1) i \quad (3.2.70)$$

The imaginary unit in Eq. (3.2.70) denotes that the radial displacements have a $\pi/2$ phase difference with respect to the axial displacements. Hence, a $\pi/2$ phase difference is also introduced in the solution of the axial fluid displacements (Eq. (3.2.13)) relative to the radial fluid displacements (Eq. (3.2.14)). Finally, the dimensionless pressure value P^* may be evaluated from Eq. (3.2.12) for $r = R$:

$$P^* = \frac{Rp_0}{\rho_p c_p^2 h} J_0(k_r R) = \frac{p_0}{\rho_p c_p^2 h^*} J_0 \left[\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right] \quad (3.2.71)$$

The pipe wall displacement amplitudes can, therefore, be evaluated using the frequency/phase velocity (ω^* , c^*) pairs for each propagating mode, obtained after solving the characteristic equation for the fluid-filled elastic pipe (Eq. (3.2.42)). The model predictions for the pipe wall displacements will be verified using experimentally obtained data in a Chapter 4.

3.2.6 Pressure wave work equations

The characteristic equation (Eq. (3.2.39)) defines the interaction between the contained fluid pressure and the pipe wall as a function of the wave frequency. In this thesis, the acoustic source is placed within the body of fluid; thus, an intuitive way to perceive the fluid-structure interaction is as a transfer of energy from the fluid to the pipe. This transfer of energy can be evaluated as the work done by the pressure wave to the pipe perimeter as opposed to the work done by the pressure wave across the pipe cross-section. In this subsection, the particular work expressions for the pressure wave are derived and will be used in Chapter 4 to provide with some further insights on the wave propagation mechanisms in a coupled fluid-filled waveguide.

Work done by the pressure wave on the pipe perimeter per unit time and length

By definition, work is the product of a force and the resulting displacement. To obtain the work done by the pressure wave on the pipe wall, pressure (Eq. (3.2.12)) is first multiplied by the radial fluid displacements (Eq. (3.2.20)), both evaluated at $r = R$, and then multiplied by the pipe perimeter, $2\pi R$. It is assumed that $x, t = 0$; hence, the resulting expression is independent of the x coordinate and time t and yields work done on the pipe wall per unit time and length. As discussed, the Bessel function argument ($k_r R$) that appears in the employed expressions is real when the phase velocity c_{ph} is larger than the wave speed in the fluid c_f , and purely imaginary when the opposite is true, since $k_r = (\omega^2/c_f^2 - \omega^2/c_{ph}^2)^{1/2}$. For the latter case, the modified Bessel functions are used instead, based on the mathematical identity $I_a(x) = i^{-a} J_a(ix)$. Regardless, the work expressions for both real and imaginary Bessel function arguments are provided.

i. $c_{ph} \geq c_f$

$$\begin{aligned}\dot{W}_R &= p \cdot w \Big|_{r=R} \cdot 2\pi R \\ &= -\frac{2\pi R \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2}}{\rho_f \omega} J_0 \left(R\omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) J_1 \left(R\omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right)\end{aligned}\quad (3.2.72)$$

The dimensionless terms of Table 3.2.1 are introduced, and Eq. (3.2.72) becomes:

$$\begin{aligned}\dot{W}_R^* &= p^* \cdot w^* \Big|_{r=R} \cdot 2\pi R \\ &= -\frac{2\pi R^2 \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2}}{\rho_f c_p^2 \omega^*} J_0 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right) J_1 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right)\end{aligned}\quad (3.2.73)$$

ii. $c_{ph} \leq c_f$

$$\begin{aligned}\dot{W}_R &= p \cdot w \Big|_{r=R} \cdot 2\pi R \\ &= \frac{2\pi R \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2}}{\rho_f \omega} I_0 \left(R\omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) I_1 \left(R\omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right)\end{aligned}\quad (3.2.74)$$

In the dimensionless terms of Table 3.2.1, Eq. (3.2.74) is written as:

$$\begin{aligned}\dot{W}_R^* &= p^* \cdot w^* \Big|_{r=R} \cdot 2\pi R \\ &= \frac{2\pi R^2 \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2}}{\rho_f c_p^2 \omega^*} I_0 \left(\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right) I_1 \left(\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right)\end{aligned}\quad (3.2.75)$$

Work done by the pressure wave on the pipe cross-sectional area

For the work done by the pressure wave on the cross-sectional area, the product of pressure (Eq. (3.2.12)) and axial fluid displacements ((3.2.19)) for angle θ from zero to 2π is integrated along the pipe radius R . Again, coordinate x and time t are taken equal to zero; thus, the resulting expression produces the work done by the pressure wave on the cross-sectional area of the fluid column per unit time and length. The expressions for both real (i.e., when $c_{ph} \geq c_f$) and purely imaginary (i.e., when $c_{ph} \leq c_f$) Bessel function arguments are presented.

i. $c_{ph} \geq c_f$

$$\begin{aligned}
\dot{W}_{\text{Area}} &= \int_0^R pu2\pi r dr \\
&= \int_0^R \frac{2\pi r}{\rho_f \omega c_{ph}} J_0 \left(r \omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) J_0 \left(r \omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) dr \\
&= \frac{2\pi}{\rho_f \omega c_{ph}} \int_0^R r J_0^2 \left(r \omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) dr
\end{aligned} \tag{3.2.76}$$

The definite integral on the right-hand side of Eq. (3.2.76) may be calculated according to the identity:

$$\int_0^x t_i J_0^2(t_i) dt_i = \frac{x^2}{2} [J_0^2(x) + J_1^2(x)] \tag{3.2.77}$$

where t_i herein is the variable of integration, and x is the upper bound. Accordingly, Eq. (3.2.76) becomes:

$$\begin{aligned}
\dot{W}_{\text{Area}} &= \int_0^R pu2\pi R dr \\
&= \frac{\pi R^2}{\rho_f \omega c_{ph}} \left[J_0^2 \left(R \omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) + J_1^2 \left(R \omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) \right]
\end{aligned} \tag{3.2.78}$$

Equation (3.2.78) in the dimensionless terms of Table 3.2.1 becomes:

$$\dot{W}_{\text{Area}}^* = \frac{\pi R^3}{\rho_f c_p^2 \omega^* c^*} \left[J_0^2 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right) + J_1^2 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right) \right] \tag{3.2.79}$$

ii. $c_{ph} \leq c_f$

$$\begin{aligned}
\dot{W}_{\text{Area}} &= \int_0^R pu2\pi r dr \\
&= \int_0^R \frac{2\pi r}{\rho_f \omega c_{ph}} I_0 \left(r \omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) I_0 \left(r \omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) dr \\
&= \frac{2\pi}{\rho_f \omega c_{ph}} \int_0^R r I_0^2 \left(r \omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) dr
\end{aligned} \tag{3.2.80}$$

To calculate the integral on the right-hand side of Eq.(3.2.80) the following identity is employed:

$$\int_0^x t_i I_0^2(t_i) dt_i = \frac{x^2}{2} [I_0^2(x) - I_1^2(x)] \quad (3.2.81)$$

where t_i is the variable of integration, and x is the upper bound. Therefore, Eq. (3.2.80) becomes:

$$\begin{aligned} \dot{W}_{\text{Area}} &= \int_0^R pu 2\pi R dr \\ &= \frac{\pi R^2}{\rho_f \omega c_{ph}} \left[I_0^2 \left(R\omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) - I_1^2 \left(R\omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) \right] \end{aligned} \quad (3.2.82)$$

Expressed in the dimensionless terms of Table 3.2.1, Eq. (3.2.82) becomes:

$$\dot{W}_{\text{Area}}^* = \frac{\pi R^3}{\rho_f c_p^2 \omega^* c^*} \left[I_0^2 \left(\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right) - I_1^2 \left(\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right) \right] \quad (3.2.83)$$

Work ratio

Most useful to identify the distribution of the work done by the pressure wave between the pipe wall and cross-section is to take the ratio of the two for a given mode and frequency (i.e., a (ω, c_{ph}) pair). Therefore, Eq. (3.2.72) is divided by Eq. (3.2.78) to obtain:

i. $c_{ph} \geq c_f$

$$\frac{\dot{W}_R}{\dot{W}_{\text{Area}}} = - \frac{2c_{ph} \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2}}{R} \frac{J_0 \left(R\omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) J_1 \left(R\omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right)}{\left[J_0^2 \left(R\omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) + J_1^2 \left(R\omega \left(\frac{1}{c_f^2} - \frac{1}{c_{ph}^2} \right)^{1/2} \right) \right]} \quad (3.2.84)$$

The dimensionless parameters of Table 3.2.1 are substituted into Eq. (3.2.84) to obtain:

$$\frac{\dot{W}_R^*}{\dot{W}_{\text{Area}}^*} = - \frac{2c^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2}}{R} \frac{J_0 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right) J_1 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right)}{\left[J_0^2 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right) + J_1^2 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right)^{1/2} \right) \right]} \quad (3.2.85)$$

For an imaginary Bessel function argument, the work ratio is obtained after dividing Eq. (3.2.73) by Eq. (3.2.82):

ii. $c_{ph} \leq c_f$

$$\frac{\dot{W}_R}{\dot{W}_{Area}} = \frac{2c_{ph} \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2}}{R} \frac{I_0 \left(R\omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) I_1 \left(R\omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right)}{\left[I_0^2 \left(R\omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) - I_1^2 \left(R\omega \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)^{1/2} \right) \right]} \quad (3.2.86)$$

Using the dimensionless parameters defined in Table 3.2.1, Eq. (3.2.86) is expressed as:

$$\frac{\dot{W}_R^*}{\dot{W}_{Area}^*} = \frac{2c^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2}}{R} \frac{I_0 \left(\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right) I_1 \left(\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right)}{\left[I_0^2 \left(\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right) - I_1^2 \left(\omega^* \left(\frac{1}{c^{*2}} - \frac{1}{\gamma^2} \right)^{1/2} \right) \right]} \quad (3.2.87)$$

At the low frequency limit, the approximations for the Bessel functions given in Eq. (3.2.44) may be applied to Eq. (3.2.84) and obtain:

$$\frac{\dot{W}_R}{\dot{W}_{Area}} = \left| \frac{4\omega c_{ph} \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right)}{i \left[4 - R^2 \omega^2 \left(\frac{1}{c_{ph}^2} - \frac{1}{c_f^2} \right) \right]} \right| \quad (3.2.88)$$

3.2.7 Pressure wave solution

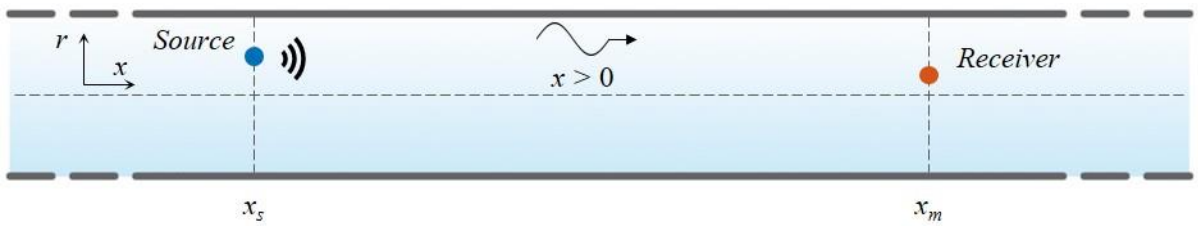


Figure 3.2.1: An acoustic point source is introduced within a fluid-filled elastic pipe at $\mathbf{x}_s$ and emits an impulse. The response of the system is evaluated at $\mathbf{x}_m$ and is given by Eq. (3.2.89).

Conveniently, the impedance condition at the pipe/fluid interface developed in this Chapter can be used to obtain the response of the fluid-filled elastic pipe system to an arbitrary acoustic source. For example, consider an acoustic point source, positioned within the pipe boundaries at $\mathbf{x}_s = (x_s, r_s)$, that emits an impulse that propagates along the waveguide ($x > 0$). In that case,

the response of the system is evaluated at point $\mathbf{x}_m = (x_m, r_m)$, and is given in the frequency domain by Green's function [80]:

$$\hat{G}(\mathbf{x}_m|\mathbf{x}_s, \omega) = \frac{i}{2\pi} \sum_n \frac{J_0(k_{r,n}r_s)J_0(k_{r,n}r_m)}{k_{x,n}R \left(1 - \frac{\omega^2 \rho_0^2}{k_{r,n}^2 Z_s^2(\omega, k_{r,n})}\right) J_0^2(k_{r,n}R)} e^{-ik_{x,n}(x_m - x_s)} \quad (3.2.89)$$

As seen in this chapter, high-frequency waves propagating in the fluid reflect off the pipe wall boundaries, follow different paths and may also interfere. At different parts of the waveguide and for a specific frequency and angle, this interference can be either constructive or destructive. The specific angles (or wavenumbers) at which the interference is constructive constitute solutions to the pressure wave equation and are referred to as modes. This phenomenon, commonly observed in bounded waveguides, is described as multipath wave propagation, and causes the input signal to disperse. The fact is acknowledged in Eq. (3.2.89), since the system response is comprised of the sum of multiple propagating modes n , each described by a modal axial and radial wavenumber, $k_{x,n}$ and $k_{r,n}$, respectively, for a given frequency ω .

The inclusion of the boundaries is imperative to describe the modal characteristics of the waveguide. As such, the effect of the pipe wall elasticity is incorporated in term $Z_s(\omega)$ of Eq. (3.2.89), and corresponds to the impedance of the pipe wall. The impedance condition has already been defined according to the approximations of the shell theory and is given in Eq. (3.2.38). However, in Eq. (3.2.89), the impedance condition is defined as the pressure over the radial fluid particle velocity at $r = R$, while in Eq. (3.2.38) as the pressure over the radial fluid particle displacement. Hence, the impedance expression of Eq. (3.2.38) must be adjusted accordingly before applied to Eq. (3.2.89). First, the solution for the radial fluid particle velocity is the time derivative of Eq. (3.2.20):

$$v_r(x, r, t) = \frac{\partial w}{\partial t} = -\frac{i\omega p_0 k_r}{\rho_0 \omega^2} J_1(k_r r) \exp[i(k_x x - \omega t)] \quad (3.2.90)$$

Therefore, the impedance condition defined as the ratio of pressure to radial fluid displacements at $r = R$ is written as:

$$Z_s(\omega, k_r) = \frac{p(r)}{v_r(r)} \Big|_{r=R} = -i\rho_0 \omega \frac{J_0(k_r R)}{k_r J_1(k_r R)} \quad (3.2.91)$$

Equation (3.2.21) is divided by Eq. (3.2.91) to obtain the relationship between $Z(\omega)$ and $Z_s(\omega)$:

$$Z_s(\omega, k_r) = \frac{i}{\omega} Z(\omega, k_r) \quad (3.2.92)$$

where the expression for $Z(\omega)$ is given in Eq. (3.2.38). Consequently, the expression for $Z_s(\omega)$ is:

$$Z_s(\omega, c_{ph}) = \frac{i\rho_p c_p^2 h}{\omega R^2} \left\{ \begin{array}{l} \text{Hoop stresses} \\ 1 \\ \text{Inertia forces} \\ - \frac{\omega^2 R^2}{c_p^2} \\ \text{Poisson effect} \\ \text{Axial stresses} \\ - \frac{c_p^2 v_p^2}{c_p^2 - c_{ph}^2} \\ \text{Flexure with shear strain} \\ + \frac{h^2 \omega^4 R (c_p^2 - c_{ph}^2)}{12 c_{ph}^2 \left[c_{ph}^2 c_p^4 + \frac{h^2 \omega^2}{6(1-\nu)K} (c_p^2 - c_{ph}^2) \right]} \end{array} \right\} \quad (3.2.93)$$

In the dimensionless terms of Table 3.2.1, $Z_s(\omega)$ becomes:

$$Z_s^*(\omega^*, c^*) = \frac{i\rho_p c_p h}{R\omega^*} \left\{ 1 - \omega^{*2} - \frac{v_p^2}{1 - c^{*2}} + \frac{\xi \omega^{*4} (1 - c^{*2})}{c^{*2} [c^{*2} + \xi \eta \omega^{*2} (1 - c^{*2})]} \right\} \quad (3.2.94)$$

Finally, the Green's function for the fluid-filled elastic pipe in the dimensionless terms becomes:

$$\hat{G}^*(\mathbf{x}_m | \mathbf{x}_s, \omega^*, c^*) = \frac{i}{2\pi} \sum_{\mu} \frac{J_0 \left[r_s^* \omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right) \right] J_0 \left[r_m^* \omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right) \right]}{\frac{\omega^*}{c^*} \left(1 - \frac{c_p^2 \rho_0^2}{R^2 \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right) Z_s^*(\omega^*, c^*)} \right) J_0^2 \left(\omega^* \left(\frac{1}{\gamma^2} - \frac{1}{c^{*2}} \right) \right)} e^{-i \frac{\omega^*}{c^*} (x_m^* - x_s^*)} \quad (3.2.95)$$

Instead of an impulse, we consider an arbitrary input signal defined in time domain, $f(t)$. The response of the system to the input signal $f(t)$ emanating from an acoustic point source at $\mathbf{x}_s = (x_s, r_s)$ is now determined at $\mathbf{x}_m = (x_m, r_m)$. The received acoustic field at the measurement location in the frequency domain is given by:

$$\hat{h} = \hat{f}(\omega) G(x_m, r_m | x_s, r_s, \omega) \quad (3.2.96)$$

where $\hat{f}(\omega)$ is the Fourier transform of the input signal $f(t)$. Also, $\hat{G}$ is the Green's function given in Eq. (3.2.89) and corresponds to the impulse response of the system recorded at the

receiver. The process, as analytically described in Eq. (3.2.96), is the forward wave propagation performed in a fluid-filled elastic pipeline.

Chapter 4

Model verification

4.1 Introduction

The derived impedance-based model is approximate in nature; thus, its appropriateness to describe the pipe waveguide must be verified experimentally. For the purpose, a novel experimental setup is developed in the hydraulic lab at the Hong Kong University of Science and Technology. Urban water supply networks typically employ pipes with a great variety of materials and cross-sectional dimensions and testing the analytical model against all would be impractical. Instead, the lab setup implements two pipes: (i) a thick-walled DN80 High Density PolyEthylene (HDPE) pipe, and (ii) a thin-walled DN80 copper pipe that represent extreme cases in terms of material (with copper being much stiffer) and wall thickness to radius ratio. The nominal diameter of the two pipes is chosen to save lab space, while still managing to excite high-order radial modes given the specifications of the selected acoustic projector. Besides, the acoustic projector is placed in the water and along the pipe centreline, while the coupled pipe/fluid waveguides are subjected to a wideband linear frequency modulated (LFM) pressure wave signal with a bandwidth ranging from 10 to 70 kHz for the DN80 HDPE pipe and 10 to 50 kHz for the DN80 copper pipe. The frequency bandwidth is primarily dictated by the capabilities of the utilized acoustic projector, closely related to its physical dimensions. Regardless, given that the attainable range is inversely proportional to the probing wave frequency, the selected upper frequency limit is a balance between the desired resolution and range. Also, the bandwidth is wider for the case of the thicker HDPE pipe as it represents an extreme case for the approximate impedance-based model assumptions.

For both pipes, their response to the wideband LFM signal is evaluated at multiple equidistant locations along the pipe length using sensors mounted both within the fluid (i.e., hydrophone) and on the outer surface of the pipe wall (i.e., accelerometers). This full suite of measurements provides a complete characterization of the wave propagation in each waveguide and allows for the construction of the experimental dispersion curves from both pressure and acceleration measurements. To verify the validity of the impedance-based model, the experimental dispersion curves are then compared to the pertinent analytical waveguide model solutions (Eq. (3.2.42)), obtained for the nominal pipe characteristics. Considering that the adopted nominal values for the geometrical and mechanical characteristics of the pipes are possibly inaccurate, the dispersion relationship from the approximate impedance-based model is also compared to that from a general, a general, state-of-the-art, model [59], solved for both pipes. One of the novelties of the developed experimental setup is the installation of accelerometers on the pipe wall to evaluate the response of the pipe wall to a wideband source positioned in the water. The experimental displacement measurements are compared to the pertinent analytical expression (Eq. (3.2.70)) for the radial to axial displacement ratio per mode for each pipe as a function of frequency, and further validate the impedance-based model. Interestingly, consideration of the pipe wall displacement results along with Snell's law explains the shape of the obtained dispersion curves and elucidates the mechanisms of wave propagation in fluid-filled elastic pipes.

In Chapter 3, it is demonstrated that the complete characteristic equation of the impedance-based model affords simplifications that produce simpler, well-known models, such as Korteweg's wave speed equation. To exactly identify the frequency intervals and modes where each of the simplified is applicable to, the predictions of the simplified models are compared to these of the complete expression of the impedance-model, as well as these of the general model [59] for a DN80 copper and DN80 acrylic pipe. Interestingly, it is found that the fundamental mode (that corresponds to the plane water hammer wave at very low frequencies) propagates in the pipe wall for excitation frequencies higher than the natural ring frequency, f_{ring} . Indeed, both analytical and numerical results suggest that frequencies higher than f_{ring} most of the energy of the fundamental mode is contained close to the pipe wall and a surface wave forms. Finally, the pressure wave solution for a cylindrical duct that incorporates the impedance condition and is given in Eq. (3.2.96) is verified experimentally for a narrowband, Gaussian modulated pulse in the DN80 HDPE pipe of the lab setup.

4.1.1 Dispersion relationship

The composition of propagating waves in a waveguide varies with space x and time t or – in terms of the reciprocal domain parameters – axial wavenumber k_x and frequency f , respectively. The fact is easily understood considering the phase of the wave equation solution, $\varphi = k_x x - 2\pi f t$. It is usually difficult to extract useful information about modes using signals in the time domain. Fourier transform analysis of signals allows for a better physical insight and mode identification by evaluating the frequency content of the obtained response. Besides, to better understand mode propagation in space and extract information on their individual phase velocities, measurements at multiple locations must be performed. A complete description of wave propagation for a specific waveguide is best represented using dispersion curves, usually in terms of axial wavenumber k_x and frequency f . To better interpret dispersion curves, consider the following intuitive example: An in-pipe acoustic projector emits a pulse and excites pressure waves in the waveguide that propagate to all directions. The propagating waves in the fluid bounce off the pipe wall boundaries, travel along different paths and interfere with each other. At different parts and points of the waveguide and for a specific frequency and angle (wavenumber), this interference can be either constructive or destructive. The frequency and axial wavenumber pair values where the waves constructively interfere produce the dispersion curves and correspond to modes that are solutions to the wave propagation problem. Each waveguide has its own unique set of solutions, or equivalently, its own unique dispersion curves [79]. Hence, if the analytical dispersion curves are validated with the pertinent experimental curves, the fact indicates that the analytical model offers a good representation of the actual waveguide.

Analytical dispersion curves

The solutions obtained from the characteristic equation (Eq. (3.2.42)) constitute pairs of dimensionless frequency ω^* and phase velocity c^* . In this thesis, the analytical characteristic equation is solved numerically using a self-coded bisection root-finding method in MATLAB. Thus, for a given frequency ω^* , multiple phase velocity values c^* are found that satisfy the equation, each corresponding to a different propagating mode. The smallest of these values is associated to the fundamental or 0th Mode. If all valid (ω^*, c^*) are plotted in a frequency/phase velocity graph, the dispersion curves will emerge. Each curve corresponds to an individual propagating mode and denotes its phase velocity variance with frequency. An equivalent

representation of the dispersion relationship for a waveguide is a frequency/axial wavenumber plot since the axial wavenumber is given by $k_x = \omega/c_{ph} = \omega^*/(Rc^*)$, or a frequency/group velocity curve since $c_{gr} = d\omega/dk_x$. It should be noted that solving the characteristic equation produces all possible propagating modes for the waveguide in question. In practice, however, whether a particular mode is excited or not depends on the characteristics of the source and its placement in the waveguide.

Experimental dispersion curves

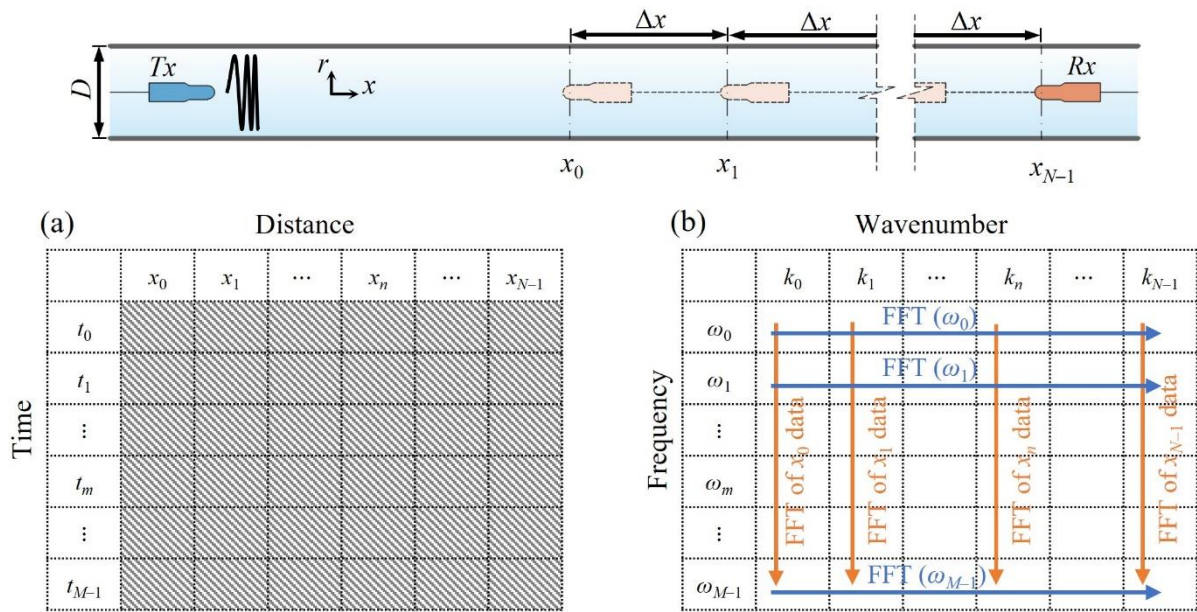


Figure 4.1.1: Illustration of the experimental process of obtaining the experimental dispersion curves for a water filled elastic pipe. (a) A wideband acoustic source is introduced in the pipe, and the system response is sampled in time at multiple equidistant locations along the pipe to eventually form a (x, t) matrix. (b) The space/time (x, t) matrix is then transformed to the reciprocal frequency/axial wavenumber/frequency (k_x, ω) domain, following a 2-Dimensional Fast Fourier Transform (FFT) algorithm. Plotting the values of the (k_x, ω) essentially produces the experimental dispersion curves.

Since wave propagation in a fluid-filled elastic pipe is determined by multiple modes μ , the pressure p at a point in the fluid is the result of multiple coinciding modes:

$$p = \sum_n p_{0,n} J_0(k_{r,n} r) \exp[i(k_{x,n} x - \omega t)] \quad (4.1.1)$$

To make the link between the pressure wave and the dispersion curves more apparent, a spatial Fourier transform is applied to Eq. (4.1.1) with respect to the spatial x coordinate (note that the exponential term $e^{-i\omega t}$ is ignored):

$$\bar{p} = \sum_n p_{0,n} J_0(k_r r) \otimes \delta(k - k_{x,n}) \quad (4.1.2)$$

In Eq. (4.1.2), δ is the Dirac delta function, $\bar{p}$ is the pressure in the axial wavenumber domain and $\otimes$ denotes the convolution operation. Evidently, the response of the system is maximized when $k = k_{x,n}$, subsequently appearing as a region of high energy around that value in the dispersion plot and signifying a point of constructive interference.

To evaluate the dispersion relationship experimentally, it is required that the response of the fluid-filled pipe to a wideband pressure wave signal (e.g., a linear frequency modulated pulse) is evaluated at multiple equidistant locations along the pipe. Naturally, the experimental measurements are performed in the distance/time domain (x, t) , but the dispersion relationship is given in the reciprocal terms of axial wavenumber and frequency (k_x, ω) . Therefore, to obtain the experimental dispersion curves, a 2-Dimensional Fast Fourier Transform (2D FFT) is applied to the (x, t) data. The 2D FFT operation is defined as:

$$\hat{p}(k_x, \omega) = \int p(x, t) \exp(-i(k_x x + \omega t)) dt dx \quad (4.1.3)$$

or alternatively:

$$\hat{p}(k_x, \omega) = \int h(x, \omega) \exp(-i(k_x x)) dx \quad (4.1.4)$$

where $h(x, \omega) = \int p(x, t) \exp(-i\omega t) dt$. That is, the Fourier transform in time is applied to the system response at each measurement location first. Then, the spatial Fourier transform is applied at each occurring frequency to finally obtain the experimental dispersion curves. Therefore, the aim is to experimentally verify the dispersion relationship from the impedance-based analytical model over a frequency interval of interest for a representative set of pipe waveguides. Hence, the analytically obtained dispersion curves are compared to experimental for two pipes – a DN80 HDPE and a DN80 copper pipe – that differ in their geometrical and mechanical characteristics. The response of the coupled system to a wideband excitation is experimentally evaluated in the fluid along the pipe axis, as well as the pipe wall and the dispersion curves are produced for both domains. In the next section, the particulars of the experimental setup are discussed.

4.2 Experimental setup

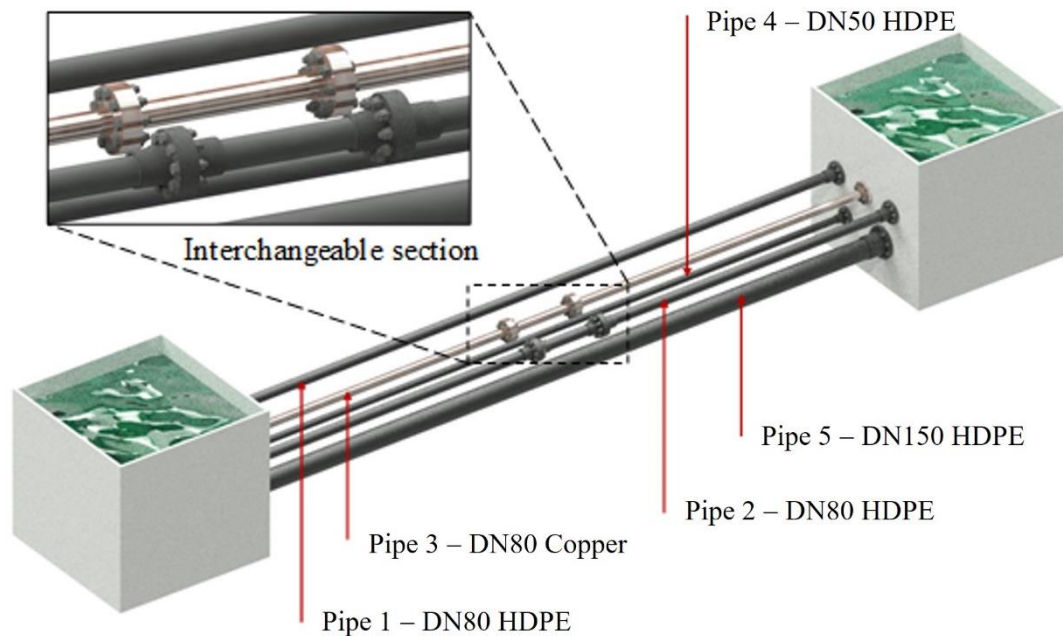


Figure 4.2.1: Schematic of the high-frequency wave experimental setup at the hydraulics lab at HKUST.

The propagation of high-frequency waves in elastic flexible pipes is experimentally tested in a purpose-built facility in the hydraulic lab at HKUST. The setup comprises of 5 pipes in total, 6.5 m in total length, that connect two water-filled tanks at their ends. Each pipe present in the setup, as illustrated in Figure 4.2.1, differs by either geometrical or mechanical characteristics. The scope of the setup is broad, as it is also used to examine the interaction of high frequency waves with defects (such as leaks and blockages) and in-line junctions (see Chapter 7). For that reason, a 0.5 m interchangeable segment is introduced in the middle of pipes 2 & 3, that could be replaced, for example, with a pipe of smaller diameter to simulate a blockage. Regardless, the work presented in this chapter focuses on the study of high-frequency acoustic wave propagation in elastic fluid-filled pipes without any discontinuities along their length. Specifically, this chapter is based on the results from 2 of the pipes in the system, namely the DN80 HDPE (i.e., pipe 1 in Figure 4.2.1) and DN80 copper (i.e., pipe 3 in Figure 4.2.1) pipes. The experimental setup is filled with water from the right-hand side reservoir. Air cavities in the water have the potential to radically change the behaviour of the waveguide by reducing the expected wave speed and introducing unwanted wave scatterers. Hence, to avoid air entrainments, the pipes are inclined with respect to the horizontal axis by 1% with the highest

point at the filling reservoir. Since there is no flow between the two tanks, however, and to ensure that pipes are devoid of any air content, a piston-like block of fabric is dragged multiple times along the pipe. The water level inside the side tanks is positioned about 1 m above the highest mounted pipe centreline and remains quiescent. Finally, the pipes are supported by aluminium frames at approximately 1 m intervals. An actual picture of the experimental setup, where the two pipes of interest – the DN80 HDPE and DN80 Copper pipes – are denoted with arrows, is presented in Figure 4.2.2. Also visible in Figure 4.2.2 are the two reservoirs and the other pipes that comprise the setup.

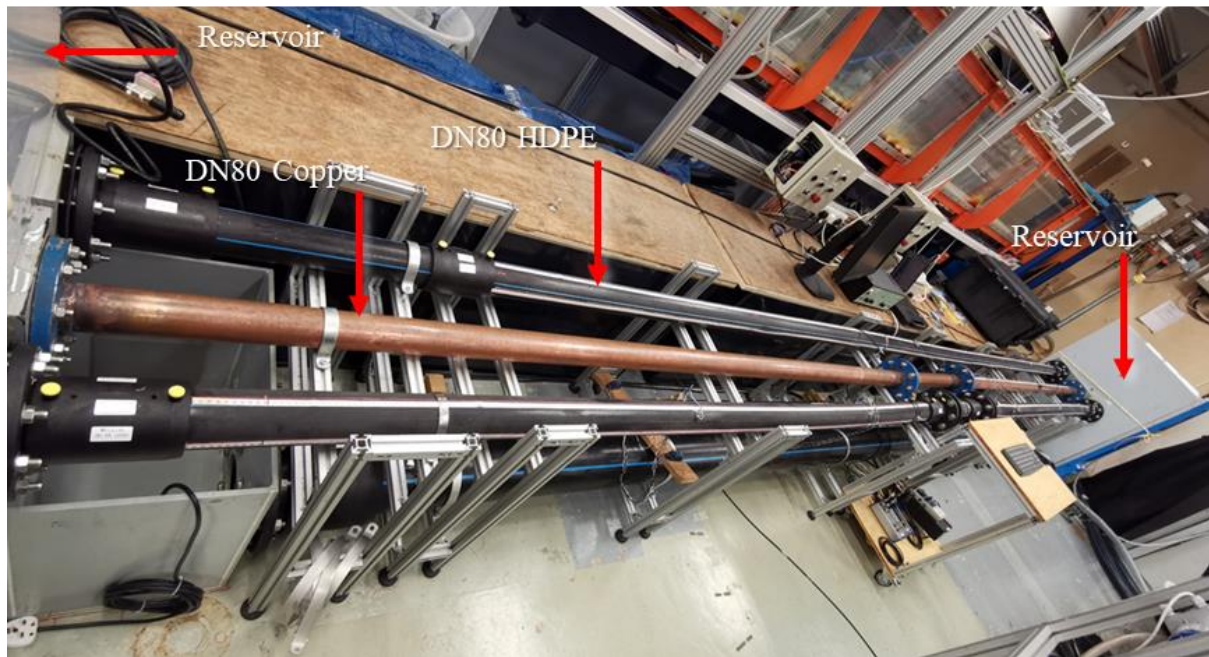


Figure 4.2.2: Actual picture of the experimental setup, where the tested DN80 HDPE and copper pipes as well as the reservoirs at their ends are discernible.

In this thesis, emphasis is placed on the study of acoustic waves with a wavelength comparable to the transverse pipe dimensions. In such case, pressure distribution changes sign along the pipe radius, typical of high-order radial modes. As shown in Eqns. (3.2.25), (3.2.33), and (3.2.49), the frequencies where radial modes begin to propagate – called cut-off frequencies – are dependent on the cross-sectional dimensions of the pipe. Indeed, the cut-off frequencies are inversely proportional to the pipe radius R : the smaller the pipe diameter, the higher the cut-off frequency values. The cross-sectional dimensions of the HDPE and copper pipes chosen (i.e., DN80), position the radial mode cut-off frequencies within the operating frequency range of the acoustic projector. With regards to the material selection, HDPE exhibits increased dissipative characteristics when compared to conventional materials, such as ductile iron, that

could hinder wave propagation. Moreover, HDPE pipes are usually manufactured with thick pipe walls that challenge the approximations adopted by the shell theory-based model at higher frequencies where probing wavelengths are of comparable length. On the other hand, copper is a conventional material, widely available in the market. Compared to HDPE, copper is stiffer and presents an increased Young's modulus and density that naturally yields a larger wave speed value in the pipe wall, c_p . In summary, the two pipes chosen present two extreme cases to validate the impedance-based model with.

Table 4.2.1: Mechanical and geometrical parameters of the employed water filled pipe waveguides

• Pipe Parameters	HDPE pipe	Copper pipe
Internal Radius, R (mm)	39.3	36.6
Wall Thickness, h (mm)	5.7	1.5
Young's modulus, E (GPa)	1.1	117
Density, ρ_p (kg/m ³)	959	8940
Poisson's Ratio, ν	0.45	0.33
Wave speed, c_p (m/s)	1200	3830
Ring frequency, f_{ring} (1/s)	4856	16440
• Water parameters		
Bulk modulus, K_f (GPa)		2.20
Density, ρ (kg/m ³)		1000
Wave speed, c_f (m/s)		1483
• Waveguide dimensionless parameters		
$\alpha [= (\rho R / \rho_p h)^{0.5}]$	2.77	1.44
$\gamma (= c_f / c_p)$	1.22	0.38
h/R	0.15	0.04

To better understand the characteristics of the water-filled pipe as a waveguide, it is useful to compare the mechanical characteristics of the pipe material to these of contained water. The pipe geometrical dimensions and mechanical parameters are retrieved from the manufacturers' specification sheets and are presented in Table 4.2.1. Also included are the calculated dimensionless parameters α and γ for each pipe, that indicate the stiffness and inertia distribution in the coupled waveguide and are originally introduced in the dimensionless characteristic equation (Eq. (3.2.42)). For the case of the HDPE pipe, in particular, water inertia is more substantial compared to that of the pipe wall. Furthermore, a value of γ larger than 1

indicates a rather flexible pipe wall, comparable to water in this aspect. In contrast, copper is much stiffer (small value of γ), and the system is less influenced by the mass of the contained fluid. The two coupled waveguides, therefore, represent two antithetic examples, worthwhile to examine separately. Note that the calculated wave speeds shown in Table 4.2.1 are only relevant for waves with wavelengths much larger compared to the lateral dimensions of the waveguide. At increased frequencies, where wavelengths are of order of the pipe radius, propagating waves “bounce off” the pipe wall and follow a longer zigzag path that effectively reduces their wave speed. Thus, in a multipath wave propagation environment, it is preferable to utilize the group velocity of a particular mode at a given frequency as a measure of the wavespeed.

4.2.1 General testing procedure

The first step is to define the pressure wave signal to be introduced to the waveguide. The task is performed in the digital domain on a computer that executes a self-developed National Instruments™ (Austin, USA) LabVIEW code (front panel is shown in Figure 4.2.3). The goal is to evaluate the response of the pipe system over a range of frequencies; hence the probing signal of preference is a Linear Frequency Modulated (LFM) pulse. For reference, the mathematical description of an LFM signal using discrete points in time may be written as:

$$y_i = A_c \sin \left(\left(\frac{\pi (f_2 - f_1)}{n} i + 2\pi f_1 \right) i \Delta t \right) \quad (4.2.1)$$

where $i = 0, 1, 2, \dots, n - 1$ denotes the discrete time samples, f_1 is the starting frequency, f_2 is the ending frequency, A_c is the wave amplitude and Δt is the selected time step. Herein, the frequency bandwidth ($f_1 - f_2$) of the LFM signal is set from 10 to 70 kHz for the DN80 HDPE pipe and 10 to 50 kHz for the DN80 copper pipe, as primarily dictated by the operating characteristics of the acoustic projector and the stated goals. To reiterate, validating the analytical model for the thicker DN80 HDPE pipe and higher frequencies presents the more disadvantageous case of the two. Regardless, given the cross-sectional dimensions of the pipes tested, many radial modes are expected to cut-on within the selected frequency interval.

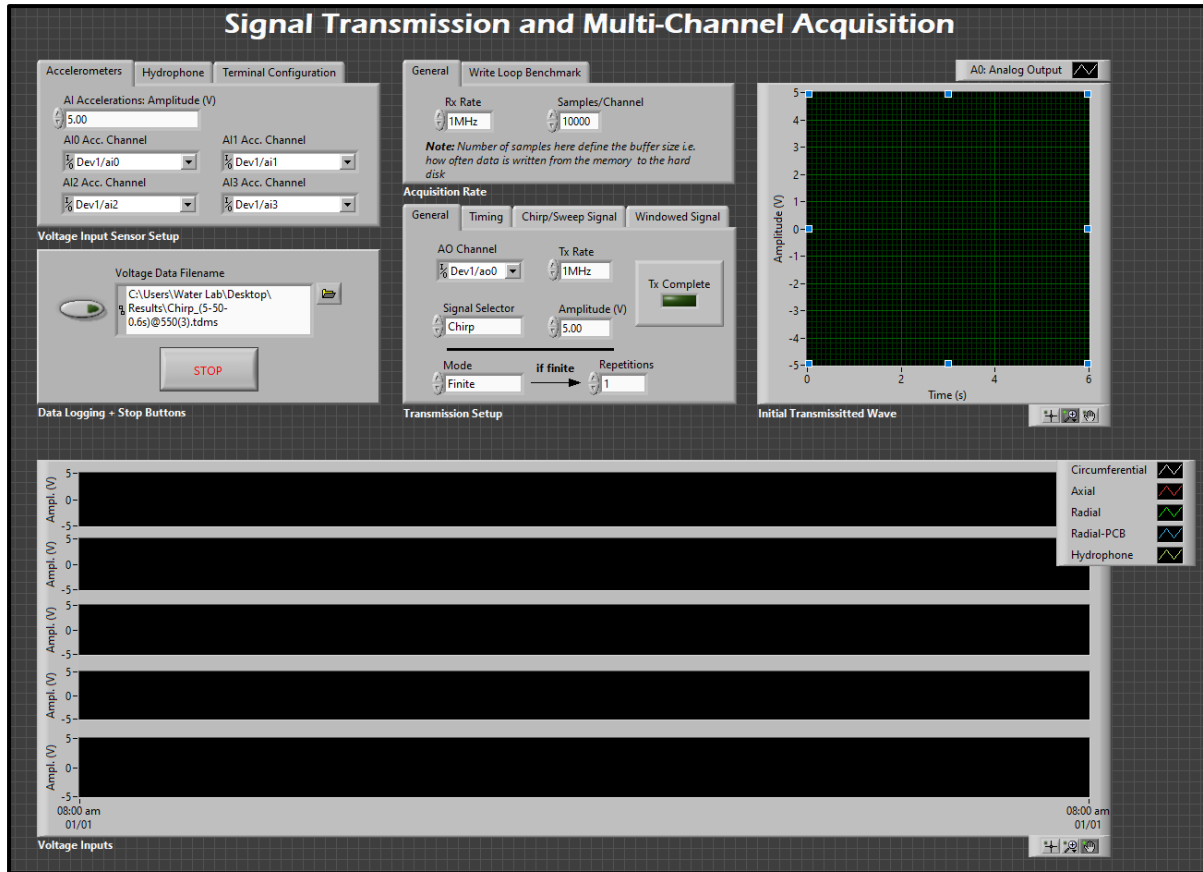


Figure 4.2.3: Control panel of the self-developed National Instruments™ LabVIEW code, that controls and synchronizes all signal transmission and data acquisition tasks for the experimental setup.

According to the Nyquist-Shannon theorem, the sampling rate must be at least twice the maximum expected frequency of the signal to avoid aliasing phenomena. As a rule of thumb, however, a good representation of the signal involves at least 10 sampling points per wavelength, translating to a minimum of 0.7 MSamples/s when the upper frequency value of the defined LFM signal is considered. Regardless, the time step Δt in Eq. (4.2.1) is set to a slightly higher 10^{-6} s, equating to transmission rate of 1M Samples/s. The total number of samples ($n-1$) is selected equal to 10^7 , for a total signal duration of 10 s. Typically, the longer the probing signal duration, the better the resulting resolution from the FFT algorithm when analysing the obtained system response (i.e., assuming an equal signal acquisition sampling rate). However, a very large number of samples (e.g., $> 10^7$) places great demands on computer memory and hard disk storage space. The selected probing signal duration of 10 s is found to be a reasonable compromise between the attainable resolution and computer requirements. The amplitude of the chirp signal A_c is defined equal to 5 V, and the discrete digital signal is

converted to a continuous analog signal using the integrated digital-to-analog converter (DAC) of a National Instruments™ USB-6356 Data Acquisition (DAQ) board with 16-bit resolution.

The output voltage of the National Instruments™ USB-6356 I/O board is inadequate to drive a piezoelectric transducer. Hence, a Brüel & Kjær (Nærum, Denmark) Type 2713 power amplifier is used to apply a 27 dB gain to the 5 V LFM signal, thus raise its amplitude to about ~100 V, a value that corresponds to the maximum amplifier capacity. The amplified signal is then fed to a Brüel & Kjær Type 8104 hydrophone, that acts as an acoustic projector, placed within the pipe boundaries and at the centre of the pipe cross-section. This is achieved via a custom-designed 3D printed resin epoxy base that positions the acoustic centre of the transducer at approximately the pipe centre. The top surface of the base is contoured to the internal pipe wall curvature, while the transducer is attached to the base via a clip holder and secured with plastic cable ties. An embedded magnet within the base forms a pair with an identical magnet mounted on the outer surface of the pipe wall. Magnetic attraction presses the curved surface of the base to that of the pipe wall and consequently places the projector at the desired radial position. The transducer is mounted centrally with regards to the radial coordinate as to excite axisymmetric modes as much as possible (where the pressure variation in the pipe cross section is independent of the azimuthal coordinate, θ). By limiting the number of excited modes to the axisymmetric ones only, the results are simplified and comparable to the predictions of the impedance-based model. The transmitting voltage response (TVR) of the transducer varies with frequency. At 70 kHz, for instance, the TVR of the transducer is rated equal to 132 dB re $\mu\text{Pa}/1 \text{ V @ } 1 \text{ m}$ by its manufacturer [84], while its input voltage V_{rms} is limited to 100 V. Utilizing these values, the maximum sound pressure level (SPL) at 1 m from the source may be calculated from:

$$\text{SPL} = V_{rms}(\text{dB}) + \text{TVR}(f) \quad (4.2.2)$$

where the V_{rms} (dB) scale is relative (re) to 1 V, thus:

$$\text{SPL} = 20\log_{10}(100) + 132 = 172 \text{ dB re } \mu\text{Pa} \quad (4.2.3)$$

or 4 cm of pressure head. The result of Eq. (4.2.3) indicates that the probing signal injected to the system is minute in magnitude compared to typical water-hammer waves and, thus, not a factor that could compromise the structural integrity of the pipeline.

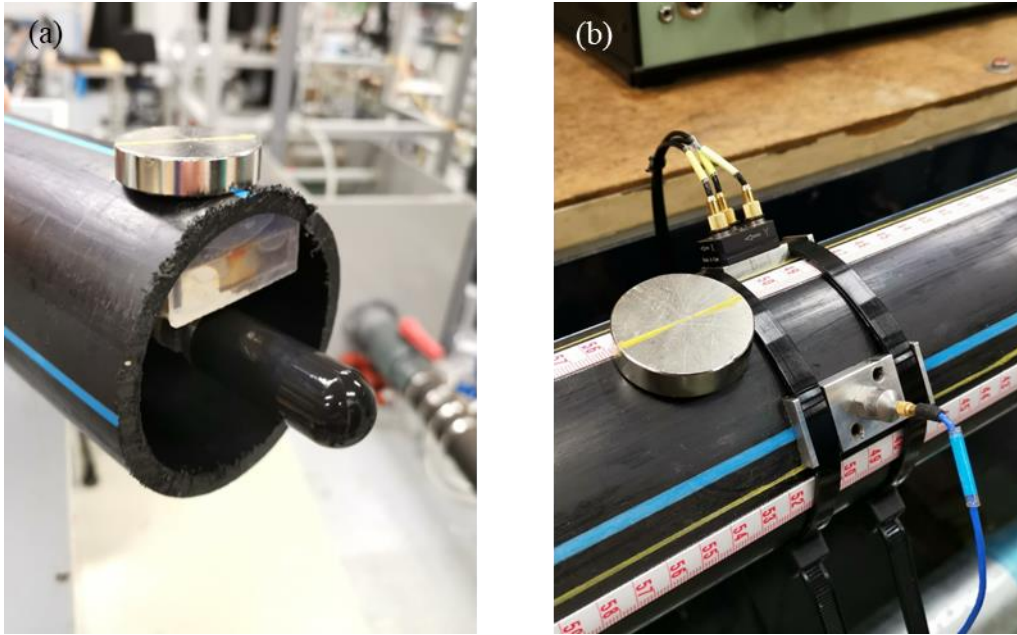


Figure 4.2.4: (a) A bespoke 3D-printed base positions the in-pipe pressure transducers at about the pipe centre. The base embeds a magnet that forms a pair with another on the outer pipe surface. Sliding the outer magnet along the pipe also translates the longitudinal position of the transducer. (b) The motion of the pipe wall is evaluated by a pair of accelerometers mounted to the pipe wall via custom stainless-steel bases that conform to the pipe curvature. To ensure that the accelerometers are firmly mounted, the bases are tied to the pipe wall with two plastic cable ties. In the particular picture, all sensors (in-pipe hydrophone and accelerometers) are at the same longitudinal position.

At the exact moment of the signal emission, the sensing circuit commences with the data acquisition. A transducer identical to the projector (i.e., a Brüel & Kjær Type 8104 hydrophone) measures the occurring pressure field in the fluid domain. Using the same mounting base as the acoustic projector, the hydrophone is also placed at about the pipe centre, initially 1 m from the opposing acoustic projector. Note that the longitudinal position of the receiver or, equivalently, the distance from the projector, may be adjusted by dragging the magnet mounted on the outer surface of the pipe wall along the pipe. The receiving response of the transducer is, according to the manufacturer [84], flat up to 80 kHz and equal to -205 dB re $1\text{ V } \mu\text{Pa}^{-1}$.

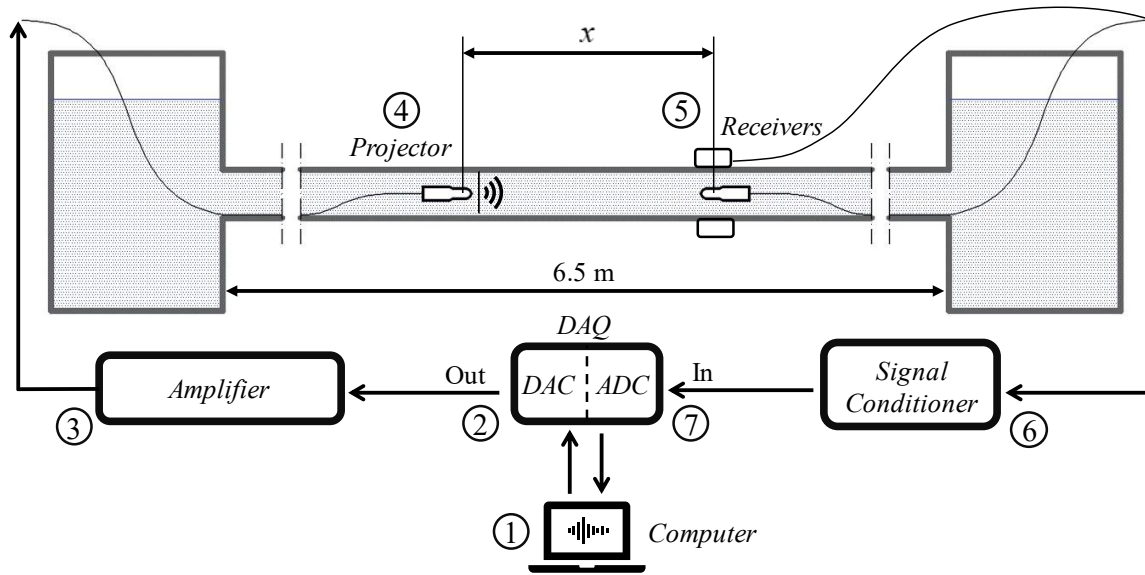


Figure 4.2.5: The probing LFM signal is defined at the computer ①. Then, the computer feeds the data to a National Instruments™ USB-6356 I/O board through a USB connection ②. The signal is converted from the digital to the analog domain, using the embedded DAC of the board. A power amplifier ③ applies a gain to the signal before it reaches the projector ④. The projector – positioned within the pipe – emits the signal, and the response of the system is evaluated at both the fluid and structure domain through a hydrophone that measures the pressure oscillations in the fluid, and two accelerometers that assess the pipe wall vibrations in all three dimensions ⑤. A charge amplifier ⑥ conditions and amplifies the signal. Finally, the analog output from the charge amplifier is converted to digital through the Analog-to-Digital Converter (ADC) of the same National Instruments™ USB-6356 I/O board ⑦ before it is stored in the computer ①. The entire process is orchestrated via a self-developed code written in National Instruments™ LabVIEW.

One of the novel aspects of the research presented, is the evaluation of the pipe wall vibrations due to an acoustic source within the fluid domain, achieved through accelerometers mounted on the outer surface of the pipe wall. Two accelerometers of different types are used concurrently: a Brüel & Kjær Type 4326 A Triaxial Charge Accelerometer and a PCB (Depew, NY, USA) Model 352A60 Uniaxial ICP®. The frequency response of the former is flat up to 30 kHz and exponentially increasing at higher values [85]. The usable frequency range of the triaxial accelerometer only partially covers the bandwidth of the probing pulse but offers the convenience of measuring on all three axes simultaneously in addition to a user-configurable sensitivity. The uniaxial accelerometer, on the other hand, has a flat frequency response across

the entirety of the frequency range excited, yet has a fixed sensitivity and can only measure along a single axis [86]. Ultimately, however, the attainable frequency response range of the accelerometers depends on the fixing conditions to the pipe wall. It is important, especially at high frequencies close to the resonant frequency of the transducer, that the connection to the waveguide is as rigid as possible. This is particularly challenging given the curved surface of the pipe wall and the requirement that the longitudinal position of the receiving sensors along the pipe should be adjustable. The proposed approach is to design and manufacture stainless steel adaptors – one for each accelerometer and pipe tested (four in total) – that conform to the curved surface of the pipe wall on one side and are flat on the other. The accelerometers are mounted on the adaptors via threaded connections, and a thin layer of petro wax is applied at the interface with the pipe wall. Before mounting, the surface of the pipe wall is lightly sanded and thoroughly cleaned to ensure optimal traction conditions. Finally, the transducers are secured to the pipe with 9 mm wide plastic cable ties passing through specially designed slots on the stainless-steel adaptors. The triaxial accelerometer measures along the longitudinal axis of the pipe, as well as the normal and tangential directions to the pipe wall. The uniaxial accelerometer is measuring the radial response of the pipe wall. The relative azimuthal positions of the accelerometers remain unchanged throughout the experiments and are at 90 degrees from each other. Regardless, the system is only subjected to axisymmetric vibrations.

Signals from five different channels are captured in total: one corresponding to pressure measurements in the fluid, while the other four to vibration measurements from the pipe wall. The receiving transducers (i.e., both the hydrophone and accelerometers) convert the mechanical movement due to pressure fluctuations and pipe wall accelerations into electric charge by exploiting the piezoelectric phenomenon. However, the electric charge produced is in the order of 10^{-1} of pC per Pa or g of excitation, thus rather small and may attenuate due to the inherent capacitance of the cables connecting the transducer to the data acquisition device. It is, therefore, necessary to condition and amplify the signals, as well as to nullify the degrading capacitance effect of the cable connecting the hydrophone and triaxial accelerometer to the rest of the circuit; thus, a Brüel & Kjær Type 2692-A Nexus Charge Amplifier is introduced to the signal chain. Besides amplifying the signals, the specific charge amplifier also allows for signal frequency filtering and user-selectable transducer sensitivity. The uniaxial accelerometer, on the other hand, has a pre-amplifier embedded and the signal is unaffected by the length of the connecting cable. The sensor, however, requires a constant current excitation in order to operate, provided by a PCB Model 480E09 battery powered signal

conditioner. The sensitivity of all accelerometer channels is set at the same value, and equal to 100 mV g^{-1} . For the receiving hydrophone, the sensitivity is set to $100 \text{ } \mu\text{V Pa}^{-1}$. In the final stage, the five analog signals from the pertinent channels are recorded in synchronicity using the same National InstrumentsTM USB-6356 DAQ board that created the analog probing signal at a sampling rate of 1 MS s^{-1} and 16-bit resolution. The total duration of the acquired signals is equal to 11 s (i.e., 11×10^6 samples for a sampling rate equal to 1 MHz), to ensure that the response of the system to the 10 s long LFM pulse is captured in full. Note that the transmission and acquisition tasks are *exactly* synchronized, given that they are operated from the same device, and both use its internal precision clock. Figure 4.2.5 summarizes the signal path from the initial definition of the probing signal to acquiring the system response at a single location. Once a measurement is complete, the array of receiving transducers is moved by 0.02 m, increasing the distance from the projector, and the measurement is repeated. A distance equal to 2 m along the pipe is covered by 2 cm increments, that amounts to 101 measurements in total. It is noted that the 0.02 m translation step allows for a maximum wavenumber value equal to $100\pi \text{ rad/m}$ [87].

A linear frequency modulated (LFM) signal (where its frequency increases linearly with time) is employed to sweep over the frequency range of interest (in this case from 10 to 70 (or 50) kHz). A cross-correlation of the LFM signal $s(t)$ that is broadcast from the source and the system response $r(t)$ yields the system impulse response. Widely used in radar applications [88], the method essentially transforms the input signal to a matched filter. The impulse response is:

$$IR(t) = \int_{T_{tot}} s^*(t) r(t + \tau) d\tau \quad (4.2.4)$$

where T_{tot} is the signal duration, and the asterisk ($\cdot^*$) denotes the complex conjugate. Note that the measured time domain signals are real, so the complex conjugate is not needed in Eq. (4.2.4), but Eq. (4.2.4) describes the cross-correlation operation in its general form that also accommodates for the case the signal may be complex. $IR(t)$ is a band limited impulse response or Green's function between the source and the receiver. As explained, to obtain the dispersion curves one must sample the $IR(t)$ at equidistant locations along the waveguide (i.e., water filled pipe) and apply the 2D Fast Fourier Transform (FFT) algorithm. These calculations are performed on a 1100000-by-101 element matrix, where the columns correspond to the measurement locations and the rows to the discrete sampling time increments. First, the $IR(t)$

at each location are calculated, then FFT is applied on the waveform data (along the columns), followed by the spatial FFT (along the rows). The final matrix is expressed in (k_x, f) instead of the initial (x, t) domain. A contour plot of the matrix produces the desired dispersion curves given in dB scale relative to the highest spectral amplitude observed. In total, five such plots are developed, each corresponding to the individual receiving channels. It should be noted that the obtained responses are appropriately compensated for the frequency varying TVR of the acoustic projector. The analytical dispersion curves and their comparison to the pertinent experimental curves for both the DN80 HDPE and copper pipes are presented in the following subsections.

4.3 DN80 HDPE pipe

4.3.1 Dispersion curves

Analytical dispersion curves

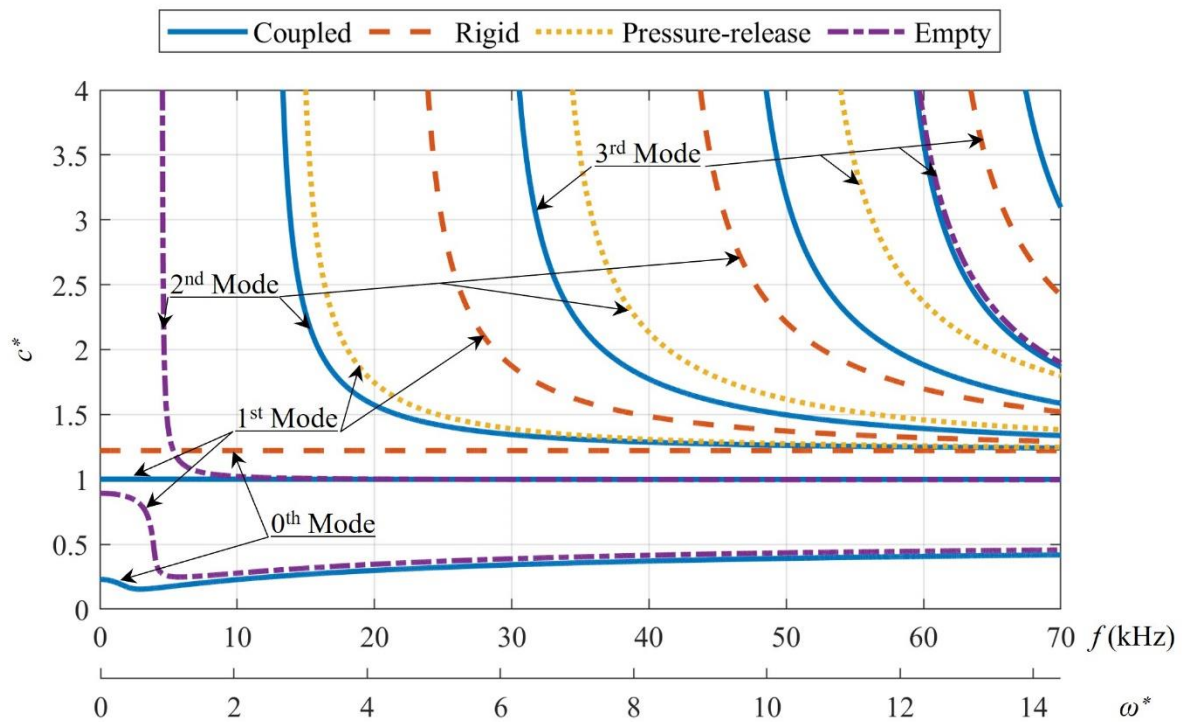


Figure 4.3.1: Analytically obtained dispersion curves (phase velocity c^* vs frequency ω^*) for the DN80 HDPE pipe when (i) filled with water (blue solid line), and (ii) empty (magenta

dashed-dotted line). Also plotted are the extreme case curves of a rigid (orange dashed line) and pressure release pipe (yellow dotted line) for a water column of the same diameter.

At low frequencies (i.e., much lower than the pipe ring frequency, $\omega^* \ll 1$), there are two modes that can possibly propagate along the coupled waveguide in total, namely the 0th and 1st Modes. There, the phase velocity of the 0th Mode may be obtained from Korteweg's formula (Eq. (3.2.57)) that describes a plane wave propagating in the fluid that is only reduced by the pipe wall elasticity, present in the denominator. In particular, Eq. (3.2.57) for the parameters of the DN80 HDPE pipe yields $c_{ph}/c_p = 0.23$, precisely matching the value from the impedance-based model presented in Figure 4.3.1. The 1st Mode, on the other hand, propagates with a phase velocity $\sim c_p$, slightly increased compared to that of the corresponding empty pipe mode due to the added stiffness of the coupled system. Indeed, the 1st Mode at low frequencies describes a pipe wall that performs purely axial vibrations, with the phase velocity being that of longitudinal waves in an infinite HDPE medium. It should also be noted that since the wave speed in the HDPE pipe material c_p is smaller compared to that in the water c_f , the phase velocity for both modes is less than the compressional wave speed in fluid, c_f .

At slightly increased frequencies that approach that of the pipe ring (i.e., $\omega^* \sim 1$), the interaction between the fluid and pipe wall becomes more apparent. The pipe wall executes a radial “breathing” motion, and the 1st Mode of the empty pipe now describes a flexural wave that propagates along the pipe wall, while an additional empty pipe radial mode appears (i.e., 2nd Mode). The dispersion curve of the coupled system 0th Mode closely follows that of the empty pipe mode throughout the remaining frequency interval shown in Figure 4.3.1. Starting from f_{ring} , the propagating wave in the fluid is no longer planar but is confined close to the fluid/pipe wall interface to essentially create a “surface wave” [77]. In fact, ignoring the flexural term in the characteristic equation forces the 0th Mode to stop propagating. The particulars of this observation are discussed in subsections 4.6 and 4.7, respectively. Also, note that at very high frequencies, the phase velocity of the 1st Mode of the empty pipe $c^* = 1/\eta^{0.5}$ (Eq. (3.2.52)). Since the curves of both the coupled system 0th Mode and empty pipe 1st Mode asymptotically converge to $c^* \sim 1/\eta^{0.5} = 0.49$ at high frequencies, the water filled pipe essentially behaves as empty (i.e., $p = 0$).

Table 4.3.1: Cut-off frequencies (in ω^*) for the pressure-release, rigid, and elastic boundary DN80 HDPE-sized water column

Mode	Pressure Release	Rigid	Coupled
0 th Mode	-	At all frequencies	At all frequencies
1 st Mode	2.97	4.73	At all frequencies
2 nd Mode	6.82	8.67	2.64
3 rd Mode	10.70	12.58	6.03
4 th Mode	-	-	9.65
5 th Mode	-	-	13.33

Besides the two coupled modes (0th and 1st) that propagate at all frequencies, there are an infinite number of propagating modes with a phase velocity c_{ph} that is greater than c_f (i.e., $c^* = \gamma$). Each of these modes presents a cut-off frequency, and only propagates at frequencies greater than that. At the cut-off frequencies, the phase velocity tends to infinity, and the motion of the pipe resembles that of a radially oscillating cylinder. This becomes clear considering that the axial wavenumber $k_x = \omega/c_{ph} \rightarrow 0$, thus $k_r = \omega/c_f$. The cut-off frequencies of all radial modes can be evaluated from Eq. (3.2.49). It is possible, however, to identify the frequency intervals where the cut-off frequencies of the coupled system lay relative to these of the theoretical cases of a water column with pressure-release and rigid walls. The cut-off frequencies for a water column with pressure release boundaries are found by equating the numerator of the left-hand side of Eq. (3.2.49) to zero i.e., $J_0(\omega^*/\gamma) = 0$. At these frequencies ω , the argument of the Bessel function of the first kind and 0th order is equal to the (known) function roots. Similarly, the cut-off frequencies of a water column with rigid boundaries are obtained after equating the denominator of the left-hand side of Eq. (3.2.49) to zero i.e., $J_1(\omega^*/\gamma) = 0$. The cut-off frequencies for a water column of cross-sectional dimensions equal to these of the HDPE pipe with pressure-release and rigid boundaries in the frequency interval of interest are presented in Table 4.3.1. Regarding the cut-off frequencies of the coupled system modes, these must be found in a region where both sides of Eq. (3.2.49) are of the same sign. For instance, the right-hand side is negative for $\omega < 1$ (i.e., for frequencies below f_{ring}), while the left-hand side (J_0/J_1) is positive at these frequencies. It follows, therefore, that the cut-off frequency of the coupled system 2nd Mode lays between $1 < \omega^* < 2.97$ where *both* sides of Eq. (3.2.49) are positive. Accordingly, the cut-off frequency of the coupled system 3rd Mode must be in the next frequency interval where (J_0/J_1) is also positive i.e., $4.73 < \omega^* < 6.82$, the 4th Mode in the interval $6.82 < \omega^* < 8.67$, etc. Subsequently, all coupled system modes that have a cut-off

frequency and propagate at a phase velocity greater than c_f , are positioned between a rigid pipe mode to the left and pressure release pipe mode to the right. Exceptions are the coupled system 2nd Mode that is positioned between an empty pipe and pressure release mode, and the coupled system 5th Mode that corresponds to the 3rd Mode of the empty pipe. Equation (3.2.49) shows that at higher frequencies, the cut-off frequencies of the coupled system modes approach the roots of $J_1(\omega^*/\gamma)$ i.e., the rigid pipe cut-offs. Besides, the approach is more rapid for a smaller value of α (i.e., for a system where the fluid presents smaller inertia relative to that of the pipe).

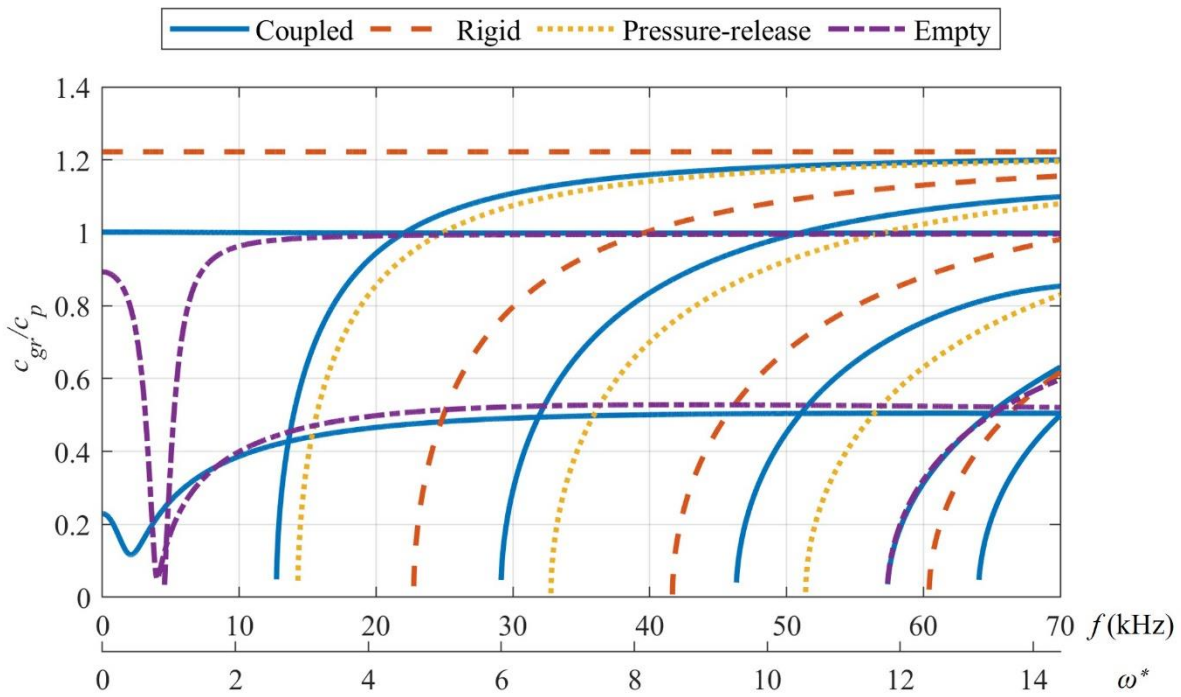


Figure 4.3.2: Analytically obtained group velocity curves (dimensionless group velocity c_{gr}/c_p vs frequency ω^*) for the DN80 HDPE pipe when (i) filled with water (in blue), and (ii) empty (in magenta). Also plotted are the extreme case curves of a rigid (in orange) and pressure release pipe (in yellow) of the same diameter.

Beyond the cut-off frequencies, the coupled system radial mode curves asymptotically tend to the 0th Mode of the rigid pipe, i.e., $k_x \rightarrow k$. Finally, Figure 4.3.2 presents the dispersion relationship as in Figure 4.3.1, but in terms of normalized group velocity c_{gr}/c_p for each mode as a function of frequency ω^* . It is reminded that the group velocity is more closely related to the wavespeed in the waveguide, as it represents the speed at which the energy propagates. Note that at high frequencies all coupled system modes asymptotically converge to but never

exceed γ , i.e., the dimensionless wave speed in a rigid pipe, c_f . That can be understood due to the fact that $c_p < c_f$ or $\gamma > 1$.

Experimental vs analytical dispersion curves

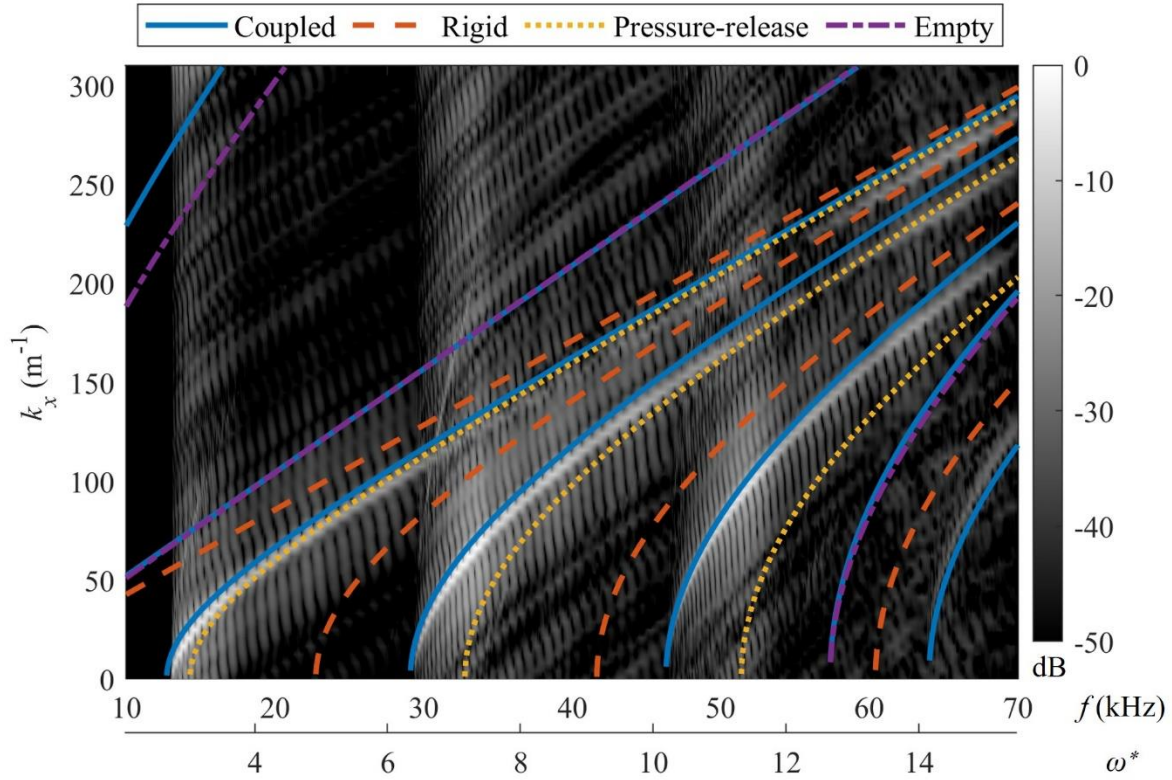


Figure 4.3.3: Comparison between the analytically (in blue) and experimentally (in 2D spectrum) obtained dispersion curves (axial wavenumber k_x vs frequency ω^*) for the water filled DN80 HDPE pipe. Also included are the analytical curves for an empty pipe (in magenta), rigid (in orange), and pressure release (in yellow) cases.

Figure 4.3.3 presents a comparison between the analytical and experimental dispersion curves in terms of axial wavenumber k_x vs frequency ω^* . The experimental dispersion curves are obtained following the process discussed in subsection 4.2.1, using the data from the hydrophone. The valid (k_x, ω^*) pairs in the experimental 2D spectrum appear as bright, white regions. Note that the 2D spectrum has been normalized with respect to the largest recorded value in the signal, and it is presented in dB scale. It is apparent that the analytical curves for the coupled system (in blue) that correspond to the high order radial modes and, therefore, exhibit cut-off frequencies match well with the experimental 2D spectrum. Specifically, the analytical cut-off frequencies of the coupled 2nd, 3rd, 4th, and 6th Modes are found at 12.7, 29.1, 46.3, and 64 kHz, respectively, and coincide with the experimental cut-off frequencies.

Exceptions are the 0th, 1st and 5th Modes that are not excited in the lab setup, since the 2D spectrum remains dark in these regions. Note that these modes, however, correspond to empty pipe modes (shown in magenta). Since the acoustic source is placed in the water, it is reasonable to assume that little to no wave energy propagates through these aforementioned modes. Besides, as noted in [60], the size of the source relative to the pipe cross section is the deciding factor for the excitation of the plane-like 0th Mode. In particular, the smaller the source compared to the pipe cross section, the lesser the energy carried from the plane-like wave. For our particular case, the transducer accounts for about 0.1 the diameter of the DN80 HDPE pipe; thus, it effectively acts as a point source that fails to excite the plane-like wave mode to detectable levels.

It is noteworthy that there appears a small deviation between the analytical and experimental curves for the excited modes at higher axial wavenumber k_x values. Take for example the analytical curve for the 3rd mode of the coupled system: at 70 kHz, the axial wavenumber predicted by the analytical model is $k_x = 274$ 1/m, while the experimental value shows $k_x = 260$ 1/m. Regardless, the experimental curve is also crossed by the curve for the 2nd mode of the pressure release pipe, that presents an extreme limit for the coupled system 3rd Mode curve. It would, therefore, be impossible for the analytical model to match the experimental curves unless the assumed physical characteristics of the system are modified. An interesting future topic for research is to investigate the sensitivity of the model to physical property values. In conclusion, the experimental dispersion curves offer a first validation for the analytical model. In the upcoming sections, the approximate analytical model is put against a state-of the art general model and is also checked for its ability to describe pipe wall displacements and reproduce the response of a real system to a narrowband pulse.

4.3.2 Shell theory-based vs general model

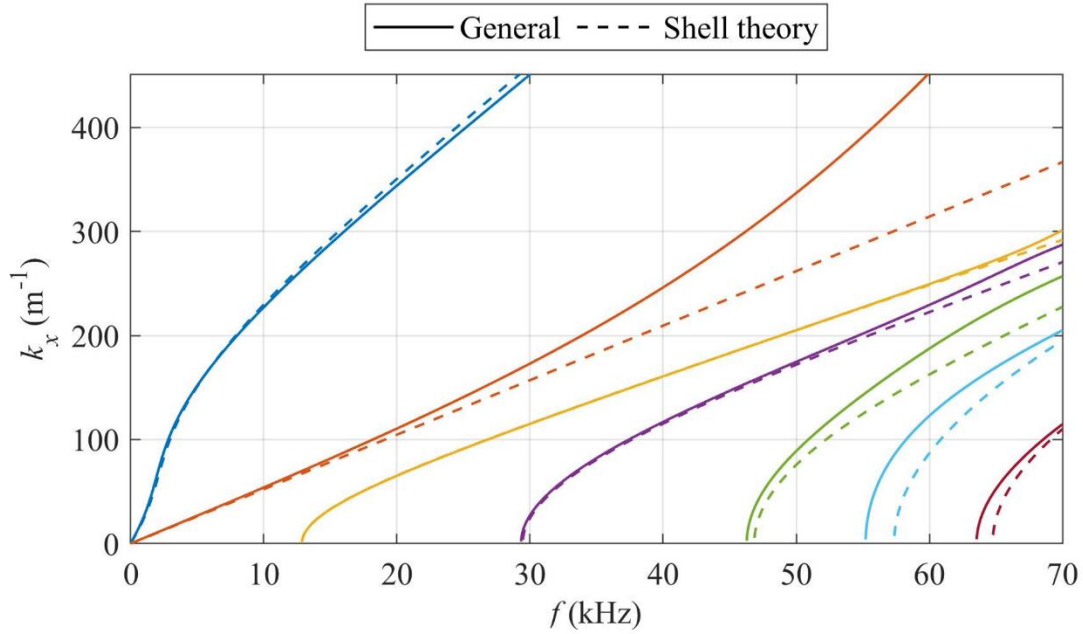


Figure 4.3.1: The adopted, approximate, shell theory-based model (dashed curves) is compared to a general, state-of-the-art model (solid curves) for the case of the water filled DN80 HDPE pipe.

In Figure 4.3.1, the dispersion relationship given by a general, state-of-the-art model that is based on the full set of equations of elastodynamics [59] is solved for the parameters of the DN80 HDPE pipe (Table 4.2.1), and compared to that of the approximate shell theory-based model adopted herein. The general model is formulated as a 6-by-6 matrix (that is simplified to a 4-by-4 matrix for an axisymmetric pressure field assumption) after the definition of the displacement and stress continuity at the fluid/pipe wall interface. For the system of four equations to have non-trivial solutions, the determinant of the matrix must be equal to zero. The condition forms the characteristic equation that, when solved, yields the valid (k_x, ω^*) pairs. However, finding these roots is a mathematically involving process, prone to instabilities; hence, the use of the more robust shell-theory based model is preferred. However, it is useful to identify the in-between error.

It appears in Figure 4.3.1 that the two models produce almost identical curves for the case of the 0th (in blue), 2nd (in yellow), and 3rd Modes, and are close for the 4th and 6th Modes. Some discrepancies are found at high frequencies (> 45 kHz), and the largest are observed in the 1st and 5th Modes that correspond to structural modes (see Figure 4.3.2). Interestingly, the general

model at 70 kHz and for the 5th Mode, for example, predicts a larger wavenumber compared to the shell theory model, thus widening the gap to the experimental curves (that demonstrate a smaller k_x than the shell theory model for the same mode and frequency). Consequently, it is possible that (i) the mechanical and geometrical property values of Table 4.2.1 for the DN80 HDPE pipe are inaccurate or, (ii) there are errors in the solutions of the general model. Nevertheless, the two models behave very close for the modes excited experimentally, therefore strengthening the case of the approximate shell-theory based model.

4.3.3 Pipe wall displacements

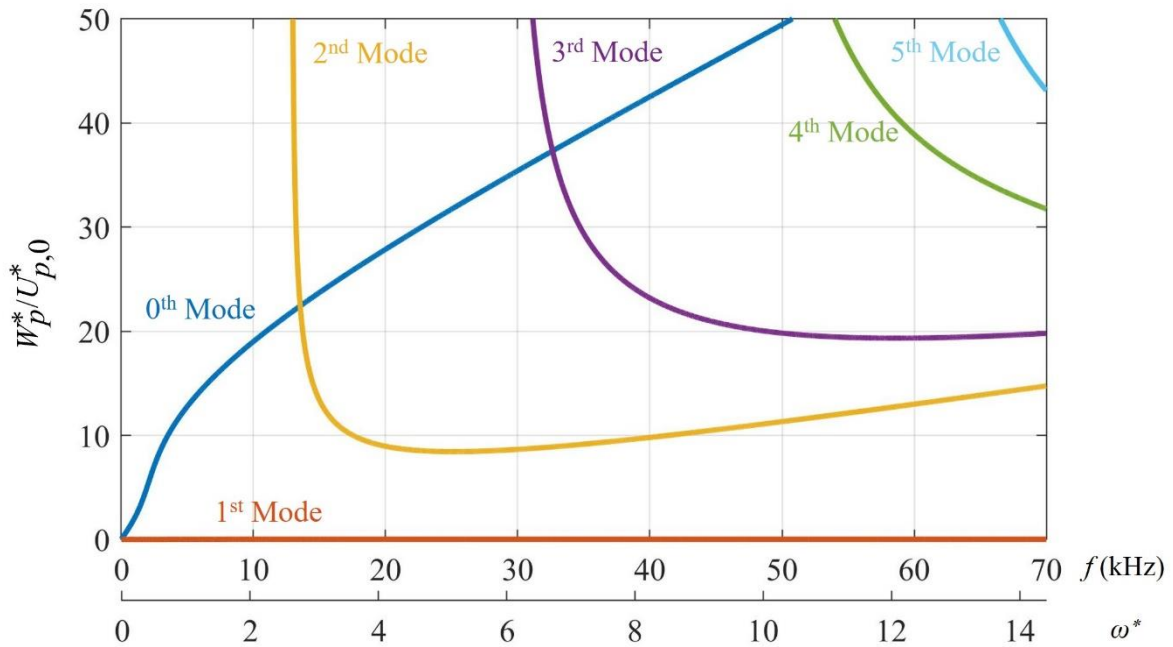


Figure 4.3.5: The analytical radial to axial displacements ratio ($W_p^*/U_{p,0}^*$) (Eq. (3.2.70)), are plotted for each Mode as a function of frequency for the water filled DN80 HDPE pipe.

In this thesis, the solutions for the pipe wall displacements are produced on the basis of the approximate shell-based theory for the first time. Figure 4.3.2 plots the analytical radial to axial displacements ratio ($W_p^*/U_{p,0}^*$) for the case of the DN80 HDPE pipe. The DN80 HDPE pipe presents an interesting case, since the compressional wave speed in the pipe wall c_p , is smaller than the compressional wave speed in the water c_f . Snell's law defines that, for two media that share an interface, the incident angle θ_1 to refraction angle θ_2 is equal to the ratio of the wave speeds in the two media:

$$\sin \theta_2 = \frac{c_p}{c_f} \sin \theta_1 \quad (4.3.1)$$

where $\sin \theta_1 = k_x/k = c_f/c_{ph}$. Therefore,

$$\sin \theta_2 = \frac{c_p}{c_{ph}} = \frac{1}{c^*} \quad (4.3.2)$$

As seen in Figure 4.3.1, however, for the case of the radial modes ($\geq 2^{\text{nd}}$ Mode), $c^* \geq \gamma > 1$. In other words, the refracted angles are always less than the incident angle. At the cut-off frequencies of radial modes, the phase velocity c^* tends to infinity, hence $\theta_2 = 0$ and the pipe executes a purely radial motion. The fact is replicated in Figure 4.3.2, where the ratio $W_p^*/U_{p,0}^*$ also tends to infinity. At high frequencies, the phase velocity c^* of radial modes tends to γ , therefore $\theta_2 = 55^\circ$ and the propagating wave in the pipe wall presents both an axial and radial component. Regarding the structural 1st mode that propagates at $c^* = 1$ at all frequencies, $\theta_2 = 90^\circ$; hence, it always corresponds to a compressional wave that propagates axially in the pipe wall. Finally, the 0th mode always presents a phase velocity $c^* < 1$, thus the propagating wave is completely reflected at the interface.

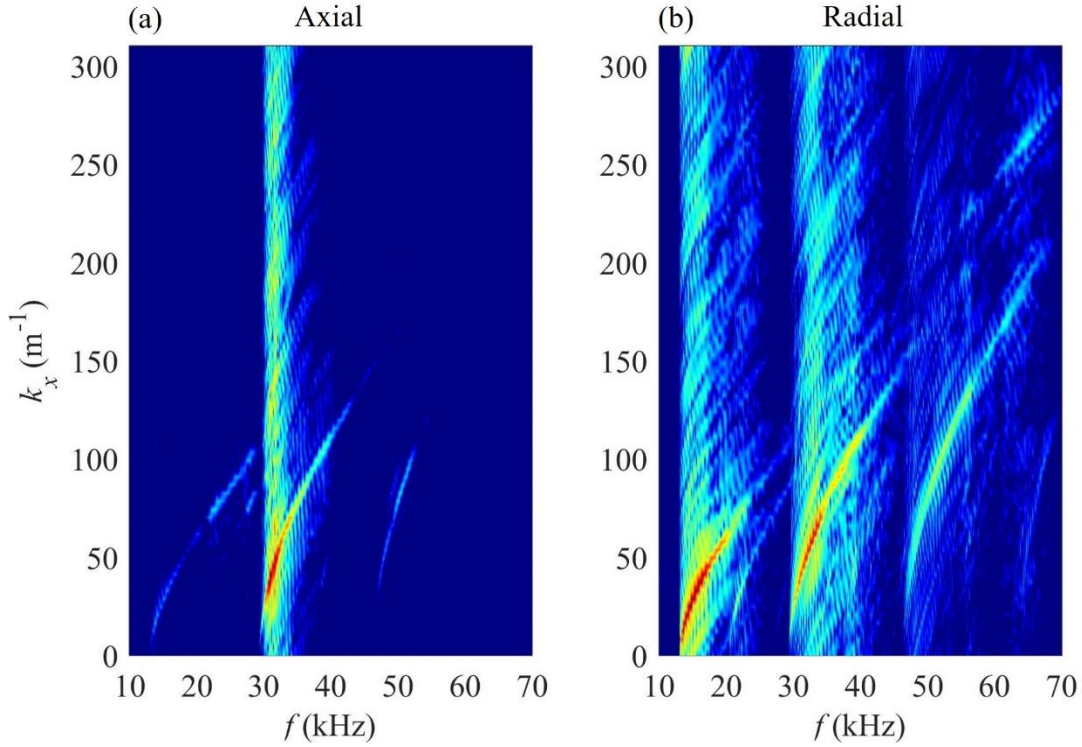


Figure 4.3.2: The experimental dispersion curves are plotted using the data collected from the accelerometers in the (a) axial and (b) radial directions for the DN80 HDPE pipe.

The analytical results of Figure 4.3.2 are verified using the data collected from the deployed accelerometers on the DN80 HDPE pipe of the lab setup. In particular, Figure 4.3.3a presents the dispersion curves from the accelerometer in the axial direction, while Figure 4.3.3b in the radial direction. Evidently, the power amplitude of the radial modes is higher at their cut-off frequencies, as indicated by the deep red contour, where the pipe oscillates radially as indicated by Figure 4.3.2. On the other hand, the most prominent feature in Figure 4.3.3a is the vertical “column” around 30 kHz. This corresponds to the resonant frequency of the accelerometer in the axial direction and should be ignored. In fact, at frequencies higher than 30 kHz, the frequency response of the utilized accelerometer is no longer flat. Otherwise, the contour traces in Figure 4.3.3a are rather weak in comparison to these in Figure 4.3.3b. Axial displacements appear the most significant for the 2nd Mode in the frequency interval from 20 to 30 kHz, where the pertinent $W_p^*/U_{p,0}^*$ ratio presents a minimum, thus the two plots are in agreement.

4.4 DN80 copper pipe

4.4.1 Dispersion curves

Analytical dispersion curves

In addition to the DN80 HDPE pipe, the approximate analytical model is also checked for the DN80 copper pipe of the lab setup. Note that the thin wall thickness of the DN80 copper pipe fits better the description of the pipes that the analytical model was developed for. In Figure 4.4.1, the analytically obtained (ω^*, c^*) pairs for each propagating mode are presented for the cases of: (i) the coupled water/copper pipe (in blue), (ii) the empty copper pipe (in magenta), as well as the extreme cases of a water column of the same inner cross-sectional dimensions of the DN80 copper pipe, but with (iii) rigid (in orange) and (iv) pressure release boundaries.

From Figure 4.4.1 it can be seen that two modes propagate at all frequencies for the water filled DN80 copper pipe, the 0th and 1st Modes. At low frequencies ($\omega^* \ll 1$), the 0th Mode corresponds to a water hammer wave whose phase velocity can be retrieved from Korteweg’s equation (Eq. (3.2.57)). To verify, Eq. (3.2.57) yields $c^* = 0.23$ for the parameters of the system (

As the frequency ω^* approaches 1 ($= f_{ring}$), the coupling between the fluid and pipe increases. That is because the pipe wall starts oscillating radially, in a “breathing” motion. Hence, for

larger ω^* the coupled 0th Mode resembles that of flexural vibrations in an empty pipe. Subsequently, starting from f_{ring} , the 0th Mode describes a wave that propagates at the water/pipe interface or a “surface wave”. At very high frequencies, the phase velocity of the 1st Mode of the empty pipe $c^* = 1/\eta^{0.5}$ (Eq. (3.2.52)). Given that the curve of the coupled system 0th Mode follows that of the empty pipe 1st Mode, it may be concluded that at high frequencies the water filled pipe can be considered unloaded (i.e., $p = 0$).

In addition to the coupled system 0th and 1st Modes, there is an infinite number of high order radial modes that only propagate following a cut-off frequency. Their phase velocity c^* is always greater than γ , and their cut-off frequencies can be found solving Eq. (3.2.49). At the cut-off frequencies of these radial modes, the phase velocity tends to infinity, thus $k_x \rightarrow 0$ and the coupled system is described as a radially oscillating cylinder. Table 4.4.1 presents the cut-off frequencies for the modes of the coupled system, as well as the extreme cases of a rigid and pressure release pipe. The cut-off frequencies for the pressure release pipe are retrieved by equating the numerator on the left-hand side of Eq. (3.2.49) to zero and finding the ω^* values that satisfy the equation, i.e., $J_0(\omega^*/\gamma) = 0$. On the other hand, the cut-off frequencies for the rigid pipe are found by equating the denominator on the left-hand side of Eq. (3.2.49) to zero, i.e., $J_1(\omega^*/\gamma) = 0$. For the coupled system radial modes, their cut-off frequencies are found in the frequency interval where both sides of Eq. (3.2.49) are of the same sign. The right-hand side of Eq. (3.2.49) is negative for $\omega^* < 1$, while the left-hand side (i.e., J_0/J_1) is positive for $0 < \omega^* < 0.89$ and negative for $0.89 < \omega^* < 1.43$. Consequently, the cut-off frequency of the coupled system 2nd Mode is found within the interval $0.89 < \omega^* < 1.43$. Following a similar approach, the cut-off frequency of the coupled 3rd Mode is in the interval $1.43 < \omega^* < 2.06$ ($\omega^* = 1.67$) and that of the 4th Mode in the interval $2.62 < \omega^* < 3.23$ ($\omega^* = 2.74$). Evidently, the cut-off frequencies of the coupled system radial modes are bounded between that of a rigid pipe mode on the left and a pressure release pipe mode on the right, with the exception of the coupled 2nd Mode that is found between a pressure release and empty pipe mode. Note that at higher frequencies, the cut-off frequencies of the coupled system modes tend to these of rigid pipe.

Table 4.4.1), a value that matches that in Figure 4.4.1 at $\omega^* = 0$. Contrary, the 1st Mode is a compressive wave that propagates at the pipe wall, as revealed by its phase velocity that is $\sim c_p$. For the case of the DN80 copper pipe $c_p > c_f$, or $\gamma < 1$, hence the phase velocity c^* of the 0th mode is below γ ($= 0.38$), while that of the 1st Mode is above γ .

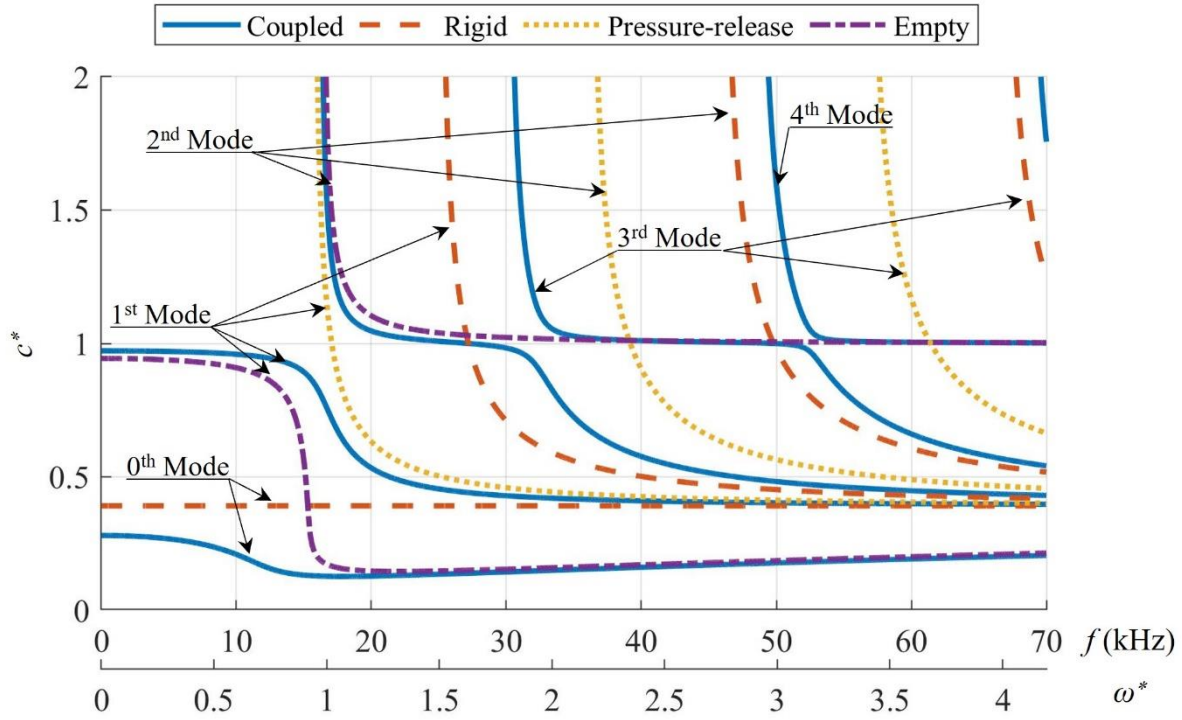


Figure 4.4.1: Analytically obtained dispersion curves (phase velocity c^* vs frequency ω^*) for the DN80 copper pipe when (i) filled with water (blue solid line), and (ii) empty (magenta dashed-dotted line). Also plotted are the extreme case curves of a rigid (orange dashed line) and pressure release pipe (yellow dotted line) for a water column of the same diameter.

As the frequency ω^* approaches 1 ($= f_{ring}$), the coupling between the fluid and pipe increases. That is because the pipe wall starts oscillating radially, in a “breathing” motion. Hence, for larger ω^* the coupled 0th Mode resembles that of flexural vibrations in an empty pipe. Subsequently, starting from f_{ring} , the 0th Mode describes a wave that propagates at the water/pipe interface or a “surface wave”. At very high frequencies, the phase velocity of the 1st Mode of the empty pipe $c^* = 1/\eta^{0.5}$ (Eq. (3.2.52)). Given that the curve of the coupled system 0th Mode follows that of the empty pipe 1st Mode, it may be concluded that at high frequencies the water filled pipe can be considered unloaded (i.e., $p = 0$).

In addition to the coupled system 0th and 1st Modes, there is an infinite number of high order radial modes that only propagate following a cut-off frequency. Their phase velocity c^* is always greater than γ , and their cut-off frequencies can be found solving Eq. (3.2.49). At the cut-off frequencies of these radial modes, the phase velocity tends to infinity, thus $k_x \rightarrow 0$ and the coupled system is described as a radially oscillating cylinder. Table 4.4.1 presents the cut-off frequencies for the modes of the coupled system, as well as the extreme cases of a rigid and pressure release pipe. The cut-off frequencies for the pressure release pipe are retrieved by

equating the numerator on the left-hand side of Eq. (3.2.49) to zero and finding the ω^* values that satisfy the equation, i.e., $J_0(\omega^*/\gamma) = 0$. On the other hand, the cut-off frequencies for the rigid pipe are found by equating the denominator on the left-hand side of Eq. (3.2.49) to zero, i.e., $J_1(\omega^*/\gamma) = 0$. For the coupled system radial modes, their cut-off frequencies are found in the frequency interval where both sides of Eq. (3.2.49) are of the same sign. The right-hand side of Eq. (3.2.49) is negative for $\omega^* < 1$, while the left-hand side (i.e., J_0/J_1) is positive for $0 < \omega^* < 0.89$ and negative for $0.89 < \omega^* < 1.43$. Consequently, the cut-off frequency of the coupled system 2nd Mode is found within the interval $0.89 < \omega^* < 1.43$. Following a similar approach, the cut-off frequency of the coupled 3rd Mode is in the interval $1.43 < \omega^* < 2.06$ ($\omega^* = 1.67$) and that of the 4th Mode in the interval $2.62 < \omega^* < 3.23$ ($\omega^* = 2.74$). Evidently, the cut-off frequencies of the coupled system radial modes are bounded between that of a rigid pipe mode on the left and a pressure release pipe mode on the right, with the exception of the coupled 2nd Mode that is found between a pressure release and empty pipe mode. Note that at higher frequencies, the cut-off frequencies of the coupled system modes tend to these of rigid pipe.

Table 4.4.1: Cut-off frequencies (in ω^*) for the pressure-release, rigid, and elastic boundary DN80 copper-sized water column

Mode	Pressure Release	Rigid	Coupled
0 th Mode	-	At all frequencies	At all frequencies
1 st Mode	0.89	1.43	At all frequencies
2 nd Mode	2.06	2.62	0.95
3 rd Mode	3.23	3.8	1.67
4 th Mode	-	-	2.74

A feature of the radial mode curves of the water filled DN80 copper pipe that is absent from the pertinent DN80 HDPE pipes are the crossings with the empty pipe 2nd Mode curve as frequencies increase from the cut-off. For instance, the coupled 3rd Mode has a cut-off at $\omega^* = 1.67$ and, as the frequency increases, the phase velocity c^* drops. At $c^* = 1$ and $\omega^* \sim 2$, the coupled 3rd Mode presents an inflection point and follows the empty pipe 2nd Mode until $\omega^* \sim 3.1$, or until it meets the curve of rigid pipe 2nd Mode. Hence, the coupled system 3rd Mode represents an empty pipe in the interval $2 < \omega^* < 3.1$, and then behaves closely to a rigid pipe mode at higher frequencies. Similar observation can be made for the remaining radial modes.

In subsection 4.4.3, consideration of the Snell's law condition (Eq. (4.3.2)) will further elucidate the phenomenon. Figure 4.4.2 presents the dispersion curves in terms of dimensionless group velocity c_{gr}/c_p (that is indicative of the speed that the wave energy propagates in the waveguide) and frequency ω^* . Since $c_{gr} = d\omega/dk_x$, at the cut-off frequencies $c_{gr} = 0$ (i.e., the mode practically does not propagate).

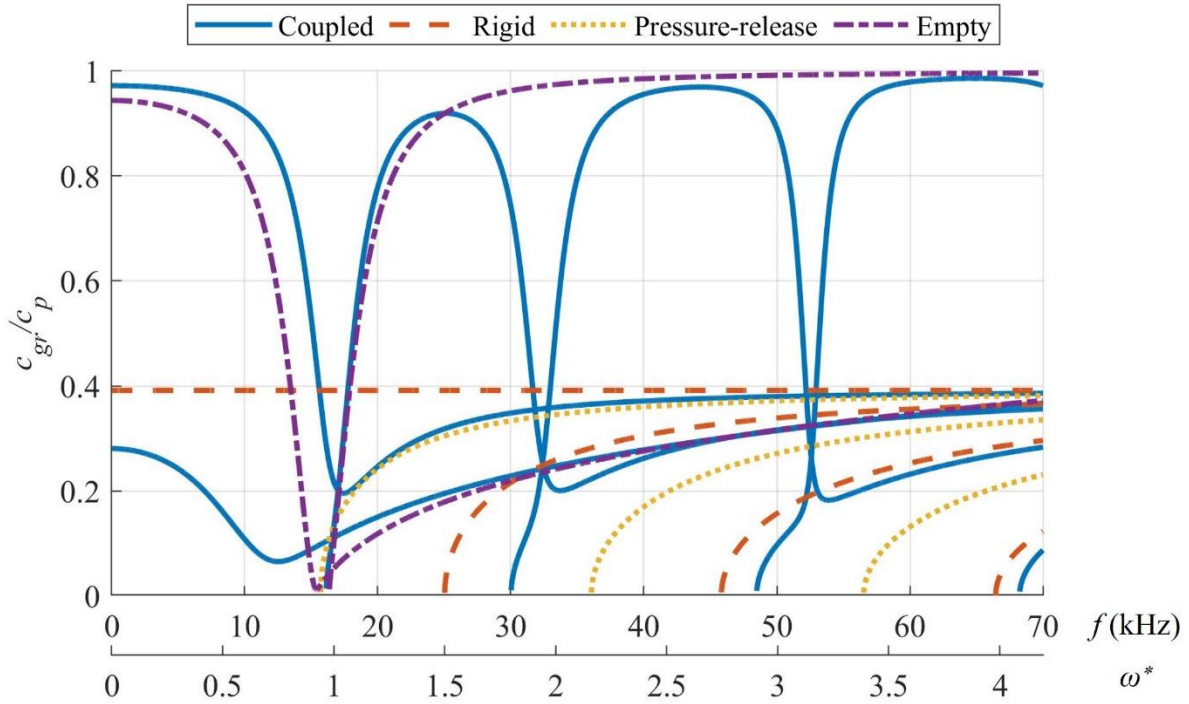


Figure 4.4.2: Analytically obtained group velocity curves (dimensionless group velocity c_{gr}/c_p vs frequency ω^*) for the DN80 copper pipe when (i) filled with water (in blue), and (ii) empty (in magenta). Also plotted are the extreme case curves of a rigid (in orange) and pressure release pipe (in yellow) of the same diameter.

Experimental vs analytical dispersion curves

Figure 4.4.3 presents a comparison between the analytical and experimental dispersion curves in terms of axial wavenumber k_x and frequency ω^* . In particular, the experimental dispersion curves (2D spectrum) are generated using hydrophone data. The 2D spectrum has been normalized with respect to the largest signal amplitude, and the bright white regions correspond to the valid (ω^*, k_x) values. Evidently, the analytical curves predict the experimental radial mode cut-off frequencies (where $k_x = 0$) very well. These are found at 16.1, 30, and 48.2 kHz for the 2nd, 3rd, and 4th Modes of the coupled system, respectively. Absent from the experimental dispersion curves is the 0th Mode, that is not found in the propagating in the water.

Contrary to the DN80 HDPE pipe the 1st Mode of the coupled system is detected. Note, however, that for the water filled DN80 copper pipe, the 1st Mode does not coincide with an empty pipe mode. Interestingly, the experimental dispersion curves for the coupled system radial modes appear to have gaps in the frequency intervals that their curves coincide with the 2nd Mode of the empty pipe (e.g., $1.3 < \omega^* < 1.7$ for the 2nd Mode, $2.1 < \omega^* < 3$ for the 3rd Mode). That is because at these intervals the wave propagates at the pipe wall, thus cannot be detected in the fluid. Besides the cut-off frequencies, the analytical model accurately traces the experimental curves throughout the frequency range investigated (i.e., 10 to 50 kHz). These results offer a first validation for the analytical model in the case of the water filled DN80 copper pipe. Next, the approximate analytical model is compared to a general, state-of-the-art model that forgoes any simplifications.

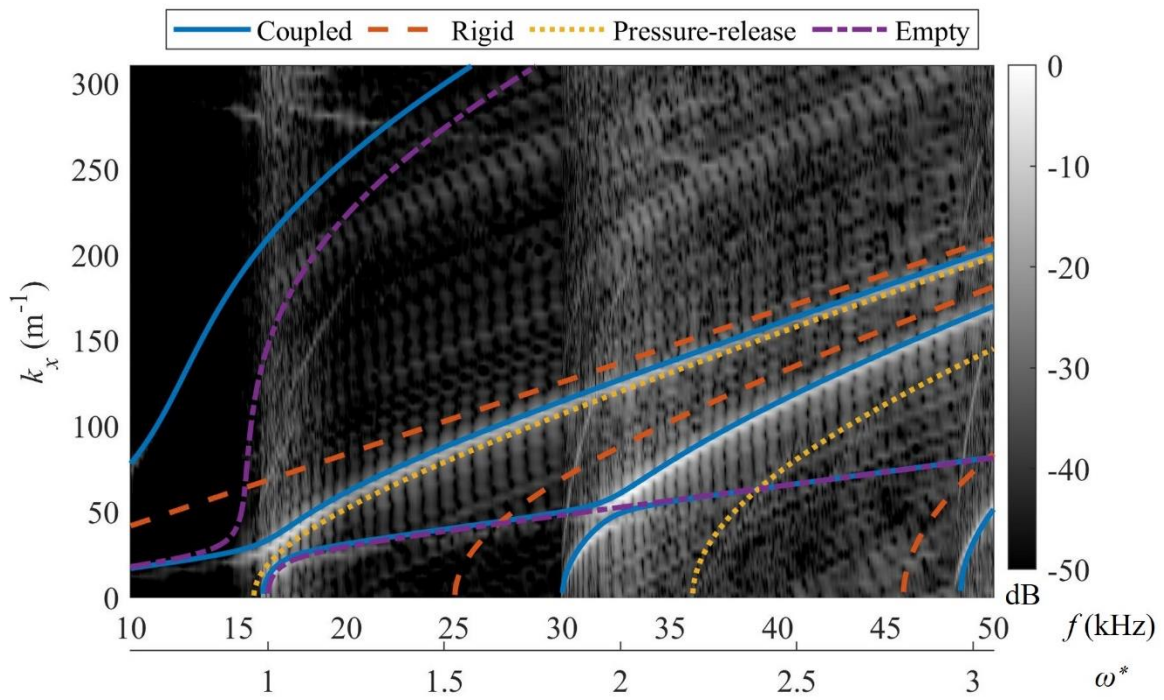


Figure 4.4.3: Comparison between the analytically (in blue) and experimentally (in 2D spectrum) obtained dispersion curves (axial wavenumber k_x vs frequency ω^*) for the water filled DN80 copper pipe. Also included are the analytical curves for an empty pipe (in magenta), rigid (in orange), and pressure release (in yellow) cases.

4.4.2 Shell theory-based vs general model

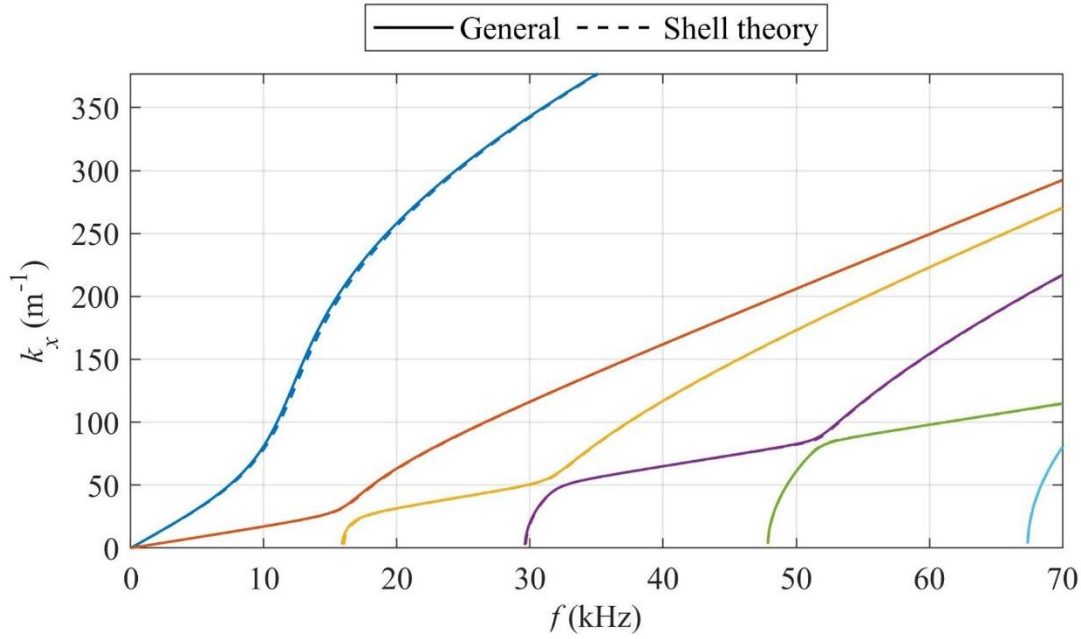


Figure 4.4.4: The adopted, approximate, shell theory-based model (dashed curves) is compared to a general, state-of-the-art model (solid curves) for the case of the water filled DN80 copper pipe.

Figure 4.4.4 compares the dispersion relationship obtained from the approximate shell theory-based model to that from a general model [59] for the case of the water filled DN80 copper pipe (Table 4.2.1). Therein, the approximate model dispersion curves are plotted using dashed lines, while the general model curves are plotted with solid lines. Evidently, the two models produce nearly indistinguishable results for all propagating modes in the DN80 HDPE pipe in the frequency interval from 10 to 70 kHz. The demonstrated results in Figure 4.4.4 show that the two models are better matching for the case of the DN80 copper than the DN80 HDPE pipe. This could be attributed to two reasons: (i) the increased wall thickness of the DN80 HDPE pipe whose deformations are not accurately described by the approximate analytical model, and/or (ii) the uncertainty regarding the mechanical properties of the HDPE material; in comparison, copper is more standardized, and also behaves purely elastically. As a final note for point (i), the wavelength at 70 kHz (i.e., the upper bound of the frequency interval examined) is equal to $\lambda_{\min} = c/f \sim 2$ cm that is almost one order of magnitude larger than the wall thickness of the HDPE pipe. Hence, reason (ii) appears more plausible.

4.4.3 Pipe wall displacements

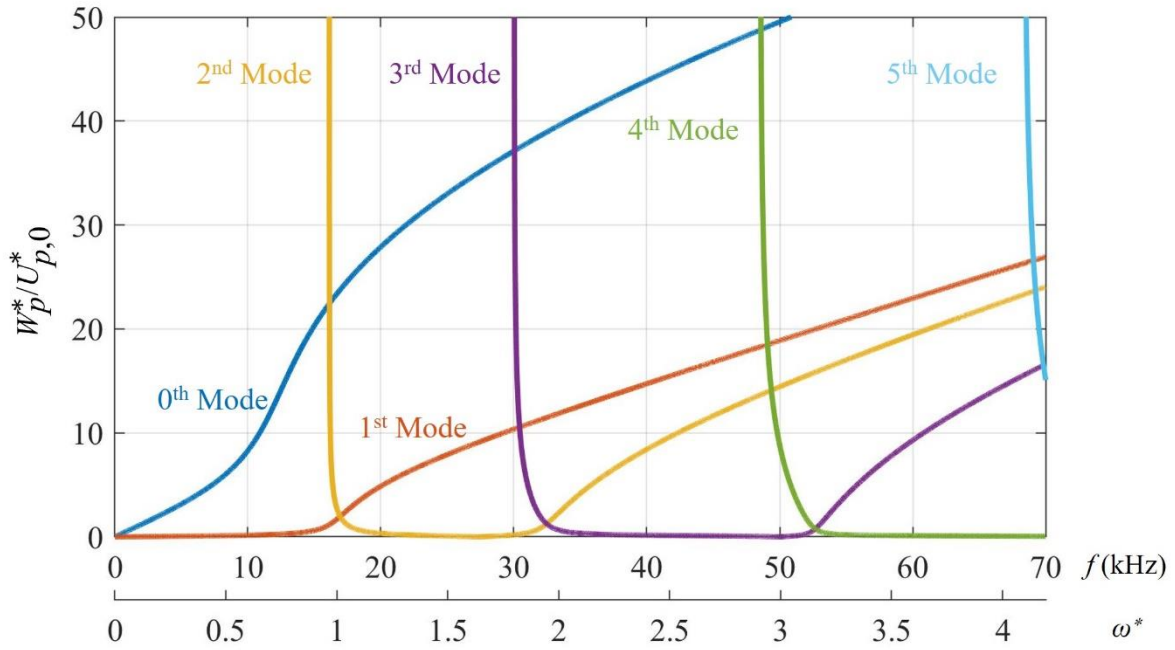


Figure 4.4.5: The analytical radial to axial displacements ratio ($W_p^*/U_{p,0}^*$) (Eq. (3.2.70)), are plotted for each Mode as a function of frequency for the water filled DN80 copper pipe.

Figure 4.4.5 presents the analytical radial to axial displacements ratio ($W_p^*/U_{p,0}^*$) for each propagating mode in the water filled DN80 copper pipe as a function of frequency. Starting with the radial modes, at their cut-off frequencies $W_p^*/U_{p,0}^* \rightarrow \infty$. This is to be expected since there the system behaves as a radially oscillating cylinder. At slightly higher frequencies, however, $W_p^*/U_{p,0}^* \rightarrow 0$, and remains close to zero over a considerable frequency interval. That interval is $1.3 < \omega^* < 1.7$ for the 2nd Mode, $2.1 < \omega^* < 3$ for the 3rd Mode, and $\omega^* > 3.2$ for the 4th Mode. Note that at these frequency intervals, the phase velocity of these modes (Figure 4.4.1) is $c^* = 1$. From Eq. (4.3.2), this yields $\sin\theta_2 = 1$ or $\theta_2 = 90^\circ$; that is, the refracted wave propagates axially in the pipe wall, therefore agreeing with the plots of Figure 4.4.5 and explaining why these intervals are absent from the experimental dispersion curves developed using pressure data. For frequencies beyond the discussed intervals, $W_p^*/U_{p,0}^*$ gradually increases. Besides, from Figure 4.4.1, the phase velocity c^* drops from 1. Now, Snell's law dictates that $\sin\theta_2 > 1$ which is impossible to satisfy, and the wave is completely reflected. That also explains why the coupled system radial mode curves trace these of rigid pipe modes following the characteristic plateaus where $c^* = 1$ (Figure 4.4.1). Regarding the 1st Mode, at

small frequencies ($\omega^* < 1$), it propagates as an axial compressional wave in the pipe wall since $W_p^*/U_{p,0}^* \rightarrow 0$ and $c_{ph} \sim 1$. At higher frequencies, the dispersion curve of the 1st Mode follows that of the 1st mode of the pressure-release water column, and $W_p^*/U_{p,0}^*$ is increased. Put differently, the coupled system 1st Mode from structural borne at low frequencies becomes fluid borne. Finally, the $W_p^*/U_{p,0}^*$ ratio for the 0th Mode increases rapidly as the frequency approaches $\omega^* = 1$, but the rate drops as $\omega^* > 1$. The reader is reminded that for $\omega^* > 1$, the 0th Mode represents a flexural/surface wave.

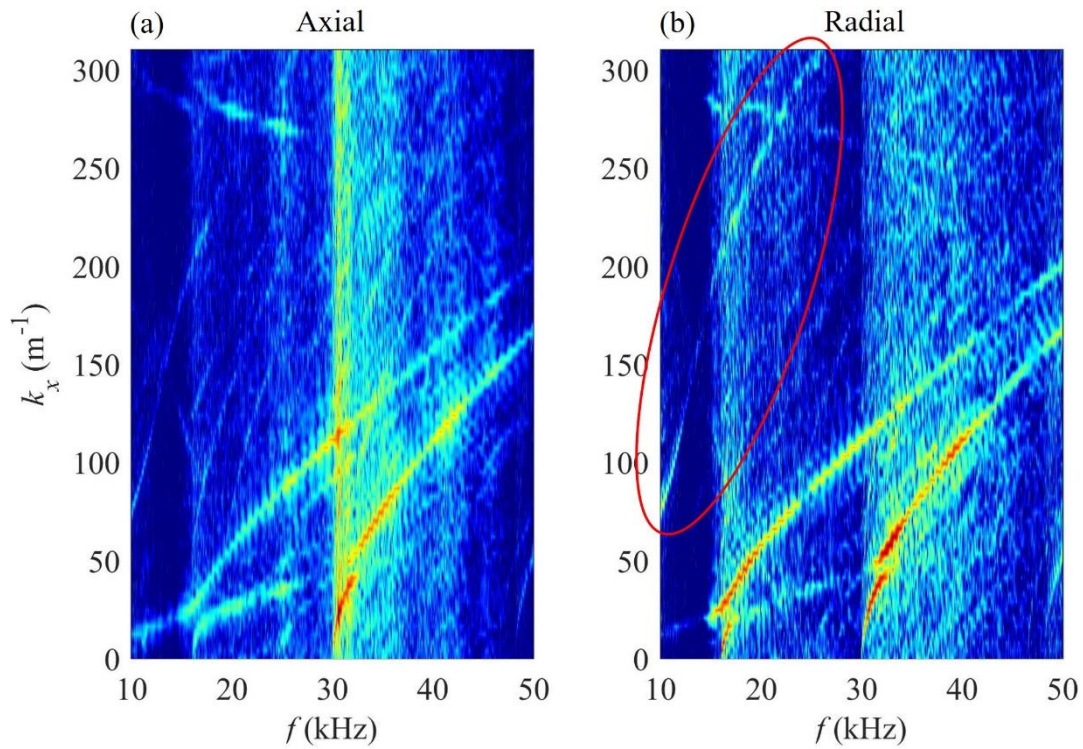


Figure 4.4.6: The experimental dispersion curves are plotted using the data collected from the accelerometers in the (a) axial and (b) radial directions for the DN80 copper pipe. Encircled is the trace that corresponds to the 0th Mode.

Some of the results presented in Figure 4.4.5 are verifiable from the experimental dispersion curves developed using the accelerometer data. Figure 4.4.6a shows the 2D spectrum obtained from the axial accelerometer channel, while Figure 4.4.6b from the radial accelerometer channel. The plots are normalized with respect to the pertinent maximum signal amplitude and are given in dB scale. Note that red colour corresponds to a stronger signal. The 1st Mode trace is found in both Figure 4.4.6a and b. In Figure 4.4.6a (axial displacements), however, it appears stronger for $f < 16$ kHz when compared to Figure 4.4.6b. On the other hand, the 1st Mode trace

more pronounced in Figure 4.4.6b as opposed to Figure 4.4.6a for $f > 16$ kHz. This observation is in accordance to the analytical results (Figure 4.4.5), showing that the 1st Mode propagates axially for $f < 16$ kHz and radially for $f > 16$ kHz. Moving to the 2nd Mode, at the cut-off frequency the pipe wall displaces radially, a fact that is reflected in the strong trace at $f = 16$ kHz in Figure 4.4.6b. Comparatively, the axial channel presents a weak signal at the 2nd Mode cut-off frequency. As the frequency increases, the 2nd Mode is no longer discernible in Figure 4.4.6b (radial) but is visible in Figure 4.4.6b (axial). Again, that observation agrees with the analytical curves of Figure 4.4.5 for radial modes that propagate axially in the pipe wall for a frequency interval starting slightly higher than their cut-off. At the end of this frequency interval (~ 31 kHz), the 2nd Mode displays primarily radial displacements, as corroborated by Figure 4.4.6b. The vertical bright region in Figure 4.4.6a at about 30 kHz represents the resonant frequency of the accelerometer, following which its response is no longer flat. Therefore, the discussion here is limited to the modes that propagate below 30 kHz. It is interesting to note that the 0th Mode is detectable in the radial accelerometer channel (Figure 4.4.6b, encircled) despite not being present in the water (Figure 4.4.3).

4.5 Pressure wave solution

In Chapter 3, the adopted impedance equation is introduced to an appropriate impulse response function that may be used to retrieve the response of the system to any probing signal (see Eq. (3.2.95)). Herein, the efficacy of the analytical pressure wave is investigated experimentally for the DN80 HDPE pipe. For a probing signal, a narrowband Gaussian modulated sine pulse is used since it allows for the precise definition of the excited frequency bandwidth. Essentially, a Gaussian modulated sine pulse is a sine wave – in the frequency domain – “multiplied” by a Gaussian window. Its mathematical discreet form is:

$$y_i = A_G e^{-k(i \cdot \Delta t - d)^2} \cos(2\pi f_c (i \cdot \Delta t - d)) \quad (4.5.1)$$

where:

$$k = \frac{5\pi^2 b^2 f_c^2}{q \ln(10)} \quad (4.5.2)$$

and $i = 0, 1, 2, \dots, n - 1$ are the sampling points. Furthermore, f_c is the central frequency (in Hz), Δt is the sampling interval, A_G is the maximum amplitude and d is the shift of the peak of the Gaussian window. Parameter b is the normalized frequency bandwidth and sets the bandwidth of the pulse, measured at a level of q dB from the peak value of the power spectrum

density. In absolute terms, the frequency bandwidth may be calculated using the expression $f_c \pm bf_c/2$. The attenuation value q is usually set to 40 dB, denoting that the frequency bandwidth is set at 1/100 of the power spectrum density peak value. The continuous form of Eq. (4.5.1) in both the time and frequency domain is given in the Appendix C.

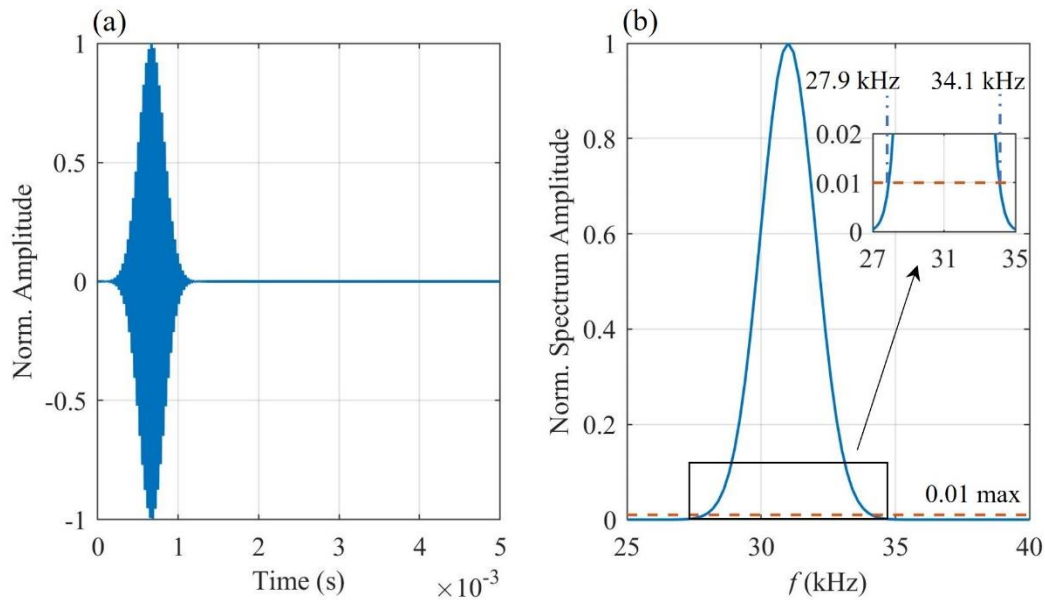


Figure 4.5.1: (a) A Gaussian modulated sine pulse with a central frequency $f_c = 31$ kHz and total duration $t_{signal} = 5$ ms is plotted in time; (b) The frequency bandwidth is defined by the points where the power spectrum curve crosses the horizontal line that corresponds to 0.01 of the maximum amplitude.

The particular Gaussian modulated pulse tested has a central frequency equal to 31 kHz, $b = 0.2$, $A = 3$ V, $d = t_{pulse}/2 = 4 \sqrt{2} \ln(10)/(\pi f_c b)$ where t_{pulse} is the duration of the pulse, $\Delta t = 10^{-6}$ s and 5000 samples for a total signal duration $t_{signal} = \Delta t \times \text{samples} = 5 \times 10^{-3}$ s. The result is plotted in Figure 4.5.1a. On the other hand, Figure 4.5.1b represents the signal in the frequency domain where the power spectrum peaks at f_c , as designed. Moreover, the horizontal line that represents 1/100 of the peak value – a threshold defined by parameter q – crosses the power spectrum curve at exactly 27900 and 34100 Hz, the frequency bandwidth defined by parameter b , since $f_c \pm bf_c/2 = 31000 \pm 0.2 \times 31000/2 = 27900$ and 34100 Hz.

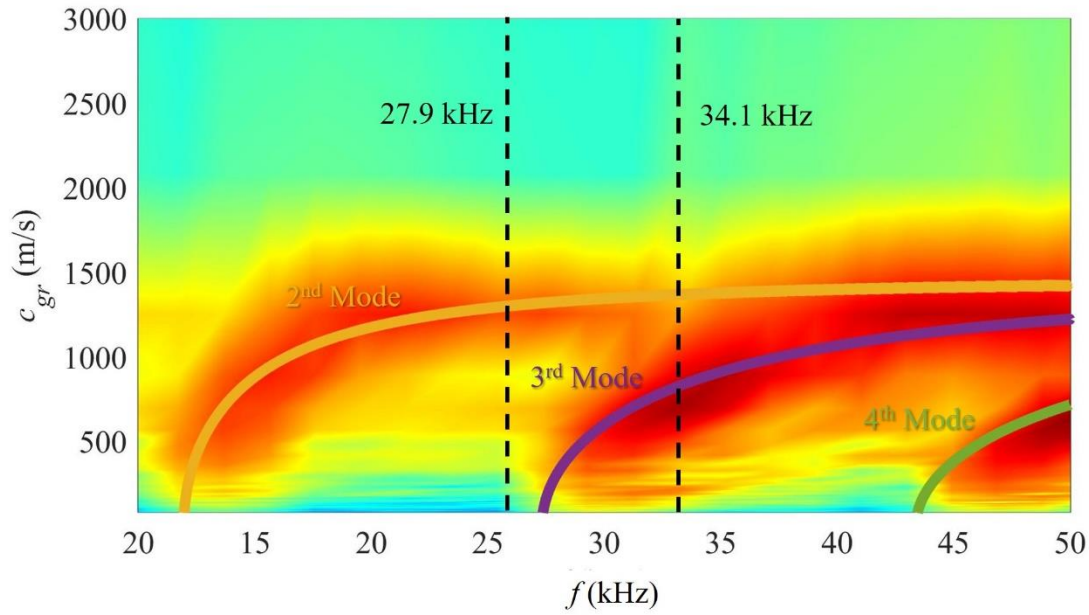


Figure 4.5.2: Group velocity curves obtained from the impedance-based analytical model for the case of the water filled DN80 HDPE pipe. The red lines define the frequency bandwidth excited from the Gaussian modulated pulse of Figure 4.5.1.

Experimental group velocity plots can be obtained by a Short-Time Fast Fourier Transform (STFFT) of the recorded system response to a chirp signal at any measurement location along the pipe. Figure 4.5.2 depicts a comparison of the analytically derived group velocity curves with these obtained experimentally for the DN80 HDPE pipe. The experimental 2D contour is shown in dB scale, calculated relative to the maximum amplitude. Also included are two vertical lines that define the bandwidth excited by the Gaussian modulated sine pulse of Figure 4.5.1. Within the excited bandwidth, therefore, there are two modes excited: the 2nd and 3rd Modes. Note that in the excited bandwidth, the two propagating modes present drastically different group velocities, with the 2nd Mode being faster. Also, the group velocity of the third mode varies significantly: at the cut-off frequency, $c_{gr} = 0$, but quickly increases to about 900 m/s.

Two tests are performed in total. In the first one, B&K Type 8104 transducer pair is placed along the DN80 HDPE pipe centreline with the in-between distance equal to 1.6 m. The projecting transducer emits the pulse shown in Figure 4.5.1a, and the receiver records the response depicted in Figure 4.5.3a (in blue). The single wavepacket of Figure 4.5.1a is split into two distinct wavepackets at the receiver, the fastest of which corresponds to the 2nd Mode. Notice that the trailing wavepacket (that corresponds to the 3rd Mode) appears distorted and

elongated in time. As noted, the group velocity of the 3rd Mode in the excited frequency interval shows great variation, therefore enhancing the dispersion effects. Also, note that the amplitude of the trailing wavepacket is much larger than the first. Indeed, as shown in Figure 4.5.2, the contour of the experimental group velocity curves that corresponds to the 3rd Mode is much more intense than that of the 2nd Mode. Later, the transducer in-between distance is adjusted to 2.2 m, and the test is repeated. The obtained response is presented in Figure 4.5.3b (in blue). Indicative of the increased range is the fact that the two wavepackets are further apart, and their amplitude is also reduced.

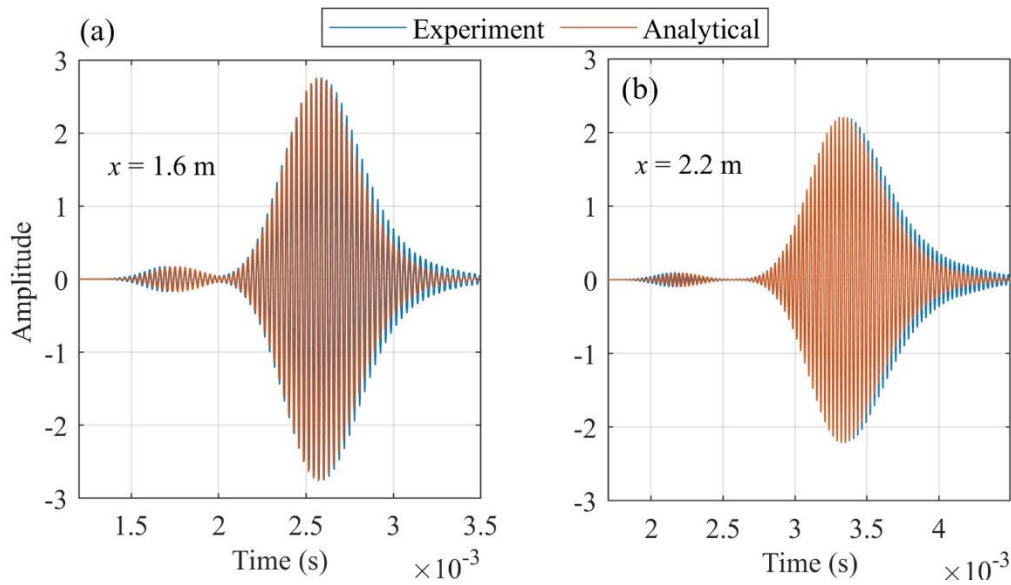


Figure 4.5.3: Comparison between the analytical and experimental response of the water filled DN80 HDPE pipe to the narrowband band Gaussian signal of Figure 4.5.1 at a distance (a) 1.6 m, and (b) 2.2 m from the source.

The analytical system response is calculated in the frequency domain using Eq. (3.2.96). For the mathematical expression of the Gaussian modulated pulse in the frequency domain, the interested reader is referred to Appendix C. Then, an inverse Fourier transform is applied to obtain the analytical response in the time domain, shown in Figure 4.5.3a and b in orange. The analytical response shows a very good match to the experimental response both in terms of signal phase and relative amplitude. In summary, the results presented in subsections 4.3 to 4.5 demonstrate that the adopted approximate analytical model is undoubtedly capable to describe high frequency wave propagation in pipes, even for the case of the more challenging thick-walled DN80 HDPE pipe. In the upcoming sections, the validity of the simplified dispersion

relationships (subsection 3.2.4) with regards to frequency is investigated, and a closer look on the occurrence of the “surface wave” for the 0th Mode is taken.

4.6 Validity of the simplified model expressions with frequency

4.6.1 Low frequency range ($f \ll f_{ring}$)

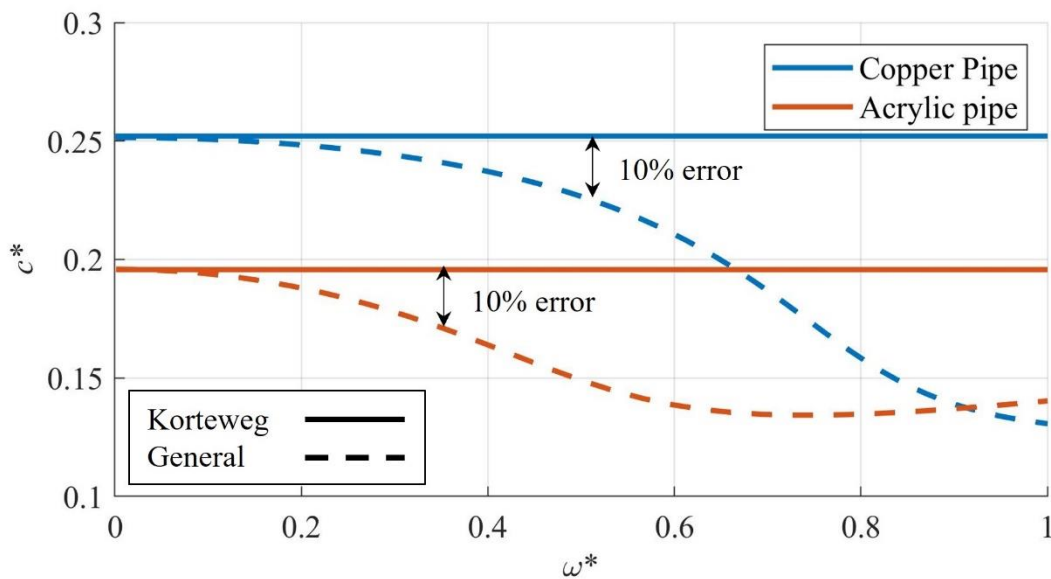


Figure 4.6.1: Phase velocity c^* vs frequency ω^* - Korteweg’s model predictions are compared to those of the general model

First, it should be mentioned that the analysis presented in this subsection is part of a conference paper [77], published in 2018. At the time, a DN65 acrylic pipe was installed in the hydraulic lab, thus part of the results is relevant to said pipe. The mechanical and geometrical characteristics of the DN65 acrylic pipe are presented in Table 4.6.1. Figure 4.6.1 presents a comparison between the predictions of Korteweg’s equation (Eq. (3.2.56)) and those of the general model [59] for the phase speed of the 0th Mode. The dispersion curves are presented in terms of phase velocity c^* and frequency ω^* for two water filled pipes, (i) the DN80 copper and (ii) the DN65 acrylic pipe. Expectedly, Eq. (3.2.56) (Korteweg’s model) produces a constant line that is frequency independent. This approach is accurate at low frequencies of up to $0.2f_{ring}$ (where f_{ring} is obtained for $\omega^* = 1$) (~ 1.5 to 3 kHz) where Korteweg’s assumptions give an error of about 1.5% for the copper pipe and 5% for the acrylic pipe compared to the general solution. For increasing frequencies, the error becomes significantly larger. At $0.6f_{ring}$

(~5 to 9 kHz), the deviation is equal to 19% and 41% for the copper pipe and the more flexible acrylic pipe, respectively. It may be concluded that Korteweg's model is adequate up to 0.3 to $0.5f_{ring}$ (~2 to 8 kHz) where the error is equal to 10% at most depending on the pipe material. To put in a defect detection resolution perspective, this frequency is sufficient to resolve defects of about 15 cm in size.

Table 4.6.1: Mechanical and geometrical parameters of the DN65 acrylic pipe

• Pipe Parameters	Acrylic pipe
Internal Radius, R (mm)	30
Wall Thickness, h (mm)	5
Young's modulus, E (GPa)	3.14
Density, ρ_p (kg/m ³)	1190
Poisson's Ratio, ν	0.39
Wave speed, c_p (m/s)	1750
Ring frequency, f_{ring} (1/s)	7955

4.6.2 Intermediate frequency range ($0 < f \leq f_{ring}$)

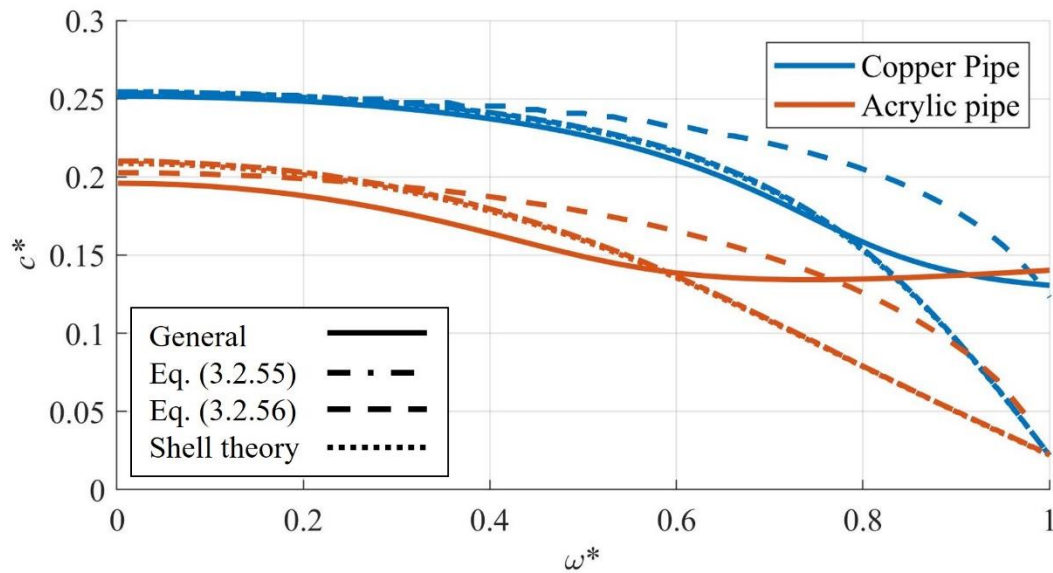


Figure 4.6.2: Phase velocity c^* vs frequency ω^* - Models that account for the pipe inertia only are compared to the general model

To further improve the resolution, the use of higher frequencies is required. At these frequencies, the model assumptions must be reassessed and include the effect of pipe inertia. In Figure 4.6.2, the predictions of Eq. (3.2.55), (3.2.54) and the adopted shell theory-based model are compared with those of the general model [59]. Since it is of interest to solely evaluate the effect of pipe inertia, the flexural terms are ignored from the characteristic equation of the shell-theory based model. It may be observed that the inclusion of pipe inertia alone improves the estimation of the phase velocity of the plane wave compared to the assumptions of Korteweg.

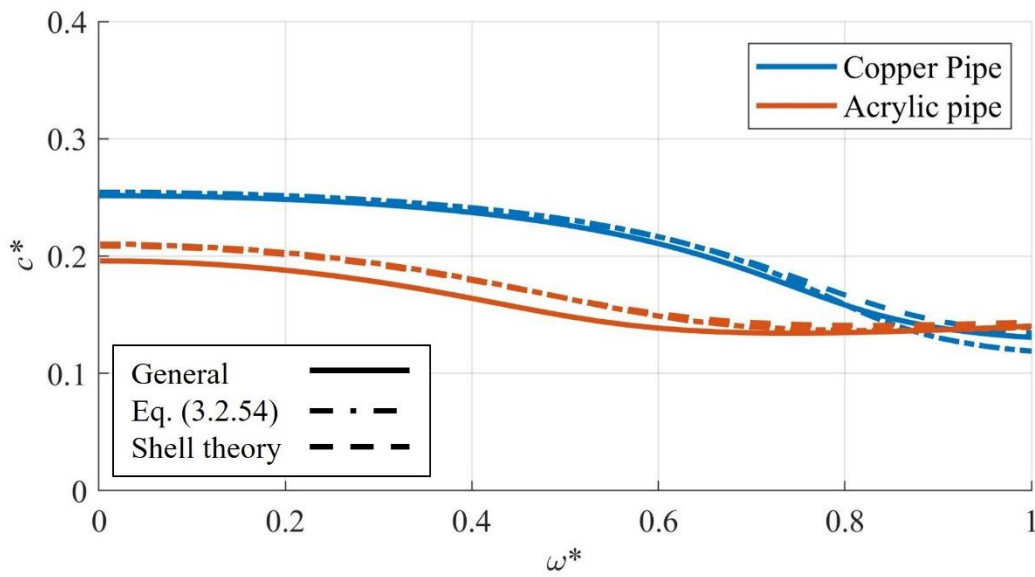


Figure 4.6.3: Phase velocity c^* vs frequency ω^* - Models that account for the pipe inertia and flexure are compared to the general model

In particular, for the copper pipe, all approximate models coincide up to $0.4f_{ring}$ (~ 6 kHz) where the predictions of the proposed model start to deviate and lead to larger errors for greater values of frequency. In contrast, the Eq. (3.2.54) and shell theory-based models, give a good estimate of the phase velocity for frequencies up to $0.8f_{ring}$ (~ 12 kHz). In fact, the curves of these two models are practically identical and can be hardly discerned from one another for the whole range of frequencies examined. However, as the frequency approaches f_{ring} , the error of the approximate models increases dramatically as they predict a phase velocity equal to zero. This is because near f_{ring} the pipe walls expand and contract in what could better be described as a “breathing motion”. This motion, however, is not contained in the cross-section plane and leads to the development of a flexural wave that propagates along the pipe. This phenomenon cannot be captured in the approximate models, as the pertinent terms were ignored from the dispersion

relationships. In contrast, the general benchmark model shows an inflection point at $0.8f_{ring}$ and the curve slope gradually decreases. Similar observations can be made for the case of the acrylic pipe, although the frequency range where the approximate models are applicable without much error is limited to $0.6f_{ring}$ (~ 4.7 kHz). That frequency attains a resolution around 5 – 6 cm.

To further improve the resolution, the next step is to consider the flexural terms in the characteristic equations of the Eq. (3.2.54) and shell theory-based models. Figure 4.6.3 depicts the results and compares them to the – benchmark – general model [59]. Their phase velocity curves follow closely the general model for the entire frequency range. At f_{ring} the error remains well below 10% for both pipe materials and, hence, the resolution is improved to 5 cm. For even higher frequencies (and, thus, resolutions), the plane wave mode is not the only mode that gets excited. In subsection 4.6.3, the higher-order radial modes is discussed. The models that accurately describe the plane wave propagation are re-evaluated based on their capacity to describe these higher-order modes.

4.6.3 High frequency range ($f > f_{ring}$)

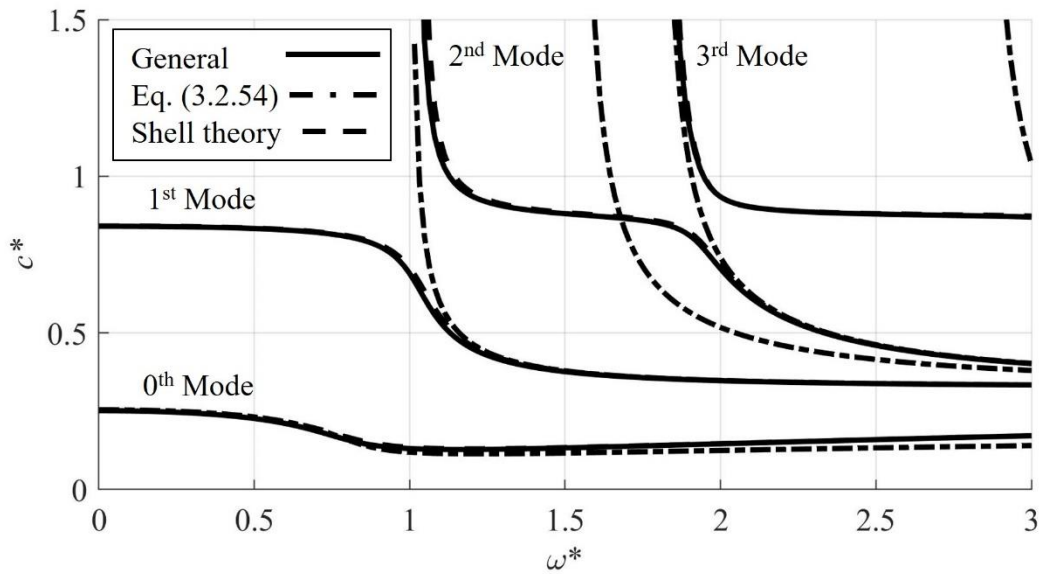


Figure 4.6.4: Phase velocity c^* vs frequency ω^* - The approximate models are compared to the general model for the case of the copper pipe at high frequencies

As the frequency becomes higher than the natural ring frequency, higher-order radial modes are excited. The model of Fay et al. [78] (Eq. (3.2.54)) and the adopted shell theory-based model that were previously verified for the plane wave can be readily applied to study high-

order axisymmetric modes. Figure 4.6.4 contains the dispersion curves for the case of the copper pipe and for frequencies up to $3f_{ring}$ (~ 46 kHz). In the examined frequency range, 4 modes are found in total. Both approximate models can predict the phase velocity variation of the 0th Mode with adequate precision. However, only the adopted shell theory-based model exactly matches the dispersion curves of the general model across all modes. The model by Fay et al. [78] completely ignores axial forces in the pipe wall and that fact alone can explain the mismatch in the prediction for the 1st Mode. Only after f_{ring} , the curve of the Eq. (3.2.54) model coincides with the rest. This shows that the character of Mode 1 changes from a compressional wave to a radial “breathing” mode in the pipe wall.

Considering the higher order modes of Figure 4.6.4, there are frequency intervals at which the phase velocity is about equal to that of Mode 1 at low frequencies. As also discussed in subsection 4.4, for the 2nd Mode this interval is defined from about 1.3 to $1.8f_{ring}$, and for the 3rd Mode this range starts at $2.1f_{ring}$ and extends to the plot limit. At these intervals the phase velocity is close to the speed of the compressional wave in the pipe material, and wave propagation happens axially through the pipe wall. Consequently, the Eq. (3.2.54) model is unable to exhibit these peculiarities. Therefore, we may conclude that a model that ignores axial stresses cannot adequately describe modes other than the 0th Mode.

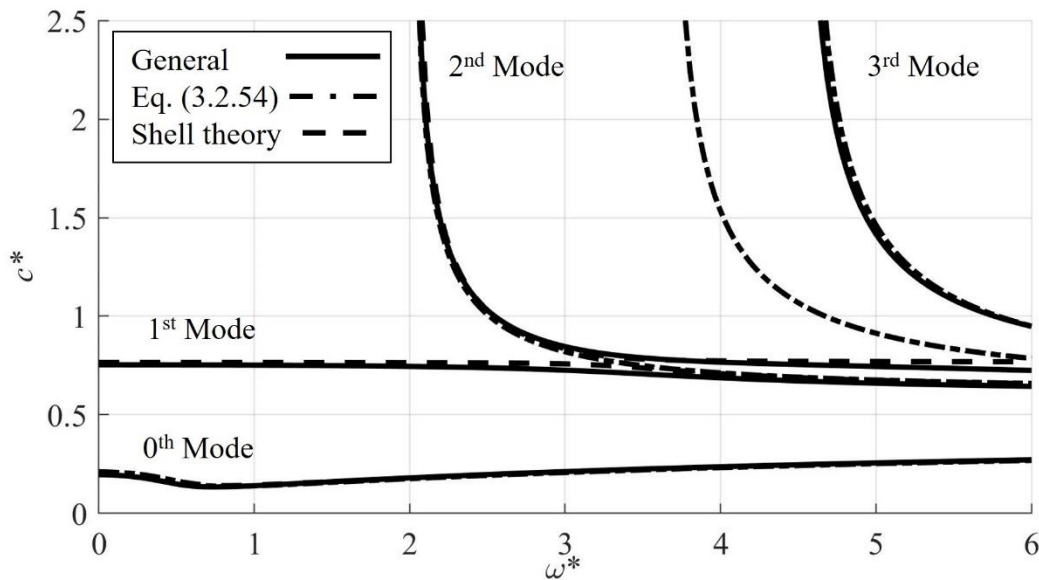


Figure 4.6.5: Phase velocity c^* vs frequency ω^* - The approximate models are compared to the general model for the case of the acrylic pipe at high frequencies

Figure 4.6.5 presents the dispersion curves for the case of the more flexible and less dense acrylic pipe (as compared to the copper pipe). In the range of frequencies examined (up to $6f_{ring}$

(~ 48 kHz)), there are 3 modes appearing in total. As is the case for the copper pipe, while both approximate models can sufficiently describe the propagation of 0th Mode, only the adopted shell theory-based model agrees with the general model for all modes and across all frequencies. As stated, the model of Fay et al. [78] (Eq. (3.2.54)) ignores the axial stresses in the pipe wall, thus cannot describe the 1st Mode and only follows the curves of the other models at frequencies larger than $3.5f_{ring}$ because after that point the wave propagation happens radially and not axially. Below that frequency, the curve from Eq. (3.2.54) follows that of the 2nd Mode, similarly to the case of the copper pipe. It may be assumed that, since modes propagating axially in the pipe wall are completely ignored, the final curve from the Eq. (3.2.54) is a mixture of two successive modes, the 1st and 2nd Modes, that can only be described in the intervals where the wave propagation happens purely radially. Between the cut-off frequencies of the 2nd and 3rd Modes, Eq. (3.2.54) predicts the cut-off frequency of a mode that goes unaccounted for by the two other models. At high frequencies, this mode asymptotically joins the curve of 2nd Mode. Apparently, this curve belongs to the 2nd Mode, but Eq. (3.2.54) cannot capture the axial-radial wave interaction that happens in this frequency interval. The 3rd Mode, on the other hand, can be truthfully described by both approximate models, the reason being it is a purely radial mode in the interval shown.

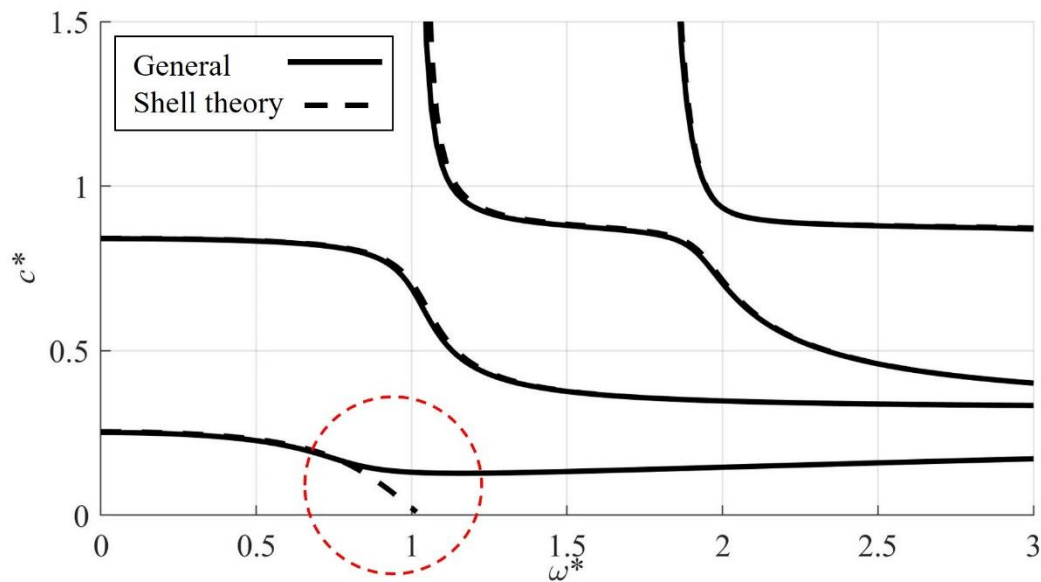


Figure 4.6.6: Phase velocity c^* vs frequency ω^* – Shell theory without the flexural term against the general model for the case of the copper pipe at high frequencies

It would also be interesting to examine how well the adopted shell theory-based model compares to the general model [59] at high frequencies when the flexural term is ignored.

Figure 4.6.6 portrays such a comparison. The behaviour of the 0th Mode when the flexure term is not considered is already discussed in subsection 4.6.2. As the frequency approaches f_{ring} , a model that neglects the flexural forces predicts a phase velocity equal to zero (the exact point is encircled in Figure 4.6.6). However, it appears that flexure has no impact on describing higher-order modes as the 1st to 3rd Mode curves coincide with those of the general model [59]. Similar results may be observed for the case of the acrylic pipe but are skipped for brevity.

4.7 Surface wave

4.7.1 Analytical study

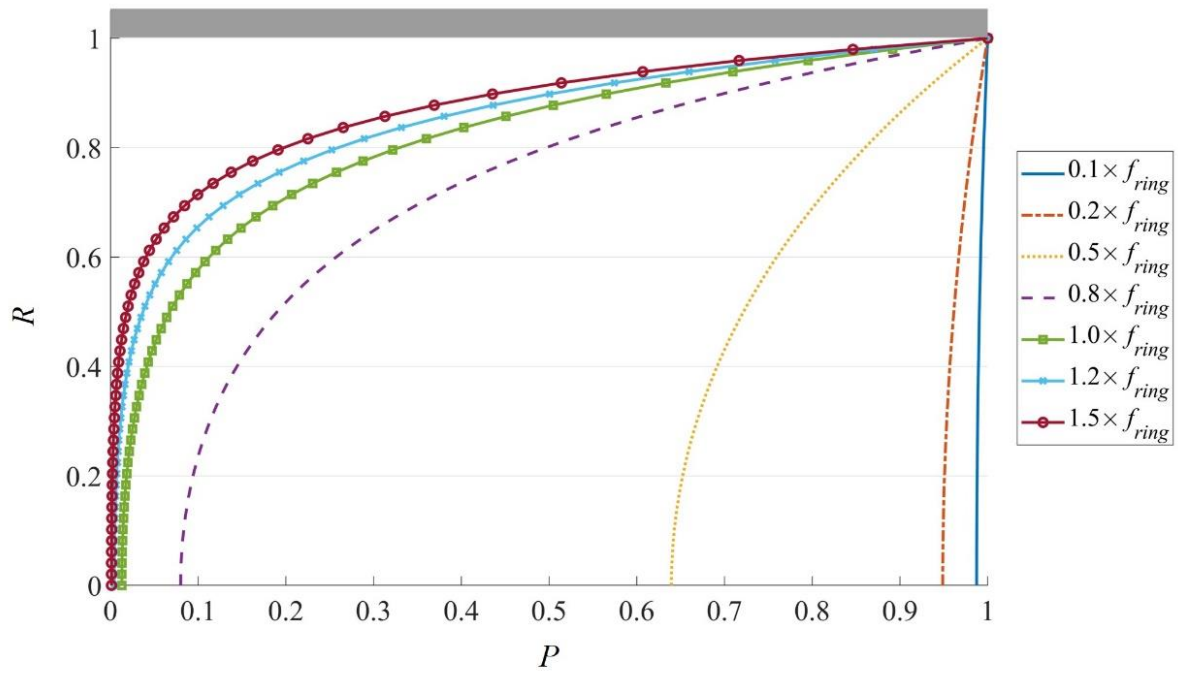


Figure 4.7.1: Pressure distribution of the 0th Mode in the contained water for increasing frequency

Since the adopted approximate model is experimentally verified successfully in subsections 4.3 to 4.5, it may be used confidently for analysis of wave propagation phenomena in fluid-filled pipes. Herein, the behaviour of 0th Mode is investigated for frequencies close to f_{ring} . It is evident that as the pipe wall starts vibrating radially, a transfer of energy from the fluid to the pipe wall takes place. For brevity, the discussion is limited to the case of the DN80 copper pipe. Figure 4.7.1 shows the pressure distribution (Eq. (3.2.12)) along the pipe radius for 0th Mode

and for different frequencies ranging from $0.1f_{ring}$ to $1.5f_{ring}$. The pressure is normalized with respect to the maximum pressure along the radius at each frequency, and the horizontal axis coincides with the centreline of the pipe. At low frequencies, the pressure is evenly distributed along the pipe radius and the wave front remains plane. As the frequency increases, however, the plane wave assumption is no longer valid. At $0.8f_{ring}$, that is the threshold below which models that do not account for the pipe flexure may be used, the pressure at the centre of the pipe is 12.5 times smaller than the pressure at the pipe wall. For even higher frequencies, the pressure of 0th Mode is close to zero at most part of the cross-sectional area. The energy is contained within an annulus from $0.7R$ to the pipe wall and a surface wave emerges.

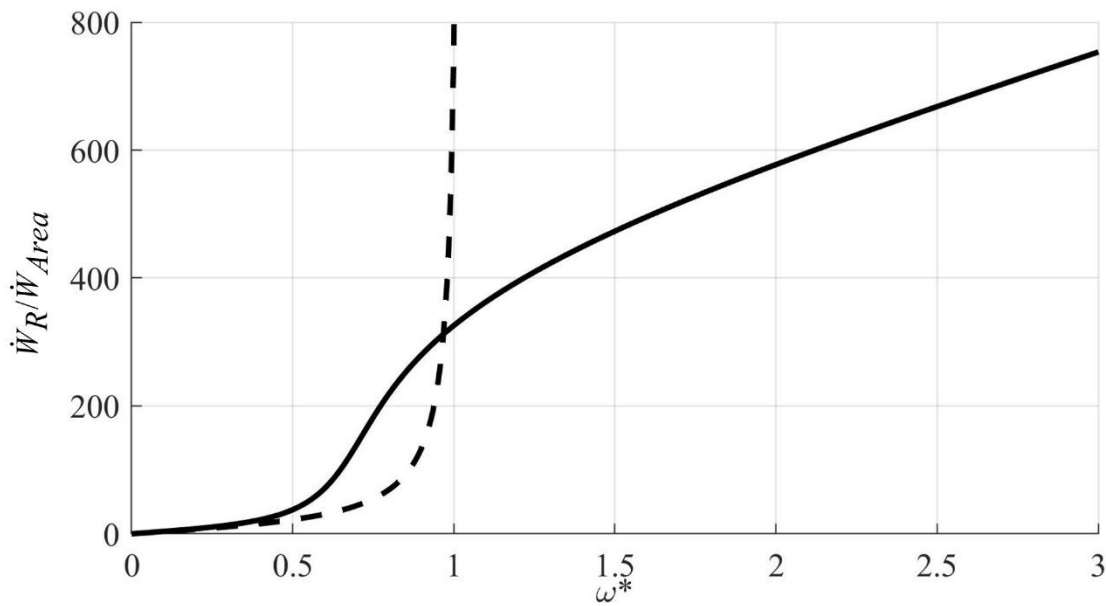


Figure 4.7.2: Work ratio $\dot{W}_R / \dot{W}_{Area}$ vs frequency ω^* for the 0th Mode

This behaviour of the 0th Mode can also be expressed using the ratio of the work done by the pressure wave on the pipe wall $\dot{W}_R$ over that done in the cross-sectional area $\dot{W}_{Area}$ (Eq. (3.2.87)). In Figure 4.6.2, a distinct point at $0.5f_{ring}$ can be identified at which $\dot{W}_R / \dot{W}_{Area}$ starts increasing dramatically. After that point most of the vibrational energy is located near the pipe wall. This characteristic of the 0th Mode could be exploited for detecting defects on the pipe wall. Since the pressure magnitude is higher close to the wall, the chances of receiving reflections from defects is increased. At the low frequency limit, appropriate simplifications can be utilized to obtain Eq. (3.2.88). If the wave speed c_{ph} calculated by Korteweg's equation is replaced in Eq. (3.2.88), the result is the dashed curve in Figure 4.7.2. At frequencies below

$0.5f_{ring}$, where the work relationship appears linear, the approximate expression of Eq. (3.2.88) compares well with the shell theory result (solid curve) for the work ratio. Thus, the interval at which Korteweg's formula holds, is validated here using work expressions. An interesting observation is that Eq. (3.2.88) for $c_{ph} = c_f$ (i.e., rigid pipe walls) yields a work ratio $\dot{W}_R/\dot{W}_{Area}$ equal to zero. The wave speed decrease at low frequencies is only due to the elasticity of the pipe.

4.7.2 Numerical analysis

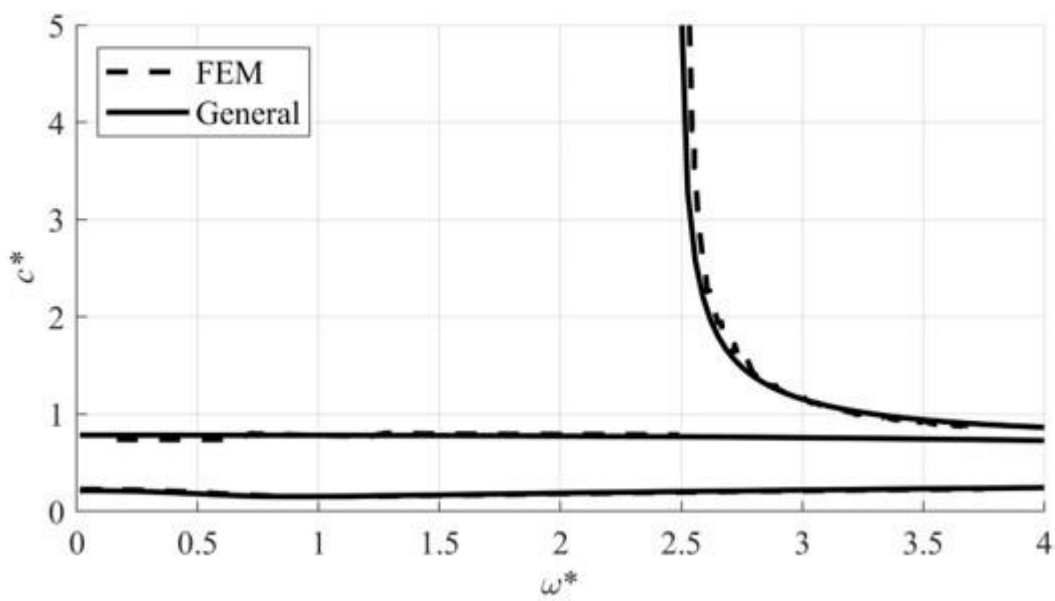


Figure 4.7.3: Phase velocity c^* vs frequency ω^* - FEM results are compared to the general analytical model for the case of the acrylic pipe

As an additional method to study the modes of a coupled fluid-pipe system, a 3D Finite Element (FE) model is devised in ABAQUS. Specifically, a modal analysis using Lanczos algorithm is performed on a 2 m acrylic pipe of Table 4.6.1 filled with water. Although analytical models assume an infinite pipe length, comparatively, the cross-sectional dimensions are small enough to minimize the finite length effect. For the fluid domain, 8-node brick linear elements are used with 1 degree-of-freedom per node, that of pressure. The pipe wall is modelled using linear 4-node shell elements with 6 degrees-of-freedom per node, fully describing translations and rotations. At the solid-fluid interface, displacements for the two media are tied together. For an adequate accuracy, at least 10 nodes should be contained within a wavelength. The maximum

frequency of interest f_{max} is set equal to 30 kHz and, therefore, the element size is equal to 4.75×10^{-3} m.

Analysing the FE results is not trivial; in the frequency interval from 0 to 30 kHz, a total of 4601 mode shapes exist. To categorize the mode shapes into Modes, each one must be examined individually. If done manually, this process demands considerable time and effort. For that reason, an automated categorization algorithm is designed that analyses the waveform characteristics along some predefined paths on the model. The algorithm evaluates the wavenumber along radial, circumferential and longitudinal paths. Radial paths identify radial modes while circumferential paths allow to distinguish axisymmetric modes from asymmetric modes. Finally, longitudinal paths yield the axial wavenumber. The result is the dispersion curves of Figure 4.7.3. In the same plot, the FE results are compared favourably to those obtained from the benchmark model.

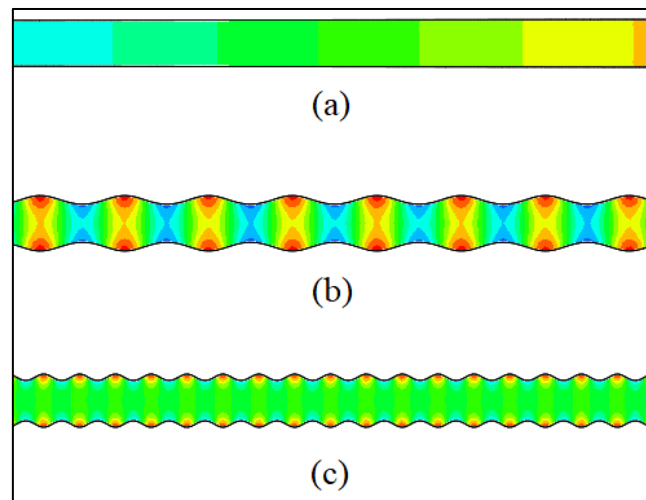


Figure 4.7.4: Pressure contour plots with pertinent pipe wall deformations for the 0th Mode
(a) $0.01f_{ring}$, (b) $0.5f_{ring}$ and (c) f_{ring}

The advantage of the FE approach is the visualization capabilities it offers. Using the analytical model, it was shown that as the frequency increases, most of the energy of the 0th Mode is contained within a narrow area close to the pipe wall. FE allows to observe the pertinent pipe wall deformations and, specifically, the development of the flexural wave along the pipe length. In Figure 4.7.4, the pressure contour plot of the 0th Mode is presented for an increasing frequency. At low frequency, the pressure wave remains plane (Figure 4.7.4a). The pipe wall participation is minimal. At $0.5f_{ring}$, the flexural wave on the pipe wall becomes apparent and the pressure wave is no longer plane, and the pressure magnitude is larger away from the

centreline (Figure 4.7.4b). At f_{ring} , the surface wave that propagates at the pipe-fluid boundary is clear. The pressure across the pipe centreline is zero, thus a receiver located at that point will not be able to detect the 0th Mode (Figure 4.7.4c).

4.8 Conclusions

In this chapter, the dispersion relationship of the adopted approximate shell theory-based model is experimentally verified. For the purpose, a novel experimental setup is developed in the hydraulic lab at HKUST. There, two water filled pipes – a DN80 HDPE and a DN80 copper pipe – that present similar cross-sectional geometries but different mechanical properties are subjected to wideband, linear frequency modulated (LFM) signals and their response is measured at both the fluid and the pipe wall to create the experimental dispersion curves. Subsequently, the analytical dispersion curves are checked against both the experimentally derived dispersion curves and the predictions of a state-of-the-art general model [59] and compares satisfactorily. Besides, the wall displacement solutions derived in Chapter 3 are used conjunction with the experimental measurements in the structure domain to explain features of the dispersion curves and further validate the model. In addition, the impedance condition of the adopted model is implemented in a pressure wave solution and is successfully tested against the experimental system response to narrowband probing pulses, hence substantiating its versatility.

Appropriate simplifications to the characteristic equation of the impedance-based model yield well-known wave speed relations (Korteweg’s equation and others) that are accurate only under particular frequency ranges. For instance, analysis shows the water hammer wave to be plane only at frequencies 20% of the pipe ring frequency f_{ring} . At frequencies that approach f_{ring} and beyond, the wavefront is no longer plane. Instead, the pipe wall resonates and executes a “breathing” motion, giving rise to a surface wave in the fluid that propagates along a small region adjacent to the pipe wall. Thus, at high frequencies, the wave is, in fact, undetectable along the pipe centerline. In short, the proposed analytical model is validated through a comprehensive experimental process. while its simple formulation allows for some profound insights on HFW propagation in elastic fluid filled pipes. In the next chapter, wave time reversal as a feasible inverse technique in pipelines is investigated for the first time.

Chapter 5

Time reversal in hydraulics

5.1 Introduction

There is no unique definition of time reversal. Here, we limit ourselves to processes that can experimentally be proven time reversible in the following sense: a source located at $\mathbf{x}_s$ emits a pulse at time $t = 0$ that propagates through a medium. Receiving transducers, positioned at a distance from the source, sample the incoming wave for a total time duration T_{tot} . The received signals are time reversed (last in, first out) and reemitted. If the wave refocuses at the original source location after time T_{tot} , then the process is deemed time reversible. A fascinating experiment that fits the stated definition is a classic problem in hydraulics and involves the time reversal of gravity waves [89] (a video of this experiment can be found here: <https://www.youtube.com/watch?v=XDbLi2YGQn8>). There, an object is dropped on the calm surface of the pool, creating concentric ripples that propagate from its entry point. The pool walls comprise of multiple paddles that each can independently sample the height of the water surface. During this particular experiment and until waves die out, only the paddles positioned along two opposing pool walls are in operation. In the time reversal stage, the same paddles actively transmit the time reversed waveform acquired during the initial (forward) stage. At first, the time-reversed wave field on the surface of the pool appears random and chaotic; with time, however, the circular ripple pattern re-emerges, and the waves refocus at the location of the original disturbance. Another ingenious series of experiments that visualize time reversal is presented in [90] where the authors employ elastic waves propagating in an aluminium plate. In one set of experiments, a number of Lego figurines is placed on top of the plate during the

time reversal stage. As the vertical displacement field refocuses, only the figurine located above a preselected focal point is overturned, while the others remain unaffected.

Time reversal is far from a theoretical endeavour; in fact, it forms the foundation of many technologies. If you happen to be reading this manuscript on an electronic device equipped with a touch screen, you are likely to be experiencing time reversal at work. Indeed, some touch and tactile screens are based on the ability of time reversal to refocus wave energy at specific points to recognize touch inputs and provide haptic feedback [68]. Another practical application of time reversal mirrors is pulse-echo defect detection employed in several fields, including the medical imaging and treatment of brain tumours [62] and kidney stones [63], ultrasonic non-destructive testing of solid materials [64], and underwater acoustics for locating targets [91]. Considering the scenario of waveguide containing a defect, an acoustic pulse is injected to “illuminate” the domain. The defect is subsequently turned into a passive acoustic source as the pulse is scattered off its surface due to the apparent impedance discontinuity. By capturing that reflection, time reversal techniques are employed to uniquely focus the distorted signal at the location of the defect. In pipe systems, the methods applied in [92, 93] implicitly show that the time reversed response is key for the precise localization and characterization of defects.

The degree of temporal compression of the recreated pulse at the location of the original source increases with the number of transducers that capture the signal, as well as in the presence of reflective boundaries and inhomogeneities. For instance, in [94, 95, 96] the performance of time reversal in a fluid waveguide bounded by non-absorbing symmetric walls was found superior to that in free field. Multiple works, e.g., [97, 98] have also shown that the time reversal process is enhanced under the presence of multiple scatterers or in non-dissipating reverberating chaotic cavities, where a single-channel time reversal mirror (i.e., a single receiving and reemitting transducer) was shown to be adequate. Theoretically, if the energy emitted from the source is acquired at its entirety, the source is then placed in a spot not smaller than half the probing wavelength, a value that corresponds to the diffraction limit [99]. Nonetheless, researchers have been able to overcome this limit by including sub-wavelength scatterers (so-called metamaterials) in the propagating media and attain super-resolution [100, 101, 102].

Recently, the Smart Urban Water Supply Systems (SUWSS) project (Smart UWSS Team, 2017) proposed the use of acoustic waves for the condition assessment of pipe systems.

Pressure waves in the low frequency regime with wavelengths much larger compared to the pipe diameter propagate over long distances (in the order of km) and provide with a rough assessment of the network condition. Once a problematic segment is identified, high frequency pressure waves are used to pinpoint the location of the defect within a few centimetres. In the previous chapters of this thesis, it is candidly demonstrated that the propagation of high frequency waves in water supply pipes is a dispersive phenomenon. For example, when a high-frequency point-like acoustic source is introduced inside a pipe, the transmitted waves propagate in all directions and bounce off the pipe boundaries. As a result, a receiver located at a distance records a signal that contains multiple reflections of the source emanating from the pipe walls. Waves that reflect at steeper angles from the pipe wall follow a longer path to the receiver, ultimately arriving at later times. The number of modes created depends on the properties of the waveguide, the frequency bandwidth of the probing signal, as well as the radial location and physical dimensions of the source. Moreover, for high frequency waves, it is imperative that a description of the deformable pipe wall is also included (see Chapters 3 & 4). Ultimately, such environment greatly complicates the identification of a wave source or defect (i.e., a passive wave source).

Time reversal is a powerful tool that could resolve such complexities; nevertheless, its validity in the waveguide of interest must be proven. Jin, Ying, & Zhao [103] used piezoelectric transducers mounted on the outer pipe surface to show that ultrasonic guided waves in elastic steel pipes are time reversible. This thesis presents a novel experimental proof-of-concept of time reversal in water-filled viscoelastic HDPE pipes, where the acoustic source is placed within the water. The signals employed in this thesis have wavelengths comparable to the pipe radius and are, therefore, dispersed due to the radial boundaries giving rise to multiple propagating modes. Regardless, pure multipath propagation effects can successfully be compensated for by the time-reversal process [95]. In contrast, an important condition for the full time-invariance property to hold is that the propagation medium is non-dissipative [104]. However, fluid pressure (acoustic) waves propagating in HDPE pipes interact with the boundaries and are significantly attenuated due to the viscous properties of the material [105]. Besides, HDPE displays a non-linear stress-strain relationship that depends on the applicable strain and strain rate, as well as the loading history under an oscillating load [106]. In effect, both wave speed and damping of waves propagating in fluid-filled HDPE pipes are frequency and mode dependent [107]. In addition to the dissipative characteristics of the HDPE pipe wall, the effect of the fluid viscosity has to also be considered.

In fact, it is demonstrated that the fundamental equations describing the viscous fluid-filled HDPE pipe are inherently not time reversible. Nonetheless, the obtained experimental results indicate that time reversal under such setting is surprisingly robust. Moreover, the spatial focusing properties of time reversal are also experimentally verified on the same HDPE pipe setup. Specifically, the time reversed pressure field is evaluated at multiple locations and is shown to uniquely refocus at the location of the original source. In light of these remarkable findings, the two mechanisms that hinder wave time reversal, namely wave path instabilities and damping, are examined. The TR refocusing error due to instabilities is found to grow only as $\sim t^{1/2}$ and, thus, is not critical. A comprehensive scale analysis is performed based on the fundamental waveguide equations to elucidate the reasons time reversal holds despite the presence of damping. The outcome of this analysis is the time reversal parameter (TR), that corresponds to the ratio of viscous forces to the temporal acceleration and reveals that damping effects grow linearly in time $\sim t$. As $TR \sim 1$, the viscous losses in the considered system become significant, and the time reversibility of waves no longer holds. Subsequently, it is possible to estimate the applicable range for various time reversal applications given the characteristic scales of the system. More importantly, the validity of the time reversal parameter TR is general and, thus, not limited to pipeline applications. The fact is demonstrated by revisiting already published experimental results on time reversal of acoustic waves in the ocean and gravity waves on free-water surfaces, both documented in the non-hydraulics literature.

5.2 Time reversal experiments in HDPE pipes

5.2.1 Experimental setup

The particular experiment is performed on the intact DN80 HDPE pipe of the high frequency pipe setup presented in Chapter 4. It is reminded that the pipe runs uninterrupted along its length with its cross-section constant throughout, and water is quiescent. To remove any air entrapments, all necessary steps are taken as described in Chapter 4. Due to the prospect of utilizing time reversal techniques for defect identification in water supply pipelines, the portion of the pipe cross-section covered by transducers must be minimal to avoid obstructing the flow. In addition, the inaccessibility of water supply systems makes it difficult, if not impossible, to introduce multiple transducers. For these reasons, the experimental setup is limited to a single-channel time reversal mirror. It must also be noted that the tested waveguide is semi-infinite

(thus far from a closed cavity). In such setting, a single-channel time reversal setup is at a disadvantage since a large part of the emitted wave energy escapes and cannot be reversed, with possible severe implications to the successful application of time reversal.

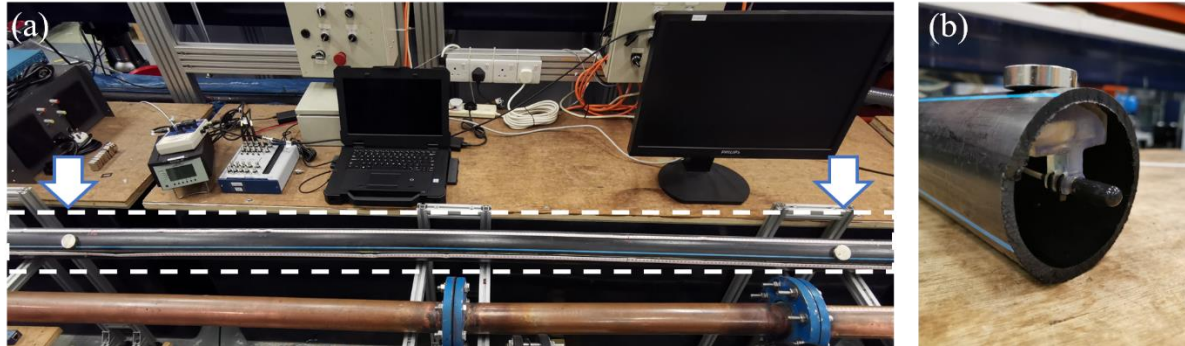


Figure 5.2.1: (a) The white dashed frame encloses part of the DN80 HDPE pipe used in the experiment. (b) A bespoke 3D printed magnetic base is used to mount the transducer inside the pipe at approximately the pipe centre.

The point-like acoustic source is placed in the water and within the boundaries of the pipe wall at one end of the pipe, by means of a Brüel & Kjær (Nærum, Denmark) Type 8103 transducer [84]. To evaluate the response of the system to the injected signal, an identical receiving transducer is positioned at a distance equal to $x = 2$ m from the source, corresponding to approximately 25 times the pipe internal diameter ($x/D = 25.3$). The specific distance is selected so that the various modes propagating at different group velocities along the waveguide have adequate time to separate and become easily discernible at the receiver. As regards to the radial direction, the emitting transducer (i.e., acoustic projector) is mounted at approximately the pipe centre (Figure 5.2.1b). The aim is to keep the excited pressure wave field axisymmetric. Subsequently, the number of excited modes is manageable, and the obtained response is easier to interpret. It is stressed that the validity of time reversal is by no means depending on the symmetry, or lack thereof, of the pressure field. As discussed in Chapter 4, the magnetic mounting mechanism also allows for the precise longitudinal repositioning of the transducers, making it possible to examine the spatial focusing of the time reversal pressure field (to be discussed in subsection 5.2.3). The single receiving transducer (i.e., hydrophone) is also positioned at the pipe centre, although its placement in the radial coordinate is, again, not critical to the time reversal process. A picture of the DN80 HDPE pipe tested (excluding the reservoirs) is presented in Figure 5.2.1a, where the two white arrows pinpoint the location of the transducers.

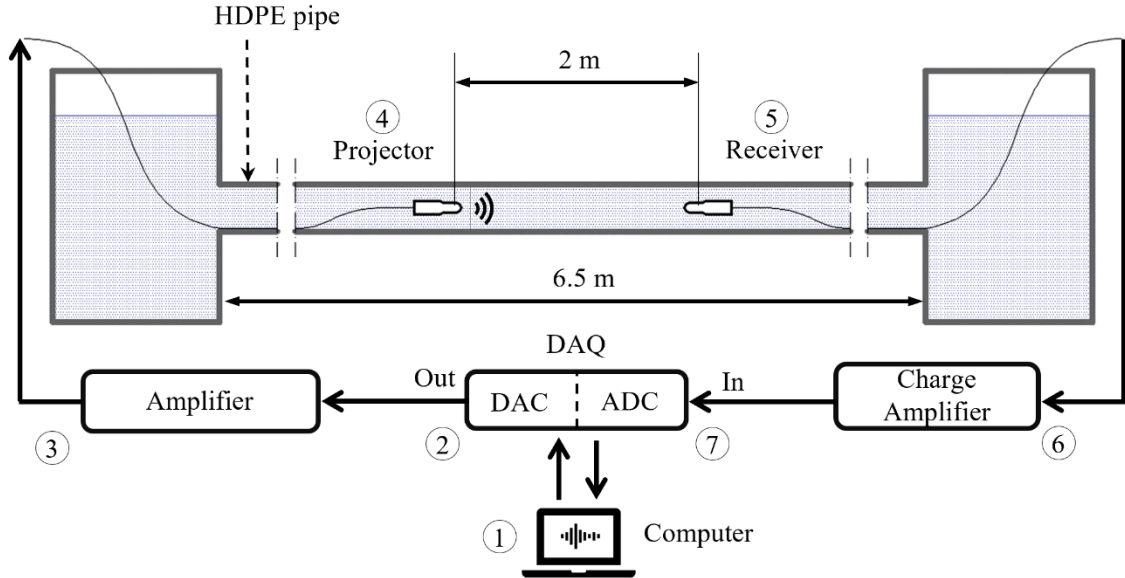


Figure 5.2.2: Signal path schematic of the experimental setup for TR: For the forward test, the narrowband probing signal is defined at the computer (1), converted from digital-to-analog in the DAQ board (2), amplified (3) and emitted from the projector (4). Concurrently, a receiver 2 m away (5) captures the system response, the signal is conditioned (6), converted from analog-to-digital (7), and stored in the computer (1). For the second – time-reversed – part of the test, the transducer roles are interchanged: the receiver becomes the projector and vice versa, while the emitted signal is the stored response but time-reversed (first in, last out).

Figure 5.2.2 depicts the signal path. For the forward wave propagation, a narrowband probing signal of 3 V amplitude is defined at the computer and fed to a National Instruments (Austin, USA) USB-6356 DAQ I/O board, where it is converted from the digital to the analog domain with 16-bit accuracy. The transmission sampling rate is set to 1 MHz, adequate to resolve probing signals with a frequency content of up to 100 kHz when considering a minimum of 10 sampling points per wave period, T_s . Before reaching the transmitting transducer, a Brüel & Kjær Type 2713 Power Amplifier applies a 30 dB gain to the analog signal (hence the output amplitude for a 3 V input signal is ~ 95 V). At the exact same time that transmission occurs, the sensing circuit commences with the signal acquisition. The pulse propagates along the pipe and interacts with the boundaries before reaching the receiving transducer. A Brüel & Kjær Type 2692-A Nexus Charge Amplifier conditions the received analog signal, that is then is converted to digital on the National Instruments USB-6356 DAQ I/O board, with the same resolution and sampling rate as that of the digital-to-analog conversion. In the final step, the

system response is stored in the computer. For the second part of the experiment, the roles of the transducers are interchanged by rerouting their connections: the receiver becomes the source, and the transmitted signal is the time-reversed system response recorded during the forward propagation. The pressure field generated in the waveguide by the time-reversed signal is evaluated at the location of the original source. The utilized pressure signals are Gaussian modulated sine pulses (described in Chapter 4) that enable the precise definition of the excited bandwidth and, hence, number of modes.

5.2.2 Results

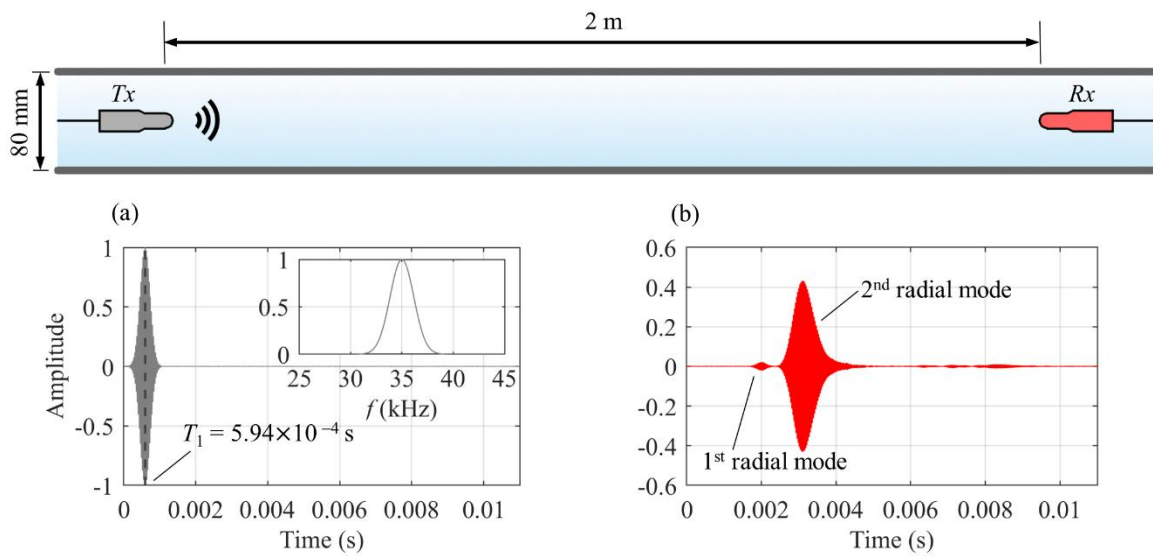


Figure 5.2.3: (a) A Gaussian modulated pulse is emitted from the source at one end of the pipe, with its amplitude spectrum shown in the inset. (b) The response of the system 2 m away from the source is characteristic of multipath propagation and comprises of two wavepackets, with the fastest attributed to the 1st radial, and the trailing to the 2nd radial mode.

For the first trial, we probe the DN80 HDPE pipe setup with a Gaussian modulated sine pulse of central frequency $f_c = 35$ kHz and 20% relative bandwidth ($b = 0.2$), peaking at time T_1 . The wavelength associated with the central frequency of the pulse is about 4 cm, and the total duration of the signal is 5×10^{-3} s, accounting to 5000 samples at 1 MHz sampling rate. Figure 5.2.3 shows the results of the forward wave propagation. Specifically, Figure 5.2.3a depicts the waveform of the pulse emanating from the projector, as evaluated by a receiver in contact with the source in initial testing, and Figure 5.2.3b shows the acquired system response. Both

waveforms presented in Figure 5.2.3 are normalized with respect to the source amplitude. The obtained response in Figure 5.2.3b is characteristic of the multipath wave propagation that occurs in a bounded waveguide. Experimental dispersion curves from the DN80 HDPE pipe presented in Chapter 4 indicate that in the excited frequency interval (depicted in the inset of Figure 5.2.3a) two modes propagate in total. These two modes are discernible as individual wavepackets in Figure 5.2.3b, with the fastest reaching the receiver at $t \approx 0.002$ s. To illustrate, consider each mode as a vector/ray that stems from the acoustic source at different angles with respect to the pipe centreline. The ray with the steeper angle “bounces” off the pipe walls more often compared to the other and eventually follows a longer path to the receiver. Thus, this particular mode appears “slow” and reaches the receiver at a later time than the “fast” mode, despite the fact that the wave speed, c_f , is constant. The faster mode is identified as the 1st radial, while the trailing mode is the 2nd radial; absent is the plane-like wave mode that is expected in typical water hammer as the acoustic projector diameter is only 0.1 the cross-sectional diameter and fails to excite the 0th Mode [60]. It is noteworthy that the dispersion effects are more prevalent for the case of the 2nd radial mode, as revealed by the certain elongation (or “tail”) of the pertinent wavepacket in time. This is due to the relatively larger variation in the group velocity of the 2nd radial mode in the excited frequency bandwidth compared to the 1st. Signals of negligible amplitude in the range $t > 0.06$ s are due to reflections from the pipe/reservoir boundaries and pipe supports. Consequently, the setup is akin to a semi-infinite waveguide problem, and reverberations from the boundaries at both ends of the pipe that would otherwise assist in the time reversal process are precluded. The total duration of the captured signal is chosen equal to $T_{tot,1} = 0.011$ s, as longer measurements indicate no further reflections beyond that point in time.

In the selected frequency bandwidth, it is apparent from Figure 5.2.3b that the 2nd radial mode carries more energy in comparison to the 1st. Regardless, the accumulated amplitude of the two modes is much reduced compared to that of the injected pulse (Figure 5.2.3a). This is partially expected due to the dispersion effects and the use of a single receiving transducer that fails to capture the transmitted wave energy in its entirety. Most crucially, however, these attenuation levels could be indicative of excessive damping due the viscous properties of the waveguide. Therefore, the validity of the time reversal process under such conditions is not certain and warrants further experimental investigation.

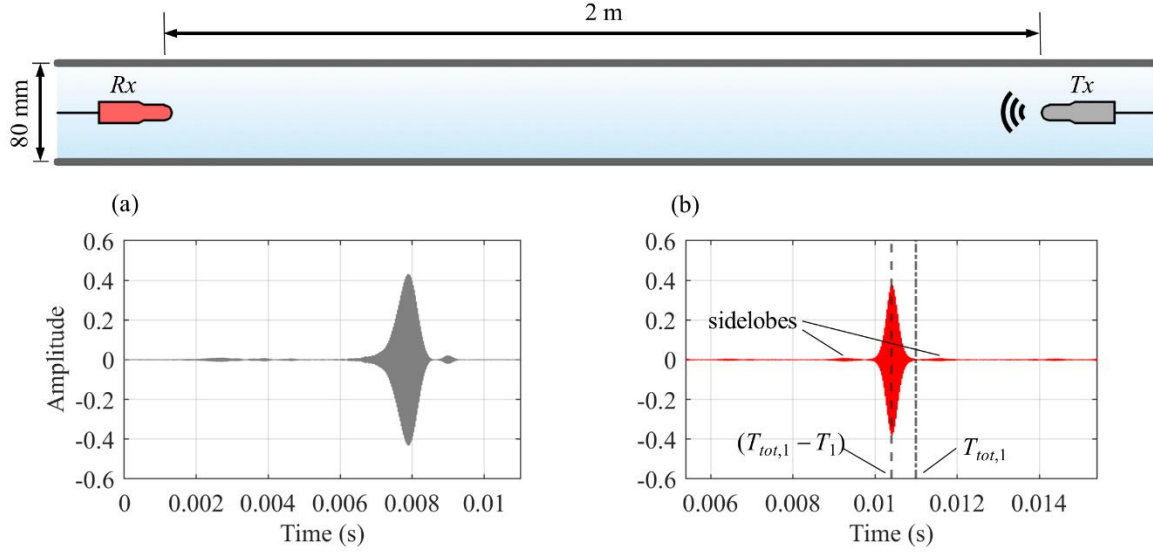


Figure 5.2.4: (a) The response of the system from the forward propagation test is time reversed and reemitted from the location of the original receiver. (b) The original pulse is successfully recreated at the location of the original source, albeit with the appearance of sidelobes.

For the time-reversal test, the system response from the forward propagation (Figure 5.2.3b) is reversed in time (first in, last out – Figure 5.2.4a) and emitted from the position of the original receiver. Consequently, the slow propagating mode (2nd radial) is released first, with the fast one (1st radial) following. The response of the system to the time-reversed pulse is evaluated at the position of the original source. Figure 5.2.4 presents the results, with the source signal and pertinent system response plotted in Figure 5.2.4a & b, respectively. All waveforms are normalized to the amplitude of the original source (Figure 5.2.3a). Surprisingly, the originally injected signal (Figure 5.2.3a) is recreated with great veracity (Figure 5.2.4b), indicating that multipath effects are successfully compensated. The peak amplitude of the recreated pulse occurs at $t \approx T_{tot,1} - T_1$, coinciding with that of the original source with a relative error of 96.1 ppm, thus demonstrating sufficient temporal compression.

A trait of the time reversal process is the appearance of symmetric sidelobes with respect to the main pulse (indicated in Figure 5.2.4b). Sidelobes occur due to the inability of the single transducer time reversal mirror to capture a portion of the wave energy and are suppressed if additional transducers/points are included to form a time reversal mirror (TRM) array [95]. However, the experimental setup offers limited cross-sectional space for the deployment of multiple transducers across the pipe diameter.

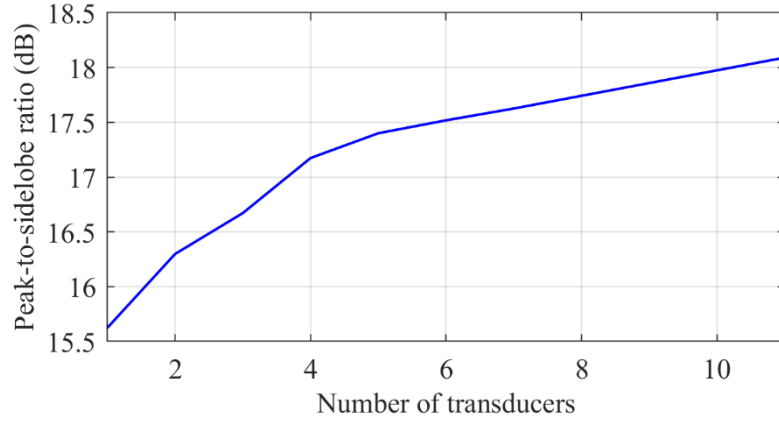


Figure 5.2.5: Analytically obtained peak-to-sidelobe ratios against the number of transducers in the TR array using the model described in subsection 3.2.7.

An alternative approach is to utilize the analytical model of Chapter 3 (subsection 3.2.7) to simulate additional transducers and study their effect to sidelobes. Figure 5.2.5 presents the results in terms of main peak to sidelobe ratio in the recreated signal against the number of transducers/points that comprise the TR array. Indeed, the peak-to-sidelobe ratio increases from about 15.6 dB for the case of a single transducer to 18.1 dB for an array of eleven.



Figure 5.2.6: (a), (b) The time reversal test is repeated at a field-scale facility that employs a 250 m long DN150 HDPE pipe. The transducer set is placed 36.5 m apart, as dictated by the two access points “A” and “B” along a straight segment of the setup.

The presented results prove that time reversal of high frequency pressure waves is feasible when employed in water-filled viscoelastic HDPE pipes. The experimental setup in the hydraulic lab of HKUST, however, features a rather small diameter pipe not found in typical water supply network installations. In addition, the distance between transducers is short and, therefore, unlikely to be implemented in practice considering the scarcity of access points in such systems. Aiming to further explore time reversal in a more realistic setting, the test is repeated at a field-scale experimental facility located at Beacon Hill, Kowloon, Hong Kong. The setup comprises of a 250 m long (in total) DN150 HDPE pipe ($D = 147.2$ mm, $h = 16.4$ mm), constructed above ground and along the perimeter of the site. The pipeline is gravity-fed from the Beacon Hill freshwater reservoir, positioned about 61 m above the site level. Figure 5.2.6a depicts the outline of the complete pipeline in red. To perform the time reversal test, a straight segment along the pipe is selected, where the same transducer pair is placed $x = 36.5$ m apart. The distance is dictated by the access points (“A” and “B” in Figure 5.2.6) along the pipe and corresponds to approximately 250 times the diameter ($x/D = 248$). In addition, since these entry points are rather confined, it is not possible to position the transducers along the pipe centreline. As a result, the generated pressure field is not axisymmetric; still, as already remarked, this has no effect on the application of time reversal. During the test, there is no flow in the pipe and ambient water pressure is held constant. Note that the transducer magnetic mounts could be dislocated under flow conditions, eventually changing the relative distance between the transducer set and/or their position radially. Such changes, especially during the time interval between the forward and time reversal tests, render any obtained results questionable. Nonetheless, a flow scenario is experimentally investigated in Chapter 7. In the Beacon Hill setup, however, it is chosen that water remains quiescent. To purge any potential pockets of air, the system is flushed for about an hour prior to executing the test. In conclusion, this setup provides with the opportunity to test the viability of time reversal in an extreme environment which is more akin to real systems, as defined by the dimensions and the viscous characteristics of the waveguide.

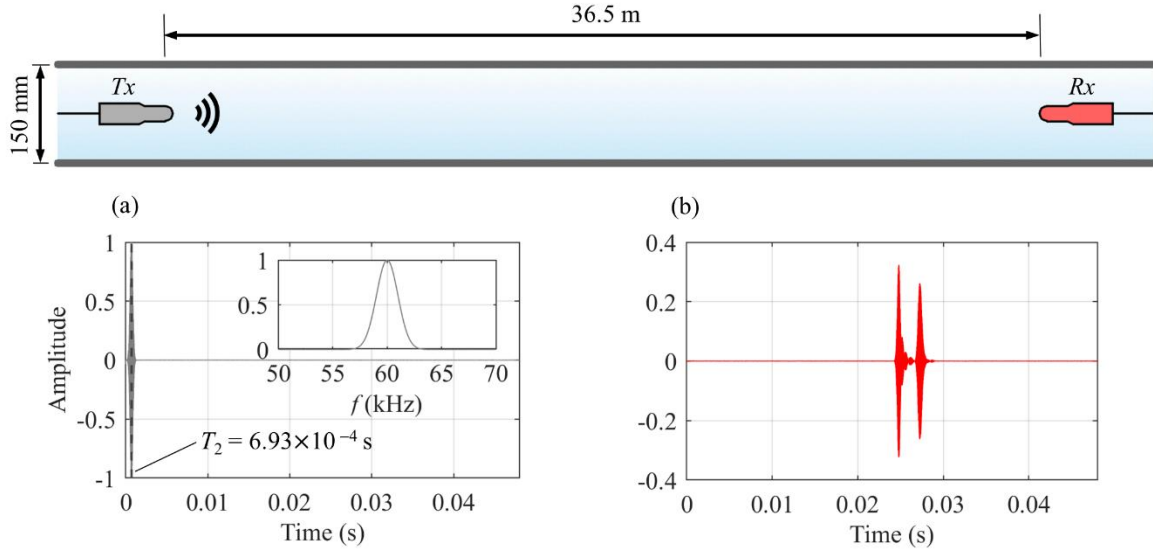


Figure 5.2.7: The experiment is repeated for a much larger waveguide with relative dimensions $x/D = 248$. (a) The source signal is a Gaussian modulated pulse of central frequency $f_c = 60$ kHz, with its frequency content presented in the inset. (b) The system response is acquired at 36.5 m range and comprises of multiple wavepackets.

For the forward test, a Gaussian modulated pulse with central frequency equal to 60 kHz and 20% relative bandwidth is selected to match the frequency response peak of the projector where the transmitting voltage response (TVR) is 125 dB re $\mu\text{Pa}/1 \text{ V @ } 1 \text{ m}$, thus better exploit its capabilities. Note that the increased bandwidth results in a shorter pulse in time as compared to that implemented in the DN80 HDPE pipe setup. In particular, the probing pulse is plotted in Figure 5.2.7a and peaks at time T_2 , while the response of the system is shown in Figure 5.2.7b. Again, the presented waveforms in Figure 5.2.7 are normalized by the source amplitude. The probing pulse reaches the receiver at a later time compared to the DN80 HDPE pipe setup due to the increased range. An estimate for the wavespeed may be calculated from the arrival time of the pulse at the receiver in the order of $c_f \sim 1400 \text{ m/s}$. No reflections are observed past $t \approx 0.03 \text{ s}$; thus, the system is semi-infinite with regards to its boundaries. The response comprises of multiple wavepackets, each associated to an axisymmetric or asymmetric mode. Additional modes in comparison to the DN80 HDPE pipe setup appear, since (i) the probing signal is of smaller wavelength while the lateral dimensions of the waveguide are larger, and (ii) the excited pressure field is asymmetric due to the eccentricity of the source from the pipe centre. Besides, the mode amplitudes relative to the input signal appear reduced compared to the DN80 HDPE pipe setup, possibly due to the increased range and accumulated viscous

losses. In addition, a larger portion of the propagating wave energy is unaccounted for at the receiver in comparison to the DN80 HDPE pipe setup, given that the cross-sectional dimensions are increased while the transducer used is the same. It is, therefore, of great interest to verify time reversal in the presence of multiple modes and under the effect of the overall enhanced attenuation characteristics of the field-scale pipe.

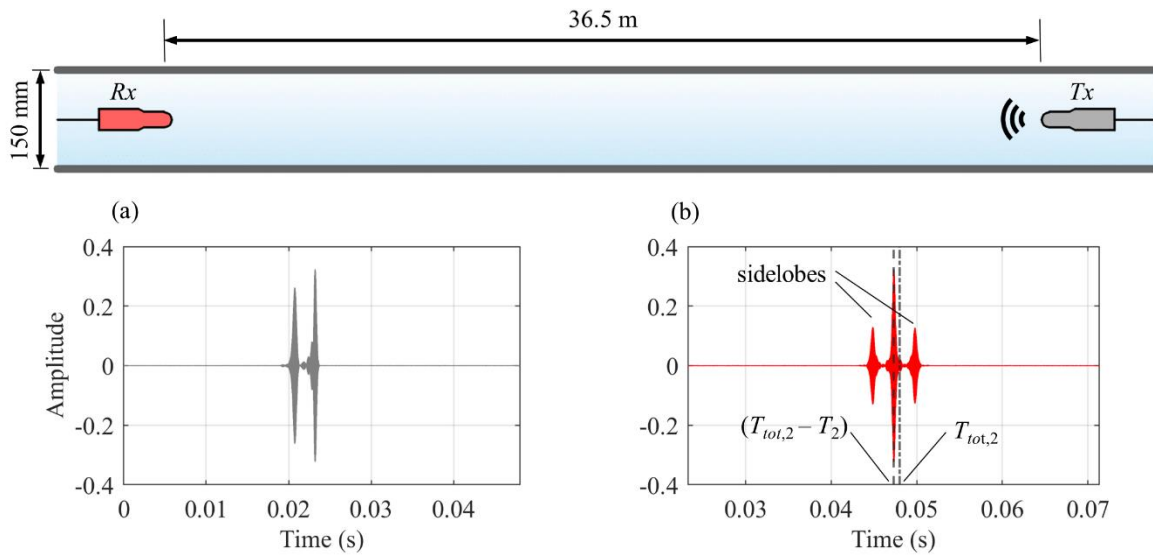


Figure 5.2.8: (a) The response of the DN150 HDPE pipe setup from the forward test is time-reversed and emitted from the location of the original receiver. (b) The original pulse is recreated at the location of the original source and time $t \approx T_{tot,2} - T_2$ displaying sufficient temporal compression.

Figure 5.2.8 depicts the results of the time-reversal step, where the system response from the forward stage (Figure 5.2.7b) is time reversed (Figure 5.2.8a) and re-emitted from the location of the original receiver. Thereafter, the time reversed pressure field is evaluated at the position of the original source. The captured signal (Figure 5.2.8b) reproduces the original Gaussian modulated pulse (Figure 5.2.8a), and peaks at approximately time $t \approx T_{tot,2} - T_2$, with a relative error equal to 359.35 ppm. Evidently, temporal compression is not as accurate as in the DN80 HDPE pipe setup, since the estimated time error is comparatively almost 4 times larger. Furthermore, the normalized amplitude of the recreated pulse for the case of the DN150 HDPE pipe (Figure 5.2.8b) is smaller than that of the DN80 HDPE pipe (Figure 5.2.4b). As already discussed, this is to be expected due to the increased range (with an accompanying increase of viscous losses) and the narrow aperture area of TRM. Also noteworthy is that the sidelobes of the recreated signal in Figure 5.2.8b are far more pronounced than that in the DN80 HDPE pipe

setup (Figure 5.2.4b), possibly because the modes contributing to the formation of the said sidelobes carry more energy (and/or are not attenuated as much). Ultimately, however, the presented experimental results prove the validity of time reversal in an adverse environment that involves (i) viscoelastic boundaries with considerable damping, (ii) a single-channel time reversal mirror of limited size relative to the cross-section of the waveguide, and (iii) semi-infinite boundaries.

5.2.3 Spatial focusing

As discussed in [108], the time-reversal process produces the optimal filter to match the transfer function of the waveguide. The transfer function is defined as the response of the fluid-filled pipe at a point $\mathbf{x}_m$ and time t to an impulse injected at the source location $\mathbf{x}_s$ and $t = 0$ and is expressed through Green's function $G(\mathbf{x}_m, t | \mathbf{x}_s, 0)$. The spatial reciprocity theorem dictates that the occurring pressure field remains unchanged when the source and receiver positions are interchanged, that is: $G(\mathbf{x}_m, t | \mathbf{x}_s, 0) = G(\mathbf{x}_s, t | \mathbf{x}_m, 0)$ [109]. For the time-reversed step, the initial response at $\mathbf{x}_m$ is time-reversed and used as a source. In that case, the response of the waveguide at a point $\mathbf{x}$ is given by:

$$h(\mathbf{x}, t) = G(\mathbf{x}, t | \mathbf{x}_m, 0) \otimes G(\mathbf{x}_m, -t | \mathbf{x}_s, 0) = \int_{-\infty}^{\infty} G(\mathbf{x}, \tau | \mathbf{x}_m, 0) G(\mathbf{x}_m, t + \tau | \mathbf{x}_s, 0) d\tau \quad (5.2.1)$$

In Eq. (5.2.1), the Green's function of the waveguide is convolved with the time-reversed signal. The maximum response is $\int_{-\infty}^{\infty} G^2(\mathbf{x}_s, \tau | \mathbf{x}_m, 0) d\tau$ and occurs at $\mathbf{x} = \mathbf{x}_s$ and $t = 0$.

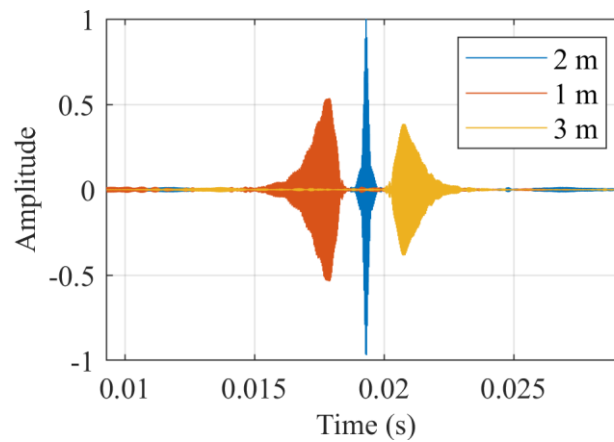


Figure 5.2.9: The time reversed pressure field is evaluated at multiple locations along the DN80 HDPE pipe at the HKUST hydraulic lab. Here, the waveform at 3 locations is

presented: 2 m (in blue), 1 m (in orange), and 3 m (in yellow) away from the time reversal source. The signal compresses and obtains its maximum amplitude 2 m from the time reversed source, coinciding with the location of the original source.

The experimental results introduced so far merely demonstrate the time compression properties of the time reversal process. It is also important to validate that the time reversed signal refocuses only at the location of the initial source. This time reversal attribute is particularly important for pulse-echo detection methods where the objective is to uniquely identify a target e.g., pipe leak or blockage. Here, spatial focusing is tested experimentally in the DN80 HDPE pipe setup at the HKUST hydraulic lab (Figure 5.2.1) by performing several measurements at wavelength multiples along the pipe length, where $x \in [1,3]$ m. To intensify the effects of dispersion in the limited available range of the setup, a signal of wider bandwidth than the previous tests is chosen where $f_c = 35$ kHz and $b = 0.6$. By doing so, the 2nd radial mode “spreads” more in time compared to the response observed in Figure 5.2.3b and time compression at the location of the original source is emphasized. In particular, the forward test is performed first, placing the transducer pair at 2 m distance. Then, the transducer roles are interchanged but instead of keeping the distance fixed at 2 m, the time reversal receiver is moved longitudinally along the pipe. Figure 5.2.9 presents a selection of 3 measurements at 1, 2 and 3 m away from the time reversal source. It is evident that the time reversed pressure field is time compressed and maximized only at the location of the original source i.e., 2 m away from the source.

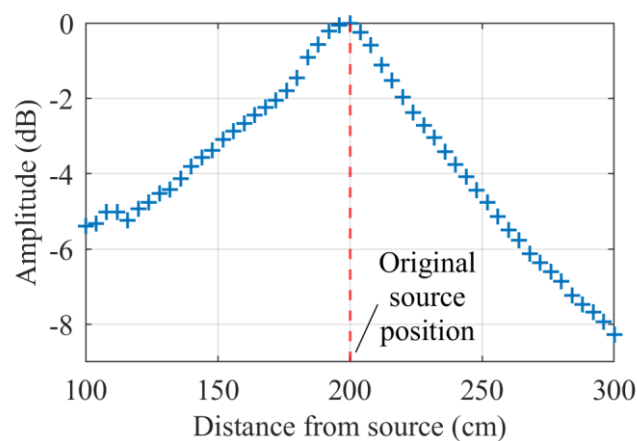


Figure 5.2.10: Multiple measurements are performed along the DN80 HDPE pipe of the HKUST hydraulic lab at wavelength multiples to demonstrate spatial focusing. Maximum amplitude for the time reversed pressure field is observed at the location of the original source (at 200 cm).

Figure 5.2.10 presents the maximum amplitude of the time reversed response at additional locations. Still, the peak of the plot is observed 2 m away from the time reversal source, coinciding with the location of the original source, thus verifying the spatial focusing properties of the time reversed pressure field. It should be noted that measurement intervals smaller than λ do not yield substantial differences due to the diffraction limit. Similar results were demonstrated in [95] for a bounded waveguide. However, unlike [95], herein: (i) the boundaries are viscoelastic instead of steel, thus, manifesting damping, (ii) the relative dimensions of the current waveguide are significantly larger, and (iii) a single-channel time reversal mirror is used. In the next section, some insight is provided into the reasons time reversal is successful in the presented experiments despite the presence of viscous damping in both the fluid and pipe wall.

5.3 Time reversal invariance in a viscous waveguide through scaling analysis

Wave propagation is robust to time reversal, but particle motion is not [110]. In fact, it is the irreversibility of particle motion that prevents, say, a broken egg from becoming whole again by reversing the velocity of its shattered pieces. Waves, on the other hand, are found to be time reversible as exemplified by the experiments of our own and others (to be discussed in following sections of this thesis). Therefore, when contemplating time reversal, it is important to distinguish whether the process one intends to reverse relies on *particle motion* or *wave propagation*. It is possible that time reversal is often dismissed altogether because it is thought in the context of particles and not waves.

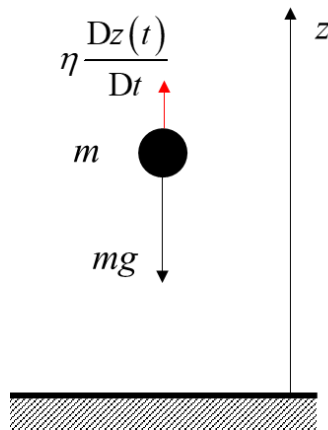


Figure 5.3.1 The free-body diagram of a particle in free-fall motion.

Why is the motion of particles not time reversible, but the propagation of waves is? The two mechanisms that limit or even prevent time reversibility are damping and stability [111]. To understand the effect of damping on limiting the time reversibility of particle motion, consider a mass m in free fall (Figure 5.3.). The forces on this particle are gravity and air resistance. Solving for the particle vertical position z as a function of time gives:

$$z(t) = \frac{m^2 \mathbf{g}}{\eta_{air}^2} \left[1 + \frac{\eta_{air} t}{m} - e^{-\frac{\eta_{air} t}{m}} \right] = -\frac{1}{2} \mathbf{g} t^2 - \frac{\eta_{air} t^3}{m} \frac{\mathbf{g}}{3!} + \dots \quad (5.3.1)$$

where η_{air} is a parameter that characterizes air resistance, and $\mathbf{g}$ is the acceleration of gravity. Applying the transformation $t \rightarrow -t$ gives:

$$z(-t) = \frac{m^2 \mathbf{g}}{\eta_{air}^2} \left[1 - \frac{\eta_{air} t}{m} - e^{-\frac{\eta_{air} t}{m}} \right] = -\frac{1}{2} \mathbf{g} t^2 + \frac{\eta_{air} t^3}{m} \frac{\mathbf{g}}{3!} + \dots \quad (5.3.2)$$

Although $z(t) \neq z(-t)$, it is clear that the difference between the forward and time reversed paths is negligible when $(\eta t^3 g / 6m) / (gt^2 / 2) \ll 1$. That is, in the presence of friction, time reversibility approximately holds if:

$$TR = \frac{\eta t}{3m} \ll 1 \quad (5.3.3)$$

where TR is a time reversibility parameter. Equation (5.3.3) points out the important fact that the relative effect of damping grows linearly in time ($\propto t$). That is, a particle loses its ability to retrace its forward path with time in a linear manner. The elapsed time t for which time-reversibility is expected to be lost is $\eta t / 3m \sim 1$ or larger, which leads to $t \sim 3m / \eta$ or larger. We will show later in the manuscript that an analogous expression to Eq. (5.3.3) arises when we study the time reversibility of the Navier-Stokes and the viscoelastic pipe wall equations.

Now the effect of instability on the time reversal of particle motion is introduced. Consider a cloud of interacting particles. At time $t = T_{tot}$, the position $\mathbf{x}_i(T_{tot})$ and velocity $\mathbf{v}_i(T_{tot})$ of each particle i is recorded and the particles are captured. What is the path of the captured particles if each is released with initial velocity equal to $-\mathbf{v}_i(T_{tot})$ (i.e., velocity is time reversed) plus a small perturbation? Snieder [111] showed that the perturbed system is unstable in the sense that the path traced by the particles in the reversed stage diverges exponentially in time from the path traced in the forward stage (i.e., $\exp(\beta t)$, where β is parameter that depends on length scale and particle density). Note that the contribution of instabilities grows exponentially in

time, while that of damping only linearly. Therefore, it is this instability that prevents the time reversibility of particles.

Regarding the time reversibility of the waves, Snieder & Scales [110] proved theoretically and numerically that waves are far more stable than particles to time reversal. Consider n scatterers situated between the wave source and receiver. The time reversed (phase conjugated) pressure field has the following form:

$$p_{tr}(\mathbf{r}) \sim \frac{\sin k_{\max} [|\mathbf{r}_R - \mathbf{r}_S| - |\mathbf{r} - \mathbf{r}_R|]}{|\mathbf{r}_R - \mathbf{r}_S| - |\mathbf{r} - \mathbf{r}_R|} + \sum_{i=1}^n \alpha_i \frac{\sin \lambda_{\min} [|\mathbf{r}_R - \mathbf{r}_i| + |\mathbf{r}_i - \mathbf{r}_S| - |\mathbf{r}_R - \mathbf{r}_i'| - |\mathbf{r}_i' - \mathbf{r}|]}{|\mathbf{r}_R - \mathbf{r}_i| + |\mathbf{r}_i - \mathbf{r}_S| - |\mathbf{r}_R - \mathbf{r}_i'| - |\mathbf{r}_i' - \mathbf{r}|} \quad (5.3.4)$$

where p_{tr} is the time reversal pressure field value, α_i is a measure of the scattering strength of scatterer i , $\mathbf{r}_i$ and $\mathbf{r}_i'$ are the unperturbed and perturbed locations of the i^{th} scatterer, $\mathbf{r}_S$ and $\mathbf{r}_R$ are the locations of the source and receiver during the forward propagation stage, while $\mathbf{r}$ is the receiver location during the time reversal stage. Moreover, $\lambda_{\min}$ is the wavelength associated with the largest frequency in the bandwidth of the probing signal. Equation (5.3.4) describes the time reversed pressure field as a superposition of the unperturbed wave that follows a direct path from the time reversal source at $\mathbf{r}_R$ to the measurement location $\mathbf{r}$, and the scattered waves that track an indirect path from $\mathbf{r}_R$ to the scatterer locations $\mathbf{r}_i$ and then to $\mathbf{r}$. The former path is expressed in the first term, while the latter in the second term on the right-hand side of Eq. (5.3.4). Let $\mathbf{r}_i' - \mathbf{r}_i = \mathbf{e}_i$ be the perturbation of the location of the i^{th} scatterer. Using the fact that $\sin x/x \rightarrow 1$ as $x \rightarrow 0$, the time reversed wave refocuses at the source $\mathbf{r}_S$ when $\mathbf{r} = \mathbf{r}_S$ (i.e., at the original source location), and $\mathbf{e}_i = 0$. What is the effect to time reversal when the scatterers are perturbed from their forward propagation positions? We now consider the time reversal field at $\mathbf{r} = \mathbf{r}_S$ when $\mathbf{e}_i$ is a random perturbation. Taylor's series expansion leads to:

$$p_{tr}(\mathbf{r} = \mathbf{r}_S) \sim 1 - \frac{1}{3!} \sum_{i=1}^n \alpha_i k_{\max}^2 |\mathbf{e}_i|^2 \quad (5.3.5)$$

The rate of degradation of time reversal due to a perturbation error $|\mathbf{e}_i|$ is:

$$\frac{\partial p_{tr}}{\partial |\mathbf{e}_i|} \sim \frac{\alpha_i}{3} k_{\max}^2 |\mathbf{e}_i| \quad (5.3.6)$$

The standard deviation of the refocusing degradation is:

$$\left[\sum_{i=1}^n E \left[\left(\frac{\partial p_{tr}}{\partial |\mathbf{e}_i|} \right)^2 \right] \right]^{0.5} \sim \frac{\sqrt{n}}{\sqrt{3}} k_{\max} \sigma_0 \sqrt{\sum \alpha_i^2} \quad (5.3.7)$$

The number of scatterers a wave meets during time t is [110]:

$$n \sim \frac{ct}{d} \quad (5.3.8)$$

where c is the wave speed, and d is the average distance between scatterers. As a result, the degradation of the wave refocusing is given by the following dimensionless standard deviation:

$$\sigma_{tr}(t) = \sqrt{\sum_{i=1}^n \alpha_i^2} \frac{2\pi f_{\max}}{c} \sigma_0 \sqrt{\frac{ct}{d}} \quad (5.3.9)$$

which shows that the degradation of the quality of the time reversal refocusing grows as $(t^{1/2})$. This result concurs with the theoretical and experimental findings of Snieder [110]. The following statement from Snieder & Scales [111] succinctly contrasts the time reversibility of particle and waves vis-à-vis instabilities:

“Physically there are two reasons that explain this difference [between time reversibility of particles and waves]. The particles are point-like objects with no scale, whereas the waves are associated with a wavelength. The wave field “feels” its environment on a scale that only depends on the wavelength; hence the wavelength determines a natural scale for the sensitivity of scattered waves for perturbations of the scatterer locations. In addition, the waves travel along all possible scattering paths, whereas the particles each travel along a unique path. When the scatterer locations are perturbed, the waves still travel along all possible scattering paths, whereas a particle may suddenly follow a fundamentally different trajectory.”

We proceed to investigate the effect of damping on the time reversibility of waves, where the pertinent viscous equations of motion for the waveguide are now considered. The point form of the momentum conservation equation for the fluid is (Eq. (2.2.70)):

$$\rho_0(t) \frac{d^2 \mathbf{r}(t)}{dt^2} + \rho_0(t) \frac{d\mathbf{r}(t)}{dt} \cdot \nabla \frac{d\mathbf{r}(t)}{dt} + \nabla p(t) - \nabla \cdot \mathbf{\tau}(t) + \rho_0(t) \mathbf{g} = 0 \quad (5.3.10)$$

where $\mathbf{\tau}$ is the viscous stress tensor, p is the fluid pressure, and $\mathbf{g}$ is the acceleration of gravity. Note that for clarity and ease of presentation of the time reversal concept we follow the tradition

of writing the equation in terms of position $\mathbf{r}(t)$, and not velocity $\mathbf{v}(t)$ as in Eq. (2.2.70). Also, 0 corresponds to a zero vector. The general expression of the viscous stress forces $\nabla \cdot \mathbf{\tau}$ for a compressible Newtonian fluid is:

$$\nabla \cdot \mathbf{\tau}(t) = \mu \left\{ \nabla^2 \frac{d\mathbf{r}(t)}{dt} + \frac{1}{3} \nabla \left[\nabla \cdot \frac{d\mathbf{r}(t)}{dt} \right] \right\} + \kappa \left\{ \nabla \left[\nabla \cdot \frac{d\mathbf{r}(t)}{dt} \right] \right\} \quad (5.3.11)$$

where μ , κ are the dynamic and bulk fluid viscosities, respectively. As regards to the pipe wall viscous equations, the argument is limited to the radial momentum component (Eq. (2.3.142)) for brevity:

$$\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2} = \frac{\partial S(t)}{\partial x} - \frac{1}{J(t)} \frac{h}{(1-\nu_p^2)} \left(\frac{w_p(t)}{R^2} + \frac{\nu_p}{R} \frac{\partial u_{p,0}(t)}{\partial x} \right) + p(t) \quad (5.3.12)$$

where w_p , $u_{p,0}$ are the radial (r) and axial (x) displacements of the centroid of a pipe wall element, respectively, R is the pipe radius, while ρ_p is the density, and ν_p is Poisson's ratio of the pipe wall material. Note that the fluid pressure $p(t)$ at the pipe/fluid interface is inserted in Eq. (5.3.12) as forcing to the pipe wall. In addition, $S(t)$ represents the shear forces developing in the pipe wall per unit length (Eq. (2.3.145)), given by:

$$S(t) = \frac{Kh}{2J(t)(1+\nu_p)} \left[\frac{\partial w_p(t)}{\partial x} + \alpha(t) \right] \quad (5.3.13)$$

where K is the correction factor for the distribution of the shear forces across the pipe thickness, and α is the angular rotation of a radial line about an axis that crosses the pipe wall element centroid and is normal to the xr -plane (x is along the length of the pipe, and r along the pipe radius). Conversely to Chapter 2, pipe wall equations of motion in this chapter are written in terms of $J(t)$ that corresponds to the time dependent creep compliance, assumed to follow the Generalized Kelvin-Voigt (GKV) model. Accordingly,

$$J(t) = J_0 + \sum_{j=1}^{N_{kv}} J_j \left[1 - \exp\left(\frac{-t}{\tau_j}\right) \right] \quad (5.3.14)$$

where N_{kv} is the number of GKV elements, each described by a creep coefficient J_j and relaxation time τ_j . Next, the time reversal transformation of the Eqns. (5.3.10) to (5.3.14) is performed. For the interested reader, the time reversal transformation for the full set of

equations of motion for the pipe wall, as well as for the inviscid waveguide equations are provided in the Appendix (B).

The time reversal transformation $t \rightarrow -t$ for the viscous fluid momentum conservation equation (Eq. (5.3.10)) is:

$$\rho_0(-t) \frac{d^2 \mathbf{r}(-t)}{d(-t)^2} + \rho_0(-t) \frac{d\mathbf{r}(-t)}{d(-t)} \cdot \nabla \frac{d\mathbf{r}(-t)}{d(-t)} + \nabla p(-t) - \nabla \cdot \boldsymbol{\tau}(-t) + \rho_0(-t) \mathbf{g} = 0 \quad (5.3.15)$$

Considering $d^2/d(-t)^2 = d^2/dt^2$, and $d/d(-t) = -d/dt$, Eq. (5.3.15) becomes:

$$\rho_0(-t) \frac{d^2 \mathbf{r}(-t)}{d(-t)^2} + \rho_0(-t) \frac{d\mathbf{r}(-t)}{d(-t)} \cdot \nabla \frac{d\mathbf{r}(-t)}{d(-t)} + \nabla p(-t) - \nabla \cdot \boldsymbol{\tau}(-t) + \rho_0(-t) \mathbf{g} = 0 \quad (5.3.16)$$

It is interesting to note that the nonlinear operator term in Eq. (5.3.16) which is responsible for the fluid turbulence retains its form; thus, it is time reversible. This fact has also been experimentally demonstrated in [112] where the mixing between two fluids of different colour was undone by performing the opposite, time reversed, motion. On the other hand, the form of the viscous term in the time reversal stage is:

$$\begin{aligned} \nabla \cdot \boldsymbol{\tau}(-t) &= \mu \left\{ \nabla^2 \frac{d\mathbf{r}(-t)}{d(-t)} + \frac{1}{3} \nabla \left[\nabla \cdot \frac{d\mathbf{r}(-t)}{d(-t)} \right] \right\} + \kappa \left\{ \nabla \left[\nabla \cdot \frac{d\mathbf{r}(-t)}{d(-t)} \right] \right\} \\ &= -\mu \left\{ \nabla^2 \frac{d\mathbf{r}(-t)}{dt} + \frac{1}{3} \nabla \left[\nabla \cdot \frac{d\mathbf{r}(-t)}{dt} \right] \right\} - \kappa \left\{ \nabla \left[\nabla \cdot \frac{d\mathbf{r}(-t)}{dt} \right] \right\} = -\nabla \cdot \boldsymbol{\tau}(t) \end{aligned} \quad (5.3.17)$$

Combining Eqns. (5.3.16) and (5.3.17) gives:

$$\rho_0(-t) \frac{d^2 \mathbf{r}(-t)}{dt^2} + \rho_0(-t) \frac{d\mathbf{r}(-t)}{dt} \cdot \nabla \frac{d\mathbf{r}(-t)}{dt} + \nabla p(-t) + \nabla \cdot \boldsymbol{\tau}(-t) + \rho_0(-t) \mathbf{g} = 0 \quad (5.3.18)$$

which differs in form from Eq. (5.3.10) due to the change in sign of the viscous term, $\nabla \cdot \boldsymbol{\tau}(t)$. Thus, the viscous fluid flow equations are not time reversible.

The momentum equation along the radial coordinate for a viscous pipe wall element under time reversal transformation ($t \rightarrow -t$) becomes:

$$\rho_p(-t) h \frac{\partial^2 w_p(-t)}{\partial(-t)^2} = \frac{\partial S(-t)}{\partial x} - \frac{1}{J(-t)} \frac{h}{(1-\nu_p^2)} \left(\frac{w_p(-t)}{R^2} + \frac{\nu_p}{R} \frac{\partial u_{p,0}(-t)}{\partial x} \right) + p(-t) \quad (5.3.19)$$

with

$$S(-t) = \frac{Kh}{2J(-t)(1+\nu_p)} \left[\frac{\partial w(-t)}{\partial x} + \alpha(-t) \right] \quad (5.3.20)$$

Since $\partial^2 \cdot / \partial (-t)^2 = \partial^2 \cdot / \partial t^2$, Eq. (5.3.19) becomes:

$$\rho_p(-t)h \frac{\partial^2 w_p(-t)}{\partial t^2} = \frac{\partial S(-t)}{\partial x} - \frac{1}{J(-t)} \frac{h}{(1-\nu_p^2)} \left(\frac{w_p(-t)}{R^2} + \frac{\nu_p}{R} \frac{\partial u_{p,0}(-t)}{\partial x} \right) + p(-t) \quad (5.3.21)$$

First, $p(-t) \neq p(t)$ since the viscous flow equations are not time reversible as can be seen by comparing Eq. (5.3.10) and Eq. (5.3.18). In addition, the creep function $J(t)$ (Eq. (5.3.14)) under time reversal is expressed as:

$$J(-t) = J_0 + \sum_{j=1}^{N_{kv}} J_j \left[1 - \exp\left(-\frac{t}{\tau_j}\right) \right] \quad (5.3.22)$$

Hence, the exponential decay in Eq. (5.3.14) gives way to an exponential growth in Eq. (5.3.22) and, as such, $J(-t) \neq J(t)$. Consequently, the differential operator that governs $\mathbf{r}(t)$, $u_{p,0}(t)$, $w_p(t)$, $\alpha(t)$, $\rho_0(t)$, $\rho_p(t)$, $S(t)$, and $p(t)$ is dissimilar to that which governs $\mathbf{r}(-t)$, $u_{p,0}(-t)$, $w_p(-t)$, $\alpha(-t)$, $\rho_0(-t)$, $\rho_p(-t)$, $S(-t)$, and $p(-t)$. This breakdown of time reversal is due to the viscous terms appearing in both the fluid and pipe wall equations. Nonetheless, the experiments presented in the preceding section demonstrate that acoustic waves propagating in water filled viscoelastic pipes are time reversible over a frequency bandwidth ranging from 24.5 to 66 kHz. In addition, time reversal has been verified in the lab and field in many systems where damping is present (e.g., gravity waves and acoustic waves in the ocean which are discussed later in the manuscript). How can this apparent contradiction between theory and experiment be reconciled?

A plausible explanation requires that the viscous terms in the governing equations be much smaller than the pertinent temporal acceleration terms. To investigate such hypothesis, we perform a scale analysis for both viscous fluid and pipe wall equations of motion. Consider a disturbance in the fluid medium that generates a propagating wave of length scale λ . Commencing with the fluid domain, the ratio of the viscous forces term to the temporal acceleration term i.e., the inverse Reynolds number $1/\text{Re}$ is:

$$TR_f = \frac{\nabla \cdot \boldsymbol{\tau}}{\rho_0(t) \frac{\partial \mathbf{v}(t)}{\partial t}} \sim \frac{v \frac{\|\mathbf{v}\|}{\lambda^2}}{\frac{\|\mathbf{v}\|}{t}} = v \frac{t}{\lambda^2} \approx \frac{1}{\text{Re}} Ma \frac{t}{T} \quad (5.3.23)$$

where $v (= \mu/\rho_0)$ is the kinematic viscosity, $T = \lambda/c_f$, t/T represents the dimensionless elapsed time from the moment the wave disturbance is introduced, c_f is the wave speed in the fluid, and Ma is the Mach number. Equation (5.3.23) defines the time reversibility parameter of waves in the fluid domain. Note that the well-known fact since the mid-19th century that viscous effects grow with frequency [113] is evident in Eq. (5.3.23).

For the case of the pipe wall, there are in total three components of the momentum equation. A complete scale analysis is given in the Appendix (B.2); here, the discussion is limited to the most unfavourable scenario (i.e., the component of the momentum equation with dominant viscosity). Such case is recovered from the radial momentum component (Eq. (B.2.4)) when the ratio of the viscous radial component of the hoop stresses to the pertinent inertial term is considered. The ratio of damping to inertia over time t in the pipe wall obtained from Eq. (B.2.4) is:

$$\frac{\sum_{j=1}^{N_{kv}} \frac{J_j}{J_0^2} \left(\frac{t}{\tau_j} \right) \frac{h}{(1-\nu_p^2)} \frac{w_p(t)}{R^2}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim \frac{t}{t_v J_0 f^2 R^2 \rho_p (1-\nu_p^2)} = TR_p \quad (5.3.24)$$

Equation (5.3.24) defines the time reversibility parameter of waves propagating in the pipe wall, TR_p . Moreover, Eq. (5.3.24) introduces the pipe viscous time scale t_v , that is retrieved from the GKV model expression and corresponds to the time scale viscous effects in the pipe wall become significant (Eqn. (B.1.13)). Unlike Eq. (5.3.23), Eq. (5.3.24) shows that the viscous damping from the pipe wall diminishes at high frequencies. This is due to the fact that pipe viscous damping is only significant when the time scale of the wave is of the same order of magnitude to the time scale of the viscous damping and reduces quickly when these time scales become different [114].

The relative order of magnitude of the viscous force to temporal acceleration terms for the combined fluid-pipe wall system involves adding Eq. (5.3.23) to (5.3.24) and the result is:

$$t \frac{v f^2}{c_f^2} + \frac{t}{t_v J_0 f^2 R^2 \rho_p (1-\nu_p^2)} = TR \quad (5.3.25)$$

where TR is the time reversibility parameter for the whole system (i.e., fluid and pipe wall). It is chosen to add the viscous effect from the two domains instead of taking the maximum, since the former is a more unfavourable scenario. Equation (5.3.25) shows that the contribution of damping towards the time irreversibility of waves in a viscoelastic pipeline grows linearly with time t . In contrast, the contribution of the instability to the wave propagation and, thus, its time irreversibility grows only as $(t^{1/2})$ [111]. Note that the time scale here is defined as the ratio of the elapsed time from the initiation of the disturbance over the wave period and, for high frequency waves, $t/T \gg 1$. Therefore, the reversibility (or irreversibility) of waves is governed by damping. The ratio of the viscous to temporal acceleration terms can also be written in terms of wave energy ratio as follows:

$$TR \sim \ln \frac{E_a(t)}{E_{a,0}} \sim \ln \frac{p^2(t)}{p_0^2} \quad (5.3.26)$$

where $E_{a,0}$ is the energy of the wave at the source, and $E_a(t)$ is the energy of the wave after it has propagated over time t . The scaling can be extended to pressure amplitudes since $E_a \propto p^2$.

5.4 Discussion of time reversal experiments

The time reversal experiments presented in Section 5.2 are now reviewed in light of the time reversal parameter, TR . An elapsed time $t = x/c$ is considered, where x is the distance between the transducer set in the DN150 HDPE pipe setup ($x = 36.5$ m). From Figure 5.2.7b we obtain $t \sim 2.5 \times 10^{-3}$ s; thus, the wavespeed $c_f \sim 1.5 \times 10^3$ m/s. According to the experiments presented in Section 5.2.2, the probing wave frequency $f \sim 10^5$ Hz. Since there is no flow, it is not necessary to use a turbulent viscosity value; thus, we apply the kinematic water viscosity $\nu \sim 10^{-6}$ m²/s. The viscoelastic parameters of the GKV elements, J_j and τ_j , are adopted from Wang, et al. [115] who experimentally calibrated these parameters for three GKV elements ($j = 3$) based on a setup that employs an identical HDPE pipe to the one presented in our experiments. Thus, the same values are employed (Appendix, Table B.1.1) to obtain $t_v \sim 10^{-1}$ s. The remaining pipe mechanical parameters (e.g., density, Young's modulus) for the HDPE pipes can be found in Chapter 4, Table 4.2.1. Substituting these values in Eq. (5.3.23) yields $TR_f = 10^{-5}$, while Eq. (5.3.24) returns $TR_p = 10^{-2}$.

The combined effect of pipe and fluid viscosity given by Eq. (5.3.25) is $TR = 10^{-2} + 10^{-5} \cong 10^{-2}$; thus, revealing that even in the wider range of our DN150 HDPE pipe

setup, the temporal acceleration terms are still some orders of magnitude larger than the pertinent viscous terms in both the fluid and HDPE pipe wall domains. Note that the viscous losses in the HDPE pipe are comparatively more significant than the viscous losses in the water, as demonstrated by the three-order difference between $TR_p = 10^{-2}$ and $TR_f = 10^{-5}$.

Now that we confirmed that viscous effects are small in the range x of the experiments prescribed in Section 5.2, a natural question is: What is the range $x = ct$ where viscous terms become significant so that time reversal breaks down? At said range, the viscous terms are comparable to the temporal acceleration terms (i.e., $TR \sim 1$). Setting the right-hand side of Eq. (5.3.25) to 1 and solving for x gives:

$$x = \frac{c_f^3 c_p t_v J_0 f^2 R^2 \rho_p (1 - \nu_p^2)}{\nu f^4 c_p t_v J_0 R^2 \rho_p (1 - \nu_p^2) + c_f^3} \quad (5.4.1)$$

The pertinent parameter values from the experimental setups are substituted in Eq. (5.4.1), and the estimated achievable range $x \sim 110$ m. Therefore, from Eq. (5.3.26) we may estimate that pipe wall viscosity in our experiment reduces the energy of the wave by about 60% every 110 m. This distance, being three times 36.5 m (i.e., the distance between the projector and receiver in our experiment), further supports the experimental findings that high-frequency waves ($f \sim 10^5$ Hz) are time reversible.

As mentioned previously, the water in the experimental setups is quiescent. In practice, however, water supply flows are generally turbulent. To take such situation into account, the eddy viscosity ν_t is used instead of the kinematic viscosity ν in estimating the time reversal parameter. Experimental measurements on fully developed turbulent flows in pipes [116] indicate a dimensionless eddy viscosity $\nu_t/(V^*R)$ of at most 0.08. In the expression of the dimensionless eddy viscosity, V^* is the friction velocity defined as $V^* = \bar{V} (f_{DW}/8)^{1/2}$ with $\bar{V}$ being the cross-sectional average flow velocity, and f_{DW} is the Darcy friction factor. Assuming a typical roughness factor for the DN150 HDPE pipe $k = 1.5 \times 10^{-6}$ m and an extreme case for the cross-sectional average flow velocity $\bar{V} = 3$ m/s, the Colebrook-White equation is used to estimate $f_{DW} = 0.0136$. The end result is $V^* = 0.12$ m/s and $\nu_t = 1.45 \times 10^{-3}$ m²/s. The calculated eddy viscosity is several orders of magnitude larger than the kinematic viscosity ν already considered. The combined effect of turbulence and pipe wall viscosities reduces the energy of the wave by about 60% every 80 m when the frequency of the wave source is $\sim 10^5$ Hz. That is, the turbulence damping reduces the range from 110 m to 80 m (i.e., 30 m range reduction).

An expression for the pressure wave energy decay rate in a viscoelastic pipe is derived in the Appendix (B.3) for the interested reader.

The range calculated in Eq. (5.4.1) for our experimental setup is dominated by the viscous losses in the pipe wall. However, if the pipe walls were elastic, the achievable range would only be limited by the flow viscosity resulting in $x \sim 33$ km (using the fluid viscosity ν for quiescent water). In particular, the experiments we conducted test time reversibility under the most extreme condition. That is, proving that waves in HDPE pipes are time reversible guarantees that such waves are also time reversible over a wide range of other pipe materials, such as ductile iron, steel, and concrete.

The presented analysis confirms that the viscous terms in both the fluid and pipe wall equations of motion are small when compared to the appropriate temporal acceleration terms, when considering the characteristic scales of our experimental setups. Thus, we show that contrary to the apparent invalidity of the equations of motion when $t \rightarrow -t$ due to the presence of the viscous terms, time reversal is practically feasible provided that the dimensionless parameter $TR \ll 1$. Moreover, the TR parameter can be applied on a plethora of environments to estimate the effect of viscosity to wave propagation. To our knowledge, the presented work is the first-ever direct proof of the time reversibility of pressure waves in viscoelastic pipes. As noted by Kuperman [117]:

“An interesting take on time reversal is that when time reversal works, it can be thought of as a kind of proof of existence that the medium supports the localization goal of beamforming (BF) and/or matched field processing (MFP)”.

The developed scales have important practical implications. For example, to achieve a successful defect detection in pipelines using the MFP method, it is recognized that a Signal-to-Noise Ratio (SNR) at least equal to -5 dB is necessary [93]. Knowledge of the minimum SNR for defect detection in a particular system allows the analyst to determine the required strength of the probing signal for a desired range x . Alternatively, given the power of the source, the maximum range for successful defect detection can be estimated. For the purpose, the derived scale of viscous over the inertial forces can be applied. Equation (5.3.26) is rewritten and time is set $t = x/c$ to obtain:

$$x \left[\frac{\nu f^2}{c_f^3} + \frac{1}{c_p t_v J_0 f^2 R^2 \rho_p (1 - \nu_p^2)} \right] = C \ln \frac{E_a(x)}{E_{a,0}} \sim C \ln \frac{p^2(x)}{p^2(0)} \quad (5.4.2)$$

where C is a proportionality factor that could be estimated from experimental results. The right-hand side of Eq. (5.4.2) may be expressed in terms of SNR as:

$$C \ln \frac{p^2(x)/n^2}{p_0^2/n^2} = C \frac{\ln(10)}{10} [\text{SNR}(x) - \text{SNR}(0)] \quad (5.4.3)$$

where n corresponds to the system noise, and $\text{SNR}(x) = 10 \log_{10}(p^2(x)/n^2)$. Thus, the relationship between the SNR at the source, range x , and the desired SNR at x is written as:

$$x = \frac{C \ln(10)}{10 \left[\frac{\nu_t f^2}{c_f^3} + \frac{1}{c_p t_v J_0 f^2 R^2 \rho_p (1 - \nu_p^2)} \right]} [\text{SNR}(x) - \text{SNR}(0)] \quad (5.4.4)$$

As an example, the pressure measurements from two different locations x_1 and x_2 along the DN80 HDPE pipe (see Section 5.2.3) are used to establish $C \sim -0.013$. For the source levels used in this setup, $\text{SNR}(0) \sim 31$ dB. Thus, if we target a $\text{SNR}(x) = -5$ dB (i.e., the minimum required for defect detection using long waves), the range $x \sim 12$ m. The measurements and calculations are repeated for the DN150 pipe setup to retrieve a proportionality factor $C \sim -0.02$ and $\text{SNR}(0) \sim 24$ dB. For a $\text{SNR}(x) = -5$ dB, the estimated range is $x \sim 170$ m. In Section 5.2.3, experimental results from multiple measurements along the DN80 HDPE pipe show an effective attenuation for the propagating modes in the frequency bandwidth of about 3 dB/m. In contrast, similar measurements in the DN150 HDPE pipe setup show an effective attenuation of about 0.2 dB/m. The reader is reminded that the excited frequency bandwidth in the latter pipe setup is higher (about double) than in the former. Equation (5.3.24) reveals that viscous forces in the pipe wall are inversely proportional to the probing frequency f . Also remarkable is the fact that pipe radius R appears in the denominator of Eq. (5.3.24), denoting that viscous losses are decreased for the case of the larger pipe. Therefore, the reduced attenuation in the DN150 HDPE pipe (in comparison to the DN80 HDPE pipe) could be attributed to both the higher frequency f of the probing signal, as well as the increased pipe radius R . Note that if the target $\text{SNR}(x)$ is increased (for a given source strength), the estimated range x is reduced.

Equation (5.4.4) is also applicable to hydraulic systems where the conduits are elastic by simply omitting the second term in the denominator that accounts for the viscous pipe damping:

$$x = \frac{C \ln(10)}{10 \left(\frac{\nu_t f^2}{c_f^2} \right)} [\text{SNR}(x) - \text{SNR}(0)] \quad (5.4.5)$$

Equation (5.4.5) shows that the range of waves in elastic pipes is inversely proportional to probing frequency f . For this reason, low frequency pressure transients have been successfully employed in pipeline defect detection over long distances. For example, [118] performed low-frequency ($f \sim 10$ Hz) transient tests by pump shutdown in a real 1322 m steel transmission line in Trento, Italy to identify a simulated leak. The authors reported that the transmission line was a steel DN500 pipe, with $\bar{V} = 0.2$ m/s, and estimated roughness $k = 8 \times 10^{-4}$ m. Hence, an applicable eddy viscosity is estimated in the order of $\nu_t \sim 10^{-4}$ m²/s. Author-provided data from initial transient tests on the intact system indicate a $\text{SNR}(0) \sim 51$ dB at the source and are used to obtain a proportionality factor $C = -3.1 \times 10^{-3}$. Now, assuming a range x equal to the overall transmission pipe length ($x = 1322$ m), Eq. (5.4.5) estimates the $\text{SNR}(x) \sim 51$ dB. The calculated $\text{SNR}(x)$ is much larger than the minimum of -5 dB required for defect detection in pipes, which supports the findings of [118] and proves that low-frequency waves have a very long range in an elastic transmission line. In [54], similar tests were conducted along an even longer 6301.4 m water supply pipe in the city of Milan, Italy. According to the authors, the largest part of the pipeline comprised of a steel DN800 pipe, and the system was rated for a maximum cross-sectional average velocity of about $\bar{V} = 3$ m/s. Assuming a pipe roughness $k = 4.5 \times 10^{-5}$ m, an appropriate eddy viscosity is $\nu_t \sim 4 \times 10^{-3}$ m²/s. Again, from the experimental data made available from the authors, $\text{SNR}(0) \sim 43$ dB at the source. At a range x equal to the pipe length ($x = 6300$ m), we find $\text{SNR}(x) \sim 42$ dB; thus, verifying once more the applicability of low-frequency waves for long-range defect detection. For the Milan tests, a number of side branches were inactive making the system more akin to a transmission line. In contrast, a water distribution system usually presents numerous branches to consumers that effectively increase damping and, therefore, reduce the attainable range. Despite that, [119] successfully introduced low-frequency transients by means of rapid valve closure in a segment of the public water supply network in Ngau Tau Kok, Hong Kong to characterize a wide range of features in the system, ranging from leaks to hydraulic devices. The pipeline was a 500 m DN200 steel pipe with cross-sectional averaged flow velocity $\bar{V} = 0.4$ m/s. The calculated eddy viscosity in this case is $\nu_t \sim 10^{-4}$ m²/s, and Eq. (5.4.5) estimates a drop in SNR from one end of the pipeline to the other equal to only ~ 1 dB. In short, low-frequency waves are demonstrably effective for

long-range defect detection in pipelines through both field-scale experiments and the developed time reversal scale.

5.5 Time reversal applications in pipeline hydraulics

The time reversal property of waves in pipelines is the backbone of the Transient-Based Defect Detection Methods (TBDDMs) developed within the framework of the SUWSS project [12]. These methods have met great success in the lab and field environments identifying defects, such as leaks and blockages, and bear attractive characteristics such as being robust in the presence of noise, and optimal in terms of signal processing. Specifically, time reversal-based methods are applicable to probing frequencies of a few Hz up to 70 kHz and proven to maximize the attainable SNR. In addition, time reversal-based methods exploit the entirety of the response spectrum, while their resolving ability is directly proportional to the probing wavelength and up to the diffraction limit ($\lambda/2$) set by wave physics.

Time reversal in the frequency domain is a phase conjugation process and leads to the Matched Field Processing (MFP) technique. MFP was initially proposed by Bucker [120] for the localization of sound sources in shallow water, and since then has been an indispensable tool for scientists and engineers in ocean acoustics (further discussed in the following section), as well as several other fields. In geophysics, for instance, MFP was applied by Cros, et al. [121] to identify the source of a hydrothermal Geyser, and Harris & Kvaerna [122] to accurately localize 549 mine explosions over a 410 km range in Northern Russia. Moreover, MFP has been used for defect detection in plate structures [123], and localization of vibration sources in beams with extensions to non-destructive testing [124]. Wang & Ghidaoui [93] have realized MFP for transient pipe signals, and the method has demonstrated superior performance to pre-existing techniques for defect detection in pipes. For instance, MFP is more efficient than e.g., Inverse Transient Analysis (ITA) since the leak localization problem is decoupled from that of leak characterization, is more robust in noisy conditions, can be used for multiple leaks [125], and is more accommodating to modelling errors [126]. In fact, MFP outperformed classical TBDDMs in an experimental HDPE pipe setup, being more accurate in leak localization [127]. More recently, MFP was used as base to a factorized model for leak localization in a tree network [128], and is currently being extended to looped network systems [129].

Area Reconstruction Method (ARM) is also a time reversal-based technique developed within the SUWSS project. ARM was initially applied for the localization of pipe blockages without the need of prior knowledge on their number [92]. Later, the method was extended to pipe networks [130], as well as pipe leak detection applications [131]. For pipe blockage identification, ARM utilizes a single measurement of the system to construct the pertinent Impulse Response Function (IRF). The IRF is then used to determine a special flow input that induces a constant pressure head in the pipe, in a process that is in essence an application of time reversal. The validity of this approach has been demonstrated with experimental data in [132].

Waqar, Louati, & Ghidaoui [133] proposed yet another time reversal-based method that employs the experimentally obtained impulse response of the system, time reverses, and then convolves it with an analytically obtained IRF from a 1D transient model. The convolution product is maximized only when the two signals are identical, i.e., when the signals contain the same information on the leak location. Recently, Waqar, Louati, & Ghidaoui [55] performed an extensive analytical and numerical analysis to evaluate the effect of damping in the refocusing quality of time-reversed water-hammer waves for two simple pipe systems. It should be noted that the algorithms developed in the SUWSS framework have all been automated and are currently employed in several test beds (including a 125 m looped HDPE pipeline at the hydraulic lab at HKUST) for real-time defect localization.

As noted, the achievable defect detection resolution is proportional to the probing wavelength λ and is limited in practice by the diffraction limit ($\lambda/2$), unless super-resolution techniques are employed. For example, in [54] possible leaks were detected to be somewhere along an 816 m pipe segment, a distance that corresponds to the probing wavelength $\lambda = c/f \approx 900/1 \approx 900$ m. In such case, an MFP technique is capable of achieving a resolution $\lambda/2 = 450$ m. Allen, et al. [53] conducted simulated burst tests and were only able to pinpoint the location to within 45 m, even though the system response was sampled 20 m away from the actual burst. In that case, the probing wavelength was $\lambda = c/f \approx 900/10 \approx 90$ m, thus explaining the 45 m accuracy by consideration of the diffraction limit ($= \lambda/2$). A higher defect detection accuracy requires signals of increased frequency that can only be excited using piezoelectric transducers, similar to the experimental setups presented in subsection 5.2.1. Such transducers have limited power, while fluid viscous losses are more prominent at higher frequencies, resulting in a much-reduced propagation range. The SUWSS project [12] proposes a defect detection system in pipelines that combines the relative strengths of low and high frequency wave techniques. Low

frequency waves (LFW) have a wide range but relatively low spatial resolution, while high frequency waves (HFW) have the opposite attributes. In practice, LFW are deployed to obtain an initial “blurred” image of a large part of the pipe system and identify zones with potential problems. HFW are then be applied locally to these zones and provide high resolution ($\sim 10^{-2}$ to 1 m) images. As explained, their range is of the order of 100 m to a few kilometres. Therefore, there is a trade-off between resolution which is proportional to frequency f and range which is proportional to the inverse of frequency $1/f$.

5.6 Time reversal of gravity waves: Experimental evidence and applications

The time reversibility of water waves was first experimentally proven by Prasadka, et al. [134] in a laboratory tank that also acted as a reverberating cavity. Water waves are dispersive and exhibit nonlinearities, but these properties do not break the time reversal invariance. According to the authors, the principal mechanism that has the potential to hamper the time reversibility of water waves is damping. The dimensions of the rectangular tank used were 53×38 cm, and the depth of the water was $H_w = 10$ cm. A single conical oscillator was used to excite the wave using a pulse of frequency $f = 5$ Hz, and the water elevation was evaluated across the entire surface using Fourier transform profilometry. The forward propagating wave field was measured for a total time $t = 20$ s, so as to include reverberations from the walls of the tank that assist in the time reversal process. For the time reversal stage, the recorded response at a single point of the surface was time reversed and reemitted using the same, but relocated, oscillator. The time reversed field refocused and recompressed at the location of the original source, thus providing the first experimental proof for the time reversibility of gravity waves. The beneficial effect of multiple time reversal sources to the refocusing quality was also demonstrated by simply adding the time reversed fields from separate tests where the time reversal source was placed at different locations. de Mello, Pérez, Adamowski, & Nishimoto [89] repeated the experiment in a much larger pool and provided perhaps the best visualization of the time reversal refocusing process. The pool was squared-shaped, with a side length equal to 14.04 m and the water depth was 4.1 m. The walls of the pool comprised of 148 movable flaps (37 per side) that could be employed to both generate waves and record the height of an incident wave height via embedded ultrasonic sensors. In their forward test, the authors generated a disturbance on the water surface by means of a cylindrical object (0.1 m in diameter and 18 cm

tall) that was dropped in the centre of the tank, and the response was measured at two opposing walls over time $t = 60$ s. A frequency spectrum analysis on the recorded signals revealed a peak frequency of 2 Hz. In the time reversal stage, the flaps generated the time reversed recorded signal and the time reversal field refocused at the entry point of the cylindrical object after about $t = 60$ s, yet again proving the time reversibility of water waves. The robustness of time reversal in water waves was further demonstrated in additional successful tests where the authors placed a physical obstacle close to the source, thus altering the shape of wave propagation field in the tank.

Chabchoub & Fink [135] experimentally demonstrated the time-reversibility of rogue waves that appear in the ocean. The experiments were conducted in a 15 m flume, where the water depth was equal to 1 m. At one end of the flume was a flap that generated the waves, while at the other end the authors constructed a wave absorbing boundary in form of an artificial beach. The source signals employed are analytical solutions to the considered governing equation, and the response of the system was measured 9 m away. Then, the obtained signals were reversed and fed to the wave maker flap. The original signals were reconstructed at the original measurement location, thus confirming that time reversal has the ability to reproduce nonlinear (rogue) water waves. Ducrozet, Bonnefoy, Mori, Fink, & Chabchoub [136] used the same methodology, but instead of analytical solutions, they considered three different recorded signals from real occurrences of rogue waves. The rogue waves were replicated experimentally using time reversal in a unidirectional 140 m long and 5 m wide wave flume, with the water depth equal to 3 m. Since the towing tank had limited dimensions, it was crucial to consider the Froude similarity values from the full-scale phenomena for the lab experiments to be relevant. Therefore, the characteristics of the wave signals such as amplitude and frequency were adapted to the model scale accordingly. The obtained results demonstrated the suitability of time reversal to accurately reconstruct real rogue waves, even in the occurrence of wave breaking during the tests.

The time reversal parameter developed in this paper (Eq. (5.3.23)) is of general applicability and is used here to shed light on the effect of viscosity on the time reversibility of gravity waves. For instance, for the time reversal experiments in Prasadka, et al. [134] the wave propagation velocity is $c_f = (gT/2\pi)\tanh(2\pi H_w/\lambda) \sim 0.3$ m/s. Given the reported dimensions of the utilized tank, the elapsed time t for the wave to reach the receiver is $t \sim 0.5 \text{ m}/c_f = 1.6$ s. To estimate an appropriate value for eddy viscosity ν_t for the tests in [134] it is first required that we find the average fluid flow velocity $\bar{V}$ in the tank due to the oscillating source is found.

Given that open channel and transient pipe flows are analogous, we employ Joukowski's equation $A_w = (c_f/g)\bar{V}$ (where A_w is the water wave amplitude) to that end, and then Manning's formula to calculate the friction velocity V^* . The maximum value for the dimensionless eddy viscosity $\nu_t/(V^*R)$ for the case of fully developed pipe flows is previously taken equal to 0.08. Since the surfaces of the tank have similar features to these of pipes, we consider $\nu_t/(V^*H_w) = 0.08$ where the pipe radius R is now replaced with the water depth H_w . Subsequently, from the reported setup details in [134], we compute an eddy viscosity $\nu_t \sim 10^{-4}$. Substituting the pertinent values to Eq. (5.3.23), we obtain $TR_f \sim 4 \times 10^{-2}$, hence, the viscous forces to temporal acceleration ratio is small and the success of the time reversal experiment justified. It is noted in Przada, et al. [134] that the total duration of the acquired signal during the forward propagation was about $t = 20$ s, after which the waves were substantially attenuated. Using $t = 20$ s, the time reversal parameter expression returns $TR_f \sim 0.6$; hence, at this time scale damping starts to become significant, and that explains the noticed attenuation. The authors also observed that in the open sea waves have a much larger period and viscous damping becomes irrelevant. This agrees with the time reversal parameter which shows that $TR_f \rightarrow 0$ as λ becomes larger. Therefore, it is not surprising that the time reversal experiments with water waves in a much larger tank that employed longer waves in de Mello, Pérez, Adamowski, & Nishimoto [89] were also effective. In fact, the calculated TR_f for this experiment is $TR_f \sim 8 \times 10^{-2}$, about half the TR_f value for the experiments in Przada, et al. [134]. Indicatively, we also investigate the New Year rogue wave case tested in Ducrozet, Bonnefoy, Mori, Fink, & Chabchoub [136] using the time reversal parameter TR . The New Year wave had the highest frequency ($f = 0.62$ Hz) among all trials, hence presents the most severe damping effects. From the reported setup characteristics, the wave propagation velocity is $c_f = g/2\pi f \sim 2.5$ m/s. Moreover, given that the overall length of the towing tank was 140 m, an applicable time $t = 140/c_f \sim 54$ s. From the stated experiment details, we estimate an eddy viscosity value $\nu_t \sim 5 \times 10^{-3}$ m²/s, and the time reversal parameter for the case at hand is $TR_f = 2 \times 10^{-2}$. Consequently, the time reversal parameter also verifies the effectiveness of time reversal in the rogue wave experiments.

The fact that gravity waves are time reversible shows a great potential to harness this property in practice. M. Fink, a pioneer in the field of time reversal and co-author of [134], in one of his interviews provided with prospective ideas [137]:

“For example, if in a harbour there is a zone where there are usually high-amplitude waves due to constructive interference and to the geometry of the shallow water structure, you can imagine controlling this phenomenon by using a set of vibrators immersed in the sea that focuses waves out of phase of these patterns to allow the wave field. (...) There can also be interesting applications by time-reversing the wake radiated by a moving boat. The anti-wake created by this process can be an efficient way to remotely push a boat or a solid structure along the water surface.”

Research is required to explore such intriguing potential applications and to assess the energy requirement to reverse gravity waves in practice. Regardless, the experimental validation of time reversal for water waves has allowed for the numerical study of large-scale phenomena. For example, Hossen, Cummins, Dettmer, & Baba [138] on the basis of time reversibility of water waves used wave recordings from fixed positions in the ocean to recreate the spatio-temporal evolution of the 2011 Tōhoku Tsunami and image its source. As the authors note, contrary to existing inversion techniques and tsunami source models used to estimate the distribution of the sea surface due to a tsunami event, time reversal does not require any assumptions on the geometry of the source, its location, or its intrinsic characteristics. The authors time-reversed the real observations from 17 stations and used them as sources in numerical experiments. These time reversal numerical experiments are based on the shallow water equations and simulated a two hour-long wave backpropagation. The results showed that the waves refocused at the source location at the correct time. Moreover, the numerical investigation in Hossen, Cummins, Roberts, & Allgeyer [139] demonstrated that the time reversal process benefits from multiple “wave mirror” points that reduce the focal size (i.e., the identification of the source location is more accurate). What is truly remarkable and perhaps even surprising is the fact that these waves are robust to time reversal over hundreds of kilometres despite the presence of damping. It is instructive to apply the time reversal parameter TR_f on the data from measurement point DART 21418, provided in [138, 140]. The wave peak frequency at that location was measured equal to $f = 7.57 \times 10^{-4}$ Hz, and the wavelength $\lambda = 360 \times 10^3$ m; thus, the wavespeed is calculated equal to $c_f = f\lambda = 272$ m/s. In addition, the depth is equal to $H = 5662$ m, and a Manning’s coefficient for the sea bottom is assumed in the order of 10^{-2} . Therefore, the eddy viscosity is estimated equal to $\nu_t \sim 10^{-1}$ m²/s. In this case, the time reversal parameter $TR_f \sim 2 \times 10^{-8}$, asserting that for long-period tsunami waves viscous losses are negligible and, therefore, supporting the assumption of a lossless

medium made in [138]. In a follow-up work, Hossen, Cummins, Roberts, & Allgeyer [139] calculated the Green's functions from a grid of possible source locations to the known observation points. Then, the observed waveforms are time reversed and convolved with the precalculated Green's functions. The convolution product is maximum at the source location, thus allowing for the selection of the most appropriate source point from the grid. Since this technique does not require real time simulation for the backpropagation of waves, it is more computationally efficient. Thus, a fast imaging of the tsunami source from near-field data could be implemented as an early warning system. More importantly, it is evident that time reversal has the potential of providing better understanding and more efficient modelling of tsunami waves. Finally, Ransley, et al. [141] employed a series of numerical models to simulate the interaction between waves and moored floating structures. One of the examined models used time reversal to recreate the source that resulted in a particular water elevation at a floating structure position.

5.7 Time reversal of acoustic waves in free-surface water

Some of the earliest experiments on the time reversibility of acoustic waves were conducted in the ocean environment. Parvulescu & Clay [142] employed two ships, each holding a submerged acoustic source and receiver, and anchored at the two sides of the Tongue of the Ocean strait in the Bahamas, with their in-between distance equal to 37 km. Both transducers were placed at 150 m depth; the source was simply suspended from the ship, while the receiver (that was closer to the coast) was fixed at the bottom of the ocean. The source signal was a 400 Hz pulse, and the response at the receiver was recorded on a tape reel that was played in reverse during the time reversal stage. The experiment succeeded in recreating the original pulse, albeit with the appearance of sidelobes. The amplitude of the sidelobes would fluctuate relative to the main pulse due to the fact that the suspended source moved during the tests up to 460 m (i.e., 131λ , since $\lambda \sim 1500/400 = 3.5$ m) from the original position, thus demonstrating the spatial focusing properties of time reversal. More recently, Hodgkiss, et al. [143] repeated the time reversal tests in the Mediterranean Sea (April 1996) but used a time reversal array instead of a single receiver. The receiving array consisted of 20 transducers spanning 77 m along a 125 m water column. The acoustic source was positioned 6.3 km from the time reversal array and emitted a 445 Hz, 5×10^{-2} s pulse. The goal of the experiments was to evaluate the focal point region; thus, the time reversed acoustic field emanating from the time reversal array was

measured by a similar vertically arranged transducer array at the location of the original source. These tests showed that the time reversed field focuses at both the depth and range of the original source (i.e., at the original source position). Moreover, in additional measurements, the time reversal stage of the test was repeated every 10 s for a total period of 2 hours following the forward propagation and produced consistent results that verified the temporal stability of time reversal in the ocean. Hodgkiss, et al. [143] presented a successful second experiment at the same location in the Mediterranean Sea, where the range between the source and the time reversal array was increased from 6.3 to 30 km. In addition, the authors further demonstrated the temporal stability of time reversal by employing 1-week old signals to reconstruct the original source through time reversal. At the same geographical location, Kim, et al. [144] repeated the time reversal tests for a source pulse of higher frequency $f = 3500$ Hz and compared the size of the attainable focal vertical width (measured along the sea depth) to that of the lower frequency $f = 445$ Hz pulse. As expected, the focal size was shown to be proportional to the wavelength, but the relationship is not linear; an observation that the authors attributed to the non-linear relationship of attenuation with frequency in the ocean bottom. Kim, et al. [145] revealed that the time reversal process when using the 3500 Hz pulse in the experiments was only valid for a short time (in the order of a few tens of minutes) following the forward propagation. In contrast, time reversal with low frequency 445 Hz pulse was proven effective even a week later (i.e., the time reversal of high frequency waves is more sensitive to large wave path fluctuations in comparison to low frequency waves, as shown in Eq. (5.3.9)).

The time reversal parameter (Eq. (5.3.23)) is now applied to the time reversal of acoustic waves in the ocean. To illustrate, we consider the experiments of Hodgkiss, et al. [143] who estimated a wavelength λ for their setup of about ~ 3.33 m, based on a depth-averaged wavespeed $c_f \sim 1500$ m/s. To apply the time reversal parameter (Eq. (5.3.23)), we adopt a reported eddy viscosity value from measurements in the ocean $\nu_t \sim 10^{-3} - 10^{-2}$ m²/s [146] and obtain a $TR_f \sim 0.6 - 6 \times 10^{-7}$. At first glance, this TR_f value might appear unreasonably small; however, there are many precedents in favour of this result. Heaney, Kuperman, & McDonald [147] acknowledged that an experiment in 1960 involving the underwater detonation of 3 explosive devices near Perth, Australia was detected halfway around the world in Bermuda (i.e., a distance of 19810 km). In another example, Shimura, Watanabe, & Ochi [148] successfully validated underwater acoustic communication in deep waters over a range ~ 600 km using passive time reversal techniques and a 450 to 550 Hz carrier signal. Interestingly enough, it is

well-known that fin whales are able to communicate over distances exceeding 6000 km by employing low-frequency pulses in the order of 20 Hz [149]. Medwin & Clay [150] showed that the absorption coefficient of sound energy in the ocean for a signal of frequency $f = 100$ Hz is in the order of 10^{-6} dB/m i.e., a distance equal to 10^6 m is required for the signal to attenuate by only 1 dB. Using a linear scale interpretation, that means that the sound energy E_a after 10^6 m is about 0.8 of the initial $E_{a,0}$. From Eq. (5.3.26), the energy attenuation in dB for a 100 Hz signal with $c_f = 1500$ m/s ($\lambda = 15$ m), and $\nu_t \sim 10^{-3} - 10^{-2}$ m²/s that propagates over a distance $x = 10^6$ m is estimated between 10^{-2} to 10^{-1} dB or, in linear scale, $E_a = 0.9 - 0.99E_{a,0}$. It should be noted, however, that Medwin & Clay [150] used a formula that, in addition to water viscosity, also accounts for the effect of various sea water chemical constituents in sound absorption. Naturally, their estimates are more conservative than the result from Eq. (5.3.26). In any event, it is apparent that viscous losses in the ocean are inconsequential to acoustic waves of adequately low frequency, that can travel over distances in the order of megametres.

The early time reversal proof of acoustic waves in the ocean has allowed for numerous applications over the past decades. In particular, acoustic techniques have been applied to infer hydro-environmental properties of the propagation medium (i.e., the ocean) that influence sound propagation. This inverse problem is known in the relevant literature as acoustic tomography and was first introduced for ocean applications by Munk & Wunsch [151] to exploit the fact that the acoustic wave field carries information about the medium such as its temperature, salinity, depth, opacity, bottom sediments, and internal layer. Some applications of acoustic tomography include mapping ocean currents [24], monitoring temperature trends and ice thickness in the Arctic [152], and defining sediment parameters in the ocean bottom [153]. MFP, which belongs to the time reversal family of methods, was first proposed by Tolstoy, Diachok, & Frazer [154], and has now become the mainstay of ocean tomography. More recently, direct time reversal has been proposed as a method to obtain ocean environmental variations from the sound speed profile [91]. It is also noteworthy that acoustic tomography has also been applied to monitor meteorological quantities in the atmosphere, image temperature and wind velocity fields, and validate LES closure models [155]. Tolstoy, Horoshenkov, & Bin Ali [156] also applied MFP with acoustic signals to detect changes (such as the appearance of leaks and blockages) in an empty pipe; an approach that has practical applications in underground unpressurized sewage pipes, a free surface water environment. Finally, Kuperman [157] mentioned that climate change-driven glacial earthquakes generate sound waves that propagate over a large range. Consequently, time reversal techniques could

potentially be applied for localizing and sizing the affected area and monitoring the extend of climate change. Larmat, Guyer, & Johnson [158] numerically applied the time reversal concept to acquired seismic waves from 165 stations around the world in order to recreate the spatio-temporal history of the 2004 Sumatra earthquake and tsunami. It is emphasized that this approach utilizes the seismic elastic waves that propagate in the Earth rather than the water waves discussed in the previous section to evaluate a tsunami source. It is noteworthy that the spherical shape of the Earth forms a reverberating cavity that enhances the global-scale time reversal refocusing of seismic waves thanks to the multitude of reflections from the boundaries. In fact, the combination of time reversal of elastic waves in Earth with the gravity waves in the ocean provides an opportunity for in-depth understanding of sources of tsunami waves.

5.8 Conclusions

This chapter presents a set of novel experiments that prove the feasibility of the time reversal of acoustic waves in a water-filled viscoelastic HDPE pipes, despite the use of a single-channel time reversal mirror. Moreover, this chapter attempts to expose the reader to the fascinating field of time reversal of waves, its essential physics, as well as its numerous current and potential applications in hydraulics.

Damping and wave path instabilities are the two main mechanisms that restrict the realization of time reversal. An analytical investigation that considers the stability of time reversal focusing to the random perturbation of scatterers' locations shows that instabilities grow as $(t^{1/2})$. In addition, the effect of damping is analysed by applying the time reversal transformation ($t \rightarrow -t$) to both the Navier-Stokes and viscous pipe wall equations, followed by an order of magnitude analysis that compares the damping to the temporal acceleration terms. The result is a time reversal parameter TR that shows the effect of damping on the time reversal of waves growing linearly with time. The fact that instabilities grow as $(t^{1/2})$ but damping as t indicates that the latter is the most critical impediment for the successful application of time reversal of waves.

Application of TR to the set of experiments conducted as part of this paper and additional experiments available in the literature shows that time reversal is valid for a $TR \sim 1$ or less. Consequently, the TR parameter could readily be used to identify the applicability of time reversal to a given system, as well as to determine the maximum achievable range over which

time reversal is feasible. In the context of viscoelastic pipelines, the *TR* parameter also shows that viscous losses sustained in the fluid naturally increase with the frequency of the probing signal, while, interestingly, the opposite is true for the viscoelastic pipe wall. Thus, in the absence of a pipe wall (or, for an elastic pipe wall) low frequency signals are shown to propagate over very long distances. The pertinence of the *TR* parameter is not restricted to pipelines but extends to a variety of hydraulic systems. A selection of experimental time reversal proofs in water waves and acoustic waves in the ocean is reviewed and examined in view of the *TR* parameter in order to substantiate the observed results. Finally, the reader is introduced to the numerous applications of time reversal that originated from these experimental proofs and have profound implications to the field of hydraulics.

In Chapter 6, the *TR* property of high frequency waves in water supply pipes is exploited to develop a pulse echo technique for defect detection. Experimental results show that the developed *TR*-based pulse echo technique has the potential to revolutionize defect detection in water supply pipes, as it delivers unprecedented defect localization accuracy. An observant reader would recognize that the experiments presented in this chapter only consider time reversal under quiescent water conditions that are far from those found in typical water supply pipes. Hence, in Chapter 7, the fidelity of *TR* is tested in unfavourable scenarios, such as under pipe flow and the presence of inline pipe junctions. Hopefully, however, the material presented in this thesis will popularize time reversal among the hydraulic community and motivate the development of similarly innovative applications apart from pipelines.

Chapter 6

Time reversal for defect detection

6.1 Introduction

One of the most critical global challenges concerning the built environment is the gradual decay of urban water supply networks. These systems are currently fraught with defects and inefficiencies, such as leaks and blockages, leading to major economic losses, disruptions, and potential health hazards [159, 160]. Recent studies [161] estimate that 126B m³ of freshwater are unaccounted for annually by water utilities around the world, for an equivalent monetary loss of US\$39B. The identification of imminent or existing faults is a formidable task since water supply networks are buried underground and demonstrate intricate topologies. Current diagnostic methods, often require the isolation of specific segments of the network, are disruptive and involve significant costs. Less invasive techniques such as acoustic correlators or ground penetrating radars have limited applicable range and are susceptible to noise [162]. To overcome the drawbacks of these existing technologies, the Smart Urban Water Supply Systems (SUWSS) project [163] advocates the use of actively generated acoustic waves in pipe systems that interact with defects to induce useful reflections. At first, pressure waves in the low frequency regime with wavelengths much larger compared to the pipe diameter will propagate over long distances (in the order of km) and provide with a rough assessment of the network condition. Once a problematic segment is identified, high frequency pressure waves (10 – 50 kHz) will pinpoint the location of the defect within a few centimetres.

The idea of utilizing pressure transients to probe the system has been explored before and the multitude of the proposed methodologies form the family of Transient-Based Defect Detection Methods (TBDDMs). The crux of TBDDMs is that the injected pressure signal is modified in accordance with both the physical characteristics of the system and potential defects as it propagates. Subsequently, the recorded response contains the defect signature(s) that, if properly analysed and processed, may identify the defect(s) in the system. The various methodologies developed in the framework of TBDDMs are grouped into (1) inverse transient methods (ITM) [36, 37, 38], (2) transient reflection-based methods (TRM) [164, 165, 41], (3) transient damping-based methods (TDM) [42], and (4) frequency response-based methods (FRM) [43, 44, 45, 46, 166, 48, 49, 50, 51]. A common trait is that these methods are based on low frequency transients (< 100 Hz) most often produced by the operation of relatively slow mechanical devices, such as valves, and may assume a simple 1D wave propagation. On the other hand, since the achievable resolution is directly proportional to the wavelength of the probing signal, low frequency transients have limited resolution capabilities in the order of meters. As such, to improve the precision of defect detection it is instrumental to employ high-frequency pressure waves using piezoelectric transducers.

As explained in Chapters 3 to 5 of this thesis, the propagation of waves with wavelengths in the order of the pipe radius R through fluid-filled elastic pipes is a dispersive phenomenon as waves of different frequencies follow different paths. In such environment, it is challenging to identify defects that act as wave scatterers. A fascinating wave property that nullifies the distortive effects of multipath propagation is time reversal (TR). However, the validity of TR in a waveguide is not a given and must be proven. Chapter 5 presents the first-ever proof of TR in water supply pipes and enables the use of TR-based pulse-echo defect detection methods in the particular environment. In a typical pulse-echo scenario, a probing pulse is injected in a defected waveguide and disperses as it propagates. Eventually, the dispersed signal meets the defect and is scattered consequently turning the defect into a passive acoustic source. If the reflection from the defect is captured and re-emitted in a model of the system, the validity of TR ensures that it will uniquely refocus at its source (i.e., the defect).

However, for this approach to be successful it is crucial that the interaction between the injected pulse and a leak in the pipe is also delineated. Prior researchers investigated the interaction between high frequency waves and leaks in rigid pipes by numerical CFD simulations [167]. In this Chapter, an analytical axisymmetric 2D solution is introduced to describe the interaction between a high-frequency probing wave and a defect in a fluid-filled elastic pipe, instead.

Following the governing equations derived in Chapter 2, the defect is simulated as mass source that constitutes a weak inhomogeneity in the system, allowing for the Born approximation when solving for the scattered field. To account for the deformable pipe walls, the pressure wave and pipe wall interaction is described by the impedance-based model discussed in Chapter 3 and experimentally verified in Chapter 4. The proposed technique leverages on the time reversibility of waves and utilizes experimentally obtained data to localize defects in pipes. The technique comprises of two distinct steps. First, the defected system is probed experimentally to extract its response. The obtained response is then time-reversed and matched to the analytical model, in a process that enables the localization of the defect.

The efficacy of the proposed TR-based methodology for defect detection is tested in practice by deliberately introducing defects along two HDPE pipes. In the first example, a 2 cm diameter orifice is bore through the pipe wall of a DN50 HDPE pipe to simulate a leak. The pipeline is then probed by a source/receiver array, and the obtained response is used to pinpoint the location of the leak with a resolution proportional to the probing wavelength. A leak and a blockage both act as wave scatterers in the waveguide. Therefore, the analytical solution is also applicable to the localization of blockages along pipes. For the second example, a small bitumen piece from a failed iron pipe lining is placed along a DN80 HDPE pipe. The frontal area of the bitumen piece only accounts for 20% of the pipe cross-sectional area. Nonetheless, a distinct reflection is still received from the defect when the system is probed with a high-frequency pulse, that allows the placement of the blockage with remarkable accuracy. Finally, experimental results demonstrate that the achievable resolution increases with the frequency of the probing pulse and scales with its wavelength.

6.2 Analytical model for the wave/leak interaction

6.2.1 Fundamental equations

Consider the elastic cylindrical pipe of Figure 5.2.1, filled with a compressible fluid (e.g., water). The pipe has a constant cross-section of radius R , lays horizontal and is longitudinally unbounded. Let x , r , and θ be the cylindrical coordinates, where the x direction coincides with the longitudinal pipe axis. As already mentioned, the pressure field is considered axisymmetric; thus, both fluid pressure variations and pipe wall vibrations are independent of the azimuthal coordinate, θ . Present within the system are an acoustic point source at $\mathbf{x}_s = (x_s, r_s)$, a leak at

$\mathbf{x}_L = (x_L, R)$, and an acoustic receiver at, say, $\mathbf{x}_m = (x_m, r_m)$ (where subscript m stands for measurement).

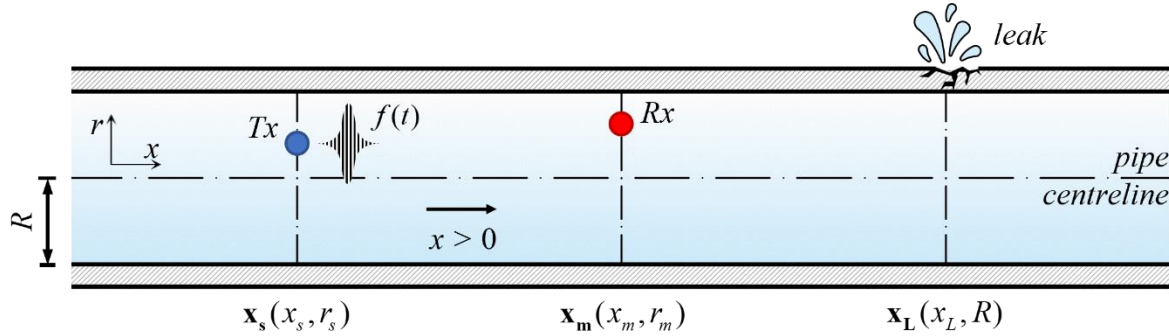


Figure 6.2.1: Water-filled pipe configuration with an acoustic source at $\mathbf{x}_s$, a leak at $\mathbf{x}_L$, and an acoustic receiver at $\mathbf{x}_m$.

From Eq. (2.2.39), the mass conservation equation for the described system is written as:

$$\frac{1}{c_f^2} \frac{\partial p(\mathbf{x}, t)}{\partial t} + \dot{m}(t) \delta_3(\mathbf{x} - \mathbf{x}_L) + \nabla [\rho_0(\mathbf{x}) \mathbf{v}(\mathbf{x}, t)] = \dot{S}(t) \delta_3(\mathbf{x} - \mathbf{x}_s) \quad (6.2.1)$$

in terms of pressure p , flow velocity $\mathbf{v}$, time t , space $\mathbf{x}$, fluid density ρ_0 , and wave speed in the fluid c_f . More importantly, $\dot{m}$ represents the mass rate due to the leak at $\mathbf{x}_L$, while $\dot{S}$ describes the same quantity for the (known) acoustic source at $\mathbf{x}_s$. Term $\dot{S}$ describes a monopole acoustic source that can be thought of as a small pulsating sphere contracting and expanding with time. Lord Raleigh [168] first described the monopole source, stating that the injection of a mass rate (i.e., $\dot{S}$) into the considered control volume leads to pressure generation. When positive, these parameters correspond to a sink where mass flows out of the system. On the other hand, negative values indicate a source that contributes mass to the system. Therefore, an acoustic source is to be understood as a mass source/sink to the system, analogous to the leak. To arrive to Eq. (3.2.1), an isentropic flow is considered (i.e., friction losses are ignored). Also, advective acceleration terms that scale to the Mach number Ma have been omitted, since in water pipe flows $Ma \sim 10^{-3}$ (see Chapter 2 for more details).

From Eq. (2.2.79), the momentum conservation equation is written as:

$$\frac{\partial (\rho_0(\mathbf{x}) \mathbf{v}(\mathbf{x}, t))}{\partial t} + \nabla p(\mathbf{x}, t) = 0 \quad (6.2.2)$$

An important observation is that Eq. (6.2.2) excludes any description of fluid viscous forces. In Chapter 2, where Eq. (6.2.2) is formally derived, it is shown that for the frequency bandwidth

of the probing signals employed in this thesis (10 to 70 kHz), viscous forces are negligible when compared to fluid inertia forces and are, therefore, omitted. This outcome also validates the assumption for isentropic flow in the mass conservation equation (Eq. (3.2.1)).

6.2.2 Wave equation under the effect of a leak and acoustic source

The pressure wave equation for the specified fluid-filled pipe system is obtained by taking the time derivative of Eq. (3.2.1) and subtracting the divergence of Eq. (6.2.2):

$$\frac{1}{c_f^2} \frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} - \nabla^2 p(\mathbf{x}, t) = \frac{\partial \dot{S}}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_s) - \frac{\partial \dot{m}}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (6.2.3)$$

The reader is reminded that $\dot{m} = \rho v_L A_L = \rho Q_L$ where v_L is the fluid flow velocity at the leak and A_L is the area of the leak; thus, the leak flowrate $Q_L = v_L A_L$. Consequently, the term $\partial \dot{m} / \partial t$ in Eq. (6.2.3) encapsulates the physical properties of the leak, such as its size. To define the leak flowrate Q_L in practice, the orifice equation from [169] may be applied:

$$Q_L = C_d A_L \sqrt{2\mathbf{g}(H_L - z_L)} \quad (6.2.4)$$

where C_d is the discharge coefficient, $\mathbf{g}$ is the acceleration of gravity, $H_L (= p/\rho\mathbf{g})$ is the pressure head at the leak, and z_L is the elevation at the orifice. The orifice equation (Eq. (6.2.4)) is developed on the promise of steady flow in the pipe. In any event, the assumptions of the orifice equation do not influence the localization of the leak that is possible through the consideration of the spatial δ_3 function. However, it may introduce inaccuracies when estimating the leak characteristics. The elevation at the leak z_L is taken equal to zero and, since $Q_L = v_L A_L$, the flow velocity at the leak v_L from Eq. (6.2.4) is:

$$v_L = C_d \sqrt{2 \frac{p}{\rho}} \quad (6.2.5)$$

At steady state, $p_L = p_{L,0}(\mathbf{x})$ and $\rho = \rho_0(\mathbf{x})$, therefore from Eq. (6.2.5) the discharge coefficient C_d is:

$$C_d = \frac{v_{L,0}}{\sqrt{2 p_{L,0} / \rho_0}} \quad (6.2.6)$$

On the other hand, for unsteady state $p = p_0(\mathbf{x}) + p'(\mathbf{x}, t)$, $\rho = \rho_0(\mathbf{x}) + \rho'(\mathbf{x}, t)$, and $v_L = v_{L,0}(\mathbf{x}) + v_L'(\mathbf{x}, t)$. Hence, Eq. (6.2.5) for the unsteady state with the discharge coefficient from Eq. (6.2.6) becomes:

$$v_L' = v_{L,0} \left\{ \left[\frac{(p_0 + p')\rho_0}{p_0(\rho_0 + \rho')} \right]^{1/2} - 1 \right\} \quad (6.2.7)$$

Differentiating Eq. (6.2.7) with p' yields:

$$\frac{\partial v_L'}{\partial p'} = \frac{v_{L,0} \sqrt{\frac{\rho_0(p_0 + p')}{p_0(\rho_0 + \rho')}}}{2(p_0 + p')} \approx \frac{v_{L,0}}{2p_0} = \frac{1}{2Z_L} \quad (6.2.8)$$

where $Z_L = p_0/v_{L,0}$ is the steady state acoustic impedance of the orifice. Note that for the simplification in Eq. (6.2.8), both oscillatory p' and ρ' are considered to be much smaller than the ambient steady-state pressure p_0 and fluid density ρ_0 , respectively. The wave equation (Eq. (6.2.3)) is written as:

$$\frac{1}{c_f^2} \frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} - \nabla^2 p(\mathbf{x}, t) = \frac{\partial \dot{S}}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_s) - \rho_0 A_L \frac{\partial v_L}{\partial p} \frac{\partial p}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (6.2.9)$$

where the term $\partial v_L / \partial p$ in Eq. (6.2.9) is substituted from Eq. (6.2.8) after dropping the prime notation to obtain:

$$\frac{1}{c_f^2} \frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} - \nabla^2 p(\mathbf{x}, t) = \frac{\partial \dot{S}}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_s) - A_L \frac{\rho_0}{2Z_L} \frac{\partial p}{\partial t} \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (6.2.10)$$

For convenience, it is preferable to express Eq. (6.2.10) in terms of angular frequency ω instead of time t . Therefore, taking the Fourier transform of Eq. (6.2.10):

$$\begin{aligned} \frac{1}{c_f^2} \int_{-\infty}^{+\infty} \left(\frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} \right) e^{-i\omega t} dt + \nabla^2 \int_{-\infty}^{+\infty} p(\mathbf{x}, t) e^{-i\omega t} dt = \\ \delta_3(\mathbf{x} - \mathbf{x}_s) \int_{-\infty}^{+\infty} \frac{\partial \dot{S}}{\partial t} e^{-i\omega t} dt - A_L \frac{\rho_0}{2Z_L} \delta_3(\mathbf{x} - \mathbf{x}_L) \int_{-\infty}^{+\infty} \frac{\partial p}{\partial t} e^{-i\omega t} dt \end{aligned} \quad (6.2.11)$$

yields:

$$\frac{\omega^2}{c_f^2} \hat{p}(\mathbf{x}, \omega) - \nabla^2 \hat{p}(\mathbf{x}, \omega) = -i\omega \hat{S} \delta_3(\mathbf{x} - \mathbf{x}_s) + A_L \frac{\rho_0}{2Z_L} i\omega \hat{p}(\mathbf{x}, \omega) \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (6.2.12)$$

where:

$$\hat{p}(\mathbf{x}, \omega) = \int_{-\infty}^{\infty} p e^{-i\omega t} dt, \text{ and } \hat{S}(\mathbf{x}, \omega) = \int_{-\infty}^{\infty} \dot{S} e^{-i\omega t} dt \quad (6.2.13)$$

Equation (6.2.12) is written in the dimensionless terms of Table 6.2.1 as:

$$\begin{aligned} \omega_f^{*2} \hat{p}^*(\mathbf{x}^*, \omega_f^*) - \nabla^2 \hat{p}^*(\mathbf{x}^*, \omega_f^*) = \\ \hat{f}(\omega_f^*) \quad S_L \\ -i\omega_f^* \hat{S}^* \delta_3(\mathbf{x}^* - \mathbf{x}_s^*) + i\omega_f^* \frac{A_L^*}{2Z_L^*} \hat{p}^*(\mathbf{x}^*, \omega_f^*) \delta_3(\mathbf{x}^* - \mathbf{x}_L^*) \end{aligned} \quad (6.2.14)$$

Note that for the dimensionless terms of Table 6.2.1, the dimensional parameters are normalized with fluid instead of the pipe wall properties as in Table 3.2.1. Since the fundamental equations in this chapter describe the pressure wave in the fluid domain, it would make little sense utilizing the same normalization as in Chapter 3, where the focus is on the pipe wall equations.

Table 6.2.1: List of dimensionless parameters used in the analytical wave/leak interaction solution.

Frequency, ω_f^*	$R\omega/c_f$
Pressure, $\hat{p}^*$	$\hat{p}/(\rho_0 c_f^2)$
Length, x^*	x/R
Mass source, $\hat{S}^*$	$\hat{S}/(\rho_0 c_f R^2)$
Leak area, A_L^*	A_L/R^2
Leak impedance, Z_L^*	$Z_L/(\rho_0 c_f)$

In Eq. (6.2.14) the term that encapsulates the physical properties of the defect $A_L^*/2Z_L^*$ is set as S_L . Note that for defect localization (discussed in later sections of the manuscript), the definition of S_L is not important; that is dictated by the argument of the δ_3 function, $(\mathbf{x} - \mathbf{x}_L)$. Also, $\hat{f}(\omega_f^*)$ ($= i\omega_f^* \hat{S}^*$) describes the pressure variation of the known acoustic source at $\mathbf{x}_s$ in time. The asterisk ($\cdot^*$) notation for the dimensionless parameters is dropped for convenience, and Eq. (6.2.14) becomes:

$$\omega_f^2 \hat{p}(\mathbf{x}, \omega_f) - \nabla^2 \hat{p}(\mathbf{x}, \omega_f) = -\hat{f}(\omega_f) \delta_3(\mathbf{x} - \mathbf{x}_s) + i\omega_f S_L \hat{p}(\mathbf{x}, \omega_f) \delta_3(\mathbf{x} - \mathbf{x}_L) \quad (6.2.15)$$

The terms residing on the right-hand side of Eq. (6.2.15) represent the pressure forcing in the waveguide as the combined effect of the acoustic point source at $\mathbf{x}_s$ and leak at $\mathbf{x}_L$. The solution to the pressure field in Eq. (6.2.15) at a point $\mathbf{x}$ may be obtained after convolving the forcing terms with the impulse response of the system, as given by an appropriate Green's function expression. For the case of a point source in a cylindrical duct with deformable walls, such expression is given in [80], and is also introduced in Chapter 3 (Eq. (3.3.88)). In the dimensionless terms of Table 6.2.1, the Green's function for the fluid-filled pipe waveguide is written as:

$$\hat{G}_f^*(\mathbf{x}|\mathbf{x}_s, \omega_f^*) = \frac{i}{2\pi} \sum_n \frac{J_0(k_{r,n}r_s)J_0(k_{r,n}r)}{k_{x,n}R \left(1 - \frac{\omega_f^{*2}}{k_{r,n}^2 Z_f^{*2}(\omega_f^*, c^*)}\right) J_0^2(k_{r,n}R)} e^{-ik_{x,n}(x-x_s)} \quad (6.2.16)$$

Note the impedance term Z_f^* in the denominator of the second term on the right-hand side of Eq. (6.2.16). This defines the impedance condition at the fluid/pipe interface and is extensively discussed in Chapter 3 to produce Eq. (3.3.93) that has been experimentally verified in Chapter 4. In the dimensionless terms of Table 6.2.1, the impedance condition Z_f^* of Eq. (6.2.16) is expressed as:

$$Z_f^*(\omega_f^*, c^*) = \frac{i}{a^2 \gamma^2 \omega_f^*} \left\{ 1 - \gamma^2 \omega_f^{*2} - \frac{v^2}{1 - c^{*2}} + \frac{\xi \gamma^4 \omega_f^{*4} (1 - c^{*2})}{c^{*2} [c^{*2} + \xi \eta \gamma^2 (1 - c^{*2})]} \right\} \quad (6.2.17)$$

To evaluate impedance Z_f^* , the characteristic equation of the waveguide (Eq. 3.2.41) needs to be solved first to retrieve the valid (ω, c_{ph}) pairs in the excited frequency bandwidth. Then, as already mentioned, the solution of the pressure field at a point $\mathbf{x}$ is produced by convolving the Green's function expression of Eq. (6.2.16) with the pressure forcing terms in the right-hand side of the wave equation (Eq. (6.2.15)):

$$\hat{p}(\mathbf{x}, \omega_f) = -\hat{f}(\omega_f) \hat{G}_f(\mathbf{x}|\mathbf{x}_s) + \int i\omega_f S_L \hat{p}(\mathbf{x}, \omega_f) \delta(\mathbf{x}' - \mathbf{x}_L) \hat{G}_f(\mathbf{x}|\mathbf{x}') d\mathbf{x}' \quad (6.2.18)$$

where $\mathbf{x}'$ is introduced as a dummy variable to describe the space where point $\mathbf{x}$ lays. Note that in Eq. (6.2.18) the asterisk notation $(\cdot^*)$ for the dimensionless parameters is dropped for simplicity.

6.2.3 Born approximation

The total pressure field $\hat{p}$ at $\mathbf{x}$ may be expressed as the superposition of an incident $\hat{p}^I$ and a reflected $\hat{p}^R$ wave:

$$\hat{p}(\mathbf{x}, \omega_f) = \hat{p}^I(\mathbf{x}, \omega_f) + \hat{p}^R(\mathbf{x}, \omega_f) \quad (6.2.19)$$

In the event where no leak is present in the system ($S_L = 0$), the effect of the reflected wave is nullified, and the solution to the pressure field is given by:

$$\hat{p}(\mathbf{x}, \omega_f) = \hat{p}^I(\mathbf{x}, \omega_f) = -\hat{f}(\omega_f) \hat{G}_f(\mathbf{x}|\mathbf{x}_s) \quad (6.2.20)$$

Following the definition of the incident pressure field, we may obtain the solution to the pressure field due to the reflected wave from Eq. (6.2.18) as:

$$\hat{p}^R(\mathbf{x}, \omega_f) = \int i\omega S_L \hat{p}(\mathbf{x}', \omega_f) \delta(\mathbf{x}' - \mathbf{x}_L) \hat{G}_f(\mathbf{x}|\mathbf{x}') d\mathbf{x}' \quad (6.2.21)$$

or alternatively:

$$\hat{p}^R(\mathbf{x}, \omega_f) = \mathcal{G}T_{op} \hat{p} \quad (6.2.22)$$

where $\mathcal{G} = i\omega_f S_L$, and T_{op} represents the integral operation against the Green's and δ functions. Substituting the result of Eq. (6.2.22) into Eq. (6.2.19) yields:

$$\hat{p}(\mathbf{x}, \omega_f) = \hat{p}^I(\mathbf{x}, \omega_f) + \mathcal{G}T\hat{p} \quad (6.2.23)$$

called the Lippmann-Schwinger equation. Following appropriate mathematical manipulations:

$$\hat{p}(\mathbf{x}, \omega_f) = \hat{p}^I(\mathbf{x}, \omega_f) (1 - \mathcal{G}T)^{-1} \quad (6.2.24)$$

Equation (6.2.24) may be expanded into a Born-Neumann series, written as:

$$\hat{p}(\mathbf{x}, \omega_f) = \hat{p}^I(\mathbf{x}, \omega_f) \left[1 + \mathcal{G}T + (\mathcal{G}T)^2 + \dots \right] \quad (6.2.25)$$

Physically, the Neumann series represents the effects of multiple scattering. Thus, the first term on the right-hand side of Eq. (6.2.25) corresponds to the incident wave, the second term to single scattering, the third to double, etc. It is assumed that a leak presents a weak inhomogeneity in the system and, as such, its effect to the total field is small compared to the incident wave ($\|\mathcal{G}T\| < 1$). Henceforth, we may apply the Born approximation to retain only the first-order term of the expansion. In such case, the total pressure field solution is written as:

$$\hat{p}(\mathbf{x}, \omega_f) = \hat{p}^I(\mathbf{x}, \omega_f) (1 + \mathcal{G}T) \quad (6.2.26)$$

Finally, the expression for the incident pressure field $\hat{p}^I(\mathbf{x}, \omega_f)$ from Eq. (6.2.20) is substituted into Eq. (6.2.26) to obtain:

$$\hat{p}(\mathbf{x}, \omega_f) = \underbrace{-\hat{f}(\omega_f) \hat{G}_f(\mathbf{x}|\mathbf{x}_s)}_{\hat{p}^{\text{intact}}} \underbrace{-i\omega_f S_L \hat{f}(\omega_f) \hat{G}_f(\mathbf{x}_L|\mathbf{x}_s) \hat{G}_f(\mathbf{x}|\mathbf{x}_L)}_{\hat{p}^{\text{leak}}} \quad (6.2.27)$$

Equation (6.2.26) presents the total pressure field within a fluid-filled elastic pipe at a location $\mathbf{x}$ for a given frequency ω due to an acoustic point source at $\mathbf{x}_s$ and a leak at $\mathbf{x}_L$. To reiterate, the first term on the right-hand side of Eq. (6.2.27) provides the response of an intact pipe $\hat{p}^{\text{intact}}$, while the second term adds the influence of the leak $\hat{p}^{\text{leak}}$.

6.3 Analytical example of wave-defect interaction

To illustrate the effect of a defect on the response of a fluid filled elastic pipe when probed under a high frequency pulse, the total field solution is analytically applied to the water filled DN80 HDPE pipe, whose mechanical properties are outlined in Table 4.2.1. The specific probing source signal, plotted in Figure 6.3.1a & Figure 6.3.2a, is a Gaussian modulated sine pulse with a central frequency equal to 35 kHz and 20% relative bandwidth (i.e., [31.5, 38.5] kHz) at 1/100 the power spectrum peak value and starts propagating at time $t = 0$. To ensure that the generated pressure field remains axisymmetric, the source is positioned at the pipe centre at $\mathbf{x}_s = (0, 0)$ m. Solving the characteristic equation (Eq. 3.2.41) for the parameters of Table 4.2.1 in the excited frequency bandwidth indicates that 4 different modes could potentially propagate in total: (i) the water-borne plane wave mode, (ii) the structural mode, corresponding to longitudinal waves in the pipe wall, (iii) the 1st, and (iv) 2nd high-order radial modes. However, due to the chosen source location as well as the assumption of a point-sized source, the wave energy is merely allocated to the 1st and 2nd high-order radial modes.

Initially, it is assumed that the water filled pipe is free any leak; thus, $\hat{p}^R = 0$. In this example, the incident pressure wave $\hat{p}^I$ due to the acoustic point source is evaluated at $\mathbf{x}_m = (2, 0)$ m. Following Eq. (6.2.20), to obtain $\hat{p}^I$ the acoustic source $\hat{f}(\omega_f)$ is convolved with the Green's function (Eq. (6.2.16)) for characteristics of the waveguide shown in Table 4.2.1. Note that the acoustic source $\hat{f}(\omega_f)$ is defined in a frequency bandwidth interval that contains the excited bandwidth. Applying the inverse Fourier transform operation to the occurring pressure response spectrum $\hat{p}^I(\mathbf{x}_m, \omega_f)$ produces the time waveform of Figure 6.3.1b, depicted normalized by the amplitude of the source (Figure 6.3.1a). The response is characteristic of the

multipath wave propagation that occurs in a bounded waveguide and comprises of two distinct wavepackets, with each corresponding to the 1st and 2nd high-order radial modes. The two modes follow a different path along the waveguide, resulting in a unique group velocity c_{gr} ($= d\omega/dk_{x,n}$). For instance, in Figure 6.3.1b the 1st radial mode propagates faster, hence, arrives earlier at $\mathbf{x}_m$. The space-time diagram of Figure 6.3.1c better illustrates the dispersive nature of waves propagating along a pipe. At increasing distances from the source, the two modes separate further. In addition, the trailing 2nd radial mode appears to spread in time, while the 1st does not. The reason may be found in the larger variability in the $k_{r,n}$ values (and by extension c_{gr}) for the 2nd radial mode when compared to the 1st in the excited frequency interval.

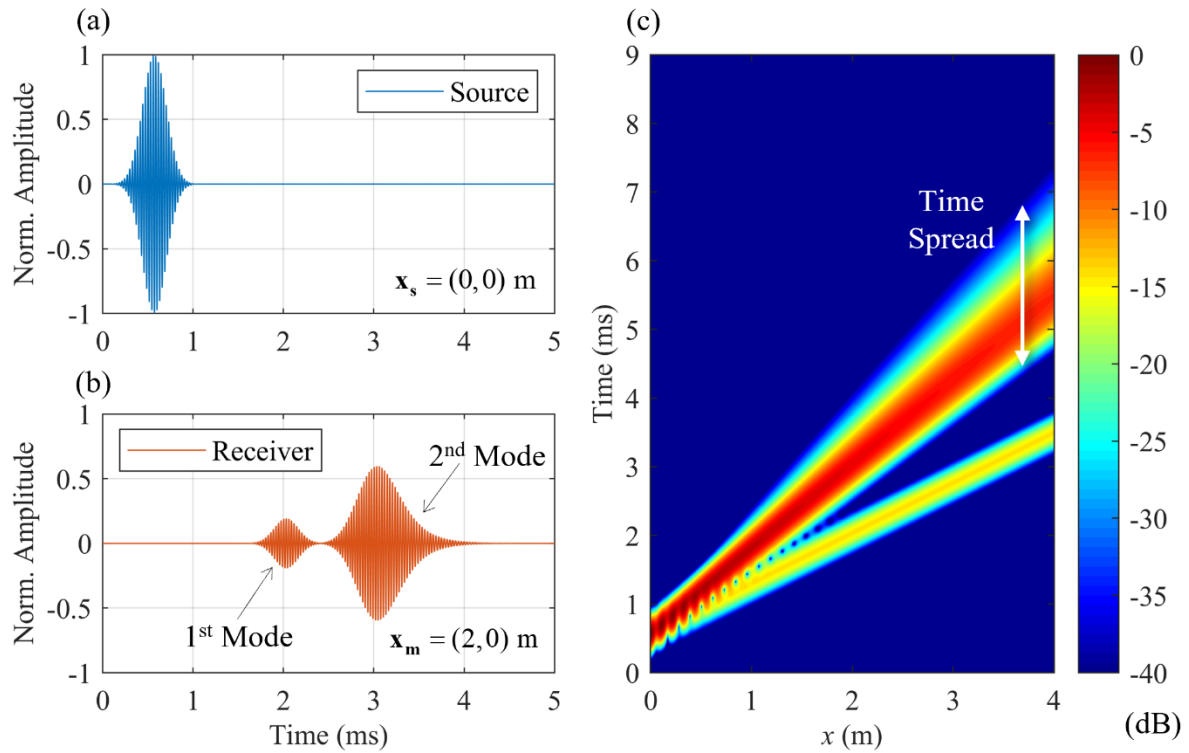


Figure 6.3.1: Analytical results for the water filled DN80 HDPE pipe in the absence of a defect. (a) A 35 kHz Gaussian modulated sine pulse is emitted at $\mathbf{x}_s = (0, 0)$ m, and (b) the response of the system is captured at $\mathbf{x}_m = (2, 0)$ m. (c) The space-time diagram displays the dispersive nature of high-frequency wave propagation in pipelines; the contour plot is shown in dB scale relative to the maximum observed value.

Next, a leak is introduced along the pipe at $\mathbf{x}_L = (2, R)$ m, while the measurement location is moved to $\mathbf{x}_m = (0, 0)$ m, now coinciding with the acoustic source location. The total wave pressure field $\hat{p}$ comprises of both the incident $\hat{p}^I$ and reflected $\hat{p}^R$ waves and is evaluated in the frequency domain using Eq. (6.2.27) after setting an arbitrary value for the leak characteristics

(e.g., $S_L = 1$). Figure 6.3.2a presents the time waveform of the probing source signal, and Figure 6.3.2b depicts the acquired system response $\hat{p}$ at $\mathbf{x}_m$ in time following an inverse Fourier transform operation. The initial pulse at $\mathbf{x}_m$ mirrors the probing pulse and is identified as the incident wave p^I . However, the pulse continues to propagate along the pipe and splits in two modes, much like in the intact case example of Figure 6.3.1. Eventually, the two modes meet the leak at $\mathbf{x}_L$ and are partially reflected. At time $t \approx 3$ ms, the 1st mode arrives back at $\mathbf{x}_m$, followed by the slow propagating 2nd mode. The two modes are associated to the reflected pressure field p^R due to the presence of the defect in the system. The space-time diagram is also provided in this case (Figure 6.3.2c) and elucidates the impact of the leak on the propagation of the pulse along the pipe. The wave energy is split at the location of the defect (indicated by the vertical white dashed line), with one part reflected and the other propagating further from the source. Note that each mode is only scattered once, in accordance to the Born approximation. If higher order terms of Eq. (6.2.25) are considered, then each mode would be scattered to two modes as dictated by the waveguide characteristics in the excited frequency bandwidth. Regardless, the results indicate that in this setting a leak constitutes a passive acoustic source that could potentially be detected. In the upcoming section, a leak localization technique is proposed based on the time-reversal property of waves in pipes proven in Chapter 5. This technique exploits the reflected pressure field to “retrace” its source, hence localize the leak.

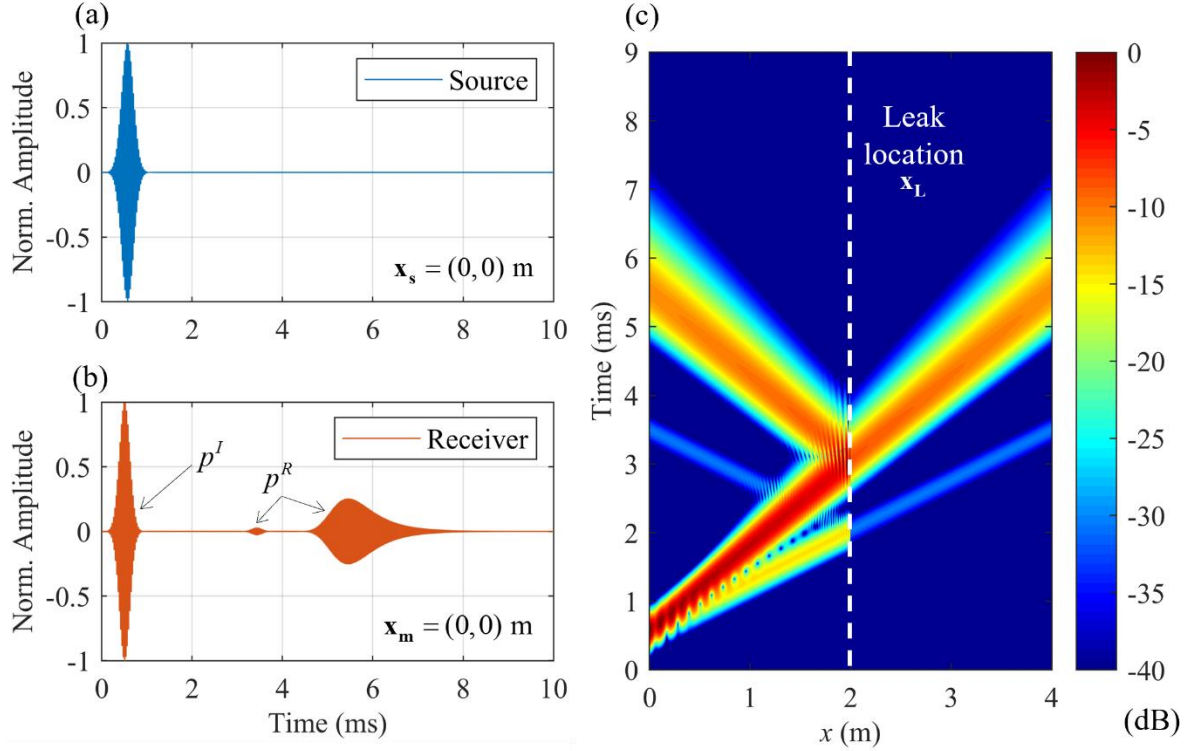


Figure 6.3.2: Analytical results for the water filled DN80 HDPE pipe with a leak at $\mathbf{x}_L = (0, 0)$ m. (a) A 35 kHz Gaussian modulated sine pulse is injected at $\mathbf{x}_s = (0, 0)$ m, and (b) the system response (evaluated at the same location) includes the effect of both the incident and reflected waves. (c) The space-time diagram shows that the propagating modes are partially reflected at the defect location.

6.4 Defect localization using TR

Assume a real-world pipe that contains a leak along its length and is probed with a short acoustic pulse. When performing pressure measurements in practice, narrowband probing pulses are employed that excite a bounded frequency interval. Moreover, the corresponding system responses contains noise. Consequently, the total pressure solution (Eq. (6.2.27)) at $\mathbf{x}$ may, therefore, be written in vector form as:

$$\hat{\mathbf{p}}(\mathbf{x}, \omega_f) = \hat{\mathbf{p}}^{\text{intact}} + \hat{\mathbf{p}}^{\text{leak}} + \mathbf{N} \quad (6.4.28)$$

where, for N frequencies, $\hat{\mathbf{p}} = [\hat{p}_1(\mathbf{x}, \omega_{f,1}), \dots, \hat{p}_N(\mathbf{x}, \omega_{f,N})]^T$, $\hat{\mathbf{p}}^{\text{intact}} = [\hat{p}_1^{\text{intact}}(\mathbf{x}, \omega_{f,1}), \dots, \hat{p}_N^{\text{intact}}(\mathbf{x}, \omega_{f,N})]^T$, $\hat{\mathbf{p}}^{\text{defect}} = [\hat{p}_1^{\text{defect}}(\mathbf{x}, \omega_{f,1}), \dots, \hat{p}_N^{\text{defect}}(\mathbf{x}, \omega_{f,N})]^T$, and $\mathbf{N}$ is the noise vector with zero-mean Gaussian distribution $N(0, \sigma^2 \mathbf{I}_N)$ with $\mathbf{I}_N$ being the N -dimensional identity vector. The

part of the experimentally measured pressure response attributed to the leak may then be expressed as a difference from the pertinent intact response as:

$$\Delta\hat{\mathbf{p}} = \hat{\mathbf{p}}^{\text{leak}} + \mathbf{N} \quad (6.4.29)$$

The goal is to localize the defect using the measured response and the developed analytical solution (Eq. (6.2.27)). To that end, the measured response $\Delta\hat{\mathbf{p}}$ is convolved with the analytically obtained response $\hat{\mathbf{p}}^{\text{leak}*}$. According to the proposed methodology, the time reversed (or phase conjugated when defined in the frequency domain) analytical response $\hat{\mathbf{p}}^{\text{leak}*}$ is the optimal filter that, when convolved with the measured response $\Delta\hat{\mathbf{p}}$, maximizes the Signal-to-Noise Ratio (SNR) at the location of the defect. The objective function A_{obj} that corresponds to the maximum SNR is defined as the inner product of the phase conjugated analytical response $\hat{\mathbf{p}}^{\text{leak}*}$ and the measured response $\Delta\hat{\mathbf{p}}$:

$$A_{obj} = |\hat{\mathbf{p}}^{\text{leak}*} \Delta\hat{\mathbf{p}}|^2 = \hat{\mathbf{p}}^{\text{leak}*} \Delta\hat{\mathbf{p}} \Delta\hat{\mathbf{p}}^* \hat{\mathbf{p}}^{\text{leak}} \quad (6.4.30)$$

The right-hand side of Eq. (6.4.30) is also identified as the ambiguity function of the Bartlett matched-field processor [170]. To apply the method in practice, the defected water filled pipe is probed from the location of a source/receiver transducer array with a known narrowband pulse $\hat{f}(\omega t)$. The pulse propagates along the pipe and interacts with the leak to produce the response difference $\Delta\hat{\mathbf{p}}$ at the receiver. This concludes the experimental part of the process.

In the next step, the analytical response of the equivalent waveguide $\hat{\mathbf{p}}^{\text{leak}}$ is evaluated from Eq. (6.2.27) for different potential defect locations, $\mathbf{x}_L = (x_L, R)$ in the computer. In particular, the vector $\mathbf{x}_L$ comprises of equally spaced locations along the pipe (i.e., the searching domain). At each location, function A (Eq. (6.4.30)) is calculated: the experimentally obtained response $\Delta\hat{\mathbf{p}}$ is convolved with the analytical response $\hat{\mathbf{p}}^{\text{leak}}$. Subsequently, the x_L value where:

$$x_L = \arg_{x_L} \max \hat{\mathbf{p}}^{\text{leak}*} \Delta\hat{\mathbf{p}} \Delta\hat{\mathbf{p}}^* \hat{\mathbf{p}}^{\text{leak}} \quad (6.4.31)$$

coincides with the location of the defect in the pipe. The operation of the complex conjugate in the frequency domain is equivalent to time reversal in the time domain; hence, the method is only valid under the condition that high-frequency waves in pipelines are time reversible. Such is the outcome of Chapter 5, where it is proven both experimentally and analytically that pressure waves in pipes are indeed time reversible hence, enabling the defect detection technique described in this subsection. The developed technique is tested experimentally on a simulated leak in the hydraulic lab at HKUST. Although it is possible to also evaluate the

physical attributes of the leak by adjusting the S_L parameter in the analytical solution to better match the experimentally obtained response, this is not the focus of this thesis. Of interest is to use the TR-based technique to localize a real leak and, thus, provide with a proof-of-concept. Interestingly, the same technique is also applicable to blockages that can also behave as quasi-passive acoustic sources. In this case, however, an alternative expression analogous to S_L for leaks is required if the goal is to characterize and not only localize the blockage.

6.5 Experimental defect localization

6.5.1 Leak test

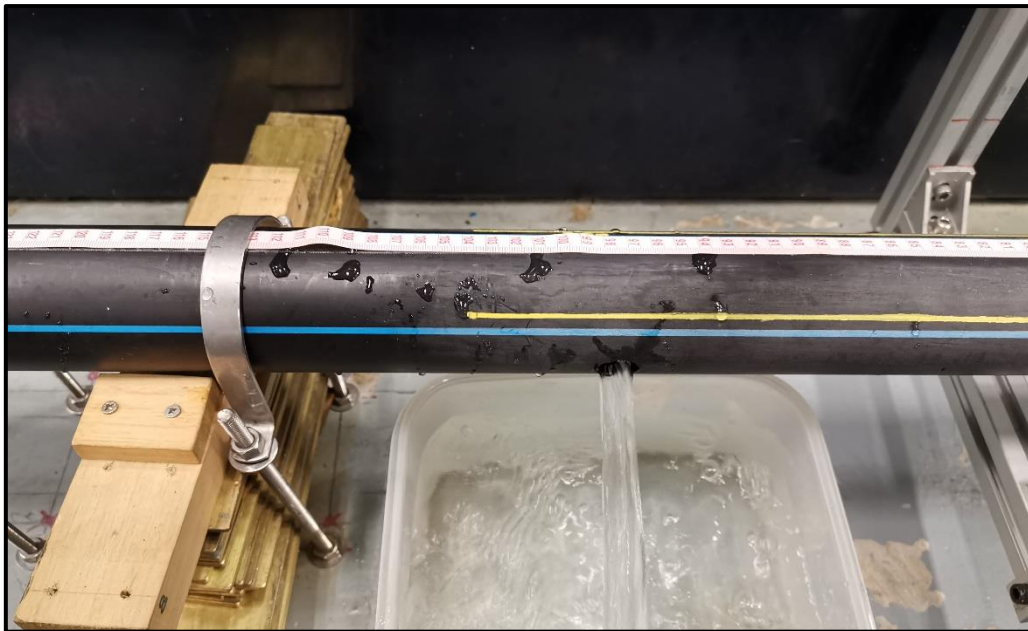


Figure 6.5.1 Close up of the DN50 HDPE leak-containing pipe. The diameter of the orifice is 2 cm, while the pipe is supported to the lab floor on wooden blocks and bolted brackets. The pressure head remains practically constant during testing.

The developed TR-based leak detection technique is tested in practice against a simulated leak. The pipe used for the purpose is the small diameter DN50 pipe of the high-frequency setup at the hydraulic lab at HKUST. The simulation of the pipe leak requires boring an orifice on the pipe wall, hence deeming the pipe unusable following the test. The DN50 HDPE is, therefore, selected, since it finds limited use compared to the larger diameter DN80 HDPE that has been the lab workhorse. Regardless, the DN50 and DN80 HDPE pipes are made from the same manufacturer and present identical mechanical properties, only differing in their cross-

sectional dimensions (DN50 HDPE pipe: $h = 3.8$ mm, $R = 27.7$ mm). The total length of the pipe is 6 m and is connected to two water filled tanks are at its ends. The water level in the tanks is about 1 m from the pipe centreline. Moreover, the pipe runs straight, does not include any junctions along its length and is supported to the lab floor on wooden blocks and bolted brackets, as seen in Figure 6.5.1. The employed transducers are the Brüel & Kjær (Nærum, Denmark) Type 8103 transducers, that are small enough to be placed side-to-side at the same position longitudinally. In essence, the two transducers form an array, where the source is placed along the pipe centreline at $\mathbf{x}_s = (0, 0)$ and the receiver at the pipe perimeter at $\mathbf{x}_m = (0, R)$. The array is realized via a slightly modified 3D printed base that houses both transducers and is designed so that the transducers are placed at their expected positions with minimal error (in the order of mm). Since the acoustic projector is aligned to the pipe centreline, the excited pressure field remains axisymmetric. Ideally, a single transducer would fulfil both roles of the projector and the receiver. Note that such layout would require for the defect reflection to arrive at the transducer after the probing signal emission is complete.

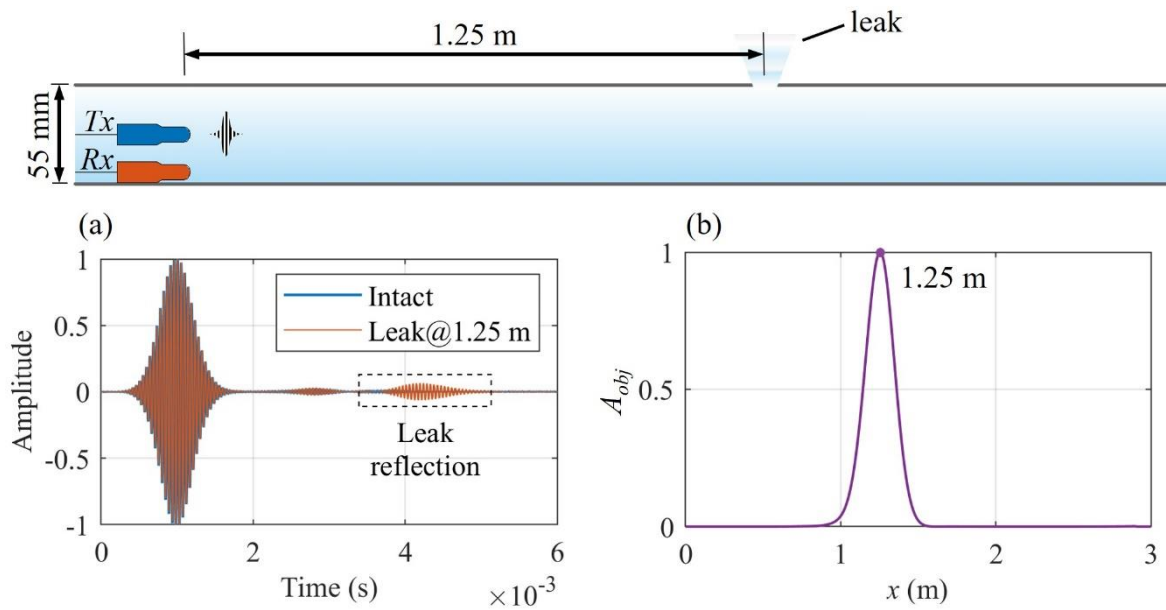


Figure 6.5.2: Experimental leak localization using HFW and the developed TR-based technique. (a) A Gaussian modulated sine pulse is injected in the intact DN50 HDPE pipe first, and the response (in blue) at $t \approx 2.5 \times 10^{-3}$ s reveals a reflector, identified as the pipe support. The measurements are repeated after a leak is created, and the updated response contains the pertinent reflection at $t \approx 4 \times 10^{-3}$ s. (b) The convolution of the isolated leak reflection with the analytical predictions in a range of axial locations produces a peak at the true leak location.

Before boring the simulated leak, the system is tested in its intact state so that a baseline is created. The water in the system is quiescent, and any entrapped air is removed by passing a piston-like block of fabric along the pipe prior to testing. The described transducer array is placed about 1 m from the end of the DN63 HDPE water-filled pipe and pointed towards the pipe midpoint. The employed probing pulse is a Gaussian modulated sine pulse with 22 kHz central frequency, 20% relative bandwidth, and 5 V amplitude. The corresponding probing wavelength is calculated equal to $\lambda = c/f_c = 6$ cm. Both the analytical model and experimentally developed dispersion curves indicate that only the 1st radial model propagates in the frequency bandwidth of the injected pulse, since it presents a cut-off frequency equal to 18.5 kHz. The signal is defined in the computer and converted to the analog domain in the National Instruments (Austin, USA) USB-6356 DAQ I/O board with 16-bit accuracy and at 1MHz sampling rate. The signal is amplified by 30 dB on the Brüel & Kjær Type 2713 Power Amplifier and is then fed to the acoustic projector for emission. Concurrently, the analog electric circuit connected to the receiving transducer starts with the signal acquisition. The pressure signal from the transducer is amplified and conditioned in the Brüel & Kjær Type 2692-A Nexus Charge Amplifier before it is stored in the computer. The acquisition sampling rate is set to 1 MHz. The received response from the intact DN50 HDPE pipe is plotted in Figure 6.5.2a using blue colour and is normalized with respect to its maximum amplitude. First at the receiver first arrives the injected pulse, and since the distance between the projector and the receiver is less than half a wavelength, no dispersion is observed. At time $t \approx 2.5 \times 10^{-3}$ s, a discernible wave packet arrives at the receiver, denoting the presence of a reflector in the system. The observed wavepacket corresponds to the 1st radial mode that is the single mode excited for the given source characteristics. Knowledge of the group velocity of the excited radial mode from the analytical model and the arrival time of the reflection, roughly pinpoints the location of the reflector to the external support of the pipe (Figure 6.5.1). Indeed, due to the nature of the HDPE material that tends to creep and sag, the pipe is locally deformed at the location of the support and the shape of the cross-section is altered. Subsequently, such discontinuity potentially constitutes a reflector to the system. The signal following the support reflection is uneventful.

Next, a circular orifice is bored on the pipe wall at a distance 1.25 m away from the transducer array. The diameter of the orifice is equal to 2 cm, and the water reservoirs at the pipe ends are large enough for the pressure to be constant during the measurement. As water flows through the leak, the system is probed with the same $f_c = 22$ kHz Gaussian modulated pulse and the

signal received by the hydrophone is plotted in Figure 6.5.2a in orange. As is the case for the intact pipe, it is the incident wave p^I that arrives first at the receiver. The reflection from the pipe support is also present, but at time $t \approx 4 \times 10^{-3}$ s an additional pulse is now recorded. The received wave packet can be probably attributed to the leak located 15 cm further to the right of the support (Figure 6.5.1). At this stage, the experimental/physical process is concluded, and the presence of a leak in the system is investigated using the TR-based analytical methodology.

As noted in Eq. (6.4.29), the experimental signal is nullified for $t < 3 \times 10^{-3}$ s so that the leak contribution is isolated, and the occurring signal is transformed to the frequency domain to obtain $\Delta \hat{p}$. Then, the system response $\hat{p}^{\text{leak}}$ is evaluated using the outmost term in Eq. (6.2.27) for different potential leak locations x_L in the excited frequency bandwidth. In the particular test, the searching domain is $x_L = [0, 3]$ m and split into 1 cm steps i.e., the analytical response $\hat{p}^{\text{leak}}(x_L)$ is evaluated 301 times in total. Note that the searching domain is defined from the transducer array position. Besides, a random value for the S_L parameter in the solution of $\hat{p}^{\text{leak}}$ is considered. That is because estimating the leak physical properties is not important at this stage; localizing the leak is. The phase conjugate of the analytical response at each of the 301 locations is calculated and its inner product with the experimentally obtained $\Delta \hat{p}$ is taken, to calculate function A (Eq. (6.4.30)). Figure 6.5.2a shows the normalized function A plotted against the searching domain. A bell-shape curve is plotted that peaks at $x = 1.25$ m from the transducer array. As denoted in Eq. (6.4.31), the maximum value of A corresponds to the leak location i.e., where the SNR is maximized. It is evident that the TR-based defect detection methodology correctly identifies the real leak location with great accuracy. In fact, the expected accuracy is of the order of the probing wavelength. The attainable resolution for the location of the real leak is defined at the full width at half maximum of A : in Figure 6.5.2a that interval is equal to 22 cm. In conclusion, the TR-based method is able to locate leaks in a fluid-filled pipe with the only requirement being prior knowledge on the pipe properties since these are required for the application of the analytical model.

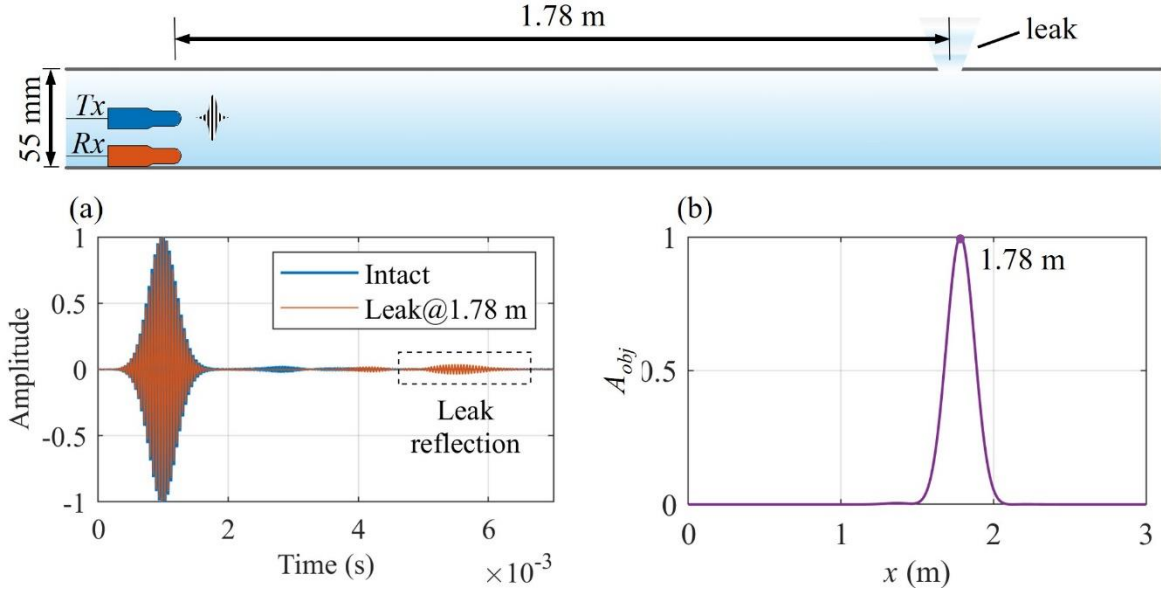


Figure 6.5.3: The transducer array is moved farther from the leak location and the measurements are repeated. (a) The intact case plot (in blue) is identical to that of Figure 6.5.2a, in lack of intact pipe data from the new transducer location. The response of the defected system at the measurement point is plotted in orange, where the reflections from the pipe support and leak arrive at a later time than in Figure 6.5.2a. (b) The true leak location is evaluated using the TR-based method.

For a final test in the same DN50 HDPE pipe system, the transducer array is moved farther from the leak location and towards the pipe end. The distance between the transducer array and the leak is now set to 1.78 m, and the measurements are repeated keeping the remaining parameters of the setup unchanged. The obtained signal from the new measurement is plotted in Figure 6.5.3a in orange colour. Also included in Figure 6.5.2a is the benchmark intact case signal (in blue), identical to that in Figure 6.5.2a. That is evident from the pipe support reflection that arrives at the same time for both plots. Unfortunately, after creating the orifice, the system cannot revert to its prior condition and evaluating the intact case from the new transducer position is impossible. With regards to the leak case, due to the increased distance, the reflections of both the pipe support and the leak arrive later compared to the leak case of Figure 6.5.2a, with the reflected mode amplitudes slightly reduced. In particular, the reflection from the support preceding the leak arrives at time $t \approx 4 \times 10^{-3}$ s, and the reflection from the leak about a ms later at time $t \approx 5 \times 10^{-3}$ s. As with the previous test, this concludes the experimental part.

The next part is performed analytically in the computer. From the experimental signal, only the part that contains the leak reflection p^R is retained. That is, the signal for time $t < 5 \times 10^{-3}$ s is equated to zero. The Fourier transform operation is applied on the formed time signal to obtain $\Delta \hat{p}$. Next, $\hat{p}^{\text{leak}}$ from Eq. (6.2.27) is analytically obtained for $x_L = [0, 3]$ m at 1 cm intervals and $S_L = 1$ over the excited frequency bandwidth. Finally, the inner product of the phase conjugated analytical responses $\hat{p}^{\text{leak}*}$ for different x_L values with $\Delta \hat{p}$ is taken to estimate function A . Figure 6.5.3b depicts the normalized function A plotted against the distance from the transducer array. Evidently, function A peaks at $x = 1.78$ m, coinciding with the real leak location. The width of the bell-shaped curve at half maximum that denotes the resolution is, again, equal to 22 cm. The effect of the probing wave frequency on the attainable resolution is investigated experimentally in a later subsection. In conclusion, the proposed TR-based leak detection methodology is proven in a real setting and provides remarkable resolution, measured in the order of cm. Despite being developed for leaks, the proposed methodology is equally applicable to blockages (i.e., cross-sectional area reductions) that also act as wave scatterers. In the next subsection, an experimental test to localize a blockage is attempted.

6.5.2 Bitumen Blockage test

The most common causes for blockages are calcifications and sediment depositions. Although not instigating water losses, blockages increase demands on pumps and lead to significant energy losses. As a representative for a blockage, however, a small piece of bitumen debris is utilized. Bitumen is used in iron water pipes as lining to prevent corrosion and premature failure of the pipe wall. Over time, however, cracks are formed in the bitumen structure that are a precursor to the eventual demise of the pipe lining. As a result, pieces of bitumen lining are detached and accumulate in the pipe. A pipe lining failure accelerates the appearance of corrosion and leaks. Another severe complication, however, relates to the fact that diluted bitumen is considered toxic and, as such, poses a health risk. It is, therefore, crucial to identify the segments of the water supply network that need to be retrofitted. The applicability of the TR-based pulse-echo method to locate such a piece obtained from an actual installation in Hong Kong. The overall dimensions of the bitumen fragment are $9 \times 6 \times 1.5$ cm (L $\times$ W $\times$ T) (Figure 6.5.4a to c). Before placed inside the pipe, a magnet is attached to the bitumen that allows the control of its longitudinal position along the pipe from an externally mounted magnet.

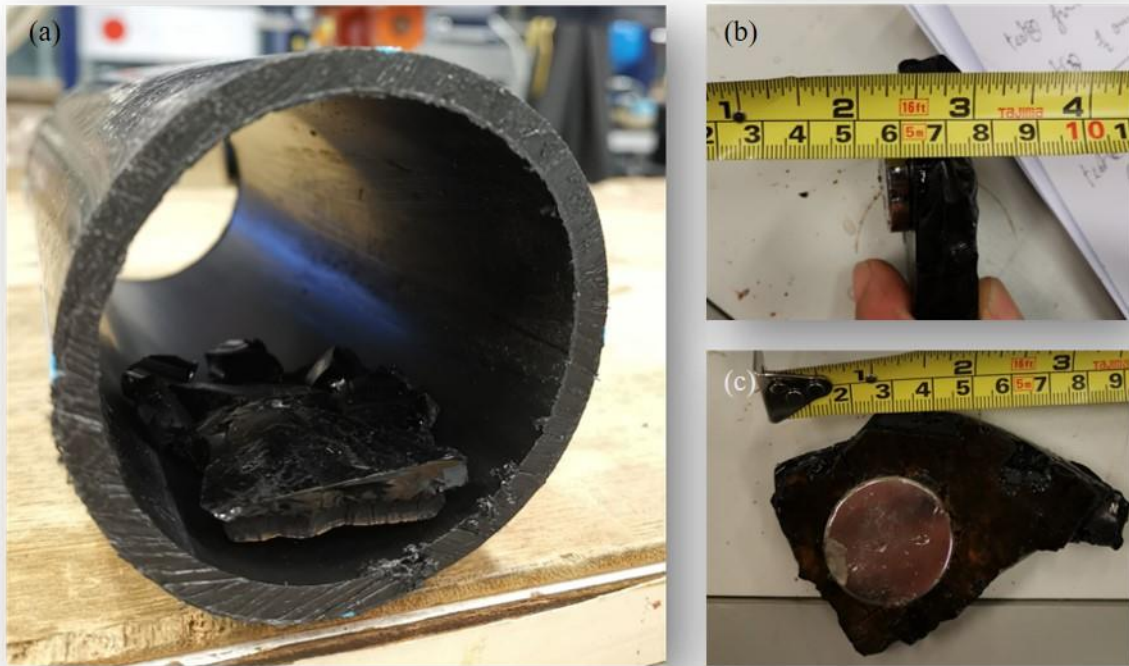


Figure 6.5.4: (a) A bitumen specimen, originating from the failed lining of an iron pipe, is placed inside the DN80 HDPE pipe to simulate a blockage. (b) At its thicker point, the bitumen specimen is about 1.5 cm. (c) The overall length is equal to 9 cm. A magnet is attached to the bitumen piece to facilitate its positioning along the pipe.

In this scenario, the intact DN80 HDPE pipe of the high frequency setup is used. The frontal area of the bitumen piece (including the attached magnet) roughly corresponds to 20% of the pipe cross-sectional area. The setup is changed from the leak test since (i) the DN50 HDPE pipe is no longer intact, and (ii) the DN80 HDPE has a larger cross-sectional area, hence it is more challenging to receive a scattered signal from a blockage that covers a smaller portion. In similar fashion to the leak test, a transducer pair is introduced at one end of the DN80 HDPE pipe. Two Brüel & Kjær (Nærum, Denmark) Type 8104 transducers are mounted on a 3D printed base that can accommodate both at the same longitudinal position. One of the transducers acts as the source and is positioned at approximately the pipe centre, $\mathbf{x}_s = (0, 0)$ m. The other takes up the role of the receiver and is arranged at the pipe perimeter, $\mathbf{x}_m = (0, R)$ m. By placing the source along the pipe centre, the pressure field remains axisymmetric and the employed impedance-based model applicable. The transducer pair is placed about 2 m away from the pipe end to establish $x = 0$ in the upcoming measurements. Prior to testing, the system

is flushed from any air entrapments by passing a piston-like block of fabrics along its length multiple times.

The first measurement is performed for an intact pipe, free from any blockages along its length. A Gaussian modulated sine pulse with 20 kHz central frequency, 20% relative bandwidth, and 5V amplitude is defined at the computer. The probing wavelength is calculated at about $\lambda = c/f_c = 7$ cm. As discussed in Chapter 4, only the 1st radial mode propagates in the selected frequency bandwidth for the DN80 HDPE pipe. The specified signal is converted to analog on a National Instruments (Austin, USA) USB-6356 DAQ I/O board with a sampling rate equal to 1 MHz and 16-bit accuracy. A 30 dB gain is applied to the signal using the Brüel & Kjær Type 2713 Power Amplifier before it reaches the acoustic projector at the pipe centre. The exact moment the signal is emitted, the receiver starts recording the system response. The acquired signal is conditioned on a Brüel & Kjær Type 2692-A Nexus Charge Amplifier and is sampled at the computer at 1 MHz rate and 16-bit accuracy (i.e., analog-to-digital conversion). Figure 6.5.5a depicts the normalized acquired response in blue that tracks the defined Gaussian modulated sine pulse. Slight dispersion of the signal is observed after time $t = 1.5 \times 10^{-3}$ s and is due to the distance between the source and the receiver. The signal remains flat after the signal transmission is complete, indicating the absence of any substantial wave scatterers in the system.

Next, the described bitumen piece is placed in front of the transducer array, at a distance equal to 1.65 m. The same signal is employed, and the system response is shown in Figure 6.5.5a in orange colour on top of the intact case waveform in blue. The two signals are identical up to time $t = 3 \times 10^{-3}$ s, both corresponding to the incident wave response p^I . After that time, however, the signal from the pipe with the bitumen blockage presents a discernible wavepacket, that could be assigned to the reflected wave response p^R . The reflection is comprised of a single wave packet since a sole propagating mode is excited. The experimental part of the defect detection process is now complete.

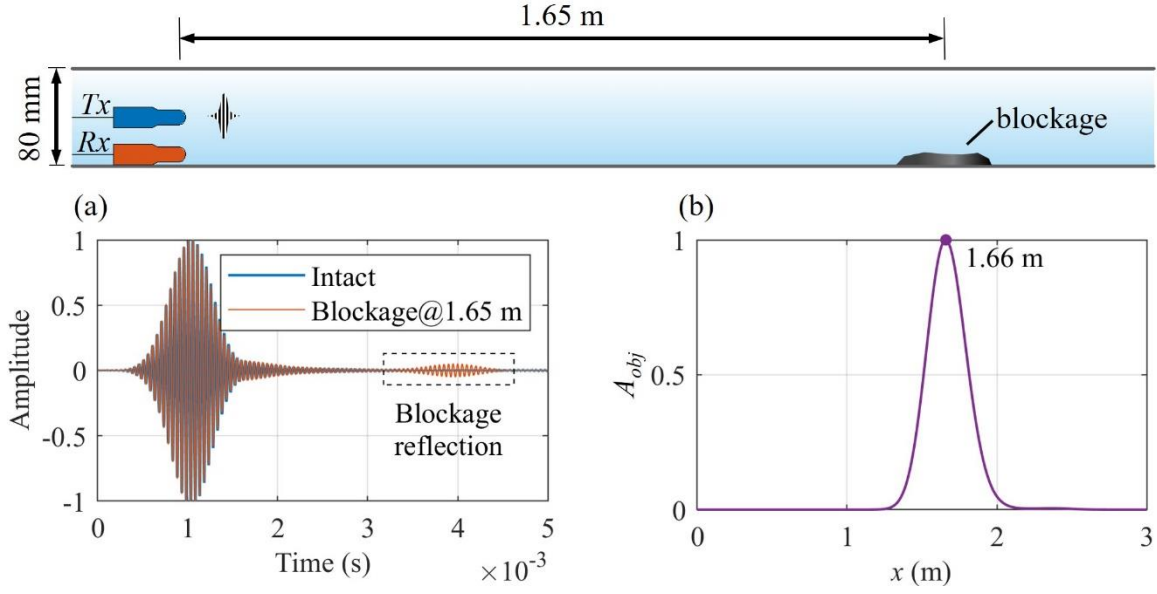


Figure 6.5.5: Experimental blockage localization in a DN80 HDPE pipe using high-frequency pressure waves and TR. (a) The case of an intact pipe is tested first and the acquired signal at the receiver is plotted in blue. A bitumen fragment 1.65 m away from the transducer array induces a reflection that arrives at the receiver at time $t = 3.2 \times 10^{-3}$ s. (b) The reflection is isolated and convolved with the analytical prediction for the blockage effect. The product of the convolution is maximized at $x_B = 1.66$ m, accurately locating the bitumen piece.

From the experimental data, the reflected part of the signal is isolated and transformed in the frequency domain to produce $\Delta \hat{\mathbf{p}}$. Note that $\Delta \hat{\mathbf{p}}$ here is defined as $\hat{\mathbf{p}}^{\text{block}} + \mathbf{n}$, since the scatterer in this case is a blockage and not a leak. Regardless, the effect of both the leak and the blockage in p^R is, in principle, identical. For instance, Eq. (6.2.27) is applicable to both a leak and a blockage, but in the latter case a new parameter S_L needs to be defined. That parameter would describe the physical properties of the blockage, such as its size and acoustic impedance at its surface. In turn, that would allow for the imaging of the blockage in addition to its localization. In this thesis, however, the priority is to localize the defect and not to verify or calibrate the S_L parameter, or any other parameter that would fit the description of a blockage. The localization of the leak/blockage depends on the δ_3 delta function. The S_L parameter, on the other hand, controls the amplitude of the reflection. The analytical response of $\hat{\mathbf{p}}^{\text{block}}$ is

$$\hat{\mathbf{p}}^{\text{block}}(\mathbf{x}_m, \omega_f) = -i\omega_f \hat{f}(\omega_f) \hat{G}_f(\mathbf{x}_B | \mathbf{x}_s) \hat{G}_f(\mathbf{x}_m | \mathbf{x}_B) \quad (6.5.32)$$

Equation (6.5.32) is essentially Eq. (6.2.27) when the second term on the right-hand side that corresponds to the scattered wave is considered only, S_L is set to 1, $\mathbf{x}$ is set to $\mathbf{x}_m$ (i.e., the

measurement location), and $\mathbf{x}_L$ is replaced by $\mathbf{x}_B$. In the particular experiment, the search domain is defined as $x_B = [0, 3]$ m at 1 cm increments. The calculated analytical response $\hat{\mathbf{p}}^{\text{block}}$ is then convolved (i.e., phase conjugated and multiplied) with the experimentally obtained $\Delta\hat{\mathbf{p}}$ to produce the objective function A (Eq. (6.4.30)) plotted in Figure 6.5.5b against the range. The maximum of function A is observed at $x = 1.66$ m that mirrors the true blockage location. The resolution, defined as the full width at half the max amplitude, is equal to 30 cm and is comparable to that of the leak cases where a probing frequency of slightly higher frequency is employed (20 vs 22 kHz). Note that at 22 kHz, the resolution is indeed better.

Finally, it should be noted that besides the bitumen piece, the technique was experimentally tested on many other simulated blockages from random objects, such as a soda can, a small piece of bamboo trunk, a concrete cube, packaging foam. In each case, the developed TR-based methodology was able to localize the object. The bitumen piece is chosen as it presents the most challenging case in terms of size, and also mimics a plausible scenario in practice (i.e., localizing pipe lining failure).

6.5.3 Frequency effect on resolution

When used for imaging, the resolving capabilities of acoustic waves are inversely proportional to their wavelength. The attainable resolution and precision in predicting the defect location is tested experimentally, by comparing the TR-based model results when probing pulses of different central frequencies f_c are used. The setup employed is based on the intact DN80 HDPE pipe, that was previously used for the bitumen blockage experiment. Instead of the bitumen blockage, however, an iron cylinder that completely covers the cross-section of the pipe. That is to remove any uncertainties concerning the interaction of the pressure field with a blockage that only partially covers the pipe cross-sectional area. For instance, different modes or even the same mode at different frequencies might produce a stronger or weaker scattering depending on the strength of the axisymmetric pressure field at the pipe perimeter where e.g., the bitumen piece would reside. The iron cylinder is placed at a distance $x = 1.95$ m from the sensor array, that comprises of two Brüel & Kjær (Nærum, Denmark) Type 8104 transducers. Of the two, one is the acoustic projector placed at about the pipe centre and the other is the receiver at R . Both share the same longitudinal position along the pipe, 2 m from its closest end. Expressed as input to the analytical model, the position of the acoustic source is at

$\mathbf{x}_s = (0, 0)$ and that of the acoustic receiver at $\mathbf{x}_m = (0, R)$. Since the source is at about the pipe centre, the pressure field remains axisymmetric.

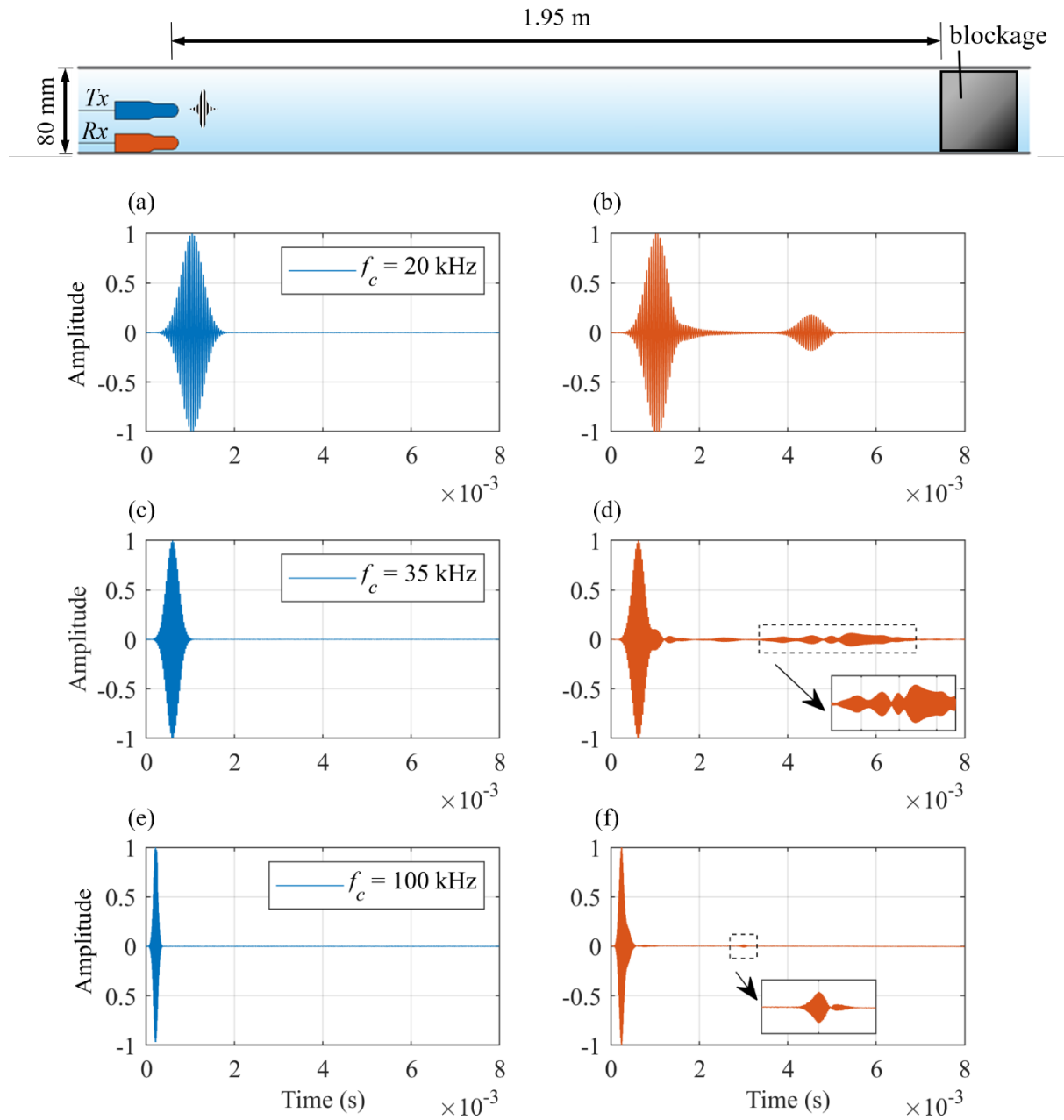


Figure 6.5.6: Pulse-echo tests utilizing probing waves of increasing central frequencies.

The measurements are repeated three times, each with a Gaussian modulated sine pulse of increasing central frequency f_c : 20 kHz, a 35 kHz, and a 100 kHz. For all cases, the relative bandwidth is set at 20% of the central frequency. Subsequently, the duration of the pulses decreases with increasing frequency. The estimated wavelengths λ are 7, 4, and 1.5 cm for the

20, 35, and 100 kHz probing pulses, respectively. The normalized source waveforms are plotted on the left-hand side of Figure 6.5.6, while on the right-hand side are the pertinent normalized recorded responses at the receiver. In particular, Figure 6.5.6b shows the response of the system to a 20 kHz pulse. First at the receiver arrives the incident wave p^I due to the configuration of the setup. The scattered pressure signal from the blockage arrives at a later time $t = 4 \times 10^{-3}$ s and comprises of a single wave packet. The reason is that at 20 kHz, only one mode (i.e., the 1st radial) is excited in the water filled DN80 HDPE pipe, as explained in Chapter 4. Conversely, in the excited bandwidth for the 35 kHz pulse, two (i.e., the 1st and 2nd radial) modes propagate in total. In Figure 6.5.6d, the incident pressure wave that reaches the receiver first appears slightly dispersed, as it differs from the $f_c = 35$ kHz probing pulse plotted in Figure 6.5.6c. That is to be expected, as the distance between the source and the receiver is equal to $R \sim \lambda$, thus enough for the wave to disperse. The reflected pressure wave from the blockage arrives slightly earlier than that of the 20 kHz (Figure 6.5.6b) and approximately at time $t \approx 3.5 \times 10^{-3}$ s, as the group velocity of the 1st mode is increased. Moreover, the scattered wave comprises of multiple wavepackets that is challenging to associate to a particular propagating mode. While a single propagating mode and the knowledge of the group velocity makes it straightforward to estimate the distance of the scatterer from the measurement location, the same task is complicated when high frequency probing pulses are used to excite multiple modes. Therefore, unless the developed analytical model is applied, estimating the defect location is practically impossible. Finally, the short 100 kHz pulse is injected to the system (Figure 6.5.6e). The response of the system at the receiver is plotted in Figure 6.5.6f. As the excited frequency increases, additional modes propagate. However, at high frequencies far from the mode cut-offs, the group velocity asymptotically reaches the wave speed in water c_f . Hence, modes that present their cut-off frequencies at lower frequencies that the excited bandwidth propagate almost together at a group velocity that approaches c_f . For that reason, in the reflected pressure signal p^R , time spreading appears less severe than, let's say, Figure 6.5.6d. The reflection signal also appears at x_m earlier than all other cases tested ($t \approx 3 \times 10^{-3}$ s), precisely because the group velocity of the propagating modes increases with frequency. The diminishing amplitude of the reflected pressure signals as the probing wave frequency increases is attributed to the reduced duration of the pulses, that consequently decreases the amount of energy injected into the system. Following the experimental measurements, the developed analytical solution is employed to localize the scatterer for each case.

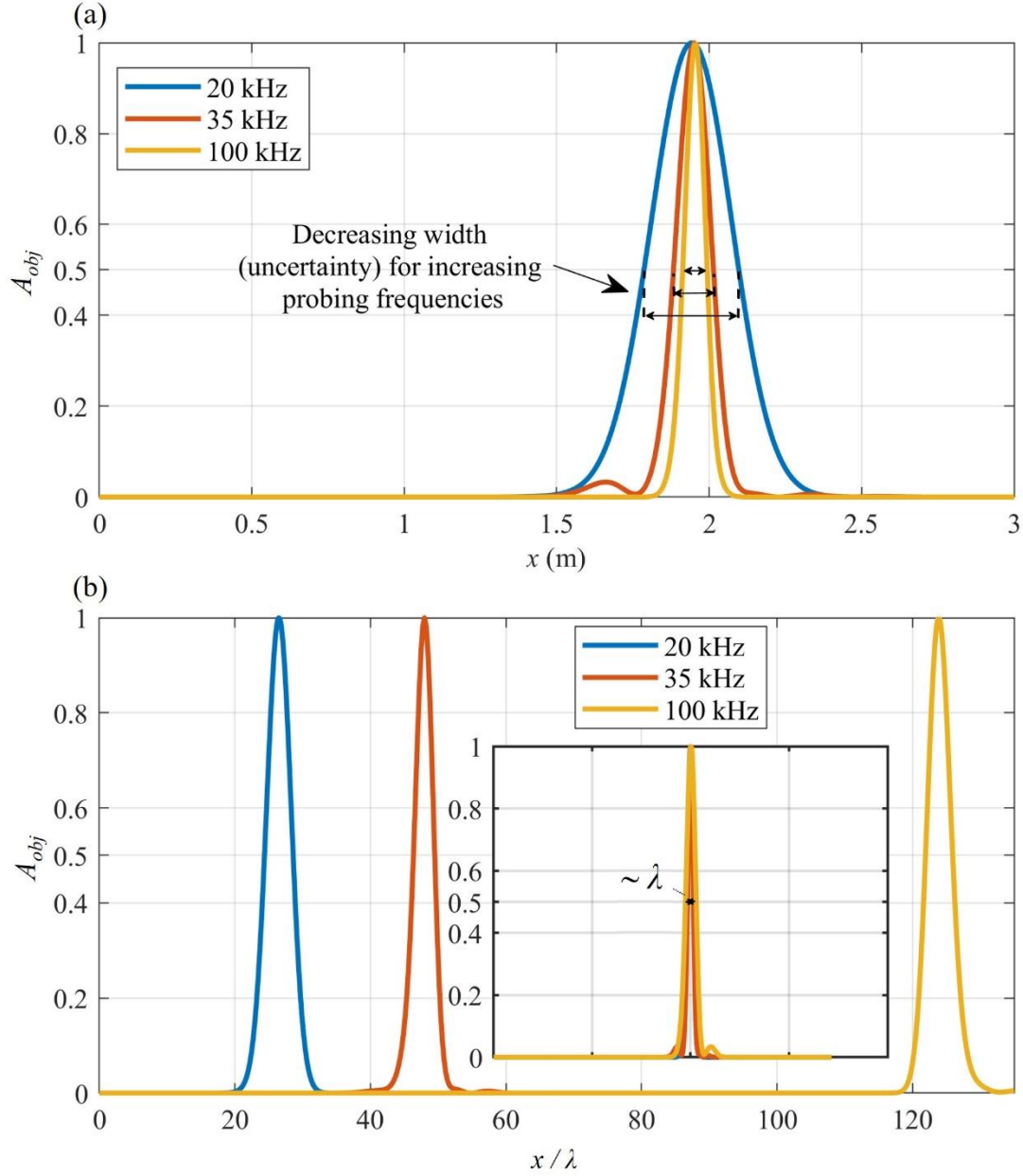


Figure 6.5.7: (a) Probing pulses of increasing central frequency provide a better resolution in the localization of the blockage. The width of the bell-shaped function at half the peak amplitude is effectively reduced at higher frequencies. (b) When range x is normalized with respect to the pertinent wavelength λ of each case, the curves become self-similar. The width of the distribution at half the maximum amplitude is for all cases is in the order of the probing wavelength λ .

In the recorded responses, the incident pressure signals p^I are nullified to isolate the reflected pressure signals p^R and the Fourier transform is applied to obtain $\Delta \hat{p}$ from each case. The analytical responses $\hat{p}^{\text{block}}$ are calculated from Eq. (6.5.32) where each case requires a different

$\hat{f}(\omega)$ to describe the probing pulse. Regardless, the search domain is $x_B = [0, 3]$ m at 1 cm increments for all scenarios. The calculated analytical response $\hat{\mathbf{p}}^{\text{block}}$ is then convolved with the experimentally obtained $\Delta\hat{\mathbf{p}}$ to calculate the objective function A for each case considered. The objective functions are plotted in Figure 6.5.7a, and all peak at $x = 1.95$ m that coincides with the true location of the cylindrical block. However, the width of the function A curves at half the maximum amplitude decreases for an increasing probing wave frequency, denoting reduced uncertainty in the localization of the defect. It is, therefore, shown that the achievable resolution is a function of the probing pulse frequency. In Figure 6.5.7b, the curves of Figure 6.5.7a are plotted, but the range x is normalized with respect to the pertinent wavelength in the fluid λ for each case. The curves appear self-similar and collapse to a single curve, further demonstrating that the achievable spatial resolution indeed scales with the frequency of the pulse. The width of the bell-shaped curve at half the maximum amplitude is in the order of the probing wavelength λ .

6.6 Conclusions

This chapter follows on from the previous to leverage time reversal in water supply pipelines for leak detection. To that end, an analytical solution to the interaction problem between a probing wave and a leak is derived. Specifically, the leak is modelled as a mass sink following from the fluid governing equations of Chapter 2, while the effect of the elastic pipe wall is described according to the impedance-based model of Chapter 3. The proposed solution involves two distinct parts. Initially, the pipe is probed experimentally with an acoustic pulse to extract the response of the leak. Subsequently, the obtained response is convolved with the results from the analytical solution. The validity of time reversal in water filled pipes guarantees that the convolution product is maximized at the location of the leak, enabling its localization.

The proposed technique is successfully tested in the hydraulic lab at HKUST to detect a 2 cm diameter leak on the pipe wall surface of a DN50 HDPE pipe, positioned about 1.8 m from the transducer array. The same analytical solution is also used to localize a small blockage in the DN80 HDPE of the setup in the form of a failed bitumen lining piece from a real-world installation. Finally, experimental tests on localizing a blockage using acoustic pulses of different central frequencies show that the attainable resolution is proportional to the probing

wavelength, Thus, the obtained results explicitly demonstrate the benefits of using high frequency waves for enhanced localization accuracy.

The objective of this thesis is to provide a novel proof-of-concept for TR-based defect detection in water pipelines using high frequency waves. Consequently, the focus is placed on successfully localizing and not characterizing the defect. The developed analytical solution can be used interchangeably on a leak or a blockage because the term that describes the location of the defect is different to that describing its characteristics (i.e., size, impedance). For the latter, the orifice equation is used in this chapter to provide a possible expression. Regardless, the verification of the orifice equation or the development of an equivalent expression that applies to blockages is a topic for future research. In this chapter, Gaussian modulated sine pulses are employed that, although allowing for the exact definition of the excited bandwidth and better understanding of wave propagation, do not carry much energy and are rather inefficient for acquiring reflections from defects. It is, therefore, a worthwhile future pursuit to apply other types of probing waves, such as Linear Frequency Modulated (LFM) pulses, that have the potential to yield a better performance when probing a defected system. The experimental tests presented herein are performed in intact pipes. In practice, however, any discontinuity along the pipe (e.g., junctions, hydraulic devices, etc.) is a probable reflector of acoustic waves. Hence, it is important that an initial benchmark state is established so that known scatterers are excluded when diagnosing the system or that a special probing signal that avoids such is designed. Finally, the acoustic projector is always placed along the pipe centreline to excite an axisymmetric pressure and comply with the impedance-based pipe wall expression. An interesting experiment would place the acoustic source along R to also excite asymmetric (azimuthal) modes and evaluate the incurred error from the analytical solution when localizing a defect.

The developed analytical solution for the wave/defect interaction is only effective under the condition that high-frequency wave time reversal in pipes holds. In the previous chapter, time reversal has only been checked against a pipe system where water is quiescent, thus different to typical water pipe installations that involve turbulent flow. Turbulent flows can potentially break wave time reversal or introduce focalization errors due to the increased viscous losses and instabilities. By extension, the analytical solution could be rendered impractical or require adjustments. In the next chapter, wave time reversal is tested against pipe flow, and useful conclusions are extracted.

Chapter 7

TR fidelity against flow and instabilities

7.1 Introduction

In previous chapters, the proof-of-concept results are demonstrated in an ideal environment, where water in the pipes is stationary and uniform, and experimental setup parameters remain constant. Consequently, the essential properties of time symmetry and spatial reciprocity that enable TR in the waveguide are satisfied, despite the presence of damping. To illustrate, consider two points A and B along a water filled pipe. A probing signal is emitted from A, and the response at B and time t is $G(B, t|A)$, where G is the Green's function of the waveguide. Then, the response at B is time reversed to yield $G(B, -t|A)$ and reemitted to complete the TR step. At A, the recorded response is $G(A, t|B) \otimes G(B, -t|A)$, where $\otimes$ denotes the convolution operator. Provided that the time symmetry (i.e., $G(t) = G(-t)$) and spatial reciprocity (i.e., $G(A|B) = G(B|A)$) properties are supported by the waveguide, the convolution product is maximized at A, and TR can form the basis of a powerful defect detection technique. However, real world UWSS incorporate processes such as flow, turbulence, vortices, transient instabilities, flow unsteadiness, dissipation, bends, junctions, and other discontinuities that violate these TR requirements. Similarly, experiments in the ocean show that TR focusing is severely affected by the time-dependent fluctuations in the medium, especially when high frequency (i.e., with wavelengths comparable to the size of internal waves) probing waves are employed [171]. In [172], it was found that even a single vortex can compromise time reversal invariance. Other fundamental studies [173, 174, 99, 175] have also shown that processes such as flow and instabilities incur predictable shifts to the focal point of TR that can be countered,

as long as the interaction of these processes with acoustic waves is known. A relevant example is provided in [176] where understanding of the interaction between the acoustic wave and vortex is leveraged to develop a TR-based imaging technique for the velocity field in closed conduit water flow of rectangular cross section. An interesting technique, named the nonreciprocal (NR) TR has been proposed and applied in the ocean to address the consequences of current flows to the TR process in [177]. This technique involves a modification to the TR step, where the TR source coincides with the original source instead of the receiver. Results from tests in the ocean against a flowing medium have proven the effectiveness of NR TR compared to traditional TR.

Nonetheless, the general problem of TR invariance in turbulent and non-uniform flow in pipes remains uncharted. This chapter presents and discusses a series of experimental tests in water filled pipes that examine TR under conditions that breach time symmetry and spatial reciprocity for the first time. Specifically, the fidelity loss of the TR process is evaluated under both steady and unsteady turbulent flow in the pipe by means of the TR field spatial focusing and the similarity of the recreated signal to the original source. In addition, the dependence of the observed phase shift to the probing signal frequency is investigated. NR TR is compared to traditional TR in terms of the temporal compression degree of the TR field for a pipe under flow conditions. Furthermore, the effectiveness of NR TR against unsteady pipe flow is tested. To identify any changes in the water filled pipe wave propagation characteristics when under flow, the experimental dispersion curves are developed for various flowrate levels. Besides, the responses of a pipe under flow to a wideband signal in space enable the study of the attenuation as a function of range and frequency.

Regarding instabilities along the wave path, the validity and accuracy of TR is tested by performing the TR step in a different waveguide than that of the forward step, in a series of tests termed “sister pipe” experiments. The different waveguide is a pipe (i) of the same nominal characteristics, (ii) that contains two junctions in-between the transducer pair, and (iii) of entirely different geometrical and mechanical properties. Finally, HF wave propagation is tested along pipe bends, and the pertinent experimental dispersion curves are compared to these from an equivalent straight pipe.

7.2 Time reversal under flow conditions

7.2.1 Forward step in stationary water and TR step under unsteady turbulent flow

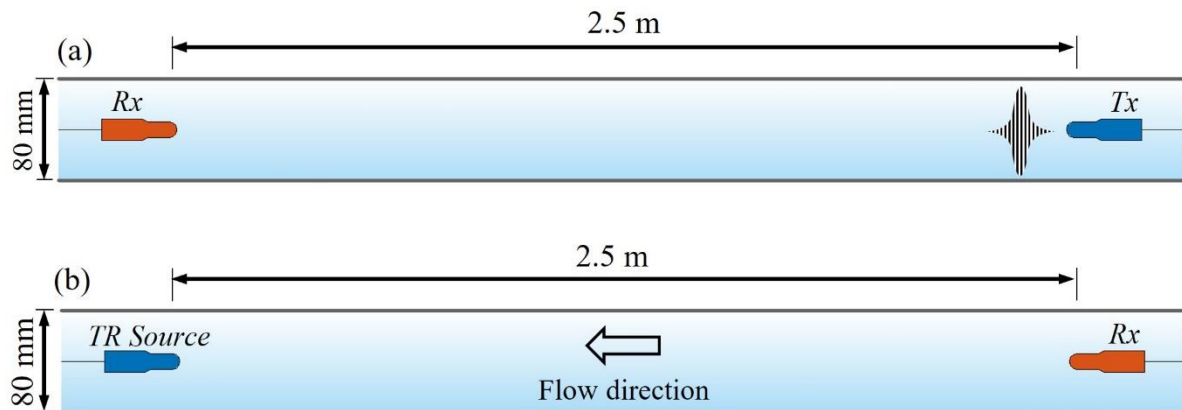


Figure 7.2.1: Schematic of the setup: (a) The forward propagation step is carried out in quiescent water, with the transducers placed 2.5 m apart. (b) The TR step uses the obtained response from the forward step, but flow in the waveguide is now included.

According to the TR test procedure described in Chapter 5, a transducer pair (two Brüel & Kjær Type 8104 transducers) is inserted and mounted along the centreline of the intact (uninterrupted) DN80 HDPE pipe. The distance between the two transducers is set to 2.5 m, and one acts as an acoustic source while the other is the receiver (Figure 2.2.1a). In the forward wave propagation step, the water contained in the pipe is quiescent, and a Gaussian modulated sine pulse of central frequency $f_c = 35$ kHz is emitted from the projector. The frequency bandwidth of the source signal at 1/100 the peak spectrum amplitude is defined in the interval [24.5, 45.5] kHz. The excited bandwidth is rather broad (i.e., 60% relative to the central frequency), so that dispersion effects are accentuated. Moreover, the minimum wavelength is equal to $\lambda_{\min} \sim 3$ cm. In the excited frequency range, two modes propagate in total: the 1st and 2nd high-order radial. The sampling rate is set equal to 1 MHz for both signal transmission and acquisition tasks. Figure 7.2.2 depicts the signal plots that comprise the forward step: (a) presents the source waveform, and (b) shows the recorded response. Each of the depicted signals are normalized with respect to their maximum values (i.e., their amplitude is “1”).

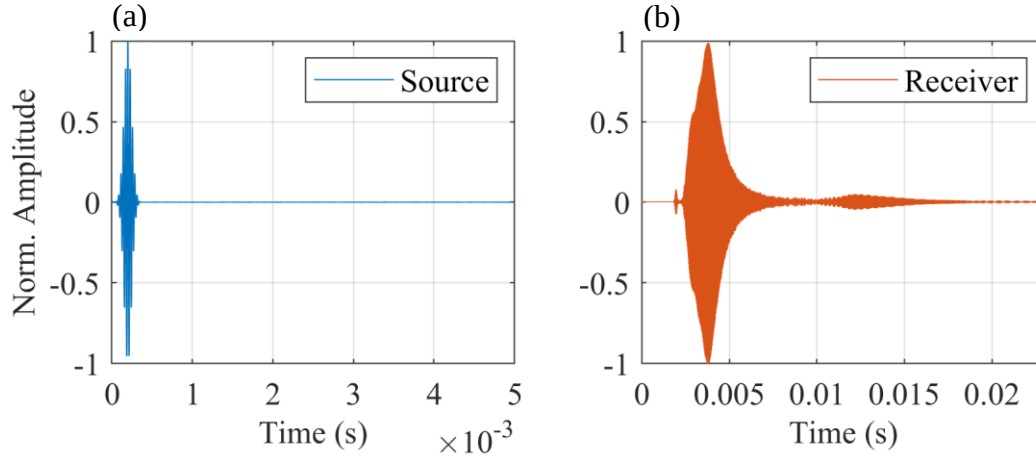


Figure 7.2.2: Forward wave propagation of a Gaussian modulated pulse of $f_c = 35$ kHz and 60% relative bandwidth over 2.5 m along the intact DN80 HDPE pipe in quiescent water. (a) The source waveform, and (b) the acquired response.

Typical for bounded waveguides probed with short wavelengths, the recorded response exhibits dispersion attributed to multipath propagation effects. As discussed in Chapter 5, to undo the effects of dispersion, the TR process is performed. However, contrary to previous tests, the time reversal step is carried out under flow conditions created via an externally attached pump to the system whose inflow and outflow are attached to the DN80 HDPE pipe ends. The flowrate Q is evaluated with an ultrasonic flowmeter installed at the downstream end of the pipe and is found to fluctuate between $0.9 \sim 1$ l/s. Given that the pipe cross-sectional area is equal to $A = 4.85 \times 10^{-3} \text{ m}^2$, the mean flow velocity ($\bar{V} = Q/A$) is equal to $0.18 \sim 0.2 \text{ m s}^{-1}$, and the Mach number $Ma (= \bar{V}/c_f)$ is $\sim 10^{-4}$. Besides, the Reynolds number is estimated equal to $Re \sim 14000$. Notably, the pump used is unable to maintain a constant flowrate; hence, the time reversal step is performed under *unsteady* turbulent flow.

In the TR step, the roles of the transducers are inverted, and the source of the forward step becomes the receiver while the receiver becomes the source (Figure 7.2.1b). The position of the transducers remains unaltered, thus the in-between distance is equal to 2.5 m. Since flow is now included in the system, it should be clarified that the TR source is at the downstream, and the receiver at the opposing upstream end of the pipe. To perform the TR step, the source emits the time-reversed system response to the forward propagation, and the TR pressure field is evaluated at the receiver whose location coincides with that of the source in the forward step. It is paramount to highlight that the Green's function of the waveguide during the backpropagation stage is different to that in the pertinent forward stage. That is because

turbulent flow, besides damping, presents turbulent vortices/structures of equivalent length scale to the probing wave and act as wave scatterers. Obviously, the presence of flow in the TR step explicitly breaks the spatial reciprocity of the waveguide since $G(A|B) \neq G(B|A)$, where A and B are the positions of the two transducers. For the Green's functions to be equal, the vortex field would have to be reversed during the TR step; that is simply not realistic. Hence, the refocusing ability of the TR process is compromised. It is of interest, however, to evaluate the refocusing error due to the flow conditions. The raw signal results of the TR step are presented in Figure 7.2.3, where each of the source and response waveforms is normalized by their pertinent maximum values.

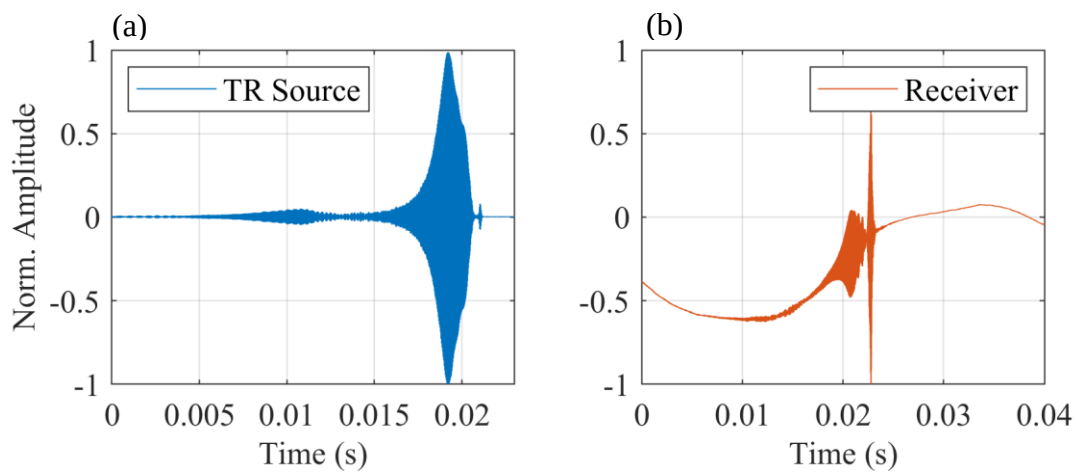


Figure 7.2.3: Time reversal step for the forward propagation step presented in Figure 7.2.2.

(a) The source signal is the time-reversed system response from the forward step (i.e., Figure 7.2.2b), and (b) the acquired response at the location of the forward step source (i.e., the recreated original Gaussian modulated sine pulse)

The product of the time reversal process (Figure 7.2.3b) includes a visible low frequency component that is to be attributed to the inconsistent pump flowrate. Unfortunately, the sampling rate (1 Hz) of the utilized flowmeter cannot resolve this low frequency component in the signal, and thus verify the hypothesis with certainty. Regardless, the result emulates the expected behaviour of a real-world water distribution system with variable consumption outflows. The observed low frequency component in the TR response may be removed with a bandpass filter conforming to the bandwidth of the injected signal and applied during post-processing using MATLAB's "bandpass" function, to obtain the waveform of Figure 7.2.4.

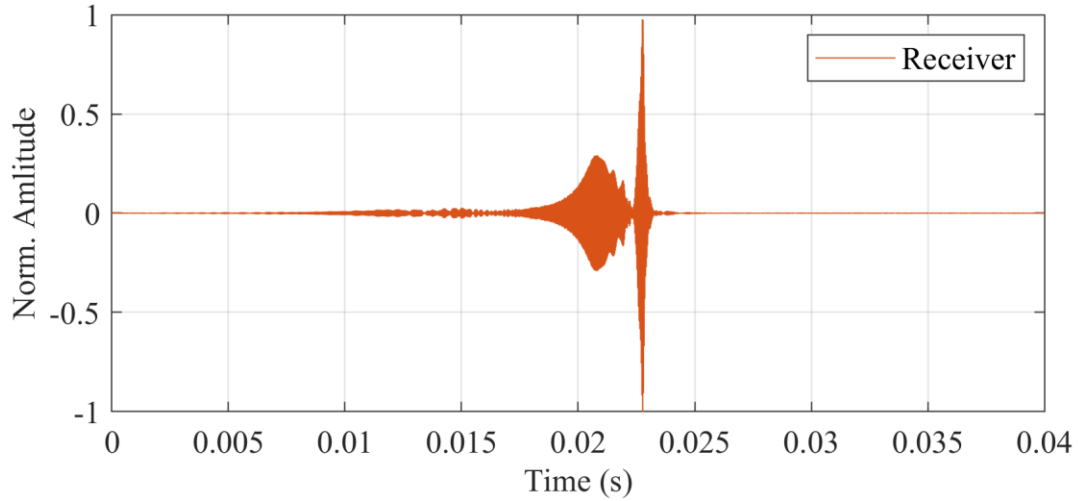


Figure 7.2.4: Filtered TR response from Figure 7.2.3b to remove the low frequency component due to the inconsistent pump.

From Figure 7.2.4, it is obvious that the characteristic symmetry in the recreated signal that is represented by the equidistant sidelobes from the main pulse is lost. However, this is not a conclusive indication that the time reversal does not hold. Prior studies indicate that fluid flows and currents cause shifts and deviations from the expected focal point [143, 176]. Therefore, another criterion to be tested in this particular setting is spatial focusing, to detect whether pulse compression occurs at a different, shifted location in the TR field than expected. Consequently, multiple measurements of the TR pressure field are performed along the DN80 HDPE pipe over a total range of 2 m along the pipe centreline, starting at 1.5 m from the TR source and repeated every 2 cm (Figure 7.2.5). The total number of individual number of measurements, that also corresponds to the total number of times the TR step is carried out, is 101. The rather short 2 cm interval is slightly less than the minimum probing wavelength $\lambda_{\min}$ so that changes of the TR pressure field in space are adequately resolved. Ideally, the sampling in space should be done at $\lambda_{\min}/2$ increments that relate to the diffraction limit. Still, 2 cm spatial resolution beats the $\lambda_{\min}$ and is, therefore, an acceptable compromise to increasing the number of measurements to double.

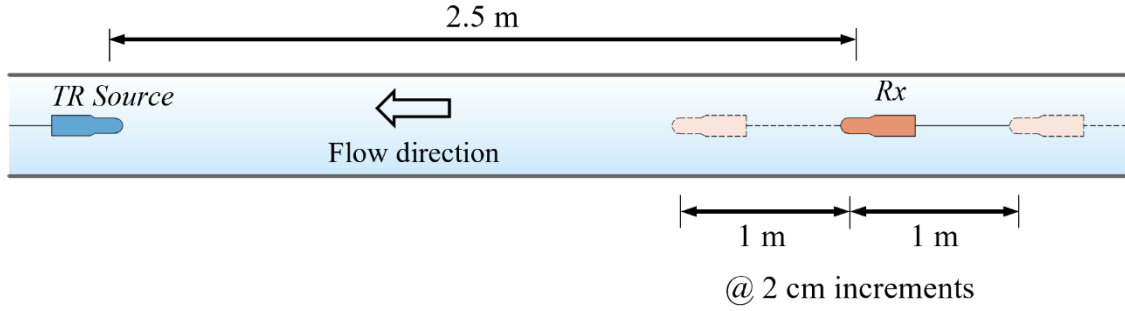


Figure 7.2.5: The TR field is scanned over a total length of 2 m (i.e., 1 m either side of the original source location) at 2 cm increments to identify the refocusing point.

Figure 7.2.6 presents the amplitude for the filtered (i.e., using a bandpass filter as in Figure 7.2.4) TR field response at each of these 101 locations in dB scale relative to the overall highest amplitude. The dashed red line denotes the location of the original source and the expected maximum point for the TR field amplitude under quiescent water conditions. Contrary to the results presented in Chapter 5, where in quiescent water the TR field response is maximized uniquely at the location of the original source of the forward step, only a local maximum is observed in the vicinity of that location. In fact, the maximum amplitude response is closest to the TR source, i.e., at $x = 150$ cm. In this particular test, however, the effect of damping due to flow is substantial and the TR field response amplitude decays at a higher rate than tests in quiescent water. As shown in Figure 7.2.6, the amplitude of the TR pressure field drops by a substantial 60 dB within 2 m of distance. In effect, the increased amplitude of recreated pulse close to the original source location is concealed due to damping, and the local maximum around $x = 250$ cm could indeed present a focus point. Besides, the local maximum is not recorder at exactly $x = 250$ cm from the TR source, but a bit farther at 252 cm: thus, indicating a potential shift of the focal point due to the fluid flow.

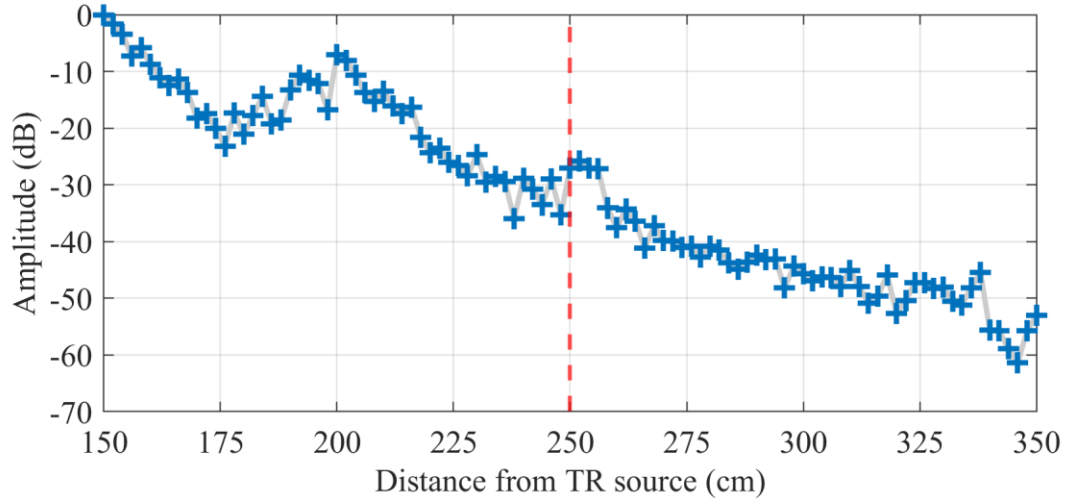


Figure 7.2.6: The maximum amplitude of the TR field is plotted against the range measured from the TR source. The red vertical dashed line indicates the location of the source in the forward step, and the expected peak for the TR field amplitude under quiescent conditions. However, the included fluid flow induces significant damping to the TR pressure field, thus masking the peak at the vicinity of the vertical dashed red line. Instead, there appears a local maximum at $x = 252$ cm from the source, indicating a shift of the focal point.

Maximum amplitude is only one metric to consider. The similarity of the recreated to the original probing pulse could also be applied. To further investigate, the normalized cross-correlation product between the original pulse (Figure 7.2.2a) and the TR field response is calculated for each of the same 101 locations. The cross-correlation coefficient is calculated from [178, 179]:

$$\hat{R}_{xy, \text{coeff}} = \max \frac{1}{\sqrt{\hat{R}_{xx}(0)\hat{R}_{yy}(0)}} \hat{R}_{xy} \quad (7.2.1)$$

where $\hat{R}$ is the cross-correlation function, subscript x denotes the time sequence of the original pulse, and y that of the TR field response at each measured location. For instance, $\hat{R}_{yy}(n=0) = y[n] * y[-n]$ corresponds to the autocorrelation (i.e., cross-correlation with itself) of signal y , at zero lag ($n=0$). The cross-correlation coefficient obtains values from 0 to 1, with 1 indicating a perfect correlation. The results for between the original Gaussian modulated sine probing pulse and the TR field are presented in Figure 7.2.7.

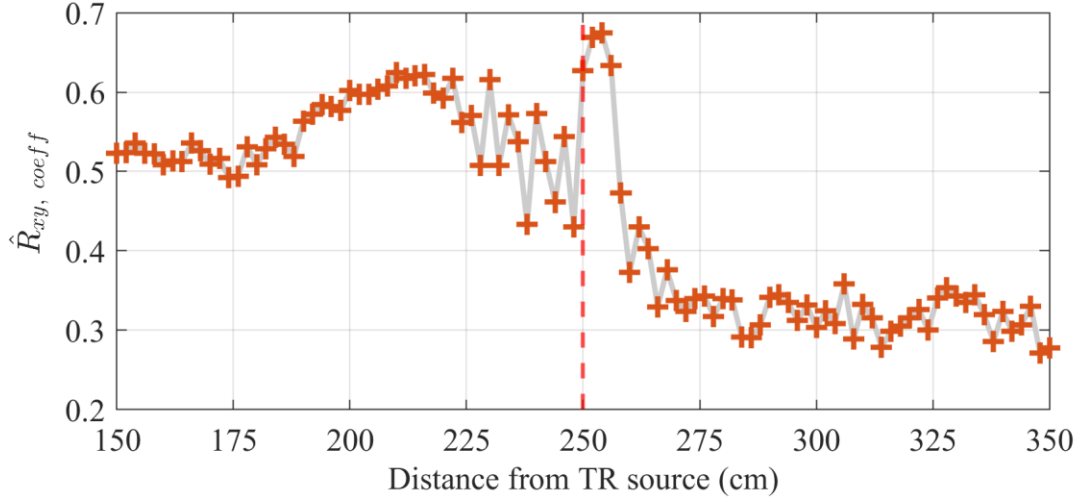


Figure 7.2.7: Cross-correlation coefficient calculated between the original probing pulse and the TR field response as recorded at 101 different locations along the DN80 HDPE pipe. The vertical red dashed line corresponds to the original source location and is the expected peak location. However, the focal point is found 4 cm farther from the original source location, indicating a spatial shift to the TR process due to the pipe flow.

The peak of the plot in Figure 7.2.7 ($\hat{R}_{xy, coeff} \sim 0.7$) is only found in the vicinity of the original source location, at about 4 cm further from it. The “shift” corresponds to roughly twice the probing wavelength λ . This result reinforces the prior observation that the pipe flow induces a deviation to the focal point of the TR from the expected location. Although the focal spot shift in this particular case appears small and inconsequential to the detection of fully-fledged defects, it could be proven crucial when HF-TR imaging is attempted. In such case, the focal spot shift should be studied under different flow conditions, probing frequencies, and ranges that can potentially exacerbate the issue. Regardless, the TR appears to refocus (even though not at the expected location) when the similarity of the TR field to the original pulse is used as metric, despite the severe flow induced damping. In this set of measurements, the forward test is performed in quiescent water, while for the backpropagation TR stage fluid flow is introduced to the system. In the next experiment, the validity and robustness of TR is investigated when both the forward and backpropagation steps are conducted under unsteady turbulent flow conditions.

7.2.2 Forward and TR steps under unsteady turbulent flow

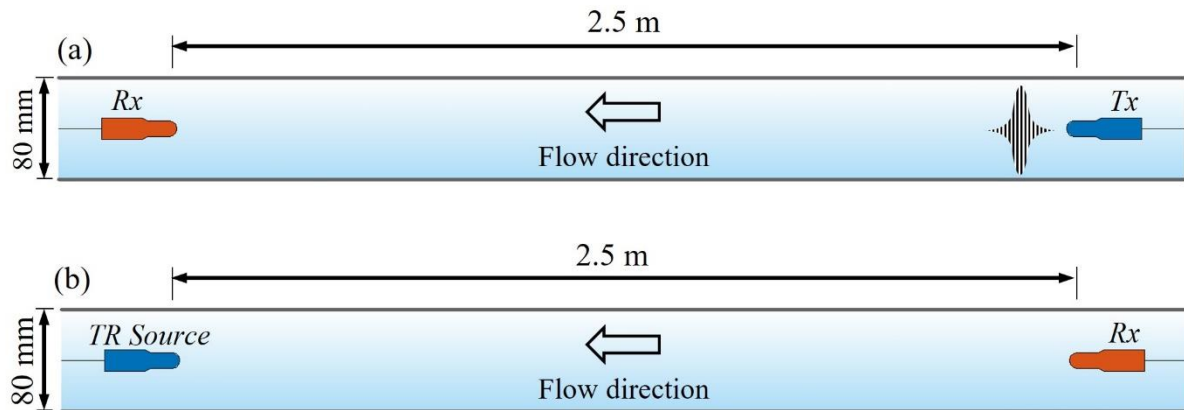


Figure 7.2.8: In this setup, both (a) the forward and (b) time reversal tests are performed under flow conditions. For the forward case, the wave propagates with the flow, while in the TR step against the flow. Naturally, the Green's functions that describe wave propagation are different between steps (a) and (b); thus, spatial reciprocity is violated.

The water flow conditions that existed during the backpropagation (i.e., TR) step of the previous set of measurements (subsection 7.2.1) are maintained, and a new experimental test is devised for the intact DN80 HDPE pipe. The flowrate, created by the externally attached pump, is still about $0.9 \sim 1$ l/s, and the same Gaussian modulated pulse of central frequency $f_c = 35$ kHz is emitted from the source location to initiate the forward wave propagation under flow this time. The bandwidth of the source signal is also the same, and defined within $[24.5, 45.5]$ kHz at 1/100 of its spectrum amplitude. The source transducer is placed at the upstream end, while the receiving transducer that records the system response is 2.5 m away towards the downstream end (Figure 7.2.8). The sampling rate for both transmission and reception tasks is set to 1 MHz. Figure 7.2.9 presents the probing and response waveforms from the forward wave propagation test.

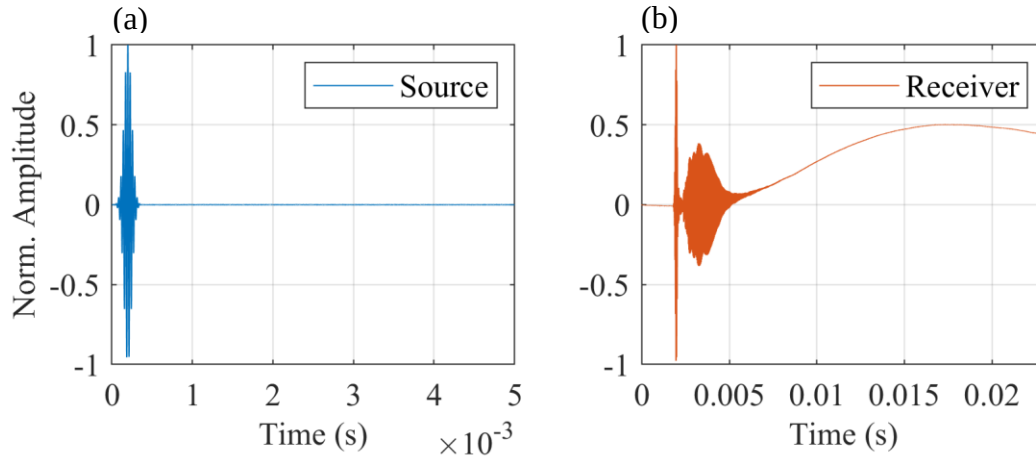


Figure 7.2.9 Forward propagation results for a Gaussian modulated sine pulse with $f_c = 35$ kHz when injected in a water filled DN80 HDPE pipe under flow conditions ($Q = 0.9$ l/s): (a) The probing pulse, and (b) the system response 2.5 m towards the downstream end of the pipe.

As in the case with the TR field response under flow in the previous trial (Figure 7.2.3b), the forward propagation response here (Figure 7.2.9b) also contains a low-frequency component that is appropriately eliminated using a bandpass filter (Figure 7.2.10). Interestingly, the response under unsteady flow (Figure 7.2.9b) does not match that recorded in stationary water (Figure 7.2.2b). Specifically, the 2nd, slower, radial mode that arrives at the receiver at a later time appears much attenuated under flow conditions, and the 1st, faster, radial mode is now dominant. That could signify mode-dependent attenuation characteristics under pipe flow conditions; however, investigating such effect is beyond the scope of this experiment.

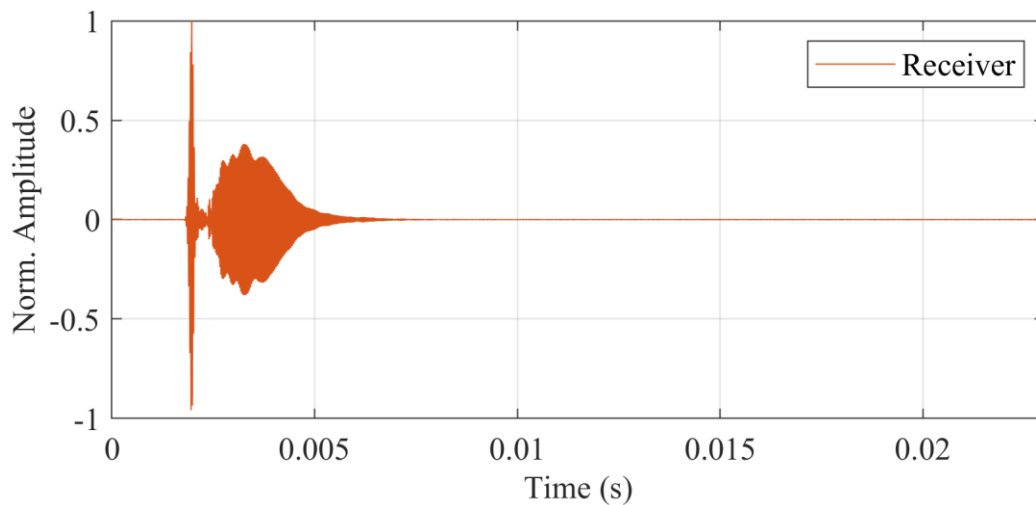


Figure 7.2.10: The response from the forward propagation test of Figure 7.2.9b is filtered to remove the low frequency component. Then, it is “flipped” in time, and re-emitted from the location of the original receiver (i.e., TR mirror).

For the TR step, the low-frequency component in the forward propagation response (Figure 7.2.9b) is removed, the signal is reversed in time, and then re-emitted from the location of the original receiver. That is, the time reversal mirror (TRM) is implemented after switching the roles of the two transducers. It is important to note that the flow direction in the TR step is not reversed compared to the forward step; hence, while during the forward step the pressure wave propagates *with* the flow, for the TR step the pressure wave propagates *against* the flow. A direct offshoot is the realization that spatial reciprocity of the waveguide is violated (i.e., $G(A|B) \neq G(B|A)$). Therefore, the goal is to estimate the error incurred to the TR process by said violation. The TR field is first measured at the location of the original source. The summary of the TR step is depicted in Figure 7.2.11.

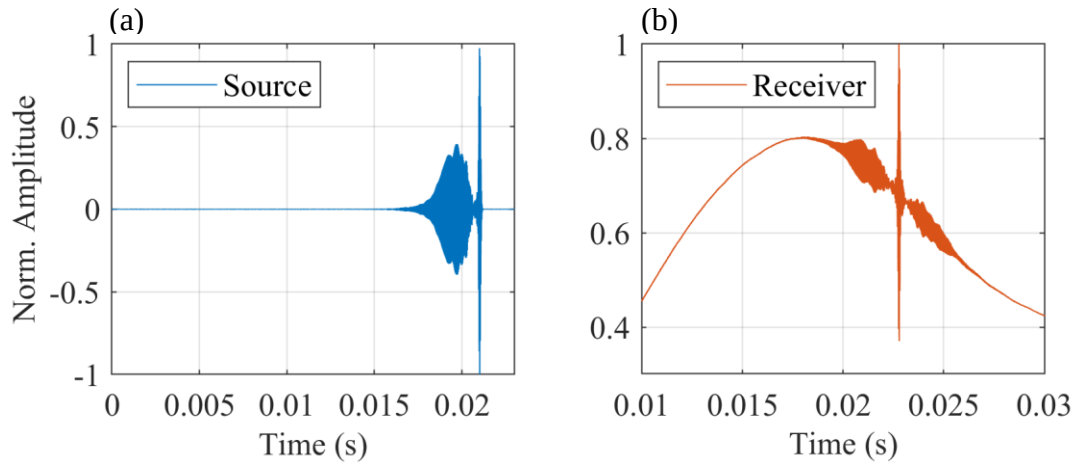


Figure 7.2.11: Time reversal step for the forward propagation step shown in Figure 7.2.9. (a) The response from the forward step is filtered of the low-frequency component, reversed, and re-emitted in the pipe system from the location of the original receiver. (b) The TR pressure field is evaluated at the location of the original source, and the low-frequency component attributed to the flow unsteadiness is also present.

The receiver, now positioned at the location of the forward step source, records the TR field response (Figure 7.2.11b), that attempts to recreate the originally injected pulse in the system (Figure 7.2.9a). Due to the flow unsteadiness, however, there exists a low frequency component in the signal, that is filtered out to yield the waveform of Figure 7.2.12. The obtained response, although close to the original probing pulse, is not perfectly symmetric, since it is clear that the

neighbouring sidelobes to the main pulse are not of the same amplitude. Therefore, there is a phase shift compared to an ideal TR experiment in pipe filled with stationary water. That phase shift, also observed in the experiment of subsection 7.2.1, is due to the pipe flow conditions that break the TR requirement of spatial reciprocity.

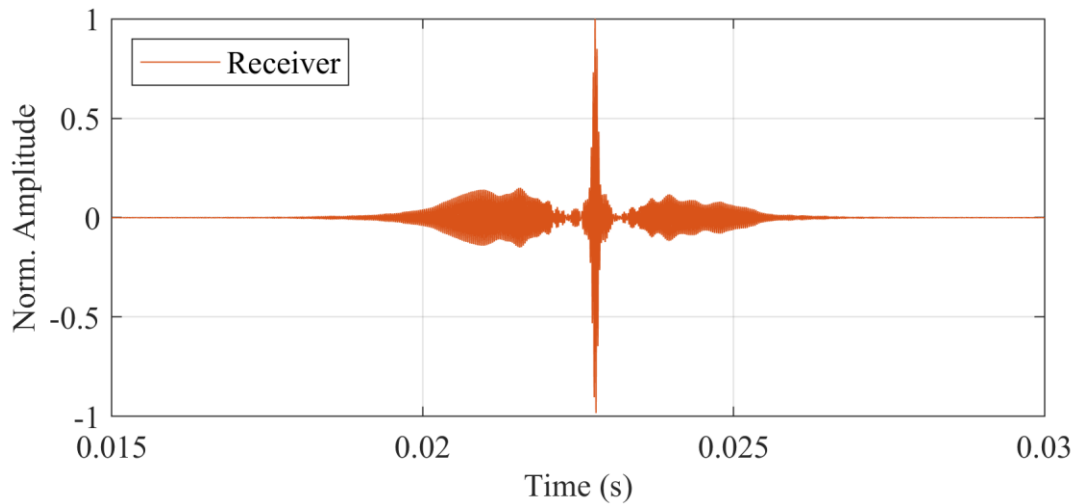


Figure 7.2.12: Filtered TR response at the location of the original receiver. The original pulse is almost recreated: the signal is not perfectly symmetric as expected in ideal TR tests, therefore demonstrating a phase shift.

Of interest is to determine the exact focal point of the TR pressure field in this case. To that end, the TR pressure field is measured at multiple positions from the TR source. In an analogous manner to the previous example presented in subsection 7.2.1, the TR pressure field is evaluated for a total of 2 m along the pipe centreline at 2 cm increments, starting from a location 1.5 m from the TR source and towards the upstream pipe end (Figure 7.2.5). In total, 101 measurements are performed, translating the transducer by 2 cm, and emitting the time reversed filtered response from the forward step (i.e., TR source) each time. As a first approach, the maximum amplitude of the recorded TR field response at each location is plotted in a dB scale, calculated relative to the maximum amplitude among all 101 measurements. The longitudinal position where the recorded response is maximum should coincide with the focal point of the TR process.

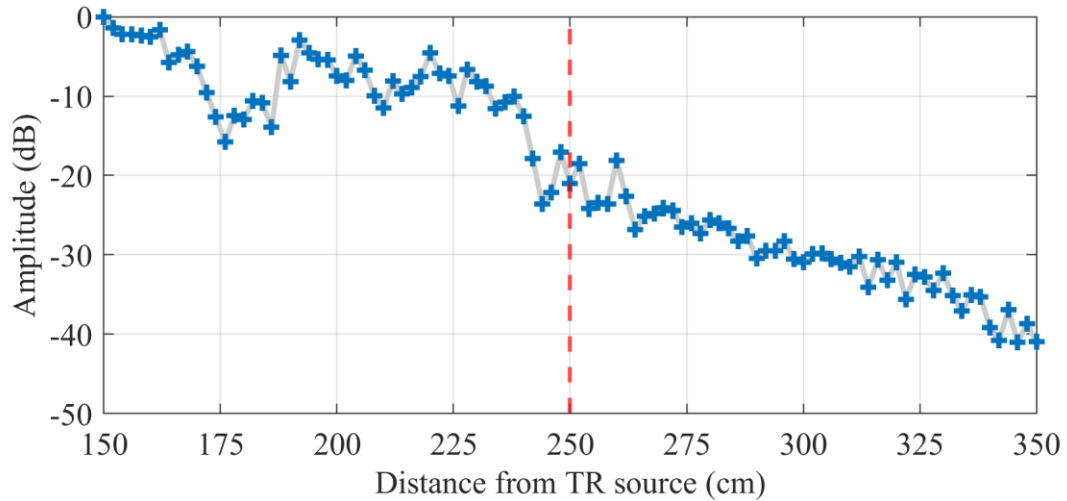


Figure 7.2.13: The maximum amplitude of the TR pressure field at each location is plotted against the range. The vertical red dotted line denotes the position of the original source, that under ideal conditions would correspond to the peak of the plot.

Figure 7.2.13 shows that despite the fact that the TR field appears to “compress” in time close to the location of the original source (Figure 7.2.12), its maximum amplitude is found the closest to the TR source. That is the effect of damping that, similarly to the results shown in Figure 7.2.6, is masking the focal point. Instead, there are some local maxima found nearby the original source location (i.e., the vertical red dashed line in Figure 7.2.13). To better verify that the original pulse is indeed recreated close to the original source location, the cross-correlation coefficient (Eq. (7.2.1)) is calculated between the responses of the TR field at each measurement location and the original pulse (Figure 7.2.9b). It is reminded that cross-correlation values closer to 1 indicate a better match. The calculated cross-correlation coefficient at each measurement location is plotted against the pertinent range from the TR source in Figure 7.2.14.

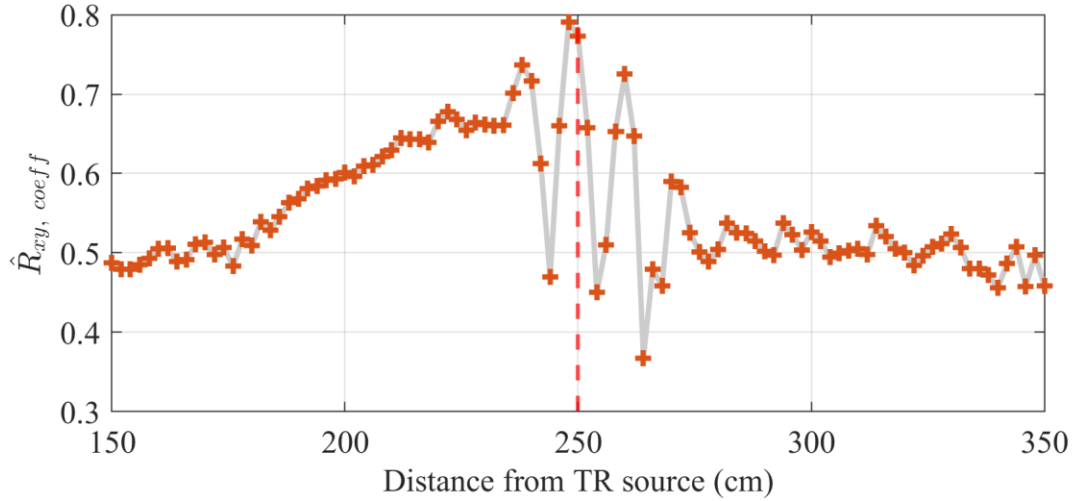


Figure 7.2.14: The maximum amplitude of the TR pressure field is plotted against the distance from the TR source. The vertical red dotted line denotes the position of the original source, that would ideally correspond to the peak of the plot.

The plot of Figure 7.2.14 reveals that the TR pressure field is most similar ($\hat{R}_{xy, coeff} \sim 0.8$) to the original pulse not exactly at the location of the original source, but one measuring increment (2 cm) closer towards the downstream end. The focal spot shift is, therefore, about equal to the minimum probing wavelength and is significant, since it is larger compared to the diffraction limit ($\lambda/2$) that defines the attainable resolution. Naturally, the focalization error, that may be directly attributed to the pipe flow condition, compromises the ability of TR to detect defects accurately. In subsections 7.2.1 and 7.2.2, the fidelity of TR is examined in the intact DN80 HDPE pipe of the HF setup under modest flow conditions ($Q \sim 1$ l/s, and $Ma \sim 10^{-4}$), and it is revealed that: (i) in agreement with studies from other fields [143, 176], pipe flow causes a temporal and spatial shift to the expected focal spot and (ii) there is significant flow-induced damping even within the limited range of the lab setup. The dependency of the shift to the probing frequency is explored in the following subsection 7.2.7.

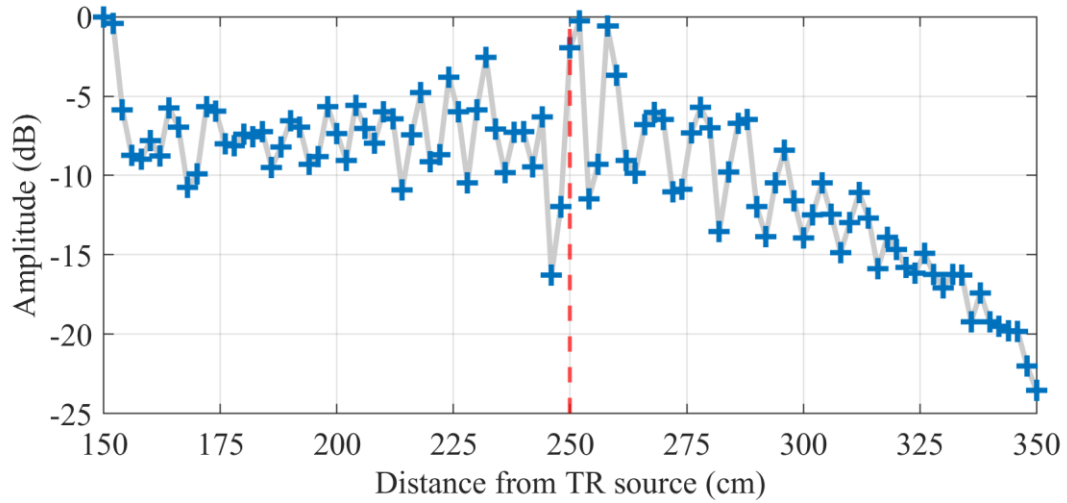


Figure 7.2.15: The maximum amplitude of the TR field for the case of the DN100 acrylic pipe is evaluated at multiple locations. In this particular test case, the forward response is evaluated under quiescent, and the TR step under flow conditions. The vertical red dashed line indicates the location of the original source. The TR field refocuses nearby the original source location, by a distance larger than the minimum probing wavelength.

Besides the DN80 HDPE pipe, the same tests are repeated on the water filled DN100 acrylic pipe of the HF setup, where the introduced pipe flow by means of an external pump is characterized by a Mach number Ma of the order of $\sim 10^{-4}$. In this case, the injected signal is a Gaussian modulated sine pulse with central frequency $f_c = 40$ kHz and excited bandwidth is defined in the interval [34, 46] kHz. The specific signal is designed to contain the cut-off frequency of the 3rd high order radial mode of the waveguide within the excited frequency interval and, thus, accentuate the dispersion effects in the received response, evaluated 2.5 m from the source. The efficacy of TR to compensate for the dispersion effects under flow conditions is tested once again using the same approach of measuring the TR pressure field at multiple points. In general, comparable results to these obtained from the DN80 HDPE pipe setup are observed. Indicatively, in Figure 7.2.15 the maximum amplitude of the TR pressure field is plotted against the pertinent measurement locations for the case where the forward step is conducted in quiescent water and the TR step under flow conditions. The plot values are presented in dB scale, relative to the overall maximum amplitude. Evidently, the overall signal attenuation in the case of DN100 acrylic pipe of is not as severe as in the case of the DN80 HDPE pipe (Figure 7.2.6), and the TR field clearly demonstrates a local maximum adjacent to the expected refocusing location that is comparable to the amplitude measured the closest to the source (i.e., 50 cm). Once again, however, the shift of the focal spot is observed: the local

maximum is found not exactly at the location of the original source, but 2 cm to the upstream end. Since the observed shift is larger than half of the minimum probing wavelength, the shift is consequential.

7.2.3 Dispersion curves under various steady turbulent flow conditions

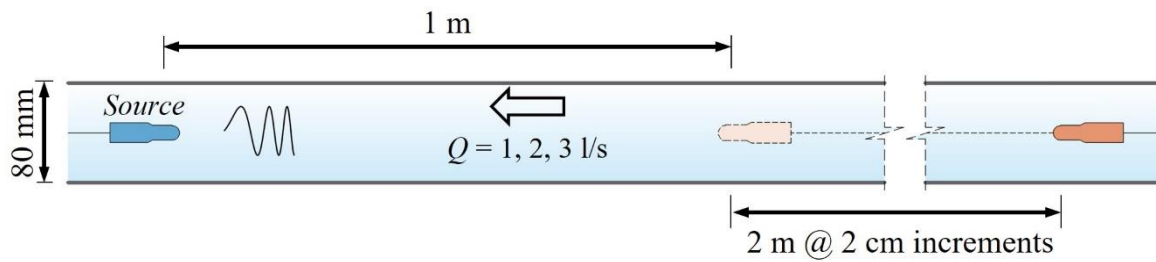


Figure 7.2.16: The experimental dispersion curves are evaluated for the DN80 HDPE pipe under three flowrates $Q = 1, 2, \text{ and } 3 \text{ l/s}$. The source emits a wideband linear frequency modulated pulse, and the response of the system is recorded over 2 m and every 2 cm by the receiver at the upstream (dimensions not to scale).

The different results of the forward wave propagation presented in subsections 7.2.1 and 7.2.2 raise the question of whether the presence of flow can potentially alter the modal characteristics of the waveguide. For instance, the response of the stationary water filled DN80 HDPE pipe to a Gaussian modulated pulse of central frequency $f_c = 35 \text{ kHz}$ and 20% relative bandwidth excites two modes, with the faster 1st being considerably attenuated compared to the 2nd, slower, mode (Figure 7.2.2b). By contrast, for an identical excitation in the same DN80 HDPE pipe but with 1 l/s flowrate, it is the slower 2nd mode that appears attenuated in comparison to the 1st (Figure 7.2.10). While this result could very well signify a mode dependency of the attenuation characteristics under flow conditions, it is also worthwhile to investigate the effect of flow to the dispersion relationship of the waveguide. Such effect could be exhibited by e.g., a shift to the cut-off frequencies of high-order radial modes, or a slope change to the dispersion curves (i.e., a change to the phase velocity, $c_{ph} = 2\pi f/k_x$).

To develop the experimental dispersion curves, the process described in Chapter 4 is followed. In short, an acoustic projector (Brüel & Kjær Type 8104) is placed along the DN80 HDPE pipe centreline at the upstream end. At the opposing downstream end, also along the pipe centreline, and 1 m from the projector, a receiving transducer (Brüel & Kjær Type 8104) is mounted. The projector emits a 10 s long linear frequency modulated (chirp) signal from 10 to 70 kHz, and

the receiver records the system response. Then, the receiver is repositioned 2 cm farther from the acoustic source, and the measurement is repeated. The receiver is translated for a total of 2 m; thus 101 measurements are obtained in total for every 2 cm (Figure 7.2.16). The sampling rate is equal to 1 MHz, and each measurement samples over 11 s. The result is a 101-by- 11×10^6 matrix, where 101 are the number of locations, and 11×10^6 are the number of samples in time at each location. To obtain the experimental dispersion curves, the 2D-FFT method is applied (see Chapter 4). The brightest regions in the developed contour plots correspond to the valid wavenumber values of the propagating modes. The entire process is repeated four times; one for stationary water, and three for a flowrate equal to 1, 2, and 3 l/s, respectively. The flowrate is measured with an ultrasonic flowmeter attached to the downstream pipe end, and the Reynolds numbers for the cases tested under flow are $Re \sim 14000$ (1 l/s), 28000 (2 l/s), and 42500 (3 l/s). However, to create the flow in this case, a more robust pump is employed compared to the one used in subsections 7.2.1 and 7.2.2 that can achieve higher flowrates. As a result, a low-frequency component in the obtained signals is no longer observed, and the flow, although turbulent, may be considered as statistically stationary. Thus, received responses no longer require the application of a bandpass filtering to remove any low-frequency pressure fluctuation component.

Figure 7.2.17 presents the obtained experimentally obtained dispersion curves for the DN80 HDPE pipe. From top (a) to bottom (d) the curves correspond to an increasing flowrate starting from quiescent water up to a flowrate of 3 l/s. The contour plot values are compensated for the frequency-dependent transmission characteristics of the acoustic projector and given in a dB scale relative to the maximum waveform amplitude observed in the measurements for the case of quiescent water. The dispersion curves between the different cases only present minute differences that relate to the mode amplitudes. The cut-off frequencies of the 1st, 2nd and 3rd radial modes are for all cases equal to about 13100, 29350, and 46600 kHz, respectively; hence, there is no shift observed. Also, the slopes of the dispersion curves are found to be identical among the different cases. Hence, it may be concluded that flow does not affect the dispersion relationship that determines the waveguide. In effect, since the impedance based adequately describes the dispersion characteristics of the waveguide, it is also suitable for a pipe under flow conditions, even for the unrealistic flowrate of 3 l/s. Besides, in the bandwidth [24.5, 45.5] kHz that is excited in the forward propagation step in subsections 7.2.1 and 7.2.2, the same two modes are propagating (i.e., 1st and 2nd radial modes), regardless of the presence of flow or not. Consequently, the different forward propagation responses in 7.2.1 and 7.2.2 cannot be credited

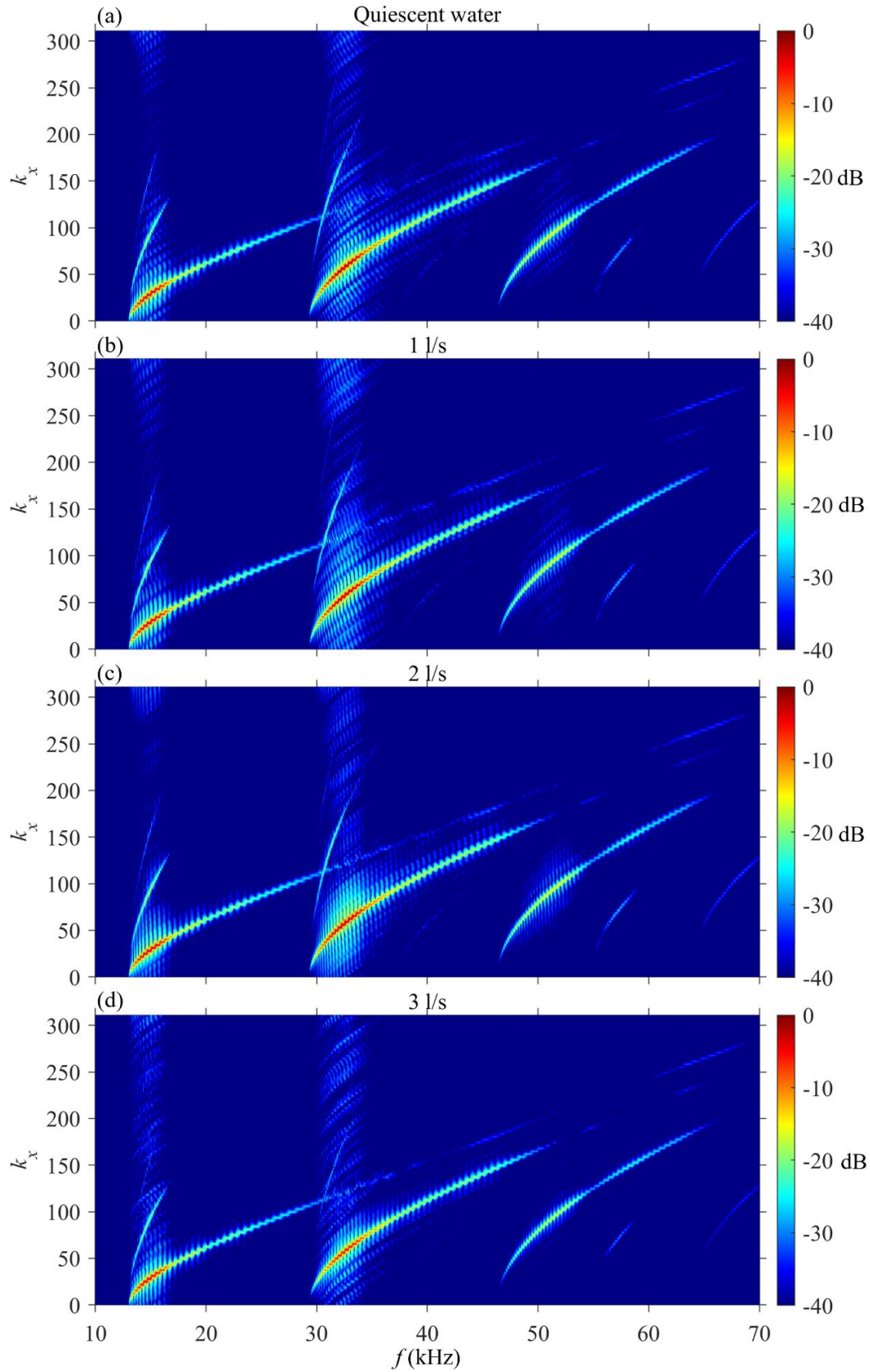


Figure 7.2.17: Experimental dispersion curves for a straight DN80 HDPE pipe from 10 to 70 kHz for (a) stationary water, (b) 1 l/s, (c) 2 l/s, and (d) 3 l/s flowrate. The contour plot values are normalized by the largest value observed in (a).

to a change to the modal characteristics of the waveguide, and the reasons have to be traced to the mode-dependent, non-linear, attenuation due to flow. In the following section 7.2.4, the measurements obtained for the development of the experimental dispersion curves under pipe flow are used to assess the change of the spectrum amplitude of the system response to a wideband excitation in space.

7.2.4 Spectral attenuation in space

For the development of the experimental dispersion curves of subsection 7.2.3, the water filled DN80 HDPE pipe is probed with a wideband chirp signal [10, 70] kHz and the response is evaluated along the pipe over a total length of 2 m under various flow conditions. It is, therefore, straightforward to calculate the amplitude spectrum at each location and for every flow scenario. That is, a fast Fourier transform (FFT) is applied on the time waveform system response to produce the spectrum amplitude. The results are presented in Figure 7.2.18(a) to (d) where, from top (a) to bottom (d), the flowrate increases from stationary water to 3 l/s. The spectrum amplitudes ($|P(f)|^2$) are normalized with respect to the maximum value found in Figure 7.2.18(a), given that quiescent water results present the least attenuation compared to these with flow.

In Figure 7.2.18 some interesting points can be made. The most prominent frequency in the system response is identified at around 14 kHz and may be attributed to the propagation of the 2nd radial mode since it is the sole mode that exists at that frequency (Figure 7.2.17). Another peak is observed at 32 kHz assigned for the most part to the 3rd radial mode that is dominant at that frequency. It appears that the attenuation for the two peaks at 13 and 32 kHz evolves linearly with distance from the source. As the flowrate increases and for a given location along the pipe, the prominent frequencies present in the system response are only slightly attenuated. The most drastic effect occurs at the extreme flowrate case of 3 l/s, where at 35 kHz, the spectrum amplitude is reduced considerably compared to the quiescent water case. In fact, flow appears to have a stronger impact to the higher frequency component (i.e., 13 vs 35 kHz), an observation that is also supported by the TR parameter, derived in Chapter 4. Finally, it should be noted that the presented attenuation characteristics cannot be strictly related to the propagating modes. In order to do that, it is necessary to extract the experimental complex wavenumber values for each mode [61]. In conclusion, the discrepancy found between the

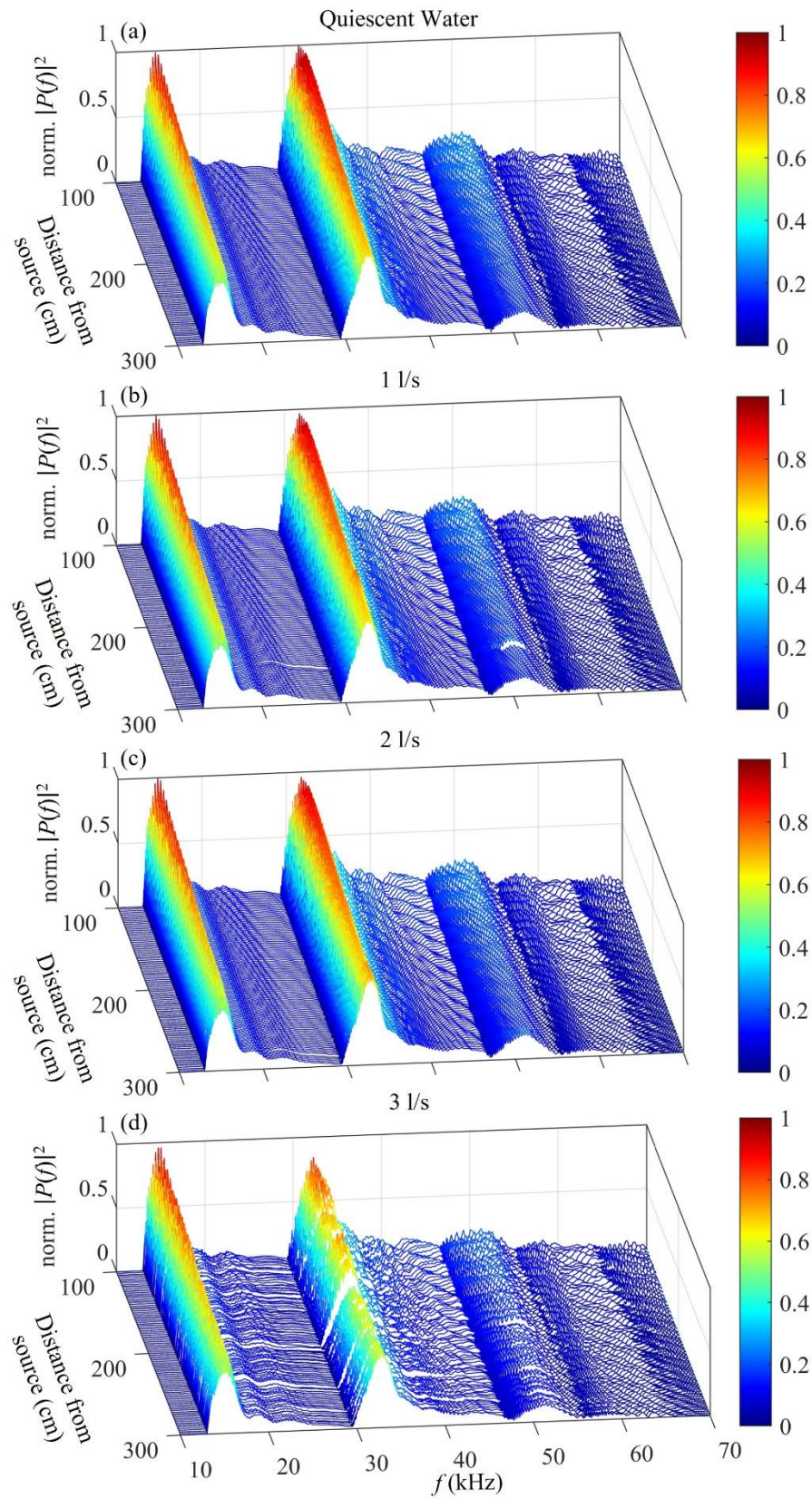


Figure 7.2.18: The amplitude spectrum of the response of the DN80 HDPE pipe in space is plotted for the cases of stationary water, and 1, 2 and 3 l/s flowrates (from (a) to (d), respectively).

forward step system responses in subsections 7.2.1 and 7.2.2 cannot be addressed using the results of Figure 7.2.18. That is because the two propagating modes have an identical frequency content, that is dictated by the probing pulse. Nevertheless, the results show that with increasing flow, higher frequency signal components are attenuated more.

7.2.5 Restoring the fidelity of TR in steady, turbulent flows

The application of the classical TR procedure in fluid-filled pipes is defined in two distinct stages. In the first – forward – stage, a point source (in the form of a piezoelectric transducer) is introduced in the pipe at A and emits a short pulse $S(A, \omega)$. A receiving transducer is located at a distance x from the source at point B and records the dispersed/scattered response of the system, defined as $S(A, \omega) \otimes G(B, \omega|A)$ where G is the Green's function of the waveguide. According to the classical TR approach, the received signal is time reversed (or phase conjugated when expressed in the frequency domain) to obtain $S^*(A, \omega) \otimes G^*(B, \omega|A)$ (where $(*)$ denotes the phase conjugate), and then re-emitted from point B. The response, now evaluated at point A, is given by $S^*(A, \omega) \otimes G^*(B, \omega|A) \otimes G(A, \omega|B)$. Provided that the spatial reciprocity in the waveguide holds, then $G(A, \omega|B)$ and $G(B, \omega|A)$ are identical, and the convolution product is maximized at point A (i.e., $\|G(A, \omega|B)\|^2$) (i.e., the original source location) essentially reconstructing the original pulse $S(A, \omega)$.

In a pipe where fluid flows from e.g., point A to point B, however, $G(A, \omega|B)$ and $G(B, \omega|A)$ are *not* equal. Due to turbulence, vortex structures are formed that interact with the propagating pressure waves. Also, since the flow is unidirectional, during the forward propagation (i.e., $G(B, \omega|A)$) the acoustic pressure signal travels with the flow, while in the backpropagation step (i.e., $G(A, \omega|B)$) against the flow. As a result, the TR refocusing at the original source location is lost. To restore the fidelity of TR in a uniform steady flow, Dowling [174] proposed reversing the flow direction in the backpropagation step. Similarly, Roux & Fink [172] demonstrated that to restore TR in a flow field that involves a single coherent vortex, the direction of the vortex rotation must be reversed for the backpropagation stage. However, none of these approaches is applicable to UWSS presenting flows that are highly turbulent and lack spatial symmetry.

A different procedure is that of non-reciprocal time reversal (NR TR) that was proposed in [177] to counter the loss of TR fidelity due to currents in the ocean. NR TR suggests a modification to classical TR and forgoes the need for spatial reciprocity in the wave propagation medium. The difference lies in the TR stage, where the time reversed forward

response is no longer emitted from the location of the receiver, but from that of the original source. That is, the transducers during the second, TR, stage retain the roles they had during the first, forward, stage. Going back to the example with the transducers placed at points A and B, NR TR utilizes the forward step response recorded at B, reverses, and re-emits not from B but A. Mathematically, the outcome of NR TR in this case is $S^*(A, \omega) \otimes G^*(B, \omega|A) \otimes G(B, \omega|A) = S^*(A, \omega) \otimes \|G(B, \omega|A)\|^2$, hence the NR TR manages to refocus. That is due to the fact that the wave path for the two stages is now identical and, thus, described by a single expression of the Green's function. This is particularly important when spatial reciprocity is violated, such as a fluid-filled pipe under turbulent flow conditions.

The NR TR procedure involves sending the recorded response from the receiver of the forward stage back to the source for time-reversal and re-transmission. This could potentially be challenging in e.g., ocean applications where the source and receiving multi-transducer arrays are placed kilometres apart, and to complete said signal transfer lasts minutes. In that interval, the medium is altered due to dynamic processes (i.e., the Green's function of the waveguide is time-dependent), and focusing quality is potentially compromised. To counter that effect, and refrain from performing the second TR step in practice, it has been suggested [177] that the numerical cross-correlation of impulse responses recorded at the receiver at slightly different times is performed instead. This approach is denoted in the literature as passive phase conjugation or passive NR TR and is related to the matched field processing (MFP) technique that the Smart UWSS group has already pioneered for pipe defect detection [93]. Although in the employed pipe setups signal transfer from receiver to source is straightforward and rapid given that the signal transmission and acquisition tasks are scheduled using the same endpoint/device, the passive NR TR technique would allow the study of TR in the fluctuating and reciprocity-breaking environment of a pipe under flow. Experiments in the (fluctuating) ocean have shown that NR TR is able to provide a better focus compared to classical TR [177], but has yet to be tested in pipes with flow.

Therefore, an experimental test is devised to test the gains of using NR versus traditional TR in pipes. A transducer pair (Brüel & Kjær Type 8104) is inserted in the water filled intact DN80 HDPE and placed 2.5 m apart along the pipe centreline. An external pump to the system is used to create flow in the pipe. The flow characteristics are evaluated via an ultrasonic flowmeter that is attached to the downstream end of the pipe. The flowrate is kept constant throughout the duration of the measurements, and equal to $Q = 1.5$ l/s. The corresponding Reynolds is calculated equal to ~ 24500 , denoting a fully developed turbulent flow. Moreover, the nominal

pipe cross-sectional area is equal to $A = 4.21 \times 10^{-3} \text{ m}^2$; thus, the flow velocity is calculated equal to $\bar{V} = Q/A = 0.36 \text{ m/s}$, and the – still negligibly small – Mach number $Ma = \bar{V}/c_f = 2.4 \times 10^{-4}$.

One of the goals set for this particular experiment is to identify the fidelity loss of TR as a function of the probing frequency. Hence, for the forward propagation step, the transducer at the upstream end is used to probe the system with Gaussian modulated sine pulses with central frequencies f_c ranging from 30 to 100 kHz, at 5 kHz intervals (Figure 7.2.19a). Consequently, 15 tests are carried out in total. The duration δ of each probing pulse is adjusted so that the excited bandwidth is equal to 5 kHz for every case. For instance, the probing signal of $f_c = 35 \text{ kHz}$, excites a bandwidth of [27.5, 32.5] kHz, while that for a probing signal of $f_c = 50 \text{ kHz}$ is [47.5, 52.5] kHz. The response of the system is evaluated at the downstream end transducer.

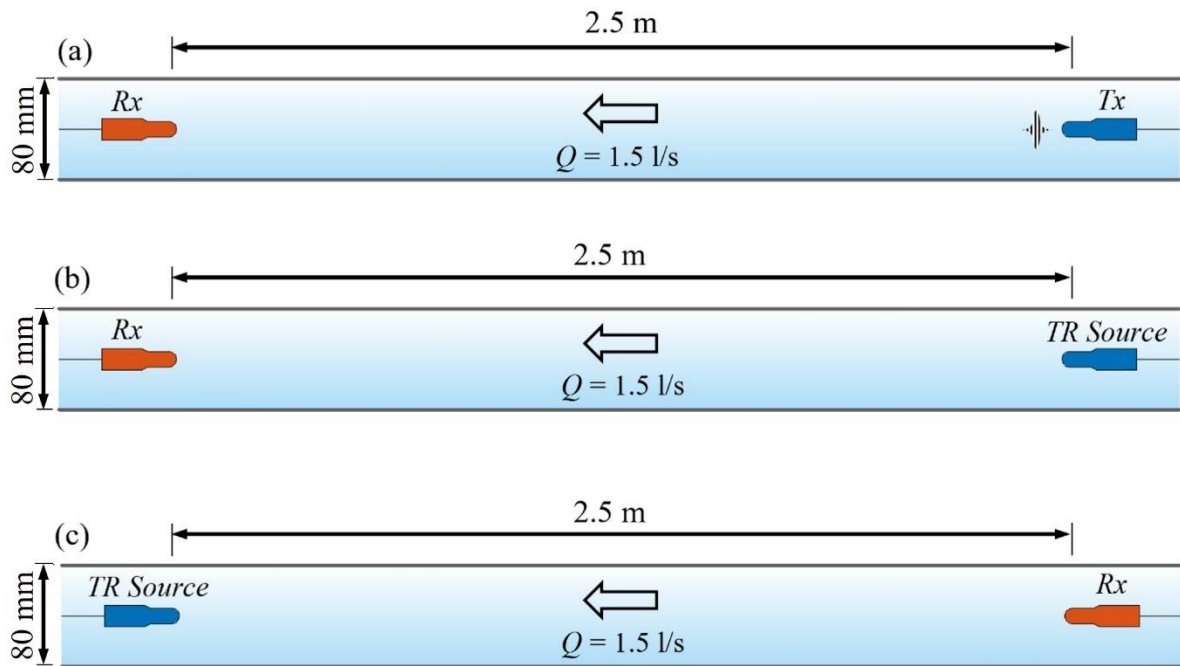


Figure 7.2.19: While (a) the forward step is identical to the traditional TR approach, for nonreciprocal TR (b) the transducers are not flipped in the TR step, but the forward source is also the TR source. For comparison, (c) for the traditional TR approach the roles of the transducers are interchanged.

The time reversal step is repeated twice for each forward test, and the flow conditions in the system remain unchanged. For convenience, the NR TR procedure is applied first since the upstream and downstream transducers remain as the source and receiver, respectively (Figure 7.2.19b). The upstream transducer emits the time-reversed response recorded from the

pertinent forward test (that was recorded at the downstream transducer), and the pressure field is evaluated at the location of the downstream transducer. Then, the time reversal step is repeated according to the “typical/traditional” procedure. The roles of the transducers are interchanged, and the downstream transducer now becomes the source, while the upstream transducer is the receiver. The downstream transducer transmits the time reversed system response from the forward test, and the pressure field is measured at the upstream transducer (the source location of the forward test.)

The objective for this set of measurements is to assess and compare the effectiveness of each of the two TR approaches in recreating the original probing pulse. To that end, two metrics are employed: (i) the degree of temporal compression, and (ii) the similarity of the recreated signal at the receiver to the original probing pulse. For the former, the peak time t_1 of the original probing pulse is compared to the peak times of the recreated pulses $T - t_{TR}$ and $T - t_{NR\ TR}$ obtained from the traditional and NR TR approaches, respectively. There, T is the response signal duration from the forward test, t_{TR} is the peak time of the recreated signal using the traditional TR approach, and t_{TR-NR} is the peak time of the recreated signal using the NR TR approach. To evaluate the temporal compression accuracy, the error percentage is calculated as follows:

$$\% \text{ Error} = \frac{|(T - t_{TR \text{ or } NR\ TR}) - t_1|}{|t_1|} \times 100 \quad (7.2.2)$$

The recreated pulse that presents the minimum error demonstrates better temporal compression and corresponds to the more effective TR approach. To gauge the similarity between each of the recreated pulses to the original probing pulse, the cross-correlation coefficient is calculated between the two waveforms (Eq. (7.2.1)). Again, a higher cross-correlation coefficient value (max = 1) represents a closer similarity. Indicatively, the test summary is presented for three different probing waves (40, 50, and 60 kHz) of the total 15. The recreated pulses using the two approaches (traditional and NR TR) are plotted together, and the calculated error and cross correlation coefficient values are given in table form. Each of the presented signal waveforms is normalized by its maximum value. For the remaining tests, the waveform results are skipped, but the calculated error and cross-correlation coefficient values are plotted against frequency at the end of this subsection.

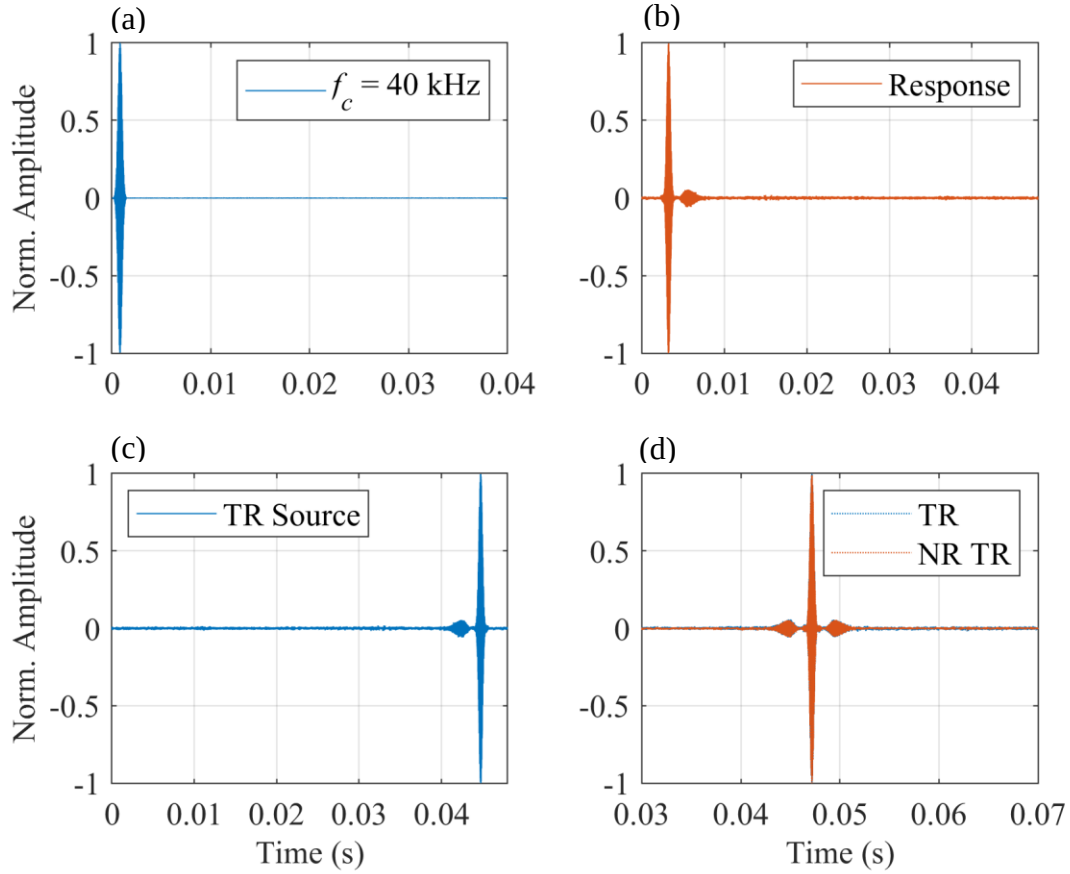


Figure 7.2.20: Comparison test between the traditional and NR TR approaches in the DN80 HDPE pipe under flow ($Q = 1.5$ l/s) for (a) the Gaussian modulated sine pulse of $f_c = 40$ kHz and bandwidth $[37.5, 42.5]$ kHz. The obtained system response (b) is time reversed (c), and sequentially reemitted from the receiver (traditional TR) and source locations (NR TR), to recreate the probing pulse (d).

To summarize Figure 7.2.20, a Gaussian modulated sine pulse of central frequency $f_c = 40$ kHz (Figure 7.2.20a) is injected at the upstream end and along the centreline of the DN80 HDPE pipe under flow conditions, and the response of the system to the particular pulse is retrieved at distance $x = 2.5$ m. In the excited frequency range $[37.5, 42.5]$ kHz, two modes propagate in total, as demoted by the two distinct wavepackets that reach the receiver at different times (Figure 7.2.20b). This concludes the forward step. The recorder response is then inverted in time to yield the waveform depicted in (Figure 7.2.20c) and proceed to the backpropagation step. The backpropagation step is performed twice: once as per the traditional TR, and then again as per the NR TR approach. According to the former, the transducer functions are interchanged, and the time reversed response is re-emitted from the location it was retrieved; hence, the transducer at the downstream operates as a wave mirror. In contrast, the NR TR

approach dictates that the recorded time-reversed response is emitted from the position of the original source, so that the wave backpropagates with, and not against, the flow. In any event, the response from each of the two approaches is recorded 2.5 m from the pertinent TR source (Figure 7.2.20d); for the traditional TR that location corresponds to the source location of the forward step, while for the NR TR approach it is the receiver location of the forward step. Visually, the recreated probing pulses from the two approaches (Figure 7.2.20d) are strikingly similar and the only differences are of the noise scale level.

Table 7.2.1: The recreated pulses from the traditional and TR approaches performed in the DN80 HDPE pipe under flow conditions are compared to the original Gaussian modulated sine pulse with $f_c = 40$ kHz in terms of the temporal compression degree and similarity.

	Source Pulse ($f_c = 40$ kHz)	TR Recreated pulses	
		Traditional	NR
Peak time (s)	8.31×10^{-4}	8.42×10^{-4}	8.41×10^{-4}
% Error	0	1.32	1.20
Correlation coef.	1	0.98	0.98

Table 7.2.1 presents results regarding the temporal compression accuracy and similarity of the recreated pulses from the two methods (i.e., traditional and NR TR) to the source signal of the forward step. The temporal accuracy of each technique is evaluated by calculating the % error between the peak time of the source pulse to that of the recreated pulse following the TR process, in accordance with Eq. (7.2.2). Obviously, when the source pulse is compared with itself the error is 0%. On the other hand, the temporal compression errors incurred by the two TR techniques are somewhat similar, with the NR TR showing only a minor advantage (1.32% for the traditional TR process versus 1.20% for the NR TR). However, the difference between the two recreated pulses is below the order of the sampling time step (i.e., $\sim 10^{-6}$ s), hence insignificant. Besides, the temporal shift in absolute terms between the source and TR recreated pulses is $\sim 10^{-5}$ s, that is approximately half a period when considering a probing frequency equal to 40 kHz. Finally, the calculated correlation coefficient when comparing both TR recreated to the source pulse are identical and approach unity, therefore denoting a high degree of similarity for both traditional and NR TR techniques.

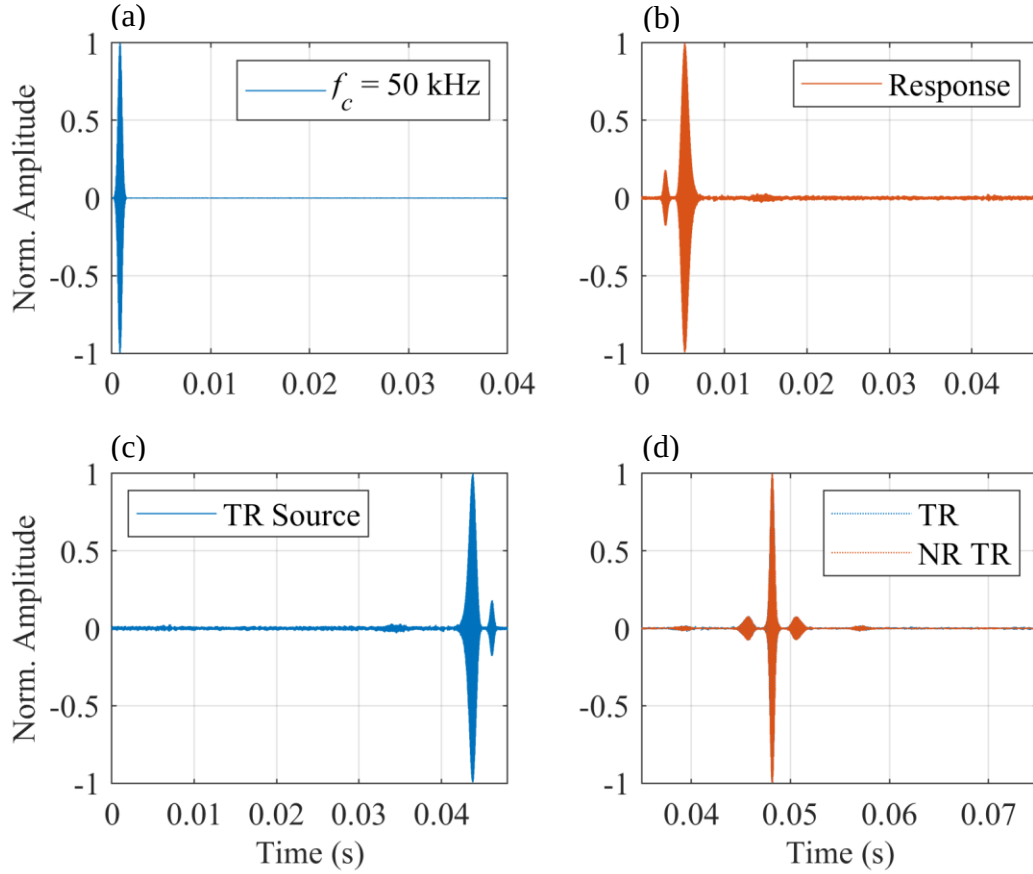


Figure 7.2.21 Comparison test between the traditional and NR TR approaches in the DN80 HDPE pipe under flow ($Q = 1.5 \text{ l/s}$) for (a) the Gaussian modulated sine pulse of $f_c = 50 \text{ kHz}$ and bandwidth $[47.5, 52.5] \text{ kHz}$. The obtained system response (b) is time reversed (c), and sequentially reemitted from the receiver (traditional TR) and source locations (NR TR), to recreate the probing pulse (d).

Figure 7.2.21 presents the results from the test where a Gaussian modulated sine pulse of slightly higher central frequency ($f_c = 50 \text{ kHz}$) is employed. In the excited frequency bandwidth $[47.5, 52.5] \text{ kHz}$, three axisymmetric radial modes are expected to propagate. However, the 1st and 2nd radial modes have a similar group velocity, as indicated by a similar slope in the experimental dispersion curves (Figure 7.2.18). Hence, the fastest of the two wavepackets that arrive at the receiver for $t < 0.01 \text{ s}$ incorporates both 1st and 2nd radial modes, while the trailing wavepacket corresponds to the 3rd radial mode. Beyond time $t = 0.01 \text{ s}$ the signal comprises of noise and minor system reflections (Figure 7.2.21b). The response is time reversed (Figure 7.2.21c) and re-transmitted twice, following the principles of the traditional and NR TR at each time, to yield the recreated pulses plotted in Figure 7.2.21d. No visible differences are

discernible between the two waveforms of Figure 7.2.21d; thus, the comparison will be performed on the basis of the two metrics presented in Table 7.2.2.

Table 7.2.2: The recreated pulses from the traditional and TR approaches performed in the DN80 HDPE pipe under flow conditions are compared to the original Gaussian modulated sine pulse with $f_c = 50$ kHz in terms of the temporal compression degree and similarity.

	Source Pulse ($f_c = 50$ kHz)	TR Recreated pulses	
		Traditional	NR
Peak time (s)	8.31×10^{-4}	8.39×10^{-4}	8.19×10^{-4}
Percent error (%)	0	0.96	1.44
Correlation coef.	1	0.91	0.92

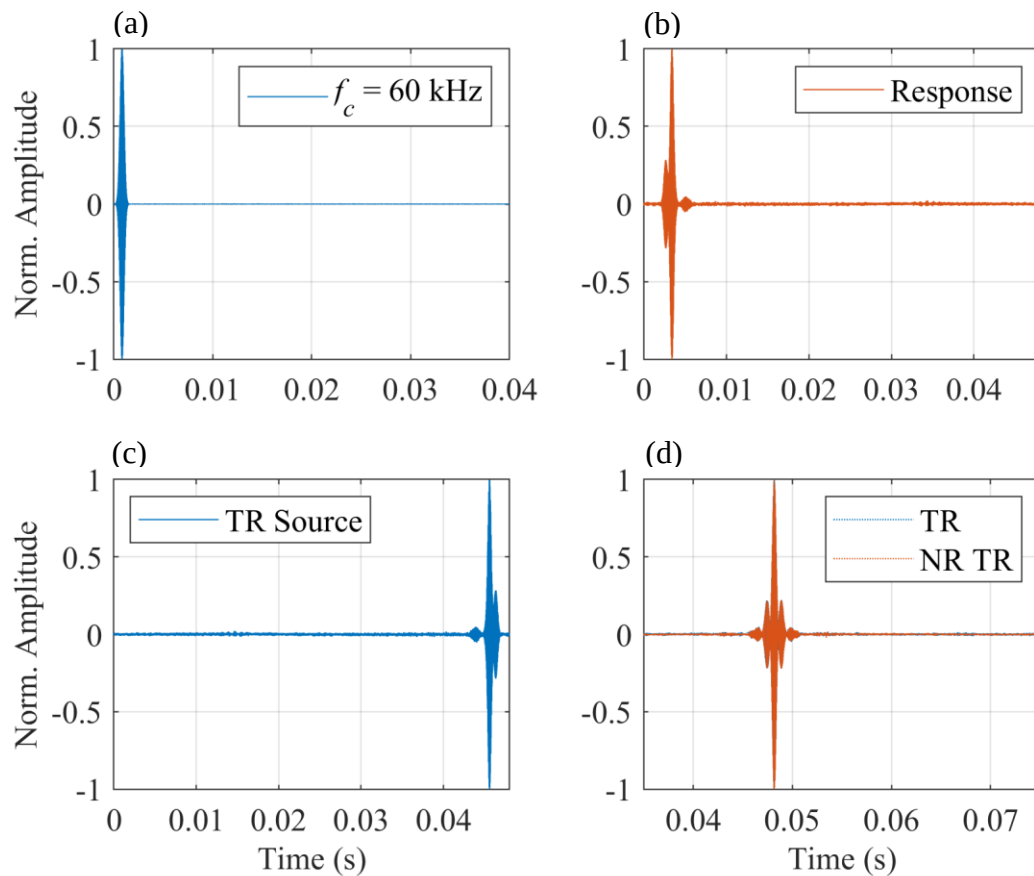


Figure 7.2.22: Comparison test between the traditional and NR TR approaches in the DN80 HDPE pipe under flow ($Q = 1.5$ l/s) for (a) the Gaussian modulated sine pulse of $f_c = 60$ kHz and bandwidth $[57.5, 62.5]$ kHz. The obtained system response (b) is time reversed (c), and sequentially reemitted from the receiver (traditional TR) and source locations (NR TR), to recreate the probing pulse (d).

Calculated data in Table 7.2.2, shows that for this particular case traditional TR manages a more accurate temporal compression accuracy, as the peak time of the recreated pulse is only off by 0.96% when compared to the source pulse. In comparison, the error for the case of the NR TR is slightly higher at 1.44%. Still, however, the difference between the two methods is small ($\sim 10^{-5}$ s in absolute terms). The estimated correlation coefficients also show that the recreated pulses from either TR method are highly similar to the source pulse.

Finally, Figure 7.2.22 plots the steps followed when a Gaussian modulated sine pulse of central frequency $f_c = 60$ kHz is injected in the DN80 HDPE pipe under turbulent flow ($Q = 1.5$ l/s). The experimental dispersion curves (Figure 7.2.17) show that in the excited frequency bandwidth [57.5, 62.5] kHz four modes propagate in total, with the 1st and 2nd propagating at the same group velocity. Therefore, the fastest discernible wavepacket in Figure 7.2.22b embodies the 1st and 2nd radial modes, and each of the following wavepackets is consistent with the 3rd and 4th radial modes. The response of the system to the probing pulse is time reversed (Figure 7.2.22c) and re-emitted using the two methods of traditional and NR TR to obtain the recreated source pulses presented in (Figure 7.2.22d).

Table 7.2.3: The recreated pulses from the traditional and TR approaches performed in the DN80 HDPE pipe under flow conditions are compared to the original Gaussian modulated sine pulse with $f_c = 60$ kHz in terms of the temporal compression degree and similarity.

	Source Pulse ($f_c = 60$ kHz)	TR Recreated pulses	
		Typical	NR
Peak time (s)	8.43×10^{-4}	8.16×10^{-4}	8.32×10^{-4}
Percent error (%)	0	3.20	1.31
Correlation coef.	1	0.93	0.93

The results presented in Table 7.2.3 show that the NR TR refocuses better, given that the peak time error of the pertinent recreated pulse to the source pulse is much smaller (1.31%) to that achieved by the traditional TR method (3.2%). Regardless, both refocused pulses match the original pulse to a great degree ($\hat{R}_{xy, \text{coeff}} \sim 0.9$).

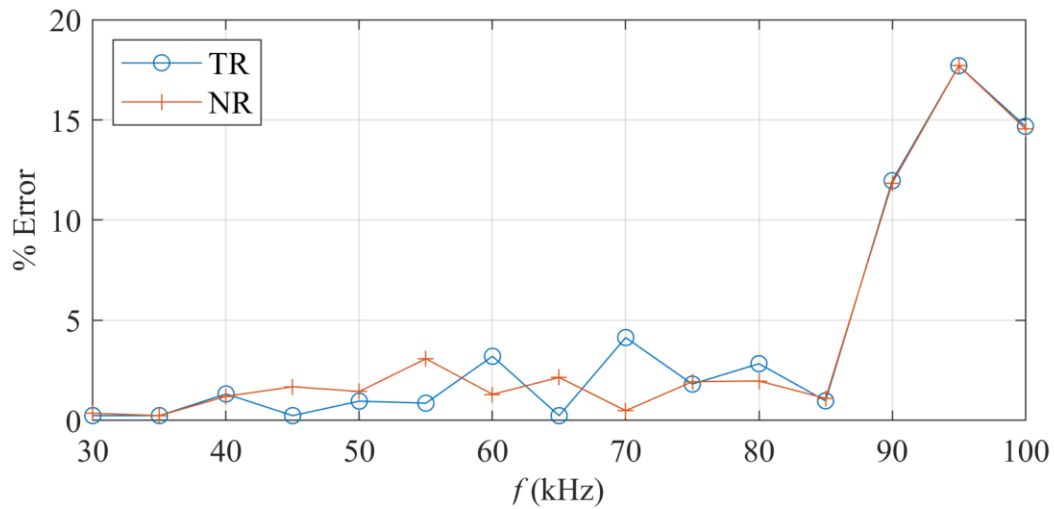


Figure 7.2.23: Percent error between the peak time of the recreated pulses using the two approaches of traditional and NR TR and the original source pulse as a function of probing frequency.

As noted at the start of this subsection, a total of 15 tests is conducted where the central frequency of the probing pulse starts at 30 kHz for the first test and increases with 5 kHz increments until the final measurement at 100 kHz. The specific tests at 40, 50, and 60 kHz are discussed in detail since they represent cases where the two methods of traditional and NR TR are either equivalent or mutually superior to each other as evaluated by the percent error and cross correlation coefficient indexes. Figure 7.2.23 presents the calculated peak time percent errors achieved by the two techniques for the entire set of measurements or, alternatively, as a function of probing frequency. The results are far from conclusive; nevertheless, it can be argued that the error appears to be increasing with frequency regardless of the followed TR approach. Besides, at higher frequencies, (> 70 kHz) NR TR displays slightly better results in comparison to traditional TR. It should be noted that the striking error increase at frequencies > 85 kHz is due to the significant drop in the attainable signal-to-noise ratio (SNR). At these frequencies, the transmission efficiency of the transducer drops appreciably, and noise becomes dominant.

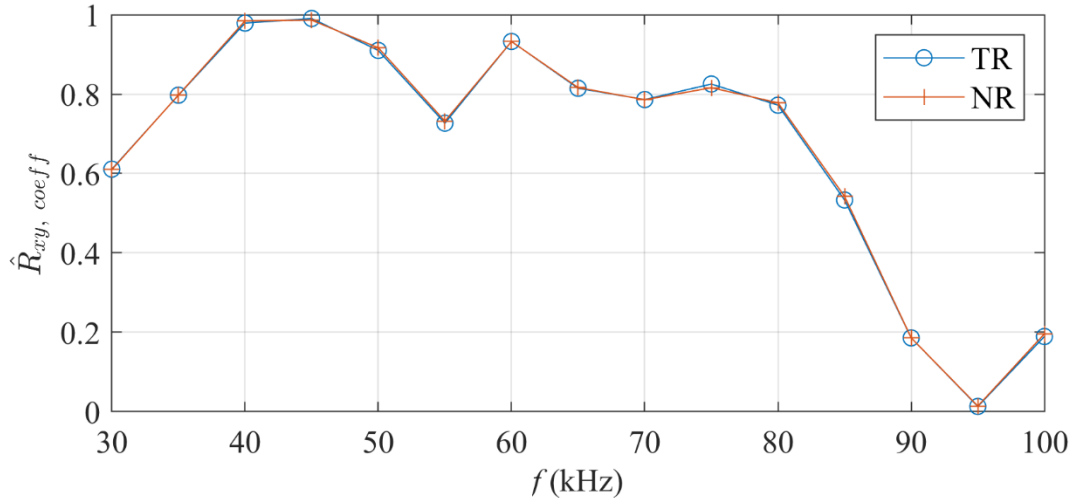


Figure 7.2.24: Maximum cross-correlation coefficient calculated between the TR recreated pulses using the two approaches of traditional and NR TR and the original source pulse as a function of probing frequency.

Figure 7.2.24 plots the cross-correlation coefficient between the refocused pulses from both traditional and NR TR methods and the original source pulse as a function of frequency. Unlike the peak time error plot of Figure 7.2.23 that shows some, albeit minor, discrepancies between the two TR methods, the calculated cross-correlation coefficient values are virtually indistinguishable. While the cross-correlation coefficient values remain above 0.8 for most of the frequency range tested, there is a sharp drop at frequencies above 80 kHz. As already noted, this is to be attributed to the reduced SNR, since the employed transducer cannot effectively project acoustic pressure signals with frequencies beyond 70 kHz. In conclusion, the performed tests cannot prove with certainty that NR TR is better than traditional TR when applied in a DN80 HDPE pipe under turbulent flow conditions. It is expected that conducting the same test over longer ranges will potentially yield more significant results. It would also be worthwhile to conduct each measurement multiple times and average the results. Unfortunately, at the time of writing, there is no experimental facility available that (i) is readily accessible to HF transducers, (ii) allows the accurate control of the flowrate, and (iii) offers adequate range other than the HF setup at the hydraulic lab at HKUST.

7.2.6 NR TR against time varying flowrates

Pressure and flow variations is a common scenario in typical UWSS, especially during the times of day (e.g., around 6 am) when the consumer demand changes rapidly, and pumps are

turned on to meet such demand. In the previous sections of this chapter, it is shown that pipe flow conditions induce a temporal and spatial shift to the refocusing point of TR. However, TR is only tested under steady flow i.e., the Green's function of the waveguide is independent of time. Consider two points (A and B) along a water supply pipe, where flow is from A to B. The forward step involves sending a signal $S(t)$ from point A to B at time t_0 , and the response of the system is given by $S(t) \otimes G(B, t|A, t_0)$. After time delay τ , the NR TR is applied and the time reversed response $S(-t) \otimes G(B, -t|A, t_0)$ is emitted again from A to B to obtain $S(-t) \otimes G(B, -t|A, t_0) \otimes G(B, t|A, t_0 + \tau)$. Under varying flow conditions, $G(B, -t|A, t_0) \neq G(B, t|A, t_0 + \tau)$, and TR refocusing is lost. In this subsection, an experiment is developed to test the fidelity of TR when there is a change of flow. Note that this trial does not aim to compare the traditional and NR TR methods; the TR step is solely performed in accordance to NR TR.

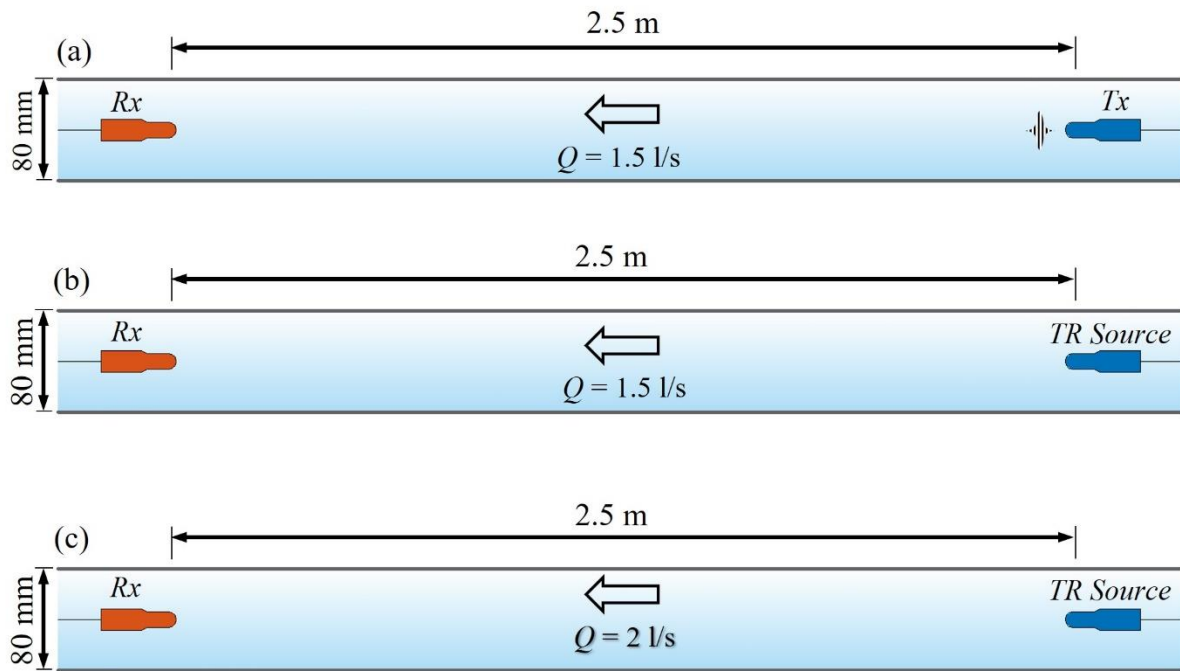


Figure 7.2.25: (a) The transducer at the upstream end emits a probing pulse and the response of the system is acquired 2.5 m away and towards the downstream end. The flowrate is measured equal to 1.5 l/s. The NR TR is applied, thus, (b) in the TR step the original source is also the TR source that reemits the time reversed response while keeping the flow constant. Finally, the TR step is repeated for an increased flowrate of $Q = 2 \text{ l/s}$, to simulate the effect of flow unsteadiness.

The main rig used in this test is the water filled straight and intact DN80 HDPE pipe of the HF setup at the HKUST hydraulic lab, where the transducer pair (two Brüel & Kjær Type 8104

transducers) is placed 2.5 m apart along the pipe centreline. An external pump creates flow through the pipe, and the flow characteristics are evaluated via an ultrasonic flowmeter that is attached to the downstream end of the pipe. For the forward propagation step, the flowrate is set to about 1.5 l/s, and the corresponding Reynolds number is calculated ~ 24500 . The pipe cross-sectional area of the DN80 HDPE pipe is equal to $A = 4.21 \times 10^{-3} \text{ m}^2$; thus, the flow velocity is $\bar{V} = Q/A = 0.36 \text{ m/s}$. Besides, the Mach number is estimated equal to $Ma = \bar{V}/c_f = 2.4 \times 10^{-4}$. The acoustic source is located at the upstream end, and the receiver at the downstream end of the pipe. The system is probed with Gaussian modulated sine pulses with central frequencies ranging from 30 to 100 kHz at 5 kHz intervals; hence, the forward step is performed 15 times in total. The excited bandwidth for each probing signal is designed to be constant and equal to 5 kHz. For instance, the probing signal of $f_c = 30 \text{ kHz}$ excites a bandwidth of [27.5, 32.5] kHz.

The time reversal step is repeated two times and NR TR is always applied. Consequently, the time reversed response from the forward step is not reemitted from the location of the receiver, but from the original source. Simply put, the roles of the transducers are not interchanged for the TR step. The TR pressure field is then evaluated at the location of the original receiver that would coincide with the TR focal point if both forward and TR steps were conducted under quiescent water conditions. Instead, for the first TR trial the flow characteristics remain unaltered from the forward and the flowrate remains equal to $Q = 1.5 \text{ l/s}$, while for the second TR trial the same TR source (that is the time reversed response from the forward propagation step) is used but the flowrate is increased to $Q = 2 \text{ l/s}$. In the latter case, the Reynolds number is calculated equal to ~ 32400 . The flow velocity is $\bar{V} = Q/A = 0.48 \text{ m/s}$, and the Mach number $Ma = \bar{V}/c_f = 3.2 \times 10^{-4}$. The remaining characteristics of the setup and signal chain during this second TR test are identical to the first. For convenience, the 15 tests (for each probing frequency) that include the forward and TR steps for $Q = 1.5 \text{ l/s}$ are performed first. Then, the flowrate is increased at $Q = 2 \text{ l/s}$ and only the TR steps are repeated.

As already explained, this experiment aims to simulate the detrimental effect of non-stationarity to the refocusing properties of TR and attempt to evaluate the error in the reconstructed pulse that occurs when the flow conditions between the forward and TR tests change. Of interest is to also compare the results of the TR process under unsteady flow conditions to these obtained from steady flow conditions, hence the TR test is repeated twice. For the comparison, two metrics that have already been discussed in the previous subsections

will be employed. To evaluate the temporal compression at the expected refocusing location, the peak time of the reconstructed pulse is contrasted against that of the original source, and the percent error is calculated. Naturally, the smaller the error, the better the temporal compression achieved by the TR process. In addition, the maximum cross-correlation coefficient (Eq. (7.2.1)) is calculated in MATLAB between the reconstructed pulse and the original source waveform. A coefficient equal to 1 denotes a perfect match between two waveforms.

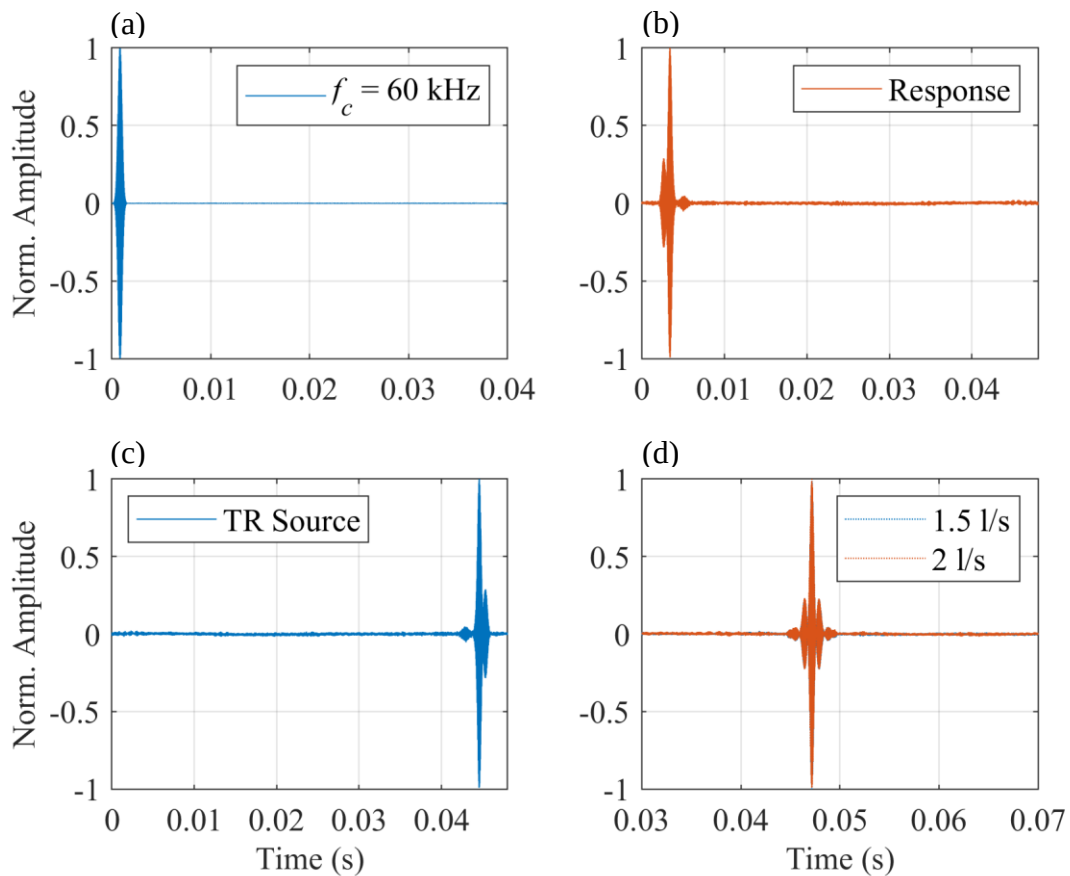


Figure 7.2.26: (a) A Gaussian modulated sine pulse of $f_c = 60 \text{ kHz}$ and bandwidth [57.5, 62.5] kHz is transmitted from the upstream end of the straight, water filled DN80 HDPE pipe under steady flow conditions ($Q = 1.5 \text{ l/s}$). The obtained system response at the downstream end (b) is time reversed (c) and reemitted from the upstream transducer (NR TR) twice for under different flow conditions each time ($Q = 1.5$ and 2 l/s). The TR pressure field from each TR test is evaluated at the location of the downstream transducer (d).

Indicatively, the waveform results from both forward and TR tests are presented for a probing signal of $f_c = 60 \text{ kHz}$. In Figure 7.2.26, subfigure (a) depicts the Gaussian modulated sine

probing pulse, subfigure (b) plots the response of the system to the forward test 2.5 m from the source, subfigure (c) shows the time reversed response from the forward step to be reemitted from the original source for the initialization of the TR step, and subfigure (d) represents the TR field response at the receiver located at the downstream pipe end for the two cases tested: (i) the TR test is performed under the same flow conditions as the forward step and the flowrate $Q = 1.5$ l/s, and (ii) the TR test is conducted after the flowrate is increased to $Q = 2$ l/s. Observing readers will notice that the first trial where NR TR is applied to a steady flow (i.e., where both forward and TR steps are performed in the pipe under a flowrate $Q = 1.5$ l/s) is identical to the NR TR process presented in subsection 7.2.5. Hence, the presented results in Figure 7.2.26 should be considered similar to Figure 7.2.21, and only differ by subfigure (d). The recreated pulses presented in Figure 7.2.26d are compared to the original probing pulse (Figure 7.2.26a) using the metrics presented in Table 7.2.4.

Table 7.2.4: The recreated pulses under the same ($Q = 1.5$ l/s) and different ($Q = 2$ l/s) flow conditions from the forward step are compared to the original Gaussian modulated sine pulse with $f_c = 60$ kHz in terms of the temporal compression degree and similarity.

	Source Pulse ($f_c = 60$ kHz)	TR Recreated pulses	
		1.5 l/s	2 l/s
Peak time (s)	8.43×10^{-4}	8.41×10^{-4}	8.33×10^{-4}
Percent error (%)	0	0.24	1.19
Correlation coef.	1	0.93	0.92

The results of Table 7.2.4 indicate a considerable temporal shift in the recreated pulse under non-stationary flow in comparison to the steady flow scenario. Specifically, the peak value in the recreated pulse under unsteady flow conditions occurs earlier to the peak value of the original pulse by $\sim 10^{-5}$ s (percent error: 1.19%). On the other hand, the NR TR under steady flow conditions shows only minimal time shift (percent error: 0.24%). Surprisingly, the attained temporal refocusing in this test is better than that presented in Table 7.2.3 of subsection 7.2.5 where NR TR is tested under matching steady flow conditions. This observation establishes a margin of error that could be controlled by increasing the number of repeated measurements for a particular case. Finally, the estimated cross-correlation coefficient between the reconstructed and source pulses shows a great degree of similarity in both cases.

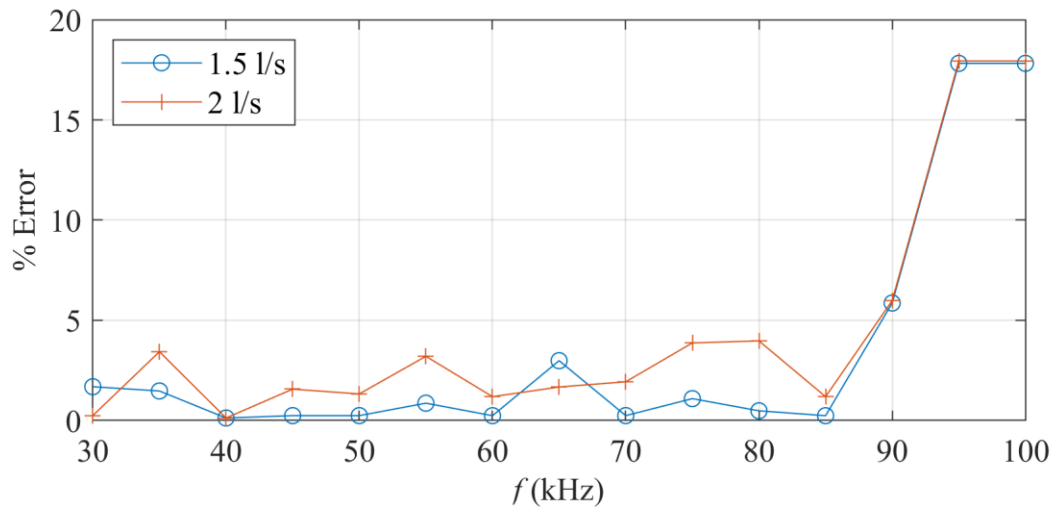


Figure 7.2.27: Percent error as a function of probing frequency between the peak time of the original and recreated pulses when the TR step is: (i) performed with an identical ($Q = 1.5$ l/s) and (ii) slightly increased ($Q = 2$ l/s) flow compared to the forward step.

The synopsis of the remaining 14 tests is shown in Figure 7.2.27 and Figure 7.2.28. In Figure 7.2.27, the percent error in the temporal compression from the steady and unsteady flow TR tests is plotted against the probing frequency. Except for the tests at 30 and 65 kHz, the NR TR process under unsteady flow conditions demonstrates a comparatively larger error in temporal refocusing. At central frequencies higher than 85 kHz, the SNR reduces rapidly, and the observed error is controlled by noise. That is due to the fact that the transducer used (Brüel & Kjær Type 8104) quickly becomes inefficient at projecting acoustic signals with a frequency content past its resonance. It is interesting to note that the plot of Figure 7.2.28 closely resembles the plot of Figure 7.2.24. That is to be partly expected, since both figures present the results for the identical case of NR TR under a steady flowrate $Q = 1.5$ l/s. In the context of TR under unsteady flow, however, the results of Figure 7.2.28 denote a temporal shift of the restructured pulse and not a loss of similarity to the source. In conclusion, it should be reiterated that the tests should be repeated multiple times in order to establish the margin of error. Also, a setup of longer range under higher frequencies and flowrates is expected to exacerbate the temporal shift effect observed in these results.

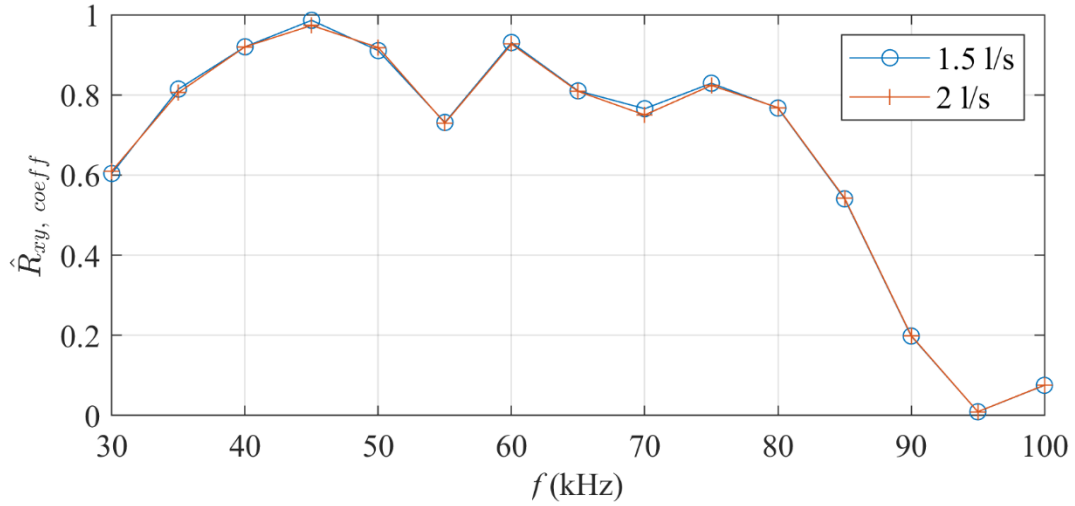


Figure 7.2.28: Maximum cross-correlation coefficient as a function of probing frequency calculated between the original and recreated pulses when the TR step is: (i) performed with an identical ($Q = 1.5$ l/s) and (ii) slightly increased ($Q = 2$ l/s) flow compared to the forward step.

7.2.7 Phase shift due to flow

Although in the previous subsections it becomes apparent that flow in the pipe introduces a temporal shift in the TR focal point, the dependence of said shift on the frequency of the probing signal is not investigated. In this set of measurements, the objective is to evaluate the phase shift between waves that travel with and against the flow. The experiment is performed on the water filled intact DN80 HDPE pipe of the HF setup at the hydraulic lab at HKUST and incorporates a series of amendments that set it apart from the typical tests presented so far. For instance, only the forward step is performed (i.e., there is no signal time reversal), and unmodulated sine waves are injected in the system.

Specifically, two transducers (Brüel & Kjær Type 8104) placed along the DN80 HDPE pipe centreline and their in-between distance is set equal to 2.5 m. An external pump is introduced to the system and creates flow along the pipe. The flowrate, measured by an ultrasonic flowmeter attached at the downstream end of the pipes, is equal to 2 l/s and the calculated Reynolds number is ~ 32400 , indicating a fully developed turbulent flow. Moreover, given that the pipe cross sectional area is $A = 4.85 \times 10^{-3} \text{ m}^2$, the mean flow velocity is $\bar{V} = Q/A = 0.48 \text{ m/s}$ and the Mach number $Ma = \bar{V}/c_f = 3.2 \times 10^{-4}$, assuming a wavespeed $c_f = 1483 \text{ m/s}$.

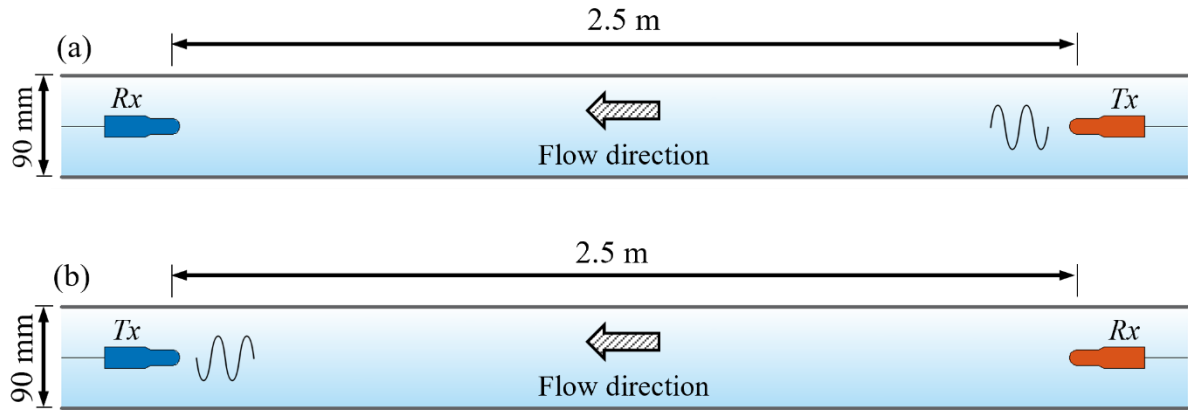


Figure 7.2.29: Experimental setup to measure phase shift in the response of the system due to flow ($Q = 2 \text{ l/s}$). (a) A sine wave pulse is emitted from the upstream transducer, and the downstream transducer records the response; the pressure wave propagates with the flow. (b) The same sine wave pulse is now emitted from the downstream transducer, and the upstream transducer records the response; the pressure wave propagates against the flow in this case.

The experiment is performed in two steps (Figure 7.2.29). First, a single frequency sine pulse of total duration equal to 0.1 s is emitted from the upstream transducer. In this case, the pressure wave propagates along the flow direction to meet the receiver 2.5 m away from the source. The receiver records the response of the system, denoted as P_1 . In the second step, the same single frequency sine pulse is emitted from the downstream transducer, and the system response P_2 is now obtained at the upstream transducer. Note that the distance between the transducers, as well as the flow characteristic do not change between the two steps. In total, 11 different single frequency sine waves are tested. The lowest frequency tested is 30 kHz, while the highest is 80 kHz. The frequency is increased at 5 kHz increments for each test. A relatively long signal duration (0.1 s) is used because abrupt changes to the signal that occur at the beginning and end of the transmission excite additional frequencies in the system and temporarily corrupt the response. If signal transmission is kept over an adequate amount of time, then it is straightforward to pick a segment of the response that is unaffected by these effects. Preliminary tests showed that a signal duration 0.1 s is sufficient. It is noted that the sampling frequency for both signal transmission and acquisition tasks is equal to 1 MHz. For comparison, both steps of the experimental process are repeated under quiescent water conditions.

In the presented results, two periods from each of the responses P_1 and P_2 are isolated, making sure that their start and end times synchronized. Of interest is to identify the phase shift between P_1 and P_2 when obtained under flow and stationary conditions over a range of frequencies. The

phase difference, or phase angle, is defined as the difference in phase between identical points (e.g., zero crossings) in two different waveforms of the same frequency. A waveform has a leading phase over another when a specific point happens earlier in time. Phase differences are demonstrated using the Lissajous pattern plots of P_1 against P_2 . In case P_1 and P_2 have a 0° or 180° phase difference, the Lissajous patterns form a perfectly straight line. Moreover, the angle of the line to the horizontal denotes the amplitude difference between the two signals, and a 45° angle corresponds to two signals of the same amplitude. In contrast, a perfect circle implies a 90° phase difference and equal amplitudes between P_1 and P_2 . Phase shift values in-between 0° and 90° appear as tilted ellipses and the exact figure is calculated by measuring the maximum vertical deflection ($P_{1\max}$) and the vertical deflection at zero horizontal deflection ($P_1(x=0)$). The phase difference $\Delta\Phi$ is then calculated from:

$$\Delta\Phi = \Phi_2 - \Phi_1 = \pm \sin^{-1} \left(P_1(x=0) / P_{1\max} \right) \quad (7.2.3)$$

given that the top of the ellipse is located in the 1st quadrant. In Eq. (7.2.3), Φ_1 and Φ_2 are the phases of P_1 and P_2 , respectively.

Figure 7.2.30 to Figure 7.2.32 present the results from three measurements with probing frequencies 40, 60, and 80 kHz in full. The layout of these figures is as follows: (a) plots the two isolated periods from the recorded signals P_1 and P_2 for the case of quiescent flow, (b) depicts the two isolated periods from the recorded responses P_1 and P_2 under flow conditions ($Q = 2$ l/s), (c) plots the Lissajous pattern (P_1 against P_2) for the waveforms in (a), and (d) plots the Lissajous pattern (P_1 against P_2) for the waveforms in (b). Then, in tabulation form, the phase difference is calculated using the values obtained from the Lissajous pattern of subfigure (d) and the expression of Eq. (7.2.3). It should be noted that P_1 and P_2 signal snippets are each normalized by the pertinent largest values in the signals. In this experiment, it is not important to distinguish amplitude differences between P_1 and P_2 . Besides, doing so would require the accurate recalibration of the employed transducers.

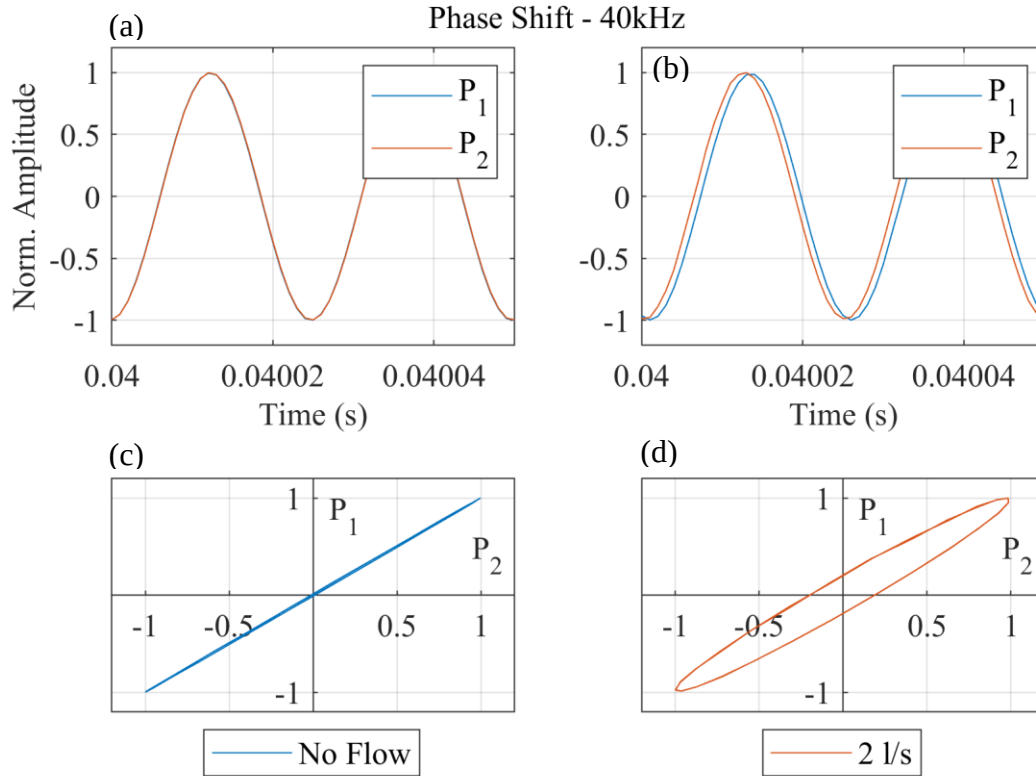


Figure 7.2.30: (a) For stationary water inside a DN80 HDPE pipe, a 40 kHz sine wave is emitted from one transducer and the system response P_1 is recorded at an opposing transducer at 2.5 m range (i.e., $G(A|B)$). The receiving transducer projects the same 40 kHz sine wave, and the response P_2 is measured at the other transducer (i.e., $G(B|A)$). Two periods from each of P_1 and P_2 are extracted and plotted together; the two curves fall on top of each other indicating no phase difference. (b) One-way flow is introduced to the DN80 HDPE pipe, and the test is repeated; P_1 and P_2 plots are now shifted, indicating a phase difference and, thus, $G(A|B) \neq G(B|A)$. (c),(d) The Lissajous patterns for the waveforms in (a) and (b), respectively.

Figure 7.2.30 displays the results obtained from all steps of the described experiment for a 40 kHz sine pulse. In particular, Figure 7.2.30a shows the results for stationary water in the pipe. To synopsise, a transducer emits a 0.1 s, 40 kHz sine wave, and a receiving transducer 2.5 m away records the system response P_1 . Then, that receiving transducer acts as a source and emits the same 0.1 s, 40 kHz sine wave, and the system response P_2 is now recorded at the location of the source transducer of the first step. Two synchronized periods from each of P_1 and P_2 are extracted from the overall waveforms and plotted in Figure 7.2.30a. The duration of these two periods for 40 kHz sine wave is equal to $2T = 2/f = 5 \times 10^{-5}$ s, as may also be seen in Figure

7.2.30a. Evidently, P_1 and P_2 in Figure 7.2.30a coincide and demonstrate almost 0° of phase difference. The fact is also supported by Figure 7.2.30b that plots the Lissajous pattern for P_1 and P_2 of Figure 7.2.30a: the blue curve is a straight line that crosses the axes at their origin. Since the amplitude of each P_1 and P_2 is normalized and equal to 1, the line has a 45° angle with the x -axis. On the other hand, Figure 7.2.30b presents the results from the same experimental procedure, but under the effect of flow in the pipe ($Q = 2$ l/s). The direction of the flow is one-way, hence, in the first and second steps the pressure wave travels with and against the flow, respectively. The pertinent results are plotted in Figure 7.2.30b as P_1 and P_2 , where the phase difference is already apparent, since the two waveforms are visibly "shifted" from each other. Accordingly, the corresponding Lissajous pattern (Figure 7.2.30d) is an ellipse, and the calculated phase shift is equal to 11.77° (Table 7.2.5). In summary, while no phase shift is detected between P_1 and P_2 under quiescent, stationary water conditions, the opposite is true under flow conditions. These findings reinforce that the Green's function of the waveguide $G(A|B) \neq G(B|A)$ and, thus, spatial reciprocity does not hold under flow conditions. Next, the results for a probing sine wave of 60 kHz (i.e., higher frequency) are presented.

Table 7.2.5: Points on the Lissajous curves to calculate the phase shift for a signal of frequency $f = 40$ kHz

40 kHz – Phase shift		
$P_1(x = 0)$	$P_{1\max}$	$\Delta\Phi$
0.188	0	11.77°

Figure 7.2.31 depicts the results from the test for a sine wave with frequency equal to 60 kHz. The P_1 and P_2 responses obtained for stationary water in the pipe are depicted in Figure 7.2.31a and, given the frequency of the signal (60 kHz), their total duration for two periods is equal to $2T = 2/f \cong 3.33 \times 10^{-5}$ s. Also evident in Figure 7.2.31a is that the phase of the two signals is almost identical. This can also be seen in the pertinent Lissajous curve (Figure 7.2.31a) that plot P_1 versus P_2 and form a straight line in this case. Also noteworthy is that the waveforms in Figure 7.2.31a appear piecewise linear at the peaks where curvature changes fast. This is to be expected since data acquisition happens at a specific sampling rate. At 1 MHz sampling rate that is applied here, there are 16 linearly distributed sample points within a period of a 60 kHz sine wave. Figure 7.2.31b plots P_1 and P_2 in the presence of unidirectional flow in the system. Now, the two waveforms "separate" from each other, and there is discernible phase shift that

is verified by the relevant Lissajous curves in Figure 7.2.31d where an ellipse appears. Points on the ellipse are sampled in Table 7.2.6, and the phase shift between P_1 and P_2 under flow is calculated equal to 16.81° . It is reminded that the phase shift between P_1 and P_2 under flow conditions for a 40 kHz probing signal is slightly less and equal to 11.77° . Therefore, there is an indication that phase shift between P_1 and P_2 scales with the frequency of the probing signal. To further investigate, the results from using an 80 kHz sine wave probing signal are shown next.

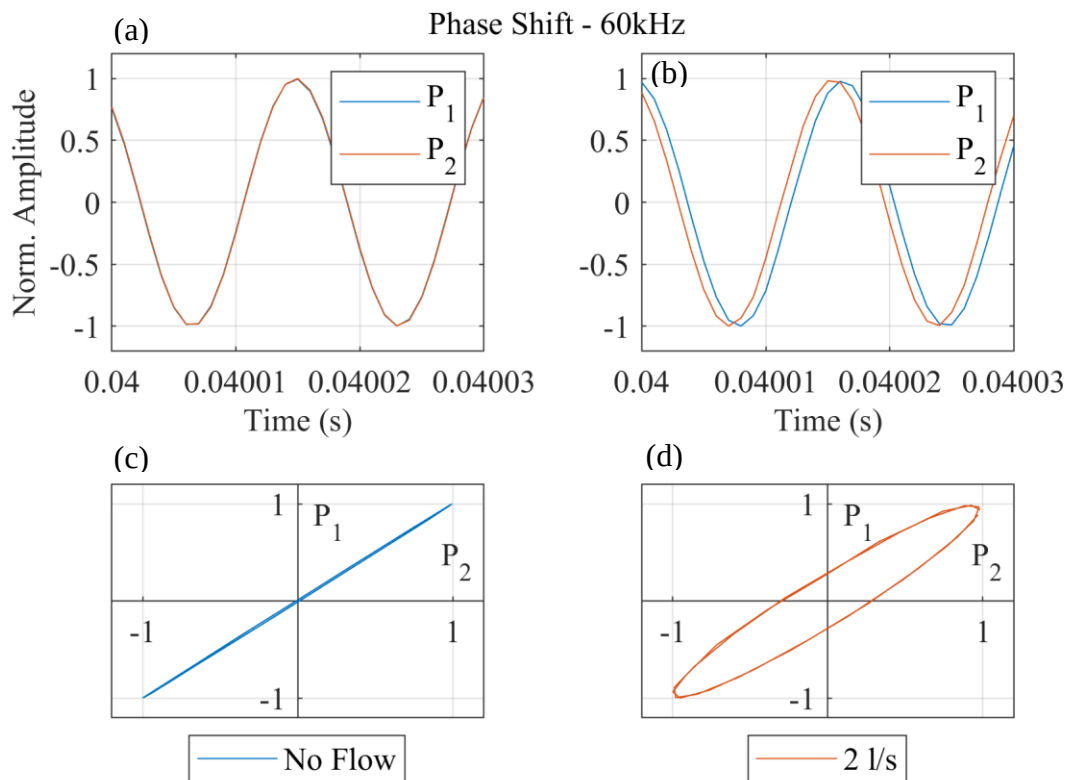


Figure 7.2.31: (a) For stationary water inside a DN80 HDPE pipe, a 60 kHz sine wave is emitted from one transducer and the system response P_1 is recorded at an opposing transducer at 2.5 m range (i.e., $G(A|B)$). The receiving transducer projects the same 60 kHz sine wave, and the response P_2 is measured at the other transducer (i.e., $G(B|A)$). Two periods from each of P_1 and P_2 are extracted and plotted together; the two curves fall on top of each other indicating no phase difference. (b) One-way flow is introduced to the DN80 HDPE pipe, and the test is repeated; P_1 and P_2 plots are now shifted, indicating a phase difference and, thus, $G(A|B) \neq G(B|A)$. (c),(d) The Lissajous patterns for the waveforms in (a) and (b), respectively

The outcomes of the test at 80 kHz are on display in Figure 7.2.32. In Figure 7.2.32a are the obtained system responses P_1 and P_2 in stationary water. The two periods displayed have a total duration of $2T = 2/f = 2.5 \times 10^{-5}$ s, and accurately match in phase. As expected, the Lissajous pattern in this case (Figure 7.2.32c) shows a straight line that crosses the axes at the origin. When measured under flow conditions (Figure 7.2.32b), P_1 and P_2 exhibit a distinct phase angle that is also substantiated from the pertinent elliptical Lissajous pattern (Figure 7.2.32d). The calculated phase angle in this case (Table 7.2.7) is $\Delta\Phi = 30.30^\circ$, that is further increased compared to that occurring for a 60 kHz probing signal.

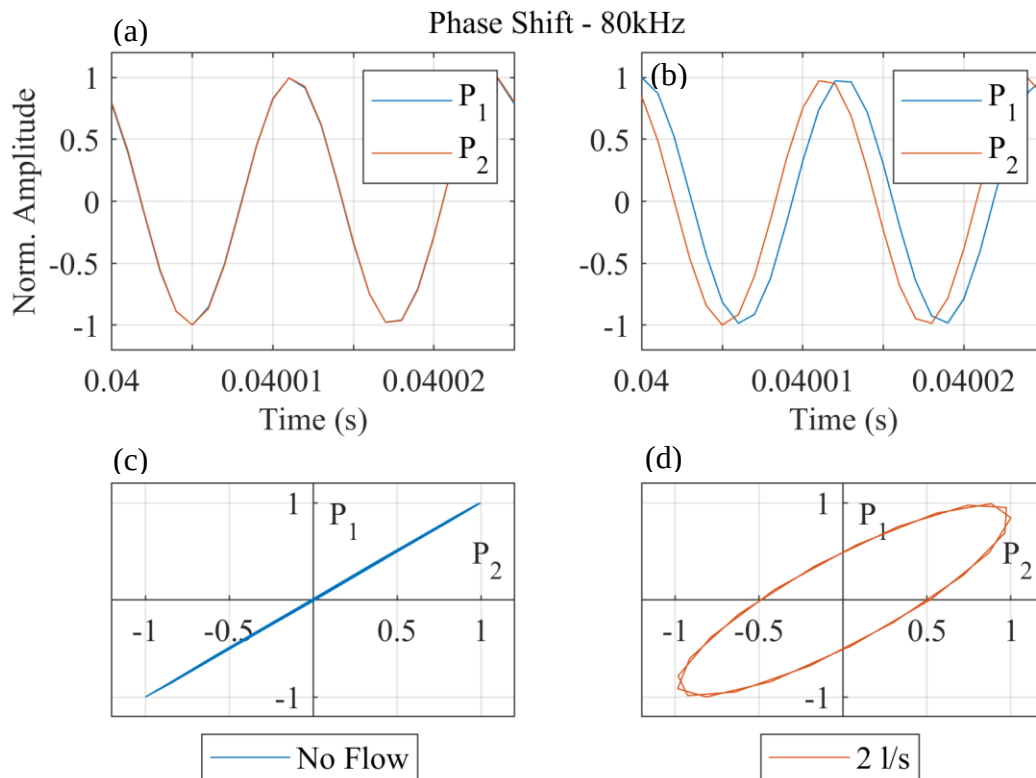


Figure 7.2.32: (a) For stationary water inside a DN80 HDPE pipe, a 80 kHz sine wave is emitted from one transducer and the system response P_1 is recorded at an opposing transducer at 2.5 m range (i.e., $G(A|B)$). The receiving transducer projects the same 80 kHz sine wave, and the response P_2 is measured at the other transducer (i.e., $G(B|A)$). Two periods from each of P_1 and P_2 are extracted and plotted together; the two curves fall on top of each other indicating no phase difference. (b) One-way flow is introduced to the DN80 HDPE pipe, and the test is repeated; P_1 and P_2 plots are now shifted, indicating a phase difference and, thus, $G(A|B) \neq G(B|A)$. (c),(d) The Lissajous patterns for the waveforms in (a) and (b), respectively.

Table 7.2.6: Points on the Lissajous curves to calculate the phase shift for a signal of frequency $f = 60$ kHz

60 kHz – Phase shift		
$P_1(x = 0)$	$P_{1\max}$	$\Delta\Phi$
0.285	0.9854	16.81°

Figure 7.2.33 combines the Lissajous curves presented already for the 40, 60 and 80 kHz sine wave probing signals. Specifically, Figure 7.2.33a plots P_1 versus P_2 for the three cases when the water in the DN80 HDPE pipe is stagnant, while Figure 7.2.33b depicts P_1 versus P_2 under flow conditions ($Q = 2$ l/s). Put differently, Figure 7.2.33a combines (c) subfigures of Figure 7.2.30 to Figure 7.2.32 and Figure 7.2.33b merges subfigures (d) of Figure 7.2.30 to Figure 7.2.32 together. It has already been discussed that in the absence of flow the pipe system is symmetric, thus there is no phase difference between P_1 and P_2 . It can also be seen in Figure 7.2.33a that the Lissajous pattern remains a straight line regardless of the probing frequency; all lines coincide. On the contrary, the ellipses formed of the Lissajous patterns for P_1 versus P_2 under flow conditions demonstrate an increasing minor axis (i.e., become wider) with the frequency of the probing signal. Thus, the phase shift under constant flow increases with frequency.

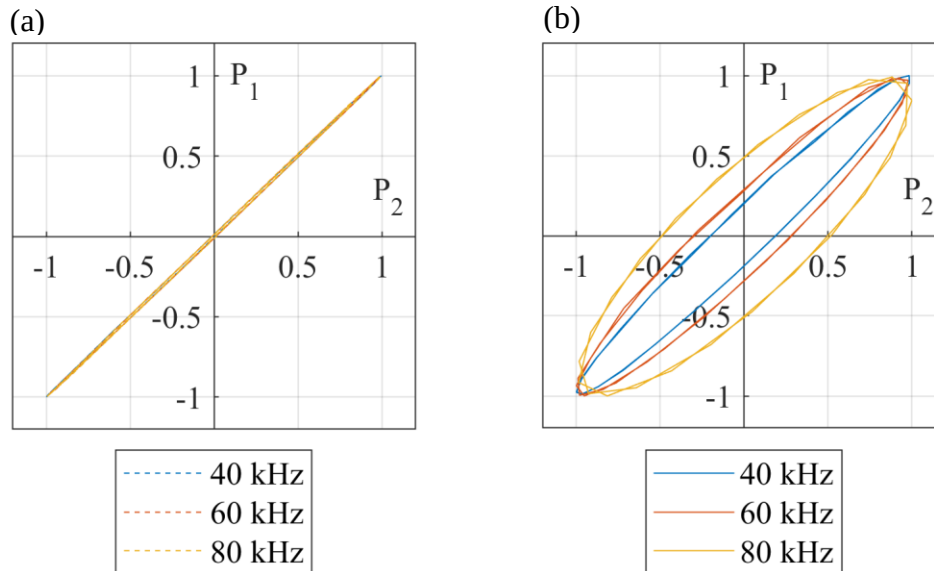


Figure 7.2.33: The P_1 against P_2 Lissajous curves for a 40, 60, and 80 kHz sine probing wave are combined together for the cases of (a) stagnant water, and (b) unidirectional water flow in the pipe.

Table 7.2.7: Points on the Lissajous curves to calculate the phase shift for a signal of frequency $f = 80$ kHz

80 kHz – Phase shift		
$P_1(x = 0)$	$P_{1\max}$	$\Delta\Phi$
0.5	0.9911	30.30°

As already noted, however, the tests are not solely performed for 3 different probing frequencies, but 11: starting from a 30 kHz up to 80 kHz sine wave at 5 kHz increments. Full display of the results is limited to the three cases of 40, 60, and 80 kHz sine waves for brevity. Regardless, the phase shift for each of the 11 under pipe flow tests is calculated and plotted against the pertinent probing wave frequency in Figure 7.2.34. There, it can be seen that the phase angle between P_1 and P_2 increases linearly in frequency. Following a linear regression analysis in MATLAB, the experimental findings appear to follow the line defined as $y = 0.49x - 9.28$, where y is the phase angle in degrees and x is the probing frequency in kHz. To conclude, the experimental results presented in this subsection not only show that pipe flow causes a phase shift on the propagating pressure waves, but also that this phase shift is linearly dependent on the probing wave frequency. In essence, the application of wave-based defect detection techniques in pipes (e.g., in-pipe pulse-echo TR), especially when high-frequency probing signals are employed, can lead to errors unless the effect of the pipe flow is also considered in the model of the system

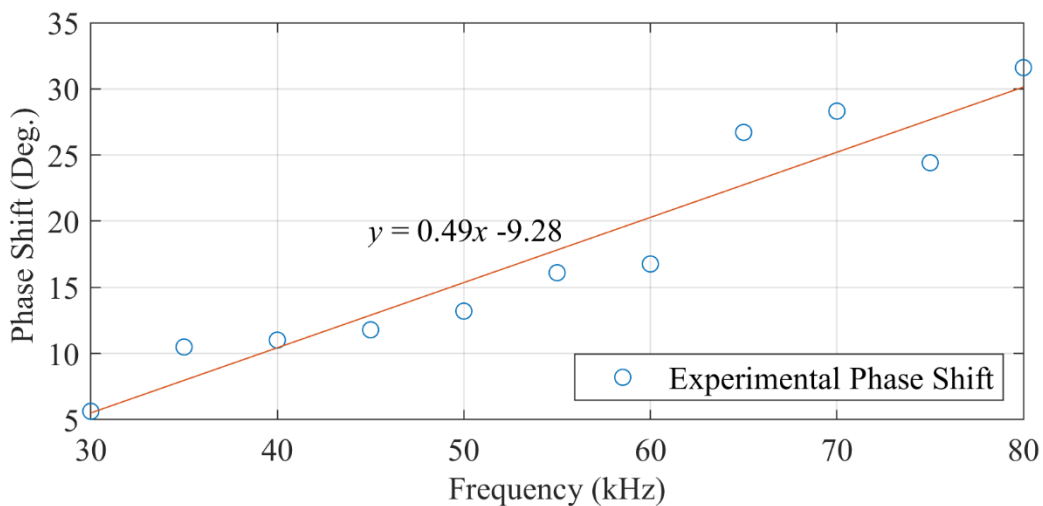


Figure 7.2.34: The phase angle between P_1 and P_2 , measured under pipe flow conditions, is evaluated from 11 tests each with a probing sine wave of increasing frequency.

7.3 Fidelity of TR under wave path instabilities

7.3.1 Inline junction effect to TR

As explained in Chapter 5, there are two mechanisms that can invalidate TR: instabilities and damping. Pipe flow that incorporates both of these mechanisms has been investigated in subsection 7.2, and has been found to induce both a temporal and spatial shift to the TR focal spot. Indeed, the impact of pipe flow to the TR process becomes more pronounced with increasing flowrates and probing wave frequencies. In this series of tests, the objective is to identify the effect of perturbations in the wave path to the ability of the TR to refocus. These perturbations that essentially break the spatial reciprocity of the waveguide, may be created in practice in different ways. For example, the forward step is applied in one water filled elastic pipe, while the backpropagation step is performed in a different water filled pipe of similar nominal characteristics. Due to manufacturing errors, two pipes are never identical both in terms of geometry and mechanical properties. Also, inconsistencies during the assembly phase can render two seemingly matching pipes dissimilar. In another example, the forward step is conducted along an uninterrupted water filled pipe, while the TR step is performed in a different pipe with an inline junction but otherwise identical to the first. Besides, the influence of an inline to the propagation of high-frequency waves in pipes has not been investigated before. In any event, the question to be answered is: If the TR step is applied on a water filled pipe that using the time reversed forward response from a different one, will the TR field still refocus? If yes, will the TR field refocus at the expected location, as defined by the distance in-between the transducer pair at the forward step, or will there be a shift?

To try and answer these questions, both aforementioned examples are realized in this subsection. The basic experimental setup is the HF setup at the hydraulic lab at HKUST, that integrates two straight DN80 HDPE pipes (aptly called “sister” pipes), the sole difference being an interchangeable 0.5 m section that is present close to the midpoint of one of the two. This interchangeable section is attached to the rest of the pipe via two bolt and nut inline junctions. The other DN80 HDPE is devoid of any junctions and runs uninterrupted over its entire length. There is no flow in the system, thus water is stationary. A schematic of the setup is presented in Figure 7.3.35.

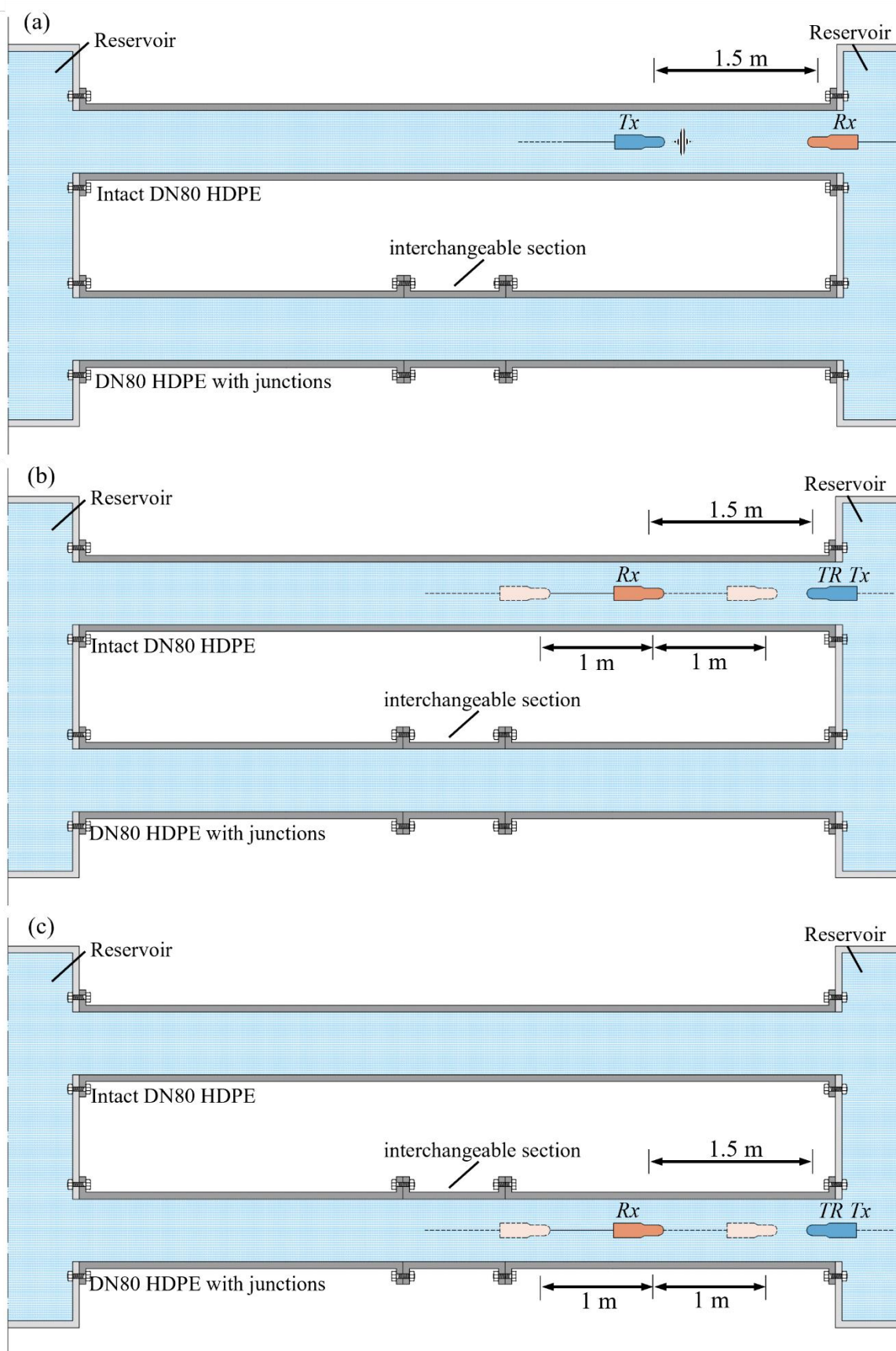


Figure 7.3.35: The HF setup at the hydraulic lab at HKUST contains two straight DN80 HDPE pipes; one that runs uninterrupted over its entire length, and another that contains two junctions around its midpoint (top view). (a) The forward test is performed in the intact DN80 HDPE pipe. (b) The TR step is first conducted in the same pipe where the TR field is sampled along a total of 2 m, and (c) the process is repeated in the DN80 HDPE pipe with the junction but on a segment of the pipe that maps that of the intact pipe and is free from any inline junctions. The aim is to investigate whether the TR process is still successful despite the fact that the TR step is performed on a different pipe to the forwards step.

For the initial experiment the junction is not considered. Here, the aim is to validate the TR process between two similarly intact “sister” pipes. Typically for TR experiments, a transducer pair (two Brüel & Kjær Type 8104 transducers) is introduced along the centreline of the wholly intact DN80 HDPE pipe (Figure 7.3.35a). For the test, a segment of the pipe neighbouring the reservoir is selected. The particular reason for doing so will be revealed later. For the forward propagation test, the receiving transducer is placed as close to the reservoir as possible, and the source transducer at a distance equal to 1.5 m from the receiver towards the pipe midpoint. A Gaussian modulated sine pulse with central frequency $f_c = 30$ kHz and bandwidth defined at 1/100 the peak spectrum amplitude equal to [24, 36] kHz is emitted from the source (Figure 7.3.36a), and the dispersed system response is recorded at the receiver by the reservoir (Figure 7.3.36b). The sampling rate for both signal transmission and acquisition tasks is set to 1 MHz. The TR step is performed in the same pipe following the role interchange of the transducers (Figure 7.3.35b). The TR source (now close to the reservoir) emits the time reversed response from the forward step (Figure 7.3.36c), and the response is measured at the location of the original source, 1.5 m away (Figure 7.3.36d) where the TR focal spot is. All waveforms in Figure 7.3.36 are normalised by their respective amplitudes. So far, the description corresponds to a typical TR experiment. However, there is an additional step that differentiates this specific experiment.

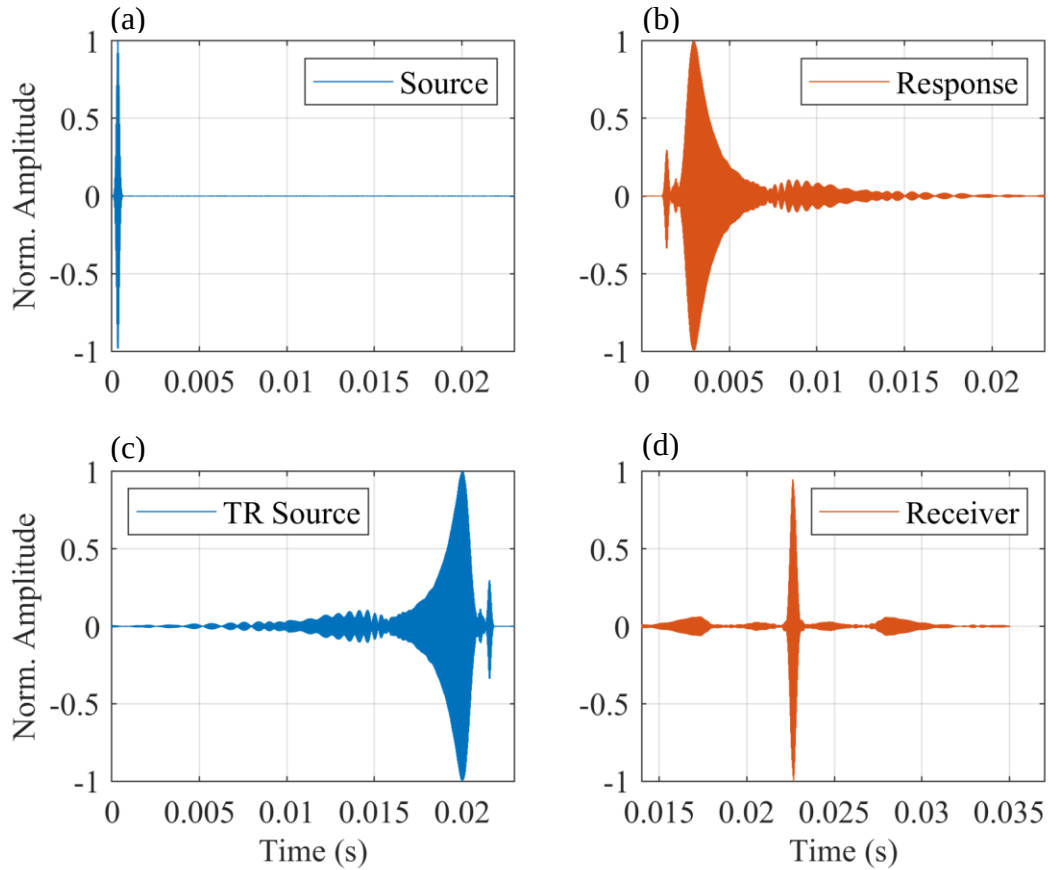


Figure 7.3.36: (a) The source transducer, placed close to the end of the intact DN80 HDPE pipe by the reservoir, emits a Gaussian modulated sine pulse of central frequency $f_c = 30$ kHz. (b) A receiving transducer, 1.5 m away from the source, records the system response to the signal in (a). (c) For the TR step, the transducer roles are interchanged, and the receiving transducer becomes the source that emits the time reversed response shown in (b). (d) The TR process reconstructs the original pulse at the receiving transducer by the reservoir.

The same transducer pair is now moved to the DN80 HDPE pipe with the junctions to repeat the TR step using the time reversed response from the forward step in the intact DN80 HDPE pipe; that is, the forward step is not repeated in the new system. However, in this test, the inline junctions are not included in the waveguide tested. For this initial test, the goal is to compare the TR results from two systems that only differ by small, immeasurable, manufacturing inconsistencies (e.g., differences in cross-sectional dimensions, mechanical properties). That is the reason the transducers are placed at the outmost right end of the two pipes (Figure 7.3.35). The transducers could not be placed in the middle of the DN80 HDPE pipe, since the junctions would be included. Also, the selected segments along the two “sister” pipes should be matching in terms of distance from the pipe ends. This way, the effect of the boundaries to the system

response (e.g., induced reflections) is equally valid for both pipes. Note that the longitudinal positions of the transducers in the DN80 HDPE with junction relative to the pipe ends matches their position in the intact DN80 HDPE pipe (Figure 7.3.35b, c), and their in-between distance is kept equal to 1.5 m. The TR source is placed by the reservoir, and the receiver is placed 1.5 m away, to match the TR focal point in the first pipe. Figure 1.31 presents the results of the TR step in the “sister” pipe: (a) is the TR source signal, and (b) shows the reconstructed signals from both pipes plotted together. Evidently, the waveforms match fairly well. To verify that the original source location (i.e., 1.5 m from the transducer by the reservoir) is indeed the refocusing point in both “sister” pipes, the TR pressure field is sampled over 2 m (i.e., 1 m either side of the original source location) at 2 cm increments (Figure 7.3.35b, c). Subsequently, the TR step is repeated for 101 times in each of the “sister” pipes, with the receiving transducer moved by 2 cm FR each measurement.

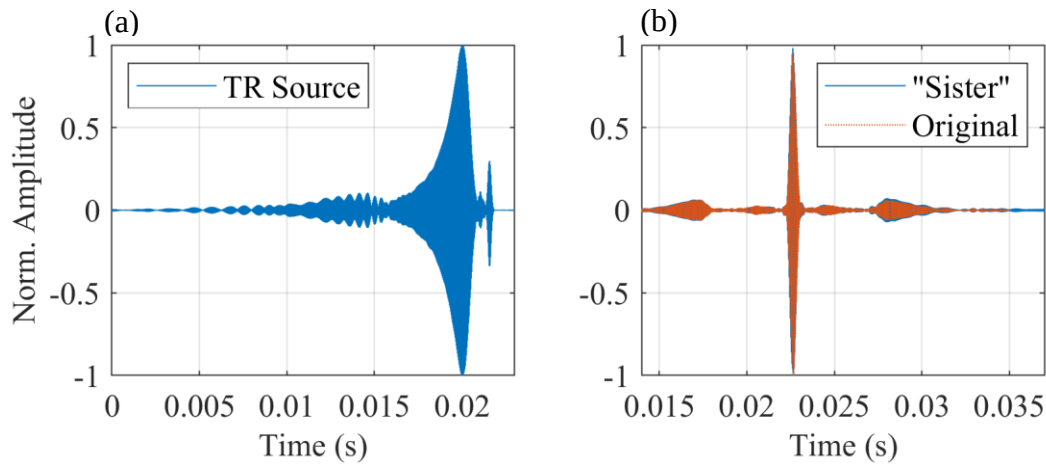


Figure 7.3.37: The TR step is repeated in the “sister” DN80 HDPE pipe, but along a segment that excludes any junctions. The specific segment matches that of the intact DN80 HDPE pipe in regards its position relative to the pipe ends. (a) The transducer closer to the reservoir emits the time reversed froward response from the intact DN80 HDPE pipe. (b) TR field is evaluated at the receiving transducer 1.5 m away, therefore mimicking the TR step performed in the intact DN80 HDPE pipe. The reconstructed pulse from the TR process in the alternative DN80 HDPE pipe matches that of the original, wholly intact DN80 HDPE pipe fairly well.

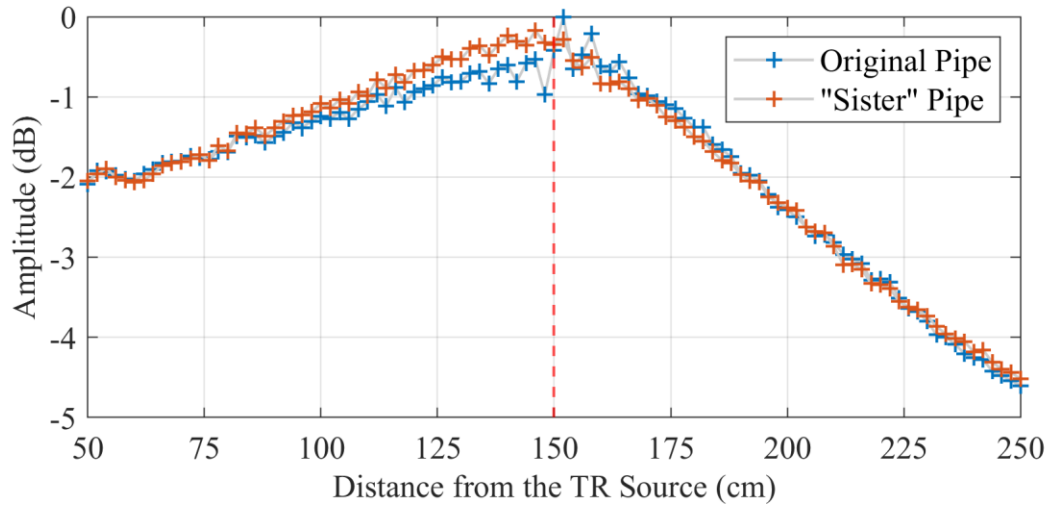


Figure 7.3.38: The amplitude of the TR field is plotted against the range from the TR source for the original and “sister” DN80 HDPE pipes. The amplitude values are given in dB scale, relative to the maximum amplitude overall. The vertical red dashed line at $x = 150$ cm, corresponds to the location of the original forward step source.

In Figure 7.3.38, the amplitude of the TR field is presented as a function of the range, measured from the TR source. The amplitudes are normalized with respect to the overall maximum amplitude of the TR field recorded from both the original and “sister” pipes. It is apparent in Figure 7.3.38 that the maximum amplitude is found in the TR field of the original, intact DN80 HDPE pipe, at approximately the location of the forward step source (blue curve). Indeed, the maximum amplitude is found 152 cm from the TR source i.e., 2 cm from the forward step source location. However, the minimum wavelength of the probing pulse is slightly larger than 4 cm; therefore, a spatial shift of $2 \text{ cm} < \lambda/2$ is not significant. It may be concluded that the TR field in the original DN80 HDPE pipe successfully refocuses at the forward source location. Regardless, the main interest in this experiment is to verify that the TR field in the “sister” pipe using the same TR source also refocuses at the location of the hypothetical forward step source location i.e., at $x = 150$ cm. Surprisingly, the orange curve in Figure 7.3.38 that corresponds to the amplitude of the TR field in the “sister” pipe also peaks in the vicinity of the supposed forward step source, at 146 cm from the TR source. Here, the spatial shift is 4 cm, thus larger than $\lambda/2$ where λ is the minimum probing wavelength. The shift is to be expected, given that the backpropagation wave path is different from the forward. Still, the differences (*instabilities*) in the wave paths are not severe enough to compromise the TR process entirely.

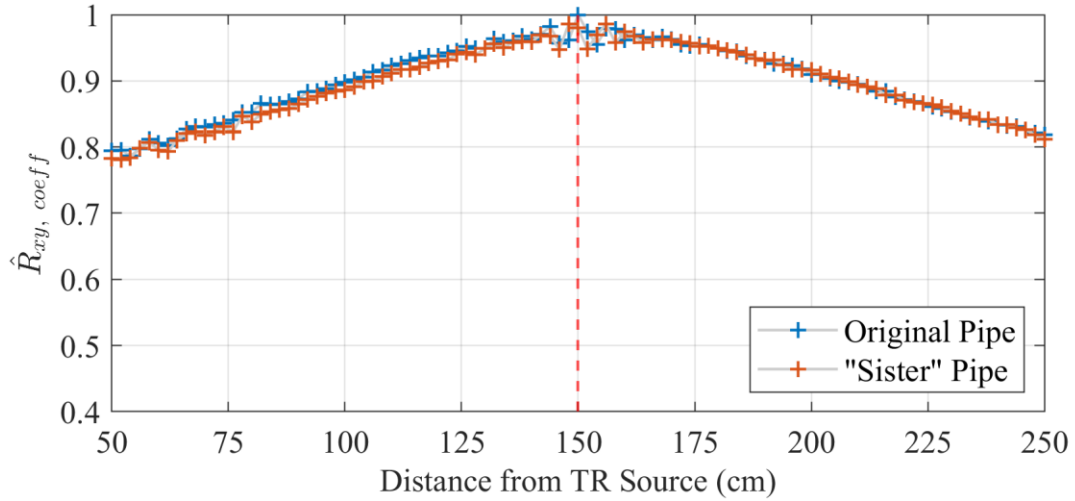


Figure 7.3.39: The maximum cross-correlation coefficients calculated between the TR field responses for both the original and sister pipes and the reconstructed pulse at $x = 150$ cm in the original pipe are plotted against the distance from the TR source. Naturally, the blue curve peaks at $x = 150$ cm.

In addition to plotting the maximum amplitude, the maximum cross-correlation coefficient calculated between the TR field response at each point and the reconstructed pulse at $x = 150$ cm in the original pipe is plotted against the range, measured from the TR source (Figure 7.3.39). Obviously, the blue point curve that represents the maximum cross-correlation coefficient calculated between the TR waveforms along the original pipe and the reconstructed pulse at the original forward step source location peaks at $x = 150$ cm (i.e., the original forward step source location). On the other hand, the orange point curve corresponds to the maximum cross-correlation coefficients between the TR field measurements along the “sister” pipe and the reconstructed pulse at $x = 150$ cm from the TR source in the original pipe. There, the peak value is found at $x = 148$ cm, hence shifted by 2 cm. The observed shift is consistent with the results presented in Figure 7.3.38. In summary, the differences in the backpropagation wave path that could be attributed to manufacturing discrepancies between the original and “sister” pipes do result in a spatial shift of the TR focal point but are not sufficient to cause a complete breakdown of the process. Next, two inline junctions are introduced to the backpropagation wave path, and the validity of TR is investigated once again.

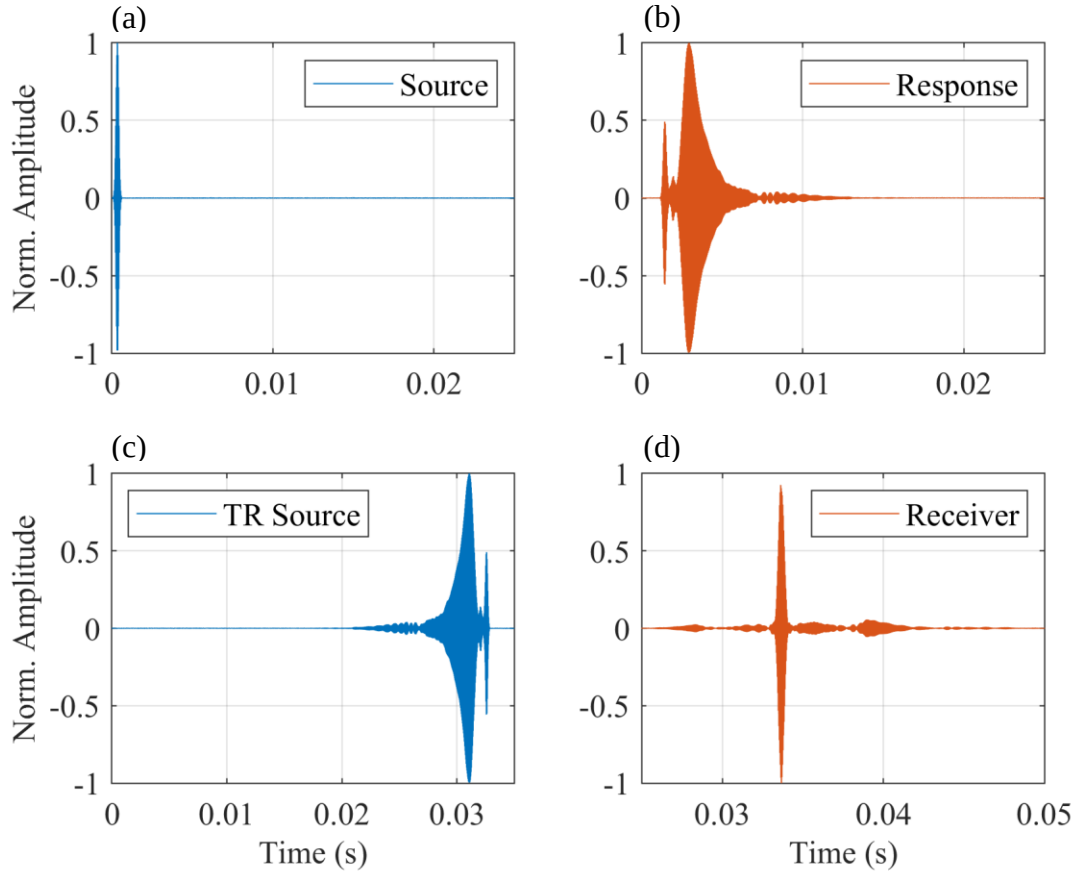


Figure 7.3.40: The forward and TR steps are repeated in the intact DN80 HDPE pipe, where the transducer pair is rearranged to the pipe middle to match the segment that includes the inline junctions in the “sister” pipe (a) A Gaussian modulated sine pulse with $f_c = 30$ kHz and 40% relative bandwidth is emitted from the source transducer, and (b) the system response is recorded 1.5 m away. For the TR step, the receiving transducer becomes the source that emits (c) the time reversed forward step response, and (d) the probing pulse is recreated at the location of the forward step source, now turned receiver.

The entire experiment, that includes both forward and TR steps, is repeated de novo. The transducer pair is now inserted at about the middle of the intact DN80 HDPE pipe and along the centreline, with the transducer in-between distance set to 1.5 m. Note that the corresponding segment along the “sister” DN80 HDPE pipe includes the two inline junctions (Figure 7.3.41a). Initially, the forward propagation step is performed. The transducer positioned to the left (Figure 7.3.41a in blue colour) is assigned the role of the acoustic projector, and emits a Gaussian modulated pulse of central frequency $f_c = 30$ kHz and 40% relative bandwidth (i.e., [24, 36] kHz) at 1/100 of the amplitude spectrum (Figure 7.3.40a). The opposing transducer (Figure 7.3.41a in blue colour), at 1.5 m distance from the source, records the response of the

system (Figure 7.3.40b). It is interesting to note that the forward step response of the system when the transducer pair is placed at about the pipe midpoint (Figure 7.3.40b) is slightly different than that obtained when the transducer pair is placed closed to the reservoir (Figure 7.3.36b), despite using an identical probing pulse and keeping the transducer in-between distance equal to 1.5 in both cases. While the excited modes are the same, the two responses differ on the basis of the reservoir boundary effect, that partially reflects the propagating wave. This realization justifies redoing the forward step to match the longitudinal positions of the two transducers in the original and sister pipes, as will be seen later. Following the forward step, the TR step is conducted in the original, intact DN80 HDPE pipe. The roles of the transducers are interchanged, and the receiver of the forward step becomes the source and vice-versa (Figure 7.3.41b). The longitudinal position of the transducers along the pipe remains unaltered. The source transducer (Figure 7.3.41b in blue colour) emits the time reversed response of the system to the forward step (Figure 7.3.40c), and the receiving transducer (Figure 7.3.41b in orange colour) records the TR pressure field at the location of the forward step source (Figure 7.3.40d), where the probing pulse re-emerges. Note that the waveforms presented in Figure 7.3.40 are normalized with respect to their amplitude.

The same transducer pair is now “transplanted” to the “sister” pipe and is placed so that the two inline junctions are in-between. In fact, the longitudinal positions of the transducers relative to the pipe ends match these occupied in the original, intact DN80 HDPE pipe. The forward step is forgone, and only the TR step is performed. The necessary TR source is the time reversed forward step response from the original DN80 HDPE pipe (Figure 7.3.40c), and is emitted by the transducer on the right, as shown in Figure 7.3.41c in blue colour. The TR field is recorded at the opposing transducer (Figure 7.3.41c in orange colour), 1.5 m away so that it coincides with the hypothetical forward step source. Provided that spatial reciprocity between the two DN80 HDPE pipes still holds, the original probing pulse (that was never introduced in the “sister” DN80 HDPE pipe with the junctions) must be recreated at exactly that location. The TR step results from the “sister” DN80 HDPE pipe under the effect of the two inline junctions are presented in Figure 7.3.42. In particular, Figure 7.3.42a plots the TR source (that is identical to the TR source in the original system, shown in Figure 7.3.40c), and Figure 7.3.42b depicts the normalized pulses recorded 1.5 m from the TR source for both the original, intact DN80 HDPE pipe (in orange) and the discontinuous, “sister” DN80 HDPE pipe (in blue).

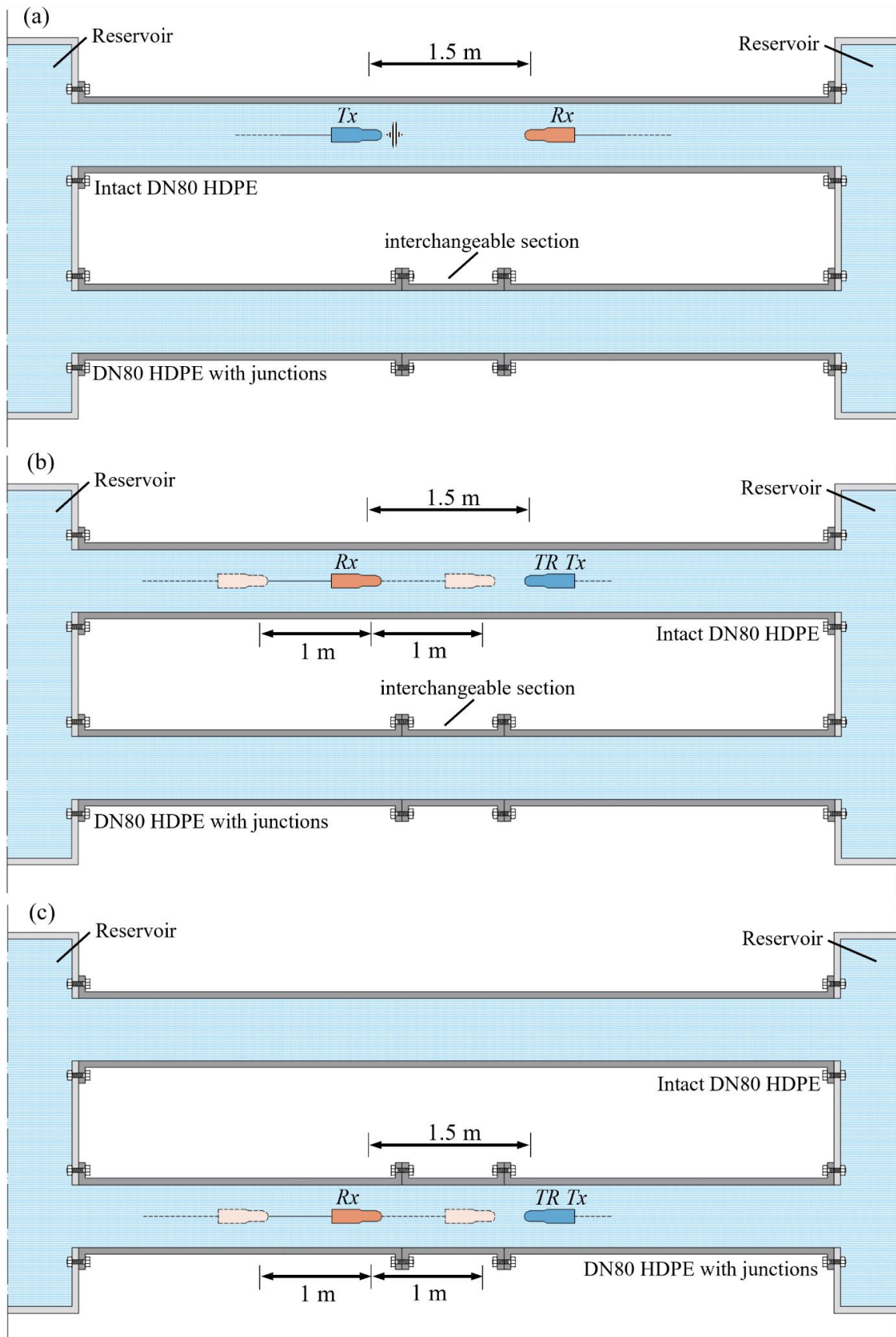


Figure 7.3.41: In this test, the objective is to evaluate the robustness of the TR process when inline junctions, that are not included in the forward step, are introduced to the backpropagation step. The two DN80 HDPE pipes of HF setup at the hydraulic lab at HKUST that only differ by the presence of two inline junctions in one are used in this test. (a) First, the forward test is performed in the intact DN80 HDPE pipe, where the transducer pair is placed at about the pipe middle to match the segment in the “sister” pipe that contains the two junctions. (b) The TR step is first conducted in the same pipe where the TR field is sampled along a total of 2 m, and (c) the process is repeated in the “sister” pipe with the two inline junctions.

The two waveforms in Figure 7.3.42 are distinguishably different. While the recreated signal in the original pipe is symmetric with respect to the main pulse and matches the original probing pulse, the recreated signal from the “sister pipe” does not exhibit similar traits. Apparently, the spatial reciprocity between the two pipes is lost as a consequence of the two inline pipe junctions. Regardless, it is possible that the TR field in the discontinuous, “sister” pipe refocuses at a different location. For that reason, the TR field is sampled in both the original and “sister” pipes over a 2 m length, 1 m from each side of the forward step source location (i.e., 1.5 m from the TR source), and at 2 cm increments. In total, 101 measurements of the TR field are performed in each of the two pipes. Note that the selected spatial resolution is smaller compared to the minimum probing wavelength.

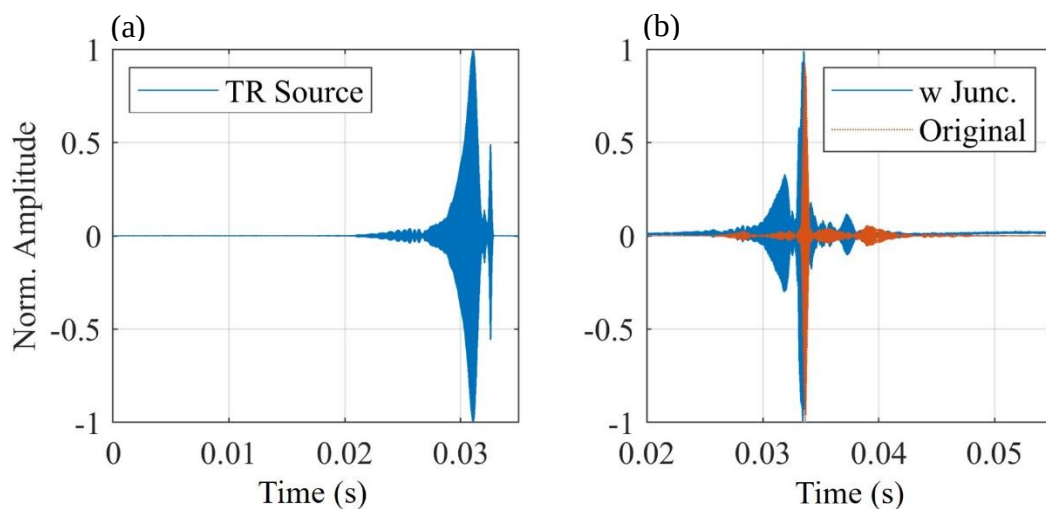


Figure 7.3.42: The transducer pair is installed along the discontinuous, “sister” pipe at the same longitudinal and radial positions as in the original pipe, and solely the TR step is conducted. (a) The TR source is the time reversed forward response of the original pipe, and (b) the TR pressure field is sampled at the supposed location of the forward step source. The

obtained signal at the same location from the TR test in the original pipe is also included. All waveforms shown are normalized by their respective amplitudes.

Figure 7.3.43 depicts the maximum amplitude at each measured location of the TR field for both the original and “sister” pipes. All values are now normalized by the maximum response amplitude overall, found at $x = 152$ cm from the TR source in the original DN80 HDPE pipe (Figure 7.3.43, blue point curve). While the forward step source location is at $x = 150$ cm, the apparent 2 cm shift is not significant since it measures smaller than the diffraction limit, $\lambda/2$. It may, therefore, be concluded that the TR field in the original pipe successfully refocuses at the location of the original source. In contrast, the TR field in the discontinuous, “sister” pipe (Figure 7.3.43, orange point curve) appears severely attenuated, and on average 30 dB less compared to that in the original pipe. In linear terms, the TR pressure field amplitude in the “sister” pipe is 31 times less than that in the original, despite the fact that an identical TR source is used. More crucially, however, the TR field in the discontinuous, “sister” pipe, does not present a clear peak i.e., it does not refocus. Essentially, the TR process fails as a result of the two inline junctions that are introduced in the backpropagation, TR step.

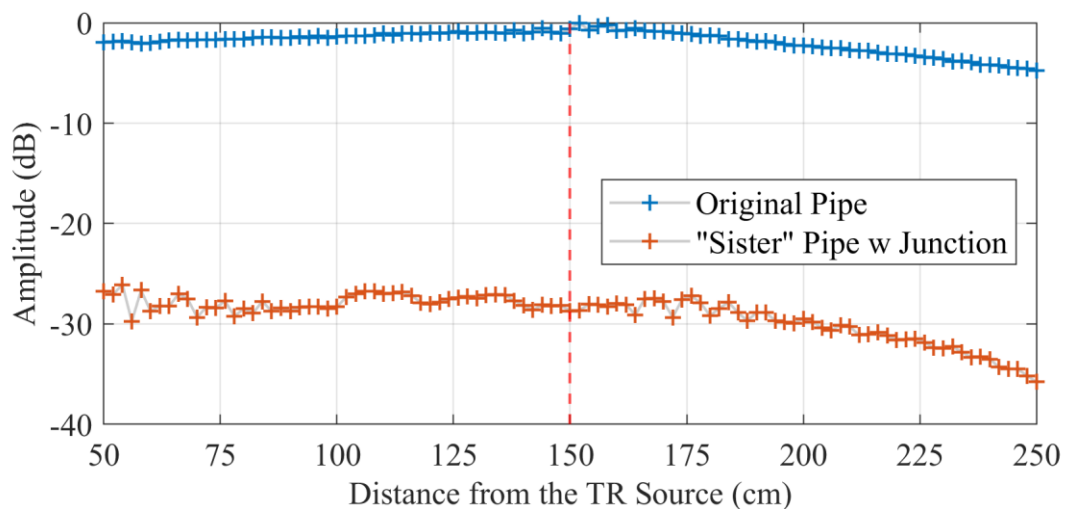


Figure 7.3.43: The TR field is sampled at multiple locations along the pipe that are spread at 2 cm increments over a total length of 2 m (1 m either side of the expected focal spot) for both the in intact and discontinuous DN80 HDPE pipe. The maximum amplitude at each location from each pipe is extracted, normalized by the overall maximum amplitude using a dB scale, and plotted against the distance from the TR source.

The maximum cross-correlation coefficient (Eq. (7.2.1)) between the each of the measurements of the TR field in each pipe and the recreated pulse at the forward step source location in the

original pipe is calculated and plotted against the range from the TR source. Naturally, the maximum value in the plot (i.e., 1) is observed in the curve for the original pipe system (Figure 7.3.44, in blue colour) at $x = 150$ cm. For increasing or decreasing values of range, the curve drops off, therefore indicating the unique focal point at the location of the original source. Conversely, for the case of the discontinuous, “sister” DN80 HDPE pipe (Figure 7.3.44, in orange colour) there is no such distinct peak. In fact, two local maxima are observed: one at $x = 54$ cm and another at $x = 192$ cm from the TR source, both in stark contrast to the supposed focal point at $x = 150$ cm. Therefore, the results show that the junction is able to break TR refocusing, and severely attenuate the signal propagating through it. Also, these exploratory results indicate that inline pipe junctions have a strong influence on the amplitude and phase of passing-through waves. In the following subsection, the rest of the pipes present in the HF setup are utilized, to investigate if the TR process is valid when the backpropagation step happens in a waveguide of dissimilar modal characteristics to that used in the forward step.

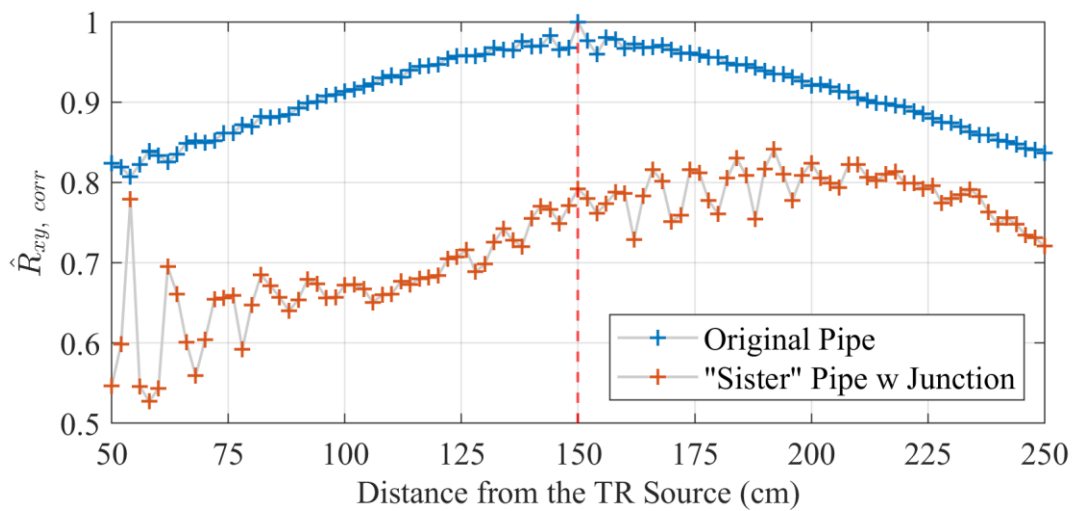


Figure 7.3.44: The maximum cross correlation coefficient is calculated between each of the TR pressure field measurements in each pipe and the recreated pulse at the forward step source location ($x = 150$ cm) in the original intact DN80 HDPE pipe and plotted against the pertinent distance from the TR source.

7.3.2 TR in pipes of different materials to the original

In the previous subsection 7.3.1 the backpropagation, TR step is attempted in a pipe that is different to that of the forward step, in order to evaluate the sensitivity of the TR process to instabilities. In particular, it is shown that while instabilities in the order of manufacturing

tolerances merely introduce a spatial shift of the TR focal point, introducing inline junctions to the backpropagation wave path nullifies the TR process. The HF setup at the hydraulic lab at HKUST, except for the two “sister” DN80 HDPE pipes, also hosts pipes made from other materials and/or cross-sectional dimensions. These include a DN80 copper pipe and a DN100 acrylic (Perspex) pipe. Hence, in this subsection the aforementioned pipes are employed to an experimental test that evaluates the fidelity of TR when the backpropagation step is performed in a waveguide of different modal characteristics to the one where the forward step is applied. Of course, the spatial reciprocity in this case is violated by default since $G(A|B) \neq G(B|A)$, where A and B represent the transducer locations. However, it is interesting to identify whether the change in the Green’s function will merely shift the focal point or cancel the TR process. In the case where a spatial shift is observed, which pipe produces the minimal error?

Much like the experiments in subsection 7.3.1, the forward step is performed along the intact DN80 HDPE pipe. Two Brüel & Kjær Type 8104 transducers are inserted along the pipe centreline, and the in-between distance is fixed to 2.5 m. The longitudinal position of the transducer pair is about the pipe midpoint. The transducer on the left is delegated the task of being the source (Figure 7.3.45a, in blue), and the one on the right is the receiver (Figure 7.3.45a, in orange). A Gaussian modulated sine pulse of $f_c = 35$ kHz is used to probe the system. The probing pulse is defined with a 60% relative bandwidth to the central frequency at full width at hundredth of maximum, or in simple terms, within [24.5, 45.5] kHz. The emitted pulse is presented in Figure 7.3.46a, and the recorded response at the receiver 2.5 m from the source is plotted in Figure 7.3.46b. That concludes the forward step. In the same intact DN80 HDPE pipe, the roles of transducers are switched (Figure 7.3.45b) and the forward step source becomes the receiver and vice-versa, so as to perform the TR step. The emitted signal is the, flipped in time, forward step response (Figure 7.3.46c). The receiving transducer, positioned at the exact same location of the forward step source, records the signal shown in Figure 7.3.45c that re-enacts the original Gaussian modulated sine pulse, albeit along with two symmetrically placed sidelobes. It should be noted that the presented signals in Figure 7.3.46 are normalized with respect to their amplitude.

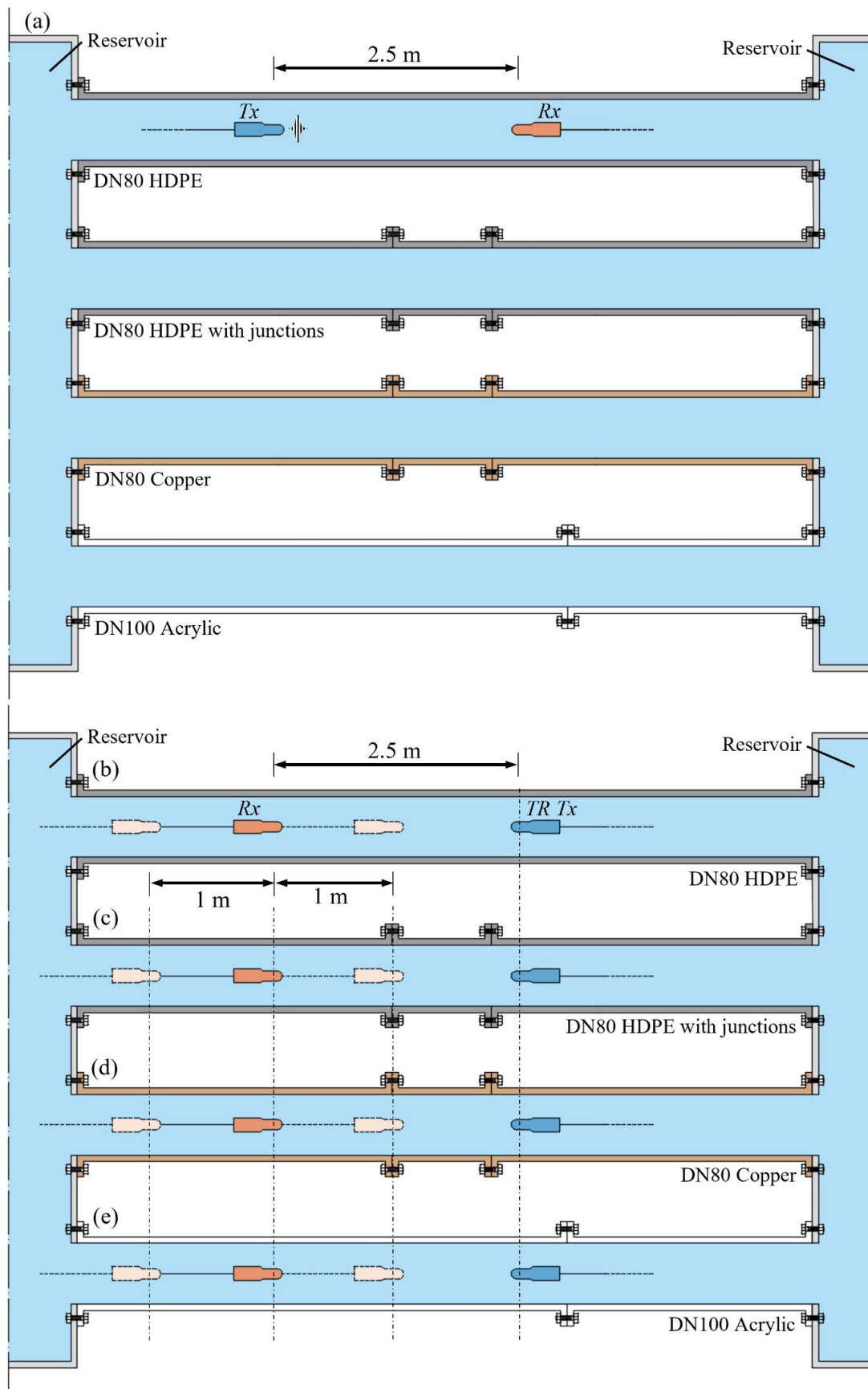


Figure 7.3.45: In this experiment, (a) the forward step is performed along the intact, uninterrupted DN80 HDPE pipe, where the distance between the two transducers is set to 2.5 m. Using the time reversed response from (a), the backpropagation step is conducted in (b) the same intact DN80 HDPE pipe, (c) the discontinuous “sister” HDPE pipe, (d) a DN80 copper pipe, and (d) a DN100 clear acrylic (Perspex) pipe. In each of these pipes, the TR field is not only evaluated at the supposed location of the forward step source, but also 1 m either side of this point at 2 cm increments.

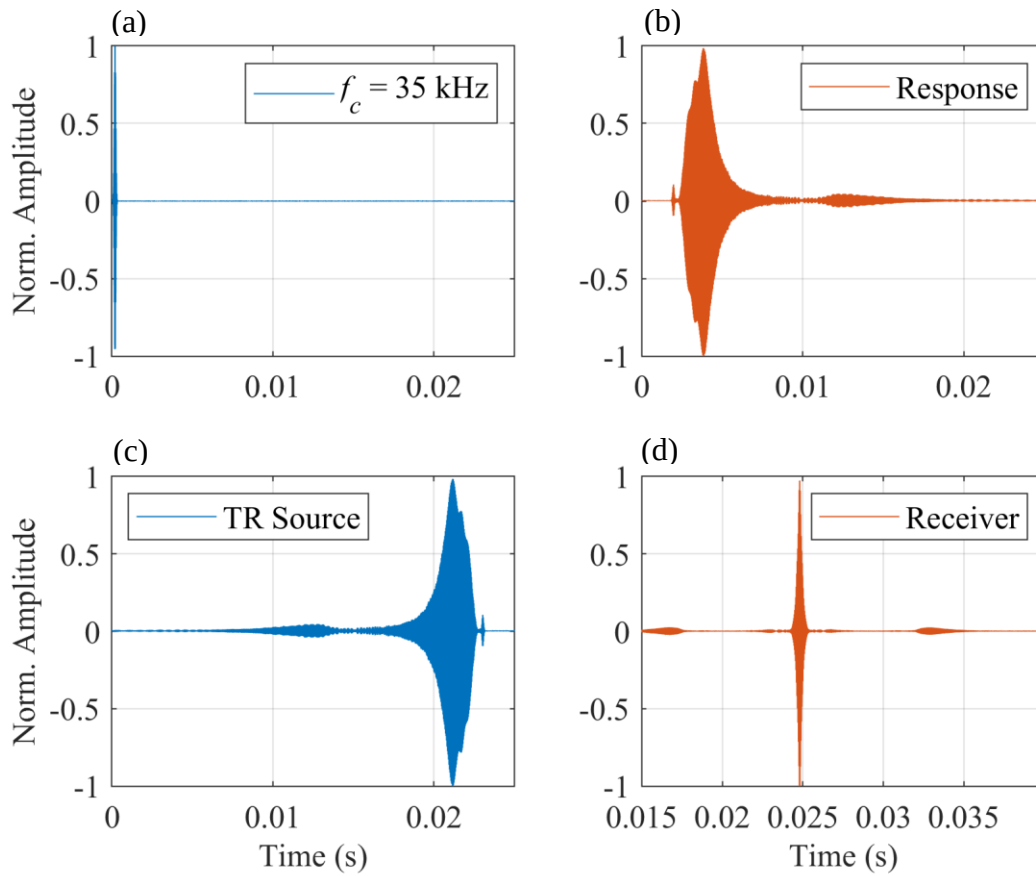


Figure 7.3.46: (a) A Gaussian modulated sine pulse of $f_c = 35 \text{ kHz}$ and bandwidth defined within the interval $[24.5, 45.5] \text{ kHz}$ is injected in the intact DN80 HDPE pipe, and (b) the system response is evaluated at a distance equal to 2.5 m to conclude the forward step. For the TR step, (c) the recorded response is time reversed, and emitted from the position of the original receiver. (d) The system response is measured at the location of the original source, where the original probing pulse is recreated.

The two transducers are then removed from the intact DN80 HDPE pipe and introduced to the other tested pipes sequentially. In each pipe, special care is taken to place the transducers at the

same longitudinal position as in the original, so that distances from the pipe ends are maintained. All pipes connect to the same reservoirs; hence their overall length is identical. Furthermore, the transducers are always kept at the pipe centres regarding their radial positioning. In each of the pipes, only the backpropagation, TR step is applied. To do so, the necessary TR source is obtained after time reversing the response of the original, intact DN80 HDPE to the forward step (Figure 7.3.47a). The signal path parameters of the experiment (e.g., signal gain, transducer sensitivity, sampling rate, and resolution) remain identical throughout.

First to test is the discontinuous, “sister” DN80 HDPE pipe that embeds two inline junctions along the wave path, absent in the original pipe (Figure 7.3.45c). Although the same pipe is discussed in subsection 7.3.1, the test is repeated here for comparison to the other pipes employed. Besides, the original probing signal employed here is different. The transducer on the right (Figure 7.3.45c, in blue colour) emits the signal shown in Figure 7.3.47a, and the receiving transducer (Figure 7.3.45c, in orange colour) samples the TR field at the location of the presumed forward step source (Figure 7.3.47b). In Figure 7.3.47b the TR response of the original DN80 HDPE at the same longitudinal position from the TR source is also plotted. Following a visual comparison of the two, the TR response from the discontinuous DN80 HDPE pipe is substantially different to the reconstructed pulse in the original pipe. Indeed, the response from the “sister” DN80 HDPE pipe lacks the typical symmetry of successfully recreated pulses, and it appears that the TR process has failed. The transducer pair is then moved to the DN80 copper pipe. Despite having a similar inner pipe radius to the original DN80 HDPE pipe, the wave propagation characteristics of the two pipes are entirely dissimilar (see Chapter 4). In addition, the DN80 copper pipe incorporates two inline junctions along the wave path. The TR step is applied and, to that end, the TR source signal of Figure 7.3.47a is emitted from the transducer on the right pipe end (Figure 7.3.45d, in blue colour). The response of the system is recorded 2.5 m away at the location of the postulated original source and is presented in Figure 7.3.47c in blue colour. Again, the pertinent response of the original DN80 HDPE pipe is also included in orange colour. The apparent dissimilarity between the two waveforms, leads to the conclusion that the TR process is ineffective in the DN80 copper pipe. Finally, the TR step is carried out in the DN100 clear acrylic pipe (Figure 7.3.45e). While there is an inline junction along the DN100 acrylic pipe, it is placed further to the right (i.e., behind) of the TR source. The recorded response to the TR source at the location of the forward step source is plotted in Figure 7.3.47d in blue colour. Again, the obtained waveform is unlike the

reconstructed pulse following the TR process in the original DN80 HDPE pipe. Hence, the TR process is proven to be unsuccessful once again.

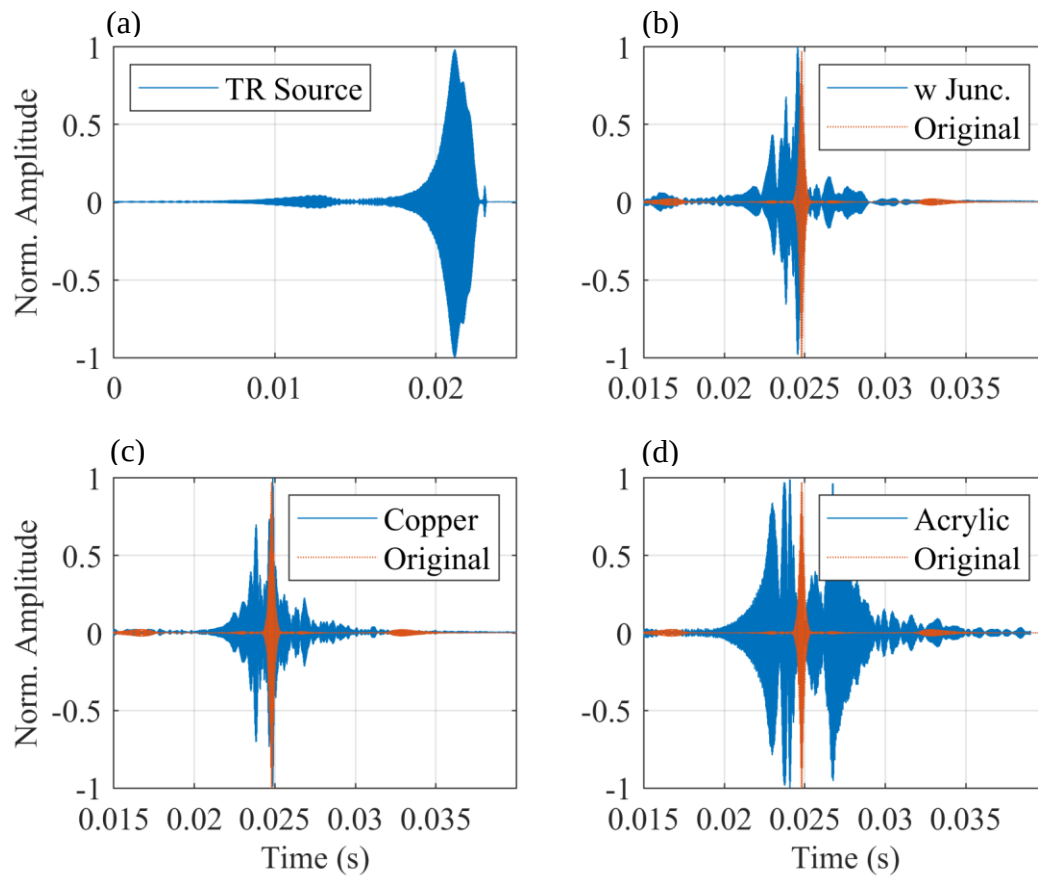


Figure 7.3.47: The TR step only is repeated in a different pipe to that of the forward step. (a) In any case, the TR source signal is the time reversed forward step response obtained from the original, intact DN80 HDPE pipe. The TR field is evaluated at the location of the supposed forward step source for the (a) the discontinuous DN80 HDPE, (b) the discontinuous DN80 HDPE pipe, and (d) the DN100 acrylic pipe. Also plotted is the recreated pulse at the same location for the original pipe.

The fact that the TR field does not refocus at the expected location of the original source in the pipes tested is not a definitive indication that TR is invalidated. Subsequently, the TR pressure field, in addition to the original source location, is also evaluated along a total length of 2 m (i.e., 1 m on each side of the original source location) at 2 cm increments for each pipe. Multiple measurements along the pipe will allow the identification of the unique focal point of the TR process in space if it exists. Figure 7.3.48 presents the amplitude of the TR pressure field at all measurement points and for all pipes tested. The values along the ordinate axis are given in dB

scale, relative to the maximum amplitude overall that is found along the original, intact DN80 HDPE pipe at the original forward step source position (i.e., at $x = 2.5$ m from the TR source). The TR pressure field amplitude for the original, intact DN80 HDPE pipe is represented by the blue curve, for the discontinuous DN80 HDPE pipe by an orange curve, for the discontinuous DN80 copper pipe by a yellow curve and, lastly, for the DN100 by a magenta-coloured curve. First and foremost, the presented curves confirm that the TR pressure field in the pipes different to the original does not exhibit a distinct peak and, by extension, no focal point. The backpropagation path in the different pipes is, therefore, altered to a degree that completely annuls the TR process. Interestingly, the TR pressure field in the discontinuous DN80 HDPE pipe is attenuated the most (25 dB on average), followed by that in the DN80 copper pipe (12 dB on average). The least attenuation, besides the original pipe, is found in the largest, DN100 acrylic pipe. It is important to note that the wave path in the DN100 acrylic pipe is uninterrupted from any inline junctions, unlike that in the discontinuous DN80 HDPE and copper pipes. Hence, there is strong evidence that inline junctions hamper the propagation of high frequency waves in general. Also, this result correlates to the *TR* parameter developed in Chapter 5, showing that range of *TR* is extended for larger diameter pipes.

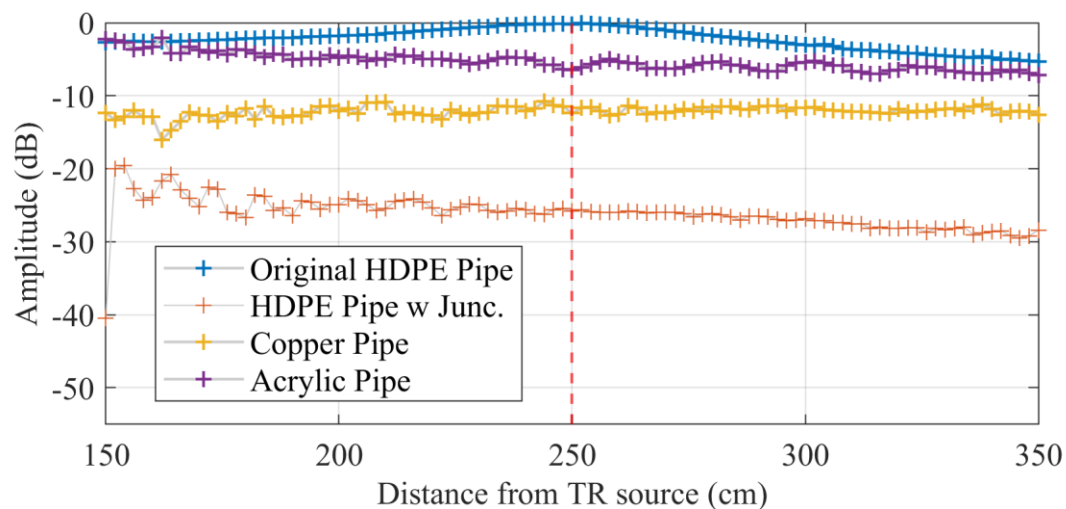


Figure 7.3.48: The TR pressure field is evaluated over a 2 m distance, starting at 1.5 m and up to 3.5 m from the TR source at 2 cm increments for every pipe tested. The amplitude of the evaluated TR response at each point is plotted against the pertinent distance from the TR source. The values are given in dB scale, relative to the maximum amplitude overall. The vertical red dashed line corresponds to the location of the original forward step source.

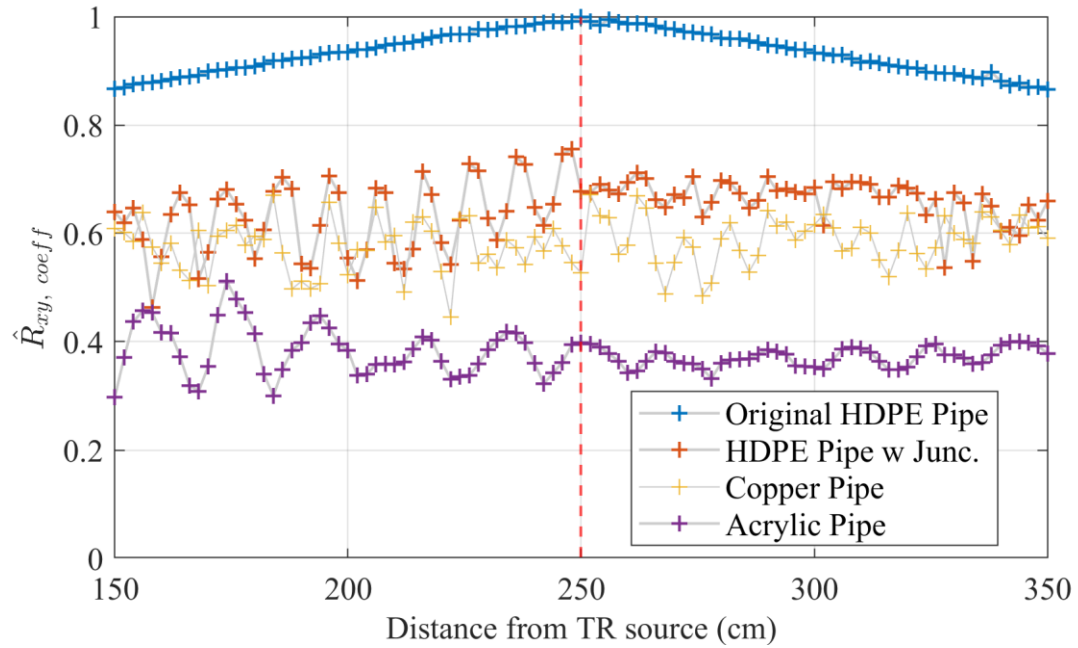


Figure 7.3.49: The maximum cross-correlation coefficient is calculated between the recorded TR pressure field response at each measurement point of each pipe tested and the reconstructed pulse at the forward step source location in the original, intact DN80 HDPE pipe. The vertical red dashed line indicates the position of the original source location, 2.5 m away from the TR source.

Finally, the maximum correlation coefficient is calculated between the measured responses of the TR pressure fields from all pipes investigated and the recreated pulse in the original DN80 HDPE pipe at the forward step source location (Figure 7.3.46d). The results are plotted in Figure 7.3.49, where the global maximum value (i.e., 1) is naturally the point on the curve for the original DN80 HDPE (Figure 7.3.49, blue point curve) pipe at $x = 250$ cm from the TR source (i.e., the original source location). Despite the fact that the TR pressure field in the discontinuous DN80 HDPE presented the most damping among all pipes, it also appears most similar to that of the original (Figure 7.3.49, orange point curve). This could be attributed to the fact that the two pipes are nominally identical, thus exhibit similar modal and wave propagation characteristics. The maximum cross correlation coefficient overall for the discontinuous DN80 HDPE pipe is found at $x = 248$ cm is shifted by 2 cm from the forward step location. Although it would be tempting to claim that the TR process succeeds, the fact is not verified by either Figure 7.3.48 or Figure 7.3.47b. In the former, there is not even a *local* maximum in the vicinity of $x = 250$ cm, while in the latter, the recreated pulse in the discontinuous DN80 HDPE pipe does not show the expected time compression or symmetry.

Consequently, the TR process in the DN80 HDPE pipe when inline junctions are present in the backpropagation wave path still cannot be considered successful. The TR field in the DN100 acrylic pipe, although it demonstrated the least amount of damping in comparison to the original DN80 HDPE pipe (Figure 7.3.48), is the least similar and shows a cross-correlation coefficient of only 0.4 on average (Figure 7.3.49, magenta point curve). Besides, there is no discernible peak that could indicate a focal point. Finally, the curve for the DN80 copper pipe (Figure 7.3.49, yellow point curve) lies in-between the discontinuous DN80 HDPE and DN100 acrylic pipe. In conclusion, all tested pipes but the original involve instabilities that render the TR process inviable.

7.4 High frequency wave propagation along pipe bends

7.4.1 Dispersion curves at the ends of a pipe bend

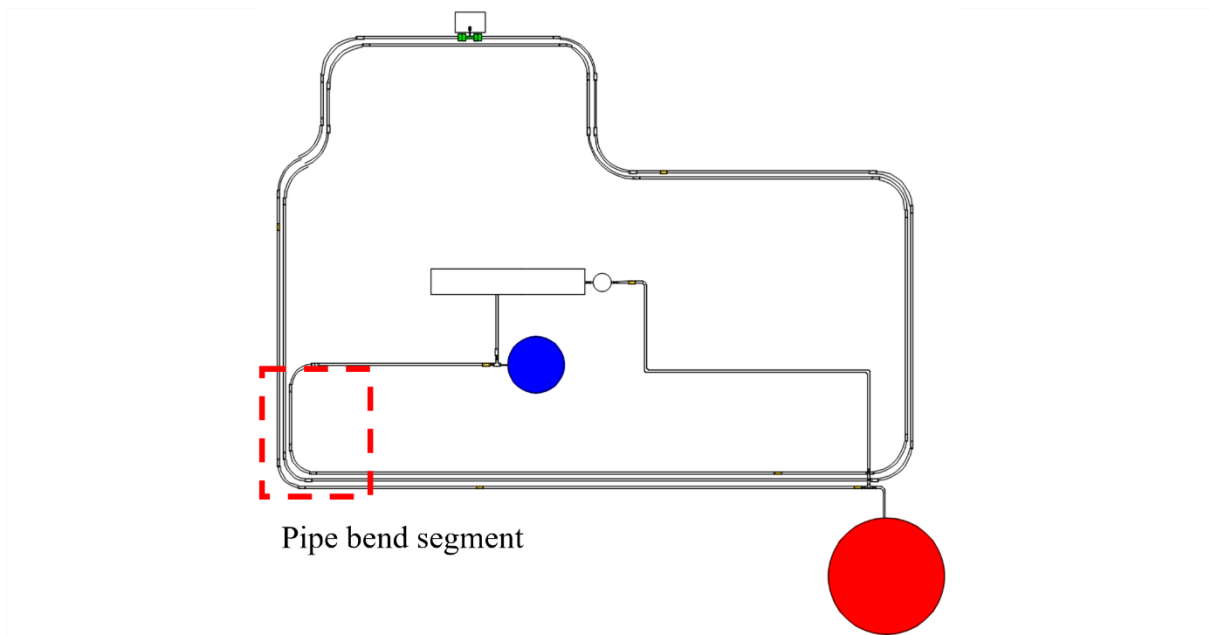


Figure 7.4.50: Top-view schematic of the LF setup at the hydraulic lab at HKUST. The segment that contains the pipe bend tested is marked with a dashed red frame. Of the three pipes shown, the inner one is used.

The intricate topologies of real world UWSS involve straight pipes, tee junctions, side branches, couplings, flanges, as well as bends, and each of these features has a particular effect on the propagation of high-frequency waves. Pipe bends in particular cause multiple wave scattering along their length, that could lead to an entirely different modal behaviour compared

to an equivalent straight segment of the same pipe. For instance, theory suggests that pipe bends result in a shift of the cut-off frequencies of high-order radial modes. In such event, the analytical wave propagation model is no longer valid, and will need to be adjusted accordingly. Hence, it is worthwhile to investigate the propagation of high-frequency waves across water-filled pipe bends and evaluate the significance of any changes compared to the straight pipe behaviour.

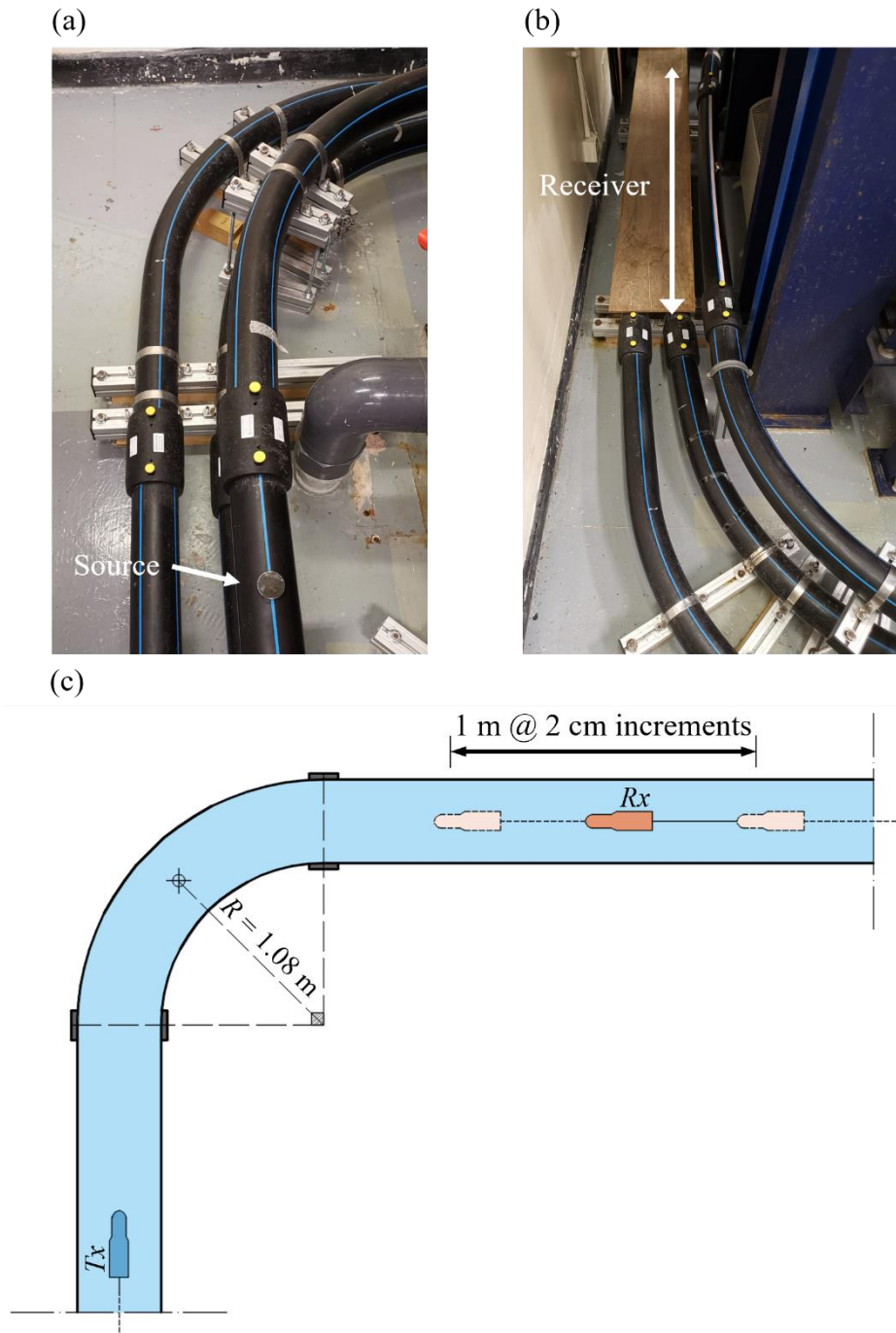


Figure 7.4.51: Real pictures of the setup are presented, where (a) shows the wideband source (i.e., the magnet is attached to the transducer) and (b) indicates the range over which the receiver is translated in order to develop the experimental dispersion curves. The schematic in (c) also illustrates the process (not to scale). The source is placed before the bend and emits a wideband signal. The response of the system is captured at the receiver placed after the bend. Then, the receiver is moved 2 cm further from the bend and the test is repeated. The test is complete when the receiver has covered a 1 m distance in total.

The HF setup at the hydraulic lab at HKUST only includes straight pipes, thus it is not suitable for testing pipe bends. Instead, the coiled LF setup is employed for this particular test, that comprises of a 145 m long DN80 HDPE pipe supported on the lab floor with anchored bolts. Only a small segment of the system is used in this series of tests that involves a 90° bend (i.e., the pipes connected to the pipe bend ends are perpendicular to each other). The arc length of the bend is measured equal to 1.7 m; thus, the corresponding radius is $R = 2 \times 1.7 / \pi = 1.08$ m. High frequency wave tests require piezoelectric transducers to be inserted in the pipe. Since there are no existing access options to do so, two access points are built before and after the bend. The standard transducer pair (two Brüel & Kjær Type 8104) is used, and each is mounted along the pipe centreline using the bespoke magnetic bases. While it is possible to create flow in the system, testing the effects of both fluid flow and pipe bends is not considered. Nevertheless, flow is only introduced for a total of an hour following the transducer mounting to flush the system from any air pockets.

The objective in this subsection is to develop the experimental dispersion curves for the pipe bend. Note that the bend is connected to the straight pipe segments via specialized sleeve couplers. The two pipe parts (e.g., curved and straight pipe) are inserted in the coupler, and an electric current is applied to create heat and permanently “fuse” the pipes together. The effect of this type of inline junction to wave propagation has not been investigated, unlike that of the typical bolted flange connections that is explored in subsection 7.3. However, given that the sleeve couplers are rather light, compact and the inner pipe surface remains flush, their impact should be minimal. The acoustic source is placed at one end of the bend, 15 cm from the first inline coupler, while the receiver is positioned at the other end, 1.2 m from the second inline coupler. Both transducers are placed at about the centre of the pipe cross section.

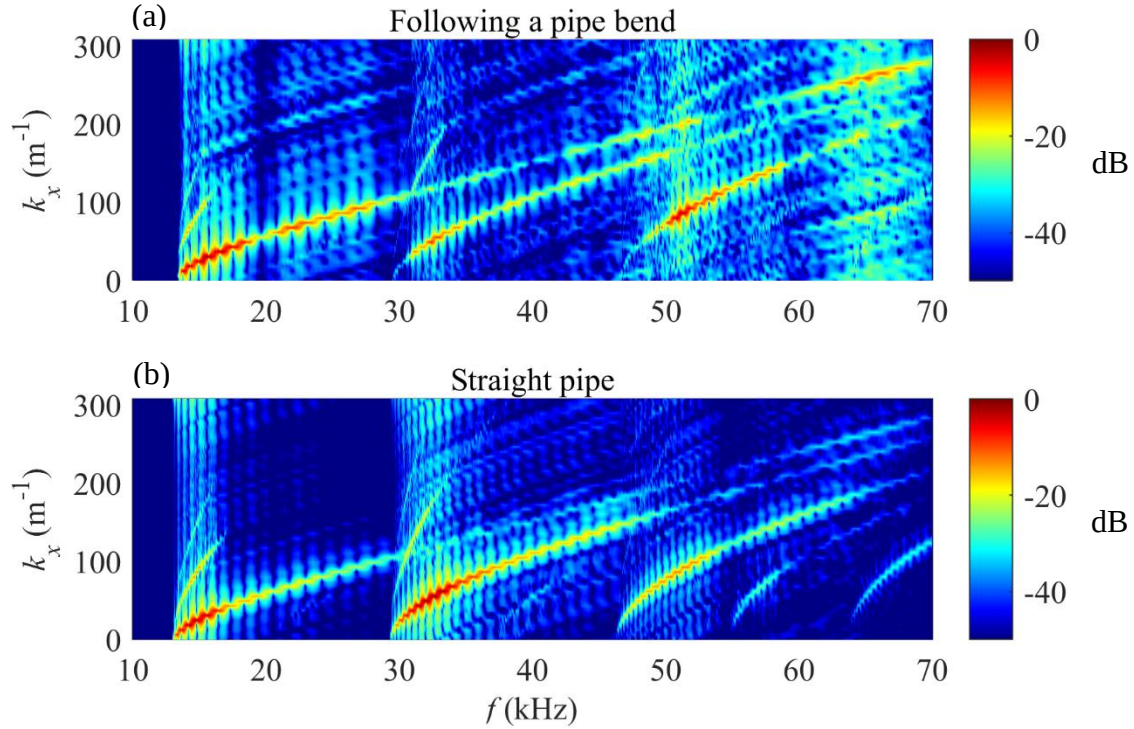


Figure 7.4.52: (a) The experimental dispersion curves are developed for the pipe bend when the source and receiver are placed before and following the ends of the pipe bend, respectively, For comparison, (b) presents the experimental dispersion curves for an equivalent but straight segment of a DN80 HDPE pipe.

To conduct the test, a linear frequency modulated wave signal from 10 to 70 kHz and 10 s in total duration is defined at the computer. The amplitude of the signal is set to 5 V and a 35 dB gain is applied, before the analog signal reaches the acoustic projector. The receiver at the other end of the bend records the system response for a total of 11 s. The transmission and acquisition sampling rates are set equal to 1 MHz. Once the system response is captured at that location, the receiving transducer is translated 2 cm closer to the bend, and the measurement is repeated. The selected step translates to a maximum spatial frequency equal to $k_x = 2\pi/0.02 \cong 314.16$. The test is complete when a 1 m total distance is covered by the receiver. In total, 51 measurements are carried out. Sampling the pressure field over a longer distance (e.g., 1.5 or 2 m instead of 1 m) would improve the attainable resolution. However, the distance along the pipe is limited.

The measurements produce a matrix with 51 columns (one for each receiver location), and 11×10^6 rows (one for each time step). The 2D-FFT algorithm is performed on the matrix, to transform the data from the space-time (x, t) domain to the reciprocal wavenumber-frequency

domain (k_x, ω) . The data is also compensated for the frequency-dependent transmission efficiency of the employed transducer. Finally, the data is plotted in dB scale relative to the maximum amplitude in Figure 7.4.52a. For comparison, the experimental dispersion curves evaluated over a 1 m segment of the identical, but straight DN80 HDPE pipe of the HF setup are plotted in Figure 7.4.52b using a dB scale relative to the maximum amplitude for the particular case. Unfortunately, the same normalization cannot be used since the amplifier gain used is different for each test. Nevertheless, some insightful observations can be made.

In Figure 7.4.52a and b, the cut-off frequencies for the first three high-order radial modes are identical and equal to 13050, 29450, and 46350 Hz, respectively. It is interesting to note, however, that the 4th mode whose cut-off is found in Figure 7.4.52b around 55 kHz and is also predicted by the analytical model, is absent from Figure 7.4.52a. In addition, the 5th radial mode that cuts-on at 63.5 kHz for the straight DN80 HDPE pipe, is irretrievable for the case of the pipe bend. It appears, therefore, that the pipe bend has a stronger influence at higher frequencies where otherwise propagating modes are not excited. Moreover, the amplitude of modes with frequency appears to be different between the straight and curved pipes. For instance, in Figure 7.4.52a, the amplitude of the 1st radial mode in the frequency intervals [13.4, 20] kHz and [22, 37] kHz is comparable. Conversely, in Figure 7.4.52b, the 1st radial mode demonstrates its maximum amplitude in the narrow frequency interval [13, 17]kHz. Furthermore, the amplitude of 2nd radial mode in Figure 7.4.52a is weaker compared to the 1st and 3rd, while the opposite is true in Figure 7.4.52b. Besides, at higher frequencies > 60 kHz, the amplitude of the 1st radial mode in a straight pipe is much attenuated in comparison to frequencies close to its cut-off (Figure 7.4.52b). For a curved pipe, however, the amplitude of the 1st radial mode for $f > 60$ kHz is comparable to that observed at frequencies close to the cut-off. Although not radically different, these subtle changes in the dispersion relationship for a curved pipe could introduce some challenges to the application of the analytical model. In the next subsection, TR is experimentally performed, and the analytical model is then applied to verify the distance between the source and receiver for probing signals of different frequencies. The analytical model considers a straight pipe; thus, it would be interesting to evaluate the error incurred in the estimated distance between the source and the receiver due to the pipe curvature.

7.4.2 TR at the ends of a pipe bend

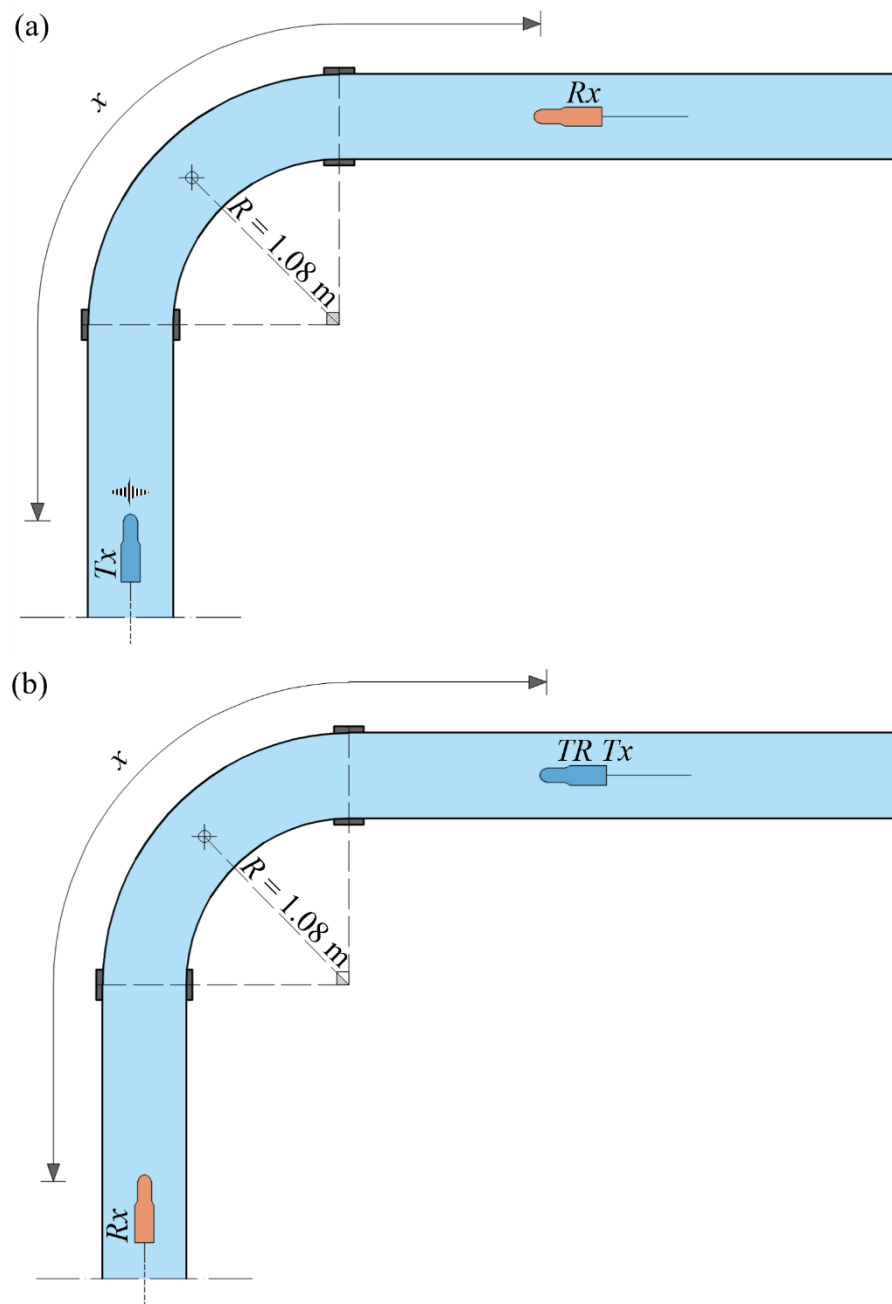


Figure 7.4.53: The TR process is performed along the pipe bend multiple times, for various distances x in-between the transducer pair and probing frequencies: (a) Forward (b) TR step.

The transducers are placed along the pipe centreline and never within the bend.

The previous subsection 7.4.1 presents the developed experimental dispersion curves for the case where the source and receiver are positioned at the two ends of a DN80 HDPE pipe bend of radius $R = 1.08$ m. The curves are compared to these from an equivalent straight pipe segment, and some interesting discrepancies are identified. Most notably, the 4th high-order

radial mode that has a cut-off frequency at about 55 kHz is absent from the dispersion curves of the pipe bend segment. Also, at frequencies above 60 kHz, the dispersion curves for the pipe bend appear “noisier” (Figure 7.4.52). To verify whether these changes in the dispersion relationship of the waveguide may potentially compromise the validity of the analytical model when applied over pipe bends, a straightforward experiment is devised. Note that the experiment is based on the fact that the analytical dispersion curves, as presented in Chapter 4, match the pertinent experimental curves for a straight DN80 HDPE pipe over 10-70 kHz.

The transducer pair used in subsection 7.4.1 to obtain the experimental dispersion curves, is employed here to perform a series of typical TR tests. Specifically, in the forward step, the transducer placed along the pipe centreline before the pipe bend emits a short, Gaussian modulated sine pulse. The wave propagates along the pipe bend, and the response of the system is captured by the receiving transducer at a distance x from the source and also along the pipe centreline. In the TR step, the roles of the transducers are interchanged so that the original source becomes the receiver and vice-versa. The TR source emits the time reversed system response from the forward step, and the TR pressure field is evaluated at the location of the TR receiver, that coincides with that of the original source. The distance x between the transducers is known at the time of the experiment. However, that distance is also evaluated by the analytical model presented in Chapter 6. In particular, the experimental forward system response is convolved with the analytical model predictions over the pipe spatial domain for the given probing pulse. Ideally, the convolution product should be maximized at the location of the receiver, hence identifying the correct distance in-between the transducers. Any potential disagreement between the real and estimated distance between the two transducers should be attributed to the effect of the pipe bend to the dispersion relationship of the waveguide, given that the analytical model is shown to accurately predict wave propagation along the DN80 HDPE pipe. The test is performed 6 times in total, for probing signals of 3 different central frequencies $f_c = 35, 55, \text{ and } 75 \text{ kHz}$ i.e., 2 tests are performed at each probing frequency. The probing pulses have a 20% relative bandwidth, while the distance x is always altered between tests and ranges between 1.7 and 2.9 m. Here, only the results from one of the two tests at each frequency are plotted, but both are discussed.

Figure 7.4.54 presents the results of the TR process applied over the pipe bend for a Gaussian modulated sine pulse with $f_c = 35 \text{ kHz}$ and bandwidth at 1/100 the spectrum amplitude equal to $[31.5, 38.5] \text{ kHz}$. Hence, the minimum wavelength of the probing pulse is $\lambda_{\min} \sim 4 \text{ cm}$, and the resolution is limited to $\lambda_{\min}/2 = 2 \text{ cm}$. The receiver is placed after the pipe bend, at a distance

equal to $x = 2.55$ m from the source. The source pulse and obtained system response for the forward step are shown is depicted in Figure 7.4.54a, and b, respectively. For the TR step, the original receiver becomes the source that emits the time reversed forward step response (Figure 7.4.54c), and the TR field is evaluated at the original source location where the probing pulse is recreated (Figure 7.4.54d). The distance between the two transducers is readjusted to $x = 2.9$ m and the test is repeated.

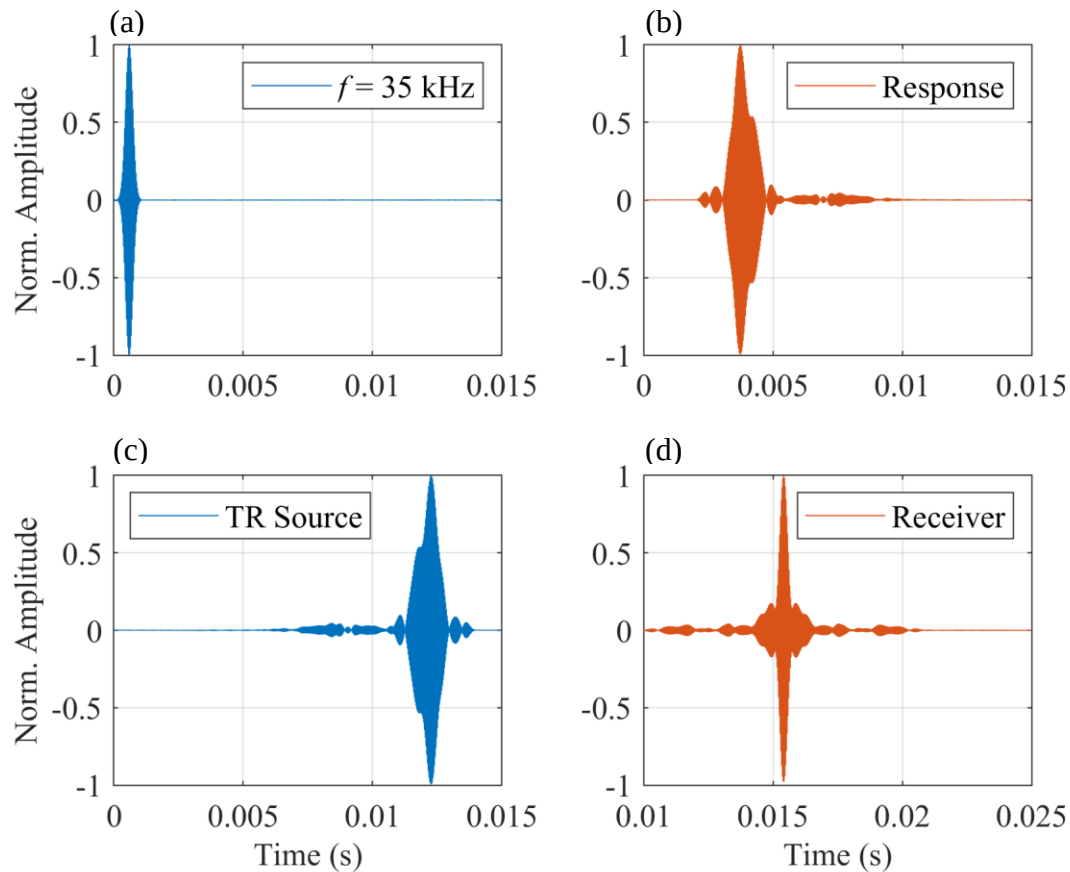


Figure 7.4.54: The forward and TR steps are performed over the DN80 HDPE pipe bend. (a)

A Gaussian modulated sine pulse with $f_c = 35$ kHz is emitted from the source, and (b) the system response is captured by the receiver at a distance $x = 2.55$ m. In the TR step, the roles of the transducers are interchanged, and the TR source emits (c) the time reversed response of the forward step. (d) The original pulse is recreated at the location of the forward step source.

The forward step responses from the two tests are convolved with the results from the analytical model for an identical $f_c = 35$ kHz probing pulse over a spatial domain of up to 6 m. The convolution product results are presented in Figure 7.4.55. For the first test (Figure 7.4.55a), the convolution product is maximized at 2.55 m, a distance that exactly matches that measured

experimentally. Hence, the analytical model successfully describes the wave propagation over the pipe bend in the excited frequency range [31.5, 38.5] kHz. The results from the second test are equally impressive since the analytical TR model accurately places the receiver 2.88 m away from the source as measured experimentally.

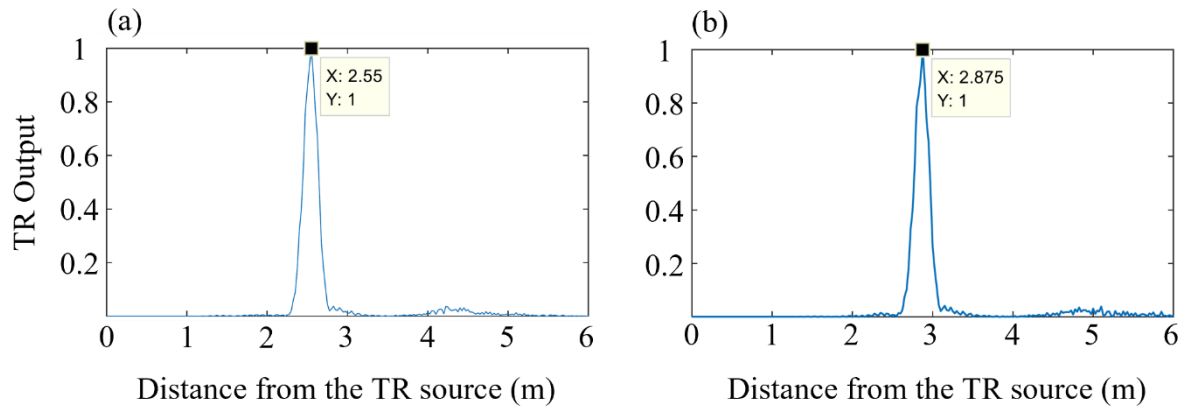


Figure 7.4.55: The forward step is performed for a Gaussian modulated pulse with $f_c = 35$ kHz twice over the pipe bend, with a different distance between the transducer pair each time.

The TR step is performed analytically, and the convolution product is maximized at the correct location of the source relative to the source for both measurements.

The problematic frequency range, however, is found around 55 kHz. There, the dispersion curves for the pipe bend (Figure 7.4.52a) ignore the 4th radial mode that is present in the dispersion curves of the straight pipe (Figure 7.4.52b). Consequently, an unfavourable scenario is to probe the system with a Gaussian modulated sine pulse of central frequency $f_c = 55$ kHz. The excited bandwidth is defined equal to [49.5, 60.5] kHz at 1/100 of the spectrum amplitude. To perform the forward step, the probing pulse is emitted (Figure 7.4.56a), and the receiver positioned at a distance $x = 2.45$ m from the source and after the pipe bend obtains the dispersed system response (Figure 7.4.56b). The forward step is followed by the TR step, where the transducer roles are inverted without changing their positions in the system, the forward step response to the original probing pulse is time reversed (Figure 7.4.56c) and remitted from the TR source. The TR pressure field created is measured at the location of the original source, where it is evident that the probing pulse is reconstructed (Figure 7.4.56d). The range between the two transducers is increased to $x = 3.3$ m, and all steps of the test are repeated. The actual obtained signals through the forward and TR steps for the second iteration are omitted for brevity.

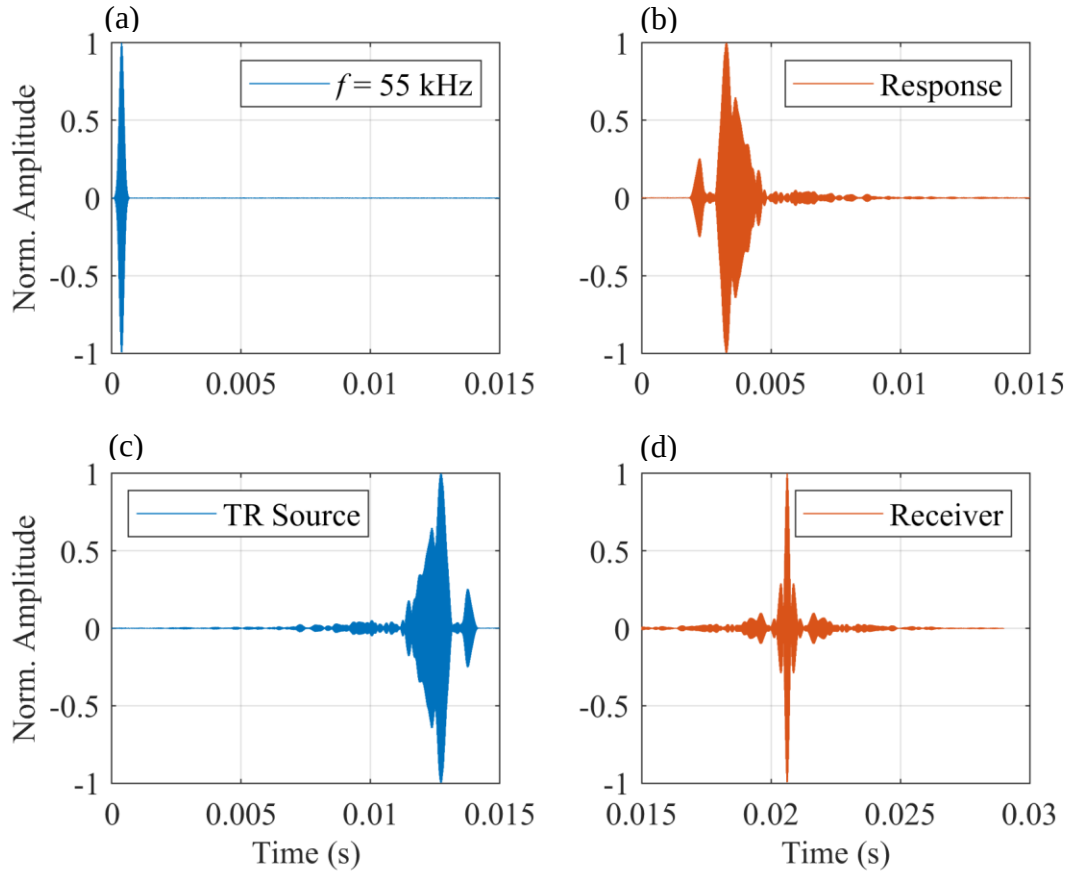


Figure 7.4.56: The forward and TR steps are performed over the DN80 HDPE pipe bend. (a) A Gaussian modulated sine pulse with $f_c = 55 \text{ kHz}$ is emitted from the source, and (b) the system response is captured by the receiver at a distance $x = 2.45 \text{ m}$. In the TR step, the roles of the transducers are interchanged, and the TR source emits (c) the time reversed response of the forward step. (d) The original pulse is recreated at the location of the forward step source.

For the two tests, the TR step is also performed analytically in the computer. The experimentally obtained forward step response is convolved with the analytically derived response over the spatial domain. Figure 7.4.55a depicts the normalized convolution product for the first test for a pulse with $f_c = 55 \text{ kHz}$, where the measured distance between the source and the receiver is $x = 2.45 \text{ m}$. Alas, the plot presents its maximum value at $x = 2.3 \text{ m}$, that is 15 cm shorter than the actual range. The convolution product for the second test for a pulse $f_c = 55 \text{ kHz}$ is presented in Figure 7.4.55b. Here, the actual distance between the two transducers is 3.3 m, but the plot in Figure 7.4.55b peaks at $x = 3.15 \text{ m}$. Evidently, the error is consistent between the actual and estimated ranges in both tests, and equal to 15 cm. It may, therefore, be concluded that the dispersion relationship in the excited frequency interval for the case of the pipe bend is substantially different to that of a straight pipe, so that the analytical

model (based on the description of the latter) produces a consistent error in localizing the source. The consequences to defect detection are, therefore, profound. One has to be careful when applying the analytical solution to curved pipe segments that demonstrate even the slightest deviation from pertinent straight pipe segments with regards to their wave propagation characteristics. Certainly, making appropriate amendments to the analytical model so as to accommodate curved pipes is preferred, and the particular problem could set the basis for future research.

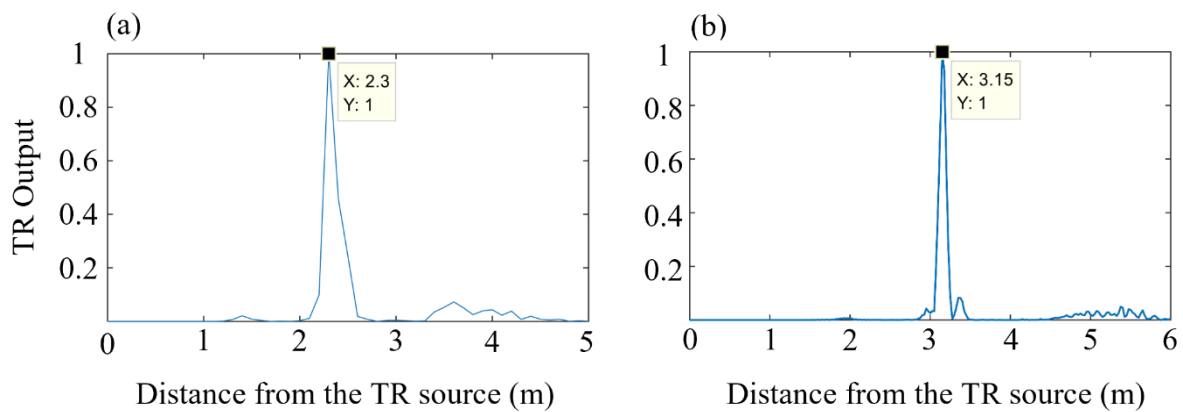


Figure 7.4.57: The forward step is performed over the pipe bend twice for a Gaussian modulated pulse with $f_c = 55$ kHz. For each iteration, the distance between the two transducers is altered. The TR step is performed analytically, and for both cases the convolution product is maximized 15 cm away from the actual source location.

This subsection is concluded with an additional test that employs a Gaussian modulated sine pulse with $f_c = 75$ kHz and 20 % relative bandwidth (i.e., [67.5, 82.5] kHz). The upper limit of the excited bandwidth exceeds the frequency range of the dispersion curves in Figure 7.4.52a that tops at 70 kHz. In the excited bandwidth, therefore, the differences in the dispersion relationship between the bend and straight pipes are not well identified. Nonetheless, the 5th radial mode that is easily distinguishable in Figure 7.4.52b for the straight pipe is not so for the pipe bend. As such, inaccuracies are expected to occur when localizing the source along the bend pipe using the analytical TR model. The probing pulse, emitted before the bend, is shown in Figure 7.4.56a, while the system response at $x = 2.25$ m from the source is plotted in Figure 7.4.56b. This completes the forward propagation step. Besides, the TR step is performed experimentally: the forward step response is inverted in time Figure 7.4.56c and reemitted into the system from the location of the forward step receiver. The transducer at the original source

location captures the signal that re-enacts the probing pulse. The entire process is repeated again for a distance $x = 1.7$ m between the two transducers.

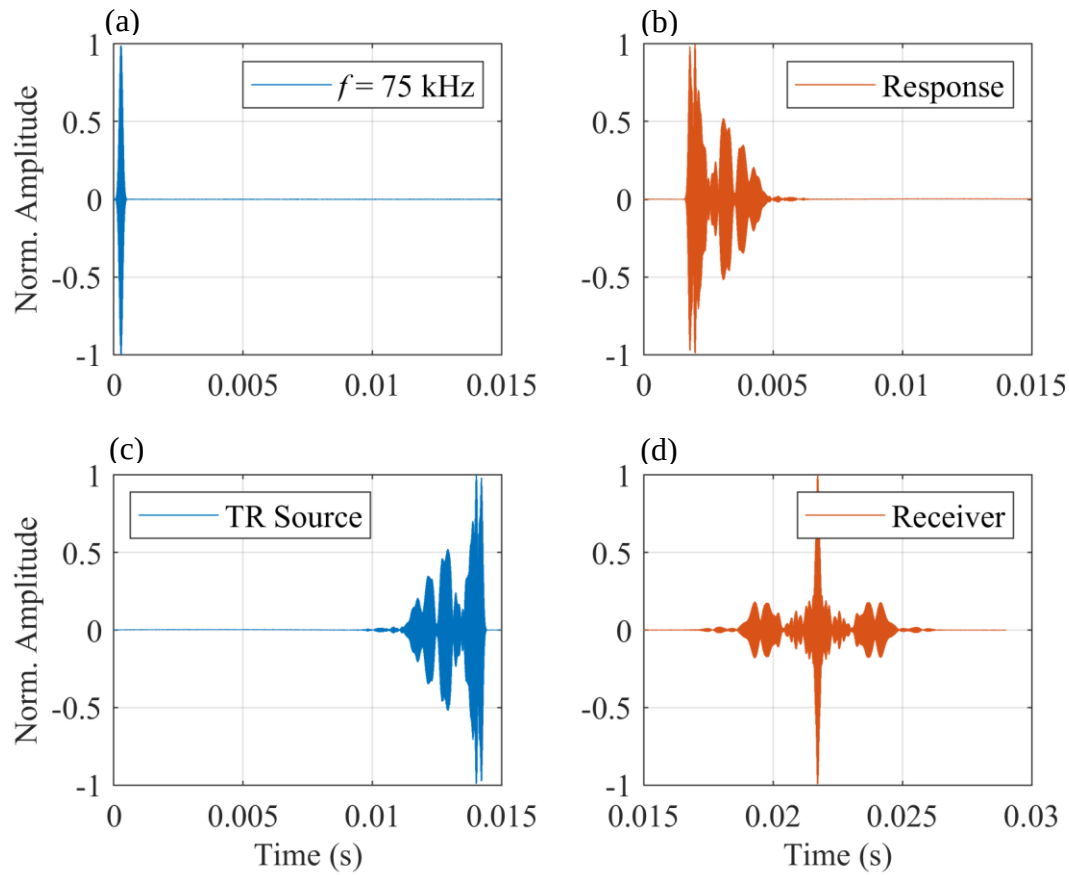


Figure 7.4.58: The forward and TR steps are performed over the DN80 HDPE pipe bend. (a)

A Gaussian modulated sine pulse with $f_c = 75$ kHz is emitted from the source, and (b) the system response is captured by the receiver at a distance $x = 2.25$ m. In the TR step, the roles of the transducers are interchanged, and the TR source emits (c) the time reversed response of the forward step. (d) The original pulse is recreated at the location of the forward step source.

The analytical TR model is employed to localize the source using the forward propagation response of the system from the two tests where a probing pulse of $f_c = 75$ kHz is used. Specifically, the forward step responses (one shown in Figure 7.4.56b) are each convolved with the analytically derived responses over a range of 4 m. The global maximum point of the convolution product corresponds to the location of the receiver relative to the source. The calculated convolution products for the two tests are plotted in Figure 7.4.59. For the first test, the actual distance of the receiver from the source is measured equal to $x = 2.25$ m. The plot in Figure 7.4.59a, however, demonstrates a global peak at $x = 1$ m and thus, inaccurately estimates

the true source location. Unlike both plots in Figure 7.4.55 where peaks are distinct, in Figure 7.4.59a there are many identifiable peaks. Considering a normalized convolution product above 0.5, these peaks are found at $x = 1.35$, 2.05 , and 2.25 m. Evidently, the secondary peak at $x = 2.25$ m where $y \sim 0.55$ corresponds to the correct distance of the source to the receiver. For the second test, Figure 7.4.59b peaks at $x = 1.7$ m that is exactly the true distance between the two transducers. Still, there are additional smaller peaks to be found; for instance, at $x = 1.1$ and 0.7 m. One factor that could explain the reason the analytical model is able to correctly identify the actual distance when at close range ($x = 1.7$ m), but not farther ($x = 2.25$ m) is the increased damping and, thus, reduced SNR experienced by the propagating wave at higher frequencies. In any case, since the analytical TR model correctly localized the source at one of the two tests, it may be concluded that the dispersion relationship between the straight and bend pipes in the excited frequency does not vary significantly. Finally, from all tests, only the two that employed a Gaussian modulated sine pulse with $f_c = 55$ kHz displayed a consistent localization error. Hence, it is possible that the pipe bend can selectively affect particular propagating modes. In any event, this is a particularly interesting research topic, that should be experimentally investigated in pipes of various mechanical and geometrical properties, bend radii, and probing frequencies.

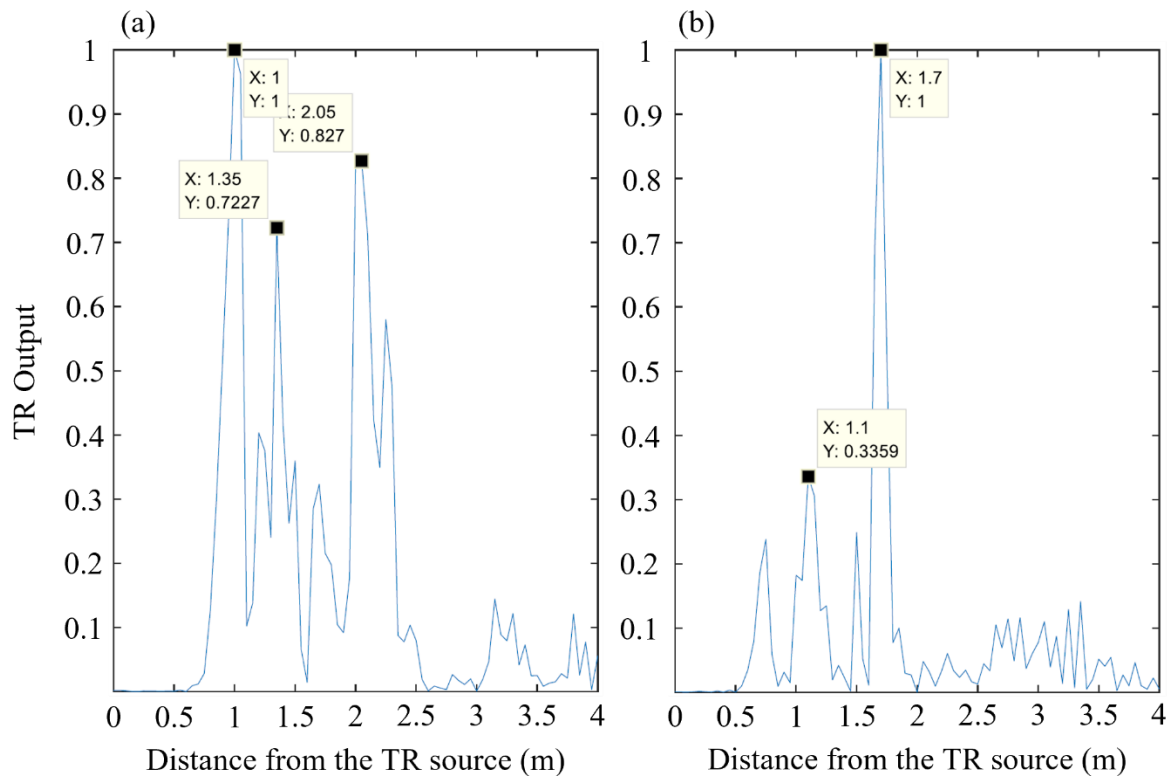


Figure 7.4.59: The forward step is performed over the pipe bend twice for a Gaussian modulated pulse with $f_c = 75$ kHz. For each iteration, the distance between the two transducers is altered. The TR step is performed analytically, and in both cases, the convolution product exhibits many peaks, hence uncertainties.

7.4.3 Dispersion curves along a pipe bend

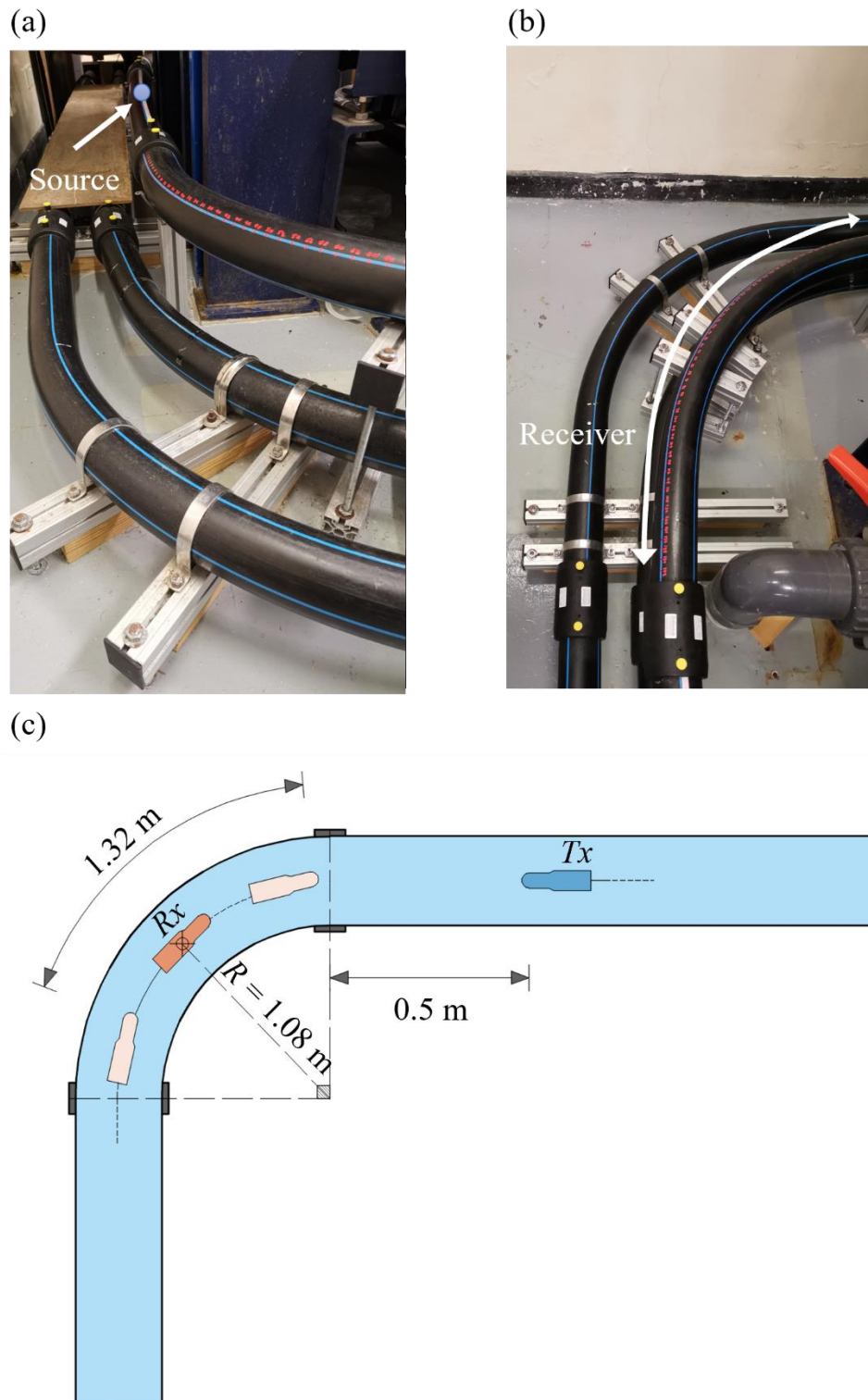


Figure 7.4.60: The experimental dispersion curves for the pipe bend are developed again, but this time the receiver is translated along the arc as opposed to along the straight pipe segments following the bend. (a) The source transducer is fixed at a distance of 0.5 m from

the pipe bend end, and (b) the receiving transducer is translated along the pipe bend at 2 cm for a total range of 1.32 m. (c) The schematic of the setup that shows the distances and receiving transducer trajectories (not to scale).

In subsection 7.4.1, the experimental dispersion curves are constructed when the source and receiving transducers are placed at the ends of a DN80 HDPE pipe bend of radius $R = 1.08$ m. Compared to the experimental dispersion curves for an equivalent straight DN80 HDPE pipe segment, the cut-off frequencies of high order radial modes remain identical. This finding contradicts the outcome of previous studies that suggest a shift of the radial mode cut-off frequencies. In particular, Rostafinski [180] suggests that for the case of a pipe bend with rigid walls the cut-off frequencies of high-order radial modes are increased, and the effect is more pronounced at high frequencies and for smooth bends that present a ratio of the outer over the inner bend wall radius R_2/R_1 close to 1. Indeed, the pipe bend tested has an outer radius $R_2 = 1.125$ m and an inner radius $R_1 = 1.035$ m that yields a ratio $R_2/R_1 \sim 1.05$. Hence, if past research is valid, large deviations in the cut-off frequencies should be expected. However, these results are relevant along the bend, not for measurements following the bend. To validate the theory, the setup is appropriately modified, and the measurements to develop the experimental dispersion curves are repeated.

The source transducer (Brüel & Kjær Type 8104) is placed along the pipe centreline following and facing towards the pipe bend. The distance of the source transducer to the end of the pipe bend is fixed to 0.5 m. On the other hand, the receiving transducer (Brüel & Kjær Type 8104) is positioned along the pipe bend centreline and opposing the source. While the pipe bend arc is 1.7 m long, the two inline couplers at its ends restrict the available range of motion to 1.32 m. For the first measurement, the receiver is placed at its farthest position from the source that emits a linear frequency modulated sine wave from 10 to 70 kHz for 10 seconds. The amplitude of the signal is set to 5 V, and a power amplifier applies a 35 dB gain before it reaches the source transducer. The receiver records the system response for 11 s, at a sampling rate set to 1 MHz. Once the measurement is complete, the transducer is moved by 2 cm along the arc and the test is repeated until the arc length is covered at its entirety. In total, the receiving transducer is repositioned 67 times. A schematic and pictures of the setup are presented in Figure 7.4.60.

The acquired data is expressed in the space/time domain and are transformed to the reciprocal wavenumber/frequency domain following the 2D-FFT algorithm. The spectrum amplitude is compensated appropriately for the frequency dependent transmission characteristics of the

transducer, and then plotted to produce the experimental dispersion curves for the wave propagation along the pipe bend. The amplitude of the dispersion curves is plotted in a dB scale, relative to the maximum amplitude overall. Finally, the occurring experimental dispersion curves along the pipe bend are juxtaposed with these from an equivalent straight pipe in Figure 7.4.61. By equivalent, it is also implied that a same number of measurements along the straight pipe is considered.

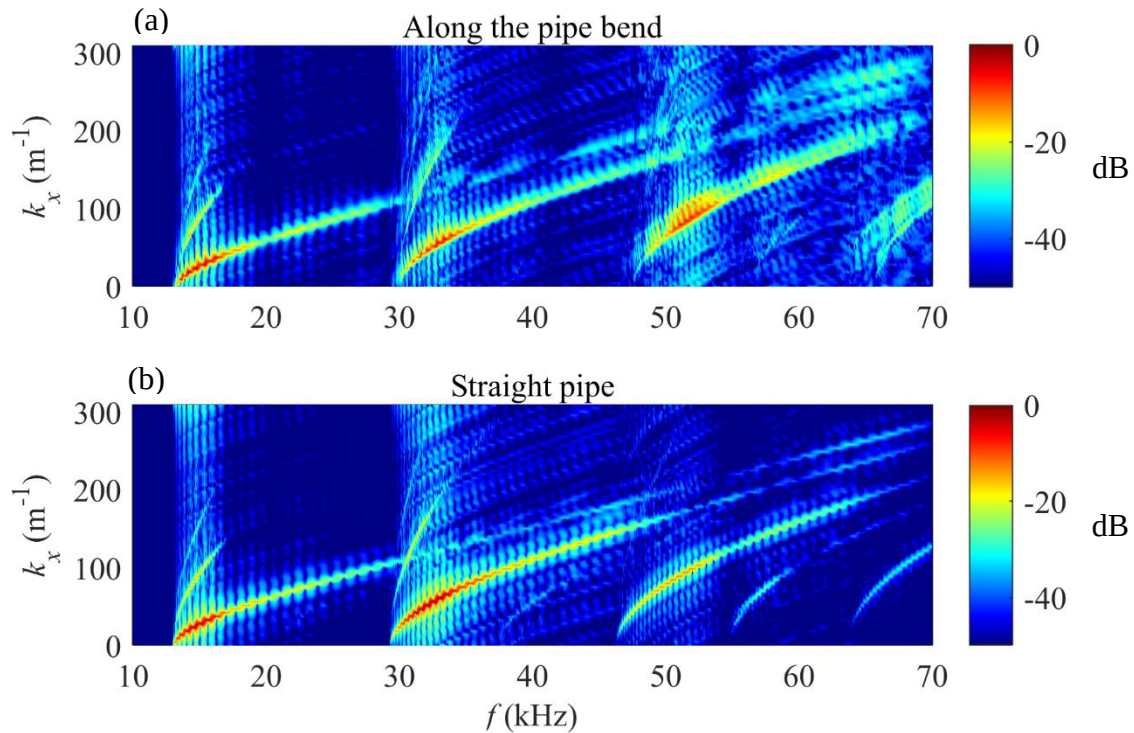


Figure 7.4.61: (a) The experimental dispersion curves along the DN80 HDPE pipe bend with radius $R = 1.08 \text{ m}$ (i.e., for a receiver that is moved along the bend) for a frequency range from 10 to 70 kHz. (b) The experimental dispersion curves for a straight DN80 HDPE pipe segment of equivalent length. The contours are shown in dB scale, relative to the maximum amplitude for each plot.

At the lower end of the frequency range tested, the cut-off frequency for the 1st radial mode in the straight DN80 HDPE pipe is at 13050 Hz. Conversely, along the pipe bend the cut-off for the 1st radial mode occurs at the – slightly higher – frequency of 13250 Hz. Moreover, the cut-off frequencies for the 2nd, 3rd, and 4th radial modes in the straight pipe are 29450, 46350, and 63600 Hz, respectively. The same modes along the pipe bend appear at 30050, 47600, and 65900 Hz. The difference corresponds to a 2% increase in the cut-off frequency of the 1st radial mode, 2.7% increase for the 2nd, and 3.6% for the 3rd. Apparently, the shift is more pronounced

for higher order modes that present their cut-offs at higher frequencies. That is consistent with the findings in existing studies for rigid pipes, nonetheless. In addition, the bright regions in the dispersion curves for the curved pipe seem to “spread” in k_x (and that spread increases for higher order modes). The upper range of the spread can be associated with the straight pipe dispersion curves. To better illustrate the differences between the dispersion curves of the straight and curved pipes, points along the brightest regions of the contour plots in Figure 7.4.61 are selected to construct line plots. In Figure 7.4.62, the dispersion curves of the straight pipe are plotted with solid lines, while these of the curved pipe with dashed lines: Note that the naming of the modes in Figure 7.4.62 is consistent with the convention established in Chapter 4, where the 1st radial mode corresponds to the 2nd axisymmetric mode. In Figure 7.4.62 is yet again clear that the pipe curve shifts the cut-offs for radial modes to higher frequency values. Consequently, when applying the analytical TR model over pipe bends, such effects should be considered. Finally, these findings set the stage for future research on the topic that will enable the technique to real world systems.

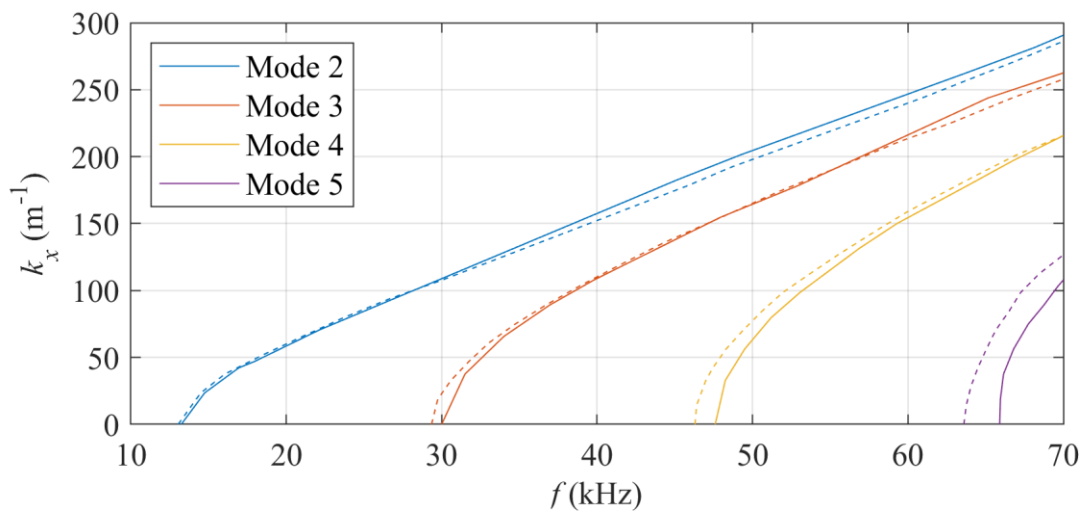


Figure 7.4.62: To highlight the discrepancies between the radial mode cut-off frequencies for a curved and straight DN80 HDPE pipe segment, points from the bright regions of the experimental dispersion curves are selected and plotted together. The solid lines and dashed lines correspond to the curved and straight pipe, respectively.

7.5 Conclusions

As noted, typical real-world UWSS present mechanisms that contravene the TR requirements of spatial reciprocity and time symmetry. Such mechanisms, that include both steady and unsteady turbulent flows, damping, and instabilities in the wave path have the potential to degrade, or even break, TR and hamper the propagation of HF waves in water filled pipes. In the present chapter, a series of experimental tests are performed that aim towards identifying the effect of these mechanisms to TR fidelity and HF wave propagation. For instance, experiments in a DN80 HDPE pipe under turbulent flow demonstrate significant damping (~ 30 dB/m) and a spatial shift of the TR focal spot equal to about two times the probing wavelength for a probing signal of central frequency $f_c = 35$ kHz. In fact, additional measurements that produce the Lissajous curves for waves that propagate with and against the flow concur that the observed phase shift is a linear function of frequency. This phase shift is only present under turbulent flow conditions and disappears when the same measurements are performed in stationary water. To counter the effects of flow in the propagating medium and restore the fidelity of TR, researchers have proposed the alternative technique of nonreciprocal (NR) TR, with applications in the shallow water environments. Herein, NR TR is tested for the first time in water filled pipes under steady turbulent flow and compared to traditional TR for its robustness against phase shifts. Although there appears to be some benefit of NR TR at higher probing frequencies (where phase shift is more significant), the results are somewhat inconclusive. It is, therefore, proposed that NR TR is tested in an experimental setup of longer range to emphasize any occurring phase shift; however, such setup is not currently available. NR TR is also tested under unsteady turbulent flow, that is implemented by increasing the flowrate between the forward and TR steps. Compared to the results of the same test under steady flow at various probing wave frequencies, phase shift is increased hence showcasing the detrimental effect of flow unsteadiness to TR. To better understand HF wave propagation in a dynamic medium, a straight DN80 HDPE under different levels of turbulent flow is probed with linear frequency modulated wideband signals in the $[10, 70]$ kHz frequency range to develop the experimental dispersion curves. The obtained results are compared to the case of stationary water and do not yield any tangible differences in the propagating modes in regard to both the cut-off frequencies of radial modes and curve slopes (i.e., group velocity values). Regardless, amplitude spectrum plots in space show that increasing flowrate in a DN80 HDPE

pipe has a stronger attenuating effect on high frequency signal components, in support of the *TR* parameter.

The effect of instabilities in the backpropagation wave path to the success of *TR* is also investigated. The HF setup at the hydraulic lab at HKUST incorporates pipes of various mechanical and geometrical properties. Therefore, it enables the “sister” pipes experiment where the forward step is conducted along an intact DN80 HDPE pipe, while the *TR* step is performed along (i) a different but still intact DN80 HDPE pipe, (ii) a discontinuous DN80 HDPE pipe with two inline junctions along the backpropagation wave path, (iii) a DN80 copper pipe again with two inline junctions, and (iv) a DN100 acrylic pipe. Only in the case of the “sister” intact DN80 HDPE pipe the *TR* field manages to refocus, albeit with significant spatial shift due to the small, immeasurable, manufacturing deviations between the two seemingly similar pipes. In all other cases tested, *TR* fails to refocus entirely. Specifically, the experiments reveal that inline junctions substantially reduce the amplitude of HF waves, and also change their phase. Therefore, it is imperative that future research efforts focus on a transformation scheme that will compensate for damping in HF waves, similarly to [181, 182] for LF waves. Furthermore, HF wave propagation is tested in pipe bends under two different settings: (i) the acoustic source and receiver are placed at the ends of the pipe bend, and (ii) the receiver is placed along the pipe bend and the source at its end. Experimental dispersion curves for case (i) are dissimilar to the ones obtained from an equivalent straight pipe at specific frequency intervals. Although the cut-off frequencies of propagating radial modes are unaltered, the 3rd radial mode is suppressed in the pipe bend. Consequently, multiple iterations of analytical *TR* performed on experimentally acquired forward step responses of the pipe bend in the pertinent bandwidth yield a consistent error in localizing the source. On the other hand, dispersion curves along the pipe bend demonstrate a considerable shift in the cut-off frequencies of radial modes that increases with frequency, therefore validating previous studies in rigid pipe bends [180]. Ergo, the application of the analytical model as is on pipe bends can cause significant deviations. In conclusion, this chapter examines the effect of pipe flow, damping, and instabilities to the fidelity of *TR*, explores NR *TR* as a counteracting method, and defines the direction of future research towards enabling the fascinating technique of HF *TR* to real world UWSS.

Chapter 8

Conclusions and future work

8.1 Summary and conclusions

Transient Based Defect Detection Methods (TBDDMs) in pipes suggest the use of pressure transients to extract a useful response from a defected system that contains the defect signature. By solving this inverse problem, it is possible to localize and characterize the defect(s). According to the status quo, TBDDMs employ slow mechanical devices (i.e., valves, pressure vessels) to generate long waves that provide limited defect detection resolution measured in the order of tenths or hundredths of meters. This thesis attempts paradigm shift and proposes the use of short waves induced by in-pipe transducers for high resolution defect detection. At these frequencies, the occurring pressure field is three-dimensional thus classical waterhammer models are not applicable. Moreover, wave propagation is dispersive (i.e., the wavespeed is frequency dependent), and fluid-pipe interaction is significant. Hence, the realization of high-resolution defect detection using high-frequency waves requires an appreciable leap in terms of knowledge over existing TBDDMs.

To contribute a comprehensive approach to the stated problem, this thesis answers a series of important questions: (1) What is the most appropriate mathematical model to describe the propagation of high frequency waves in fluid filled elastic pipes that is simple yet effective, minimizes modelling errors, and may be adapted to an inverse solution? (2) Is Time Reversal (TR) of pressure waves a valid method to tackle the convoluted inverse problem of defect detection in pipes using short waves and what are its limitations? (3) How can wave TR be

applied to a defect detection method in pipes? (4) How does the proposed TR-based defect detection method fare in an unfavourable environment that involves turbulent flow, inline junctions, bends, etc. and, thus, violates its necessary conditions?

High-frequency wave propagation in water filled elastic pipes

To address the first question, the governing equations of the two domains that synthesize the coupled fluid filled elastic pipe waveguide are derived following a thorough control volume (C.V.) approach. Subsequently, the equations of motion for the pipe wall are simplified by consideration of the shell theory approximations, as described in [69]. The two media – the pipe wall and contained fluid – are combined through the stress and displacement continuity at their interface. The produced dimensionless characteristic equation, as opposed to general state-of-the-art model expressions, is explicit, manageable, easily solvable, and clearly delineates the different mechanisms that contribute to the wave propagation in the coupled waveguide.

A novel experimental setup is developed at the hydraulic lab of HKUST to investigate the efficacy of the shell theory-based model for the intended use case and probing frequencies. The verification tests are performed on thick-walled DN80 High Density Polyethylene (HDPE) and a thin-walled DN80 copper pipe, so as a sample of two pipes of substantially different properties, in the frequency range from 10 to 70 kHz. The system response to the in-pipe wideband acoustic source is sampled at multiple location along the pipes in both the fluid and structural domains using a hydrophone and accelerometers, respectively. This novel approach enables the construction of experimental dispersion curves for both media (i.e., fluid and pipe wall), and understanding of their in-between wave energy distribution as a function of the probing frequency. In both cases of the water filled DN80 HDPE and DN80 copper pipe, the analytical dispersion relationship obtained from the proposed shell-theory based model for the nominal values of the system properties matches well with the experimental throughout the frequency range considered. For the water filled DN80 copper pipe in particular, the experimental dispersion curves from hydrophone data demonstrate discontinuities at specific frequency intervals where the pertinent modes propagate as an axial wave along the pipe wall. These observations are also supported by the accelerometer measurements, as well as the pipe wall displacement solutions from the analytical model and can afford an intuitive interpretation by consideration of Snell's law. Additional validation for the approximate analytical model is obtained following the successful comparison to a state-of-the-art general model for the two

pipes. Some noted discrepancies between the models concern structural modes at the upper limit of the frequency interval that cannot be evaluated experimentally since the setup cannot excite said modes. The conclusion is that the approximate analytical model is shown to be effective, and adequately describes wave propagation in both a thick and thin-walled pipe.

This thesis shows that the characteristic equation of the proposed analytical model can be used to retrieve other well-known models that are only valid at low frequencies. To define the frequency intervals that these models are usable, the phase speed of the fundamental mode is calculated as a function of frequency and compared to the predictions of the complete expression for a DN60 acrylic and DN80 copper pipe. For instance, the Korteweg equation is accurate up to $0.2f_{\text{ring}}$ (where f_{ring} is the pipe ring frequency). If pipe wall inertia is also considered, the model is valid up to $0.8f_{\text{ring}}$. Only by including the effects of pipe wall flexure the model is accurate beyond f_{ring} . In fact, at these frequencies, the pressure wavefront (that is plane at frequencies approaching zero) forms a surface wave that propagates at the pipe/fluid interface, thus undetected along the pipe centreline. The existence of the peculiar surface wave is verified further from (i) plots that demonstrate the ratio of the work done by the pressure wave to the cross-sectional pipe area to that in the pipe perimeter, and (ii) FEM analysis. Finally, the proposed analytical model is adapted to a pressure wave solution for a cylindrical duct with impedance walls. The analytical response of a DN80 HDPE pipe system to a narrowband signal is obtained and juxtaposed favourably to the pertinent experimental response.

Time reversal of waves in hydraulics

Wave TR is an innovative technique that is most effective in multi-scattering environments and can be used to trace wave sources and infer the properties of the propagating medium. However, wave TR is susceptible to damping and wave path instabilities, and water pipes, especially those made of viscous HDPE, exert considerable (and frequency dependant) damping to pressure transients, while turbulent vortex structures create possible instabilities in the medium. Hence, time symmetry and spatial reciprocity – the two necessary conditions for wave TR – are not guaranteed. To address the second question, this thesis provides the first-ever proof of the time-reversibility of pressure waves in water pipes, despite using a disadvantageous single-channel transducer setup. A transducer pair is introduced in the DN80 HDPE pipe at the hydraulic lab of HKUST and placed 2 m apart. For the forward wave propagation step, a high-frequency narrowband pulse is emitted from the acoustic source, and

the system response is recorded at the receiver. For the TR step, the dispersed response is flipped in time and re-emitted from the receiver (hence TR mirror). As it backpropagates, the signal uniquely refocuses at the location of the original source, thus establishing the TR property. The test is repeated successfully in a larger DN180 HPDE pipe setup where the same transducer pair is placed 36.5 m apart.

To explain the experimental findings, the two mechanisms that restrict wave TR, namely damping and instabilities, are studied analytically. Errors due to instabilities in the medium only evolve in time as $\sim t^{1/2}$ and, hence, are not critical. With regards to damping, an extensive magnitude analysis on the non-reversible terms appearing in the fundamental equations of the coupled waveguide yields the non-dimensional *TR* parameter that grows linearly in time ($\sim t$). One of the pivotal outcomes of this thesis is to show that wave TR only holds for $TR \sim 1$ or less, thus the derived parameter provides with a general criterion to estimate the range over which TR-based wave methodologies are applicable in hydraulics. To illustrate, the TR parameter is checked against both the demonstrated pipe experiments as well as several TR proofs in ocean acoustics and physics. Within the strict context of this thesis, these findings enable TR-based pulse echo techniques for defect detection in pipes. In the wider sense, however, this thesis introduces the fascinating wave TR method in the field of hydraulics and, hopefully, shall stimulate the development of applications such as wave control in the sea, early warning systems against floods, etc.

TR-based pulse echo method for defect detection in pipes

To tackle the third question, a pressure wave solution for a fluid-filled elastic pipe is derived that describes a leak as a mass source, and also involves a probing acoustic source. The total pressure field in the defected pipe is estimated as a superposition of an incident wave (i.e., from the acoustic source to the measurement point) and scattered waves (i.e., from the acoustic source to the leak and then to the measurement point), expressed as a Neumann series sum. To simplify, the Born approximation is applied, and solely the first-order term of the series is retained. Consequently, the pressure response of a defected pipe to a probing signal is expressed as the sum of two terms: (i) the response of an equivalent intact pipe and, (ii) the leak response.

In practice, the response of the defected system also contains measurement noise, while the model involves epistemic uncertainties. The validity of wave TR ensures that the convolution

product (signal-to-noise ratio) between the analytical and experimental signals is maximized at the leak location in the presence of white noise. The TR-based method is successfully tested experimentally on a DN63 HDPE pipe with a simulated leak, probed with a narrowband signal. Using the quasi-steady state assumption, the leak characteristics are represented using the orifice equation [169]. The experimental results support the use of this model since the leak is localized successfully with an accuracy of the probing wavelength order on two occasions, where the distance of the transducer array to the leak is (i) 1.25 m and (ii) 1.78 m, respectively. The analytical solution is adaptable to localizing other wave scatterers besides leaks, such as blockages. An additional experiment places a small bitumen debris from a failed pipe lining in a water filled DN80 HDPE, that covers its cross-sectional area by 20%, that is accurately localized by the TR-based pulse echo method 1.65 m from the transducer array. To demonstrate the dependency of the probing frequency to the attainable resolution, a blockage is localized using probing signals of decreasing wavelength. The results show that the resolution indeed scales with the probing frequency, and explicitly demonstrate the gains of using high frequency probing signals.

TR fidelity under flow conditions and wave path instabilities

Typical water supply pipe networks incorporate processes such as vortex structures, dissipation, bends, junctions, and other discontinuities that inherently violate the TR requirements of time symmetry and spatial reciprocity. Thus, it is crucial that the performance of TR is experimentally evaluated under unfavourable conditions, as suggested by the fourth question. TR tests under flow conditions reveal a – predictable – shift in the focal point, as well as substantial damping. To further elucidate, developed experimental Lissajous curves expose that the phase shift due to flow is linearly dependent on the probing signal frequency. Non-Reciprocal TR (NR TR) involves a slight modification to the traditional TR and has been found to be effective against focal shifts due to ocean currents [177]. The efficacy of NR TR is tested herein in a turbulent pipe environment, and is compared against traditional TR. Despite some apparent benefit of using NR TR at higher probing frequencies (where phase shift is also greater), the results are rather ambiguous due to the limited extend of the pipe setup. Additional TR tests under unsteady turbulent flow are performed by varying the pipe flowrate between the forward and TR steps and show an increase in the phase shift as opposed to keeping the flowrate constant. To better understand the effect of flow to the propagation of high frequency waves in pipes, the experimental dispersion curves under various flowrate conditions are developed for

the DN80 HDPE pipe. In comparison to quiescent water conditions, there are no appreciable differences to the Green's function under flow. Amplitude spectrum plots, however, indicate that high frequency signal components are attenuated more under flow, as supported by the TR parameter.

The effect of instabilities to TR is also explored experimentally. For instance, the forward step is performed in the intact DN80 HDPE pipe, while the TR step is carried out in a “sister” DN80 HDPE pipe. Initially, the test is successful when both pipes are continuous along their length and the TR process still manages to refocus at the expected location despite the anticipated differences due to manufacturing tolerances. However, the introduction of an inline junction in the backpropagation path in the “sister” DN80 HDPE pipe breaks wave TR, and results in a severe attenuation of the pressure signal. The TR step is repeated in the remainder of the setup pipes that include a DN80 copper pipe with an inline junction, and a DN100 acrylic pipe where, in both cases, wave TR fails. Hence, the tested waveguides present such discrepancies that are sufficient to nullify TR. Finally, high frequency wave propagation is experimentally tested along pipe bends, and substantial discrepancies are found in the dispersion relationship compared to an equivalent straight pipe. In particular, the cut-off frequencies of higher order radial modes are increasingly shifted with frequency, in support of previous theoretical studies in curved rigid ducts [180].

8.2 Future work

Although this thesis makes considerable and novel advancements, there are still many challenges in way of rendering the proposed TR-based defect detection method readily applicable in engineering practice. Considering the Technology Readiness Level³ (TRL) that qualifies the maturity of a technology on a scale from 1 to 10, the developed methodology is currently at a 3, as merely an experimental proof of concept has been established. Undoubtedly, the proposed method must be further enhanced to broaden its relevancy to the majority of water supply systems.

The presented explorative TR experiments under pipe flow conditions revealed considerable damping and a shift from the expected focusing point. As a result, the effective range of the

³The technology readiness level (TRL) scale was introduced in the 70s by the National Aeronautics and Space Administration (NASA) and is used in many industries today to describe the evolution level of a technology. A level 1 denotes that the basic principles of the technology have been observed, while a level 9 corresponds to a technology that is commercially available

method is reduced, and resolution degrades. Damping compensation techniques have been implemented successfully for low-frequency water hammer waves [12] and viscous media [13], but their applicability to high frequency waves has yet to be explored. To counter the phase shift due to flow instabilities, NR TR is briefly tested. Regardless, the available setup has limited extend and is, therefore, inappropriate for evaluating NR TR. It is proposed that NR TR is revisited in a longer pipe setup and/or turbulent flow conditions. Finally, a more thorough investigation of the effect of inline pipe junctions to pressure wave propagation is imperative, given their potential to break TR.

The adopted analytical model currently supposes axisymmetric pressure and displacement fields in the contained fluid and pipe wall, respectively. Therefore, the placement of the acoustic source in the pipe is restricted at the pipe centre so as to create such an axisymmetric field. To remove such limitation, the analytical model could be extended to include the solutions in the azimuthal direction. Alternatively, an experimental setup that places the acoustic source at a place different than the centre could be used to evaluate the introduced error. Another limitation of the model is the assumption that the coupled waveguide is surrounded by vacuum. However, water pipes are buried in practice, and wave energy radiates in the surrounding soil with detrimental effects to the attainable range [183, 184]. Although researchers have already attempted to study the effect of the surrounding soil to pressure wave propagation [185, 186], the focus has been on the low frequency regime ($< f_{ring}$). Accordingly, it may be necessary to devise new pipe types that implement an isolation layer from the soil or internal metamaterial structures to reduce losses and focus energy in the pipe. Recently, it has been proposed that probing wavelengths smaller than the pipe wall thickness are employed to enable the detection of incipient defects. However, the assumptions of the shell theory model for a linear distribution of the axial displacements and invariable pipe wall thickness falter at said frequencies. To counter that inefficiency, higher order terms in the Taylor series expansion of the axial and radial displacements could be considered, and the improved model be verified experimentally.

A particularly interesting outcome of the carried research is the identification of the surface wave at frequencies that approach the natural pipe ring frequency f_{ring} . Given that the surface wave propagates at the fluid/pipe wall interface where defects are found, there is great potential in exploiting that characteristic to extract a high SNR response. The employed transducers herein, however, are small in size in comparison to the cross-sections of the setup pipes, while also inefficient at low frequencies close to the f_{ring} ; thus, not suitable for exciting the

analytically identified surface wave. A targeted lab setup would utilize acoustic projectors of larger size, that enable the excitation and, hence, study of surface waves, as well as their effectiveness of in identifying leaks.

Applications of wave TR typically combine multiple transducers to form an array of increased aperture, with evident gains in the attainable refocusing compared to single-channel setups. Such approach is difficult to apply in water supply pipes of limited cross-sectional area and accessibility options. Regardless, the aperture of the array could be seemingly increased by performing multiple measurements at different radial and axial positions in the pipe.

Regarding the proposed TR-based pulse echo leak detection, the characteristics of the leak are described by an orifice equation based on a steady state assumption. Indeed, the focus of this thesis is on localizing the leak as opposed to its full characterization. Therefore, future research could put effort to better calibrate the adopted orifice equation model. Besides, different leak sizes could be tested to define the minimum required SNR for their detection. An interesting observation from the test results on defects is that the response signatures of both the leak and blockage are indistinguishable. Hence, further work is needed to characterize the different types of wave scatterers. Besides, the presented experiments utilize short narrowband probing signals that better illustrate propagating modes but carry little energy. Instead, defect detection with long linear frequency modulated (LFM) signals could be attempted, similar to radars [88]. Finally, in the presence of multiple features in the waveguide, time reversal could be used to selectively focus at the largest among the scatterers, in a process known as iterative TR [187]. Although already tested in other fields [188], iterative TR could be used in water pipes to ensure the presence of a defect in the system and avoid false positives.

Appendix A

Supplementary material for governing equations

A.1. Linear momentum equations for an infinitesimal pipe wall element

The general form of the momentum conservation equation for a material control volume, such as the pipe wall element (shown in Figure 8.2.1) is written as:

$$\frac{D}{Dt} \int_{\mathcal{V}_p} \rho_p \mathbf{v}_p d\mathcal{V}_p = \int_{\mathcal{V}_p} \rho_p \mathbf{g} d\mathcal{V}_p + \int_A \mathbf{n} \cdot \mathbf{\tau}_p dA \quad (\text{A.1.1})$$

It is useful to calculate the volume of the infinitesimal curved pipe wall element as:

$$\Delta\mathcal{V}_p = \int_x^{x+\Delta x} \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \int_{\theta}^{\theta+\Delta\theta} r d\theta dr dx = \int_x^{x+\Delta x} \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} r \Delta\theta dr dx = \int_x^{x+\Delta x} R \Delta r \Delta\theta dx = R \Delta r \Delta\theta \Delta x \quad (\text{A.1.2})$$

Therefore, the first term on the right-hand side of the momentum conservation equation for the pipe wall element (Eq. (A.1.1)) becomes:

$$\frac{D}{Dt} \int_{\mathcal{V}_p} \rho_p \mathbf{v}_p d\mathcal{V}_p = \frac{D}{Dt} \left[\bar{\rho}_p \bar{\mathbf{v}}_p (R \Delta r \Delta\theta \Delta x) \right] \quad (\text{A.1.3})$$

where $\bar{\rho}_p$ is the average pipe material density evaluated over the pipe wall C.V. According to the mean value theorem, there is a point in the infinitesimal volume ($r + \kappa_r \Delta r$, $\theta + \kappa_\theta \Delta\theta$, $x + \kappa_x \Delta x$) with $\kappa_r, \kappa_\theta, \kappa_x \in [0, 1]$ where:

$$\int_{\mathcal{V}} \rho_p d\mathcal{V}_p = \rho_p (r + \kappa_r \Delta r, y + \kappa_\theta \Delta \theta, x + \kappa_x \Delta x) \mathcal{V}_p \quad (\text{A.1.4})$$

It follows that the average density $\bar{\rho}_p$ is:

$$\bar{\rho}_p = \frac{\int_{\mathcal{V}} \rho d\mathcal{V}}{\int_{\mathcal{V}} d\mathcal{V}} = \rho_p (r + \kappa_r \Delta r, y + \kappa_\theta \Delta \theta, x + \kappa_x \Delta x) \quad (\text{A.1.5})$$

Moreover, $\bar{\mathbf{v}}$ is the mass-weighted average velocity of the C.V. By the mean value theorem, there is a point in the C.V. $(r + \varsigma_r \Delta r, \theta + \varsigma_\theta \Delta \theta, x + \varsigma_x \Delta x)$ with $\varsigma_r, \varsigma_\theta, \varsigma_x \in [0, 1]$ where:

$$\int_{\mathcal{V}} \rho_p \mathbf{v}_p d\mathcal{V}_p = \bar{\rho}_p \int_{\mathcal{V}} \mathbf{v}_p d\mathcal{V} = \bar{\rho}_p \mathbf{v}_p (r + \varsigma_r \Delta r, y + \varsigma_\theta \Delta \theta, x + \varsigma_x \Delta x) \mathcal{V}_p \quad (\text{A.1.6})$$

Thus, the mass-weighted average velocity $\bar{\mathbf{v}}_p$ is expressed as:

$$\bar{\mathbf{v}}_p = \frac{\int_{\mathcal{V}} \rho_p \mathbf{v}_p d\mathcal{V}_p}{\int_{\mathcal{V}} \rho_p d\mathcal{V}_p} = \mathbf{v}_p (r + \varsigma_r \Delta r, y + \varsigma_\theta \Delta \theta, x + \varsigma_x \Delta x) \quad (\text{A.1.7})$$

The first term on the left-hand side of Eq. (A.1.1) represents the effect of gravity forces on the pipe wall element, and may be evaluated as:

$$\int_{\mathcal{V}} \rho_p \mathbf{g} d\mathcal{V}_p = [\bar{\rho}_p \mathbf{g} (R \Delta r \Delta \theta \Delta x)] \quad (\text{A.1.8})$$

The second term on the left-hand side of Eq. (A.1.1) represents the effect of surface forces on the pipe wall element momentum. The forces acting on the C.V. surfaces are:

$$\begin{aligned} \int_A \mathbf{n} \cdot \mathbf{T}_p dA = & \mathbf{i} \cdot \left[\left(\bar{\mathbf{T}}_p R \Delta \theta \Delta r \right) \Big|_{x+\Delta x} - \left(\bar{\mathbf{T}}_p R \Delta \theta \Delta r \right) \Big|_x \right] + \mathbf{e}_\theta \cdot \left[\left(\bar{\mathbf{T}}_p \Delta r \Delta x \right) \Big|_{\theta+\Delta \theta} - \left(\bar{\mathbf{T}}_p \Delta r \Delta x \right) \Big|_\theta \right] \\ & + \mathbf{e}_r \cdot \left\{ \left[\bar{\mathbf{T}}_p \left(R + \frac{\Delta r}{2} \right) \Delta \theta \Delta x \right] \Big|_{R+\frac{\Delta r}{2}} - \left[\bar{\mathbf{T}}_p \left(R - \frac{\Delta r}{2} \right) \Delta \theta \Delta x \right] \Big|_{R-\frac{\Delta r}{2}} \right\} \end{aligned} \quad (\text{A.1.9})$$

where $\mathbf{i}$, $\mathbf{e}_\theta$, $\mathbf{e}_r$ are the unit vectors along the x , θ and r directions, respectively, and $\mathbf{T}_p$ is the stress tensor applied on the pipe wall C.V. defined as:

$$\mathbf{T}_p = \begin{bmatrix} \sigma_{rr} & \tau_{r\theta} & \tau_{rx} \\ \tau_{\theta r} & \sigma_{\theta\theta} & \tau_{\theta x} \\ \tau_{xr} & \tau_{x\theta} & \sigma_{xx} \end{bmatrix} \quad (\text{A.1.10})$$

or, in index notation:

$$\mathbf{T}_p = \sigma_{rr} \mathbf{e}_r \mathbf{e}_r + \tau_{r\theta} \mathbf{e}_r \mathbf{e}_\theta + \tau_{rx} \mathbf{e}_r \mathbf{i} + \tau_{\theta r} \mathbf{e}_\theta \mathbf{e}_r + \sigma_{\theta\theta} \mathbf{e}_\theta \mathbf{e}_\theta + \tau_{\theta x} \mathbf{e}_\theta \mathbf{i} + \tau_{xr} \mathbf{i} \mathbf{e}_r + \tau_{x\theta} \mathbf{i} \mathbf{e}_\theta + \sigma_{xx} \mathbf{i} \mathbf{i} \quad (\text{A.1.11})$$

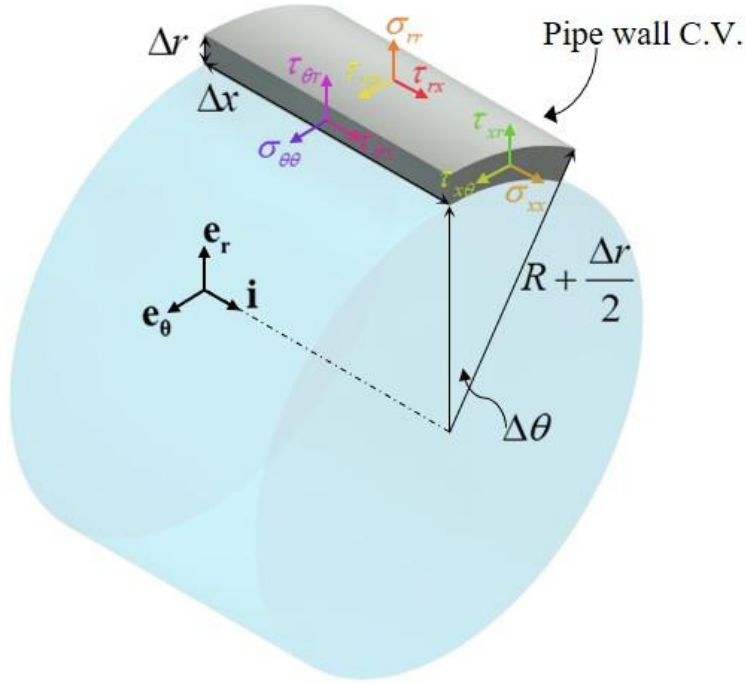


Figure 8.2.1: Pipe wall element control volume with the appropriate stress components marked on the pertinent surfaces.

In Eq. (A.1.9) $\bar{\mathbf{T}}_p$ denotes the average stresses over a particular integration surface. As an indicative example, we consider the face of the C.V. on the $r\theta$ -plane at x . By the mean value theorem, there is a point on the surface $(r + \varphi_r \Delta r, \theta + \varphi_\theta \Delta \theta, x)$ with $\varphi_r, \varphi_\theta \in [0, 1]$ where:

$$\int_A \mathbf{T}_p dA \Big|_x = \bar{\mathbf{T}}_p (r + \varphi_r \Delta r, \theta + \varphi_\theta \Delta \theta, x) R \Delta \theta \Delta r = \bar{\mathbf{T}}_p \Big|_x R \Delta \theta \Delta r \quad (\text{A.1.12})$$

It follows that the average stresses $\bar{\mathbf{T}}$ over the assumed surface are:

$$\bar{\mathbf{T}}_p \Big|_x = \frac{\int_A \mathbf{n} \cdot \mathbf{T}_p dA}{\int_A dA} \Big|_x = \mathbf{T}_p (r + \varphi_r \Delta r, \theta + \varphi_\theta \Delta \theta, x) \quad (\text{A.1.13})$$

The results from Eqns. (A.1.3), (A.1.8) and (A.1.9) are substituted in Eq. (A.1.1) to obtain:

$$\begin{aligned}
\frac{D}{Dt} [\bar{\rho}_p \bar{\mathbf{v}}_p (R \Delta r \Delta \theta \Delta x)] = & \\
& [\bar{\rho}_p \mathbf{g} (R \Delta r \Delta \theta \Delta x)] + \mathbf{i} \cdot \left[(\bar{\mathbf{T}}_p R \Delta \theta \Delta r) \Big|_{x+\Delta x} - (\bar{\mathbf{T}}_p R \Delta \theta \Delta r) \Big|_x \right] \\
& + \mathbf{e}_\theta \cdot \left[(\bar{\mathbf{T}}_p \Delta r \Delta x) \Big|_{\theta+\Delta \theta} - (\bar{\mathbf{T}}_p \Delta r \Delta x) \Big|_\theta \right] \\
& + \mathbf{e}_r \cdot \left\{ \left[\bar{\mathbf{T}}_p \left(R + \frac{\Delta r}{2} \right) \Delta \theta \Delta x \right] \Big|_{R+\frac{\Delta r}{2}} - \left[\bar{\mathbf{T}}_p \left(R - \frac{\Delta r}{2} \right) \Delta \theta \Delta x \right] \Big|_{R-\frac{\Delta r}{2}} \right\}
\end{aligned} \tag{A.1.14}$$

Equation (A.1.14) is divided by the volume of the C.V. (Eq. (A.1.2)) and yields:

$$\begin{aligned}
\frac{D(\bar{\rho}_p \bar{\mathbf{v}}_p)}{Dt} = & (\bar{\rho}_p \mathbf{g}) + \mathbf{i} \cdot \frac{[(\bar{\mathbf{T}}_p) \Big|_{x+\Delta x} - (\bar{\mathbf{T}}_p) \Big|_x]}{\Delta x} + \mathbf{e}_\theta \cdot \frac{[(\bar{\mathbf{T}}_p) \Big|_{\theta+\Delta \theta} - (\bar{\mathbf{T}}_p) \Big|_\theta]}{R \Delta \theta} \\
& + \mathbf{e}_r \cdot \frac{\left[\left[\bar{\mathbf{T}}_p \left(R + \frac{\Delta r}{2} \right) \right] \Big|_{R+\frac{\Delta r}{2}} - \left[\bar{\mathbf{T}}_p \left(R - \frac{\Delta r}{2} \right) \right] \Big|_{R-\frac{\Delta r}{2}} \right]}{R \Delta r}
\end{aligned} \tag{A.1.15}$$

We take the limit as Δr , $\Delta \theta$, and $\Delta x \rightarrow 0$:

$$\bar{\rho}_p \rightarrow \rho_p(r, \theta, x), \quad \bar{\mathbf{v}}_p \rightarrow \mathbf{v}_p(r, \theta, x), \quad \text{and} \quad \bar{\mathbf{T}}_p \rightarrow \mathbf{T}_p(r, \theta, x) \tag{A.1.16}$$

and Eq. (A.1.15) becomes:

$$\frac{D(\rho_p \mathbf{v}_p)}{Dt} = (\rho_p \mathbf{g}) + \mathbf{i} \cdot \frac{\partial \mathbf{T}_p}{\partial x} + \mathbf{e}_\theta \cdot \frac{1}{R} \frac{\partial \mathbf{T}_p}{\partial \theta} + \mathbf{e}_r \cdot \left(\frac{\partial (\mathbf{T}_p R)}{R \partial r} \right) \tag{A.1.17}$$

Furthermore, we adopt vector notation to express the divergence operator and Eq. (A.1.17) becomes:

$$\frac{D(\rho_p \mathbf{v}_p)}{Dt} - \rho_p \mathbf{g} - \nabla \cdot \mathbf{T}_p = 0 \tag{A.1.18}$$

The momentum conservation equation (Eq. (A.1.18)) may further be simplified if we consider the mass conservation equation for the material pipe wall C.V. In such case, the general form of the mass conservation equation is:

$$\frac{D}{Dt} \int_V \rho_p dV_p = 0 \tag{A.1.19}$$

For our particular C.V., Eq. (A.1.19) becomes:

$$\frac{D}{Dt} \int_{\mathcal{V}} \rho_p d\mathcal{V}_p = \frac{D}{Dt} [\bar{\rho}_p (R\Delta r \Delta \theta \Delta x)] \quad (\text{A.1.20})$$

Equation (A.1.20) is divided by the volume of the considered C.V., and taking the limit as Δr , $\Delta \theta$, and $\Delta x \rightarrow 0$ yields:

$$\frac{D\rho_p}{Dt} = 0 \quad (\text{A.1.21})$$

We combine the mass conservation (Eq. (A.1.21)), and the momentum conservation (Eq. (A.1.18)) equations for the pipe wall C.V. to obtain:

$$\rho_p \frac{D\mathbf{v}_p}{Dt} - \rho_p \mathbf{g} - \nabla \cdot \mathbf{T}_p = 0 \quad (\text{A.1.22})$$

where $\mathbf{v}$ is the velocity vector of the C.V.:

$$\mathbf{v}_p = v_{p,r} \mathbf{e}_r + v_{p,\theta} \mathbf{e}_\theta + v_{p,x} \mathbf{i} \quad (\text{A.1.23})$$

The material derivative is expanded, and Eq. (A.1.22) becomes:

$$\rho_p \left(\frac{\partial \mathbf{v}_p}{\partial t} + \mathbf{v}_p \cdot \nabla \mathbf{v}_p \right) - \rho_p \mathbf{g} - \nabla \cdot \mathbf{T}_p = 0 \quad (\text{A.1.24})$$

We shall perform an order analysis on the terms that appear in Eq. (A.1.24) to identify possible simplifications on the basis of the application presented in this thesis. The individual terms of Eq. (A.1.24) scale to:

$$\begin{aligned} \text{i.} \quad & \frac{\partial \mathbf{v}_p}{\partial t} \sim \frac{\|\mathbf{v}_p\|}{T} = \frac{\|\mathbf{v}_p\|}{L} c_p & \text{ii.} \quad & \mathbf{v}_p \cdot \nabla \mathbf{v}_p \sim \frac{\|\mathbf{v}_p\|^2}{L} \\ \text{iii.} \quad & \mathbf{g} \cdot \mathbf{e}_r \sim \|\mathbf{g}\| & \text{iv.} \quad & \nabla \cdot \mathbf{T}_p \sim \frac{\mathbf{T}_p}{L} \end{aligned}$$

Thus, the relative magnitude of the advective over that of the temporal acceleration term is:

$$\frac{\mathbf{v}_p \cdot \nabla \mathbf{v}_p}{\frac{\partial \mathbf{v}_p}{\partial t}} \sim \frac{\frac{\|\mathbf{v}_p\|^2}{L}}{\frac{\|\mathbf{v}_p\|}{L} c_p} = \frac{\|\mathbf{v}_p\|}{c_p} = Ma \quad (\text{A.1.25})$$

Mach number for the case of a wave propagating in a solid medium is negligibly small; thus, the advective acceleration terms are excluded from further consideration. The gravity

acceleration term is also compared to the temporal acceleration term, and their relative magnitude is:

$$\frac{\mathbf{g}}{\frac{\partial \mathbf{v}_p}{\partial t}} \sim \frac{\frac{\|\mathbf{g}\|}{L}}{\frac{\|\mathbf{v}_p\|}{c_p}} = \frac{\|\mathbf{g}\| L}{c_p \|\mathbf{v}_p\|} \quad (\text{A.1.26})$$

For the applications that define the subject of this thesis, the order of magnitude for the length scale L is 10^{-2} m, whereas for the wave speed c_p is 10^3 m/s. Particle velocity under excitation is measured experimentally using accelerometers mounted on the surface of the pipe and is of the order of 10 m/s. Therefore, the relative magnitude of gravity over temporal acceleration is evaluated equal to:

$$\frac{\mathbf{g} \cdot \mathbf{e}_r}{\frac{\partial v_r}{\partial t}} \sim \frac{\|\mathbf{g}\| L}{c_p \|\mathbf{v}_p\|} = 10^{-5} \quad (\text{A.1.27})$$

The gravity acceleration term is multiple orders smaller than the temporal acceleration term and is, therefore, omitted. After performing the scale analysis, the momentum conservation equation (Eq. (A.1.24)) is reduced to:

$$\rho_p \frac{\partial \mathbf{v}_p}{\partial t} - \nabla \cdot \mathbf{T}_p = 0 \quad (\text{A.1.28})$$

The divergence of the stress tensor $\mathbf{T}_p$ produces a vector, that in cylindrical coordinates is written as:

$$\begin{aligned} \nabla \cdot \mathbf{T}_p = & \left[\frac{\partial \sigma_{rr}}{\partial r} + \frac{1}{R} \frac{\partial \tau_{\theta r}}{\partial \theta} + \frac{\partial \tau_{xr}}{\partial x} + \frac{1}{R} (\sigma_{rr} - \sigma_{\theta\theta}) \right] \mathbf{e}_r \\ & + \left[\frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{R} \frac{\partial \sigma_{\theta\theta}}{\partial \theta} + \frac{\partial \tau_{x\theta}}{\partial x} + \frac{1}{R} (\tau_{r\theta} + \tau_{\theta r}) \right] \mathbf{e}_\theta \\ & + \left[\frac{\partial \tau_{rx}}{\partial r} + \frac{1}{R} \frac{\partial \tau_{\theta x}}{\partial \theta} + \frac{\partial \sigma_{xx}}{\partial x} + \frac{\tau_{rx}}{R} \right] \mathbf{i} \end{aligned} \quad (\text{A.1.29})$$

The momentum conservation (Eq. (A.1.28)) constitutes a vector equation; thus, taking its dot product with the unit vectors in cylindrical coordinates, gives 3 components:

i. Dot product with $\mathbf{e}_r$ (r -component)

$$\frac{\partial v_{p,r}}{\partial t} - \frac{1}{\rho_p} \left[\frac{\partial \sigma_{rr}}{\partial r} + \frac{1}{R} \frac{\partial \tau_{\theta r}}{\partial \theta} + \frac{\partial \tau_{xr}}{\partial x} + \frac{1}{R} (\sigma_{rr} - \sigma_{\theta\theta}) \right] = 0 \quad (\text{A.1.30})$$

ii. Dot product with $\mathbf{e}_\theta$ (θ -component)

$$\frac{\partial v_{p,\theta}}{\partial t} - \frac{1}{\rho_p} \left[\frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{R} \frac{\partial \sigma_{\theta\theta}}{\partial \theta} + \frac{\partial \tau_{x\theta}}{\partial x} + \frac{1}{R} (\tau_{r\theta} + \tau_{\theta r}) \right] = 0 \quad (\text{A.1.31})$$

iii. Dot product with $\mathbf{i}$ (x -component)

$$\frac{\partial v_{p,x}}{\partial t} - \frac{1}{\rho_p} \left[\frac{\partial \tau_{rx}}{\partial r} + \frac{1}{R} \frac{\partial \tau_{\theta x}}{\partial \theta} + \frac{\partial \sigma_{xx}}{\partial x} + \frac{\tau_{rx}}{R} \right] = 0 \quad (\text{A.1.32})$$

The displacements along the r -axis are denoted as $w_p(r, \theta, x, t)$, the x -axis as $u_p(r, \theta, x, t)$, and the θ -axis as $v_p(r, \theta, x, t)$. Therefore, the components of the velocity vector in terms of displacements are written as:

$$v_{p,r} = \frac{\partial w_p}{\partial t}, \quad v_{p,\theta} = \frac{\partial v_p}{\partial t}, \quad \text{and} \quad v_{p,x} = \frac{\partial u_p}{\partial t} \quad (\text{A.1.33})$$

Given that only axisymmetric vibrations are considered in this thesis, the azimuthal displacements v_p , and derivatives with respect to θ are equal to zero; thus, by extension, the velocity component $v_{p,\theta}$ is also zero. Besides, from Eq. (2.3.115), $\tau_{r\theta} = \tau_{\theta r} = 0$. As a result, the momentum conservation component in the θ -direction (Eq. (A.1.31)) leads to $(0 = 0)$ and the remaining two component equations (Eqns. (A.1.30) & (A.1.32)) become:

i. r -component

$$\rho_p \frac{\partial^2 w_p}{\partial t^2} - \left[\frac{\partial \sigma_{rr}}{\partial r} + \frac{\partial \tau_{xr}}{\partial x} + \frac{1}{R} (\sigma_{rr} - \sigma_{\theta\theta}) \right] = 0 \quad (\text{A.1.34})$$

ii. x -component

$$\frac{\partial^2 u_p}{\partial t^2} - \frac{1}{\rho_p} \left[\frac{\partial \tau_{rx}}{\partial r} + \frac{\partial \sigma_{xx}}{\partial x} + \frac{\tau_{rx}}{R} \right] = 0 \quad (\text{A.1.35})$$

We proceed to integrate Eqns. (A.1.34) and (A.1.35) across the thickness of the pipe wall element and construct the equations of motion for the pipe wall element. We start with the momentum component along r (Eq. (A.51)), where the radial displacements w_p are substituted by the expression of Eq. (2.3.127):

$$\rho_p \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \frac{\partial^2 w_{p,0}}{\partial t^2} dr - \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \left[\frac{\partial \sigma_{rr}}{\partial r} + \frac{\partial \tau_{xr}}{\partial x} + \frac{1}{R} (\sigma_{rr} - \sigma_{\theta\theta}) \right] dr = 0 \quad (\text{A.1.36})$$

As shown in Eq. (2.3.127), radial displacements w_0 are not a function of r , and are treated as constant:

$$\rho_p \frac{\partial^2 w_0}{\partial t^2} \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} dr - [\sigma_{rr}]_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} - \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \frac{\partial \tau_{xr}}{\partial x} dr - \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \frac{\sigma_{rr}}{R} dr + \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \frac{\sigma_{\theta\theta}}{R} dr = 0 \quad (\text{A.1.37})$$

The outmost left term in Eq. (A.1.37) is evaluated as:

$$\rho_p \frac{\partial^2 w_{p,0}}{\partial t^2} \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} dr = \rho_p \frac{\partial^2 w_{p,0}}{\partial t^2} \Delta r \quad (\text{A.1.38})$$

It is assumed that the pipe is surrounded by vacuum, while the contained fluid exerts a pressure $-p$ to the inner pipe surface. Therefore, the second term from the left in Eq. (A.1.37) is:

$$[\sigma_{rr}]_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} = p \quad (\text{A.1.39})$$

The third term from the left in Eq. (A.1.37) after substituting the τ_{rx} expression (Eq. (2.3.138)) becomes:

$$\int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \frac{\partial \tau_{xr}}{\partial x} dr = \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} G \left(\frac{\partial^2 w_{p,0}}{\partial x^2} + \frac{\partial \alpha}{\partial x} \right) dr = KG\Delta r \left(\frac{\partial^2 w_{p,0}}{\partial x^2} + \frac{\partial \alpha}{\partial x} \right) \quad (\text{A.1.40})$$

The average radial stresses σ_{rr} are negligible in comparison to the hoop stresses $\sigma_{\theta\theta}$ for the case of a thin-walled pipe, as discussed in subsection 2.3.3. Thus, the radial stress term in Eq. (A.1.37) is omitted, and the hoop stress term that remains is:

$$\begin{aligned} \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \frac{\sigma_{\theta\theta}}{R} dr &= \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \frac{\sigma_{\theta\theta}}{R} dr \\ &= \frac{E}{(1-\nu_p^2)R} \int_{R-\frac{1}{2}\Delta r}^{R+\frac{1}{2}\Delta r} \left\{ \frac{w_{p,0}}{r} + \nu_p \left[\frac{\partial u_{p,0}}{\partial x} + (r-r_0) \frac{\partial \alpha}{\partial x} \right] \right\} dr \\ &= \frac{E\Delta r}{(1-\nu_p^2)R} \left[\frac{w_{p,0}}{\Delta r} \ln \left(\frac{2R+\Delta r}{2R-\Delta r} \right) + \nu_p \frac{\partial u_{p,0}}{\partial x} + \frac{\nu_p \Delta r^2}{12R} \frac{\partial \alpha}{\partial x} \right] \end{aligned} \quad (\text{A.1.41})$$

We combine the results of Eqns. (A.1.38) to (A.1.41), and Eq. (A.1.37) becomes:

$$\begin{aligned} \rho_p \frac{\partial^2 w_{p,0}}{\partial t^2} \Delta r &= \\ KG\Delta r \left(\frac{\partial^2 w_{p,0}}{\partial x^2} + \frac{\partial \alpha}{\partial x} \right) &- \frac{E\Delta r}{(1-\nu_p^2)R} \left[\frac{w_{p,0}}{\Delta r} \ln \left(\frac{2R+\Delta r}{2R-\Delta r} \right) + \frac{\nu_p}{R} \frac{\partial u_{p,0}}{\partial x} + \frac{\nu_p \Delta r^2}{12R^2} \frac{\partial \alpha}{\partial x} \right] + p \end{aligned} \quad (\text{A.1.42})$$

For an additional simplification of Eq. (A.1.42), we retain only the first term in the logarithm expansion and eliminate the final term contained within the square brackets since $R \gg \Delta r$, to obtain:

$$\boxed{\rho_p \frac{\partial^2 w_{p,0}}{\partial t^2} \Delta r = KG \Delta r \left(\frac{\partial^2 w_{p,0}}{\partial x^2} - \frac{\partial \alpha}{\partial x} \right) - \frac{E \Delta r}{(1-\nu^2)} \left[\frac{w_{p,0}}{R^2} \ln \frac{(2R + \Delta r)}{(2R - \Delta r)} + \frac{\nu_p}{R} \frac{\partial u_{p,0}}{\partial x} \right] + p} \quad (\text{A.1.43})$$

Equation (A.1.43) is the differential equation of motion in the radial direction. A similar expression shall be derived in the axial direction, as well. We start by replacing the expression for the axial displacements (Eq. (2.3.137)) into the x-component of the momentum conservation (Eq. (A.1.35)), and then integrating over the pipe thickness:

$$\rho_p \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \left(\frac{\partial^2 u_{p,0}}{\partial t^2} + (r-r_0) \frac{\partial^2 \alpha}{\partial t^2} \right) dr - \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \left[\frac{\partial \tau_{rx}}{\partial r} + \frac{\partial \sigma_{xx}}{\partial x} + \frac{\tau_{rx}}{R} \right] dr = 0 \quad (\text{A.1.44})$$

We evaluate Eq. (A.1.44) in a term-by-term basis:

$$\begin{aligned} \rho_p \left(\frac{\partial^2 u_{p,0}}{\partial t^2} \right) \left[r \right]_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} + \rho_p \left(\frac{\partial^2 \alpha}{\partial t^2} \right) \left[\frac{r^2}{2} - Rr \right]_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \\ - \left[\tau_{rx} \right]_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} - \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \frac{\partial \sigma_{xx}}{\partial x} dr + \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \frac{\tau_{rx}}{R} dr = 0 \end{aligned} \quad (\text{A.1.45})$$

For the second term, r_0 is replaced with R (Eq. (2.3.70), for $R \gg h$) and the evaluated quantity in square brackets is equal to zero; thus, the term is omitted. Moreover, for an inviscid fluid contained within (an assumption proven valid for our application in Eq. (2.2.76)), the shear stresses τ_{rx} at the inner surface of the pipe are zero. The same holds for the outer surface of the pipe, hence, the third term in Eq. (A.1.45) is removed. It also follows that for an inviscid, frictionless fluid, normal stresses σ_{xx} are much larger than the shearing stresses, τ_{rx} , and in effect, the outmost right term in Eq. (A.1.45) is also omitted. The outmost left term in Eq. (A.1.45) represents the inertial forces in the axial direction and is calculated as:

$$\rho_p \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \frac{\partial^2 u_{p,0}}{\partial t^2} dr = \rho_p \left(\frac{\partial^2 u_{p,0}}{\partial t^2} \right) \left[r \right]_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} = \rho_p \left(\frac{\partial^2 u_{p,0}}{\partial t^2} \right) \Delta r \quad (\text{A.1.46})$$

The fourth term from the left in Eq. (A.1.45) is evaluated after replacing the expression for σ_{xx} (Eq. (A.81)) as:

$$\begin{aligned} \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \frac{\partial \sigma_{xx}}{\partial x} dr &= \frac{E}{(1-\nu_p^2)} \int_{R-\frac{\Delta r}{2}}^{R+\frac{\Delta r}{2}} \left[\frac{\partial^2 u_{p,0}}{\partial x^2} + (r-r_0) \frac{\partial^2 \alpha}{\partial x^2} + \frac{\nu_p}{r} \frac{\partial w_{p,0}}{\partial x} \right] dr \\ &= \frac{E \Delta r}{(1-\nu_p^2)} \left[\frac{\partial^2 u_{p,0}}{\partial x^2} + \frac{\nu_p \Delta r^2}{12 R^2} \frac{\partial^2 \alpha}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_{p,0}}{\partial x} \right] \end{aligned} \quad (A.1.47)$$

Finally, Eqns. (A.1.46) & (A.1.47) are combined to yield the differential equation of motion in the axial direction:

$$\rho_p \frac{\partial^2 u_{p,0}}{\partial t^2} \Delta r = \frac{E \Delta r}{(1-\nu_p^2)} \left[\frac{\partial^2 u_{p,0}}{\partial x^2} + \frac{\nu_p \Delta r^2}{12 R^2} \frac{\partial^2 \alpha}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_{p,0}}{\partial x} \right] \quad (A.1.48)$$

For a thin-walled assumption ($R \gg \Delta r$), Eq. (A.1.48) becomes:

$$\boxed{\rho_p \frac{\partial^2 u_{p,0}}{\partial t^2} \Delta r = \frac{E \Delta r}{(1-\nu_p^2)} \left[\frac{\partial^2 u_{p,0}}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_{p,0}}{\partial x} \right]} \quad (A.1.49)$$

Appendix B

Time reversal transformation

B.1. Time-reversal transformation for inviscid fluid and pipe wall equations

The point mass balance equation for an inviscid fluid flow is:

$$\overbrace{\frac{D\rho_0(t)}{Dt}}^{\text{Mass rate of change}} + \overbrace{\rho_0(t)\nabla \cdot \frac{d\mathbf{r}(t)}{dt}}^{\text{Net outward flow rate of mass}} = 0 \quad (\text{B.1.1})$$

where ρ_0 is the fluid density, and $\mathbf{r}$ is the fluid particle position vector. The point form of conservation of momentum for an inviscid fluid flow is:

$$\overbrace{\rho_0(t)\frac{d^2\mathbf{r}(t)}{dt^2}}^{\text{Temporal acceleration}} + \overbrace{\rho_f(t)\frac{d\mathbf{r}(t)}{dt}\nabla \cdot \frac{d\mathbf{r}(t)}{dt}}^{\text{Advective acceleration}} + \overbrace{\nabla p(t)}^{\text{Pressure forces}} + \overbrace{\rho_0(t)\mathbf{g}}^{\text{Gravity forces}} = \overbrace{\delta(t)}^{\text{Source}} \quad (\text{B.1.2})$$

where p is the pressure, $\mathbf{g}$ the acceleration of gravity, and $\delta(t)$ corresponds to a source.

To describe the motions of the pipe wall, the equations developed in Chapter 2 are employed. Apart from offering a simple formulation, this model is successfully validated against experimental data over the frequency range of interest and pipes employed in this thesis (see Chapter 4). The equations of motion on the axisymmetric xr -plane, including an angular momentum equation to describe flexural deformations on said plane, are written as:

$$\overbrace{\rho_p(t)h \frac{\partial^2 u_{p,0}(t)}{\partial t^2}}^{\text{Axial inertia}} = \frac{1}{J(t)} \frac{h}{(1-\nu_p^2)} \left(\overbrace{\frac{\partial^2 u_{p,0}(t)}{\partial x^2}}^{\text{Axial stresses}} + \overbrace{\frac{\nu_p}{R} \frac{\partial w_p(t)}{\partial x}}^{\text{Axial stresses: Poisson Effect}} \right) \quad (\text{B.1.3})$$

$$\overbrace{\rho_p(t)h \frac{\partial^2 w_p(t)}{\partial t^2}}^{\text{Radial inertia}} = \overbrace{\frac{\partial S(t)}{\partial x}}^{\text{Shear force}} - \frac{1}{J(t)} \frac{h}{(1-\nu_p^2)} \left(\overbrace{\frac{w_p(t)}{R^2}}^{\text{Hoop stresses}} + \overbrace{\frac{\nu_p}{R} \frac{\partial u_{p,0}(t)}{\partial x}}^{\text{Hoop stresses: Poisson effect}} \right) + \overbrace{p(t)}^{\text{Applied pressure}} \quad (\text{B.1.4})$$

$$\overbrace{\rho_p(t) \frac{h^3}{12} \frac{\partial^2 \alpha(t)}{\partial t^2}}^{\text{Rotatory inertia}} = \overbrace{\frac{h^3}{12J(t)(1-\nu_p^2)} \frac{\partial^2 \alpha(t)}{\partial x^2}}^{\text{Bending stresses}} + \overbrace{S(t)}^{\text{Shear force}} \quad (\text{B.1.5})$$

Equations (B.1.3) and (B.1.4) are the momentum components in the axial (x) and radial (r) directions, respectively, while Eqn. (B.1.5) is the angular momentum equation about an axis normal to the xr -plane. There, $u_{p,0}$ and w_p are the axial and radial displacements of the centroid of a pipe wall element, respectively, and $J(t)$ is the creep function describing the time dependent behaviour of the viscoelastic material. Furthermore, α is the rotational angle about an axis that crosses the pipe wall element centroid and is normal to the xr -plane, h is the pipe wall thickness, ν_p is the Poisson's ratio of the pipe wall material, and $S(t)$ corresponds to the shear forces per unit length developing in the pipe wall, given by:

$$S(t) = \frac{Kh}{2J(t)(1+\nu_p)} \left[\overbrace{\frac{\partial w_p(t)}{\partial x}}^{\text{Deflection}} + \alpha(t) \right] \quad (\text{B.1.6})$$

In Eq. (B.1.6) K is the correction factor to account for the non-uniform distribution of the shear stresses across the pipe wall thickness and is taken almost equal to unity (8/9).

An advantage of the simple model expression is that the physical mechanisms that participate in the wave propagation in the pipe wall are easily identifiable. For instance, the left-hand side in Eq. (B.1.4) corresponds to inertial forces in the radial direction, while on the right-hand side are the contributions from the shear force, the hoop stresses due to radial and axial displacements, as well as the pressure originating from the contained fluid. Note that axial displacements result in hoop stresses due to the Poisson effect, as revealed by the leading

Poisson ratio parameter, ν_p . The physical description for each of the terms appearing in Eqns. (B.1.3) to (B.1.5) is given in overbraces for completeness.

For an inviscid pipe material, $J = J_0 = 1/E$ and the mechanical properties are independent of time. In viscoelastic terms, however, J_0 is solely the instantaneous creep-compliance that represents the immediate response of the material. The complete viscoelastic behaviour according to the Generalized Kelvin-Voigt (GKV) model is described by:

$$J = J(t) = J_0 + \sum_{j=1}^{N_{kv}} J_j \left[1 - \exp\left(\frac{-t}{\tau_j}\right) \right] \quad (\text{B.1.7})$$

The outmost left term represents the time-dependent viscous material behaviour and is added to the elastic part J_0 . Moreover, in Eq. (B.1.7) N_{kv} is the number of employed GKV elements, each defined by its creep coefficient J_j and relaxation time τ_j . In the equations of motion of the pipe wall appears the inverse of the creep function $1/J(t)$, that may be written as:

$$\frac{1}{J(t)} = \frac{1}{J_0 + \sum_{j=1}^{N_{kv}} J_j \left[1 - \exp\left(\frac{t}{\tau_j}\right) \right]} = \frac{1}{J_0} \frac{1}{\left\{ 1 + \sum_{j=1}^{N_{kv}} \frac{J_j}{J_0} \left[1 - \exp\left(\frac{t}{\tau_j}\right) \right] \right\}} \quad (\text{B.1.8})$$

A Taylor series expansion is applied to the denominator of the outmost left term of Eq. (B.1.8), and the linear terms are retained to obtain:

$$\frac{1}{J(t)} \approx \frac{1}{J_0} \left\{ 1 - \sum_{j=1}^{N_{kv}} \frac{J_j}{J_0} \left[1 - \exp\left(\frac{-t}{\tau_j}\right) \right] \right\} \quad (\text{B.1.9})$$

Similarly, a Taylor series expansion for the exponential function in Eq. (B.1.9) yields:

$$\exp\left(\frac{-t}{\tau_j}\right) \approx 1 - \frac{t}{\tau_j} \quad (\text{B.1.10})$$

Employing the outcomes of Eqns. (B.1.9) and (B.1.10) to Eq. (B.1.8) gives:

$$\frac{1}{J(t)} = \frac{1}{J_0} - \sum_{j=1}^{N_{kv}} \frac{J_j}{J_0^2} \left(\frac{t}{\tau_j} \right) \quad (\text{B.1.11})$$

The pipe wall behaves elastically when the second term in the right-hand side of Eq. (B.1.11) that represents the viscous behaviour is insignificant compared to the first i.e.:

$$\frac{1}{J_0} \gg \sum_{j=1}^{N_{kv}} \frac{J_j}{J_0^2} \left(\frac{t}{\tau_j} \right), \text{ or } t \ll \frac{J_0}{\sum_{j=1}^{N_{kv}} \frac{J_j}{\tau_j}} = t_v \quad (\text{B.1.12})$$

Equation (B.1.12) denotes that the pipe wall may be assumed elastic if the elapsed time t from the onset of the disturbance is much smaller than the viscous time scale of the pipe t_v . Put differently, it is expected that the pipe viscous effects become apparent after time t_v following the pulse injection to the system. For the viscoelastic parameters J_j and τ_j , the calibrated values from Wang, et al. [115] are adopted, where the authors performed experiments on an identical HDPE pipe at the Water Resources Research Laboratory of HKUST. Therein, the viscous properties of the pipe were successfully modelled with three GKV elements, whose parameters are presented in Table B.1.1. The viscous time scale t_v is then calculated from Eq. (B.1.12) using the values of Table B.1.1:

$$t_v = \frac{J_0}{\sum_{j=1}^{N_{kv}} \frac{J_j}{\tau_j}} = 0.13 \text{ s} \quad (\text{B.1.13})$$

As a sidenote, it may be shown that the contribution of the 2nd and 3rd GKV elements to the viscous time scale is insignificant compared to that of the 1st element. That is because the quotient of J_j/τ_j for $j = 2, 3$ is 1 and 3 orders of magnitude smaller, respectively, than that for $j = 1$. In essence, the calculations could be simplified by assuming a single GKV element ($j = 1$) without introducing significant error.

Table B.1.1 HDPE pipe viscoelastic parameters for the GKV model from [115].

$J_1 = 1.957 \times 10^{-10} \text{ Pa}^{-1}$	$\tau_1 = 0.038 \text{ s}$
$J_2 = 1.0954 \times 10^{-10} \text{ Pa}^{-1}$	$\tau_2 = 0.6 \text{ s}$
$J_3 = 9.05 \times 10^{-12} \text{ Pa}^{-1}$	$\tau_3 = 1.5 \text{ s}$

Applying the time reversal transformation to the Euler equations (Eqns. (B.1.1) and (B.1.2)) gives:

$$\frac{D\rho_0(-t)}{D(-t)} + \rho_0(-t) \nabla \cdot \frac{d\mathbf{r}(-t)}{d(-t)} = 0 \quad (\text{B.1.14})$$

$$\rho_f(-t) \frac{d^2 \mathbf{r}(-t)}{d(-t)^2} + \rho_f(-t) \frac{d\mathbf{r}(-t)}{d(-t)} \nabla \cdot \frac{d\mathbf{r}(-t)}{d(-t)} + \nabla p(-t) + \rho_f(-t) \mathbf{g} = \delta(-t) \quad (\text{B.1.15})$$

Both terms on the left-hand side of Eq. (B.1.14) change sign since $d/d(-t) = -d/dt$. In Eq. (B.1.15) the second order time derivative at the outmost left term is invariant to the transformation $t \rightarrow -t$ since $d^2/d(-t)^2 = d^2/dt^2$. Moreover, for the second term of Eq. (B.1.15) the minus signs originating from $d/d(-t) = -d/dt$ cancel, proving that the nonlinear advection term is time reversal invariant. In effect, Eqns. (B.1.14) and (B.1.15) have an identical form to Eqns. (B.1.1) and (B.1.2). Therefore, if $\mathbf{r}(t)$, $\rho_f(t)$, and $p(t)$ are a solution of Eqns. (B.1.1) and (B.1.2), then $\mathbf{r}(-t)$, $\rho_f(-t)$, and $p(-t)$ are a solution of Eqns. (B.1.14) and (B.1.15). One important implication is that, for a particle positioned at $\mathbf{r}$, if $\mathbf{r}(t)$ is a solution to the fundamental equation, then so is $\mathbf{r}(-t)$. Intuitively, this signifies that a particle during time reversal retraces its forward path.

Applying the transformation $t \rightarrow -t$ to Eqns. (B.1.3), (B.1.4), (B.1.5), and (B.1.6) yields:

$$\rho_p(-t)h \frac{\partial^2 u_{p,0}(-t)}{\partial(-t)^2} = \frac{1}{J_0} \frac{h}{(1-\nu_p^2)} \left(\frac{\partial^2 u_{p,0}(-t)}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_p(-t)}{\partial x} \right) \quad (\text{B.1.16})$$

$$\begin{aligned} \rho_p(-t)h \frac{\partial^2 w_p(-t)}{\partial(-t)^2} = \\ \frac{\partial S(-t)}{\partial x} - \frac{1}{J_0} \frac{h}{(1-\nu_p^2)} \left(\frac{w_p(-t)}{R^2} + \frac{\nu_p}{R} \frac{\partial u_{p,0}(-t)}{\partial x} \right) + p_p(-t) \end{aligned} \quad (\text{B.1.17})$$

$$\rho_p(-t) \frac{h^3}{12} \frac{\partial^2 \alpha(-t)}{\partial(-t)^2} = \frac{h^3}{12J_0(1-\nu_p^2)} \frac{\partial^2 \alpha(-t)}{\partial x^2} + S(-t) \quad (\text{B.1.18})$$

$$S(-t) = \frac{Kh}{2J_0(1+\nu_p)} \left[\frac{\partial w_p(-t)}{\partial x} + \alpha(-t) \right] \quad (\text{B.1.19})$$

Again, the second order time derivatives in Eqns. (B.1.16) to (B.1.19) remain unaltered when $t \rightarrow -t$. The only external force appearing in Eqns. (B.1.16) to (B.1.19) is due to the fluid pressure (i.e., $p(-t)$). However, the Euler equations are time reversible; thus, $p(t) = p(-t)$. Consequently, the form of the pipe wall equations of motion remains identical under time reversal transformation. Since $u_{p,0}(t)$, $w_p(t)$, $\alpha(t)$, $S(t)$, and $\rho_p(t)$ are solutions of Eqns. (B.1.3), (B.1.4), (B.1.5), and (B.1.6), then $u_{p,0}(-t)$, $w_p(-t)$, $\alpha(-t)$, $S(-t)$, and $\rho_p(-t)$ are solutions of Eqns. (B.1.16), (B.1.17), (B.1.18), and (B.1.19). Ultimately, in the absence of viscous forces, both the flow and pipe wall equations are time reversible.

B.2. Scale analysis of the viscous terms in the pipe wall equations

Although the time reversal transformation is applied to complete set of the equations of axisymmetric motion for the pipe wall in B, the analysis in subsection B.2 is limited to the radial and axial components. As noted in subsection 5.2.2, only high-order radial modes are identified with the plane mode wave missing for the excited frequency bandwidth. Higher order radial modes, unlike the plane wave mode, are demonstrably unaffected by the consideration of rotatory inertia (Eq. (B.1.5)) [57, 189]. In fact, higher order modes appear in the pipe wall as a sole combination of radial and axial waves, not flexural. Consequently, Eq. (B.1.5) and terms that involve angle α are excluded from further consideration.

Regarding the viscous pipe wall equations of motion in each of the radial and axial directions, the aim is to evaluate the ratio of the terms affected by viscosity over the inertial term in orders of scale. Note that in this section a wavelength distance λ , or equivalently an elapsed time $t = \lambda/c_p$ is considered to calculate said ratios.

i. r -direction

First is the equation of motion in the radial direction. Substituting Eq. (B.1.11) into Eq. (B.1.4) and introducing the pipe viscous time scale t_v (Eq. (B.1.13)) yields:

$$\rho_p(t)h \frac{\partial^2 w_p(t)}{\partial t^2} = \frac{\partial S(t)}{\partial x} - \frac{1}{J_0} \left[1 - \frac{t}{t_v} \right] \frac{h}{(1-\nu_p^2)} \left(\frac{w_p(t)}{R^2} + \frac{\nu_p}{R} \frac{\partial u_{p,0}(t)}{\partial x} \right) + p(t) \quad (\text{B.2.1})$$

The viscous radial component of the hoop stresses is isolated, and its ratio to the radial inertial forces is written as:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{w_p(t)}{R^2}}{\rho_p(t)h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim \frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\|w_p\|}{R^2}}{\frac{\rho_p h \|w_p\|}{T^2}} \quad (\text{B.2.2})$$

where the wave period T is the time scale of the disturbance equivalently expressed as $1/f$ i.e., the inverse of the probing wave frequency f . Further simplification of Eq. (B.2.2) produces:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{w_p(t)}{R^2}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim \frac{t}{t_v J_0 f^2 R^2 \rho_p (1-\nu_p^2)} \quad (\text{B.2.3})$$

The pipe radius is in the order of $R \sim 10^{-2}$ m, while the values of the remaining mechanical parameters appearing in Eq. (B.2.3) are presented in Table 4.2.1. The signals employed in the two presented experimental TR setups in Chapter 5 are of frequencies in the order of $f \sim 10^5$ Hz. Finally, the pipe viscous time scale t_v is calculated in Eq. (B.1.13) in the order of $\sim 10^{-1}$ s, and the elapsed time $t = \lambda/c_p = 1/f$ i.e., the viscous to inertia forces ratio is calculated as the wave propagates a wavelength distance. For the above values, Eq. (B.2.3) yields:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{w_p(t)}{R^2}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim 10^{-5} \quad (\text{B.2.4})$$

Next is the term that represents the viscous hoop stresses from axial displacements due to the Poisson's effect. Taking its ratio to the inertial term on the left-hand side of Eq. (B.2.1) gives:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\nu_p}{R} \frac{\partial u_{p,0}(t)}{\partial x}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim \frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\nu_p}{R} \frac{\|u_{p,0}\|}{L}}{\frac{\rho_p h \|w_p\|}{T^2}} \quad (\text{B.2.5})$$

where L is the characteristic length scale for the system. Simplifying Eq. (B.2.5) gives:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\nu_p}{R} \frac{\partial u_{p,0}(t)}{\partial x}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim \frac{t \nu_p}{t_v J_0 f^2 L R \rho_p (1-\nu_p^2)} \frac{\|u_{p,0}\|}{\|w_p\|} \quad (\text{8.2.6})$$

A closed-form formula for the amplitude ratio of axial to radial pipe wall displacements is given in Eq. (3.2.69) as a function of the probing wave frequency and the mode-specific radial wavenumber. From the latter, it is understood that to estimate $u_{p,0}/w_p$, the characteristic equation for the coupled pipe/fluid waveguide must be solved beforehand. Such analysis for the experimentally tested coupled waveguides shows that the maximum value of axial to radial displacement amplitude ratio for the excited radial modes in the frequency bandwidth of

interest is of the order of 10^{-1} (Figure 4.3.5). Moreover, the length scale L is equal to the probing wavelength $\lambda (= c_p/f)$. Accordingly, the ratio of Eq. (8.2.6) yields:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\nu_p}{R} \frac{\partial u_{p,0}(t)}{\partial x}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim 10^{-6} \quad (\text{B.2.7})$$

Last viscous term to be compared to the radial inertia is the one attributed to the shear forces developing in the pipe wall. As already explained, the effect of flexure (and, by extension, rotation α) to the observed high-order modes is negligible. Hence, the shear force expression (Eq. (B.1.6)) is reduced to:

$$S(t) = \frac{Kh}{2J(t)(1+\nu_p)} \frac{\partial w_p(t)}{\partial x} \quad (\text{B.2.8})$$

Substituting the expression for the inverse of the creep function $1/J(t)$ from Eq. (B.1.11) into Eq. (B.2.8) gives:

$$S(t) = \frac{1}{J_0} \left[1 - \frac{t}{t_v} \right] \frac{Kh}{2(1+\nu_p)} \frac{\partial w_p(t)}{\partial x} \quad (\text{B.2.9})$$

Therefore, the ratio of the viscous shear forces (per unit length) to the radial inertia is:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{Kh}{2(1+\nu_p)} \frac{\partial^2 w_p(t)}{\partial x^2}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim \frac{\frac{1}{J_0} \frac{t}{t_v} \frac{Kh}{2(1+\nu_p)} \frac{\|w_p\|}{L^2}}{\frac{\rho_p h \|w_p\|}{T^2}} \quad (\text{B.2.10})$$

After simplifications, Eq. (B.2.10) becomes:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{Kh}{2(1+\nu_p)} \frac{\partial^2 w_p(t)}{\partial x^2}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim \frac{t}{t_v J_0 f^2 L^2 \rho_p 2(1+\nu_p)} \quad (\text{B.2.11})$$

Evaluated for the experimental setup values, the ratio of Eq. (B.2.11) is placed in the order of:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{Kh}{2(1+\nu_p)} \frac{\partial^2 w_p(t)}{\partial x^2}}{\rho_p(t) h \frac{\partial^2 w_p(t)}{\partial t^2}} \sim 10^{-5} \quad (\text{B.2.12})$$

Equation (B.2.12) concludes the scale analysis for the radial momentum component.

ii. x-direction

A comprehensive analysis must also consider the axial momentum equation (Eq. (B.1.3)). Again, replacing the approximate expression for the inverse of the creep function $1/J(t)$ from Eq. (B.1.11) into Eq. (B.1.3) while considering the pipe viscous time scale t_v (Eq. (B.1.13)) leads to:

$$\rho_p(t) h \frac{\partial^2 u_{p,0}(t)}{\partial t^2} = \frac{1}{J_0} \left[1 - \frac{t}{t_v} \right] \frac{h}{(1-\nu_p^2)} \left(\frac{\partial^2 u_{p,0}(t)}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_p(t)}{\partial x} \right) \quad (\text{B.2.13})$$

We proceed to assess the ratio of the viscous axial stresses to the pertinent inertial term, thus:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\partial^2 u_{p,0}(t)}{\partial x^2}}{\rho_p(t) h \frac{\partial^2 u_{p,0}(t)}{\partial t^2}} \sim \frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\|u_{p,0}\|}{L^2}}{\frac{\rho_p h \|u_{p,0}\|}{T^2}} \quad (\text{B.2.14})$$

Following simplifications, Eq. (B.2.14) becomes:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\partial^2 u_{p,0}(t)}{\partial x^2}}{\rho_p(t) h \frac{\partial^2 u_{p,0}(t)}{\partial t^2}} \sim \frac{t}{t_v J_0 f L^2 \rho_p (1-\nu_p^2)} \quad (\text{B.2.15})$$

For the experimental system values, as discussed for the case of the radial momentum equation terms, we calculate:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\partial^2 u_{p,0}(t)}{\partial x^2}}{\rho_p(t) h \frac{\partial^2 u_{p,0}(t)}{\partial t^2}} \sim 10^{-5} \quad (\text{B.2.16})$$

Finally, the ratio of the axial stresses due to the Poisson effect to the axial inertia is:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\nu_p}{R} \frac{\partial w_p(t)}{\partial x}}{\rho_p(t) h \frac{\partial^2 u_{p,0}(t)}{\partial t^2}} \sim \frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\nu_p}{R} \frac{\|w_p\|}{L}}{\frac{\rho_p h \|u_{p,0}\|}{T^2}} \quad (\text{B.2.17})$$

After performing the manipulations in Eq. (B.2.17):

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\nu_p}{R} \frac{\partial w_p(t)}{\partial x}}{\rho_p(t) h \frac{\partial^2 u_{p,0}(t)}{\partial t^2}} \sim \frac{t \nu_p}{t_v J_0 f^2 R L \rho_p (1-\nu_p^2)} \frac{\|w_p\|}{\|u_{p,0}\|} \quad (\text{B.2.18})$$

Replacing the pertinent system values in Eq. (B.2.18) returns:

$$\frac{\frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \frac{\nu_p}{R} \frac{\partial w_p(t)}{\partial x}}{\rho_p(t) h \frac{\partial^2 u_{p,0}(t)}{\partial t^2}} \sim 10^{-7} \quad (\text{B.2.19})$$

Therefore, the most (equally) unfavourable scenarios are retrieved from Eq. (B.2.4) and Eq. (B.2.16). The former (that is discussed in Chapter 5) corresponds to the ratio of the viscous radial component of the hoop stresses over the radial inertial forces, while the latter to the ratio of viscous axial stresses to the axial inertial forces.

B.3. Pipe wall energy equation

The temporal acceleration terms on the left-hand side of the radial and axial momentum component equations (Eqns. (B.1.4) & (B.1.3)) could be equivalently expressed in terms of particle velocities since:

$$v_{p,r}(x,t) = \frac{\partial w_p(x,t)}{\partial t}, \text{ and } v_{p,x}(x,t) = \frac{\partial u_{p,0}(x,t)}{\partial t} \quad (\text{B.3.1})$$

Thus, the radial momentum equation for the pipe wall (Eq. (B.2.1)) is rewritten as:

$$\begin{aligned}
\rho_p h \frac{\partial v_{p,r}(x,t)}{\partial t} = & \frac{h}{J_0(1+\nu_p)} \left\{ \frac{K}{2} \frac{\partial^2 w_p(x,t)}{\partial x^2} - \frac{1}{(1-\nu_p)} \left[\frac{w_p(x,t)}{R^2} + \frac{\nu_p}{R} \frac{\partial u_{p,0}(x,t)}{\partial x} \right] \right\} \\
& - \frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1+\nu_p)} \left\{ \frac{K}{2} \frac{\partial^2 w_p(x,t)}{\partial x^2} - \frac{1}{(1-\nu_p)} \left[\frac{w_p(x,t)}{R^2} + \frac{\nu_p}{R} \frac{\partial u_{p,0}(x,t)}{\partial x} \right] \right\} \\
& + p(x,t)
\end{aligned} \tag{B.3.2}$$

while the axial momentum equation (Eq. (B.2.13)) becomes:

$$\begin{aligned}
\rho_p h \frac{\partial v_{p,x}(x,t)}{\partial t} = & \frac{h}{J_0(1-\nu_p^2)} \left[\frac{\partial^2 u_{p,0}(x,t)}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_p(x,t)}{\partial x} \right] \\
& - \frac{1}{J_0} \frac{t}{t_v} \frac{h}{(1-\nu_p^2)} \left[\frac{\partial^2 u_{p,0}(x,t)}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_p(x,t)}{\partial x} \right]
\end{aligned} \tag{B.3.3}$$

Equations (B.3.2) and (B.3.3) are added together to produce the mechanical energy rate:

$$\begin{aligned}
\frac{\partial}{\partial t} \left\{ \frac{\rho_p}{2} [v_{p,x}^2(x,t) + v_{p,r}^2(x,t)] \right\} = & \frac{1}{J_0(1-\nu_p^2)} \left\{ v_{p,x}(x,t) \left[\frac{\partial^2 u_{p,0}(x,t)}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_p(x,t)}{\partial x} \right] \right. \\
& \left. + v_{p,r}(x,t) \left[\frac{K(1-\nu_p)}{2} \frac{\partial^2 w_p(x,t)}{\partial x^2} - \frac{w_p(x,t)}{R^2} - \frac{\nu_p}{R} \frac{\partial u_{p,0}(x,t)}{\partial x} \right] \right\} \\
& - \frac{1}{J_0} \frac{t}{t_v} \frac{1}{(1-\nu_p^2)} \left\{ v_{p,x}(x,t) \left[\frac{\partial^2 u_{p,0}(x,t)}{\partial x^2} + \frac{\nu_p}{R} \frac{\partial w_p(x,t)}{\partial x} \right] \right. \\
& \left. + v_{p,r}(x,t) \left[\frac{K(1-\nu_p)}{2} \frac{\partial^2 w_p(x,t)}{\partial x^2} - \frac{w_p(x,t)}{R^2} - \frac{\nu_p}{R} \frac{\partial u_{p,0}(x,t)}{\partial x} \right] \right\} \\
& + v_r \frac{p(x,t)}{h}
\end{aligned} \tag{B.3.4}$$

In the frequency range explored in the experiments, radial displacements in the pipe wall are one order of magnitude larger than the pertinent axial displacements. Hence, Eq. (B.3.4) may be simplified to:

$$\begin{aligned} \frac{\partial}{\partial t} \left\{ \frac{\rho_p}{2} [v_{p,r}^2(x,t)] \right\} = \\ - \frac{v_{p,r}(x,t)}{J_0(1-v_p^2)} \frac{w(x,t)}{R^2} + \frac{1}{J_0} \frac{t}{t_v} \frac{v_{p,r}(x,t)}{(1-v_p^2)} \frac{w_p(x,t)}{R^2} + v_{p,r} \frac{p(x,t)}{h} \end{aligned} \quad (\text{B.3.5})$$

From Eq. (B.3.1), we obtain:

$$\begin{aligned} \frac{\partial}{\partial t} \left\{ \frac{\rho_p}{2} [v_{p,r}^2(x,t)] \right\} = - \frac{\partial}{\partial t} \frac{1}{R^2 J_0 (1-v_p^2)} \left(\frac{w_p^2(x,t)}{2} \right) \\ + \frac{1}{J_0} \frac{t}{t_v} \frac{1}{R^2 (1-v_p^2)} \frac{\partial}{\partial t} \left(\frac{w_p^2(x,t)}{2} \right) + v_{p,r} \frac{p(x,t)}{h} \end{aligned} \quad (\text{B.3.6})$$

The left-hand side of Eq. (B.3.4) corresponds to the kinetic energy of the system and, to simplify, we set $e_k = \rho_p v_{p,r}^2 / 2$. On the other hand, the first and second terms on the right-hand side of Eq. (B.3.4) represent the (potential) elastic strain energy and viscous strain energy, respectively. For compactness, we prescribe $e_p = w_p^2 / [2R^2 J_0 (1 - v_p^2)]$. Besides, the last term on the right-hand side denotes the work done by the fluid pressure wave. Following the aforementioned expression assignments, Eq. (B.3.4) is written as:

$$\frac{\partial}{\partial t} (e_k + e_p) = \frac{t}{t_v} \frac{\partial e_p}{\partial t} + v_r \frac{p(x,t)}{h} \quad (\text{B.3.7})$$

Applying the energy equipartition principle [190], Eq. (B.3.7) becomes:

$$\left[1 - \frac{t}{2t_v} \right] \frac{\partial e}{\partial t} = v_r \frac{p(x,t)}{h} \quad (\text{B.3.8})$$

where $e = e_p + e_k$. Letting $E = eh$, Eq. (B.3.8) becomes:

$$\frac{\partial E}{\partial t} = p(x,t) \left[\frac{v_{p,r}}{1 - \frac{t}{2t_v}} \right] \cong p(x,t) v_{p,r} + p(x,t) v_{p,r} \frac{t}{2t_v} \quad (\text{B.3.9})$$

Note that the right-hand side of Eq. (B.3.9) represents the work of the pressure force on the pipe wall.

The energy dissipation rate in the fluid due to viscosity may be written as [191]:

$$E_a(t) \sim E_{a,0} e^{-\frac{v_t}{\lambda^2} t} \quad (\text{B.3.10})$$

and is proportional to pressure p_p squared. Hence, we may write:

$$p(x, t) \sim p_0(t) e^{-\frac{vx}{2\lambda^2 c_f}} \quad (\text{B.3.11})$$

and the amplitude of the pressure wave decays over distance from the source due to the viscosity of the fluid. Combining Eqns. (B.3.9) and (B.3.11) allows for the evaluation of the combined effect of the viscoelastic pipe wall and the viscous fluid on the energy decay in the system:

$$\frac{\partial E_a}{\partial t} = \frac{v_{p,r}}{h} \frac{p_0(t) e^{-\frac{vx}{2\lambda^2 c_f}}}{\left[2 - \left(\frac{x}{t_v c_p} \right) \right]} \quad (\text{B.3.12})$$

The mechanical energy equation of the pipe wall Eq. (B.3.12) sheds some light on the energy decay in the coupled fluid/pipe system. Specifically, the denominator on the right-hand side of Eq. (B.3.12) corresponds to the strain energy in the pipe wall and increases with range x . The numerator, on the other hand, represents the energy in the fluid that decays with range x . Overall, Eq. (B.3.12) represents the transfer of energy from the fluid to the pipe wall as range x increases.

Appendix C

Gaussian modulated sine pulse

C.1. Mathematical expression

The employed acoustic signals in the presented experimental tests are Gaussian-modulated sinusoidal pulses that, although not efficient as carriers of wave energy compared to e.g., linear frequency-modulated signals, allow for the precise definition of the excited frequency bandwidth and, by extension, provide the ability to the experimenter to preselect the excited modes. The mathematical expression of a Gaussian pulse in the time domain is given by:

$$f(t) = A_G e^{-\frac{(t-\delta)^2}{2\sigma^2}} \sin[\omega_c(t-\delta)] \quad (\text{C.1.1})$$

where $\omega_c = 2\pi f_c$ is the angular central frequency (with f_c being the normal central frequency), A_G is the pulse amplitude, σ is the standard deviation (that is related to the frequency bandwidth), and δ corresponds to the duration of the pulse (in seconds). Time duration δ is specified equal to the full width of the waveform at tenth of maximum (FWTM):

$$\delta = \text{FWTM} = 2\sqrt{2\ln(10)}\sigma \quad (\text{C.1.2})$$

Determining the bandwidth of the pulse requires Eq. (C.1.1) to be expressed in the frequency domain; hence, the Fourier transform operation is applied to yield:

$$\hat{f}(\omega) = A_G \frac{\sigma\sqrt{2\pi}}{2i} e^{-i\omega\delta} \left[e^{-\frac{\sigma^2(\omega-\omega_c)^2}{2}} - e^{-\frac{\sigma^2(\omega+\omega_c)^2}{2}} \right] \quad (\text{C.1.3})$$

We specify the excited frequency bandwidth relative to the central frequency as $\text{FBW} = b\omega_c$, evaluated at $1/\gamma_{att}$ of the spectrum amplitude peak $\hat{f}_{\max}$ (that occurs at $\omega = \omega_c$), with $b, \gamma_{att} > 0$. Assessing Eq. (C.1.3) at $\omega_c + \text{FBW}/2$ (where the amplitude is equal to $\hat{f}_{\max}/\gamma$) returns:

$$\sigma = \frac{2\sqrt{2\ln(\gamma_{att})}}{\text{FBW}} \quad (\text{C.1.4})$$

Combining Eqns. (C.1.2) and (C.1.4) produces the expression for the pulse duration δ as a function of the selected frequency bandwidth FBW and attenuation factor γ_{att} :

$$\delta = \frac{8\sqrt{\ln(10)\ln(\gamma_{att})}}{\text{FBW}} \quad (\text{C.1.5})$$

It is apparent from Eq. (C.1.4) that the pulse duration is inversely proportional to the selected frequency bandwidth. As such, to excite a wide band of frequencies, a short (or “sharp”) pulse is required. In the experimental work presented herein, an attenuation factor $\gamma_{att} = 100$ (that corresponds to a 40 dB attenuation) is always used; thus, Eq. (C.1.4) becomes:

$$\delta = \frac{8\sqrt{2}\ln(10)}{\text{FBW}} \quad (\text{C.1.6})$$

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